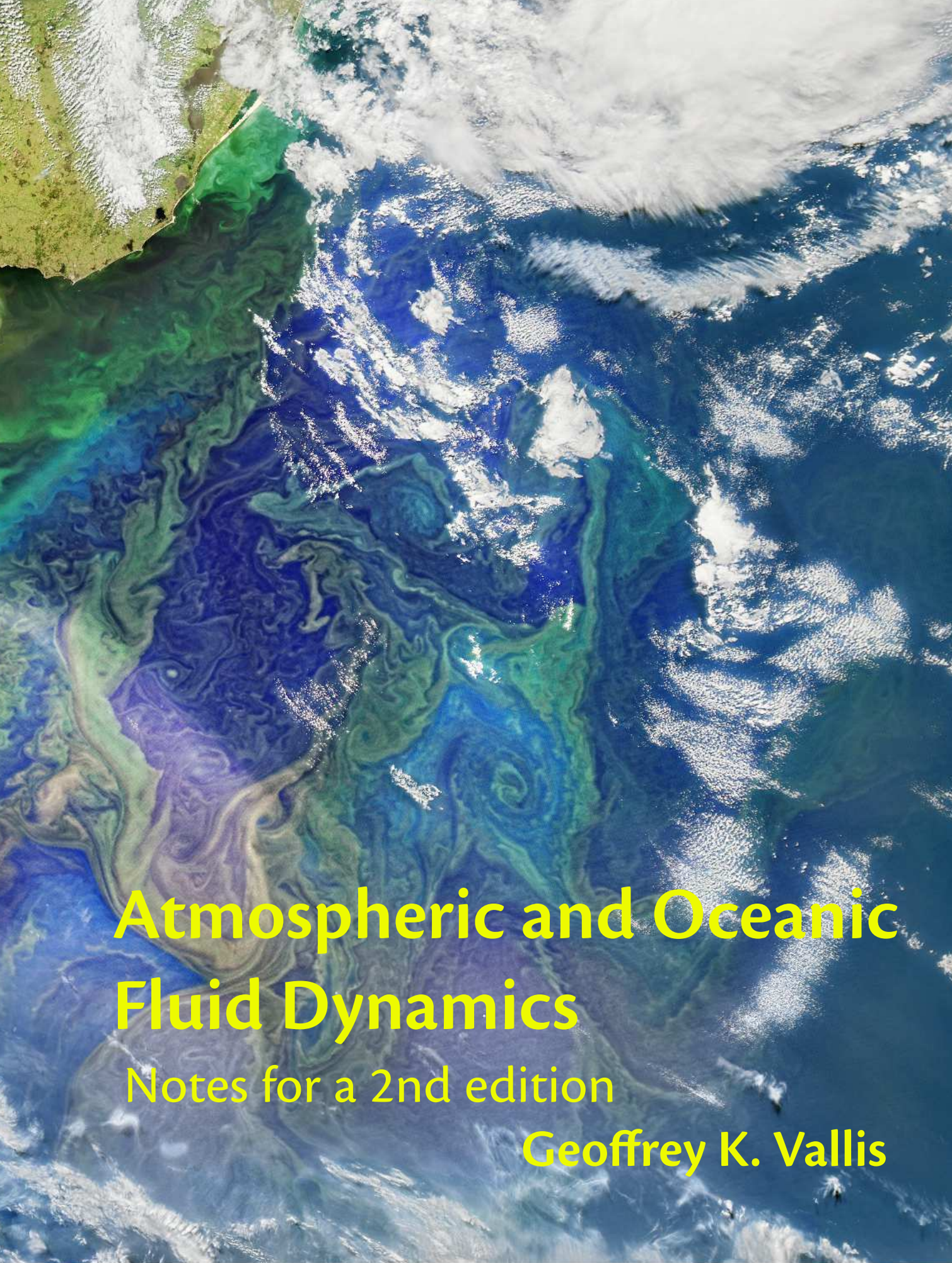




Atmospheric and Oceanic Fluid Dynamics

Notes for a 2nd edition

Geoffrey K. Vallis

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In the main text, sections that are more advanced or that contain material that is peripheral to the main narrative are marked, like ski slopes, with a black diamond, $\blacklozenge$. Sections that contain material that is still not settled or are at the edge of knowledge are marked with a dagger, $\dagger$.

Look again at that dot. That's here. That's home. That's us... Our planet is a lonely speck in the great enveloping cosmic dark. In our obscurity, in all this vastness, there is no hint that help will come from elsewhere to save us from ourselves... Like it or not, for the moment the Earth is where we make our stand.
Carl Sagan, *Pale Blue Dot: A Vision of the Human Future in Space*, 1994.

Preface

June 27, 2016.

The preface is yet to be written,

Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

Part II

WAVES, INSTABILITIES AND TURBULENCE

*Below, a myriad, myriad waves hastening, lifting up their necks,
Tending in ceaseless flow toward the track of the ship,
Waves of the ocean bubbling and gurgling, blithely prying,
Waves, undulating waves, liquid, uneven, emulous waves,
Toward that whirling current, laughing and buoyant.*
Walt Whitman, *After the Sea-Ship*, in *Leaves of Grass*, 1900.

CHAPTER 6

Wave Fundamentals

WAVES ARE EVERYWHERE: on the beach, on piano wires, in football stadiums, and filling the space between the distant stars and Earth. This chapter provides an introduction to their properties, paying particular attention to a wave that is especially important to the large scale flow in both ocean and atmosphere — the Rossby wave. We start with an elementary introduction to wave kinematics, discussing such basic concepts as phase speed and group velocity. Then beginning with section 6.4, we discuss the dynamics of Rossby waves, and this part may be considered to be the natural follow-on from the geostrophic theory of the previous chapter. Finally, in section 6.7, we return to group velocity in a more general way and illustrate the results using Poincaré waves, with applications to gravity and Rossby waves in later chapters.

The reason for such an ordering of topics is that wave kinematics without a dynamical example is jejune and dry, yet understanding wave dynamics of any sort is hardly possible without appreciating at least some of its formal structure, and readers should flip pages back and forth through the chapter as needed. Those readers who wish to cut to the chase may skip the first few sections and begin at section 6.4, referring back as needed. Other readers may wish to skip the sections on Rossby waves altogether and, after absorbing the sections on the wave theory move on to chapter 7 on gravity waves, returning to Rossby waves (or not) later on. The Rossby wave and gravity wave discussions are largely independent of each other, although they both require that the reader is familiar with the such ideas as group velocity and phase speed. Rossby waves and gravity waves can, of course, co-exist, and close to the equator the two kinds of waves become more intertwined and we deal with the ensuing waves in chapter 8. We also extend our discussion of Rossby waves in a global atmospheric context in chapter 16.

6.1 FUNDAMENTALS AND FORMALITIES

6.1.1 Definitions and kinematics

A wave may be more easily recognized than defined. Loosely speaking, as wave is a propagating disturbance that has a characteristic relationship between its frequency and size, and a linear wave may be defined as a disturbance that satisfies a *dispersion relation*. (Nonlinear waves exist, but the curious reader must look elsewhere to learn about them.¹) In order to see what all this means, and what a dispersion relation is, suppose that a disturbance, $\psi(\mathbf{x}, t)$ (where ψ might be velocity,

streamfunction, pressure, etc.), satisfies the equation

$$L(\psi) = 0, \quad (6.1)$$

where L is a linear operator, typically a polynomial in time and space derivatives; one example is $L(\psi) = \partial \nabla^2 \psi / \partial t + \beta \partial \psi / \partial x$. If (6.1) has constant coefficients (if β is constant in this example) then harmonic solutions may often be found that are a superposition of *plane waves*, each of which satisfy

$$\psi = \text{Re } \tilde{\psi} e^{i\theta(\mathbf{x},t)} = \text{Re } \tilde{\psi} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (6.2)$$

where $\tilde{\psi}$ is a complex constant, θ is the phase, ω is the wave frequency and $\mathbf{k}$ is the vector wave-number (k, l, m) (also written as (k^x, k^y, k^z) or, in subscript notation, k_i). The prefix Re denotes the real part of the expression, but we will drop it if there is no ambiguity.

Earlier, we said that waves are characterized by having a particular relationship between the frequency and wavevector known as the *dispersion relation*. This is an equation of the form

$$\omega = \Omega(\mathbf{k}) \quad (6.3)$$

where $\Omega(\mathbf{k})$ [or $\Omega(k_i)$, and meaning $\Omega(k, l, m)$] is some function determined by the form of L in (6.1) and which thus depends on the particular type of wave — the function is different for sound waves, light waves and the Rossby waves and gravity waves we will encounter in this book (peak ahead to (6.60) and (7.57), and there is more discussion in section 6.1.3). Unless it is necessary to explicitly distinguish the function Ω from the frequency ω , we will often write $\omega = \omega(\mathbf{k})$.

If the medium in which the waves are propagating is inhomogeneous then (6.1) will probably not have constant coefficients (for example, β may vary with y). Nevertheless, if the medium is varying sufficiently slowly, wave solutions may often still be found with the general form

$$\psi(\mathbf{x}, t) = a(\mathbf{x}, t) e^{i\theta(\mathbf{x}, t)}, \quad (6.4)$$

where $a(\mathbf{x}, t)$ varies slowly compared to the variation of the phase, θ . The frequency and wavenumber are then defined by

$$\mathbf{k} \equiv \nabla \theta, \quad \omega \equiv -\frac{\partial \theta}{\partial t}. \quad (6.5)$$

The example of (6.2) is clearly just a special case of this. Equation (6.5) implies the formal relation between $\mathbf{k}$ and ω :

$$\frac{\partial \mathbf{k}}{\partial t} + \nabla \omega = 0. \quad (6.6)$$

6.1.2 Wave propagation and phase speed

A common property of waves is that they propagate through space with some velocity (which in special cases might be zero). Waves in fluids may carry energy and momentum but not normally, at least to a first approximation, fluid parcels themselves. Further, it turns out that the speed at which properties like energy are transported (the group speed) may be different from the speed at which the wave crests themselves move (the phase speed). Let's try to understand this statement, beginning with the phase speed.

Phase speed

Let us consider the propagation of monochromatic plane waves, for that is all that is needed to introduce the phase speed. Given (6.2) a wave will propagate in the direction of $\mathbf{k}$ (Fig. 6.1). At a given instant and location we can align our coordinate axis along this direction, and we write

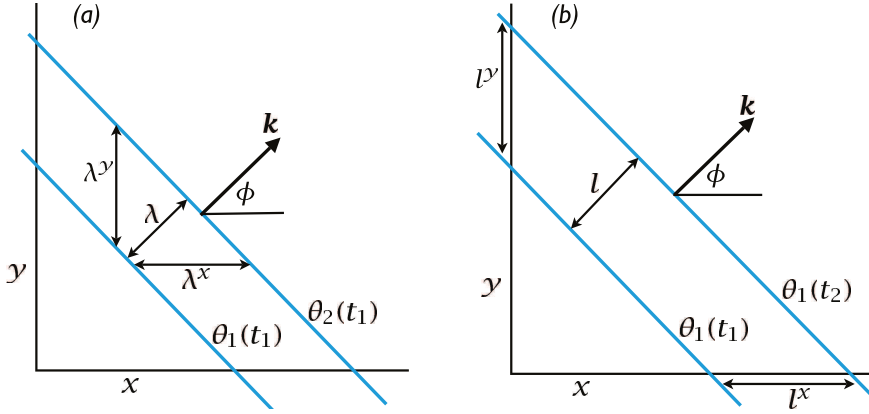


Fig. 6.1 The propagation of a two-dimensional wave. (a) Two lines of constant phase (e.g., two wavecrests) at a time t_1 . The wave is propagating in the direction $\mathbf{k}$ with wavelength λ . (b) The same line of constant phase at two successive times. The phase speed is the speed of advancement of the wavecrest in the direction of travel, and so $c_p = l/(t_2 - t_1)$. The phase speed in the x -direction is the speed of propagation of the wavecrest along the x -axis, and $c_p^x = l^x/(t_2 - t_1) = c_p / \cos \phi$.

$\mathbf{k} \cdot \mathbf{x} = Kx^*$, where x^* increases in the direction of $\mathbf{k}$ and $K^2 = |\mathbf{k}|^2$ is the magnitude of the wavenumber. With this, we can write (6.2) as

$$\psi = \text{Re } \tilde{\psi} e^{i(Kx^* - \omega t)} = \text{Re } \tilde{\psi} e^{iK(x^* - ct)}, \quad (6.7)$$

where $c = \omega/K$. From this equation it is evident that the phase of the wave propagates at the speed c in the direction of $\mathbf{k}$, and we define the *phase speed* by

$$c_p \equiv \frac{\omega}{K}. \quad (6.8)$$

The wavelength of the wave, λ , is the distance between two wavecrests — that is, the distance between two locations along the line of travel whose phase differs by 2π — and evidently this is given by

$$\lambda = \frac{2\pi}{K}. \quad (6.9)$$

In (for simplicity) a two-dimensional wave, and referring to Fig. 6.1, the wavelength and wave vectors in the x - and y -directions are given by,

$$\lambda^x = \frac{\lambda}{\cos \phi}, \quad \lambda^y = \frac{\lambda}{\sin \phi}, \quad k^x = K \cos \phi, \quad k^y = K \sin \phi. \quad (6.10)$$

In general, lines of constant phase intersect both the coordinate axes and propagate along them. The speed of propagation along these axes is given by

$$c_p^x = c_p \frac{l^x}{l} = \frac{c_p}{\cos \phi} = c_p \frac{K}{k^x} = \frac{\omega}{k^x}, \quad c_p^y = c_p \frac{l^y}{l} = \frac{c_p}{\sin \phi} = c_p \frac{K}{k^y} = \frac{\omega}{k^y}, \quad (6.11)$$

using (6.8) and (6.10), and again referring to Fig. 6.1 for notation. The speed of phase propagation along any one of the axis is in general *larger* than the phase speed in the primary direction of the wave. The phase speeds are clearly *not* components of a vector: for example, $c_p^x \neq c_p \cos \phi$. Analogously, the wavevector $\mathbf{k}$ is a true vector, whereas the wavelength λ is not.

Wave Fundamentals

- A wave is a propagating disturbance that has a characteristic relationship between its frequency and size, known as the dispersion relation. Waves typically arise as solutions to a linear problem of the form

$$L(\psi) = 0, \quad (\text{WF.1})$$

where L is, commonly, a linear operator in space and time. Two examples are

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0 \quad \text{and} \quad \frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0. \quad (\text{WF.2})$$

The first example is so common in all areas of physics it is sometimes called ‘the’ wave equation. The second example gives rise to Rossby waves.

- Solutions to the governing equation are often sought in the form of plane waves that have the form

$$\psi = \text{Re } A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (\text{WF.3})$$

where A is the wave amplitude, $\mathbf{k} = (k, l, m)$ is the wavevector, and ω is the frequency.

- The dispersion relation connects the frequency and wavevector through an equation of the form $\omega = \Omega(\mathbf{k})$ where Ω is some function. The relation is normally derived by substituting a trial solution like (WF.3) into the governing equation (WF.1). For the examples of (WF.2) we obtain $\omega = c^2 K^2$ and $\omega = -\beta k / K^2$ where $K^2 = k^2 + l^2 + m^2$ or, in two dimensions, $K^2 = k^2 + l^2$.
- The phase speed is the speed at which the wave crests move. In the direction of propagation and in the x , y and z directions the phase speeds are given by, respectively,

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k}, \quad c_p^y = \frac{\omega}{l}, \quad c_p^z = \frac{\omega}{m}. \quad (\text{WF.4})$$

where $K = 2\pi/\lambda$ where λ is the wavelength. The wave crests have both a speed (c_p) and a direction of propagation (the direction of $\mathbf{k}$), like a vector, but the components defined in (WF.4) are not the components of that vector.

- The group velocity is the velocity at which a wave packet or wave group moves. It is a vector and is given by

$$\mathbf{c}_g = \frac{\partial \omega}{\partial \mathbf{k}} \quad \text{with components} \quad c_g^x = \frac{\partial \omega}{\partial k}, \quad c_g^y = \frac{\partial \omega}{\partial l}, \quad c_g^z = \frac{\partial \omega}{\partial m}. \quad (\text{WF.5})$$

Many physical quantities of interest are transported at the group velocity.

- If the coefficients of the wave equation are not constant (for example if the medium is inhomogeneous) then, if the coefficients are only slowly varying, approximate solutions may sometimes be found in the form

$$\psi = \text{Re } A(\mathbf{x}, t) e^{i\theta(\mathbf{x}, t)}, \quad (\text{WF.6})$$

where the amplitude A is also slowly varying and the local wavenumber and frequency are related to the phase, θ , by $\mathbf{k} = \nabla \theta$ and $\omega = -\partial \theta / \partial t$. The dispersion relation is then a *local* one of the form $\omega = \Omega(\mathbf{k}; \mathbf{x}, t)$.

To summarize, the phase speed and its components are given by

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k^x}, \quad c_p^y = \frac{\omega}{k^y}. \quad (6.12)$$

Phase velocity

Although it is not particularly useful, there is a way of defining a phase speed so that is a true vector, and which might then be called phase velocity. We define the phase velocity to be the velocity that has the magnitude of the phase speed and direction in which wave crests are propagating; that is

$$\mathbf{c}_p \equiv \frac{\omega}{K} \frac{\mathbf{k}}{|\mathbf{k}|} = c_p \frac{\mathbf{k}}{|\mathbf{k}|}, \quad (6.13)$$

where $\mathbf{k}/|\mathbf{k}|$ is the unit vector in the direction of wave-crest propagation. The components of the phase velocity in the x - and y -directions are then given by

$$c_p^x = c_p \cos \phi, \quad c_p^y = c_p \sin \phi. \quad (6.14)$$

Defined this way, the quantity given by (6.13) is a vector velocity. However, the components in the x - and y -directions are manifestly not the speed at which wave crests propagate in those directions, and it is a misnomer to call these quantities phase speeds. Still, it is helpful to ascribe a direction to the phase speed the quantity given by (6.13) can be useful.

6.1.3 The dispersion relation

The above description is kinematic, in that it applies to almost any disturbance that has a wavevector and a frequency. The particular *dynamics* of a wave are determined by the relationship between the wavevector and the frequency; that is, by the *dispersion relation*. Once the dispersion relation is known a great many of the properties of the wave follow in a more-or-less straightforward manner. Picking up from (6.3), the dispersion relation is a functional relationship between the frequency and the wavevector of the general form

$$\omega = \Omega(\mathbf{k}). \quad (6.15)$$

Perhaps the simplest example of a linear operator that gives rise to waves is the one-dimensional equation

$$\frac{\partial \psi}{\partial t} + c \frac{\partial \psi}{\partial x} = 0. \quad (6.16)$$

Substituting a trial solution of the form $\psi = \text{Re } A e^{i(kx - \omega t)}$, where Re denotes the real part, we obtain $(-i\omega + cik)A = 0$, giving the dispersion relation

$$\omega = ck. \quad (6.17)$$

The phase speed of this wave is $c_p = \omega/k = c$. A few other examples of governing equations, dispersion relations and phase speeds are:

$$\frac{\partial \psi}{\partial t} + \mathbf{c} \cdot \nabla \psi = 0, \quad \omega = \mathbf{c} \cdot \mathbf{k}, \quad c_p = |\mathbf{c}| \cos \theta, \quad c_p^x = \frac{\mathbf{c} \cdot \mathbf{k}}{k}, \quad c_p^y = \frac{\mathbf{c} \cdot \mathbf{k}}{l}, \quad (6.18a)$$

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0, \quad \omega^2 = c^2 K^2, \quad c_p = \pm c, \quad c_p^x = \pm \frac{cK}{k}, \quad c_p^y = \pm \frac{cK}{l}, \quad (6.18b)$$

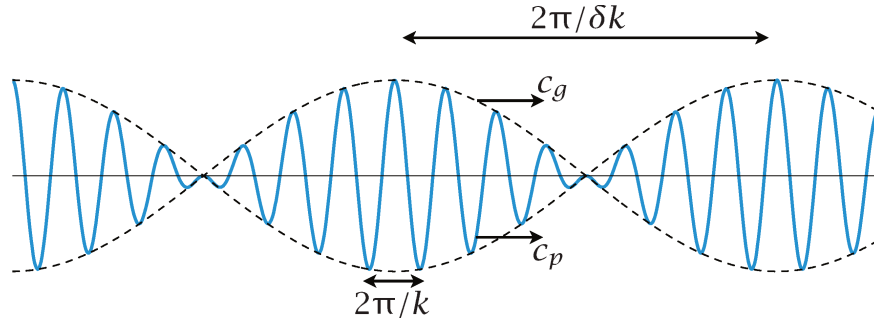


Fig. 6.2 Superposition of two sinusoidal waves with wavenumbers k and $k + \delta k$, producing a wave (solid line) that is modulated by a slowly varying wave envelope or wave packet (dashed line). The envelope moves at the group velocity, $c_g = \partial\omega/\partial k$ and the phase of the wave moves at the phase speed $c_p = \omega/k$.

$$\frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0, \quad \omega = \frac{-\beta k}{K^2}, \quad c_p = \frac{\omega}{K}, \quad c_p^x = -\frac{\beta}{K^2}, \quad c_p^y = -\frac{\beta k/l}{K^2}, \quad (6.18c)$$

where $K^2 = k^2 + l^2$ and θ is the angle between $\mathbf{c}$ and $\mathbf{k}$, and the examples are all two-dimensional, with variation in x and y only.

A wave is said to be *nondispersive* if the phase speed is independent of the wavelength. This condition is satisfied for the simple example (6.16) but is manifestly not satisfied for (6.18c), and these waves (Rossby waves, in fact) are *dispersive*. Waves of different wavelengths then travel at different speeds so that a group of waves will spread out — disperse — even if the medium is homogeneous. When a wave is dispersive there is another characteristic speed at which the waves propagate, the group velocity, and we come to this shortly.

Most media are inhomogeneous, but if the medium varies sufficiently slowly in space and time — and in particular if the variations are slow compared to the wavelength and period — we may still have a *local* dispersion relation between frequency and wavevector,

$$\omega = \Omega(\mathbf{k}; \mathbf{x}, t). \quad (6.19)$$

where $\mathbf{x}$ and t are slowly varying parameters. We'll resume our discussion of this topic in section 6.3, but before that we must introduce the group velocity.

6.2 GROUP VELOCITY

Information and energy do not, in general, propagate at the phase speed. Rather, most quantities of interest propagate at the *group velocity*, a quantity of enormous importance in wave theory.² Roughly speaking, group velocity is the velocity at which a packet or a group of waves will travel, whereas the individual wave crests travel at the phase speed. To introduce the idea we will consider the superposition of plane waves, noting that a truly monochromatic plane wave already fills all space uniformly so that there can be no propagation of energy from place to place.

6.2.1 Superposition of two waves

Consider the linear superposition of two waves. Limiting attention to the one-dimensional case for simplicity, consider a disturbance represented by

$$\psi = \text{Re } \tilde{\psi}(e^{i(k_1 x - \omega_1 t)} + e^{i(k_2 x - \omega_2 t)}). \quad (6.20)$$

Let us further suppose that the two waves have similar wavenumbers and frequency, and, in particular, that $k_1 = k + \Delta k$ and $k_2 = k - \Delta k$, and $\omega_1 = \omega + \Delta\omega$ and $\omega_2 = \omega - \Delta\omega$. With this, (6.20) becomes

$$\begin{aligned}\psi &= \operatorname{Re} \tilde{\psi} e^{i(kx - \omega t)} [e^{i(\Delta k x - \Delta\omega t)} + e^{-i(\Delta k x - \Delta\omega t)}] \\ &= 2 \operatorname{Re} \tilde{\psi} e^{i(kx - \omega t)} \cos(\Delta k x - \Delta\omega t).\end{aligned}\quad (6.21)$$

The resulting disturbance, illustrated in Fig. 6.2 has two aspects: a rapidly varying component, with wavenumber k and frequency ω , and a more slowly varying envelope, with wavenumber Δk and frequency $\Delta\omega$. The envelope modulates the fast oscillation, and moves with velocity $\Delta\omega/\Delta k$; in the limit $\Delta k \rightarrow 0$ and $\Delta\omega \rightarrow 0$ this is the *group velocity*, $c_g = \partial\omega/\partial k$. Group velocity is equal to the phase speed, ω/k , only when the frequency is a linear function of wavenumber. The energy in the disturbance must move at the group velocity — note that the node of the envelope moves at the speed of the envelope and no energy can cross the node. These concepts generalize to more than one dimension, and if the wavenumber is the three-dimensional vector $\mathbf{k} = (k, l, m)$ then the three-dimensional envelope propagates at the group velocity given by

$$\mathbf{c}_g = \frac{\partial\omega}{\partial\mathbf{k}} \equiv \left(\frac{\partial\omega}{\partial k}, \frac{\partial\omega}{\partial l}, \frac{\partial\omega}{\partial m} \right). \quad (6.22)$$

The group velocity is also written as $\mathbf{c}_g = \nabla_{\mathbf{k}}\omega$ or, in subscript notation, $c_{gi} = \partial\omega/\partial k_i$, with the subscript i denoting the component of a vector.

6.2.2 Superposition of many waves

Now consider a slight extension of the above arguments to the case in which many waves are excited. The essential assumption of this derivation is that the wavenumber distribution is sufficiently narrow so that the dispersion relation can be approximated as

$$\omega(k) \approx \omega(k_0) + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0). \quad (6.23)$$

We treat only the one-dimensional case but the three dimensional generalization is possible.

A superposition of plane waves, each satisfying some dispersion relation, can be represented by the Fourier integral

$$\psi(x, t) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i(kx - \omega t)} dk. \quad (6.24a)$$

The function $\tilde{A}(k)$ is given by the initial conditions:

$$\tilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(x, 0) e^{-ikx} dx. \quad (6.24b)$$

Note that if the waves are dispersionless and $\omega = ck$ where c is a constant, then

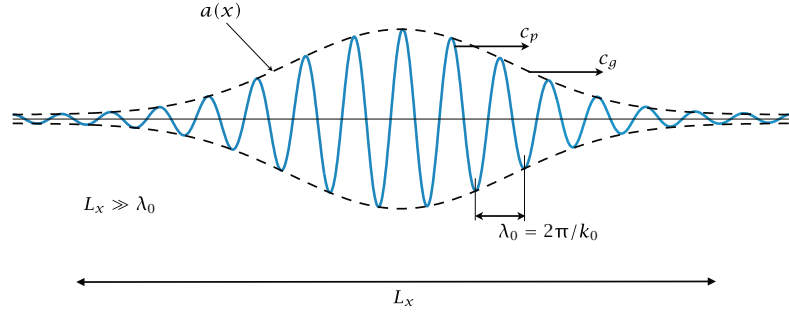
$$\psi(x, t) = \int_{-\infty}^{+\infty} \tilde{A}(k) e^{ik(x - ct)} dk = \psi(x - ct, 0), \quad (6.25)$$

by comparison with (6.24a) at $t = 0$. That is, the initial condition simply translates at a speed c , with no change in structure.

Now consider the case for which the disturbance is a *wave packet* — essentially a nearly plane wave or superposition of waves but confined to a finite region of space. We will consider a case with the initial condition

$$\psi(x, 0) = a(x, 0) e^{ik_0 x} \quad (6.26)$$

Fig. 6.3 A wave packet. The envelope, $a(x)$, has a scale, L_x , much larger than the wavelength, λ_0 , of the wave embedded inside. The envelope moves at the group velocity, c_g , and the phase of the waves at the phase speed, c_p .



where $a(x, t)$, rather like the envelope in Fig. 6.3, modulates the amplitude of the wave on a scale much longer than that of the wavelength $2\pi/k_0$, and more slowly than the wave period. That is,

$$\frac{1}{a} \frac{\partial a}{\partial x} \ll k_0, \quad \frac{1}{a} \frac{\partial a}{\partial t} \ll k_0 c, \quad (6.27a,b)$$

and the disturbance is essentially a slowly modulated plane wave. We suppose that $a(x, 0)$ is peaked around some value x_0 and is very small if $|x - x_0| \gg k_0^{-1}$; that is, $a(x, 0)$ is small if we are sufficiently many wavelengths of the plane wave away from the peak, as is the case in Fig. 6.3. We would like to know how such a packet evolves.

We can express the envelope as a Fourier integral by first writing the initial conditions as a Fourier integral,

$$\psi(x, 0) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{ikx} dk \quad \text{where} \quad \tilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \psi(x, 0) e^{-ikx} dx, \quad (6.28a,b)$$

so that, using (6.26),

$$\tilde{A}(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} a(x, 0) e^{i(k_0 - k)x} dx \quad \text{and} \quad a(x, 0) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i(k - k_0)x} dk. \quad (6.29a,b)$$

We still haven't made much progress beyond (6.24). To do so, we note first that $a(x)$ is confined in space, so that to a good approximation the limits of the integral in (6.29a) can be made finite, $\pm L$ say, provided $L \gg k_0^{-1}$. We then note that when $(k_0 - k)$ is large the integrand in (6.29a) oscillates rapidly; successive intervals in x therefore cancel each other and make a small net contribution to the integral. Thus, the integral is dominated by values of k near k_0 , and $\tilde{A}(k)$ is peaked near k_0 . (Note that the finite spatial extent of $a(x, 0)$ is crucial for this argument.)

We can now evaluate how the wave packet evolves. Beginning with (6.24a) we have

$$\psi(x, t) = \int_{-\infty}^{\infty} \tilde{A}(k) \exp\{i(kx - \omega(k)t)\} dk \quad (6.30a)$$

$$\approx \int_{-\infty}^{\infty} \tilde{A}(k) \exp \left\{ i[k_0 x - \omega(k_0)t] + i(k - k_0)x - i(k - k_0) \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0} t \right\} dk \quad (6.30b)$$

having expanded $\omega(k)$ in a Taylor series about k_0 , using (6.23). We thus have

$$\begin{aligned} \psi(x, t) &= \exp \{i[k_0 x - \omega(k_0)t]\} \int \int_{-\infty}^{\infty} \tilde{A}(k) \exp \left\{ i(k - k_0) \left[x - \left. \frac{\partial \omega}{\partial k} \right|_{k=k_0} t \right] \right\} dk \\ &= \exp \{i[k_0 x - \omega(k_0)t]\} a(x - c_g t), \end{aligned} \quad (6.31)$$

using (6.29b) and where $c_g = \partial \omega / \partial k$ evaluated at $k = k_0$. That is to say, the envelope $a(x, t)$ moves at the group velocity, keeping its initial shape.

The group velocity has a meaning beyond that implied by the derivation above: it turns out to be a quite general property of waves that energy (and certain other quadratic properties) propagate at the group velocity. This is to be expected, at least in the presence of coherent wave packets, because there is no energy outside of the wave envelope so the energy must propagate with the envelope.

6.2.3 ♦ The method of stationary phase

We will now relax the assumption that wavenumbers are confined to a narrow band but (since there is no free lunch) we confine ourselves to seeking solutions at large t ; that is, we will be seeking a description of waves far from their source. Consider a disturbance of the general form

$$\psi(x, t) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i[kx - \omega(k)t]} dk = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i\Theta(k; x, t)} dk \quad (6.32)$$

where $\Theta(k; x, t) \equiv kx/t - \omega(k)$. (Here we regard Θ as a function of k with parameters x and t ; we will sometimes just write $\Theta(k)$ with $\Theta'(k) = \partial\Theta/\partial k$.) Now, a standard result in mathematics (known as the ‘Riemann–Lebesgue lemma’) states that

$$I = \lim_{t \rightarrow \infty} \int_{-\infty}^{\infty} f(k) e^{ikt} dk = 0 \quad (6.33)$$

provided that $f(k)$ is integrable and $\int_{-\infty}^{\infty} f(k) dk$ is finite. Intuitively, as t increases the oscillations in the integral increase and become much faster than any variation in $f(k)$; successive oscillations thus cancel and the integral becomes very small.

Looking at (6.32), with $\tilde{A}$ playing the role of $f(k)$, the integral will be small if Θ is everywhere varying with k . However, if there is a region where Θ does not vary with k — that is, if there is a region where the phase is stationary and $\partial\Theta/\partial k = 0$ — then there will be a contribution to the integral from that region. Thus, for large t , an observer will predominantly see waves for which $\Theta'(k) = 0$ and so, using the definition of Θ , for which

$$\frac{x}{t} = \frac{\partial\omega}{\partial k}. \quad (6.34)$$

In other words, at some space-time location (x, t) the waves that dominate are those whose group velocity $\partial\omega/\partial k$ is x/t . An example is plotted in Fig. 6.4 with a dispersion relation $\omega = -\beta/k$; the wavenumber that dominates, k_0 say, is thus given by solving $\beta/k_0^2 = x/t$, which for $x/t = 1$ and $\beta = 400$ gives $k_0 = 20$.

We may actually approximately calculate the contribution to $\psi(x, t)$ from waves moving with the group velocity. Let us expand $\Theta(k)$ around the point, k_0 , where $\Theta'(k_0) = 0$. We obtain

$$\psi(x, t) = \int_{-\infty}^{\infty} \tilde{A}(k) \exp \left\{ it \left[\Theta(k_0) + (k - k_0)\Theta'(k_0) + \frac{1}{2}(k - k_0)^2\Theta''(k_0) \dots \right] \right\} dk \quad (6.35)$$

The higher order terms are small because $k - k_0$ is presumed small (for if it is large the integral vanishes), and the term involving $\Theta'(k_0)$ is zero. The integral becomes

$$\psi(x, t) = \tilde{A}(k_0) e^{i\Theta(k_0)} \int_{-\infty}^{\infty} \exp \left\{ it \frac{1}{2} (k - k_0)^2 \Theta''(k_0) \right\} dk. \quad (6.36)$$

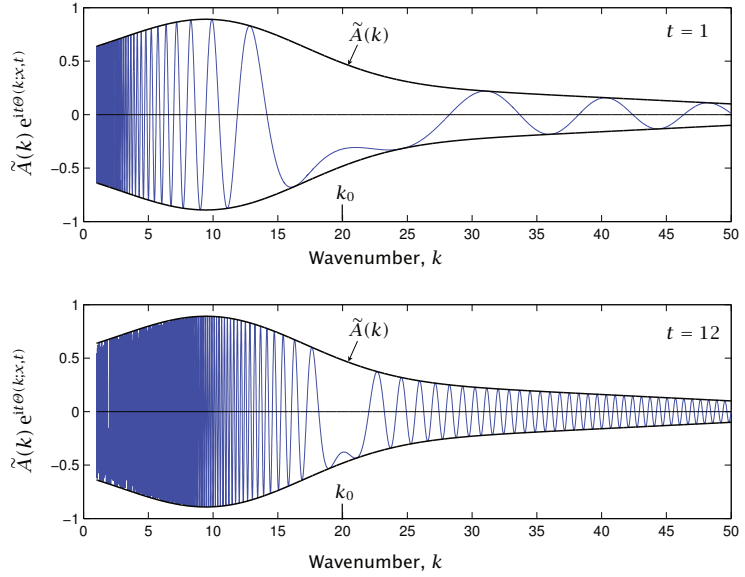
We thus have to evaluate a Gaussian, and because $\int_{-\infty}^{\infty} e^{-cx^2} dx = \sqrt{\pi/c}$ we obtain

$$\psi(x, t) \approx \tilde{A}(k_0) e^{i\Theta(k_0)t} \left[\frac{-2\pi}{it\Theta''(k_0)} \right]^{1/2} = \tilde{A}(k_0) \left[\frac{2i\pi}{t\Theta''(k_0)} \right]^{1/2} e^{i(k_0 x - \omega(k_0)t)}. \quad (6.37)$$

The solution is therefore a plane wave, with wavenumber k_0 and frequency $\omega(k_0)$, slowly modulated by an envelope determined by the form of $\Theta(k_0; x, t)$, where k_0 is the wavenumber such that $x/t = c_g = \partial\omega/\partial k|_{k=k_0}$.

Fig. 6.4 The integrand of (6.32), namely the function that when integrated over wavenumber gives the wave amplitude at a particular x and t .

The example shown is for a Rossby wave with $\omega = -\beta/k$, with $\beta = 400$ and $x/t = 1$, and hence $k_0 = 20$, for two times $t = 1$ and $t = 12$. (The amplitude of the envelope, $\tilde{A}(k)$ is arbitrary.) At the later time the oscillations are much more rapid in k , so that the contribution is more peaked from wavenumbers near to k_0 .



6.3 RAY THEORY

Most waves propagate in a media that is inhomogeneous; for example, in the Earth's atmosphere and ocean the stratification varies with altitude and the Coriolis parameter varies with latitude. In these cases it can be hard to obtain the solution of a wave problem by Fourier methods, even approximately. Nonetheless, the idea of signals propagating at the group velocity is a robust one, and we can often obtain some of the information we want — and in particular the trajectory of a wave — using a recipe known as *ray theory*, using the word theory a little generously.³

6.3.1 Introduction

In an inhomogeneous medium let us suppose that the solution to a particular wave problem is of the form

$$\psi(\mathbf{x}, t) = a(\mathbf{x}, t) e^{i\theta(\mathbf{x}, t)}, \quad (6.38)$$

where a is the wave amplitude and θ the phase, and a varies slowly in a sense we will make more precise shortly. The local wavenumber and frequency are defined by,

$$k_i \equiv \frac{\partial \theta}{\partial x_i}, \quad \omega \equiv -\frac{\partial \theta}{\partial t}. \quad (6.39)$$

where the first expression is equivalent to $\mathbf{k} \equiv \nabla \theta$ and so $\nabla \times \mathbf{k} = 0$. We suppose that the amplitude a varies slowly over a wavelength and a period; that is $|\Delta a|/|a|$ is small over the length $1/k$ and the period $1/\omega$ or

$$\frac{|\partial a / \partial x|}{a} \ll |k|, \quad \frac{|\partial a / \partial t|}{a} \ll \omega, \quad (6.40)$$

and similarly in the other directions. We will assume that the wavenumber and frequency as defined by (6.39) are the same as those that would arise if the medium were homogeneous and a were a constant. Thus, we may obtain a local dispersion relation from the governing equation by keeping the spatially (and possibly temporally) varying parameters fixed and obtain

$$\omega = \Omega(k_i; x_i, t), \quad (6.41)$$

and then allow x_i and t to vary, albeit slowly.

Let us now consider how the wavevector and frequency might change with position and time. It follows from their definitions above that the wavenumber and frequency are related by

$$\frac{\partial k_i}{\partial t} + \frac{\partial \omega}{\partial x_i} = 0, \quad (6.42)$$

where we use a subscript notation for vectors and repeated indices are summed. Using (6.42) and (6.41) gives

$$\frac{\partial k_i}{\partial t} + \frac{\partial \Omega}{\partial x_i} + \frac{\partial \Omega}{\partial k_j} \frac{\partial k_j}{\partial x_i} = 0 \quad \text{or} \quad \frac{\partial k_i}{\partial t} + \frac{\partial \Omega}{\partial x_i} + \frac{\partial \Omega}{\partial k_j} \frac{\partial k_i}{\partial x_j} = 0, \quad (6.43a,b)$$

where to get (6.43b) we use $\partial k_j / \partial x_i = \partial k_i / \partial x_j$, true because $\mathbf{k}$ has no curl. Equation (6.43b) may be written as

$$\frac{\partial \mathbf{k}}{\partial t} + \mathbf{c}_g \cdot \nabla \mathbf{k} = -\nabla \Omega \quad (6.44)$$

where

$$\mathbf{c}_g = \frac{\partial \Omega}{\partial \mathbf{k}} = \left(\frac{\partial \Omega}{\partial k}, \frac{\partial \Omega}{\partial l}, \frac{\partial \Omega}{\partial m} \right) \quad (6.45)$$

is, once more, the group velocity. The left-hand side of (6.44) is similar to an advective derivative, but uses the group velocity. Evidently, if the dispersion relation for frequency is not an explicit function of space then *the wavevector is propagated at the group velocity*.

The frequency is, in general, a function of space, wavenumber and time, and from the dispersion relation, (6.41), its variation is governed by

$$\frac{\partial \omega}{\partial t} = \frac{\partial \Omega}{\partial t} + \frac{\partial \Omega}{\partial k_i} \frac{\partial k_i}{\partial t} = \frac{\partial \Omega}{\partial t} - \frac{\partial \Omega}{\partial k_i} \frac{\partial \omega}{\partial x_i}, \quad (6.46)$$

using (6.42). Using the definition of group velocity, we may write (6.46) as

$$\frac{\partial \omega}{\partial t} + \mathbf{c}_g \cdot \nabla \omega = \frac{\partial \Omega}{\partial t}. \quad (6.47)$$

As with (6.44) the left-hand side is like an advective derivative, but uses a group velocity. Thus, if the dispersion relation is not a function of time, the frequency also propagates at the group velocity.

Motivated by (6.44) and (6.47) we define a *ray* as the trajectory traced by the group velocity, and we see that if the function Ω is not an explicit function of space or time, then *both the wavevector and the frequency are constant along a ray*.

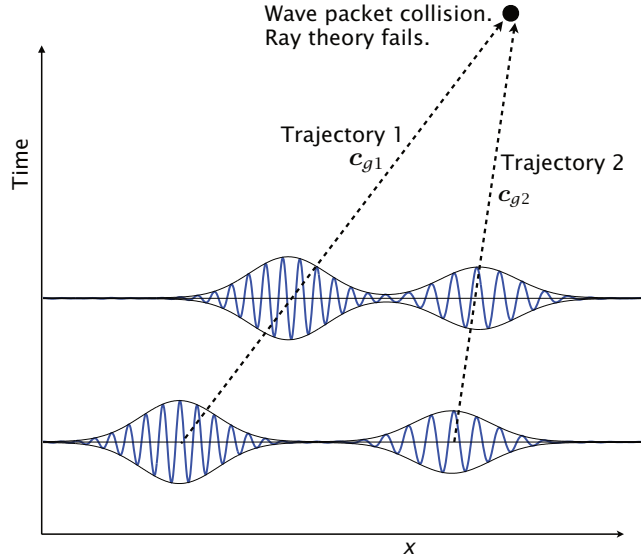
6.3.2 Ray theory in practice

What use is ray theory? The idea is that we may use (6.44) and (6.47) to track a group of waves from one location to another without solving the full wave equations of motion. We can then sometimes solve interesting problems using ordinary differential equations (ODEs) rather than partial differential equations (PDEs).

Suppose that the initial conditions consist of a group of waves at a position x_0 , for which the amplitude and wavenumber vary only slowly with position. We also suppose that we know the dispersion relation for the waves at hand; that is, we know the functional form of $\Omega(k; x, t)$. Now, the total derivate following the group velocity is given by

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{c}_g \cdot \nabla, \quad (6.48)$$

Fig. 6.5 Schema of the trajectory of two wavepackets, each with a different wavelength and moving with a different group velocity, as might be calculated using ray theory. If the wave packets collide ray theory must fail. Ray theory gives only the trajectory of the wave packet, not the detailed structure of the waves within a packet.



so that (6.44) and (6.47) may be written as

$$\frac{d\mathbf{k}}{dt} = -\nabla\Omega, \quad (6.49a)$$

$$\frac{d\omega}{dt} = -\frac{\partial\Omega}{\partial t}. \quad (6.49b)$$

These are ordinary differential equations for wavevector and frequency, solvable provided we know the right-hand sides — that is, provided we know the space and time location at which the dispersion relation [i.e., $\Omega(k; x, t)$] is to be evaluated. But the location *is* known because it is moving with the group velocity and so

$$\frac{d\mathbf{x}}{dt} = \mathbf{c}_g. \quad (6.49c)$$

where $\mathbf{c}_g = \partial\Omega/\partial\mathbf{k}|_{\mathbf{x},t}$ (i.e., $c_{gi} = \partial\Omega/\partial k_i|_{\mathbf{x},t}$). The set (6.49) is a triplet of ordinary differential equations for the wavevector, frequency and position of a wave group. The equations may be solved, albeit sometimes numerically, to give the trajectory of a wave packet or collection of wave packets as schematically illustrated in Fig. 6.5. Of course, if the medium or the wavepacket amplitude is not slowly varying ray theory will fail, and this will perforce happen if two wave packets collide.

The evolution of the amplitude of the wave packet is not given by ray theory. However, the evolution of a quantity related to the amplitude of a wave packet — specifically, the wave activity — may be calculated if the group velocity is known. In chapter 10 we will find that the wave activity, A , satisfies $\partial A/\partial t + \nabla \cdot (\mathbf{c}_g A) = 0$; that is, the flux of wave activity is along a ray. Another way to calculate the evolution of a wave and its amplitude in a varying medium is to use ‘WKB theory’ — see the appendix on page 248, with examples in section 7.5 and chapters 16 and 17. Before all that we turn our attention to a specific form of wave, Rossby waves, but the reader whose interest is more in the general properties of waves may skip forward to section 6.7.

6.4 ROSSBY WAVES

We now shift gears and consider in some detail a particular wave, namely the Rossby wave in a quasi-geostrophic system. Rossby waves are perhaps the most important large-scale wave in the atmosphere and ocean (although gravity waves, discussed in the next chapter, are as important in some contexts).⁴

6.4.1 The linear equation of motion

The relevant equation of motion is the inviscid, adiabatic potential vorticity equation in the quasi-geostrophic system, as discussed in chapter 5), namely

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0, \quad (6.50)$$

where $q(x, y, z, t)$ is the potential vorticity and $\mathbf{u}(x, y, z, t)$ is the horizontal velocity. The velocity is related to a streamfunction by $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ and the potential vorticity is some function of the streamfunction, which might differ from system to system. Two examples, one applying to a continuously stratified system and the second to a single layer system, are

$$q = f + \zeta + \frac{\partial}{\partial z} \left(S(z) \frac{\partial \psi}{\partial z} \right), \quad q = \zeta + f - k_d^2 \psi. \quad (6.51a,b)$$

where $S(z) = f_0^2/N^2$, $\zeta = \nabla^2\psi$ is the relative vorticity and $k_d = 1/L_d$ is the inverse radius of deformation for a shallow water system. (Note that definitions of k_d and L_d can vary, typically by factors of 2, π , etc.) Boundary conditions may be needed to form a complete system.

We now *linearize* (6.50); that is, we suppose that the flow consists of a time-independent component (the ‘basic state’) plus a perturbation, with the perturbation being small compared with the mean flow. The basic state must satisfy the time-independent equation of motion, and it is common and useful to linearize about a zonal flow, $\bar{u}(y, z)$. The basic state is then purely a function of y and so we write

$$q = \bar{q}(y, z) + q'(x, y, t), \quad \psi = \bar{\psi}(y, z) + \psi'(x, y, z, t) \quad (6.52)$$

with a similar notation for the other variables. Note that $\bar{u} = -\partial\bar{\psi}/\partial y$ and $\bar{v} = 0$. Substituting into (6.50) gives, without approximation,

$$\frac{\partial q'}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{q} + \bar{\mathbf{u}} \cdot \nabla q' + \mathbf{u}' \cdot \nabla \bar{q} + \mathbf{u}' \cdot \nabla q' = 0. \quad (6.53)$$

The primed quantities are presumptively small so we neglect terms involving their products. Further, we are assuming that we are linearizing about a state that is a solution of the equations of motion, so that $\bar{\mathbf{u}} \cdot \nabla \bar{q} = 0$. Finally, since $\bar{v} = 0$ and $\partial\bar{q}/\partial x = 0$ we obtain

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0. \quad (6.54)$$

This equation or one very similar appears very commonly in studies of Rossby waves. To proceed, let us consider the simple example of waves in a single layer.

6.4.2 Waves in a single layer

Consider a system obeying (6.50) and (6.51b). The equation could be written in spherical coordinates with $f = 2\Omega \sin \vartheta$, but the dynamics are more easily illustrated on Cartesian β -plane for which $f = f_0 + \beta y$, and since f_0 is a constant it does not appear in our subsequent derivations.

Infinite deformation radius

If the scale of motion is much less than the deformation scale then we make the approximation that $k_d = 0$ and the equation of motion may be written as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0 \quad (6.55)$$

We linearize about a constant zonal flow, $\bar{u}$, by writing

$$\psi = \bar{\psi}(y) + \psi'(x, y, t), \quad (6.56)$$

where $\bar{\psi} = -\bar{u}y$. Substituting (6.56) into (6.55) and neglecting the nonlinear terms involving products of ψ' to give

$$\frac{\partial}{\partial t} \nabla^2 \psi' + \bar{u} \frac{\partial \nabla^2 \psi'}{\partial x} + \beta \frac{\partial \psi'}{\partial x} = 0. \quad (6.57)$$

This equation is just a single-layer version of (6.54), with $\partial \bar{q} / \partial y = \beta$, $q' = \nabla^2 \psi'$ and $v' = \partial \psi' / \partial x$.

The coefficients in (6.57) are not functions of y or z ; this is not a requirement for wave motion to exist but it does enable solutions to be found more easily. Let us seek solutions in the form of a plane wave, namely

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly - \omega t)}, \quad (6.58)$$

where $\tilde{\psi}$ is a complex constant and Re indicates the real part of the function (a notation sometimes omitted if no ambiguity is so-caused). Solutions of this form are valid in a domain with doubly-periodic boundary conditions; solutions in a channel can be obtained using a meridional variation of $\sin ly$, with no essential changes to the dynamics. The amplitude of the oscillation is given by $\tilde{\psi}$ and the phase by $kx + ly - \omega t$, where k and l are the x - and y -wavenumbers and ω is the frequency of the oscillation.

Substituting (6.58) into (6.57) yields

$$[(-\omega + Uk)(-K^2) + \beta k] \tilde{\psi} = 0, \quad (6.59)$$

where $K^2 = k^2 + l^2$. For non-trivial solutions this implies

$$\omega = Uk - \frac{\beta k}{K^2}. \quad (6.60)$$

This is the *dispersion relation* for barotropic Rossby waves, and evidently the velocity U Doppler shifts the frequency. The components of the phase speed and group velocity are given by, respectively,

$$c_p^x \equiv \frac{\omega}{k} = \bar{u} - \frac{\beta}{K^2}, \quad c_p^y \equiv \frac{\omega}{l} = \bar{u} \frac{k}{l} - \frac{\beta k}{K^2 l}, \quad (6.61a,b)$$

and

$$c_g^x \equiv \frac{\partial \omega}{\partial k} = \bar{u} + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \quad c_g^y \equiv \frac{\partial \omega}{\partial l} = \frac{2\beta k l}{(k^2 + l^2)^2}. \quad (6.62a,b)$$

The phase speed in the absence of a mean flow is *westwards*, with waves of longer wavelengths travelling more quickly, and the eastward current speed required to hold the waves of a particular wavenumber stationary (i.e., $c_p^x = 0$) is $U = \beta / K^2$. The background flow $\bar{u}$ evidently just provides a uniform shift to the phase speed, and could be transformed away by a change of coordinate.

Finite deformation radius

For a finite deformation radius the basic state $\Psi = -Uy$ is still a solution of the original equations of motion, but the potential vorticity corresponding to this state is $q = U y k_d^2 + \beta y$ and its gradient is $\nabla q = (\beta + U k_d^2) \mathbf{j}$. The linearized equation of motion is thus

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) (\nabla^2 \psi' - \psi' k_d^2) + (\beta + \bar{u} k_d^2) \frac{\partial \psi'}{\partial x} = 0. \quad (6.63)$$

Substituting $\psi' = \tilde{\psi} e^{i(kx+ly-\omega t)}$ we obtain the dispersion relation,

$$\omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2} = Uk - k \frac{\beta + Uk_d^2}{K^2 + k_d^2}. \quad (6.64)$$

The corresponding components of phase speed and group velocity are

$$c_p^x = \bar{u} - \frac{\beta + \bar{u}k_d^2}{K^2 + k_d^2} = \frac{\bar{u}K^2 - \beta}{K^2 + k_d^2}, \quad c_p^y = \bar{u} \frac{k}{l} - \frac{k}{l} \left(\frac{\bar{u}K^2 - \beta}{K^2 + k_d^2} \right) \quad (6.65a,b)$$

and

$$c_g^x = \bar{u} + \frac{(\beta + \bar{u}k_d^2)(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \quad c_g^y = \frac{2kl(\beta + \bar{u}k_d^2)}{(k^2 + l^2 + k_d^2)^2}. \quad (6.66a,b)$$

The uniform velocity field now no longer provides just a simple Doppler shift of the frequency, nor a uniform addition to the phase speed. From (6.65a) the waves are stationary when $K^2 = \beta/\bar{u} \equiv K_s^2$; that is, the current speed required to hold waves of a particular wavenumber stationary is $\bar{u} = \beta/K^2$. However, this is *not* simply the magnitude of the phase speed of waves of that wavenumber in the absence of a current — this is given by

$$c_p^x = \frac{-\beta}{K_s^2 + k_d^2} = \frac{-\bar{u}}{1 + k_d^2/K_s^2} \neq -\bar{u}. \quad (6.67)$$

Why is there a difference? It is because the current does not just provide a uniform translation, but, if k_d is non-zero, it also modifies the basic potential vorticity gradient. The basic state height field η_0 is sloping; that is $\eta_0 = -(f_0/g)\bar{u}y$, and the ambient potential vorticity field increases with y and $q = (\beta + Uk_d^2)y$. Thus, the basic state defines a preferred frame of reference, and the problem is not Galilean invariant.⁵

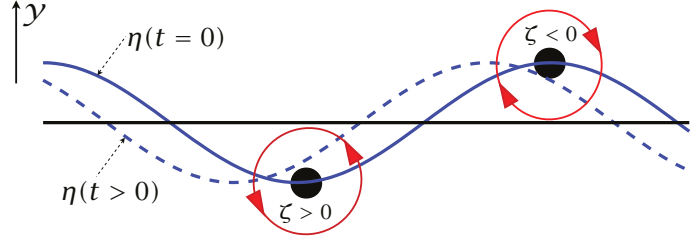
We also note that, from (6.65b), the group velocity is negative (westward) if the x -wavenumber is sufficiently small, compared to the y -wavenumber or the deformation wavenumber. That is, said a little loosely, *long waves move information westward and short waves move information eastward*, and this is a common property of Rossby waves. The x -component of the phase speed, on the other hand, is always westward relative to the mean flow.

6.4.3 The mechanism of Rossby waves

The fundamental mechanism underlying Rossby waves is easily understood. Consider a material line of stationary fluid parcels along a line of constant latitude, and suppose that some disturbance causes their displacement to the line marked $\eta(t=0)$ in Fig. 6.6. In the displacement, the potential vorticity of the fluid parcels is conserved, and in the simplest case of barotropic flow on the β -plane the potential vorticity is the absolute vorticity, $\beta y + \zeta$. Thus, in either hemisphere, a northward displacement leads to the production of negative relative vorticity and a southward displacement leads to the production of positive relative vorticity. The relative vorticity gives rise to a velocity field which, in turn, advects the parcels in material line in the manner shown, and the wave propagates westwards.

In more complicated situations, such as flow in two layers, considered below, or in a continuously stratified fluid, the mechanism is essentially the same. A displaced fluid parcel carries with it its potential vorticity and, in the presence of a potential vorticity gradient in the basic state, a potential vorticity anomaly is produced. The potential vorticity anomaly produces a velocity field (an example of potential vorticity inversion) which further displaces the fluid parcels, leading to the formation of a Rossby wave. The vital ingredient is a basic state potential vorticity gradient, such as that provided by the change of the Coriolis parameter with latitude.

Fig. 6.6 A two-dimensional (x - y) Rossby wave. An initial disturbance displaces a material line at constant latitude (the straight horizontal line) to the solid line marked $\eta(t = 0)$. Conservation of potential vorticity, $\beta y + \zeta$, leads to the production of relative vorticity, ζ , as shown. The associated velocity field (arrows on the circles) then advects the fluid parcels, and the material line evolves into the dashed line with the phase propagating westward.



6.4.4 Rossby waves in two layers

Now consider the dynamics of the two-layer model, linearized about a state of rest. The two, coupled, linear equations describing the motion in each layer are

$$\frac{\partial}{\partial t} [\nabla^2 \psi'_1 + F_1(\psi'_2 - \psi'_1)] + \beta \frac{\partial \psi'_1}{\partial x} = 0, \quad (6.68a)$$

$$\frac{\partial}{\partial t} [\nabla^2 \psi'_2 + F_2(\psi'_1 - \psi'_2)] + \beta \frac{\partial \psi'_2}{\partial x} = 0, \quad (6.68b)$$

where $F_1 = f_0^2/g'H_1$ and $F_2 = f_0^2/g'H_2$. By inspection (6.68) may be transformed into two uncoupled equations: the first is obtained by multiplying (6.68a) by F_2 and (6.68b) by F_1 and adding, and the second is the difference of (6.68a) and (6.68b). Then, defining

$$\bar{\psi} = \frac{F_1 \psi'_2 + F_2 \psi'_1}{F_1 + F_2}, \quad \tau = \frac{1}{2}(\psi'_1 - \psi'_2), \quad (6.69a,b)$$

(think ‘ τ for temperature’), (6.68) become

$$\frac{\partial}{\partial t} \nabla^2 \bar{\psi} + \beta \frac{\partial \bar{\psi}}{\partial x} = 0, \quad (6.70a)$$

$$\frac{\partial}{\partial t} [(\nabla^2 - k_d^2)\tau] + \beta \frac{\partial \tau}{\partial x} = 0, \quad (6.70b)$$

where now $k_d = (F_1 + F_2)^{1/2}$. The internal radius of deformation for this problem is the inverse of this, namely

$$L_d = k_d^{-1} = \frac{1}{f_0} \left(\frac{g'H_1H_2}{H_1 + H_2} \right)^{1/2}. \quad (6.71)$$

The variables $\bar{\psi}$ and τ are the *normal modes* for the two-layer model, as they oscillate independently of each other. [For the continuous equations the analogous modes are the eigenfunctions of $\partial_z[(f_0^2/N^2)\partial_z\phi] = \lambda^2\phi$.] The equation for $\bar{\psi}$, the *barotropic mode*, is identical to that of the single-layer, rigid-lid model, namely (6.57) with $U = 0$, and its dispersion relation is just

$$\omega = -\frac{\beta k}{K^2}. \quad (6.72)$$

The barotropic mode corresponds to synchronous, depth-independent, motion in the two layers, with no undulations in the dividing interface.

The displacement of the interface is given by $2f_0\tau/g'$ and so proportional to the amplitude of τ , the *baroclinic mode*. The dispersion relation for the baroclinic mode is

$$\omega = -\frac{\beta k}{K^2 + k_d^2}. \quad (6.73)$$

The mass transport associated with this mode is identically zero, since from (6.69) we have

$$\psi_1 = \bar{\psi} + \frac{2F_1\tau}{F_1 + F_2}, \quad \psi_2 = \bar{\psi} - \frac{2F_2\tau}{F_1 + F_2}, \quad (6.74a,b)$$

and this implies

$$H_1\psi_1 + H_2\psi_2 = (H_1 + H_2)\bar{\psi}. \quad (6.75)$$

The left-hand side is proportional to the total mass transport, which is evidently associated with the barotropic mode.

The dispersion relation and associated group and phase velocities are plotted in Fig. 6.7. The x -component of the phase speed, ω/k , is negative (westwards) for both baroclinic and barotropic Rossby waves. The group velocity of the barotropic waves is always positive (eastwards), but the group velocity of long baroclinic waves may be negative (westwards). For very short waves, $k^2 \gg k_d^2$, the baroclinic and barotropic velocities coincide and their phase and group velocities are equal and opposite. With a deformation radius of 50 km, typical for the mid-latitude ocean, then a non-dimensional frequency of unity in the figure corresponds to a dimensional frequency of $5 \times 10^{-7} \text{ s}^{-1}$ or a period of about 100 days. In an atmosphere with a deformation radius of 1000 km a non-dimensional frequency of unity corresponds to $1 \times 10^{-5} \text{ s}^{-1}$ or a period of about 7 days. Non-dimensional velocities of unity correspond to respective dimensional velocities of about 0.25 m s^{-1} (ocean) and 10 m s^{-1} (atmosphere).

The deformation radius only affects the baroclinic mode. For scales much smaller than the deformation radius, $K^2 \gg k_d^2$, we see from (6.70b) that the baroclinic mode obeys the same equation as the barotropic mode so that

$$\frac{\partial}{\partial t} \nabla^2 \tau + \beta \frac{\partial \tau}{\partial x} = 0. \quad (6.76)$$

Using this and (6.70a) implies that

$$\frac{\partial}{\partial t} \nabla^2 \psi_i + \beta \frac{\partial \psi_i}{\partial x} = 0, \quad i = 1, 2. \quad (6.77)$$

That is to say, the two layers themselves are uncoupled from each other. At the other extreme, for very long baroclinic waves the relative vorticity is unimportant.

6.5 ROSSBY WAVES IN STRATIFIED QUASI-GEOSTROPHIC FLOW

6.5.1 Setting up the problem

Let us now consider the dynamics of linear waves in stratified quasi-geostrophic flow on a β -plane, with a resting basic state. (In chapter 16 we explore the role of Rossby waves in a more realistic setting.) The interior flow is governed by the potential vorticity equation, (5.118), and linearizing this about a state of rest gives

$$\frac{\partial}{\partial t} \left[\nabla^2 \psi' + \frac{1}{\bar{\rho}(z)} \frac{\partial}{\partial z} \left(\bar{\rho}(z) F(z) \frac{\partial \psi'}{\partial z} \right) \right] + \beta \frac{\partial \psi'}{\partial x} = 0, \quad (6.78)$$

where $\bar{\rho}$ is the density profile of the basic state and $F(z) = f_0^2/N^2$. (F is the square of the inverse Prandtl ratio, N/f_0 .) In the Boussinesq approximation $\bar{\rho} = \rho_0$, i.e., a constant. The vertical boundary conditions are determined by the thermodynamic equation, (5.120). If the boundaries are flat,

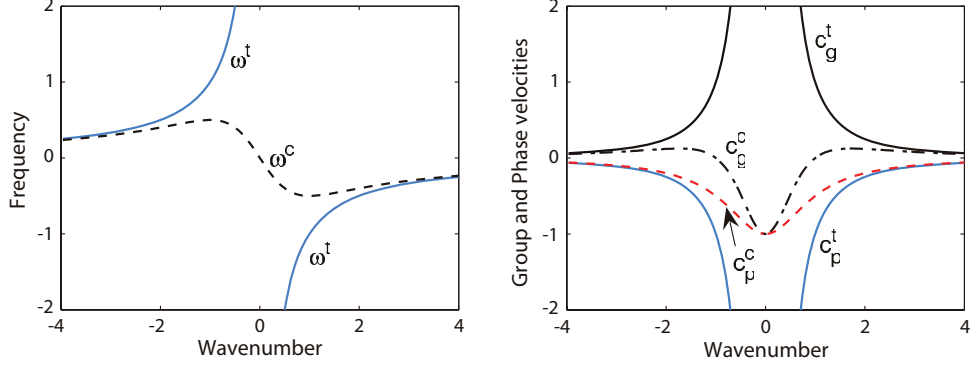


Fig. 6.7 Left: the dispersion relation for barotropic (ω^t , solid line) and baroclinic (ω^c , dashed line) Rossby waves in the two-layer model, calculated using (6.72) and (6.73) with $k^y = 0$, plotted for both positive and negative zonal wavenumbers and frequencies. The wavenumber is non-dimensionalized by k_d , and the frequency is non-dimensionalized by β/k_d . Right: the corresponding zonal group and phase velocities, $c_g = \partial\omega/\partial k^x$ and $c_p = \omega/k^x$, with superscript 't' or 'c' for the barotropic or baroclinic mode, respectively. The velocities are non-dimensionalized by β/k_d^2 .

rigid, slippery surfaces then $w = 0$ at the boundaries and if there is no surface buoyancy gradient the linearized thermodynamic equation is

$$\frac{\partial}{\partial t} \left(\frac{\partial \psi'}{\partial z} \right) = 0. \quad (6.79)$$

We apply this at the ground and, with somewhat less justification, at the tropopause: we assume the higher static stability of the stratosphere inhibits vertical motion. If the ground is not flat or if friction provides a vertical velocity by way of an Ekman layer, the boundary condition must be correspondingly modified, but we will stay with the simplest case and apply (6.79) at $z = 0$ and $z = H$.

6.5.2 Wave motion

As in the single-layer case, we seek solutions of the form

$$\psi' = \text{Re } \tilde{\psi}(z) e^{i(kx + ly - \omega t)}, \quad (6.80)$$

where $\tilde{\psi}(z)$ will determine the vertical structure of the waves. The case of a sphere is more complicated but introduces no truly new physical phenomena.

Substituting (6.80) into (6.78) gives

$$\omega \left[-K^2 \tilde{\psi}(z) + \frac{1}{\bar{\rho}} \frac{d}{dz} \left(\bar{\rho} F(z) \frac{d\tilde{\psi}}{dz} \right) \right] - \beta k \tilde{\psi}(z) = 0. \quad (6.81)$$

Now, if $\tilde{\psi}$ satisfies

$$\frac{1}{\bar{\rho}} \frac{d}{dz} \left(\bar{\rho} F(z) \frac{d\tilde{\psi}}{dz} \right) = -\Gamma \tilde{\psi}, \quad (6.82)$$

where Γ is a constant, then the equation of motion becomes

$$-\omega [K^2 + \Gamma] \tilde{\psi} - \beta k \tilde{\psi} = 0, \quad (6.83)$$

and the dispersion relation follows, namely

$$\omega = -\frac{\beta k}{K^2 + \Gamma}. \quad (6.84)$$

Equation (6.82) constitutes an eigenvalue problem for the vertical structure; the boundary conditions, derived from (6.79), are $\partial\tilde{\psi}/\partial z = 0$ at $z = 0$ and $z = H$. The resulting eigenvalues, Γ are proportional to the inverse of the squares of the deformation radii for the problem and the eigenfunctions are the vertical structure functions.

A simple example

Consider the case in which $F(z)$ and $\tilde{\rho}$ are constant, and in which the domain is confined between two rigid surfaces at $z = 0$ and $z = H$. Then the eigenvalue problem for the vertical structure is

$$F \frac{d^2 \tilde{\psi}}{dz^2} = -\Gamma \tilde{\psi} \quad (6.85a)$$

with boundary conditions of

$$\frac{d\tilde{\psi}}{dz} = 0, \quad \text{at } z = 0, H. \quad (6.85b)$$

There is a sequence of solutions to this, namely

$$\tilde{\psi}_n(z) = \cos(n\pi z/H), \quad n = 1, 2, \dots \quad (6.86)$$

with corresponding eigenvalues

$$\Gamma_n = n^2 \frac{F\pi^2}{H^2} = (n\pi)^2 \left(\frac{f_0}{NH} \right)^2, \quad n = 1, 2, \dots \quad (6.87)$$

Equation (6.87) may be used to define the deformation radii for this problem, namely

$$L_n \equiv \frac{1}{\sqrt{\Gamma_n}} = \frac{NH}{n\pi f_0}. \quad (6.88)$$

The first deformation radius is the same as the expression obtained by dimensional analysis, namely NH/f , except for a factor of π . (Definitions of the deformation radii both with and without the factor of π are common in the literature, and neither is obviously more correct. In the latter case, the first deformation radius in a problem with uniform stratification is given by NH/f , equal to $\pi/\sqrt{f_1}$.) In addition to these baroclinic modes, the case with $n = 0$, that is with $\tilde{\psi} = 1$, is also a solution of (6.85) for any $F(z)$.

Using (6.84) and (6.87) the dispersion relation becomes

$$\omega = -\frac{\beta k}{K^2 + (n\pi)^2 (f_0/NH)^2}, \quad n = 0, 1, 2, \dots, \quad (6.89)$$

and the horizontal wavenumbers k and l are also quantized in a finite domain. The dynamics of the barotropic mode are independent of height and independent of the stratification of the basic state, and so these Rossby waves are *identical* with the Rossby waves in a homogeneous fluid contained between two flat rigid surfaces. The structure of the baroclinic modes, which in general depends on the structure of the stratification, becomes increasingly complex as the vertical wavenumber n increases. This increasing complexity naturally leads to a certain delicacy, making it rare that they can be unambiguously identified in nature. The eigenproblem for a realistic atmospheric profile is further complicated because of the lack of a rigid lid at the top of the atmosphere.⁶

Essentials of Rossby Waves

- Rossby waves owe their existence to a gradient of potential vorticity in the fluid. If a fluid parcel is displaced, it conserves its potential vorticity and so its relative vorticity will in general change. The relative vorticity creates a velocity field that displaces neighbouring parcels, whose relative vorticity changes and so on.
- A common source of a potential vorticity gradient is differential rotation, or the β -effect, and *planetary waves* is the name given to this type of Rossby wave. In the presence of non-zero β the ambient potential vorticity increases northward and the phase of the Rossby waves propagates westward. In general, Rossby waves propagate pseudo-westwards, meaning to the left of the direction of the potential vorticity gradient.
- A common equation of motion for Rossby waves is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (\text{RW.1})$$

with an overbar denoting the basic state and a prime a perturbation. In the case of a single layer of fluid with no mean flow this equation becomes

$$\frac{\partial}{\partial t} (\nabla^2 + k_d^2) \psi' + \beta \frac{\partial \psi'}{\partial x} = 0, \quad (\text{RW.2})$$

with dispersion relation

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \quad (\text{RW.3})$$

- In the absence of a mean flow (i.e., $\bar{u} = 0$), the phase speed in the zonal direction ($c_p^x = \omega/k$) is always negative, or westward, and is larger for large waves. For (RW.3) the components of the group velocity are given by

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \quad c_g^y = \frac{2\beta k l}{(k^2 + l^2 + k_d^2)^2}. \quad (\text{RW.4})$$

The group velocity is westward if the zonal wavenumber is sufficiently small, and eastward if the zonal wavenumber is sufficiently large.

- Rossby waves exist in stratified fluids, and have a similar dispersion relation to (RW.3) with an appropriate vertical wavenumber appearing in place of the inverse deformation radius, k_d .
- The reflection of such Rossby waves at a wall is specular, meaning that the group velocity of the reflected wave makes the same angle with the wall as the group velocity of the incident wave. The energy flux of the reflected wave is equal and opposite to that of the incoming wave in the direction normal to the wall.

6.6 ENERGY PROPAGATION AND REFLECTION OF ROSSBY WAVES

We now consider how energy is fluxed in Rossby waves. To keep matters algebraically simple from we consider waves in a single layer and without a mean flow, but we allow for a finite radius of deformation. To remind ourselves, the dynamics are governed by the evolution of potential vorticity and the linearized evolution equation is

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \psi + \beta \frac{\partial \psi}{\partial x} = 0.$$

The dispersion relation follows in the usual way and is

$$\omega = \frac{-k\beta}{K^2 + k_d^2}, \quad (6.91)$$

which is a simplification of (6.64), and the group velocities are

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(K^2 + k_d^2)^2}, \quad c_g^y = \frac{2\beta kl}{(K^2 + k_d^2)^2}, \quad (6.92a,b)$$

which are simplifications of (6.66), and as usual $K^2 = k^2 + l^2$.

To obtain an energy equation multiply (6.90) by $-\psi$ to obtain, after a couple of lines of algebra,

$$\frac{1}{2} \frac{\partial}{\partial t} ((\nabla \psi)^2 + k_d^2 \psi^2) - \nabla \cdot \left(\psi \nabla \frac{\partial \psi}{\partial t} + \mathbf{i} \frac{\beta}{2} \psi^2 \right) = 0, \quad (6.93)$$

where $\mathbf{i}$ is the unit vector in the x direction. The first group of terms are the energy itself, or more strictly the energy density. (An energy density is an energy per unit mass or per unit volume, depending on the context.) The term $(\nabla \psi)^2/2 = (u^2 + v^2)/2$ is the kinetic energy and $k_d^2 \psi^2/2$ is the potential energy, proportional to the displacement of the free surface, squared. The second term is the energy flux, so that we may write

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{F} = 0. \quad (6.94)$$

where $E = (\nabla \psi)^2/2 + k_d^2 \psi^2/2$ and $\mathbf{F} = -(\psi \nabla \partial \psi / \partial t + \mathbf{i} \beta \psi^2)$. We haven't yet used the fact that the disturbance has a dispersion relation, and if we do so we may expect, following the derivations of section 6.2, that the energy moves at the group velocity. Let us now demonstrate this explicitly.

We assume a solution of the form

$$\psi = A(x) \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) = A(x) \cos(kx + ly - \omega t) \quad (6.95)$$

where $A(x)$ is assumed to vary slowly compared to the nearly plane wave. (Note that $\mathbf{k}$ is the wave vector, to be distinguished from $\mathbf{k}$, the unit vector in the z -direction.) The kinetic energy in a wave is given by

$$KE = \frac{A^2}{2} (\psi_x^2 + \psi_y^2) \quad (6.96)$$

so that, averaged over a wave period,

$$\overline{KE} = \frac{A^2}{2} (k^2 + l^2) \frac{\omega}{2\pi} \int_0^{2\pi/\omega} \sin^2(\mathbf{k} \cdot \mathbf{x} - \omega t) dt. \quad (6.97)$$

The time-averaging produces a factor of one half, and applying a similar procedure¹ to the potential energy we obtain

$$\overline{KE} = \frac{A^2}{4} (k^2 + l^2), \quad \overline{PE} = \frac{A^2}{4} k_d^2, \quad (6.98)$$

so that the average total energy is

$$\bar{E} = \frac{A^2}{4}(K^2 + k_d^2), \quad (6.99)$$

where $K^2 = k^2 + l^2$.

The flux, F , is given by

$$F = -\left(\psi \nabla \frac{\partial \psi}{\partial t} + \mathbf{i} \frac{\beta}{2} \psi^2\right) = -A^2 \cos^2(\mathbf{k} \cdot \mathbf{x} - \omega t) \left(\mathbf{k} \omega - \mathbf{i} \frac{\beta}{2}\right), \quad (6.100)$$

so that evidently the energy flux has a component in the direction of the wavevector, $\mathbf{k}$, and a component in the x -direction. Averaging over a wave period straightforwardly gives us additional factors of one half:

$$\bar{F} = -\frac{A^2}{2} \left(\mathbf{k} \omega + \mathbf{i} \frac{\beta}{2}\right). \quad (6.101)$$

We now use the dispersion relation $\omega = -\beta k / (K^2 + k_d^2)$ to eliminate the frequency, giving

$$\bar{F} = \frac{A^2 \beta}{2} \left(\mathbf{k} \frac{k}{K^2 + k_d^2} - \mathbf{i} \frac{1}{2}\right), \quad (6.102)$$

and writing this in component form we obtain

$$\bar{F} = \frac{A^2 \beta}{4} \left[\mathbf{i} \left(\frac{k^2 - l^2 - k_d^2}{K^2 + k_d^2} \right) + \mathbf{j} \left(\frac{2kl}{K^2 + k_d^2} \right) \right] \quad (6.103)$$

Comparison of (6.103) with (6.92) and (6.99) reveals that

$$\bar{F} = \mathbf{c}_g \bar{E} \quad (6.104)$$

so that the energy propagation equation (6.94), when averaged over a wave, becomes

$$\frac{\partial \bar{E}}{\partial t} + \nabla \cdot \mathbf{c}_g \bar{E} = 0. \quad (6.105)$$

It is interesting that the variation of A plays no role in the above manipulations, so that the derivation appears to go through if the amplitude $A(\mathbf{x}, t)$ is in fact a constant and the wave is a single plane wave. This seems hard to reconcile with our previous discussion, in which we noted that the group velocity was the velocity of a wave *packet* involving a superposition of plane waves. Indeed, the derivative of the frequency with respect to wavenumber means little if there is only one wavenumber. In fact there is nothing wrong with the above derivation if A is a constant and only a single plane wave is present. The resolution of the paradox arises by noting that a plane wave fills all of space and time; in this case there is no convergence of the energy flux and the energy propagation equation is trivially true.

6.6.1 ♦ Rossby wave reflection

We now consider how Rossby waves might be reflected from a solid boundary. The topic has an obvious oceanographic relevance, for the reflection of Rossby waves turns out to be one way of interpreting why intense oceanic boundary currents form on the western sides of ocean basins, not the east. Rossby waves also reflect off the western boundary in equatorial regions during the El Niño phenomenon. There is also an atmospheric relevance, for meridionally propagating Rossby waves may effectively be reflected as they approach a ‘turning latitude’ where the meridional wavenumber goes to zero, as considered in chapter 16. As a preliminary we give a useful graphic interpretation of Rossby wave propagation.⁷

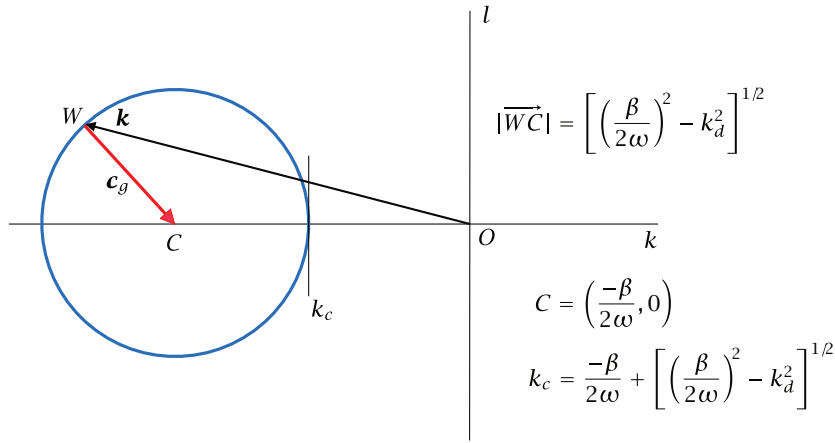


Fig. 6.8 The energy propagation diagram for Rossby waves. The wavevectors of a given frequency all lie in a circle of radius $[(\beta/2\omega)^2 - k_d^2]^{1/2}$, centered at the point C . The closest distance of the circle to the origin is k_c , and if the deformation radius is infinite k_c the circle touches the origin. For a given wavenumber k , the group velocity is along the line directed from W to C .

The energy propagation diagram

The dispersion relation for Rossby waves, $\omega = -\beta k / (k^2 + l^2 + k_d^2)$, may be rewritten as

$$(k + \beta/2\omega)^2 + l^2 = (\beta/2\omega)^2 - k_d^2. \quad (6.106)$$

This equation is the parametric representation of a circle, meaning that the wavevector (k, l) must lie on a circle centered at the point $(-\beta/2\omega, 0)$ and with radius $[(\beta/2\omega)^2 - k_d^2]^{1/2}$, as illustrated in Fig. 6.8. If k_d is zero the circle touches the origin, and if it is nonzero the distance of the closest point to the circle, k_c say, is given by $k_c = -\beta/2\omega + [(\beta/2\omega)^2 - k_d^2]^{1/2}$. For low frequencies, specifically if $\omega \ll \beta/2k$, then $k_c \approx -\omega k_d^2/\beta$. The radius of the circle is a positive real number only when $\omega < \beta/2k_d$. This is the maximum frequency possible, and it occurs when $l = 0$ and $k = k_d$ and when $c_g^x = c_g^y = 0$.

The group velocity, and hence the energy flux, can be visualized graphically from Fig. 6.8. By direct manipulation of the expressions for group velocity and frequency [equations (RW.3) and (RW.4)] we find that

$$c_g^x = \frac{-2\omega}{K^2 + k_d^2} \left(k + \frac{\beta}{2\omega} \right), \quad c_g^y = \frac{-2\omega}{K^2 + k_d^2} l. \quad (6.107a,b)$$

(To check this, it is easiest to begin with the right-hand sides and use the dispersion relation for ω .) Now, since the center of the circle of wavevectors is at the position $(-\beta/2\omega, 0)$, and referring to Fig. 6.8, we have

$$\mathbf{c}_g = \frac{2\omega}{K^2 + k_d^2} \mathbf{R} \quad (6.108)$$

where $\mathbf{R} = \overrightarrow{WC}$ is the vector directed from W to C , that is from the end of the wavevector itself to the center of the circle around which all the wavevectors lie.

Equation (6.108) and Fig. 6.8 allow for a useful visualization of the energy and phase. The phase propagates in the direction of the wave vector, and for Rossby waves this is always westward. The group velocity is in the direction of the wave vector to the center of the circle, and this can be either eastward (if $k^2 > l^2 + k_d^2$) or westward ($k^2 < l^2 + k_d^2$). Interestingly, the velocity vector is normal to the wave vector. To see this, consider a purely westward propagating wave for which $l = 0$. Then

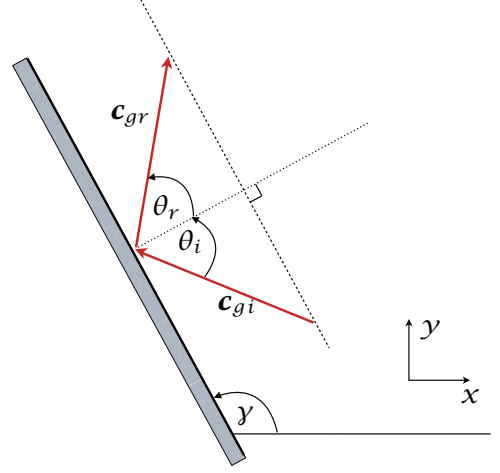


Fig. 6.9 The reflection of a Rossby wave at a western wall, in physical space. A Rossby wave with a westward group velocity impinges at an angle θ_i to a wall, inducing a reflected wave moving eastward at an angle θ_r . The reflection is specular, with $\theta_r = \theta_i$, and energy conserving, with $|c_{gr}| = |c_{gi}|$ — see text and Fig. 6.10.

$v = \partial\psi/\partial x = ik\tilde{\psi}$ and $u = -\partial\psi/\partial y = -il\tilde{\psi} = 0$. We now see how some of these properties can help us understand the reflection of Rossby waves.

Reflection at a wall

Consider Rossby waves incident on wall making an angle γ with the x -axis, and suppose that somehow these waves are reflected back into the fluid interior. This is a reasonable expectation, for the wall cannot normally simply absorb all the wave energy, and if reflection does occur it will have the following two properties.

1. The incident and reflected wave will have the same wavenumber component along the wall.
2. The incident and reflected wave will have the same frequency.

To understand these properties, consider the case in which the wall is oriented meridionally, along the y -axis with $\gamma = 90^\circ$. There is no loss of generality in this choice, because we may simply choose coordinates so that y is parallel to the wall and the β -effect, which differentiates x from y , does not enter the argument. The incident and reflected waves are

$$\psi_i(x, y, t) = A_i e^{i(k_i x + l_i y - \omega_i t)}, \quad \psi_r(x, y, t) = A_r e^{i(k_r x + l_r y - \omega_r t)}, \quad (6.109)$$

with subscripts i and r denoting incident and reflected. At the wall, which we take to be at $x = 0$, the normal velocity $u = -\partial\psi/\partial y$ must be zero so that

$$A_i l_i e^{i(l_i y - \omega_i t)} + A_r l_r e^{i(l_r y - \omega_r t)} = 0. \quad (6.110)$$

For this equation to hold for all y and all time then we must have

$$l_r = l_i, \quad \omega_r = \omega_i. \quad (6.111)$$

This result is independent of the detailed dynamics of the waves, requiring only that the velocity is determined from a streamfunction. When we consider Rossby-wave dynamics specifically, the x - and y -coordinates are not arbitrary and so the wall cannot be taken to be aligned with the y -axis; rather, the result means that the *projection* of the incident wavevector, $\mathbf{k}_i$ on the wall must equal the *projection* of the reflected wavevector, $\mathbf{k}_r$. The magnitude of the wavevector (the wavenumber) is not in general conserved by reflection. Finally, given these results and using (6.110) we see that the incident and reflected amplitudes are related by

$$A_r = -A_i. \quad (6.112)$$

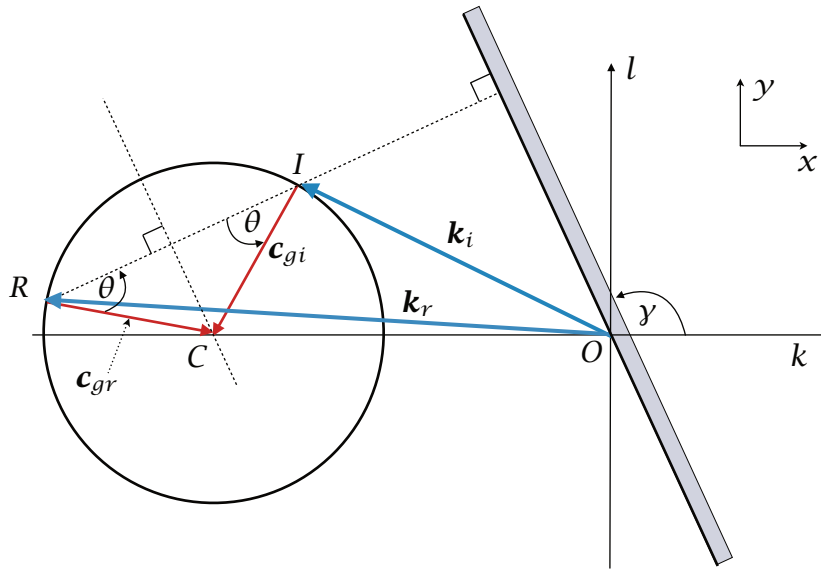


Fig. 6.10 Graphical representation of the reflection of a Rossby wave at a western wall, in spectral space. The incident wave has wavevector $\mathbf{k}_i$, ending at point I . Construct the wavevector circle of constant frequency through point I with radius $(\beta/2\omega)^2 - k_a^2)^{1/2}$ and center $C = (-\beta/2\omega, 0)$; the group velocity vector then lies along $\overline{IC}$ and is directed westward. The reflected wave has a wavevector $\mathbf{k}_r$ such that its projection on the wall is equal to that of $\mathbf{k}_i$, and this fixes the point R . The group velocity of the reflected wave then lies along $\overline{RC}$, and it can be seen that $\mathbf{c}_{gr}$ makes the same angle to the wall as does $\mathbf{c}_{gi}$, except that it is directed eastward. The reflection is therefore specular and is such that the energy flux directed away from the wall is equal to the energy flux directed toward the wall.

Now let's delve a little deeper into the problem.

Generally, when we consider a wave to be incident on a wall, we are supposing that the *group velocity* is directed toward the wall. Suppose that a wave of given frequency, ω , and wavevector, $\mathbf{k}_i$, and with westward group velocity is incident on a predominantly western wall, as in Fig. 6.9. (Similar reasoning, *mutatis mutandis*, can be applied to a wave incident on an eastern wall.) Let us suppose that incident wave, $\mathbf{k}_i$, lies at the point I on the wavenumber circle, and the group velocity is found by drawing a line from I to the center of the circle, C (so $\mathbf{c}_{gi} \propto \overline{IC}$), and in this case the vector is directed westward.

The projection of the $\mathbf{k}_i$ must be equal to the projection of the reflected wave vector, $\mathbf{k}_r$, and both wavevectors must lie in the same wavenumber circle, centered at $-\beta/2\omega$, because the frequencies of the two waves are the same. We may then graphically determine the wavevector of the reflected wave using the construction of Fig. 6.10. Given the wavevector, the group velocity of the reflected wave follows by drawing a line from the wavevector to the center of the circle (the line $\overline{RC}$). We see from the figure that the reflected group velocity is directed eastward and that it forms the same angle to the wall as does the incident wave; that is, the reflection is *specular*. Since the amplitude of the incoming and reflected wave are the same, the components of the energy flux perpendicular to the wall are equal and opposite. Furthermore, we can see from the figure that the wavenumber of the reflected wave has a larger magnitude than that of the incident wave. For waves reflecting of an eastern boundary, the reverse is true. Put simply, at a western boundary incident long waves are reflected as short waves, whereas at an eastern boundary incident short waves are reflected as long waves.

Quantitatively solving for the wavenumbers of the reflected wave is a little tedious in the case

when the wall is at angle, but easy enough if the wall is a meridional, along the y -axis. We know the frequency, ω , and the y -wavenumber, l , so that the x -wavenumber is may be deduced from the dispersion relation

$$\omega = \frac{-\beta k_i}{k_i^2 + l^2 + k_d^2} = \frac{-\beta k_r}{k_r^2 + l^2 + k_d^2}. \quad (6.113)$$

We obtain

$$k_i = \frac{-\beta}{2\omega} + \sqrt{\left(\frac{\beta}{2\omega}\right)^2 - (l^2 + k_d^2)}, \quad k_r = \frac{-\beta}{2\omega} - \sqrt{\left(\frac{\beta}{2\omega}\right)^2 - (l^2 + k_d^2)}. \quad (6.114a,b)$$

The signs of the square-root terms are chosen for reflection at a western boundary, for which, as we noted, the reflected wave has a larger (absolute) wavenumber than the incident wave. For reflection at an eastern boundary we reverse the signs.

Oceanographic relevance

The behaviour of Rossby waves at lateral boundaries is not surprisingly of some oceanographic importance, with somewhat different behaviour in the equatorial ocean and midlatitudes. Suppose that Rossby waves are generated in the middle of the ocean, for example by the wind or possibly by some fluid dynamical instability in the ocean. In brief, shorter waves will tend to propagate eastward, and be reflected back at the eastern boundary as long waves, and long waves will tend to propagate westward, being reflected back as short waves.

However, reflection at the western boundary is complicated by friction and by other waves. In mid-latitudes the reflection at a western boundary generates Rossby waves that have a short *zonal* length scale (the meridional scale of the reflected wave is the same as the incident wave if the wall is meridional), which means that their *meridional* velocity is large. Now, if the zonal wavenumber is much larger than both the meridional wavenumber l and the inverse deformation radius k_d then, using either (6.62) or (6.66) the group velocity in the x -direction is given by $c_g^x = \bar{u} + \beta/k^2$, where $\bar{u}$ is the zonal mean flow. If the mean flow is westward, so that $\bar{u}$ is negative, then very short waves will be unable to escape from the boundary; specifically, if $k > \sqrt{-\beta/\bar{u}}$ then the waves will be trapped in a western boundary layer. Even with no mean flow, the short zonal length scale means that frictional effects will be large.

In some circumstances Rossby waves incident on a western boundary also have the option of generating coastal Kelvin waves and, near the equator, reflecting back as equatorial Kelvin waves, which we introduce in the next chapter. This effect may be particularly important in the dynamics of El Niño, as we discuss in section 22.7.

6.7 GROUP VELOCITY, REVISITED

We now return to a more general discussion of group velocity. Our goal is to show that the group velocity arises in fairly general ways, not just from methods stemming from Fourier analysis or from ray theory. We first give a simple and direct derivation of group velocity that valid in the simple but important special case of a homogeneous medium. Then, in section 6.7.2, we give a rather general derivation of the *group velocity property*, namely that conserved quantities that are quadratic in the wave amplitude — that is, *wave activities* — are transported at the group velocity.

6.7.1 Group velocity in homogeneous media

Consider waves propagating in a medium in which the wave equation has the form

$$L(\psi) = \Lambda \left[\frac{\partial}{\partial t}, \frac{\partial}{\partial x} \right] \psi(x, t) = 0. \quad (6.115)$$

where Λ is a polynomial operator in the space and time derivatives, with constant coefficients, and its arguments are in square brackets. For simplicity we restrict attention to waves in one dimension, and a simple example is $\Lambda = \partial(\partial_{xx})/\partial t + \beta\partial/\partial x$ so that $L(\psi) = \partial(\partial_{xx}\psi)/\partial t + \beta\partial\psi/\partial x$. We will seek a solution of the form [c.f., (6.4)]

$$\psi(x, t) = A(x, t)e^{i\theta(x, t)}, \quad (6.116)$$

where θ is the phase of the disturbance and $A(x, t)$ is the slowly varying amplitude, so that the solution has the form of a wave packet. The phase is such that $k = \partial\theta/\partial x$ and $\omega = -\partial\theta/\partial t$, and the slowly varying nature of the envelope $A(x, t)$ is formalized by demanding that

$$\frac{1}{A} \frac{\partial A}{\partial x} \ll k, \quad \frac{1}{A} \frac{\partial A}{\partial t} \ll \omega, \quad (6.117)$$

The space and time derivatives of ψ are then given by

$$\frac{\partial \psi}{\partial x} = \left(\frac{\partial A}{\partial x} + iA \frac{\partial \theta}{\partial x} \right) e^{i\theta} = \left(\frac{\partial A}{\partial x} + iAk \right) e^{i\theta}, \quad (6.118a)$$

$$\frac{\partial \psi}{\partial t} = \left(\frac{\partial A}{\partial t} + iA \frac{\partial \theta}{\partial t} \right) e^{i\theta} = \left(\frac{\partial A}{\partial t} - iA\omega \right) e^{i\theta}, \quad (6.118b)$$

so that the wave equation becomes

$$\Lambda\psi = \Lambda \left[\frac{\partial}{\partial t} - i\omega, \frac{\partial}{\partial x} + ik \right] A = 0. \quad (6.119)$$

Noting that the space and time derivative of A are small compared to k and ω we expand the polynomial in a Taylor series about (ω, k) to obtain

$$\Lambda[-i\omega, ik]A + \frac{\partial \Lambda}{\partial(-i\omega)} \frac{\partial A}{\partial t} + \frac{\partial \Lambda}{\partial(ik)} \frac{\partial A}{\partial x} \approx 0. \quad (6.120)$$

Now, the dispersion relation for plane waves is $\Lambda[-i\omega, ik] = 0$. Taking this to be satisfied the first term in (6.120) vanishes giving

$$\frac{\partial A}{\partial t} - \frac{\partial \Lambda / \partial k}{\partial \Lambda / \partial \omega} \frac{\partial A}{\partial x} = \frac{\partial A}{\partial t} + \frac{\partial \omega}{\partial k} \frac{\partial A}{\partial x} = 0, \quad (6.121)$$

having used $(\partial \Lambda / \partial k) / (\partial \Lambda / \partial \omega) = -\partial \omega / \partial k|_{\Lambda}$. Thus, since $c_g \equiv \partial \omega / \partial k$, we have that

$$\frac{\partial A}{\partial t} + c_g \frac{\partial A}{\partial x} = 0, \quad (6.122)$$

meaning that *the envelope moves at the group velocity*.

6.7.2 ♦ The group velocity property

In section 6.6 we found that, when averaged over a wave, the energy of a Rossby wave, $\bar{E}$, obeys

$$\frac{\partial \bar{E}}{\partial t} + \nabla \cdot c_g \bar{E} = 0. \quad (6.123)$$

This equation tells us that the energy flux is equal to $c_g \bar{E}$, and it turns out to be a very general feature of waves. It is called the *group velocity property*, and it is not restricted to Rossby waves,

or energy, or homogeneous media; it holds for almost any conserved quantity that is quadratic in the wave amplitude, and we now demonstrate this in a rather general way.⁸ A quantity that is quadratic and conserved is known as a *wave activity*. (The corresponding local quantity, such as the wave activity per unit volume, is strictly called the *wave activity density*.) The group velocity property is useful because if we can determine $\mathbf{c}_g$ then we know straightaway how wave activities propagate. Energy itself is sometimes a wave activity but often not: in a growing baroclinic wave energy is drawn from the background state and is not conserved. However, we will see in chapter 10 that even in a growing baroclinic disturbance it is possible to define a conserved wave activity.

♦ *The formal procedure*

The derivation, which is rather formal, will hold for waves and wave activities that satisfy the following three assumptions.

(i) The wave activity, A , and flux, $\mathbf{F}$, obey the general conservation relation

$$\frac{\partial A}{\partial t} + \nabla \cdot \mathbf{F} = 0. \quad (6.124)$$

(ii) Both the wave activity and the flux are quadratic functions of the wave amplitude.

(iii) The waves themselves are of the general form

$$\psi = \tilde{\psi} e^{i\theta(\mathbf{x},t)} + \text{c.c.}, \quad \theta = \mathbf{k} \cdot \mathbf{x} - \omega t, \quad \omega = \omega(\mathbf{k}), \quad (6.125\text{a,b,c})$$

where (6.125c) is the dispersion relation, and ψ is any wave field. We will carry out the derivation in case in which $\tilde{\psi}$ is a constant, but the derivation may be extended to the case in which it varies slowly over a wavelength.

Given assumption (ii), the wave activity must have the general form

$$A = b + a e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + a^* e^{-2i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (6.126\text{a})$$

where the asterisk, $*$, denotes complex conjugacy, and b is a real constant and a is a complex constant. For example, suppose that $A = \psi^2$ and $\psi = c e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + c^* e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, then we find that (6.126a) is satisfied with $a = c^2$ and $b = 2cc^*$. Similarly, the flux has the general form

$$\mathbf{F} = \mathbf{g} + \mathbf{f} e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + \mathbf{f}^* e^{-2i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (6.126\text{b})$$

where $\mathbf{g}$ is a real constant vector and $\mathbf{f}$ is a complex constant vector. The mean activity and mean flux are obtained by averaging over a cycle; the oscillating terms vanish on integration and therefore the wave activity and flux are given by

$$\overline{A} = b, \quad \overline{\mathbf{F}} = \mathbf{g}, \quad (6.127)$$

where the overbar denotes the mean.

Now consider a wave with a slightly different phase, $\theta + i \delta \theta$, where $\delta \theta$ is small compared with θ . Thus, we formally replace $\mathbf{k}$ by $\mathbf{k} + i \delta \mathbf{k}$ and ω by $\omega + i \delta \omega$ where, to satisfy the dispersion relation, we have

$$\omega + i \delta \omega = \omega(\mathbf{k} + i \delta \mathbf{k}) \approx \omega(\mathbf{k}) + i \delta \mathbf{k} \cdot \frac{\partial \omega}{\partial \mathbf{k}}, \quad (6.128)$$

and therefore

$$\delta \omega = \delta \mathbf{k} \cdot \frac{\partial \omega}{\partial \mathbf{k}} = \delta \mathbf{k} \cdot \mathbf{c}_g, \quad (6.129)$$

where $\mathbf{c}_g \equiv \partial \omega / \partial \mathbf{k}$ is the group velocity.

The new wave has the general form

$$\psi' = (\tilde{\psi} + \delta\tilde{\psi})e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}e^{-\delta\mathbf{k}\cdot\mathbf{x}+\delta\omega t} + \text{c.c.}, \quad (6.130)$$

and, analogously to (6.126), the associated wave activity and flux have the forms:

$$A' = \left[b + \delta b + (a + \delta a)e^{2i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + (a^* + \delta a^*)e^{-2i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \right] e^{-2\delta\mathbf{k}\cdot\mathbf{x}+2\delta\omega t} \quad (6.131a)$$

$$\mathbf{F}' = \left[\mathbf{g} + \delta\mathbf{g} + (\mathbf{f} + \delta\mathbf{f})e^{2i(\mathbf{k}\cdot\mathbf{x}-\omega t)} + (\mathbf{f}^* + \delta\mathbf{f}^*)e^{-2i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \right] e^{-2\delta\mathbf{k}\cdot\mathbf{x}+2\delta\omega t}, \quad (6.131b)$$

where the δ quantities are small. If we now demand that A' and $\mathbf{F}'$ satisfy assumption (i), then substituting (6.131) into (6.124) gives, after a little algebra,

$$(\mathbf{g} + \delta\mathbf{g}) \cdot \delta\mathbf{k} = (b + \delta b)\delta\omega \quad (6.132)$$

and therefore at first order in δ quantities, $\mathbf{g} \cdot \delta\mathbf{k} = b\delta\omega$. Using (6.129) and (6.127) we obtain

$$\mathbf{c}_g = \frac{\mathbf{g}}{b} = \frac{\bar{\mathbf{F}}}{\bar{A}}, \quad (6.133)$$

and using this the conservation law, (6.124), becomes

$$\frac{\partial \bar{A}}{\partial t} + \nabla \cdot (\mathbf{c}_g \bar{A}) = 0. \quad (6.134)$$

Thus, for waves satisfying our three assumptions, the flux velocity — that is, the propagation velocity of the wave activity — is equal to the group velocity.

A remark: In this section and section 6.1 we have seen various derivations of the group velocity, some more kinematic, some more physical, some more general. Why are there different ways to derive it? Which is the ‘real’ derivation? Perhaps the answer is that quantity $\partial\omega/\partial\mathbf{k}$ is fundamental, and if, for example, the energy propagates at this speed then the wave envelope must move at this speed, and vice versa. The derivations imply each other, and the karma of mathematics means they have to give the same answer.

6.8 ENERGY PROPAGATION OF POINCARÉ WAVES

In the final section of this chapter we discuss the energetics of Poincaré waves (section 3.8.2) and show explicitly that the energy propagation occurs at the group velocity. We begin with the one-dimensional problem as this shows the essential aspects and the algebra is a little simpler.

6.8.1 Energetics in one dimension

The one-dimensional (i.e., no variations in the y -direction), inviscid linear shallow-water equations on the f -plane, linearized about a state of rest, are

$$\frac{\partial u}{\partial t} - f_0 v = -g \frac{\partial h}{\partial x}, \quad \frac{\partial v}{\partial t} + f_0 u = 0, \quad \frac{\partial \eta}{\partial t} = -H \frac{\partial u}{\partial x}. \quad (6.135a,b,c)$$

To obtain the dispersion relation we differentiate the first equation with respect to t and substitute from the second and third to obtain

$$\frac{\partial^2 u}{\partial t^2} - Hg \frac{\partial u}{\partial x} + f_0^2 u = 0, \quad (6.136)$$

whence, assuming solutions of the form $u = \text{Re } \tilde{u}e^{i(kx-\omega t)}$, we obtain the dispersion relation,

$$\omega^2 = f_0^2 + Hgk^2, \quad (6.137)$$

similar to (3.122) on page 125. Differentiating with respect to k gives $2\omega \partial\omega/\partial k = 2kHg$ or

$$c_g = \frac{Hg}{c_p}, \quad (6.138)$$

where $c_g = \partial\omega/\partial k$ and $c_p = \omega/k$. Using (6.137) and (6.138) the ratio of the group and phase velocities is found to be

$$\frac{c_g}{c_p} = \frac{L_d^2 k^2}{1 + L_d^2 k^2}, \quad (6.139)$$

where $L_d = \sqrt{gH}/f$ is the deformation radius. This ratio is always less than unity, tending to zero in the long-wave limit ($kL_d \ll 1$) and to unity for short waves ($kL_d \gg 1$).

The energy equations are obtained by multiplying the three equations of (6.135) by u, v and η respectively, and adding, to give

$$\frac{\partial E}{\partial t} + \frac{\partial F}{\partial x} = 0, \quad (6.140a)$$

where

$$E = \frac{1}{2}(Hu^2 + Hv^2 + g\eta^2), \quad F = gHu\eta, \quad (6.140b)$$

are the energy density and the energy flux, respectively. Here in the linear approximation the energy is transported only by the pressure term, whereas in the full nonlinear equations there is also an advective transport.

The group velocity property for Poincaré waves

To specialize to the case of propagating waves we need to average over a wavelength and use the phase relationships between u, v and η implied by the equations of motion. Writing $u = \text{Re } \tilde{u}e^{i(kx-\omega t)}$, and similarly for v and η , we have

$$\tilde{v} = -if \frac{\tilde{\eta}}{Hk}, \quad \tilde{u} = \omega \frac{\tilde{\eta}}{Hk}. \quad (6.141a,b)$$

The kinetic energy, averaged over a wavelength, is then

$$KE = \frac{1}{2}H(\overline{u^2} + \overline{v^2}) = \frac{1}{4}(\omega^2 + f^2) \frac{\overline{\tilde{\eta}^2}}{Hk^2} = \frac{1}{4} \frac{\omega^2 + f^2}{\omega^2 - f^2} g\overline{\tilde{\eta}^2} \quad (6.142)$$

using (6.141) and the dispersion relation, with the extra factor of one half arising from the averaging over a wavelength. Similarly, the potential energy of the wave is

$$PE = \frac{1}{2}g\overline{\eta^2} = \frac{1}{4}g\overline{\tilde{\eta}^2} \quad (6.143)$$

Thus, the ratio of kinetic to potential energy is just

$$\frac{KE}{PE} = \frac{\omega^2 + f^2}{\omega^2 - f^2} = 1 + \frac{2}{k^2 L_d^2} \quad (6.144)$$

using the dispersion relation, and where $L_d = \sqrt{gH}/f$ is the deformation radius. Thus, the kinetic energy is always *greater* than the potential energy (there is no equipartition in this problem), with the ratio approaching unity for small scales (large k).

The total energy (kinetic plus potential) is then

$$KE + PE = \frac{1}{4} \left(\frac{\omega^2 + f^2}{\omega^2 - f^2} + 1 \right) g \tilde{\eta}^2 = \frac{1}{2} \frac{\omega^2}{k^2 H} \tilde{\eta}^2 = \frac{1}{2} \frac{c_p^2}{H} \tilde{\eta}^2, \quad (6.145)$$

again using the dispersion relation. The energy flux, F , averaged over a wavelength, is

$$F = g H \overline{u \eta} = \frac{1}{2} \frac{g \omega}{k} \tilde{\eta}^2 = \frac{1}{2} g c_p \tilde{\eta}^2. \quad (6.146)$$

From (6.145) and (6.146) the flux and the energy are evidently related by

$$F = \frac{H g}{c_p} E = c_g E, \quad (6.147)$$

using (6.138). That is, the energy flux is equal to the group velocity times the energy itself. Note that in this problem there is no flux in the y direction, because v and η are exactly out of phase from (6.141a).

6.8.2 ♦ Energetics in two dimensions

The derivations of the preceding section carry through, *mutatis mutandis*, in the full two-dimensional case. We will give only the key results and allow the reader to fill in the algebra.

As derived in section 3.8.2 the dispersion relation is

$$\omega^2 = f_0^2 + gH(k^2 + l^2). \quad (6.148)$$

The relation between the components of the group velocity and the phase speed is very similar to the one-dimensional case, and in particular we have

$$c_g^x = \frac{\partial \omega}{\partial k} = gH \frac{k}{\omega} = \frac{gH}{c_p^x}, \quad c_g^y = \frac{\partial \omega}{\partial l} = gH \frac{l}{\omega} = \frac{gH}{c_p^y}. \quad (6.149)$$

The magnitude of the group velocity is $c_g \equiv |\mathbf{c}_g| = (c_g^{x2} + c_g^{y2})^{1/2}$. The magnitude of the phase speed, in the direction of travel of the wave crests, is $c_p = \omega/(k^2 + l^2)^{1/2}$ (note that in general this is *smaller* than the phase speed in either the x or y directions, ω/k or ω/l). Thus, we have

$$c_g^2 = (gH)^2 \frac{k^2 + l^2}{\omega^2} = \frac{(gH)^2}{c_p^2}, \quad \mathbf{c}_g = \left(\frac{gH}{c_p K} \right) \mathbf{k}, \quad (6.150)$$

which is analogous to (6.138). The ratio of the magnitudes of the group and phase velocities is, analogously to (6.139),

$$\frac{c_g}{c_p} = \frac{gH}{c_p^2} = \frac{L_d^2 K^2}{1 + L_d^2 K^2}, \quad (6.151)$$

where $K^2 = k^2 + l^2$. As in the one-dimensional case the group velocity is large for short waves, in which rotation plays no role, and small for long waves.

The energy equation is found to be

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{F} = 0 \quad (6.152a)$$

with

$$E = \frac{1}{2}(Hu^2 + Hv^2 + g\eta^2), \quad F = gH(u\mathbf{i} + v\mathbf{j})\eta. \quad (6.152b)$$

From the equations of motion the phase relations between the fields are found to be

$$\tilde{v} = \frac{\omega l - ikf}{HK^2}\tilde{\eta}, \quad \tilde{u} = \frac{\omega k - ilf}{HK^2}\tilde{\eta}, \quad (6.153)$$

so that the kinetic energy is given by, similar to (6.142),

$$KE = \frac{1}{2}H(\overline{u^2} + \overline{v^2}) = \frac{1}{4}(\omega^2 + f^2)\frac{\tilde{\eta}^2}{Hk^2} = \frac{1}{4}\frac{\omega^2 + f^2}{\omega^2 - f^2}g\tilde{\eta}^2, \quad (6.154)$$

and the potential energy by

$$PE = \frac{1}{2}g\overline{\eta^2} = \frac{1}{4}g\tilde{\eta}^2 \quad (6.155)$$

The ratio of the kinetic and potential energies is given by

$$\frac{KE}{PE} = \frac{\omega^2 + f^2}{\omega^2 - f^2} = 1 + \frac{2}{K^2 L_d^2} \quad (6.156)$$

The total (kinetic plus potential) energy is given by

$$E = KE + PE = \frac{1}{4}\left(\frac{\omega^2 + f^2}{\omega^2 - f^2} + 1\right)g\tilde{\eta}^2 = \frac{1}{2}\frac{\omega^2}{K^2 H}\tilde{\eta}^2 = \frac{1}{2}\frac{c_p^2}{H}\tilde{\eta}^2, \quad (6.157)$$

The energy flux, F , averaged over a wavelength, is

$$\mathbf{F} = gH\overline{\mathbf{u}\eta} = \frac{1}{2}\frac{g\omega}{k^2 + l^2}\tilde{\eta}^2\mathbf{k} = \frac{1}{2}\frac{g\omega}{K^2}\tilde{\eta}^2\mathbf{k}, \quad (6.158)$$

using (6.153) and where $\mathbf{k} = k\mathbf{i} + l\mathbf{j}$ is the wavevector of the wave.

From (6.157) and (6.158), and using (6.150), the flux and the energy are found to be related by

$$\mathbf{F} = \mathbf{c}_g E. \quad (6.159)$$

That is, the energy flux is equal to the group velocity times the energy itself.

6.A APPENDIX: THE WKB APPROXIMATION FOR LINEAR WAVES

The WKB method (after Wentzel, Kramers and Brillouin, the last people to discover the technique⁹) is a way of finding approximate solutions to certain linear differential equations in which the term with the highest derivative is multiplied by a small parameter. The theory for such equations is quite extensive but our interests are modest, being mainly in dispersive waves, and WKB theory can be used to find approximate solutions in cases in which the coefficients of the wave equation vary slowly in space or time.

Consider an equation of the form

$$\frac{d^2\xi}{dz^2} + m^2(z)\xi = 0, \quad (6.160)$$

where $m^2(z)$ is positive for wavelike solutions. If m is constant the solution has the harmonic form

$$\xi = \text{Re } A_0 e^{imz} \quad (6.161)$$

where A_0 is a complex constant. If m varies only ‘slowly’ with z — meaning that the variations occur on a scale much longer than $1/m$ — one might reasonably expect that the harmonic solution above would provide a reasonable first approximation; that is, we expect the solution to locally look like a plane wave with local wavenumber $m(z)$. However, we might also expect that the solution would not be *exactly* of the form $\exp(im(z)z)$, because the phase of ξ is $\theta(z) = mz$, so that $d\theta/dz = m + z dm/dz \neq m$. Thus, in (6.161) m is not the wavenumber unless m is constant. Nevertheless, this argument suggests that we seek solutions of a similar form to (6.161), and we find such solutions by way of a perturbation expansion below. The condition that variations in m , or in the wavelength m^{-1} , occur only slowly may be expressed as

$$\frac{m}{|\partial m / \partial z|} \gg m^{-1} \quad \text{or} \quad \left| \frac{\partial m}{\partial z} \right| \ll m^2. \quad (6.162)$$

This condition will generally be satisfied if variations in the background state, or in the medium, occur on a scale much longer than the wavelength.

6.A.1 Solution by perturbation expansion

To explicitly recognize the rapid variation of m we rescale the coordinate z with a small parameter ϵ ; that is, we let $\hat{z} = \epsilon z$ where $\hat{z}$ varies by $\mathcal{O}(1)$ over the scale on which m varies. Equation (6.160) becomes

$$\epsilon^2 \frac{d^2 \xi}{d\hat{z}^2} + m^2(\hat{z}) \xi = 0, \quad (6.163)$$

and we may now suppose that all variables are $\mathcal{O}(1)$. If m were constant the solution would be of the form $\xi = A \exp(m\hat{z}/\epsilon)$ and this suggests that we look for a solution to (6.163) of the form

$$\xi(z) = e^{g(\hat{z})/\epsilon}, \quad (6.164)$$

where $g(\hat{z})$ is some as yet unknown function. We then have, with primes denoting derivatives,

$$\xi' = \frac{1}{\epsilon} g' e^{g/\epsilon}, \quad \xi'' = \left(\frac{1}{\epsilon^2} g'^2 + \frac{1}{\epsilon} g'' \right) e^{g/\epsilon}. \quad (6.165a,b)$$

Using these expressions in (6.163) yields

$$\epsilon g'' + g'^2 + m^2 = 0, \quad (6.166)$$

and if we let $g = \int h d\hat{z}$ we obtain

$$\epsilon \frac{dh}{d\hat{z}} + h^2 + m^2 = 0. \quad (6.167)$$

To obtain a solution of this equation we expand h in powers of the small parameter ϵ ,

$$h(\hat{z}; \epsilon) = h_0(\hat{z}) + \epsilon h_1(\hat{z}) + \epsilon^2 h_2(\hat{z}) + \dots. \quad (6.168)$$

Substituting this in (6.167) and setting successive powers of ϵ to zero gives, at first and second order,

$$h_0^2 + m^2 = 0, \quad 2h_0 h_1 + \frac{dh_0}{d\hat{z}} = 0. \quad (6.169a,b)$$

The solutions of these equations are

$$h_0 = \pm im, \quad h_1(\hat{z}) = -\frac{1}{2} \frac{d}{d\hat{z}} \ln \frac{m(\hat{z})}{m_0}. \quad (6.170a,b)$$

where m_0 is a constant. Now, ignoring higher-order terms, (6.164) may be written in terms of h_0 and h_1 as

$$\xi(\hat{z}) = \exp\left(\int h_0 d\hat{z}/\epsilon\right) \exp\left(\int h_1 d\hat{z}\right), \quad (6.171)$$

and, using (6.170) and with z in place of $\hat{z}$, we obtain

$$\xi(z) = A_0 m^{-1/2} \exp\left(\pm i \int m dz\right), \quad (6.172)$$

where A_0 is a constant, and this is the wkb solution to (6.160). Explicitly, the solution is

$$\xi(z) = A_0 m^{-1/2} \exp\left(i \int m dz\right) + A_0^* m^{-1/2} \exp\left(-i \int m dz\right). \quad (6.173)$$

or, in terms of real quantities,

$$\xi(z) = B_0 m^{-1/2} \cos\left(\int m dz\right) + C_0 m^{-1/2} \sin\left(\int m dz\right). \quad (6.174)$$

where B_0 , and C_0 are real constants.

A property of (6.172) is that the derivative of the phase is just m ; that is, m is indeed the local wavenumber. Note that a crucial aspect of the derivation is that m varies slowly, so that there is a small parameter, ϵ , in the problem. Having said this, it is often the case that wkb theory can provide qualitative guidance even when there is little scale separation between the variation of the background state and the wavelength. Asymptotics often works when it seemingly should not.

6.A.2 Alternate derivation

Another, rather quicker but less systematic, way to obtain the same result is to seek solutions of (6.160) in the form

$$\xi = A(z) e^{i\theta(z)}, \quad (6.175)$$

where $A(z)$ and $\theta(z)$ are both presumptively real. Using (6.175) in (6.160) yields

$$i \left[2 \frac{dA}{dz} \frac{d\theta}{dz} + A \frac{d^2\theta}{dz^2} \right] + \left[A \left(\frac{d\theta}{dz} \right)^2 - \frac{d^2A}{dz^2} - m^2 A \right] = 0. \quad (6.176)$$

The terms in square brackets must each be zero. The wkb approximation is to assume that the amplitude varies sufficiently slowly that $|A^{-1} d^2A/dz^2| \ll m^2$, and hence that the term involving d^2A/dz^2 may be neglected. The real and imaginary parts of (6.176) become

$$\left(\frac{d\theta}{dz} \right)^2 = m^2, \quad 2 \frac{dA}{dz} \frac{d\theta}{dz} + A \frac{d^2\theta}{dz^2} = 0. \quad (6.177a,b)$$

These two equations are very similar to (6.169). The solution of the first one is

$$\theta = \pm \int m dz, \quad (6.178)$$

and substituting this into (6.177b) gives

$$2 \frac{dA}{dz} m + A \frac{dm}{dz} = 0, \quad (6.179)$$

the solution of which is

$$A = A_0 m^{-1/2}. \quad (6.180)$$

Using (6.178) and (6.180) in (6.175) recovers (6.172).

Notes

- 1 For example *Linear and Nonlinear Waves* by G. B. Whitham or *Nonlinear Dispersive Waves* by M. J. Ablowitz.
- 2 Group velocity seems to have been first articulated in about 1841 by the Irish mathematician and physicist W. R. Hamilton (1806–1865), who is also remembered for his formulation of ‘Hamiltonian mechanics’. Hamilton may have been motivated by optics, and it was George Stokes, Osborne Reynolds and Lord Rayleigh who further developed and generalized the idea in a more hydrodynamic context.
- 3 More detailed treatments of ray theory and related matters are given by Whitham (1974), Lighthill (1978) and LeBlond & Mysak (1980).
- 4 What are now called Rossby waves were probably first discovered in a theoretical context by Hough (1897, 1898). He considered the linear shallow water equations on a sphere (i.e., Laplace’s tidal equations) expanding the solution in powers of the sine of latitude, and obtained two classes of waves: long, rotationally modified, gravity waves and a balanced wave dependent on variations in Coriolis parameter. However, his work was mainly aimed at understanding ocean tides and it was not until the topic was revisited by Rossby (1939) that the meteorological relevance was appreciated. Rossby used the beta-plane approximation in Cartesian co-ordinates, and the simplicity of the presentation along with the meteorological context led to the work attracting significant notice.
- 5 This non-Doppler effect also arises quite generally, even in models in height coordinates. See White (1977) and problem 5.??.
- 6 See Chapman & Lindzen (1970).
- 7 Following Longuet-Higgins (1964).
- 8 The form of this derivation was originally given by Hayes (1977) in the context of wave energy. See Vanneste & Shepherd (1998) for generalizations.
- 9 A description of the wkb method, also called the jwkb method and sometimes the geometrical optics approximation, can be found in many books on perturbation methods, for example Simmonds & Mann (1998), Holmes (2013) and Bender & Orszag (1978). Developments in perturbation theory and multiple scale analysis have generalized the method to the extent that ‘wkb’ does not appear as a separate topic, or in the index, in the well-known book by Kevorkian & Cole (2011). Wentzel, Kramers and Brillouin separately presented the technique in 1926 as a way to find approximate solutions of the Schrödinger equation. Harold Jeffreys, a mathematical geophysicist, had proposed a similar technique in 1924, and Rayleigh in 1912 had already addressed some aspects of the theory. A mathematical treatment of the topic was in fact given by Joseph Liouville and George Green in 1837, with even earlier relevant work by Francesco Carlini, an Italian astronomer and director of an observatory in Milan, in 1817. The story thus affirms the hypothesis that methods are named after the last people to discover them...

Further reading

Waves

- Andrews, D. G., Holton, J. R. & Leovy, C.B., 1987. *Middle Atmosphere Dynamics*.
Provides a discussion of a number of topics in wave dynamics and wave–mean-flow interaction, mainly in the context of stratospheric dynamics.
- Bühler, O. 2009. *Waves and Mean Flows*.
A modern, advanced, discussion of waves, mean flows and their interaction, including the Transformed Eulerian Mean, the Generalized Lagrangian Mean, and more.
- LeBlond, P. H. & Mysak, L. A. 1980. *Waves in the ocean*
A comprehensive review of the dynamics of many kinds of waves in the ocean.
- Pedlosky, J., 2003 *Waves in the Ocean and Atmosphere: Introduction to Wave Dynamics*.
A compact, informal introduction to the main waves to be found in the atmosphere and ocean, in the style of lecture notes.

Instabilities

The following two books cover most forms of hydrodynamic instability except, alas, baroclinic instability.

Chandrasekhar, S., 1961. *Hydrodynamic and Hydromagnetic Stability*.

This book has become an enduring classic, but has no discussion of baroclinic instability at all.

Drazin P. & Reid, W. H., 1981. *Hydrodynamic Stability*.

A standard text on hydrodynamic instability theory. It has straightforward and extensive discussions of most of the classic cases, but only a brief treatment of baroclinic instability.

Turbulence

There are numerous books on turbulence, but one place to start is

Tennekes, H. & Lumley, J., 1972. *A First Course in Turbulence*.

The book remains a classic introduction to the subject.

Another book that has stood the test of time is

Monin & Yaglom 1966. *Statistical Fluid mechanics*.

It is encyclopædic in content and contains a wealth of information, and particularly strong on turbulent diffusion.

Two more modern general introductions and references are

Davidson, P. D., 2015. *Turbulence: An Introduction for Scientists and Engineers*.

Pope, S. B., 2000. *Turbulent FLOws*.

Both of these are written at about the graduate student level.

The following two books deal mainly with the statistical aspects of turbulence.

Frisch, U., 1995. *Turbulence: the Legacy of A. N. Kolmogorov*.

Lesieur, M., 1997. *Turbulence in Fluids*.

Two books containing review articles covering a range of topics related to jets and turbulence on both large and small scales, in atmospheres and oceans, are

Galperin, B. & Read, P. A. Eds., 2016. *Jets*

Baumert, H. Z., Simpson, J. & Sündermann, J. Eds., 2003. *Marine Turbulence: Theories, Observations and Models*.

The more applied side of prediction and predictability is discussed by

Kalnay, E., 2003. *Atmospheric Modeling, Data Assimilation and Predictability*.

*It mounts at sea, a concave wall,
Down-ribbed with shine,
And pushes forward, building tall,
Its steep incline.*

Thom Gunn, *From the Wave*, 1971.

CHAPTER 7

Gravity Waves

GRAVITY WAVES ARE, UNSURPRISINGLY, waves in a fluid in which gravity provides the restoring force. (Gravitational waves occur in general relativity.) For gravity to have an effect the fluid density must vary, and thus the waves must either exist at a fluid interface or stratification must be present — and a fluid interface is just an abrupt form of stratification. It is thus common to think of gravity waves as being either internal waves or surface waves: the former being in the interior of a fluid where the density changes may be continuous, and the latter at a fluid interface. Naturally enough the two waves have many similarities — indeed surface waves (also called interfacial waves) are a limiting form of internal waves, when the density variations in the vertical become discontinuous. We considered such interfacial waves in the hydrostatic, shallow water case in chapter 3, and we will first extend that to the nonhydrostatic case. We then consider internal waves in the continuously stratified equations and that constitutes the bulk of the chapter.¹

In most of the chapter we will restrict attention to the Boussinesq equations, because in making the incompressibility approximation sound waves are eliminated, greatly simplifying the treatment. In the atmosphere the Boussinesq equations are not a quantitatively good approximation except for motions of a small vertical extent; the anelastic equations improve matters in allowing for a vertical variation of the basic state density, an effect important when considering the vertical propagation of gravity waves high into the atmosphere. Nevertheless, no truly new types of waves are introduced in this way, and so we leave the details to the original literature. If, on the other hand, the fluid is truly compressible then sound waves make themselves heard, and we consider the algebraically complex case of *acoustic-gravity* waves at the end of this chapter. We begin with the simpler case of surface gravity waves atop a constant density fluid.

7.1 SURFACE GRAVITY WAVES

Consider an incompressible fluid with a free surface and a flat bottom that obeys the three-dimensional momentum and mass continuity equations, namely

$$\frac{D\mathbf{v}}{Dt} = -\nabla\phi - g\mathbf{k}, \quad \nabla \cdot \mathbf{v} = 0, \quad (7.1)$$

using our standard notation where $\phi = p/\rho_0$. The free surface at the top of the fluid is at $z = \eta(x, y, t)$, the mean position of the free surface is at $z = 0$ and the bottom of the fluid, assumed flat, is at $z = -H$ — refer to Fig. 3.1 on page 106.

In a state of rest the pressure, ϕ_0 say, is given by hydrostatic balance and so $\phi_0 = -gz$. If we write $\phi = -gz + \phi'$ the momentum equation becomes, without approximation,

$$\frac{D\mathbf{v}}{Dt} = -\nabla\phi'. \quad (7.2)$$

Linearizing the equations of motion about such a resting state straightforwardly yields

$$\frac{\partial \mathbf{v}'}{\partial t} = -\nabla\phi', \quad \nabla \cdot \mathbf{v}' = 0, \quad (7.3a,b)$$

where a prime denotes a perturbation quantity in the usual way. We now proceed by expressing the problem solely in terms of pressure. (An equivalent alternative would be to use a velocity potential, ξ say, such that $\mathbf{v} = \nabla\xi$. Such a procedure is possible because, from (7.3a) the flow is irrotational.) Taking the divergence of (7.3a) and using (7.3b) gives us Laplace's equation for the pressure, namely

$$\nabla^2\phi' = 0. \quad (7.4)$$

This equation has no explicit time dependence, but the boundary conditions are time dependent and that is how we will obtain the dispersion relation.

7.1.1 Boundary conditions

Since (7.4) is an equation for pressure we seek boundary conditions on pressure. At the bottom of the fluid ($z = -H$) the condition that $w = 0$ may be turned into a condition on pressure using (7.3a), namely that

$$\frac{\partial\phi'}{\partial z} = 0 \quad \text{at } z = -H. \quad (7.5)$$

At the top surface, $z = \eta$, the pressure must equal that of the atmosphere above. We will take this to be a constant, and in particular zero, so that $\phi = 0$ at $z = \eta$. Now, the perturbation pressure is given by $\phi = -gz + \phi'$, so that at $z = \eta$ we obtain

$$\phi' = g\eta \quad \text{at } z = \eta. \quad (7.6)$$

A second boundary condition at the top is the kinematic condition that a fluid parcel in the free surface must remain within the fluid, and therefore that (with full nonlinearity)

$$\frac{D}{Dt}(z - \eta) = 0. \quad (7.7)$$

If we linearize this equation and use the definition of w we obtain $w' = \partial\eta/\partial t$ at $z = \eta$, which using (7.6) becomes $w' = g^{-1}\partial\phi'/\partial t$. Using the vertical component of the momentum equation, (7.3a), we obtain the pressure boundary condition

$$\frac{1}{g} \frac{\partial^2\phi'}{\partial t^2} = -\frac{\partial\phi'}{\partial z} \quad \text{at } z = \eta. \quad (7.8)$$

The value of η is in fact unknown without solving the problem itself, and in the general (nonlinear) case we have to solve the whole problem in a self-consistent fashion. However, in the linear problem η is presumptively small (remember we are linearizing the free surface about $z = 0$) and we will apply this boundary condition at $z = 0$ rather than at $z = \eta$, for the error will only be second order.

Having established the equations and the boundary conditions, and noting that we will be dealing exclusively with linear equations in the rest of this section (and for most of this chapter), we'll now drop the primes on perturbation quantities unless needed.

7.1.2 Wave solutions

We now seek solutions to (7.4) in the form

$$\phi = \text{Re } \Phi(z) \exp(i[\mathbf{k} \cdot \mathbf{x} - \omega t]) \quad (7.9)$$

where $\mathbf{x} = ix + jy$ and $\mathbf{k} = ik + jl$ and Re denotes that the real part is to be taken, a notation that we subsequently drop unless it causes ambiguity. Equation (7.4) becomes

$$\frac{d^2 \Phi}{dz^2} - K^2 \Phi = 0, \quad (7.10)$$

where $K^2 = k^2 + l^2$ and the boundary conditions are that $d\Phi/dz = 0$ at $z = -H$ and $d^2\Phi/dz^2 = -gd\Phi/dz$ at $z = 0$. The bottom boundary condition is satisfied by a solution of the form

$$\Phi = A \cosh K(z + H). \quad (7.11)$$

Substituting into the top boundary condition, (7.8) at $z = 0$, we obtain

$$-\omega^2 \cosh KH = gK \sinh KH = 0, \quad (7.12)$$

or

$$\omega = \pm \sqrt{gK \tanh KH}. \quad (7.13)$$

This is the dispersion relation for surface gravity waves. The corresponding phase speed is given by

$$c_p = \frac{\omega}{K} = \pm \sqrt{gH} \left(\frac{\tanh KH}{KH} \right)^{1/2}. \quad (7.14)$$

Using (7.9) and (7.11) the full solution for the pressure field is

$$\phi = \text{Re } \Phi_0 \cosh K(z + H) \exp(i[\mathbf{k} \cdot \mathbf{x} - \omega t]) \quad (7.15)$$

with ω given by (7.13) and the amplitude Φ_0 being set by the initial conditions. It is convenient to write the amplitude Φ_0 in terms of the amplitude of the free surface elevation, η_0 , using the upper boundary condition that $\phi = g\eta$, so that $\eta_0 = \Phi_0/g$. The other field variables may be found from (7.3a) and are given by

$$u = \eta_0 \frac{k}{\omega} g C \cosh K(z + H), \quad v = \eta_0 \frac{l}{\omega} g C \cosh K(z + H), \quad w = -i\eta_0 \frac{K}{\omega} g C \sinh K(z + H) \quad (7.16a,b,c)$$

where $C = \exp(i[\mathbf{k} \cdot \mathbf{x} - \omega t]) / \cosh KH$, and only the real parts of each expression that should be taken. Thus, if we take η_0 to be real then u and v vary like $\cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$ and w varies as $\sin(\mathbf{k} \cdot \mathbf{x} - \omega t)$, and this is what we will assume.

7.1.3 Properties of the solution

First, from (7.13) we see that for each wavevector amplitude there are two waves propagating in opposite directions, with a frequency and phase speed that depend only on the wavelength K and not the orientation of the wave vector. Second, the waves are *dispersive*. That is, similar to Rossby waves but unlike light waves in a vacuum or shallow water waves, the phase speed is different for waves of different wavelengths. Since the frequency is a function only of K (and not of k or l individually) the group velocity is parallel to the wave vector itself and is given by

$$\mathbf{c}_g = \nabla_{\mathbf{k}} \omega = \frac{\partial \omega}{\partial K} \frac{\mathbf{k}}{K}, \quad (7.17)$$

where $\mathbf{k} = k\mathbf{i} + l\mathbf{j}$ and $\mathbf{k}/K$ is the unit vector in the direction of propagation. Using the dispersion relation $\omega^2 = gK \tanh KH$ we obtain

$$2\omega \frac{\partial \omega}{\partial K} = g \left(\tanh KH + \frac{KH}{\cosh^2 KH} \right) \quad (7.18)$$

so that

$$\mathbf{c}_g = \frac{g}{2c_p K} \left(\tanh KH + \frac{KH}{\cosh^2 KH} \right) \mathbf{k} \quad (7.19)$$

and, now using (7.14), the ratio of the group speed (i.e., the magnitude of the group velocity) to the phase speed is given by

$$\frac{c_g}{c_p} = \frac{1}{2} \left(1 + \frac{2KH}{\sinh 2KH} \right), \quad (7.20)$$

having used the relation $2 \sinh x \cosh x = \sinh 2x$.

We may note two important limiting cases.

1. The long wavelength or shallow water limit, $KH \ll 1$. It is this limit that is relevant to large-scale flow in the ocean and atmosphere. In this limit the wavelength is much greater than the depth of the fluid and the dispersion relation (7.13) reduces to $\omega = K\sqrt{gH}$ (since for small x , $\tanh x \rightarrow x$) and $c_p = c_g = \sqrt{gH}$, and the waves are nondispersive. This result is apparent from (7.20) in the limit of $KH \ll 1$. As expected, this is the same dispersion relation as was previously derived *ab initio* for shallow water waves in chapter 3. This limit is appropriate as water waves approach the shore and start feeling the bottom, and for long waves such as tides and tsunamis.

The pressure field in this limit is given, using (7.15),

$$\phi = \eta_0 g \exp(i[\mathbf{k} \cdot \mathbf{x} - \omega t]). \quad (7.21)$$

This is the *perturbation* pressure associated with the wave, and evidently it does not depend on depth. The total pressure at a given point in the fluid is given by the static pressure plus perturbation pressure and this is, including the density ρ_0 ,

$$p = -\rho_0 g z + \rho_0 \phi = \rho_0 g(\eta - z). \quad (7.22)$$

Evidently, the pressure in the shallow water limit is hydrostatic. If $1/k > 20H$ the error in this approximation less than 3%.

2. The short wavelength or deep water limit, $KH \gg 1$. For large KH , $\tanh KH \rightarrow 1$ so that the dispersion relation becomes $\omega^2 = gK$ and $c_p^2 = g/K$. These waves are dispersive, with long waves travelling faster than short waves. A familiar manifestation of this arises when a rock is thrown into a pool; initially, waves of all wavelengths are excited (for the initial disturbance is like a delta function), but the long waves propagate away faster than the short waves and reach distance objects first. The group speed in this case is given by

$$c_g = \frac{\partial \omega}{\partial k} = \frac{g}{2\omega} = \frac{1}{2} \sqrt{\frac{g}{K}} = \frac{c_p}{2}. \quad (7.23)$$

This result is also apparent from (7.20) in the limit of short waves, $KH \gg 1$, and it has an interesting consequence for wave packets. Consider a packet of short waves moving in the positive x -direction; the envelope moves with the group speed and the individual crests move with the phase speed, so that individual crests enter the packet from the rear and travel through the packet, exiting at the front.

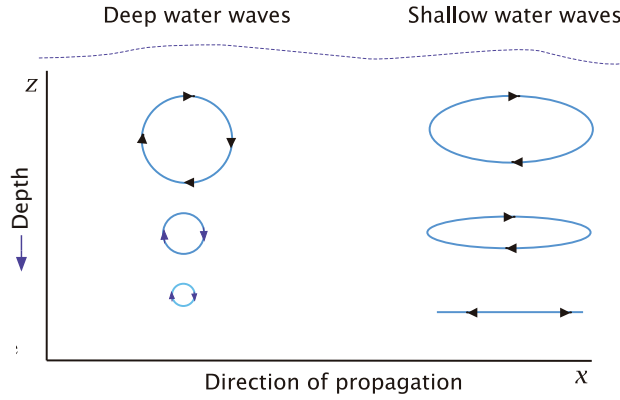


Fig. 7.1 Parcel motion for deep and shallow water waves. The motion is circular for deep water waves, with an amplitude that decreases exponentially with depth. The motion is elliptical for shallow water waves, but the horizontal excursion is independent of depth and the vertical excursion decays linearly with depth.

Parcel motion

The trajectories of water parcels is rather interesting in water waves. It turns out that in deep water the parcels make circular orbits with an amplitude diminishing with depth, whereas shallow water waves trace elliptic paths, as illustrated in Fig. 7.1 and as we now explain.

We obtain the parcel excursions using the expressions for velocity (7.16), taking $v = 0$ without loss of generality. For shallow water waves ($KH \ll 1$) u is depth independent and the velocity and the excursion in the x direction, which we denote as X , are given by

$$u = \eta_0 \frac{kg}{\omega} \cos(kx - \omega t), \quad X = -\eta_0 \frac{gk}{\omega^2} \sin(kx - \omega t), \quad (7.24a)$$

and this is independent of z . The excursion in the z direction, Z , is given by

$$w = \eta_0 \frac{k^2}{\omega} (z + H) \sin(kx - \omega t), \quad Z = \eta_0 \frac{gk^2}{\omega^2} (z + H) \cos(kx - \omega t). \quad (7.24b)$$

where $\omega = k\sqrt{gH}$. Note that at $z = 0$ $Z = \eta_0$, as expected. The above expressions for X and Z are, at some fixed location x and z , parametric representations of an ellipse. As z varies the horizontal amplitude of the ellipses remains constant whereas the vertical amplitude decreases linearly from the top $z = 0$ to a zero amplitude at the bottom, $z = -H$. The vertical amplitude is also generally much less than the horizontal amplitude, by the ratio

$$\frac{|Z|}{|X|} = \frac{|w|}{|u|} \sim kH \ll 1. \quad (7.25)$$

That is, the fluid motion is mostly horizontal.

In the deep water limit, $kH \gg 1$, the horizontal and vertical velocities and excursions are given by

$$u = \eta_0 \frac{kg}{\omega} \exp kz \cos(kx - \omega t), \quad X = -\eta_0 \frac{kg}{\omega^2} \exp kz \sin(kx - \omega t), \quad (7.26a)$$

$$w = \eta_0 \frac{kg}{\omega} \exp kz \sin(kx - \omega t), \quad Z = \eta_0 \frac{kg}{\omega^2} \exp kz \cos(kx - \omega t), \quad (7.26b)$$

where $\omega^2 = gk$, and again we have that $Z = \eta$ at $z = 0$. The expressions for X and Z , having the same amplitude, are now parametric representations of circles whose amplitudes diminish exponentially with depth. Evidently, all the dynamical variables decrease exponentially with depth, with an e-folding scale of the wavelength itself. The wave field cannot feel the bottom of the fluid container and all the expressions become independent of the water depth H .

♦ *Energy propagation*

For our final discussion on this topic we look at the energy and energy propagation of surface waves. The kinetic energy per unit horizontal area is given by

$$KE = \int_{-H}^0 \frac{1}{2} \rho_0 \mathbf{v}^2 dz. \quad (7.27)$$

The upper limit on the integration is taken to be $z = 0$, rather than $z = \eta$, because using the latter would lead to a term of order $\eta \mathbf{v}^2$, which is third order in perturbation quantities. The potential energy per unit horizontal area is

$$PE = \int_{-H}^{\eta} \rho_0 g z dz = \frac{\rho_0 g}{2} (\eta^2 - H^2). \quad (7.28)$$

The integral now must be over the complete depth of the fluid in order to calculate the potential energy to quadratic order. The term in H^2 is a constant and so is largely irrelevant to the problem of energy propagation. Also, since ρ_0 is a constant we will set its value to unity.

The kinetic energy equation is obtained by taking the dot product of the linearized momentum equation, (7.3a) with $\mathbf{v}$ and integrating over the depth of the fluid to give

$$\int_{-H}^0 dz \left[\frac{\partial}{\partial t} \frac{\mathbf{v}^2}{2} + \nabla \cdot (\mathbf{u}\phi) + \frac{\partial w\phi}{\partial z} \right] = 0, \quad (7.29)$$

noting that $\mathbf{v} = \mathbf{u} + w\mathbf{k}$ and $\nabla \cdot \mathbf{v} = 0$. The boundary conditions on w are that $w = 0$ at $Z = -H$ and $w = \partial\eta/\partial t$ at $z = 0$. Further, at $z = 0$ $\phi = g\eta$, and using these results (7.29) becomes

$$\int_{-H}^0 dz \left[\frac{\partial}{\partial t} \frac{\mathbf{v}^2}{2} + \nabla \cdot (\mathbf{u}\phi) \right] + g \frac{\partial}{\partial t} \frac{\eta^2}{2} = 0, \quad (7.30)$$

which, using (7.27) and (7.28), is just

$$\frac{\partial}{\partial t} (KE + PE) + \nabla \cdot \mathbf{F} = 0. \quad (7.31)$$

where $\mathbf{F} = \int_{-H}^z \mathbf{u}\phi dz$ is the energy flux, a vector with only horizontal components. (Thus, the divergence term in (7.31) is just a horizontal divergence.)

Equation (7.31) is an energy conservation equation for the linearized equations. It is fairly general at the moment, for we have not specialized to the case of *wave* motion. Let's do that now, by using the properties of the waves derived above and averaging over a wave period. Without loss of generality we'll assume the waves are propagating in the x direction so that $v = 0$ and $K = k$; nevertheless, the calculation is rather algebraic and the trusting reader may skim it.

The kinetic energy averaged over a wave period, $\overline{KE}$ is given by

$$\begin{aligned} \overline{KE} &= \frac{\omega}{2\pi} \int dt \left(\int_{-H}^0 \frac{1}{2} \mathbf{v}^2 dz \right) \\ &= \frac{k^2 \eta_0^2 g^2}{2\omega^2 \cosh^2 kH} \frac{\omega}{2\pi} \int dt \int dz [\cosh^2 k(z+H) \cos^2(kx - \omega t) + \sinh^2 k(z+H) \sin^2(kx - \omega t)]. \end{aligned} \quad (7.32)$$

In this expression the time integrals range from 0 to $2\pi/\omega$ and the vertical integrals range from $-H$ to 0. The time averages of $\sin^2$ and $\cos^2$ produce a factor of 1/2, and noting that $\cosh^2 x + \sinh^2 x = \cosh 2x$ we obtain

$$\overline{KE} = \frac{k^2 \eta_0^2 g^2}{2\omega^2 \cosh^2 kH} \frac{1}{2} \frac{\sinh(2kH)}{2k}. \quad (7.33)$$

Using the dispersion relation $\omega^2 = gk \tanh kH$ we finally obtain the simple expression

$$\overline{KE} = \frac{g\eta_0^2}{4}. \quad (7.34)$$

The perturbation potential energy is given by

$$\overline{PE} = \frac{\omega}{2\pi} \int \frac{1}{2} g \eta^2 dt = \frac{g\eta_0^2}{2} \frac{\omega}{2\pi} \int \cos^2(kx - \omega t) dt = \frac{g\eta_0^2}{4}. \quad (7.35)$$

Evidently, from (7.34) and (7.35) there is equipartitioning of energy time-averaged potential and kinetic energy components. Such equipartitioning is not, however, a universal property of wave motion.

The time averaged energy flux, which is in the x direction, is given by

$$\overline{F} = \frac{\omega}{2\pi} \int dt \int u' \phi' dz. \quad (7.36)$$

Using the wave expressions (7.15) we obtain, after a couple of lines of algebra,

$$\overline{F} = \frac{1}{2} \eta_0^2 \frac{g^2}{2c} \frac{1}{\cosh^2 kH} \left[\frac{\sinh 2kH}{2k} + H \right]. \quad (7.37)$$

Using (7.19) and the fact that $\sinh 2hK = 2 \sinh kH \cosh kH$ we obtain

$$\overline{F} = \frac{\eta_0^2 g}{2} c_g = (\overline{KE} + \overline{PE}) c_g. \quad (7.38)$$

Thus, using (7.34), (7.35) and (7.38), and generalizing the direction of propagation, we have that

$$\frac{\partial \overline{E}}{\partial t} + \nabla \cdot \mathbf{c}_g \overline{E} = 0, \quad (7.39)$$

where $\overline{E} = \overline{KE} + \overline{PE}$. Thus, the flux of energy is equal to the energy times the group velocity, or equivalently the energy in the wave propagates with the group velocity. As we established in chapter 6, this property is a rather general one for wave motion and it is satisfying to see how it applies to surface waves.

7.2 SHALLOW WATER WAVES ON FLUID INTERFACES

Let us now generalize our treatment of surface gravity waves to waves that exist on the interface between *two* moving fluids of different densities. The ensuing waves are a simple model of gravity waves that exist in the interior of the atmosphere and, perhaps especially, the ocean, in which we idealize the continuous stratification of the real fluid by supposing that the fluid comprises two (or conceivably more) layers of immiscible fluids of different densities stacked on top of each other. We will consider only the hydrostatic case in which case the layers form a ‘stacked shallow water’ system. We further limit ourselves to two moving layers; an extension to multiple layers is conceptually if not algebraically straightforward, but it soon becomes easier to treat the continuously stratified case which we do in later sections.

7.2.1 Equations of motion

Consider a two-layer shallow water model as illustrated in Fig. 3.5 on page 114. From section 3.3 the nonlinear momentum equations are, for the upper layer,

$$\frac{D\mathbf{u}_1}{Dt} + \mathbf{f} \times \mathbf{u}_1 = -g\nabla\eta_0, \quad (7.40a)$$

and in the bottom layer

$$\frac{D\mathbf{u}_2}{Dt} + \mathbf{f} \times \mathbf{u}_2 = -\frac{\rho_1}{\rho_2} (g\nabla\eta_0 + g'_1\nabla\eta_1). \quad (7.40b)$$

where $g'_1 = g(\rho_2 - \rho_1)/\rho_1$ (we will henceforth drop the subscript 1 and denote this as g'), and in the Boussinesq case we take $\rho_1/\rho_2 = 1$. We will only consider the non-rotating case, and after linearization about a resting state we have for the upper and lower layers respectively

$$\frac{\partial \mathbf{u}'_1}{\partial t} = -g\nabla\eta'_0, \quad (7.41a)$$

$$\frac{\partial \mathbf{u}'_2}{\partial t} = -g\nabla\eta'_0 - g'\nabla\eta'_1. \quad (7.41b)$$

The equations of motion are completed by the mass continuity equations for each layer, namely

$$\frac{D}{Dt}(\eta_0 - \eta_1) + h_1\nabla \cdot \mathbf{u}_1 = 0 \quad \longrightarrow \quad \frac{\partial}{\partial t}(\eta'_0 - \eta'_1) + H_1\nabla \cdot \mathbf{u}'_1 = 0 \quad (7.42a,b)$$

and

$$\frac{D\eta_1}{Dt} + h_2\nabla \cdot \mathbf{u}_2 = 0 \quad \longrightarrow \quad \frac{\partial \eta'_1}{\partial t} + H_2\nabla \cdot \mathbf{u}'_2 = 0, \quad (7.43a,b)$$

where the two rightmost expressions follow after linearization and we assume that the bottom is flat; that is $\eta_b = 0$. Henceforth we will also omit any primes on the perturbed quantities.

7.2.2 Dispersion relation

We first eliminate the velocity from (7.41a) and (7.42b) to give

$$\frac{\partial^2}{\partial t^2}(\eta_0 - \eta_1) - gH_1\nabla^2\eta_0 = 0, \quad (7.44)$$

and similarly for the lower layer:

$$\frac{\partial^2 \eta_1}{\partial t^2} - H_2(g\nabla^2\eta_0 + g'\nabla^2\eta_1) = 0. \quad (7.45)$$

Equations (7.44), and (7.45) form a complete set and in the usual fashion we may look for solutions of the form $\eta_i = \text{Re } \tilde{\eta}_i \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$. We obtain

$$(\omega^2 - gH_1K^2)\tilde{\eta}_0 - \omega^2\tilde{\eta}_1 = 0 \quad (7.46a)$$

$$-gH_2K^2\tilde{\eta}_0 + (\omega^2 - g'H_2K^2)\tilde{\eta}_1 = 0. \quad (7.46b)$$

where $K^2 = k^2 + l^2$. For these equations to have non-trivial solutions we must have

$$(\omega^2 - gH_1K^2)(\omega^2 - g'H_2K^2) - \omega^2gH_2K^2 = 0, \quad (7.47)$$

which, for small $g'/g \ll 1$ gives, after a couple of lines of algebra,

$$\omega^2 = \frac{1}{2}K^2 gH \pm \frac{1}{2}K^2 gH \sqrt{1 - 4 \frac{g'}{g} \frac{H_1 H_2}{H^2}} \approx \frac{1}{2}K^2 gH \pm \frac{1}{2}K^2 gH \left(1 - 2 \frac{g'}{g} \frac{H_1 H_2}{H^2}\right). \quad (7.48)$$

where $H = H_1 + H_2$. If $g' = 0$ we recover the familiar single-layer dispersion relation, $\omega = K\sqrt{gH}$ (as well as $\omega = 0$). In the more general case there are two distinct modes:

(i) A fast mode with phase speed given by

$$c_p^2 = \left(\frac{\omega}{k}\right)^2 = gH \left(1 - \frac{g'}{g} \frac{H_1 H_2}{H^2}\right), \quad (7.49)$$

where, for algebraic simplicity (and, in fact, without loss of generality, since it amounts only to an alignment of our coordinate system), we take $l = 0$. Using (7.46a) we then find that

$$\frac{\eta_0}{\eta_1} \approx \frac{H}{H_2}. \quad (7.50)$$

That is, since $H > H_2$, the displacement of the upper surface is larger than that of the lower. This mode is sometimes called the 'barotropic' mode, for the oscillations are vertically coherent (the phase on the interior surface is the same as that at the surface), and virtually the same oscillation would exist even in the absence of a density jump in the interior.

(ii) A slower mode with phase speed given by

$$c_p^2 \approx g' \frac{H_1 H_2}{H}, \quad (7.51)$$

and vertical structure

$$\frac{\eta_0}{\eta_1} \approx \frac{g' H_2}{gH} \ll 1. \quad (7.52)$$

In this case the displacement of the upper surface is smaller than the interior displacement by the ratio of g' to g ; in the ocean, where density differences are small, the ratio might well be of order 1/100. Furthermore, the internal displacement is *out of phase* with that at the surface. Often, in oceanic situations the interface may be taken as representing the thermocline, in which case $H_2 \gg H_1$ (i.e., the abyss has a greater depth than the thermocline) and $H \approx H_2$. In this case $c_p^2 \approx g' H_1$, and internal waves on the thermocline behave rather like surface waves, but with a weaker restoring force (and consequently a larger amplitude) because the density difference between the two layers of seawater is much smaller than the density difference between the seawater and air above it.

7.3 INTERNAL WAVES IN A CONTINUOUSLY STRATIFIED FLUID

We now turn our attention to *internal gravity waves*, namely waves that are internal to a given fluid and that owe their existence to the restoring force of gravity. Interfacial waves are, of course, a model of internal waves with a discontinuous jump in density within the fluid. Surface waves might even be thought of as internal waves if one supposes that part of the fluid has zero density, although this stretches the definition of the word internal somewhat. In this section we will consider the simplest and most fundamental case, that of internal waves in a Boussinesq fluid with constant stratification and no background rotation.

Reprising and extending the material of section 2.10.4, let us consider a continuously stratified Boussinesq fluid, initially at rest, in which the background buoyancy varies only with height and

so the buoyancy frequency, N , is a function only of z . Linearizing the equations of motion about this basic state gives the linear momentum equations,

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi' \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (7.53a,b)$$

the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.54a,b)$$

Our notation is such that $\mathbf{u} \equiv u\mathbf{i} + v\mathbf{j}$, $\mathbf{v} \equiv u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$, and the gradient operator is two-dimensional unless noted. Thus, $\nabla \equiv \mathbf{i}\partial_x + \mathbf{j}\partial_y$ and $\nabla_3 \equiv \mathbf{i}\partial_x + \mathbf{j}\partial_y + \mathbf{k}\partial_z$.

A little algebra gives a single equation for w' ,

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + N^2 \nabla^2 \right] w' = 0. \quad (7.55)$$

This equation is evidently *not* isotropic. If N^2 is a constant — that is, if the background buoyancy varies linearly with z — then the coefficients of each term are constant, and we may then seek solutions of the form

$$w' = \text{Re } \tilde{w} e^{i(kx + ly + mz - \omega t)}, \quad (7.56)$$

where Re denotes the real part, a denotation that will frequently be dropped unless ambiguity arises, and other variables oscillate in a similar fashion. Using (7.56) in (7.55) yields the dispersion relation:

$$\omega^2 = \frac{(k^2 + l^2)N^2}{k^2 + l^2 + m^2} = \frac{K^2 N^2}{K_3^2}, \quad (7.57)$$

where $K^2 = k^2 + l^2$ and $K_3^2 = k^2 + l^2 + m^2$. The frequency (see Fig. 7.2) is thus always less than N , approaching N for small horizontal scales, $K^2 \gg m^2$. If we neglect pressure perturbations, as in the parcel argument, then the two equations,

$$\frac{\partial w'}{\partial t} = b', \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (7.58)$$

form a closed set, and give $\omega^2 = N^2$.

If the basic state density increases with height then $N^2 < 0$ and we expect this state to be unstable. Indeed, the disturbance grows exponentially according to $\exp(\sigma t)$ where

$$\sigma = i\omega = \pm \frac{K\tilde{N}}{K_3}, \quad (7.59)$$

where $\tilde{N}^2 \equiv -N^2$ and $K_3 = \sqrt{K^2 + m^2}$. Most convective activity in the ocean and atmosphere is, ultimately, related to an instability of this form, although of course there are many complicating issues — water vapour in the atmosphere, salt in the ocean, the effects of rotation and so forth.

7.3.1 Hydrostatic internal waves

Let us now suppose that the fluid satisfies the hydrostatic Boussinesq equations, and for simplicity assume that $l = 0$. The linearized two-dimensional equations of motion become

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi', \quad 0 = -\frac{\partial \phi'}{\partial z} + b', \quad (7.60a)$$

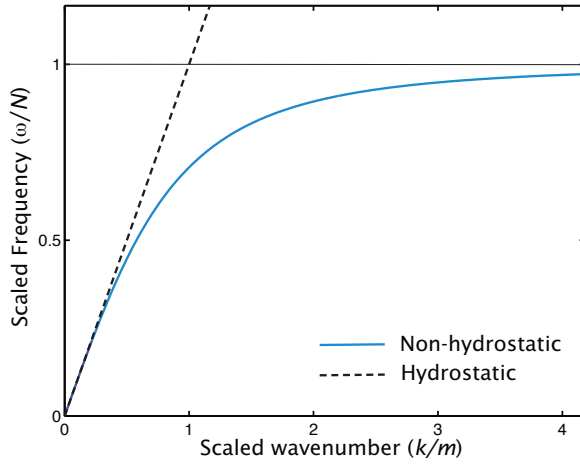


Fig. 7.2 Scaled frequency, ω/N , plotted as a function of scaled horizontal wavenumber, k/m , using the full dispersion relation of (7.57) with $l = 0$ (solid line, asymptoting to unit value for large k/m), and with the hydrostatic dispersion relation (7.61) (dashed line, tending to ∞ for large k/m).

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (7.60b)$$

where these are the horizontal and vertical momentum equations, the mass continuity equation and the thermodynamic equation respectively. A little algebra gives the dispersion relation,

$$\omega^2 = \frac{K^2 N^2}{m^2}. \quad (7.61)$$

The frequency and, if N^2 is negative, the growth rate, is unbounded as $K^2/m^2 \rightarrow \infty$, and the hydrostatic approximation thus has quite unphysical behaviour for small horizontal scales. Many numerical models of the large-scale circulation in the atmosphere and ocean do make the hydrostatic approximation. In these models convection must be *parameterized*; otherwise, it would simply occur at the smallest scale available, namely the size of the numerical grid, and this type of unphysical behaviour should be avoided. Of course in non-hydrostatic models convection must also be parameterized if the horizontal resolution of the model is too coarse to properly resolve the convective scales.

7.3.2 Some Properties of Internal Waves

Internal waves have a number of interesting and counter-intuitive properties so let's point a few of them out.

The dispersion relation

We can write the dispersion relation, (7.57), as

$$\omega = \pm N \cos \vartheta, \quad (7.62)$$

where $\cos^2 \vartheta = K^2/(K^2 + m^2)$ so that ϑ is the angle between the three-dimensional wave-vector, $\mathbf{k} = k\mathbf{i} + l\mathbf{j} + m\mathbf{k}$, and the horizontal. The frequency is evidently a function only of N and ϑ , and, if this is given, the frequency is not a function of wavelength. This has some interesting consequences for wave reflection, as we see below.

We can also write the dispersion relation, (7.57), as

$$\frac{\omega^2}{N^2 - \omega^2} = \frac{K^2}{m^2}. \quad (7.63)$$

Thus, and consistently with our first point, given the wave frequency the ratio of the vertical to the horizontal wavenumber is fixed.

Polarization relations

If the pressure field is oscillating like $\phi' = \tilde{\phi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] = \tilde{\phi} \exp[i(kx + ly + mz - \omega t)]$ then, using (7.53a), the horizontal velocity components have the phases

$$(\tilde{u}, \tilde{v}) = (k, l)\omega^{-1}\tilde{\phi} \quad (7.64a)$$

As the frequency is real, the velocities are in phase with the pressure. A little algebra also reveals that the buoyancy perturbation is related to the pressure perturbation by

$$\tilde{b} = \frac{imN^2}{N^2 - \omega^2}\tilde{\phi} = \frac{iN^2K^2}{m\omega^2}\tilde{\phi} = \frac{iK_3^2}{m}\tilde{\phi}, \quad (7.64b)$$

using the dispersion relation, so that the buoyancy and pressure perturbations are $\pi/2$ out of phase.

The vertical velocity is related to the pressure perturbation by

$$\tilde{w} = \frac{-\omega m}{N^2 - \omega^2}\tilde{\phi} = \frac{-K^2}{m\omega}\tilde{\phi}, \quad (7.64c)$$

where the second expression uses (7.63). The vertical velocity is in phase with the pressure perturbation, and for regions of positive m (and so with upward phase propagation) regions of high relative pressure are associated with downward fluid motion.

The pressure, buoyancy and velocity fields are all real fields and we can write the above phase relationships in terms of sines and cosines as follows.

$$\phi = \Phi_0 \cos(kx + ly + mz - \omega t), \quad (7.65a)$$

$$(u, v) = (k, l) \frac{\Phi_0}{\omega} \cos(kx + ly + mz - \omega t), \quad (7.65b)$$

$$w = \left(\frac{-\omega m}{N^2 - \omega^2} = \frac{-K^2}{m\omega} \right) \Phi_0 \cos(kx + ly + mz - \omega t) \quad (7.65c)$$

$$b = \left(\frac{mN^2}{N^2 - \omega^2} = \frac{N^2K^2}{m\omega^2} \right) \Phi_0 \sin(kx + ly + mz - \omega t), \quad (7.65d)$$

where Φ_0 is a constant. We might equally well have chosen ϕ to have a sine dependence in (7.65a), in which case (7.65b,c,d) should be changed appropriately. The relations between pressure, buoyancy and velocity in (7.64) and (7.65) are known as *polarization relations*.

Relation between wave vector and velocity

$$\mathbf{k} \cdot \tilde{\mathbf{v}} = 0. \quad (7.66)$$

This means that, at any instant, the wave vector is perpendicular to the velocity vector, and the velocity is therefore aligned *along* the direction of the troughs and crests, along which there is no pressure gradient. If the wave vector is purely horizontal (i.e., $m = 0$), then the motion is purely vertical and $\omega = N$.

The vertical and horizontal velocities are related to the wave wavenumbers. Suppose for simplicity (and with little loss of generality) that the motion is all in the x - z plane, with $l = 0$ and $v = 0$. Then

$$\frac{\tilde{u}}{\tilde{w}} = -\frac{m}{k}. \quad (7.67)$$

Furthermore, from (7.56) with $l = 0$, at any given instant all of the perturbation quantities in the wave are constant along the lines $kx + mz = \text{constant}$. Thus, *all fluid parcel motions are parallel to the*

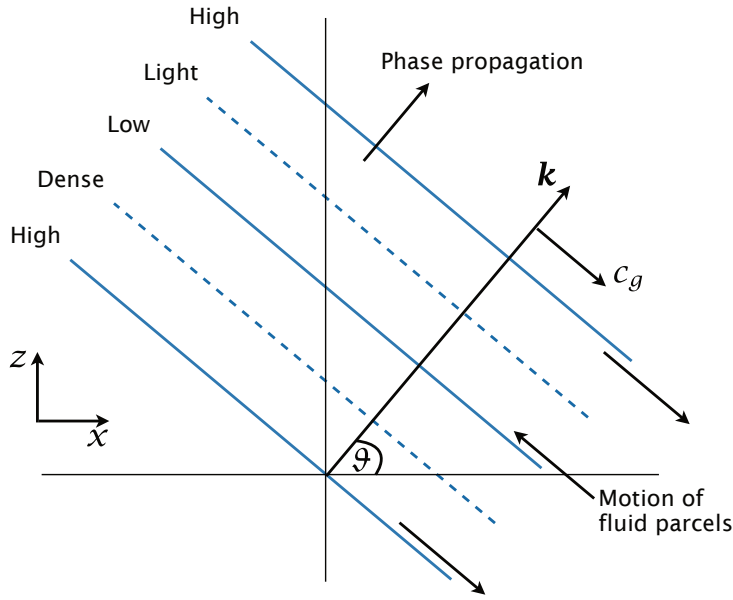


Fig. 7.3 An internal wave propagating in the direction k . Both k and m are positive for the wave shown. The solid lines show crests and troughs of constant pressure, and the dashed lines the corresponding crests and troughs of buoyancy (or density). The motion of the fluid parcels in along the lines of constant phase, as shown, and is parallel to the group velocity and perpendicular to the phase speed.

wave fronts. Now, since the wave frequency is related to the background buoyancy frequency by $\omega = \pm N \cos \vartheta$, it follows that the fluid parcels oscillate along lines that are at an angle $\vartheta = \cos^{-1}(\omega/N)$ to the vertical. The polarization relations and the group and phase velocities are illustrated in Fig. 7.3. Let us now discuss them, and the figure, in a little more detail.²

7.3.3 A parcel argument and some physical interpretation

Let us consider first the dispersion relation itself and try to derive it more physically, or at least heuristically. Let us suppose there is a wave propagating in the (x, z) plane at some angle ϑ to the horizontal, with fluid parcels moving parallel to the troughs and crests, as in Fig. 7.3. In general the restoring force on a parcel is due to both the pressure gradient and gravity, but along the crests there is no pressure gradient. Referring to Fig. 7.4, for a total displacement Δs the restoring force, F_{res} , in the direction of the particle displacement is

$$\begin{aligned} F_{res} &= g \cos \vartheta \times \Delta \rho = g \cos \vartheta \times \frac{\partial \rho}{\partial z} \Delta z \\ &= g \cos \vartheta \times \frac{\partial \rho}{\partial z} \Delta s \cos \vartheta = \rho_0 \frac{\partial b}{\partial z} \cos^2 \vartheta \Delta s, \end{aligned} \quad (7.68)$$

noting that $\Delta z = \cos \vartheta \Delta s$. The equation of motion of a parcel moving along a trough or crest is therefore

$$\rho_0 \frac{d^2 \Delta s}{dt^2} = -\rho_0 N^2 \cos^2 \vartheta \Delta s, \quad (7.69)$$

which implies a frequency $\omega = N \cos \vartheta$, as in (7.62). One of the $\cos \vartheta$ factors in (7.69) comes from the fact that the parcel displacement is at an angle to the direction of gravity, and the other comes from the fact that the restoring force that a parcel experiences is proportional to $N \cos \vartheta$. (The reader may also wish to refer ahead to Fig. 7.14 and section 7.6.1 for a similar argument.)

Now consider the wave illustrated in Fig. 7.3. For this wave both k and m are positive, and the frequency is assumed positive by convention to avoid duplicative solutions. The slanting solid and dashed lines are lines of constant phase, and from (7.64b) the buoyancy and pressure are 1/4 of a

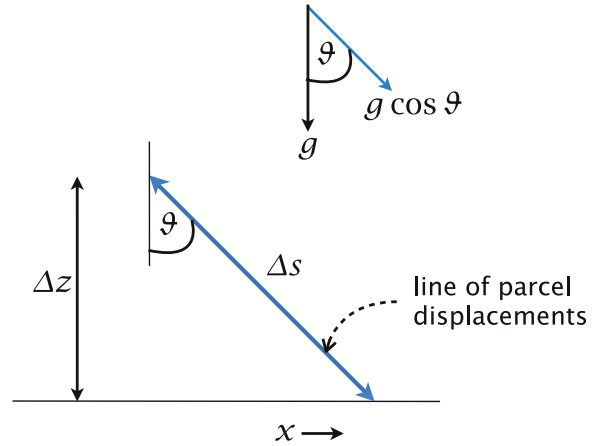


Fig. 7.4 Parcel displacements and associated forces in a internal gravity wave in which the parcel displacements are occurring at an angle ϑ to the vertical, as in Fig. 7.3.

wavelength out of phase. When k and m are both positive the extrema in the buoyancy field lag the extrema in the vertical velocity by $\pi/2$, as illustrated. The perturbation velocities are zero along the lines of extreme buoyancy. This follows because the velocities are in phase with the pressure, which as we noted is out of phase with the buoyancy.

Given the direction of the fluid parcel displacement in Fig. 7.3, the direction of the phase propagation c_p up and to the right may be deduced from the following argument. Buoyancy perturbations arise because of vertical advection of the background stratification, $w' \partial b_0 / \partial z = w' N^2$. A local maximum in rising motion, and therefore a tendency to increase the fluid density, is present along the 'Low' line $1/4$ wavelength upward and to the right of the 'Dense' phase line. Thus, the density of fluid along the 'Low' phase line increases and the 'Dense' phase line moves upward and to the right. If the fluid parcel motion were reversed the pattern of 'High–Dense–Low–Light–High' in Fig. 7.3 would remain the same. However, the downward fluid motion along the 'Low' line would cause the fluid to lose density, and so the phase lines would propagate downward and to the left. Evidently, the wave fronts, or the lines of constant phase, move at right angles to the fluid-parcel trajectories. In the figure we see that the group velocity is denoted as being at right angles to the phase speed, so let's discuss this.

7.3.4 Group velocity and phase speed

As we noted above, the frequency of internal waves is given by $\omega = N \cos \vartheta$, where ϑ is the angle the wave vector makes with the horizontal. This means that the surfaces of constant frequency are *cones*, as illustrated in Fig. 7.5.

To evaluate phase and group velocities in a useful way it is convenient to use spherical polar coordinates, as in Fig. 7.6, in which

$$k = K_3 \cos \vartheta \cos \lambda, \quad l = K_3 \cos \vartheta \sin \lambda, \quad m = K_3 \sin \vartheta, \quad (7.70)$$

so that $\mathbf{k} = K_3(\cos \vartheta \cos \lambda, \cos \vartheta \sin \lambda, \sin \vartheta)$. The angles are ϑ , the angle of the wave vector with the horizontal and λ , which determines the orientation in the horizontal plane. (The notation is similar to the spherical coordinates of chapter 2 — see Fig. 2.3 — although here ϑ is the angle with the horizontal, not the angle with the equatorial plane.) We also note that

$$\sin^2 \vartheta = \frac{m^2}{k^2 + l^2 + m^2}, \quad \cos^2 \vartheta = \frac{K_3^2}{K_3^2} = \frac{k^2 + l^2}{k^2 + l^2 + m^2}, \quad \tan \lambda = \frac{l}{k}. \quad (7.71)$$

In many problems we can align the direction of the wave propagation with the x -axis and take $l = 0$ and $\tan \lambda = 0$.

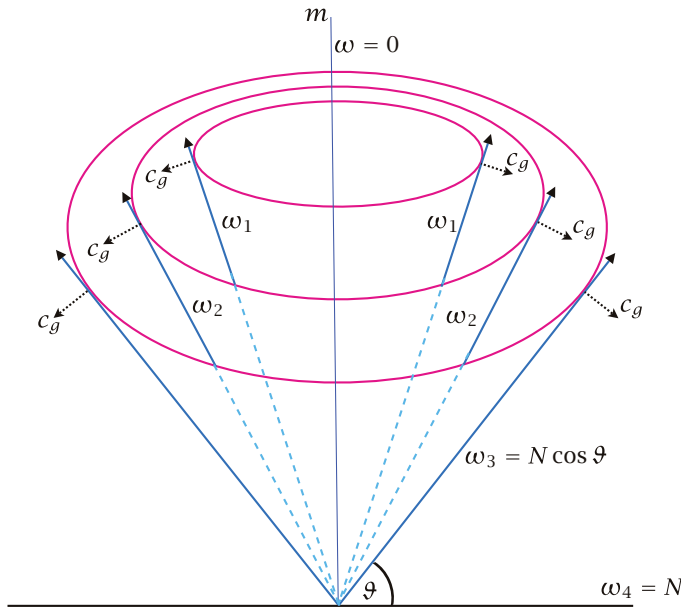


Fig. 7.5 Internal wave cones. The surfaces of constant frequency are cones, defined by the surface that has a constant angle to the horizontal. The wave vector, and so the phase velocity, points along the cone away from the origin, and the frequency of any wave with a wave vector in the cone is $N \cos \vartheta$. The group velocity is at right angles to the cone and pointed in the direction of increasing frequency, as indicated by the arrows on the dotted lines. In the vertical direction the phase speed and group velocity have opposite signs.

The phase speed of the internal waves in the direction of the wave vector (sometimes referred to as the phase velocity) is given by

$$c_p = \frac{\omega}{K_3} = \frac{N}{K_3} \cos \vartheta = \frac{NK}{K_3^2}. \quad (7.72)$$

The phase speeds (as conventionally-defined) in the x, y and z directions are

$$c_p^x \equiv \frac{\omega}{k} = \frac{N}{k} \cos \vartheta, \quad c_p^y \equiv \frac{\omega}{l} = \frac{N}{l} \cos \vartheta, \quad c_p^z \equiv \frac{\omega}{m} = \frac{N}{m} \cos \vartheta. \quad (7.73a,b,c)$$

As noted in section 6.1.2 these quantities are the speed of propagation of the wave crests in the respective directions. In general, each speed is *larger* than the phase speed in the direction perpendicular to the wave crests (that is, in the direction of the wave vector), but no information is transmitted at these speeds.

The group velocity is given by

$$\mathbf{c}_g = \left(\frac{\partial \omega}{\partial k}, \frac{\partial \omega}{\partial l}, \frac{\partial \omega}{\partial m} \right). \quad (7.74)$$

Using (7.57) we find

$$c_g^x = \frac{\partial \omega}{\partial k} = \frac{Nm}{K_3^2} \frac{km}{KK_3} = \left(\frac{N}{K_3} \sin \vartheta \right) \cos \lambda \sin \vartheta, \quad (7.75a)$$

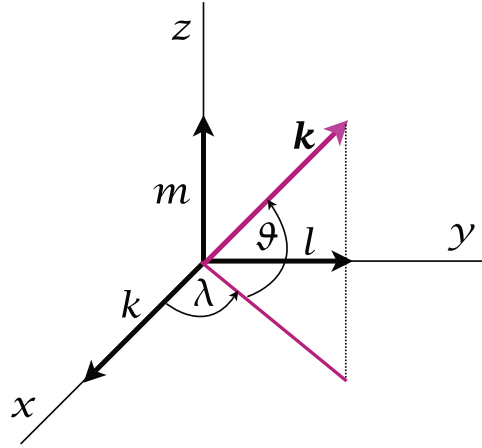
$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{Nm}{K_3^2} \frac{lm}{KK_3} = \left(\frac{N}{K_3} \sin \vartheta \right) \sin \lambda \sin \vartheta, \quad (7.75b)$$

$$c_g^z = \frac{\partial \omega}{\partial m} = -\frac{Nm}{K_3^2} \frac{K}{K_3} = -\left(\frac{N}{K_3} \sin \vartheta \right) \cos \vartheta. \quad (7.75c)$$

The magnitude of the group velocity is evidently

$$|c_g| = \frac{N}{K_3} \sin \vartheta, \quad (7.76)$$

Fig. 7.6 The spherical coordinates used to describe internal waves, as in (7.70). The angle ϑ is the angle of the wave vector with the horizontal, and λ determines the orientation in the horizontal plane. The wave vector $\mathbf{k}$ is given by $\mathbf{k} = (k, l, m)$, these being the wavenumbers in the direction of increasing (x, y, z) , respectively.



and the group velocity vector is directed at an angle ϑ to the vertical, as in Fig. 7.5. This angle is perpendicular to the cone itself; that is, the group velocity is perpendicular to the wave vector, as may be verified by taking the dot product of (7.70) and (7.75) which gives

$$\mathbf{k} \cdot \mathbf{c}_g = 0. \quad (7.77)$$

The group velocity is therefore parallel to the motion of the fluid parcels, as illustrated in Fig. 7.3. Furthermore, because energy propagates with the group velocity, and the latter is *parallel* to lines of constant phase, energy propagates perpendicular to the direction of phase propagation — very different from the case of acoustic waves or even shallow water waves. In the vertical direction we see from (7.73c) and (7.75c) that

$$\frac{\omega}{m} \frac{\partial \omega}{\partial m} = -\frac{N^2}{K_3^2} \cos^2 \vartheta < 0. \quad (7.78)$$

That is, the phase speed and the group velocity have opposite signs, meaning that if the wave crests move downward the group moves upward!

Effect of a mean flow

Suppose that there is a mean flow, U , in the x -direction, as is common in both atmosphere and ocean. The dispersion relation, (7.57), simply becomes

$$(\omega - Uk)^2 = \frac{K^2 N^2}{K^2 + m^2}. \quad (7.79)$$

The frequency is Doppler shifted, as expected, but the upward propagation of waves is affected in an interesting way. From (7.79) we find that the vertical component of the group velocity may be written as

$$\frac{\partial \omega}{\partial m} = \frac{-m(\omega - Uk)}{K^2 + m^2} = \frac{-mk(c - U)}{K^2 + m^2}. \quad (7.80)$$

where $c = \omega/k$ is the phase speed in the x -direction. If U is not constant but is varying slowly with z then (7.80) still holds, although m itself will also vary slowly with z . The point to note is that the group velocity goes to zero at the location where $U = c$, that is at a critical layer and the wave stalls. Of course m itself may become large near a critical layer (as we consider in more detail in section 17.4). In this case — which is essentially the hydrostatic one, with $m^2 \gg K^2$ — we obtain

$$\frac{\partial \omega}{\partial m} = \frac{-k(c - U)}{m} = \frac{-k^2(c - U)^2}{KN}. \quad (7.81)$$

The physical consequence of group velocity going to zero as the wave approaches a critical layer is that any dissipation that may be present has more time to act. That is, we can expect a wave to be preferentially dissipated near a critical layer, giving up its momentum to the mean flow and its energy to create mixing — the former being important in the atmosphere (for this is the mechanism producing the quasi-biennial oscillation) and the latter in the ocean.

7.3.5 Energetics of internal waves

In this section we explore the energetics of internal waves, and we first show that the linearized equations conserve a sensible form of energy. Linearized equations do not, of course, automatically conserve energy even if the original nonlinear equations from which they derive do: an unstable wave will draw energy from the background state and grow in amplitude, as we saw in chapter 6 on baroclinic instability.

Energy Conservation

To obtain an energy equation we proceed much as in the nonlinear case described in section A2.4.3. From (7.53) we obtain an equation for the evolution of kinetic energy, namely

$$\frac{\partial}{\partial t} \left(\frac{\mathbf{v}'^2}{2} \right) = b' w' - \nabla_3 \cdot (\phi' \mathbf{v}'), \quad (7.82)$$

where $\mathbf{v}'^2 = u'^2 + v'^2 + w'^2$, and from (7.54) we obtain

$$\frac{1}{N^2} \frac{\partial b'^2}{\partial t} + w' b' = 0. \quad (7.83)$$

Adding the above two equations gives

$$\frac{\partial}{\partial t} \frac{1}{2} \left(\mathbf{v}'^2 + \frac{b'^2}{N^2} \right) + \nabla_3 \cdot (\phi' \mathbf{v}') = 0. \quad (7.84)$$

This is the linear version of (A2.112). Two differences are apparent: (i) The transport of energy is only by way of the pressure term and the advective transport is absent, as expected in a linear model; (ii) the potential energy term bz of the linear model is replaced by b'^2/N^2 . It is less obvious why this should be so. However, the quantity

$$A = \frac{1}{2} \int \frac{\overline{b'^2}}{\partial \bar{b} / \partial z} dz dA = \frac{1}{2} \int \frac{\overline{b'^2}}{N^2} dz dA \quad (7.85)$$

is just the *available potential energy* of a Boussinesq fluid in which the isopycnal surfaces vary only slightly from a stable, purely horizontal, resting state.

If we integrate (7.84) over a volume such that the normal component of the velocity vanishes at the boundaries (for example, we integrate over a volume enclosed by rigid walls) then the divergence term vanishes and we obtain the integral conservation statement:

$$\hat{E} = \frac{1}{2} \int \left(\mathbf{v}'^2 + \frac{b'^2}{N^2} \right) dV, \quad \frac{d\hat{E}}{dt} = 0. \quad (7.86)$$

The quantity $\hat{E}$ is an example of a *wave activity*: a conserved quantity that is quadratic in wave amplitude. This conservation statement (7.84) is true whether or not the basic state is stably stratified; that is, whether or not N^2 is positive. However, (7.86) only provides a bound on growing

perturbations if N^2 is positive, in which case all the terms that constitute $\hat{E}$ are positive definite. If $N^2 < 0$ then both $\mathbf{v}'^2$ and b'^2 can grow without bound even as $\hat{E}$ itself remains constant.

Consider now the energy in a wave, and we will denote by $\bar{E}$ the energy density, meaning the mean perturbation energy per unit volume, averaged over a wavelength. Thus

$$2\bar{E} = \overline{\mathbf{v}'^2} + \frac{\overline{b'^2}}{N^2}. \quad (7.87)$$

If we use the polarization relations of section 7.3.2 then the kinetic and potential energy densities may be written in terms of the pressure amplitude as

$$2\overline{KE} = \left(\frac{k^2}{\omega^2} + \frac{l^2}{\omega^2} + \frac{(k^2 + l^2)^2}{m^2\omega^2} \right) |\tilde{\phi}|^2 = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2, \quad (7.88a)$$

$$2\overline{PE} = \frac{N^2 K^4}{m^2\omega^4} = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2, \quad (7.88b)$$

using also the dispersion relation, $\omega^2 K_3^2 = K^2 N^2$. Thus, there is *equipartition* between the kinetic and potential energies, a common feature of waves in non-rotating systems (although not a universal feature of waves). The total energy density is thus

$$\bar{E} = \frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2 = \frac{K_3^2}{K^2} |\tilde{w}|^2 = \frac{|\tilde{w}|^2}{\cos^2 \vartheta}, \quad (7.89)$$

using (7.65c), where $\tilde{w}$ is the amplitude of the vertical component of the velocity perturbation.

Energy propagation and the group velocity property

In section 6.7 we derived, from rather general considerations, the ‘group velocity property’ for wave activity. We showed that if a wave activity, A , and its flux, $\mathbf{F}$ obeyed a conservation law of the form $\partial A / \partial t + \nabla \cdot \mathbf{F} = 0$, and if the wave activity and its flux were both quadratic functions of the wave amplitude, then the flux is related to the wave activity by $\mathbf{F} = \mathbf{c}_g A$. The internal wave energy density and its flux do have these properties — see (7.84) — so we should expect the group velocity property to hold, and we now demonstrate that explicitly, albeit briefly.

The energy flux vector for internal waves is

$$\mathbf{F} = \overline{\phi' \mathbf{v}'} \quad (7.90)$$

and using (7.64a) and (7.64c) this is

$$\mathbf{F} = \left(\frac{k}{\omega}, \frac{l}{\omega}, -\frac{K^2}{m\omega} \right) |\tilde{\phi}|^2. \quad (7.91)$$

Using (7.75) and (7.89) the group velocity times the energy density is

$$\mathbf{c}_g^x \times \bar{E} = \left[\frac{Nm^2}{K_3^3} \frac{k}{K} \right] \times \left[\frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2 \right] = \frac{k}{\omega} |\tilde{\phi}|^2, \quad (7.92a)$$

$$\mathbf{c}_g^y \times \bar{E} = \left[\frac{Nm^2}{K_3^3} \frac{l}{K} \right] \times \left[\frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2 \right] = \frac{l}{\omega} |\tilde{\phi}|^2, \quad (7.92b)$$

$$\mathbf{c}_g^z \times \bar{E} = \left[\frac{NmK}{K_3^3} \right] \times \left[\frac{K^2 K_3^2}{m^2\omega^2} |\tilde{\phi}|^2 \right] = -\frac{K^2}{m\omega} |\tilde{\phi}|^2, \quad (7.92c)$$

which evidently is the same as (7.91), completing our demonstration.

7.4 ♦ INTERNAL WAVE REFLECTION

Suppose a propagating internal wave encounters a solid boundary — sloping topography, for example. The boundary effectively acts as a source of waves and so the original wave is reflected in some fashion. However, because of the nature of the dispersion relation for internal waves the reflection occurs in a rather peculiar way, as we now discuss.

For algebraic simplicity let us initially suppose that the wave is propagating in the x - z plane, and the equation of mass continuity $\partial_x u + \partial_z w = 0$ is then satisfied by introducing a streamfunction ψ such that

$$u = -\frac{\partial \psi}{\partial z}, \quad w = \frac{\partial \psi}{\partial x}. \quad (7.93)$$

If the incident wave is denoted ψ_1 and the reflected wave ψ_2 then the total wave field is

$$\psi = \tilde{\psi}_1 e^{i(k_1 x + m_1 z - \omega_1 t)} + \tilde{\psi}_2 e^{i(k_2 x + m_2 z - \omega_2 t)}, \quad (7.94)$$

where as usual a tilde denotes a complex wave amplitude and the real part of the expression is implied. The total streamfunction must be constant *at the boundary* — in fact without loss of generality we may suppose that $\psi = 0$ at the boundary — and this can only be achieved if

$$k_1 x + m_1 z - \omega_1 t = k_2 x + m_2 z - \omega_2 t \quad (7.95)$$

for all t and for all x and z along the boundary. This implies that

$$\omega_1 = \omega_2 \quad (7.96)$$

and

$$k_1 x + m_1 z_b(x) = k_2 x + m_2 z_b(x) \quad (7.97)$$

where $z_b(x)$ parameterizes the height of the reflecting boundary. We can view this another way: suppose that the boundary slopes at an angle γ to the horizontal, as in Fig. 7.7 or Fig. 7.8. We then have $z_b = x \tan \gamma$ and a unit vector along the boundary satisfies $\mathbf{j}_\gamma = \mathbf{i} \cos \gamma + \mathbf{j} \sin \gamma$. Equation (7.94) may be written as

$$\psi = \tilde{\psi}_1 e^{i[k_1 + m_1 \tan \gamma]x - \omega_1 t} + \tilde{\psi}_2 e^{i[(k_2 + m_2 \tan \gamma)x - \omega_2 t]}, \quad (7.98)$$

from which the wavenumber condition that must be satisfied is

$$k_1 + m_1 \tan \gamma = k_2 + m_2 \tan \gamma \quad (7.99)$$

or, and as may also be seen from (7.97),

$$\mathbf{k}_1 \cdot \mathbf{j} = \mathbf{k}_2 \cdot \mathbf{j}. \quad (7.100)$$

This means that the components of the wave vector parallel to the boundary for the incoming and outgoing wave are equal to each other. This, and the conservation of frequency expressed by (7.96), are *general* results about wave reflection; they apply to light waves, for example. However, the dispersion relation of internal waves gives rise to rather unintuitive and decidedly non-specular properties of reflection.

7.4.1 Properties of internal wave reflection

Suppose an internal wave is incident on a solid boundary, sloping at an angle γ to the horizontal, as in Fig. 7.7 or Fig. 7.8. The incident and reflected wave must satisfy the following conditions.

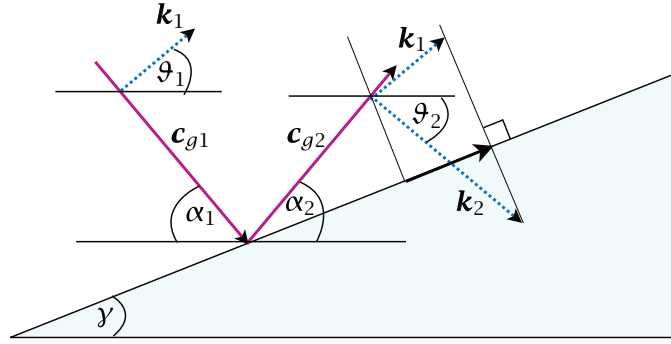


Fig. 7.7 Internal wave reflection from a shallow sloping boundary. The incoming wave vector, k_1 , makes an angle ϑ_1 with the horizontal, and the incoming group velocity, c_{g1} makes an angle $\alpha_1 = \pi/2 - \vartheta_1$. The group velocity of the reflected wave, c_{g2} is directed away from the slope, and to satisfy the frequency condition we have $\alpha_2 = \alpha_1$. The projection along the slope of the reflected wave vector, k_2 must be equal to that of the incoming wave vector (the projection is the short thick arrow along the slope), and so the magnitude of the reflected wave vector is larger than that incoming wave.

1. The frequency of the reflected wave is equal to that of the incident wave. Because the frequency is given by $\omega = N \cos \vartheta$, the angle of the reflected wave *with respect to the horizontal* is equal to that of the incident wave.
2. The components of the wave vector along the slope of the reflected wave and incident wave are equal.
3. The group velocity of the reflected wave must be directed away from the slope.

We did not derive the third of these conditions, but the reflected wave must carry energy and information away from the slope, and these are carried by the group velocity. Similarly, a wave incident on a boundary is one in which the group velocity is directed toward the slope.

Consider a wave approaching a slope as in Fig. 7.7, such that the incoming wave vector makes an angle of ϑ_1 with the horizontal, and the boundary slope is γ . The condition (7.100) states that the projections along the boundary of the the incoming and outgoing wave vectors are equal to each other, and so

$$\kappa_1 \cos(\vartheta - \gamma) = \kappa_2 \cos(\vartheta + \gamma), \quad (7.101)$$

where κ_1 and κ_2 are the magnitudes of the incoming and reflected wave vectors and $\vartheta = \vartheta_1 = \vartheta_2$, because the outgoing wave makes the same angle with the horizontal as does the incoming wave. The group velocity is perpendicular to the wave vector and makes an angle $\alpha = \pi/2 - \vartheta$ to the horizontal, and in terms of this (7.101) may be written, provided $\alpha > \gamma$,

$$\kappa_1 \sin(\alpha + \gamma) = \kappa_2 \sin(\alpha - \gamma). \quad (7.102)$$

For a sufficiently steep boundary slope we may have $\alpha < \gamma$, and in this case the wave will be back reflected down the slope, as in Fig. 7.8. A little geometry reveals that the condition (7.102) should be replaced by

$$\kappa_1 \sin(\alpha + \gamma) = \kappa_2 \sin(\gamma - \alpha). \quad (7.103)$$

The case with $\alpha = \gamma$ is plainly a critical one. In this case the group velocity of the reflected wave is directed along the slope, and the wave vector is perpendicular to the slope. The magnitude of the reflected wave vector is infinite; that is, the waves have zero wavelength, and so would in reality be subject to viscous dissipation and diffusion. Reflection of internal waves is in fact an important mechanism leading to mixing in the ocean.

we expect the solution to look locally like plane waves and we can obtain approximate solutions. This is the territory of the WKB approximation, as described in the Appendix on 248, and we will employ this technology.

7.5.1 Obtaining a WKB solution

No assumptions are made about the uniformity of N when deriving (7.55), so the equation of motion is again

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + N^2 \nabla^2 \right] w' = 0. \quad (7.104)$$

where $N^2 = N^2(z)$. Let us seek solutions in the form

$$w' = \text{Re } W(z) e^{i(kx + ly - \omega t)}, \quad (7.105)$$

whence we obtain

$$\frac{d^2 W}{dz^2} + m^2(z) W = 0, \quad (7.106)$$

where

$$m^2 \equiv \frac{(N^2 - \omega^2) K^2}{\omega^2}. \quad (7.107)$$

We will see in a moment that this is essentially the dispersion relation for the gravity waves, with m being the vertical wavenumber. The WKB solution to (7.106) is

$$W(z) = C m^{-1/2} \exp \left(\pm i \int^z m \, dz' \right). \quad (7.108)$$

where C is a constant. The phase of the wave, $\theta(z)$, is given by

$$\theta(z) = \int^z m \, dz' = \int^z \pm K \left(\frac{N^2 - \omega^2}{\omega^2} \right)^{1/2} dz'. \quad (7.109)$$

Since $m = d\theta/dz$, locally the flow behaves like a plane wave with vertical wavenumber m and with amplitude varying as $m^{-1/2}$. Re-arranging (7.107) we obtain the dispersion relation in a familiar form,

$$\omega^2 = \frac{N^2 K^2}{K^2 + m^2} = N^2 \cos^2 \vartheta(z). \quad (7.110)$$

where $\cos^2 \vartheta = K^2/(K^2 + m^2)$. Given this, we can interpret (7.107) as giving the vertical wavenumber in a medium in which the stratification is varying and the frequency and horizontal wavenumber are known. Note that N , ϑ and m are functions of z , but ω is not, essentially because the medium is time independent (c.f., the discussion in section 6.3).

7.5.2 Properties of the solution

The WKB solution above is *almost* that of a plane wave with slowly varying wavenumber. Thus, it seems that the solution (7.108) might be further approximated as

$$w \approx C m^{-1/2} \exp (\pm i m(z) z) \quad (7.111)$$

where $m(z)$ is given by (7.107). The accuracy of this solution increases as the variation of m diminishes, and in many circumstances (7.111) may be used to infer the qualitative behaviour of a wave.

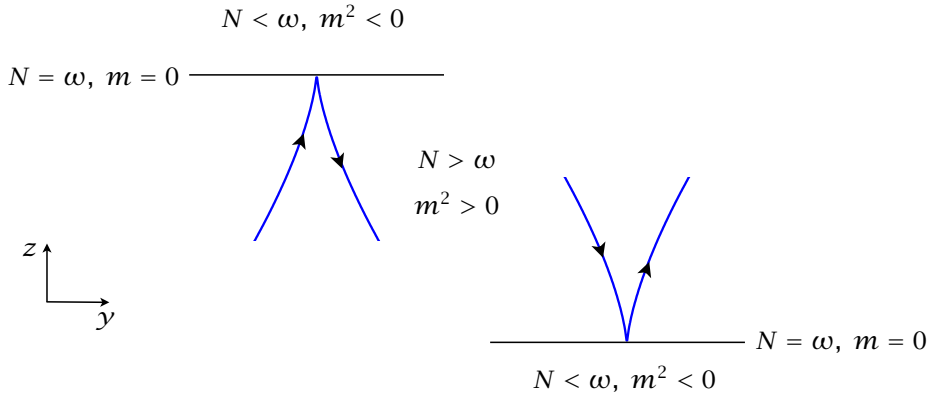


Fig. 7.10 Trajectories of internal waves approaching a turning height where $N = \omega$. The trajectory makes a cusp, as given by (7.117). If a region of high stratification is sandwiched between two regions of lower stratification then the waves may be vertically confined to a waveguide.

Nonetheless, it is an integral that appears in the phase in the solution (7.108), so the solution is not completely local, as discussed in the wkb appendix.

From (7.108) the amplitude, $A(z)$ say, varies with height as $m^{-1/2}$, so that if the stratification (N^2) increases m will increase and A will decrease. Here we have derived this result directly by solving the wave equations of motion, but the result is a consequence of the conservation of energy in internal waves: energy is a ‘wave activity’ — namely a conserved quantity, quadratic in the wave amplitude — in this problem. As discussed in section 7.3.5, the vertical component of the energy flux, F^z , is $c_g^z \bar{E}$, where $\bar{E}$ is the energy density and c_g^z is the vertical component of the group velocity, and for a wave propagating vertically this energy flux must be constant. Now, manipulating (7.75c) and using (7.89) we have

$$c_g^z = -\frac{\omega m}{K_3^2}, \quad \bar{E} = \left(\frac{A}{\cos \vartheta} \right)^2 \quad (7.112a,b)$$

so that

$$F^z = c_g^z \bar{E} = -\frac{A^2 \omega m}{K^2} = \text{constant}. \quad (7.113)$$

Thus, because the horizontal wavenumber K is preserved (since there are no inhomogeneities in the horizontal) and the frequency is constant (because the medium itself is not time varying), we must have $A \propto m^{-1/2}$, as in (7.108).

7.5.3 Wave trajectories and an idealized example

Rays

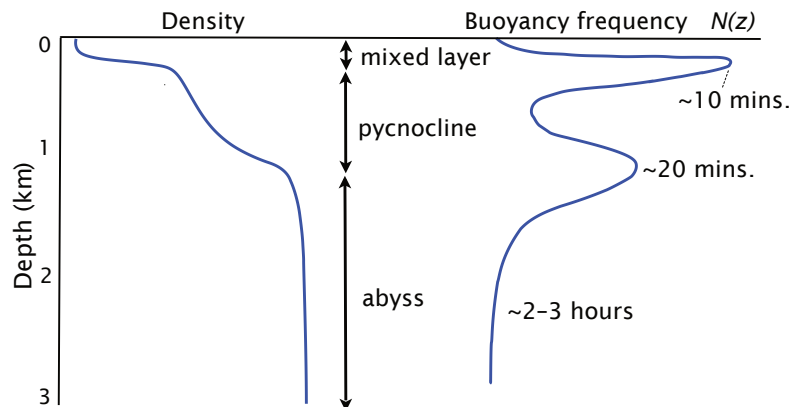
As we discussed in section 6.3 a wave packet will follow a *ray*, where a ray is simply a trajectory following the group velocity. Restricting attention to two dimensions and using (7.75) the horizontal and vertical components of the group velocity are (for $l > 0$),

$$c_g^y = \frac{N m^2}{(l^2 + m^2)^{3/2}}, \quad c_g^z = \frac{-N l m}{(l^2 + m^2)^{3/2}}. \quad (7.114a,b)$$

The path of a ray may thus be parameterized by the expression

$$\frac{dz}{dy} = \frac{c_g^z}{c_g^y} = -\frac{l}{m} = \frac{-\omega}{\sqrt{N^2 - \omega^2}}. \quad (7.115)$$

Fig. 7.11 Sketch of an ocean density profile, left, and buoyancy frequency, right, labelled with the approximate period. The pycnocline is sandwiched between two weakly stratified regions. The double peak in the buoyancy frequency is exaggerated, but the pycnocline is generally the region of highest frequency internal waves.



where the rightmost expression follows from the dispersion relation (7.57) with $k = 0$. The above expressions hold even when N varies in the vertical. Now, for there to be vertical propagation the vertical wavenumber must be positive and the wave frequency must be less than N . Suppose a wave is generated in a strongly stratified region and propagates vertically to a more weakly stratified region (with smaller N). The vertical wavenumber m becomes smaller and smaller, both the vertical and horizontal components of the group velocity tend to zero and the wave packet will stall. However, c_g^y goes to zero faster than c_g^z and the ray path turns toward the region of lower stratification.

This behaviour may be interpreted in terms of the dispersion relation $\omega = N \cos \vartheta$, where $\vartheta = \cos^{-1}[l^2/(l^2 + m^2)]$ is the angle between the three-dimensional wavevector and the horizontal (see section 7.3.2). If N decreases as we move vertically then ϑ must decrease until we reach the maximum value of $\cos \vartheta = 1$ and the wave vector is purely horizontal. The group velocity is perpendicular to the wave vector and so is then purely vertical. The wave cannot propagate into the region in which $N^2 < \omega^2$ for then m is imaginary and the disturbance will decay. Rather, the wave will tend to reflect, and the region where $N = \omega$ is often called a turning level. The trajectory can be obtained analytically in the region of the turning level as follows. Suppose that $N = \omega$ at $z = z^*$ so that, expanding N^2 around that point, we have $N^2(z) \approx N^2(z^*) + (z - z^*)dN^2(z^*)/dz$. Equation (7.115) becomes

$$\frac{dz}{dy} = \frac{-\omega}{\sqrt{(z - z^*)dN^2/dz}}. \quad (7.116)$$

which, upon integrating, yields

$$z - z^* = \frac{\omega(y^* - y)^{2/3}}{\sqrt{dN^2/dz}}. \quad (7.117)$$

This cusp-like trajectory is illustrated in Fig. 7.10.

An idealized oceanic waveguide

The stratification of the ocean is decidedly nonuniform in the vertical, as schematically illustrated in Fig. 7.11. The density is almost uniform in a layer at the top of the ocean about 50–100 m deep known as the mixed layer. The density then increases fairly rapidly over a region 500–1000 m deep known as the pycnocline, and is then fairly uniform in the abyss. The weak stratification in the abyss and in the mixed layer will inhibit the propagation of internal waves generated in the pycnocline. For example, consider a wave of frequency ω propagating downwards from the oceanic thermocline with and into the weakly stratified abyss. As soon as $N(z) < \omega$ the vertical wavenumber becomes imaginary and disturbance will vary like $e^{\pm mz}$. On physical grounds we must choose the solution that evanesces with depth. Similar behaviour will occur for a wave propagating up

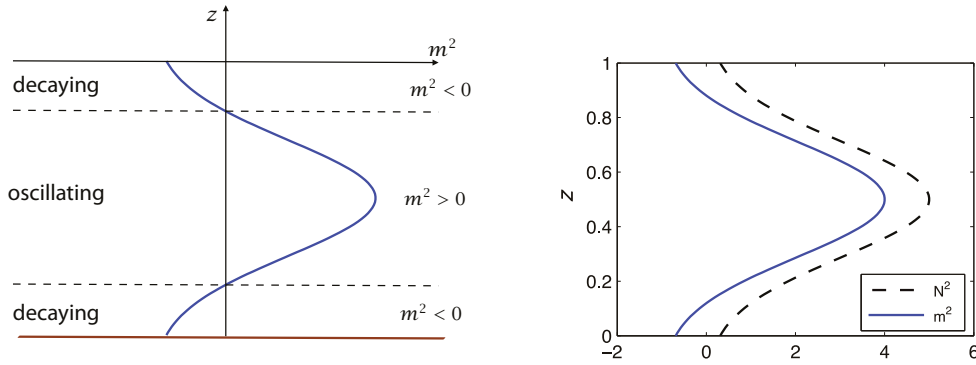


Fig. 7.12 An oceanic wave guide. The left panel shows m^2 and the regions of oscillation and decay, and the right panel also shows the value of N^2 from which m^2 is calculated using (7.107). Waves generated in the central region will propagate before evanescent in the region of negative m^2 , so confining the waves to the central wave guide.

from the thermocline into the weakly stratified mixed layer. Thus, waves are trapped in a region where $N^2 > \omega^2$, and this region forms a *wave guide*, as sketched in Fig. 7.12. Essentially the same dynamics are described again in an atmospheric context below.

The profile of N^2 is a simple exponential and the corresponding value of m^2 is calculated using (7.107) with $K = \omega = 1$. (the values are nondimensional; the reader is invited to ‘re-dimensionalize’). The value of m goes to zero near the top and the bottom of the domain, as illustrated. The corresponding group velocities are illustrated in Fig. 7.13, and can be seen to be purely vertical at the two turning heights. The amplitude of a wave becomes very large near the turning heights, but the wave itself need not break because its energy is constant and its vertical wavelength is very large. Rather, the wave will be reflected (following the trajectory illustrated in Fig. 7.10), and the wave is confined in the waveguide.

7.5.4 Atmospheric considerations

The atmosphere differs from the ocean in many ways, but for the purposes of internal waves two of these are particularly important: (i) The density diminishes in the vertical and so the Boussinesq approximation is not valid, except for small vertical displacements; (ii) there is no upper surface, so we must consider radiation conditions for large z , or require that the solutions remain bounded for $z \rightarrow \infty$, rather than conventional boundary conditions.

There are two commonly-used ways to deal with density variations — through the use of pressure coordinates or the anelastic equations. We will use pressure coordinates in chapter 17, but here we briefly consider the anelastic equations. These (see section 2.5) differ from the Boussinesq primarily in the mass continuity equation, which becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{\rho_0} \frac{\partial}{\partial z} (w \rho_0) = 0 \quad (7.118)$$

where $\rho_0 = \rho_0(z)$ is a specified profile of density. Using (7.118) instead of (7.54a) gives the equation of motion

$$\frac{\partial^2}{\partial t^2} \left(\nabla^2 w' + \frac{\partial}{\partial z} \frac{1}{\rho_0} \frac{\partial \rho_0 w'}{\partial z} \right) + N^2 \nabla^2 w' = 0, \quad (7.119)$$

in place of (7.55). Because ρ_0 is a function of z we cannot find plane wave solutions without additional approximation — for example unless we assume that ρ_0 changes only slowly with z . For this

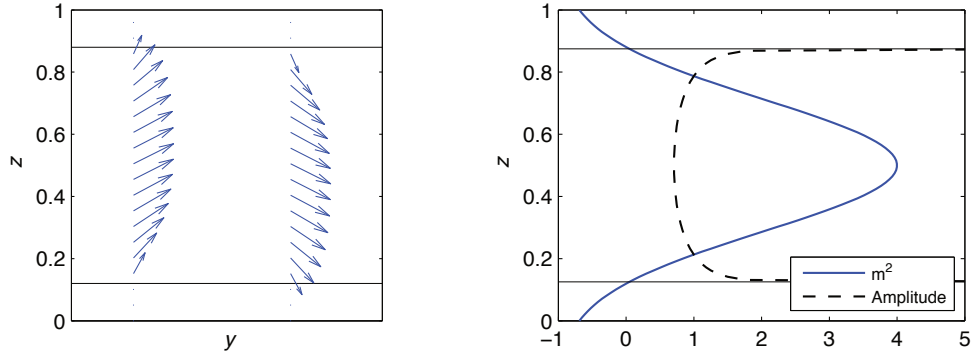


Fig. 7.13 Left panel: Group velocity vectors for upward and downward propagating gravity waves in a stratification illustrated in Fig. 7.12, calculated using (7.114). Right panel: The values of m^2 and the amplitude of the wave, the latter varying as $m^{-1/2}$. The thin horizontal lines in both panels indicate the height at which $m^2 = 0$.

reason the Boussinesq approximation is often imposed from the outset in theoretical work, even for the atmosphere; the approximation is a quantitatively poor, but the qualitative character of the waves is maintained.

The second factor (the lack of an upper surface) becomes an issue when considering gravity waves propagating high into the atmosphere, a phenomena we look at in chapter 17. In section 7.7 we consider the generation of internal waves by flow over topography, a phenomena of particular atmospheric importance. To finish this section off, let us consider an atmospheric waveguide. The dynamics are very similar to those of the oceanic waveguide discussed above but, for the sake of variety, we will treat it in a slightly different way.

An atmospheric waveguide

We suppose the atmosphere to be a semi-infinite region from the ground at $z = 0$ to infinity. If N^2 is constant then solutions, as in the bounded case, vary sinusoidally in z , for example $w' \sim \sin mz$, where m is the vertical wavenumber. These solutions remain bounded as $z \rightarrow \infty$, although they do not decay. If N varies, then other possibilities exist. Suppose that a region of small stratification, N_1 overlies a region of larger stratification, N_2 ; that is

$$N = \begin{cases} N_1 & z > H, \\ N_2 & 0 < z < H. \end{cases} \quad (7.120)$$

where $N_2 > N_1$. (This is *not* a model of the stratosphere overlying the troposphere, because the stratosphere is highly stratified. If anything, it is a model of the mesosphere overlying the stratosphere and troposphere.) The frequency in the two regions must be the same and if $\omega < N_1 < N_2$ then

$$\omega^2 = \frac{N_1^2}{K^2 + m_1^2} = \frac{N_2^2}{K^2 + m_2^2}, \quad (7.121)$$

whence

$$m_1 = m_2 \left(\frac{N_1^2 - \omega^2}{N_2^2 - \omega^2} \right)^{1/2}. \quad (7.122)$$

In contrast, if $N_1 < \omega < N_2$ then wave-like solutions are not allowed in the upper region, because the frequency must always be less than the local value of N . Rather, solutions in the upper region evanesce according to

$$w'_1 = \tilde{w}_1 e^{-\mu z} e^{i(kx + ly - \omega t)}, \quad (7.123)$$

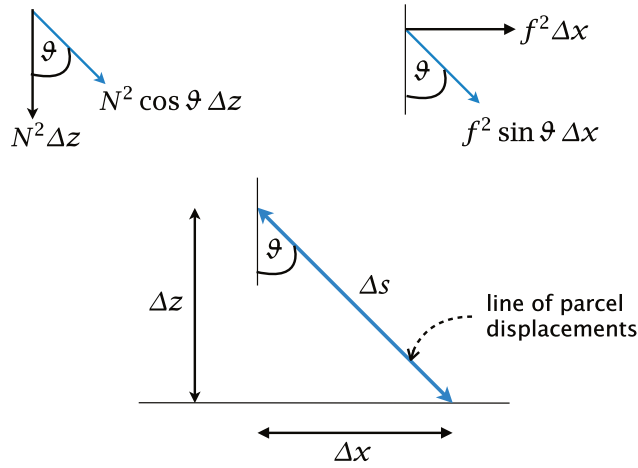


Fig. 7.14 Parcel displacements and associated forces in an inertia-gravity wave in which the parcel displacements are occurring at an angle θ to the vertical. Both Coriolis and buoyancy forces are present, and $\Delta s = \Delta z / \cos \theta = \Delta x / \sin \theta$.

where

$$\mu^2 = \frac{\omega^2 - N_1^2}{\omega^2} K^2. \quad (7.124)$$

The solutions still vary sinusoidally in the lower layer, according to

$$w'_2 = \bar{w}_2 \sin m_1 z e^{i(kx + ly - \omega t)}, \quad (7.125)$$

where m now takes on only discrete values in order to satisfy the boundary conditions that w and ϕ are continuous $z = H$, and that w vanishes at $z = 0$.

7.6 INTERNAL WAVES IN A ROTATING FRAME OF REFERENCE

In the presence of both a Coriolis force and stratification a displaced fluid will feel two restoring forces — one due to gravity and the other to rotation. The first gives rise to gravity waves, as we have discussed, and the second to inertial waves. When the two forces are together simultaneously the resulting waves are called *inertia-gravity waves*. The algebra describing them can be complicated so let us begin with a simpler parcel argument to try to lay bare the basic dynamics; the reader may also wish to first refer back to section 7.3.3.

7.6.1 A parcel argument

Consider a parcel that is displaced along a slantwise path in the x - z plane, as shown in Fig. 7.14, with a horizontal displacement of Δx and a vertical displacement of Δz . Let us suppose that the fluid is Boussinesq and that there is a stable and uniform stratification given by $N^2 = -g\rho_0^{-1}\partial\rho_0/\partial z = \partial b/\partial z$. Referring to (7.68) as needed, the component of the restoring buoyancy force, F_b say, in the direction of the parcel oscillation is given by (7.68)

$$F_b = -N^2 \cos \theta \Delta z = -N^2 \cos^2 \theta \Delta s. \quad (7.126)$$

The parcel will also experience a restoring Coriolis force, F_C , and the component of this in the direction of the parcel displacement is

$$F_C = -f^2 \sin \theta \Delta x = -f^2 \sin^2 \theta \Delta s. \quad (7.127)$$

Using (7.126) and (7.127) the (Lagrangian) equation of motion for a displaced parcel is

$$\frac{d^2 \Delta s}{dt^2} = -(N^2 \cos^2 \theta + f^2 \sin^2 \theta) \Delta s, \quad (7.128)$$

and hence the frequency is given by

$$\omega^2 = N^2 \cos^2 \vartheta + f^2 \sin^2 \vartheta. \quad (7.129)$$

Now, nearly everywhere in both atmosphere and ocean, $N^2 > f^2$. From (7.129) we then see that the frequency lies in the interval $N^2 > \omega^2 > f^2$. (To see this, put $N = f$ or $f = N$ in (7.129), and use $\sin^2 \vartheta + \cos^2 \vartheta = 1$.) If the parcel displacements approach the vertical then the Coriolis force diminishes and $\omega \rightarrow N$, and similarly $\omega \rightarrow f$ as the displacements become horizontal. The ensuing waves are then pure inertial waves.

We can write (7.129) in terms of wavenumbers if we note that, in the x - z plane,

$$\cos^2 \vartheta = \frac{k^2}{k^2 + m^2}, \quad \sin^2 \vartheta = \frac{m^2}{k^2 + m^2} \quad (7.130)$$

where k and m are the horizontal and vertical wavenumbers. The dispersion relation becomes

$$\omega^2 = \frac{N^2 k^2 + f^2 m^2}{k^2 + m^2}. \quad (7.131)$$

Let's now move on to a discussion using the linearized equations of motion.

7.6.2 Equations of motion

In a rotating frame of reference, specifically on an f -plane, the linearized equations of motion are the momentum equations

$$\frac{\partial \mathbf{u}'}{\partial t} + \mathbf{f}_0 \times \mathbf{u}' = -\nabla \phi' \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (7.132a,b)$$

and the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.132c,d)$$

These are similar to (7.53) and (7.54), with the addition of a Coriolis term in the horizontal momentum equations.

To obtain a single equation for w' we take the horizontal divergence of (7.132a) and use the continuity equation to give

$$\frac{\partial}{\partial t} \left(\frac{\partial w'}{\partial z} \right) + f_0 \zeta' = \nabla^2 \phi' \quad (7.133)$$

where $\zeta' \equiv (\partial v' / \partial x - \partial u' / \partial y)$ is the vertical component of the vorticity. We may obtain an evolution equation for that vorticity by taking the curl of (7.132a), giving

$$\frac{\partial \zeta'}{\partial t} = f_0 \frac{\partial w'}{\partial z}. \quad (7.134)$$

Eliminating vorticity between these equations gives

$$\left(\frac{\partial^2}{\partial t^2} + f_0^2 \right) \frac{\partial w'}{\partial z} = \frac{\partial}{\partial t} \nabla^2 \phi'. \quad (7.135)$$

We may obtain another equation linking pressure and vertical velocity by eliminating the buoyancy between (7.132b) and (7.132d), so giving

$$\frac{\partial^2 w'}{\partial t^2} + N^2 w' = -\frac{\partial}{\partial t} \frac{\partial \phi'}{\partial z}. \quad (7.136)$$

Eliminating ϕ' between (7.135) and (7.136) gives a single equation for w' analogous to (7.55), namely

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] w' = 0. \quad (7.137)$$

If we assume a time dependence of the form $w' = \hat{w}e^{-i\omega t}$, this equation may be written in the sometimes useful form,

$$\frac{\partial^2 \hat{w}}{\partial z^2} = \left(\frac{N^2 - \omega^2}{\omega^2 - f_0^2} \right) \nabla^2 \hat{w}. \quad (7.138)$$

7.6.3 Dispersion Relation

Assuming wave solutions to (7.137) of the form

$$w' = \tilde{w}e^{i(kx+ly+mz-\omega t)}, \quad (7.139)$$

we readily obtain the dispersion relation

$$\omega^2 = \frac{f_0^2 m^2 + (k^2 + l^2)N^2}{k^2 + l^2 + m^2}. \quad (7.140)$$

We can also write the dispersion relation as

$$\omega^2 = f_0^2 \sin^2 \vartheta + N^2 \cos^2 \vartheta, \quad (7.141)$$

or

$$\omega^2 = f_0^2 + (N^2 - f_0^2) \cos^2 \vartheta, \quad \text{or} \quad \omega^2 = N^2 - (N^2 - f_0^2) \sin^2 \vartheta, \quad (7.142)$$

where ϑ is the angle of the wavevector with the horizontal. The frequency therefore lies between N and f_0 . The waves satisfying (7.140) are sometimes called inertia-gravity and are analogous to surface gravity waves in a rotating frame — that is, Poincaré waves — discussed in section 3.8.2.

In many atmospheric and oceanic situations $f_0 \ll N$ (in fact typically $N/f_0 \sim 100$, the main exception being weakly stratified near-surface mixed layers in the ocean) and $f < \omega < N$. From (7.141) the frequency is dependent only on the angle the wavevector makes with the horizontal, and the surfaces of constant frequency again form cones in wavenumber space, although depending on the values of f and ω the frequency does not necessarily decrease monotonically with ϑ as in the non-rotating case. For reference, the group velocity is

$$c_g^x = \left[\frac{N^2 - f_0^2}{\omega K_3^4} Km \right] \frac{km}{K} = \left[\frac{N^2 - f_0^2}{\omega K_3} \cos \vartheta \sin \vartheta \right] \cos \lambda \sin \vartheta, \quad (7.143a)$$

$$c_g^y = \left[\frac{N^2 - f_0^2}{\omega K_3^4} Km \right] \frac{lm}{K} = \left[\frac{N^2 - f_0^2}{\omega K_3} \cos \vartheta \sin \vartheta \right] \sin \lambda \sin \vartheta, \quad (7.143b)$$

$$c_g^z = - \left[\frac{N^2 - f_0^2}{\omega K_3^4} Km \right] K = - \left[\frac{N^2 - f_0^2}{\omega K_3} \cos \vartheta \sin \vartheta \right] \cos \vartheta, \quad (7.143c)$$

using (7.70), and where $K_3^4 \equiv (k^2 + l^2 + m^2)^2$. These expressions reduce to (7.75) if $f = 0$, in which case $\omega = N \cos \vartheta$. Notice that the directional factors — the sin and cos terms outside of the square brackets — are the same as those in (7.75). Thus, the group velocity is, as in the non-rotating case,

at an angle ϑ to the vertical, or $\alpha = \pi/2 - \vartheta$ to the horizontal. The magnitude of the group velocity is now given by

$$|c_g| = \frac{N^2 - f_0^2}{\omega K_3^3} K m = \frac{N^2 - f_0^2}{\omega K_3} \cos \vartheta \sin \vartheta. \quad (7.144)$$

There are a few notable limits:

1. A purely horizontal wave vector. In this case $m = 0$ and $\omega = N$. The waves are then unaffected by the Earth's rotation. This is because the Coriolis force is (in the f -plane approximation) due to the product of the Coriolis parameter and the horizontal component of the velocity. If the wave vector is horizontal, the fluid velocities are purely vertical and so the Coriolis force vanishes.
2. A purely vertical wave vector. In this case $\omega = f_0$. In this case the fluid velocities are horizontal and the fluid parcels do not feel the stratification. The oscillations are then known as *inertial waves*, although they are not inertial in the sense of there being no implied force in an inertial frame of reference. This case and the previous one show that when $\omega = N$ or $\omega = f_0$ the group velocity is zero.
3. In the limit $N \rightarrow 0$ we have pure inertial waves with a frequency $0 < \omega < f_0$, and specifically $\omega = f_0 \sin \vartheta$. Similarly, as $f_0 \rightarrow 0$ we have pure internal waves, as discussed previously, with $\omega = N \cos \vartheta$.
4. The hydrostatic limit, which we discuss below.

The hydrostatic limit

Hydrostasy occurs in the limit of large horizontal scales, $k, l \ll m$. If we therefore neglect k^2 and l^2 where they appear with m^2 in (7.140) we obtain

$$\omega^2 = f_0^2 + N^2 \frac{k^2 + l^2}{m^2} = f_0^2 + N^2 \cos^2 \vartheta. \quad (7.145)$$

where the rightmost expression arises from (7.141) if we take

$$\sin^2 \vartheta = \frac{m^2}{k^2 + l^2 m^2} \rightarrow 1, \quad \cos^2 \vartheta = \frac{K^2}{k^2 + m^2} \rightarrow \frac{K^2}{m^2} \ll 1, \quad (7.146)$$

with $K^2 = k^2 + l^2$.

If we make the hydrostatic approximation from the outset in the rotating, linearized, equations of motion then we have

$$\frac{\partial u'}{\partial t} - f_0 v' = -\frac{\partial \phi'}{\partial x}, \quad \frac{\partial v'}{\partial t} + f_0 u' = -\frac{\partial \phi'}{\partial y}, \quad 0 = -\frac{\partial \phi'}{\partial z} + b', \quad (7.147a)$$

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.147b)$$

This reduces to the single equation

$$\left[\frac{\partial^2}{\partial t^2} \frac{\partial^2}{\partial z^2} + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] w' = 0, \quad (7.148)$$

and corresponding dispersion relation

$$\omega^2 = \frac{f_0^2 m^2 + K^2 N^2}{m^2} = f_0^2 + N^2 \alpha'^2, \quad (7.149)$$

so recovering (7.145). This is sometimes known as the *rapidly rotating regime*.

It is notable that the Coriolis parameter f now appears in isolation, and simply provides inertial oscillations that are independent of the wavenumber and the stratification. The group velocity is therefore independent of the background rotation.

To make the small aspect ratio limit explicit let us define

$$\alpha' \equiv \frac{\text{vertical scale}}{\text{horizontal scale}} = \frac{K}{m} = \frac{1}{\tan \vartheta} \ll 1, \quad (7.150)$$

and using (7.143) and (7.142) we find

$$\alpha' = \frac{c_g^z}{c_g^h} = \frac{\omega^2 - f_0^2}{N^2 - \omega^2}. \quad (7.151)$$

where $c_g^h = (c_g^{x2} + c_g^{y2})^{1/2}$. If f is small, then the hydrostatic limit corresponds to $N^2 \gg \omega^2$ with dispersion relation

$$\omega^2 \approx N^2 \cos^2 \vartheta \approx N^2 \frac{K^2}{m^2}. \quad (7.152)$$

This is the same as the dispersion relation in the non-rotating, hydrostatic case derived earlier *ab initio*, giving (7.61).

We return to the hydrostatic limit in section 17.2 on gravity waves in the stratosphere.

7.6.4 Polarization relations

Just as in the non-rotating case we can derive phase relations between the various fields, useful if we are trying to identify internal waves from observations. As for all waves in an incompressible fluid, the condition $\nabla_3 \cdot \mathbf{v} = 0$ gives

$$\mathbf{k} \cdot \mathbf{v}' = 0, \quad (7.153)$$

so that the fluid motion is in the plane that is perpendicular to the wave vector. The derivations of the other polarization relations are left as exccerics for the reader, and the relations are found to be

$$\tilde{u} = \frac{k\omega + i l f_0}{\omega^2 - f_0^2} \tilde{\phi}, \quad \tilde{v} = \frac{l\omega - i k f_0}{\omega^2 - f_0^2} \tilde{\phi}, \quad (7.154a,b)$$

which should be compared with (7.64a). We also have a relation between buoyancy and pressure,

$$\tilde{b} = \frac{i m N^2}{N^2 - \omega^2} \tilde{\phi} \quad (7.155)$$

and one between vertical velocity and pressure,

$$\tilde{w} = \frac{-m\omega}{N^2 - \omega^2} \tilde{\phi} = \frac{-\omega K_3^2}{(N^2 - f_0^2)m} \tilde{\phi}, \quad (7.156)$$

with the second equality following with use of the dispersion relation.

7.6.5 Geostrophic motion and vortical modes

If we seek *steady* solutions to (7.132) and (7.132c,d), the equations of motions become

$$-f_0 v = -\frac{\partial \phi'}{\partial x}, \quad f_0 u = -\frac{\partial \phi'}{\partial y}, \quad 0 = -\frac{\partial \phi'}{\partial z} + b', \quad (7.157a,b)$$

and

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad w' N^2 = 0. \quad (7.158a,b)$$

These are the equations of geostrophic and hydrostatic balance, with zero vertical velocity. What is the dispersion relation corresponding to this?

If instead of eliminating pressure between (7.135) and (7.136) we eliminate vertical velocity we obtain

$$\frac{\partial}{\partial t} \left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] \phi' = 0, \quad (7.159)$$

which is similar to (7.137), except for the extra time derivative, which allows for the possibility of a solution with $\omega = 0$. If $\omega \neq 0$ then

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] \phi' = 0, \quad (7.160)$$

and the dispersion relation is given by (7.140). If $\omega = 0$, then the quantity in square brackets in (7.159) may not be a function of time; that is

$$\left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f_0^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] \phi' = \chi(x, y, z), \quad (7.161)$$

where χ is a function of space, but not time, and so determined by the initial conditions of ϕ' . When $\omega \neq 0$, then $\chi = 0$. What is χ ? We shall see that it is nothing but the potential vorticity of the flow!

Potential vorticity

Recall the vorticity equation and the buoyancy equation, namely

$$\frac{\partial \zeta'}{\partial t} = f_0 \frac{\partial w'}{\partial z}, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.162a,b)$$

If we eliminate w' from these equations we obtain

$$\frac{\partial q}{\partial t} = 0 \quad (7.163)$$

where

$$q = \left[\zeta' + f_0 \frac{\partial}{\partial z} \left(\frac{b'}{N^2} \right) \right] \quad (7.164)$$

is the potential vorticity for this problem. In general, for adiabatic flow potential vorticity is conserved on fluid parcels and $DQ/Dt = 0$ where for a Boussinesq fluid $Q = \omega_a \cdot \nabla b$. There are two differences between this general case and ours; first, because we have linearized the dynamics the advective term is omitted, and $\partial q / \partial t = 0$. Second, q is not exactly the same as Q , but it is an approximation to it valid when the stratification is dominated by its background value, N^2 . Very informally, we have then, for constant N ,

$$\begin{aligned} Q &= (\omega + f_0) \cdot \nabla b \approx (\zeta + f_0) \left(N^2 + \frac{\partial b'}{\partial z} \right) \\ &\approx f_0 N^2 + f_0 \frac{\partial b'}{\partial z} + \zeta N^2 = N^2 \left[f_0 + \zeta + f_0 \frac{\partial}{\partial z} \left(\frac{b'}{N^2} \right) \right]. \end{aligned} \quad (7.165)$$

The first term on the right-hand side of this expression, namely $f_0 N^2$, is a constant and so dynamically unimportant, and the remaining terms are equal to q as given by (7.164).

Another way to see that (7.164) is the potential vorticity is to note that the displacement of an isentropic surface, η say, is related to the change in buoyancy by

$$\eta \approx -\frac{b'}{\partial b / \partial z} = -\frac{b'}{N^2}, \quad (7.166)$$

as illustrated in Fig. 3.13 on page 139. The thickness of an isentropic layer is the difference between the heights of two neighbouring isentropic surfaces, and so is given by

$$h = -\frac{b'_1}{N^2} + \frac{b'_2}{N^2} \approx -H \frac{\partial}{\partial z} \left(\frac{b'}{N^2} \right), \quad (7.167)$$

where H is the mean thickness between the surfaces. Thus, the expression (7.164) may be written

$$q = \left[\zeta' - \frac{f_0 h}{H} \right], \quad (7.168)$$

which is the 'shallow water' expression for the potential vorticity of a fluid layer, linearized about a mean thickness H and a state of rest (with $|\zeta'| \ll f_0$).

Let us now relate q to χ , and we do this by expressing ζ' and b' in terms of ϕ' and w' . From (7.133) and (7.132b) respectively we have,

$$f_0 \zeta' = \nabla^2 \phi' - w'_{zt}, \quad (7.169a)$$

$$\frac{f_0^2}{N^2} b'_z = \frac{f_0^2}{N^2} w_{zt} + \frac{f_0^2}{N^2} \phi'_{zz}, \quad (7.169b)$$

using subscripts to denote derivatives. Thus, f_0 times the potential vorticity is

$$f_0 q = \nabla^2 \phi' + \frac{f_0^2}{N^2} \phi'_{zz} + \frac{f_0^2}{N^2} w_{zt} - w_{zt}. \quad (7.170)$$

We now use (7.136) to express the second w_{zt} term in terms of ϕ' , giving

$$f_0 q = \nabla^2 \phi' + \frac{f_0^2}{N^2} \phi'_{zz} + \frac{f_0^2}{N^2} w'_{zt} + \frac{1}{N^2} (w'_{zttt} + \phi'_{zztt}), \quad (7.171)$$

and we then use (7.135) to eliminate w' , giving

$$f_0 q = \nabla^2 \phi' + \frac{f_0^2}{N^2} \phi'_{zz} + \frac{1}{N^2} (\nabla^2 \phi'_{tt} + \phi'_{zztt}), \quad (7.172)$$

or, re-arranging,

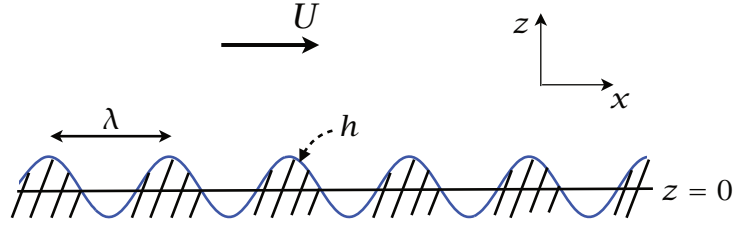
$$f_0 q = \frac{1}{N^2} \left[\frac{\partial^2}{\partial t^2} \left(\nabla^2 \phi' + \frac{\partial^2 \phi'}{\partial z^2} \right) + N^2 \nabla^2 \phi' + f_0^2 \frac{\partial^2 \phi'}{\partial z^2} \right]. \quad (7.173)$$

Comparing this with (7.161), we can see that

$$\chi = f_0 N^2 q. \quad (7.174)$$

That is to say, the conserved quantity for motions with $\omega = 0$ is nothing but a constant multiple of the potential vorticity. When $\omega \neq 0$, then χ and hence the potential vorticity are zero. In other words, *oscillating linear gravity waves, even in a rotating reference frame, have zero potential vorticity*. This is an important result, because large-scale balanced dynamics is characterized by the advection of potential vorticity, so that (in the linear approximation at least) internal waves play not direct role in the potential vorticity budget. However, they *do* play an important role in transporting and dissipating energy, as we will see later on.

Fig. 7.15 Uniform flow, U , in the x -direction flowing over sinusoidal topography, h . The vertical co-ordinate is stretched, and in reality $|h| \ll \lambda$.



7.7 TOPOGRAPHIC GENERATION OF INTERNAL WAVES

How are internal waves generated? One way that is important in both the ocean and atmosphere is by way of a horizontal flow, such as a mean wind or, in the ocean, a tide, passing over a topographic feature. This forces the fluid to move up and/or down, so generating an internal wave. In this section we illustrate this mechanism with simple examples of steady flow over idealized topography.

7.7.1 Sinusoidal mountain waves

For simplicity we ignore the effects of the Earth's rotation and pose the problem in two dimensions, x and z , using the Boussinesq approximation. Our goal is to calculate the response to a steady, uniform flow of magnitude U over a sinusoidally varying boundary $h = \tilde{h} \cos kx$ at $z = 0$, as in Fig. 7.15 with $k = 2\pi/\lambda$. The topographic variations are assumed small, so allowing the dynamics to be linearized, which would turn enable an arbitrarily shaped boundary to be considered by appropriately summing over Fourier modes. Further, because the problem is linear, the frequency of the response is equal to that of the forcing. Now, suppose we pose the problem in the frame of reference of the mean flow; the topography then has the form

$$h = h_0 \cos[k(x + Ut)]. \quad (7.175)$$

Thus, any resulting internal waves have frequency $\omega = -Uk$, because this is the only time dependence in the problem. This is also a convenient frame in which to work, because there is no mean flow advecting the fields. (An equivalent way to proceed is to stay in the stationary frame and replace each time derivative $\partial/\partial t$ with an advection term $U\partial/\partial x$.)

To proceed it is convenient to write $h = \text{Re } h_0 e^{i(kx - \omega t)}$, and the various dynamical fields in the response, such as the vertical velocity w and the pressure field $\phi(z)$ then have the form

$$w = \text{Re } \tilde{w}(z) e^{i(kx - \omega t)}, \quad \phi = \text{Re } \tilde{\phi}(z) e^{i(kx - \omega t)}, \quad (7.176)$$

where $\omega = -Uk$. The problem is to determine the form of $\tilde{w}(z)$, $\tilde{\phi}(z)$ and so on.

At the lower boundary the vertical velocity must satisfy the linearized kinematic boundary condition $w = Dh/Dt = \partial h/\partial t + U\partial h/\partial x$. In the moving frame $U = 0$ and therefore we have

$$w = w_0 e^{i(kx - \omega t)} = \frac{\partial h}{\partial t} = -i\omega h_0 e^{i(kx - \omega t)}, \quad \text{at } z = 0, \quad (7.177)$$

where $w_0 = \tilde{w}(0)$, and so the amplitude of the vertical velocity at the surface is given by

$$w_0 = -i\omega h_0 = iUkh_0. \quad (7.178)$$

The equation of motion to be satisfied by the vertical velocity above the boundary is (7.55), namely

$$\left[\frac{\partial^2}{\partial t^2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \right) + N^2 \frac{\partial^2}{\partial x^2} \right] w = 0. \quad (7.179)$$

which, given a harmonic dependence in t and x as in (7.176), becomes

$$\frac{\partial^2 \tilde{w}}{\partial z^2} = \frac{\omega^2 - N^2}{\omega^2} k^2 \tilde{w}. \quad (7.180)$$

If N^2 is constant this equation admits of a solution of the form $\tilde{w} = w_0 e^{imz}$ where

$$m^2 = \frac{k^2(N^2 - \omega^2)}{\omega^2} = \left(\frac{N}{U}\right)^2 - k^2, \quad (7.181)$$

using $\omega = -Uk$. Equation (7.181) is of course just the dispersion relation for internal gravity waves, but here we are using it to determine the vertical wavenumber since the frequency is given. This solution satisfies the boundary condition at $z = 0$ because the amplitude of the waves is given by (7.178) and the frequency of the waves is given by $\omega = -Uk$. Note that m^2 may be negative, and so m imaginary, if $N^2 < k^2 U^2$, so evidently there will be a qualitative difference between short waves and long waves.

Given the solution for w we can use the polarization relations of section 7.3.2 with $\omega = -Uk$ to obtain the solutions for perturbation horizontal velocity and pressure. In the stationary frame of reference the solutions are then

$$w = \tilde{w}(z) e^{ikx} = w_0 e^{i(kx+mz)} = iUkh_0 e^{imz} e^{ikx}, \quad (7.182a)$$

$$u = \tilde{u}(z) e^{ikx} = u_0 e^{i(kx+mz)} = -imUh_0 e^{imz} e^{ikx}, \quad (7.182b)$$

$$\phi = \tilde{\phi}(z) e^{ikx} = \phi_0 e^{i(kx+mz)} = imU^2 h_0 e^{imz} e^{ikx}, \quad (7.182c)$$

where m is given by (7.181). In the moving frame of reference we replace x by $x + Ut$; that is, the above solutions are multiplied by $\exp[-i\omega t]$. The above relationships between w, u and ϕ are the de facto polarization relations for this problem.

Having obtained the mathematical form of the solutions let us see what the solutions mean, and if and in what sense the waves propagate away from the mountains.

7.7.2 Energy Propagation

The frequency of the mountain waves is just that of internal waves; that is, $\omega = \pm N \cos \vartheta$, where ϑ is the angle between the wave vector and the horizontal. In our problem, the frequency is determined from the outset by the velocity of the mean flow and the scale of the topography, and thus so is the direction of propagation of the waves crests. The direction of energy propagation is given by the group velocity, given by (7.75), or (7.143) with $f = 0$. The group velocity is at an angle ϑ to the vertical, and two results that will be useful are that the vertical group velocity and phase speeds are given by

$$c_g^z = \frac{-\omega m}{k^2 + m^2} = \frac{Ukm}{k^2 + m^2}, \quad c_p^z = -\frac{Uk}{m} \quad (7.183a,b)$$

Short, trapped waves

Suppose that the wave frequency is sufficiently high that $\omega^2 > N^2$, which will occur if the undulations on the boundary have a sufficiently short wavelength that $k^2 > (N/U)^2$. From (7.181) m^2 is negative and m is pure imaginary. Writing $m = is$, so that $s^2 = k^2 - (N/U)^2$, the solutions have the form

$$\tilde{w} = w_0 e^{i(kx - \omega t) - sz}. \quad (7.184)$$

We must choose the solution with $s > 0$ in order that the solution decays away from the mountain, and internal waves are not propagated into the interior. (If there were a rigid lid or a density discontinuity at the top of the fluid (as at the top of the ocean) then the possibility of reflection would

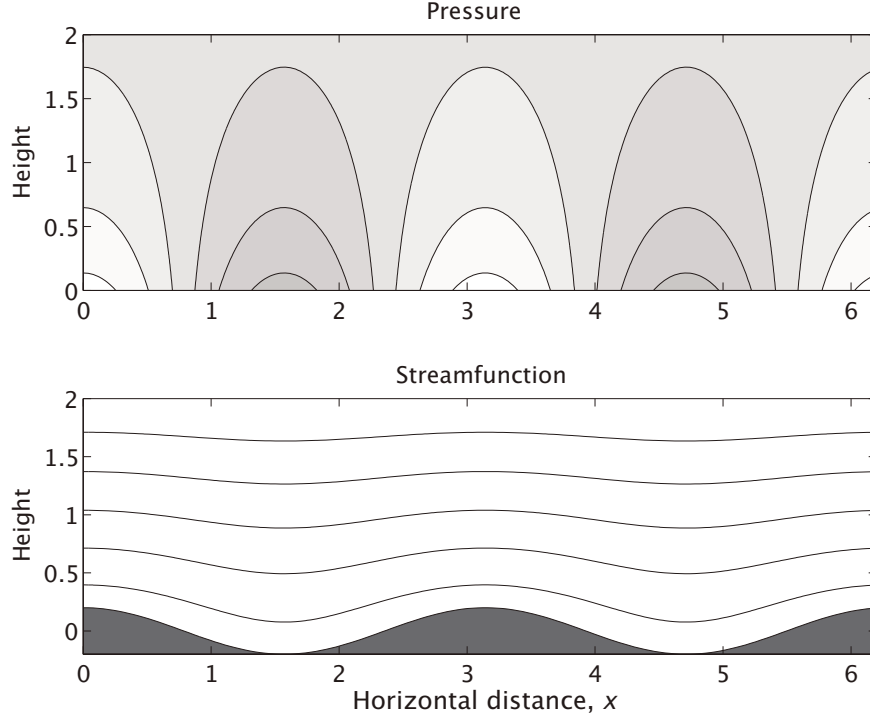


Fig. 7.16 Solutions for the flow over a sinusoidal ridge, using (7.182), in the short wave limit and with $m = i$. The top panel shows phase lines of pressure, with darker gray indicating higher pressure. The bottom panel shows contours of the total streamfunction, $\psi - Uz$, with flow coming in from the left, and the topography itself (solid). The perturbation amplitude decreases exponentially with height.

arise and we would seek to satisfy the upper boundary condition with a combination of decaying and amplifying modes.) The above result is entirely consistent with the dispersion relation for internal waves, namely $\omega = N \cos \vartheta$: because $\cos \vartheta < 1$ the frequency ω must be less than N so that if the forcing frequency is higher than N no internal waves will be generated.

Because the waves are trapped waves we do not expect energy to propagate away from the mountains. To verify this, from the polarization relation (7.182) we have

$$\bar{w} = \frac{k}{mU} \tilde{\phi} = \frac{-ik}{sU} \tilde{\phi}. \quad (7.185)$$

The pressure and the vertical velocity are therefore out of phase by $\pi/2$, and the vertical energy flux, $\overline{w\phi}$ [see section 7.3.5 and in particular (7.84)] is identically zero. This is consistent with fact that the energy flux is in the direction of the group velocity; the group velocity is given by (7.183a) and for an imaginary m the real part is zero. A solution is shown in Fig. 7.16 with $s = 1$ ($m = i$).

Long, propagating waves

Suppose now that $k^2 < (N/U)^2$ so that $\omega^2 < N^2$. From (7.181) m is now real and the solution has propagating waves of the form

$$w = w_0 e^{i(kx + mz - \omega t)}, \quad m^2 = \left(\frac{N}{U}\right)^2 - k^2 \quad (7.186)$$

Topographically Generated Gravity Waves (Mountain Waves)

- In both atmosphere and ocean an important mechanism for the generation of gravity waves is flow over bottom topography, and the ensuing waves are sometimes called mountain waves. A canonical case is that of a uniform flow over a sinusoidal topography, with constant stratification. If the flow is in the x -direction and there is no y -variation then the boundary condition is

$$w(x, z = 0) = U \frac{\partial h}{\partial x} = -iUk\tilde{h}. \quad (\text{MW.1})$$

Solutions of the problem may be found in the form $w(x, z, t) = w_0 \exp[i(kx + mz - \omega t)]$, where the boundary condition at $z = 0$ is given by (MW.1), the frequency is given by the internal wave dispersion relation, and the other dynamical fields are obtained using the polarization relations.

- One way to easily solve the problem is to transform into a frame moving with the background flow, U . The topography then appears to oscillate with a frequency $-Uk$, and this in turn becomes the frequency of the gravity waves.
- Propagating gravity waves can only be supported if the frequency is less than N , meaning that $Uk < N$. That is, the waves must be sufficiently long and therefore the topography must be of sufficiently large scale.
- When propagating waves exist, energy is propagated upward away from the topography. The topography also exerts a drag on the background flow.
- If the waves are too short they are evanescent, decaying exponentially with height. That is, they are trapped near the topography
- In the presence of rotation the wave frequency must lie between the buoyancy frequency N and the inertial frequency f . That is, waves can radiate upward if

$$f < Uk < N. \quad (\text{MW.2})$$

Thus, both very long waves and very short waves are evanescent.

Vertical propagation is occurring because the forcing frequency is less than the buoyancy frequency. The angle at which fluid parcel oscillations occur is then slanted off the vertical at an angle ϑ such that the forcing frequency is equal to the natural frequency of oscillations at that angle, namely

$$\vartheta = \cos^{-1} \left(\frac{Uk}{N} \right). \quad (7.187)$$

The angle ϑ is also the angle between the wavevector $\mathbf{k}$ and the horizontal, as in (7.62), because the wavevector is at right angles to the parcel oscillations. If $Uk = N$ then the fluid parcel oscillations are vertical and, using (7.181), $m = 0$. Thus, although the group velocity is directed vertically, parallel to the fluid parcel oscillations, its magnitude is zero, from (7.183).

Our intuition suggests that if there is vertical propagation there must be an upwards energy flux, since the energy source is at the ground. Let's confirm this. Using the polarization relations

(7.182a,c) we obtain

$$\omega_0 = \frac{k}{mU} \phi_0. \quad (7.188)$$

and the energy flux in the vertical direction is, from (7.91)

$$F^z = \frac{k}{2mU} |\phi_0|^2 = \frac{mU}{2k} |\omega_0|^2, \quad (7.189)$$

which is evidently non zero. This energy flux must be upward, away from the source (the topography), and this determines the sign of m that must be chosen by the solution. Specifically, for positive U , the group velocity must be positive so from (7.183) m must be positive. If U were negative the sign of m would be negative. Note that if $m = 0$ there is no vertical energy propagation

Because energy is propagating upward and away from the topography there must be a drag at the lower boundary. This stress at the boundary is the rate at which horizontal momentum is transported upwards and so is given by

$$\tau = -\rho_0 \overline{u\omega}. \quad (7.190)$$

where the overbar denotes averaging over a wavelength. From (7.182)

$$u_0 = -imUh_0, \quad \omega_0 = iUkh_0, \quad (7.191)$$

so that

$$\tau = -\rho_0 \overline{u\omega} = \frac{1}{2} kmU^2 h_0^2, \quad (7.192)$$

where the factor of 1/2 comes from the averaging, and note that we take the product $u_0 \omega_0^*$ where ω_0^* is the complex conjugate of ω_0 . The sign of the stress depends on the sign of m , and thus on the sign of U . For positive U , m is positive and so the stress is positive at the surface.

Solutions for flow over topography in the long wave limit and $m = 1$ are shown in Fig. 7.17. The flow is coming in from the left, and the phase lines evidently tilt upstream with height. Lines of constant phase follow $kx + mz = \text{constant}$, and in the solution shown both k and m are positive ($k = m = 1$). Thus, the lines slope back at a slope $x/z = -m/k$, and energy propagates up and to the left. The phase propagation is actually downward in this example — see (7.183). The pressure is high on the upstream side of the mountain, and this provides a drag on the flow — a topographic form drag.

♦ Frames of reference, group velocity and critical levels

For radiating waves, the group velocity seen in the resting and moving frames are different. In the moving frame we have

$$c_g^x = \frac{Um^2}{k^2 + m^2} = -\frac{Nm^2}{(k^2 + m^2)^{3/2}}, \quad (7.193)$$

$$c_g^z = \frac{Ukm}{k^2 + m^2} = \frac{-Nkm}{(k^2 + m^2)^{3/2}}, \quad (7.194)$$

noting that $U = -N/(k^2 + m^2)^{1/2}$ using the dispersion relation and $\omega = -Uk$. We see that $c_g^z/c_g^x = -k/m$, so that the group velocity is, as expected, directed parallel to the phase lines. In the resting frame the horizontal component of the group velocity is shifted by an amount $-U$, so that $c_g^z/c_g^x = -m/k$ and group velocities relative to the ground and air are perpendicular to each other.

If and as the waves propagate upwards it may encounter a *critical level* at which the phase speed of the waves equals the background flow, that is $c = U$. The location of the critical level is not dependent on the choice of frame of reference. At a critical level the wave amplitudes can be expected to become large and linear theory will break down. It is not uncommon for dissipative effects to become important, as we will see explicitly in our discussion of the quasi-biennial oscillation in chapter 17.

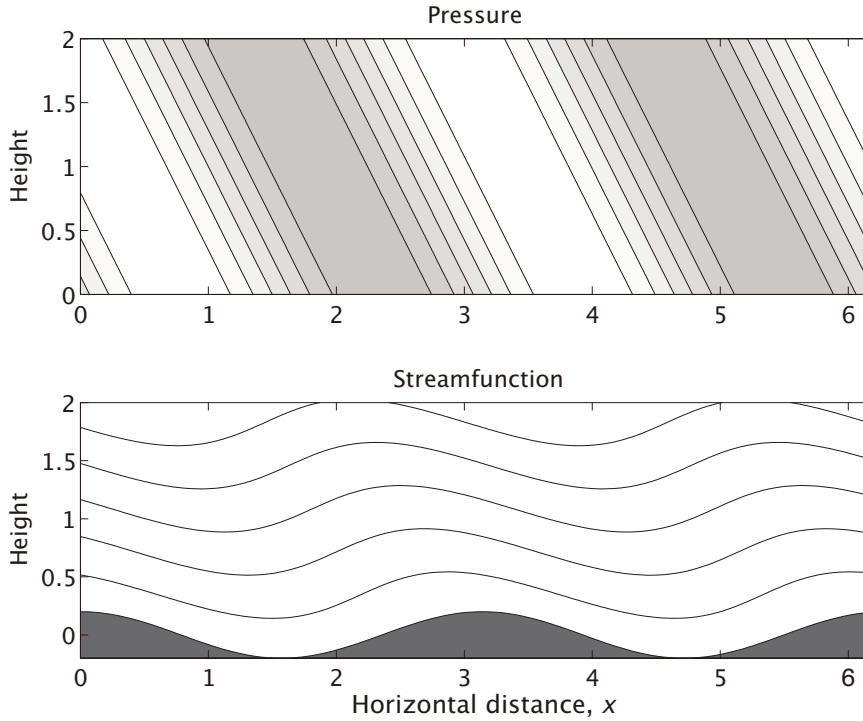


Fig. 7.17 As for Fig. 7.16, but now showing solutions using (7.182) in the long wave limit with $m = 1$. The top panel shows phase lines of pressure, with gray indicating higher pressure. The bottom panel shows contours of the total streamfunction, $\psi - Uz$, with flow coming in from the left, and the topography itself (solid). Note that pressure is high on the windward side of the topography, and phase lines tilt upstream with height for both pressure and streamfunction.

Atmospheric and oceanic parameters

A natural question to ask is whether, for typical atmospheric and oceanic parameters, evanescent or propagating gravity waves are more likely to be excited. Consider the atmosphere with a surface flow of $u = 5 \text{ m s}^{-1}$ and $N = 10^{-2} \text{ s}^{-1}$. Then the critical wavenumber separating evanescent and propagating waves is $k = N/U = 2 \times 10^{-3}$, corresponding to a wavelength of about 3000 m. This is quite large, and almost certainly at that scale rotational effects are also important. Still, large-scale topographic features like the Rockies, Andes and Himalayas do contain such large wavelengths and so we can expect them to excite upward propagating gravity waves.

In the ocean the abyssal stratification is quite weak, typically with $N \approx 10^{-4} \text{ s}^{-1}$ (although N can be as high as 10^{-3} s^{-1}) and the velocities are also weak, compared to those of the upper ocean, although they can be of order 1 cm s^{-1} in eddying regions. Using these values we find a critical wavenumber of $k = N/U = 10^{-2} \text{ m}^{-1}$ with a wavelength of 600 m. Certainly the ocean bathymetry has many scales larger than this (and for smaller values of u the critical scales are correspondingly smaller) meaning that it is relatively easy for abyssal flow to generate gravity waves that propagate upward into the ocean interior. The upper ocean is much more greatly stratified, with $N \approx 10^{-2} \text{ s}^{-1}$. Gravity waves are no longer generated by flow over topography but by the stirring effects of winds making a turbulent mixed layer. The forcing frequency must still be less than N in order to efficiently generate gravity waves, and using again a velocity of 1 cm s^{-1} we might heuristically estimate that propagating gravity waves can be generated with scales of meters, much

smaller than the gravity waves generated in the abyss.

7.7.3 Flow over an isolated ridge

Most mountains are of course not perfect sinusoids, but we can construct a solution for any given topography using a superposition of Fourier modes. In this section we will illustrate the solution for a mountain consisting of a single ridge; the actual solution must be obtained numerically and here we will just sketch the method and illustrate the results.

Sketch of the methodology

The methodology to compute a solution is as follows. Consider a topographic profile, $h(x)$, and let us suppose that it is periodic in x over some distance L . Such profile can (nearly always) be decomposed into a sum of Fourier coefficients, meaning that we can write

$$h(x) = \sum_k \tilde{h}_k e^{ikx} \quad (7.195)$$

where $\tilde{h}_k$ are the Fourier coefficients. We can obtain the set of $\tilde{h}_k$ by multiplying (7.195) by e^{-ikx} and integrating over the domain from $x = 0$ to $x = L$, a procedure known as taking the Fourier transform of $h(x)$, and there are standard computer algorithms for doing this efficiently. Once we have obtained the values of $\tilde{h}_k$ we essentially solve the problem separately for each k in precisely the same manner as we did in the previous section. Note that for *each* k there will be a vertical wavenumber given by (7.181), so that for each wavenumber we obtain a solution for pressure of the form $\tilde{\phi}_k(z)$, and similarly for the other variables. Once we have the solution for each wavenumber, then at each level we sum over all the wavenumbers to obtain the solution in real space; that is, we evaluate

$$\phi(x, z) = \sum_k \tilde{\phi}_k(z) e^{ikx}. \quad (7.196)$$

This is known as taking the inverse Fourier transform.

The solution

For specificity let us consider the bell-shaped topographic profile

$$h(x) = \frac{h_0 a^2}{a^2 + x^2}, \quad (7.197)$$

sometimes called the Witch of Agnesi.³ (Results with a Gaussian profile are qualitatively similar.) Such a profile is composed of *many* (in fact an infinite number of) Fourier coefficients of differing amplitudes. If the profile is narrow (meaning a is small, in a sense made clearer below) then there will be a great many significant coefficients at high wavenumbers. In fact, in the limiting case of an infinitely thin ridge (a delta function) all wavenumbers are present with equal weight, so there are certainly more large wavenumbers than small wavenumbers. However, if a is large, then the contributing wavenumbers will predominantly be small.

In the problem of flow over topography the natural horizontal scale is U/N . If $a \gg U/N$ then the dominant wavenumbers are small and the solution will consist of waves propagating upward with little loss of amplitude and phase lines tilting upstream, as illustrated in Fig. 7.18. If the ridge is sufficiently wide then the solution is essentially hydrostatic, with little dependence of the vertical structure on the horizontal wavenumber. That is, using (7.181) at large scales, $m^2 \approx (N/U)^2$ and the pattern repeats itself in the vertical at intervals of $2\pi U/N$, and so at any given level there can be only one wave crest in the fluid flowing over the ridge.

In the case of a narrow ridge, as illustrated in Fig. 7.19, the perturbation is largely trapped near to the mountain and the perturbation fields largely decay exponentially with height. Nevertheless,

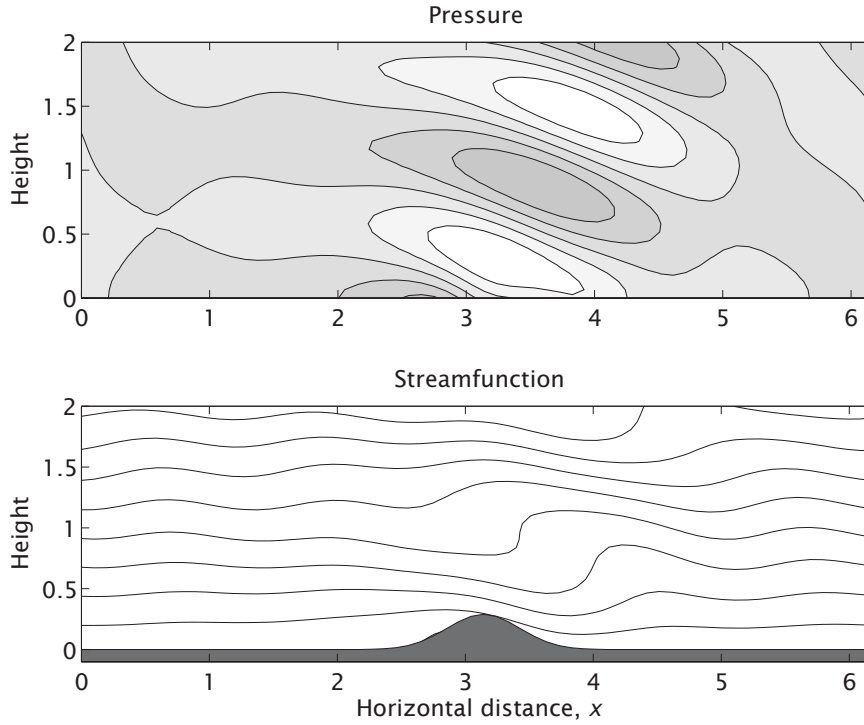


Fig. 7.18 Solutions for the flow over a bell-shaped ridge (7.197), with $a^2 = 4U^2/N^2$. High pressure is shaded darker, and the flow comes in from the left.

because the ridge *does* contain some small wavenumbers some weak, propagating large-scale disturbances are generated. The fluid acts as a low-pass filter, and the perturbation aloft consists only of large scales.

7.7.4 Effects of rotation

General considerations

We now consider, albeit briefly, the effects of a Coriolis force on mountain waves. The problem is in many ways quite similar to the non-rotating case but the dispersion relation and so the criteria for upward propagation differ accordingly. First, we note that the steady flow must be in geostrophic balance, so that if the flow is zonal there is a background meridional pressure gradient that satisfies

$$f_0 U = -\frac{\partial \Phi}{\partial y}. \quad (7.198)$$

The main difference in the solution field arises from the fact that the waves now obey the dispersion relation with rotation, namely (7.140) or, restricting attention to the x - z plane,

$$\omega^2 = \frac{f_0^2 m^2 + k^2 N^2}{k^2 + m^2}. \quad (7.199)$$

The frequency of the waves is still given by $\omega = -Uk$ (if we consider the problem in the translating frame), so that the vertical wavenumber is now given by

$$m^2 = \frac{k^2(N^2 - U^2 k^2)}{U^2 k^2 - f_0^2}. \quad (7.200)$$

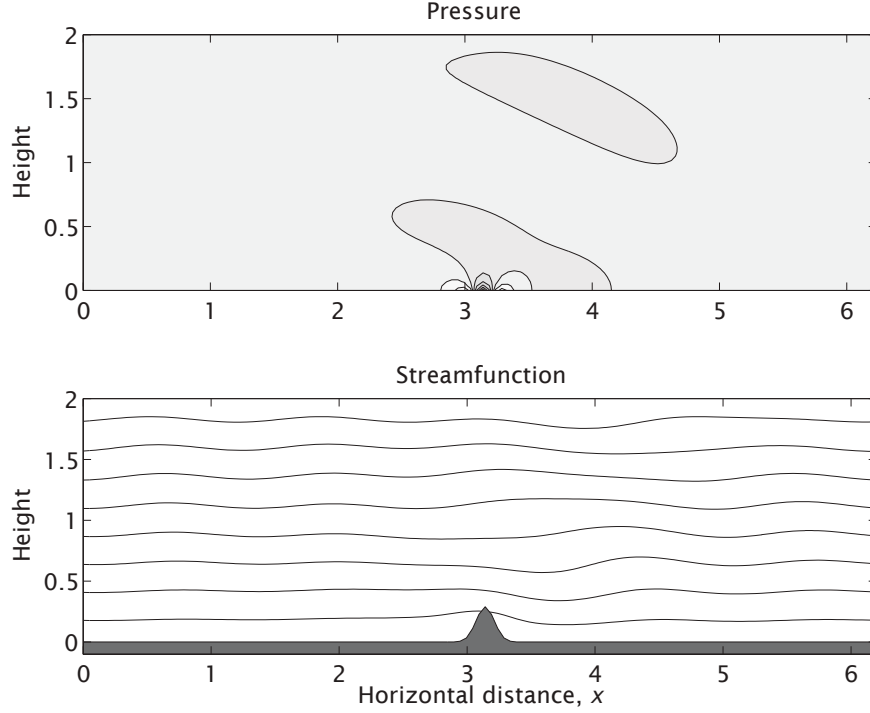


Fig. 7.19 As for Fig. 7.18 but now for a narrow ridge, with $a^2 = U^2/4N^2$.

Evanescent solutions arise when m is imaginary and, as before, such solutions arise for small scales for which $k > N/U$. However, from (7.200), evanescent solutions also arise for very large scales for which $k < f_0/U$. Propagating waves exist in the interval $N/U > k > f_0/U$ with the vertical wavenumber a real number, and these waves have frequencies between N and f_0 . In the atmosphere with $U = 10 \text{ m s}^{-1}$ and $f_0 = 10^{-4} \text{ s}^{-1}$ the large scale at which evanescence reappears is $L = 2\pi/k = 2\pi U/f \approx 600 \text{ km}$, which of course is not very large at all relative to global scales (and still smaller if we take $U = 5 \text{ m s}^{-1}$). Thus, upward propagating gravity waves exist between scales of a few kilometres (see the calculation on page 291) and several hundred kilometres. For the deep ocean, let us take $N = 10^{-3} \text{ s}^{-1}$ and $f_0 = 10^{-4} \text{ s}^{-1}$, and $U = 1 \text{ cm s}^{-1}$. Thus, very roughly, propagating waves exist between scales of a few tens of meters to a few hundred metres. If we use $N = 10^{-4} \text{ s}^{-1}$ the range of scales is further restricted.

Wave solutions and energy propagation

Obtaining a wave solution in the rotating case follows a similar path to the non-rotating case. In the resting frame vertical velocity satisfies the boundary condition $w = U\partial h/\partial x$, and in the moving frame $w = \partial h/\partial t$. Using the polarization relations appropriate for rotation we find analogous relations to (7.182), to wit

$$w = \tilde{w}(z)e^{ikx} = w_0 e^{i(kx+mz)} = iUkh_0 e^{imz} e^{ikx}, \quad (7.201a)$$

$$u = \tilde{u}(z)e^{ikx} = u_0 e^{i(kx+mz)} = -imUh_0 e^{imz} e^{ikx}, \quad (7.201b)$$

$$\phi = \tilde{\phi}(z)e^{ikx} = \phi_0 e^{i(kx+mz)} = \frac{im(U^2k^2 - f_0^2)}{k^2} h_0 e^{imz} e^{ikx}, \quad (7.201c)$$

$$v = \tilde{v}(z)e^{ikx} = v_0 e^{i(kx+mz)} = -if_0 \frac{m}{k} h_0 e^{imz} e^{ikx}, \quad (7.201d)$$

Of these, the expressions for w and u are no different from the non-rotating case, because w is set by the same boundary condition and u is given by mass continuity, $\partial u/\partial x + \partial w/\partial z = 0$, in both rotating and non-rotating cases. To obtain the expression for the pressure perturbation, (7.201c), we use (7.156). Finally, we note that the solutions now produce a meridional velocity, (7.201d), even when there is no variation in the topography in the y -direction. To obtain we use (7.154) with $l = 0$, giving $\tilde{v} = -i\tilde{u}f_0/Uk$.

As in the non-rotating case, when there are propagating waves there is high pressure on the windward (upstream) side of the topography and low pressure on leeward side, and the phase lines tilt upstream with height. The drag on the flow is equal to the rate of upward momentum transport and using (7.201a,b) we obtain

$$\rho_0 \overline{uw} = -\frac{1}{2}\rho_0 kmU^2 h_0^2 < 0. \quad (7.202)$$

This is just the same as (7.192). It is independent of height, and a momentum flux divergence will only arise in the free atmosphere if the waves break and dissipative effects become important.

The vertical flux of energy density is given by

$$\overline{\phi w} = \frac{1}{2}\rho_0 U \frac{m}{k} h_0^2 (U^2 k^2 - f_0^2) > 0 \quad (7.203)$$

Energy is propagating away from the mountain, consistent with the group velocity being directed upward.

7.8 ♦ ACOUSTIC-GRAVITY WAVES IN AN IDEAL GAS

In the final section of this chapter we consider wave motion in a stratified, *compressible* fluid such as the Earth's atmosphere. The stratification allows gravity waves to exist, and the compressibility allows sound waves to exist. The resulting problem is, not surprisingly, complicated and arcane and to make it as tractable as possible we will specialize to the case of an isothermal, stationary atmosphere and ignore the effects of rotation and sphericity. The results are not without interest, both in themselves and in illustrating the importance of simplifying the equations of motion from the outset, for example by making the Boussinesq or hydrostatic approximation, in order to isolate phenomena of interest.

In what follows we will denote the unperturbed state with a subscript 0 and the perturbed state with a prime ('), and we will omit some of the algebraic details. Because it is at rest, the basic state is in hydrostatic balance,

$$\frac{\partial p_0}{\partial z} = -\rho_0(z)g. \quad (7.204)$$

Ignoring variations in the y -direction for algebraic simplicity (and without loss of generality, in fact) the linearized equations of motion are:

$$u \text{ momentum:} \quad \rho_0 \frac{\partial u'}{\partial t} = -\frac{\partial p'}{\partial x} \quad (7.205a)$$

$$w \text{ momentum:} \quad \rho_0 \frac{\partial w'}{\partial t} = -\frac{\partial p'}{\partial z} - \rho' g \quad (7.205b)$$

$$\text{mass conservation:} \quad \frac{\partial \rho'}{\partial t} + w' \frac{\partial \rho_0}{\partial z} = -\rho_0 \left(\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} \right) \quad (7.205c)$$

$$\text{thermodynamic:} \quad \frac{\partial p'}{\partial t} - w' \frac{p_0}{H} = -\gamma p_0 \left(\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} \right). \quad (7.205d)$$

where $\gamma = c_p/c_v = 1/(1 - \kappa)$. For an isothermal basic state we have $p_0 = \rho_0 RT_0$ where T_0 is a constant, so that $\rho_0 = \rho_s e^{-z/H}$ and $p_0 = p_s e^{-z/H}$ where $H = RT_0/g$. The thermodynamic equation, (7.205c), is the linear form of (1.102) on page 24, which has the equation of state for an ideal gas built-in, and (7.205a,b,c,d) thus form a complete set with variables u' , w' , ρ' and p' .

Differentiating (7.205a) with respect to time and using (7.205d) leads to

$$\left(\frac{\partial^2}{\partial t^2} - c_s^2 \frac{\partial^2}{\partial x^2} \right) u' = c_s^2 \left(\frac{\partial}{\partial z} - \frac{1}{\gamma H} \right) \frac{\partial}{\partial x} w'. \quad (7.206a)$$

where $c_s^2 = \gamma p_0/\rho_0$ is the square of the speed of sound (equal to $\partial p/\partial \rho$ or γRT_0). Similarly, differentiating (7.205b) with respect to time and using (7.205c) and (7.205d) leads to

$$\left(\frac{\partial^2}{\partial t^2} - c_s^2 \left[\frac{\partial^2}{\partial z^2} - \frac{1}{H} \frac{\partial}{\partial z} \right] \right) w' = c_s^2 \left(\frac{\partial}{\partial z} - \frac{\kappa}{H} \right) \frac{\partial u'}{\partial x}, \quad (7.206b)$$

Equations (7.206a) and (7.206b) combine to give, after some cancellation,

$$\frac{\partial^4 w'}{\partial t^4} - c_s^2 \frac{\partial^2}{\partial t^2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{H} \frac{\partial}{\partial z} \right) w' - c_s^2 \frac{\kappa g}{H} \frac{\partial^2 w'}{\partial x^2} = 0. \quad (7.207)$$

If we set $w' = W(x, z, t) e^{z/(2H)}$, so that $W = (\rho_0/\rho_s)^{1/2} w'$, then the term with the single z -derivative is eliminated, giving

$$\frac{\partial^4 W}{\partial t^4} - c_s^2 \frac{\partial^2}{\partial t^2} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} - \frac{1}{4H^2} \right) W - c_s^2 \frac{\kappa g}{H} \frac{\partial^2 W}{\partial x^2} = 0. \quad (7.208)$$

Although superficially complicated, this equation has constant coefficients and we may seek wave-like solutions of the form

$$W = \text{Re } \widetilde{W} e^{i(kx + mz - \omega t)}, \quad (7.209)$$

where $\widetilde{W}$ is the complex wave amplitude. Using (7.209) in (7.208) leads to the dispersion relation for acoustic-gravity waves, namely

$$\omega^4 - c_s^2 \omega^2 \left(k^2 + m^2 + \frac{1}{4H^2} \right) + c_s^2 N^2 k^2 = 0, \quad (7.210)$$

with solution

$$\omega^2 = \frac{1}{2} c_s^2 K^2 \left[1 \pm \left(1 - \frac{4N^2 k^2}{c_s^2 K^4} \right)^{1/2} \right], \quad (7.211)$$

where $K^2 = k^2 + m^2 + 1/(4H^2)$. (The reader may investigate as to whether the factor $[1 - 4N^2 k^2/(c_s^2 K^4)]$ is positive.) For an isothermal, ideal-gas atmosphere $4N^2 H^2/c_s^2 \approx 0.8$ and so this may be written

$$\frac{\omega^2}{N^2} \approx 2.5 \widehat{K}^2 \left[1 \pm \left(1 - \frac{0.8 \widehat{k}^2}{\widehat{K}^4} \right)^{1/2} \right], \quad (7.212)$$

where $\widehat{K}^2 = \widehat{k}^2 + \widehat{m}^2 + 1/4$, and $(\widehat{k}, \widehat{m}) = (kH, mH)$.

7.8.1 Interpretation

Acoustic and gravity waves

There are two branches of roots in (7.211), corresponding to acoustic waves (using the plus sign in the dispersion relation) and internal gravity waves (using the minus sign). These (and the Lamb

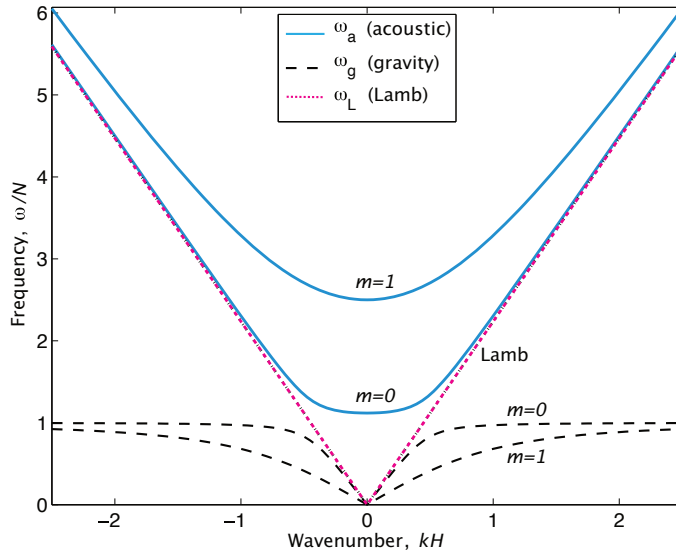


Fig. 7.20 Dispersion diagram for acoustic-gravity waves in an isothermal atmosphere, calculated using (7.212). The frequency is in units of buoyancy frequency N , and the wavenumbers are non-dimensionalized by the inverse of the scale height, H . Solid curves indicate acoustic waves, whose frequency is always higher than that of the corresponding Lamb wave (i.e., ck), and of the base acoustic frequency $\approx 1.12N$. The dashed curves are internal gravity waves, whose frequency asymptotes to N at small horizontal scales.

wave, described below) are plotted in Fig. 7.20. If $4N^2k^2/c_s^2K^4 \ll 1$ then the two sets of waves are well separated. From (7.212) this is satisfied when

$$\frac{4\kappa}{\gamma}(kH)^2 \approx 0.8(kH)^2 \ll \left[(kH)^2 + (mH)^2 + \frac{1}{4} \right]^2; \quad (7.213)$$

that is, when *either* $mH \gg 1$ or $kH \gg 1$. The two roots of the dispersion relation are then

$$\omega_a^2 \approx c_s^2 K^2 = c_s^2 \left(k^2 + m^2 + \frac{1}{4H^2} \right) \quad (7.214)$$

and

$$\omega_g^2 \approx \frac{N^2 k^2}{k^2 + m^2 + 1/(4H^2)}, \quad (7.215)$$

corresponding to acoustic and gravity waves, respectively. The acoustic waves owe their existence to the presence of compressibility in the fluid, and they have no counterpart in the Boussinesq system. On the other hand, the internal gravity waves are just modified forms of those found in the Boussinesq system, and if we take the limit $(kH, mH) \rightarrow \infty$ then the gravity wave branch reduces to $\omega_g^2 = N^2 k^2 / (k^2 + m^2)$, which is the dispersion relationship for gravity waves in the Boussinesq approximation. We may consider this to be the limit of infinite scale height or (equivalently) the case in which wavelengths of the internal waves are sufficiently small that the fluid is essentially incompressible.

Vertical structure

Recall that $w' = W(x, z, t)e^{z/(2H)}$ and, by inspection of (7.206), u' has the same vertical structure. That is,

$$w' \propto e^{z/(2H)}, \quad u' \propto e^{z/(2H)}, \quad (7.216)$$

and the amplitude of the velocity field of the internal waves increases with height. The pressure and density perturbation amplitudes fall off with height, varying like

$$p' \propto e^{-z/(2H)}, \quad \rho' \propto e^{-z/(2H)}. \quad (7.217)$$

The kinetic energy of the perturbation, $\rho_0(u'^2 + w'^2)$ is *constant* with height, because $\rho_0 = \rho_s e^{-z/H}$.

Hydrostatic approximation and Lamb waves

Equations (7.206) also admit to a solution with $w' = 0$. We then have

$$\left(\frac{\partial^2}{\partial t^2} - c_s^2 \frac{\partial^2}{\partial x^2} \right) u' = 0 \quad \text{and} \quad \left(\frac{\partial}{\partial z} - \frac{\kappa}{H} \right) \frac{\partial u'}{\partial x} = 0, \quad (7.218)$$

and these have solutions of the form

$$u' = \text{Re } \tilde{U} e^{\kappa z/H} e^{i(\kappa x - \omega t)}, \quad \omega = ck, \quad (7.219)$$

where $\tilde{U}$ is the wave amplitude. These are horizontally propagating sound waves, known as *Lamb waves* after the hydrodynamicist Horace Lamb. Their velocity perturbation amplitude increases with height, but the pressure perturbation falls with height; that is

$$u' \propto e^{\kappa z/H} \approx e^{2z/(7H)}, \quad p' \propto e^{(\kappa-1)z/H} \approx e^{-5z/(7H)}. \quad (7.220)$$

Their kinetic energy density, $\rho_0 u'^2$, varies as

$$KE \propto e^{-z/H + 2\kappa z/H} = e^{(2R - c_p)z/(c_p H)} = e^{(R - c_v)z/(c_p H)} \approx e^{-3z/(7H)} \quad (7.221)$$

for an ideal gas. (In a simple ideal gas, $c_v = nR/2$ where n is the number of excited degrees of freedom, 5 for a diatomic molecule.) The kinetic energy density thus falls away exponentially from the surface, and in this sense Lamb waves are an example of edge waves or surface-trapped waves.

Now consider the case in which we make the hydrostatic approximation ab initio, but without restricting the perturbation to have $w' = 0$. The linearized equations are identical to (7.205), except that (7.205b) is replaced by

$$\frac{\partial p'}{\partial z} = -\rho' g. \quad (7.222)$$

The consequence of this is that first term ($\partial^2 w' / \partial t^2$) in (7.206b) disappears, as do the first two terms in (7.207) [the terms $\partial^4 w' / \partial t^4 - c^2 (\partial^2 / \partial t^2) (\partial^2 w' / \partial x^2)$]. It is a simple matter to show that the dispersion relation is then

$$\omega^2 = \frac{N^2 k^2}{m^2 + 1/(4H^2)}. \quad (7.223)$$

These are long gravity waves, and may be compared with the corresponding Boussinesq result (7.61). Again, the frequency increases without bound as the horizontal wavelength diminishes. The Lamb wave, of course, still exists in the hydrostatic model, because (7.218) is still a valid solution. Thus, horizontally propagating sound waves still exist in hydrostatic (primitive equation) models, but vertically propagating sound waves do not — essentially because the term $\partial w / \partial t$ is absent from the vertical momentum equation.

Notes

- 1 As well as references given on page 251, gravity waves are treated extensively in the book by Sutherland (2010). I am also grateful to S. Legg for some unpublished lecture notes.
- 2 Here, and in our discussion of mountain waves in section 7.7, we draw from Durran (1990).
- 3 Treatments of this rather canonical profile are given by Queney (1948) and Durran (1990). The profile is named for Maria Agnesi, 1718–1799, an Italian mathematician and later a theologian, who had discussed the properties of the curve, as had Fermat and Guido Grandi earlier. The term ‘witch’ (in ‘Witch of Agnesi’) seems to be a mistranslation from Italian of *averisera*, or *versiera*, which refer to a curve and not an adversary of God or she-devil. Maria Agnesi may have been the first woman appointed to a professorship of mathematics, at the University of Bologna.

CHAPTER 8

Linear Dynamics at Low Latitudes

AT LOW LATITUDES THE ATMOSPHERE AND OCEAN take on rather different characters than in midlatitudes, and this chapter is our first taste of that. The tasting will be rather anodyne and mathematical, focussing on the linear dynamics of wave motion. We won't get into the real *phenomenology* of low latitudes: the tropical atmosphere with its humidity, its convection, and its towering cumulonimbus clouds, or the equatorial ocean with its undercurrents and counter-currents. And most certainly we don't get into low latitude atmosphere-ocean interaction and the wonderful phenomenon of El Niño—these all come later. Rather, this chapter is really just about the linear geophysical fluid dynamics of the shallow water equations at low latitudes, when the beta effect is important and the flow is not completely geostrophically balanced. Still, let us not be too deprecatory about these dynamics — they are important both in their own right and as prerequisites for these more complex phenomena that we encounter later.

Wave motion at low latitudes can be more complicated than its midlatitude counterpart. In midlatitudes there is a fairly clear separation in the time and space scales between balanced and unbalanced motion, and it is useful to recognize this by explicitly filtering out gravity wave motion and considering purely balanced motion, using for example the quasi-geostrophic equations. In equatorial regions, where the Coriolis parameter can become very small and is zero at the equator, the Rossby number may be order unity or larger and such a separation is less useful. However, even as f becomes small, β becomes large and Rossby waves remain important but the frequency separation between Rossby and gravity waves is smaller. The reader may then readily imagine the complications arising even from linear wave problems in equatorial regions and determining the dispersion relation for combined Rossby and gravity waves in a continuously stratified fluid is an algebraically complex task. The task is much simplified by posing the problem in the context of the shallow water equations, and these arise both as a physical model (e.g., of the thermocline in the ocean) or via a modal expansion, as in section 3.4.1.

Before diving into the mathematics, ask why do we talk about the 'tropical' atmosphere but the 'equatorial' ocean? It is because an essential demarcation in the atmosphere lies at the edge of the Hadley Cell, at about 30° latitude, and the dynamics are rather different poleward and equatorward of this edge. In some contrast, the dynamics of the ocean do not change their essential character until we approach quite close to the equator. At 10° latitude ocean dynamics has many of the characteristics of the mid-latitudes — the Rossby number is still very small, for example. Only until we get within a very few degrees of the equator does the dynamics change in a qualitative way.

8.1 CO-EXISTENCE OF ROSSBY AND GRAVITY WAVES

To see how Rossby waves and gravity waves co-exist, consider a single layer of shallow water on the beta plane in which the Coriolis parameter, f , is held constant except where it is differentiated, an approximation similar to that made when deriving the quasi-geostrophic equations.¹ Our treatment is thus more appropriate to a mid-latitude setting, but it provides a useful introduction to the more complex problem in which we allow both f and β to vary fully with latitude, described in section 8.2. Our equations of motion are

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \phi}{\partial x}, \quad \frac{\partial v}{\partial t} + fu = -\frac{\partial \phi}{\partial y}, \quad (8.1a,b)$$

$$\frac{\partial \phi}{\partial t} + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \quad (8.1c)$$

where, in terms of the familiar shallow water variables, $\phi = g'\eta$ and $c^2 = g'H$, where ϕ is the kinematic pressure, η is the free surface height, H is the reference depth of the fluid and g' is the reduced gravity.

After some manipulation (described more fully in section 8.2) we obtain, without additional approximation, a single equation for v :

$$\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \nabla^2 v - \beta \frac{\partial v}{\partial x} = 0. \quad (8.2)$$

In this equation the Coriolis parameter is given by the β -plane expression $f = f_0 + \beta y$; thus, the equation has a non-constant coefficient, entailing considerable algebraic difficulties. We will address these difficulties later, but for now we assume that f is constant except where differentiated, an approximation that is reasonable in mid-latitudes provided we are concerned with sufficiently small variations in latitude. Equation (8.2) then has constant coefficients and we may look for plane wave solutions of the form $v = \tilde{v} \exp [i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$, whence

$$\frac{\omega^2 - f_0^2}{c^2} - (k^2 + l^2) - \frac{\beta k}{\omega} = 0. \quad (8.3a)$$

This is a cubic equation in ω , as might be expected given (8.1). Written differently it becomes

$$\left(k + \frac{\beta}{2\omega} \right)^2 + l^2 = \left(\frac{\beta}{2\omega} \right)^2 + \frac{\omega^2 - f_0^2}{c^2}, \quad (8.3b)$$

which may be compared to (6.106). Noting that $k_d^2 = f_0^2/g'H = f_0^2/c^2$, the two equations are identical except for the appearance of a term involving frequency on last term on the right-hand side of (8.3b). The wave propagation diagram is illustrated in Fig. 8.1. The wave vectors at a given frequency all lie on a circle centered at $(-\beta/2\omega, 0)$ and with radius R given by

$$R = \left[\left(\frac{\beta}{2\omega} \right)^2 + \frac{\omega^2 - f_0^2}{c^2} \right]^{1/2}, \quad (8.4)$$

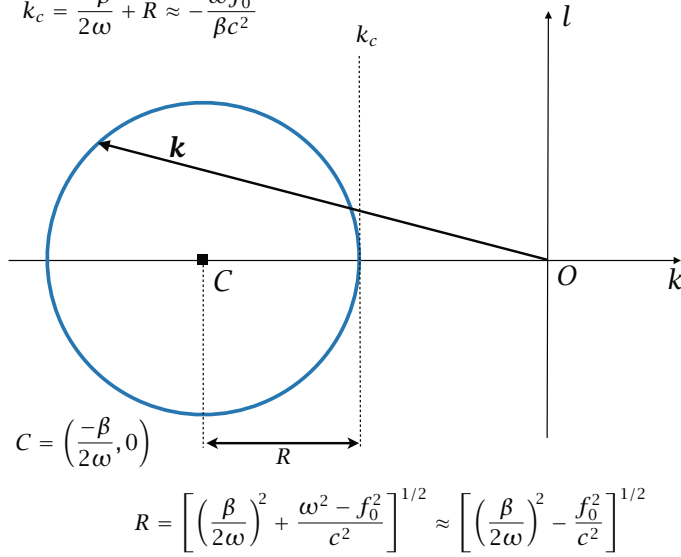
and the radius must be positive in order for the waves to exist. In the low frequency case the diagram is essentially the same as that shown in Fig. 6.8, but is quantitatively significantly different in the high frequency case. These limiting cases are discussed further in section 8.1.1 below.

To plot the full dispersion relation it is useful to nondimensionalize using the following scales for time (T), distance (L) and velocity (U)

$$T = f_0^{-1}, \quad L = L_d = k_d^{-1} = c/f_0, \quad U = L/T = c, \quad (8.5a,b)$$

Rossby waves

$$k_c = \frac{-\beta}{2\omega} + R \approx -\frac{\omega f_0^2}{\beta c^2}$$



Gravity waves

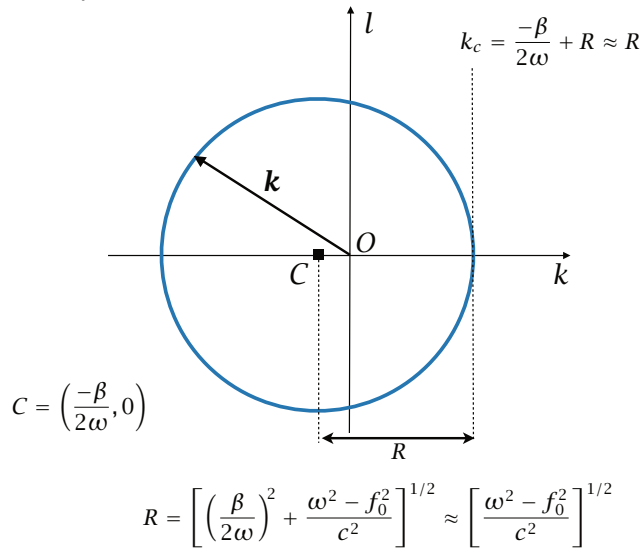


Fig. 8.1 Wave propagation diagrams for Rossby-gravity waves, obtained using (8.3). The top figure shows the diagram in the low frequency, Rossby wave limit, and the bottom figure shows the high frequency, gravity wave limit. In each case the locus of wavenumbers for a given frequency is a circle centered at $C = (-\beta/2\omega, 0)$ with a radius R given by (8.4), but the approximate expressions differ significantly at high and low frequency.

so that, denoting nondimensional quantities with a hat,

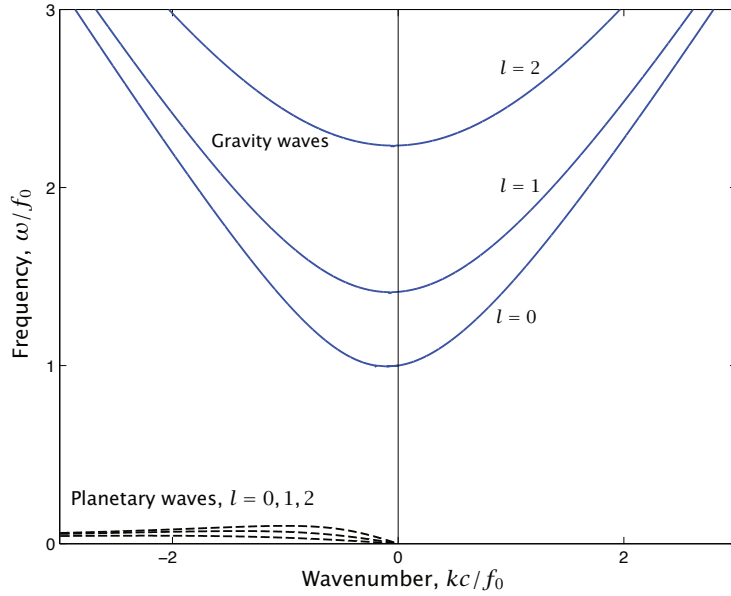
$$\omega = \hat{\omega} f_0, \quad (k, l) = (\hat{k}, \hat{l}) k_d, \quad \beta = \hat{\beta} \frac{f_0^2}{c} = \hat{\beta} \frac{f_0}{L_d} = \hat{\beta} f_0 k_d. \quad (8.6)$$

The dispersion relation (8.3) may then be written as

$$\hat{\omega}^2 - 1 - (\hat{k}^2 + \hat{l}^2) - \hat{\beta} \frac{\hat{k}}{\hat{\omega}} = 0 \quad (8.7)$$

We may expect that two of the roots correspond to gravity waves and the third to Rossby waves. The

Fig. 8.2 Dispersion relation for Rossby-gravity waves, obtained from (8.8) with $\hat{\beta} = 0.2$ for three values of l . There a frequency gap between the Rossby or planetary waves and the gravity waves. For the stratified mid-latitude atmosphere or ocean the frequency gap is in reality even larger.



only parameter in the dispersion relation is $\hat{\beta} = \beta c / f_0^2 = \beta L_d / f_0$. In the atmosphere a representative value for L_d is 1000 km, whence $\hat{\beta} = 0.1$. In the ocean $L_d \sim 100$ km, whence $\hat{\beta} = 0.01$. If we allow ourselves to consider ‘external’ Rossby waves (which are of some oceanographic relevance) then $c = \sqrt{gH} = 200 \text{ m s}^{-1}$ and $L_d = 2000$ km, whence $\hat{\beta} = 0.2$.

To actually obtain a solution we regard the equation as a quadratic in k and solve in terms of the frequency, giving

$$\hat{k} = -\frac{\hat{\beta}}{2\hat{\omega}} \pm \frac{1}{2} \left[\frac{\hat{\beta}^2}{\hat{\omega}^2} + 4(\hat{\omega}^2 - \hat{l}^2 - 1) \right]^{1/2}. \quad (8.8)$$

The solutions are plotted in Fig. 8.2, with $\hat{\beta} = 0.2$, and we see that the waves fall into two groups, labelled gravity waves and planetary waves in the figure. The gap between the two groups of waves is in fact still larger if a smaller (and generally more relevant) value of $\hat{\beta}$ is used. To interpret all this let us consider some limiting cases.

8.1.1 Special cases and properties of the waves

We now consider a few special cases of the dispersion relation.

(i) Constant Coriolis parameter

If $\beta = 0$ then the dispersion relation becomes

$$\omega [\omega^2 - f_0^2 - (k^2 + l^2)c^2] = 0, \quad (8.9)$$

with the roots

$$\omega = 0, \quad \omega^2 = f_0^2 + c^2(k^2 + l^2). \quad (8.10a,b)$$

The root $\omega = 0$ corresponds to geostrophic motion (and, since $\beta = 0$, Rossby waves are absent), with the other root corresponding to Poincaré waves, considered in chapter 3. Note that $\omega^2 > f_0^2$.

Rossby and Gravity Waves

- Generically speaking, Rossby-gravity waves are waves that arise under the combined effects of a potential vorticity gradient and stratification. Sometimes the definition is restricted to a wave on a single branch of the dispersion curve connecting Rossby and Gravity waves. In mid-latitudes Rossby waves and gravity waves are well separated with distinct physical mechanisms
- The simplest setting in which such waves occur is in the linearized shallow water equations which may be written as a single equation for v , namely

$$\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \nabla^2 v - \beta \frac{\partial v}{\partial x} = 0. \quad (\text{RG.1})$$

- If we take both f and β to be constants then the equation above admits of plane-wave solutions with dispersion relation

$$\omega^2 - \frac{\beta k c^2}{\omega} = f_0^2 + c^2(k^2 + l^2). \quad (\text{RG.2})$$

- In Earth's atmosphere and ocean it is common, especially in mid-latitudes, for there to be a frequency separation between two classes of solution. To a good approximation, high frequency waves satisfy

$$\omega^2 = f_0^2 + c^2(k^2 + l^2). \quad (\text{RG.3})$$

These are inertio-gravity waves, also known as Poincaré waves. The low frequency waves satisfy

$$\omega = \frac{-\beta k c^2}{f_0^2 + c^2(k^2 + l^2)} = \frac{-\beta k}{k_d^2 + k^2 + l^2}, \quad (\text{RG.4})$$

where $k_d^2 = f_0^2/c^2$, and these are called Rossby waves or planetary waves.

- Rossby-gravity waves also exist in the stratified equations. Solutions may be found by decomposing the vertical structure into a series of orthogonal modes, and a sequence of shallow water equations for each mode results, with a different c for each mode. Solutions may also be found if f is allowed to vary in (RG.1), at the price of some algebraic complexity, as in section 8.2.

(ii) High frequency waves

If we take the limit of $\omega \gg f_0$ then (8.3a) gives

$$\frac{\omega^2}{c^2} - (k^2 + l^2) - \frac{\beta k}{\omega} = 0. \quad (8.11)$$

To be physically realistic we should also now eliminate the β term, because if $\omega \gg f_0$ then, from geometric considerations on a sphere, $k^2 \gg \beta k/\omega$. Thus, the dispersion relation is simply $\omega^2 = c^2(k^2 + l^2)$. These waves are just gravity waves uninfluenced by rotation, and are a special case of Poincaré waves.

(iii) *Low frequency waves*

Consider the limit of $\omega \ll f_0$. The dispersion relation reduces to

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \quad (8.12)$$

This is just the dispersion relation for quasi-geostrophic Rossby waves as previously obtained — see (6.64) or (6.91). In this limit, the requirement that the radius of the circle be positive becomes

$$\omega^2 < \frac{\beta^2}{4k_d^2}. \quad (8.13)$$

That is to say, the Rossby waves have a maximum frequency, and directly from (8.12) this occurs when $k = k_d$ and $l = 0$.

The frequency gap

The maximum frequency of Rossby waves is usually much less than the frequency of the Poincaré waves: the lowest frequency of the Poincaré waves is f_0 and the highest frequency of the Rossby waves is $\beta/2k_d$. Thus,

$$\frac{\text{Low gravity wave frequency}}{\text{High Rossby wave frequency}} = \frac{f_0}{\beta/2k_d} = \frac{f_0^2}{2\beta c}. \quad (8.14)$$

If $f_0 = 10^{-4} \text{ s}^{-1}$, $\beta = 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$ and $k_d = 1/100 \text{ km}^{-1}$ (a representative oceanic baroclinic deformation radius) then $f_0/(\beta/2k_d) = 200$. If $L_d = 1000 \text{ km}$ (an atmospheric baroclinic radius) then the ratio is 20. If we use a barotropic deformation radius of $L_d = 2000 \text{ km}$ then the ratio is 10. Evidently, for most midlatitude applications there is a large gap between the Rossby wave frequency and the gravity wave frequency. Because of this frequency gap, to a good approximation Fig. 8.2 may be obtained by separately plotting (8.10b) for the gravity waves, and (8.12) for the Rossby or planetary waves. The differences between these and the exact results become smaller as $\hat{\beta}$ gets smaller, virtually indistinguishable in the plots shown.

Finally, we remark that a ‘Rossby-gravity wave’ is sometimes defined to be the wave on a single branch of the dispersion curve that connects Rossby waves and gravity waves, depending on the value of the wavenumber. The equatorial beta plane does support such a wave — the ‘Yanai wave’ derived in chapter 8 and shown in Fig. 8.6. However, in the mid-latitude system above there is no such wave; rather, there are Rossby waves and gravity waves, separated by a frequency gap.

8.1.2 Planetary geostrophic Rossby waves

A good approximation for the large-scale ocean circulation involves ignoring the time-derivatives and nonlinear terms in the momentum equation, allowing evolution only to occur in the thermodynamic equation. This is the planetary-geostrophic approximation, introduced in section 5.2 178, and it is interesting to see to what extent that system supports Rossby waves.² It is easiest just to begin with the linear shallow water equations themselves, and omitting time derivatives in the momentum equation gives

$$-fv = -\frac{\partial \phi}{\partial x}, \quad fu = -\frac{\partial \phi}{\partial y}, \quad (8.15a,b)$$

$$\frac{\partial \phi}{\partial t} + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0. \quad (8.15c)$$

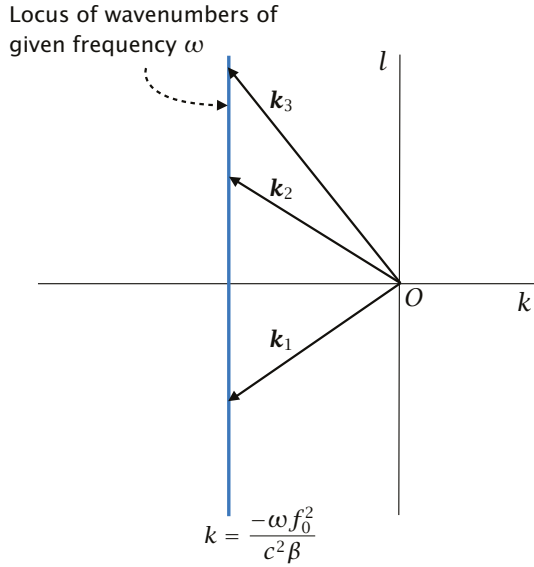


Fig. 8.3 The locus of points on planetary-geostrophic Rossby waves. Waves of a given frequency all have the same x -wavenumber, given by (8.17).

From these equations we straightforwardly obtain

$$\frac{\partial \phi}{\partial t} - \frac{c^2 \beta}{f^2} \frac{\partial \phi}{\partial x} = 0. \quad (8.16)$$

Again we will treat both f and β as constants so that we may look for solutions in the form $\phi = \tilde{\phi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)]$. The ensuing dispersion relation is

$$\omega = -\frac{c^2 \beta}{f_0^2} k = -\frac{\beta k}{k_d^2} \quad (8.17)$$

which is a limiting case of (8.12) with $k^2, l^2 \ll k_d^2$. The waves are a form of Rossby waves with phase and group speeds given by

$$c_p = -\frac{c^2 \beta}{f_0^2}, \quad c_g^x = -\frac{c^2 \beta}{f_0^2}. \quad (8.18)$$

That is, the waves are non-dispersive and propagate westward. Eq. (8.16) has the general solution $\phi = G(x + \beta c^2 / f^2 t)$, where G is any function, so an initial disturbance will just propagate westward at a speed given by (8.18), without any change in form.

Note finally that the locus of wavenumbers in k - l space is no longer a circle, as it is for the usual Rossby waves. Rather, since the frequency does not depend on the y -wavenumber, the locus is a straight line, parallel to the y -axis, as in Fig. 8.3. Waves of a given frequency all have the same x -wavenumber, given by $k = -\omega f_0^2 / (c^2 \beta) = -\omega k_d^2 / \beta$, as shown in Fig. 8.3.

Physical mechanism

Because the waves are a form of Rossby wave their physical mechanism is related to that discussed in section 6.4.3, but with an important difference: relative vorticity is no longer important, but the flow divergence is. Thus, consider flow round a region of high pressure, as illustrated in Fig. 8.4. If the pressure is circularly symmetric as shown, the flow to the south of H in the left-hand sketch, and to the south of L in the right-hand sketch, is larger than that to the north. Hence, in the left sketch the flow converges at W and diverges at E , and the flow pattern moves westward. In the flow depicted in the right sketch the low pressure propagates westward in a similar fashion.

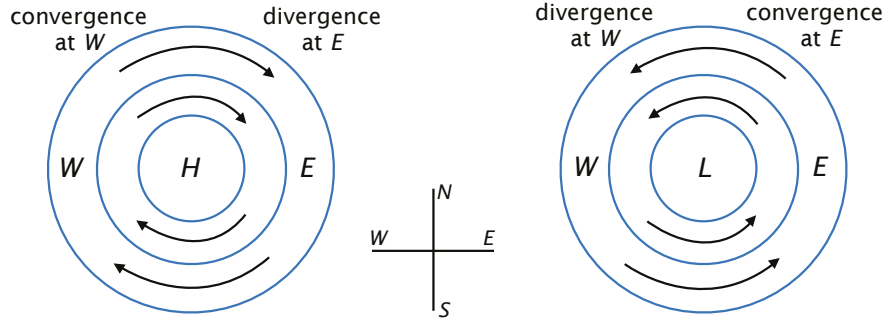


Fig. 8.4 The westward propagation of planetary-geostrophic Rossby waves. The circular lines are isobars centered around high and low pressure centres. Because of the variation of the Coriolis force, the mass flux between two isobars is greater to the south of a pressure center than it is to the north. Hence, in the left-hand sketch there is convergence to the west of the high pressure and the pattern propagates westward. Similarly, if the pressure centre is a low, as in the right-hand sketch, there is divergence to the west of the pressure centre and the pattern still propagates westward.

8.2 WAVES ON THE EQUATORIAL BETA PLANE

We now discuss the properties of shallow water waves at low latitudes, allowing the Coriolis parameter to properly vary in all terms, albeit using the β -plane approximation.³ Thus, Taylor-expanding the Coriolis parameter around a latitude ϑ_0 we obtain

$$f = 2\Omega \sin \vartheta \approx 2\Omega \sin \vartheta_0 + 2\Omega(\vartheta - \vartheta_0) \cos \vartheta_0 = f_0 + \beta y \quad (8.19)$$

where $f_0 = 2\Omega \sin \vartheta_0$, $\beta = 2\Omega \cos \vartheta_0/a$ and $y = a(\vartheta - \vartheta_0)$. For motions at low latitudes we take $\vartheta_0 = 0$, giving the *equatorial beta-plane approximation* in which $\sin \vartheta \approx \vartheta$, $\cos \vartheta \approx 1$ and $f = 2\Omega\vartheta = \beta y$. The linearized momentum and mass conservation equations are then

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \phi}{\partial x}, \quad \frac{\partial v}{\partial t} + fu = -\frac{\partial \phi}{\partial y}, \quad (8.20a,b)$$

$$\frac{\partial \phi}{\partial t} + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0. \quad (8.20c)$$

To make a connection with the conventional shallow water equations we note that $\phi = g'\eta$, where g' is the reduced gravity and η the free surface height, and $c^2 = g'H$ where H is the reference depth of the fluid. There are just two dimensional parameters in the above equations, c and β , and from these we may form the scales

$$T_{eq} = (c\beta)^{-1/2}, \quad L_{eq} = \left(\frac{c}{\beta} \right)^{1/2}. \quad (8.21a,b)$$

These are the fundamental time and length scales for equatorial dynamics (although some definitions differ by a factor of $\sqrt{2}$, as discussed later). The length scale L_{eq} is known as the equatorial radius of deformation. If we regard the above shallow water equations as coming from a modal decomposition of the primitive equations, as in section 3.4, then there is a wave speed, $c_m = \sqrt{g'H_m}$ and deformation radius for each mode. For the first baroclinic mode and for the atmosphere, if $c_1 = 25 \text{ m s}^{-1}$ and $\beta = 2.3 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$, then $L_{eq} \approx 1000 \text{ km}$ and $T_{eq} \approx 0.5 \text{ days}$; for the equatorial ocean with $c = 2 \text{ m s}^{-1}$ then $L_{eq} \approx 300 \text{ km}$ and $T_{eq} \approx 1.7 \text{ days}$.

Cross-differentiating (8.20a) and (8.20b) and using (8.20c) to eliminate the divergence we may also derive the linearized potential vorticity equation, namely

$$\frac{\partial}{\partial t} \left(\zeta - \frac{f\phi}{c^2} \right) + \beta v = 0. \quad (8.22)$$

This is the same as the familiar linearized potential vorticity equation on the f -plane, with the addition of the term $Df/Dt = \beta v$. Equation (8.22) is not independent of (8.20) but it turns out to be convenient to use it. In all the above equations, $f = \beta y$ and β is a constant.

To obtain a single equation for a single unknown, operate on (8.20a) with $(f/c^2)\partial_t$, on (8.20b) with $(1/c^2)\partial_{tt}$, on (8.20c) with $(g'/c^2)\partial_{ty}$ and on (8.22) with ∂_x , where $c^2 = g'H$. Using subscripts to denote derivatives the resulting equations are

$$\frac{f}{c^2}u_{tt} - \frac{f^2}{c^2}v_t = -\frac{f}{c^2}\phi_{xt}, \quad \frac{1}{c^2}v_{ttt} + \frac{f}{c^2}u_{tt} = -\frac{1}{c^2}\phi_{ytt}, \quad (8.23a,b)$$

$$\frac{1}{c^2}\phi_{tty} + (u_{xyt} + v_{yyt}) = 0, \quad v_{xxt} - u_{xyt} - \frac{f}{c^2}\phi_{xt} + \beta v_x = 0. \quad (8.23c,d)$$

These equations linearly combine to give a single equation for v , namely

$$\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} \right) - \beta \frac{\partial v}{\partial x} = 0. \quad (8.24)$$

This equation is third order, and has non-constant coefficients, and is thus somewhat complicated. Before proceeding, we note one common approximation, sometimes called the *longwave approximation*. If zonal scales are much greater than meridional scales then we expect the zonal wind to be in geostrophic balance with the meridional pressure gradient. In this case we replace (8.20b) by

$$fu = -g' \frac{\partial \eta}{\partial y}. \quad (8.25)$$

This equation combines with (8.20a,c) to give

$$\frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial^2 v}{\partial y^2} \right) - \beta \frac{\partial v}{\partial x} = 0. \quad (8.26)$$

This equation is first order in time and the dispersion relation may be obtained reasonably straightforwardly. This approximation is particularly useful in the forced-dissipative problem as we will see in section 8.4. In the free problem the dispersion equation can in fact be obtained easily enough in the general case, that is from (8.24), allowing us to make the longwave approximation at a later stage.

8.2.1 Dispersion Relations

In this section we explore the properties of (8.24), in particular obtaining a dispersion relation. The coefficients of (8.24) vary in the meridional direction but are constant in the zonal direction. We thus search for solutions in the form of a plane wave in the zonal direction only and we let

$$v = \tilde{v}(y)e^{i(kx - \omega t)}, \quad (8.27)$$

and assume boundary conditions of $\tilde{v}(y) \rightarrow 0$ as $y \rightarrow \pm\infty$. Substituting (8.27) into (8.24) gives

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta^2 y^2}{c^2} \right) \tilde{v} = 0. \quad (8.28)$$

Given the velocity, c , and the presence of the beta effect there is a rather obvious way to nondimensionalize the equations. However, it turns out that by introducing an additional factor of $\sqrt{2}$ into the scaling, the mathematics of one of the problems that we address later is simplified. At the risk of discussing a trivial difference, let's do both — the confident and impatient reader may choose one and skim the other.

Nondimensionalization I

Let us scale time and distance with the quantities $T_{eq} = (c\beta)^{-1/2}$, $L_{eq} = (c/\beta)^{1/2}$ as in (8.21). The non-dimensional frequency, lengthscale and wavenumber are then given by

$$\hat{\omega} = \frac{\omega}{(\beta c)^{1/2}}, \quad \hat{y} = y \left(\frac{\beta}{c} \right)^{1/2}, \quad \hat{k} = k \left(\frac{c}{\beta} \right)^{1/2}. \quad (8.29)$$

If we take $\delta\rho/\rho_0 = 0.002$, $H = 100$ m and $\beta = 2\Omega/a = 2.3 \times 10^{-11} \text{ m}^{-1} \text{ s}^{-1}$ we find

$$g' \approx 0.02 \text{ m s}^{-2}, \quad c \approx 1.4 \text{ m s}^{-1}, \quad L_{eq} \approx 250 \text{ km}, \quad T_{\beta} = 1.7 \times 10^5 \text{ s} \approx 2 \text{ days}. \quad (8.30)$$

The mid-latitude shallow-water deformation radius, L_d is usually defined as $L_d = c/f$ which differs from (8.21b) most notably in the power of f . However, if in mid-latitude expression we take $f = \beta y$, as if near the equator, and $y = L_d$, then $L_d = c/(\beta L_d)$, which is the same as (8.21b).

Substituting (8.29) into (8.28) gives the slightly simpler-looking equation

$$\frac{d^2 v}{d\hat{y}^2} + \left(\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{\hat{\omega}} - \hat{y}^2 \right) v = 0. \quad (8.31)$$

This equation may be put into a standard form⁴ by writing $v(\hat{y}) = \Psi(\hat{y}) \exp(-\hat{y}^2/2)$, whence (8.31) becomes

$$\frac{d^2 \Psi}{d\hat{y}^2} - 2\hat{y} \frac{d\Psi}{d\hat{y}} + \lambda \Psi = 0 \quad (8.32)$$

where $\lambda = \hat{\omega}^2 - \hat{k}^2 - \hat{k}/\hat{\omega} - 1$. Equation (8.32) is known as *Hermite's equation*, and it is an eigenvalue equation, with solutions if and only if $\lambda = 2m$, for $m = 0, 1, 2, \dots$. The solutions are Hermite polynomials, $\Psi(\hat{y}) = H_m(\hat{y})$, where the first few polynomials are given by

$$\begin{aligned} H_0 &= 1, & H_1 &= 2\hat{y}, & H_2 &= 4\hat{y}^2 - 2, \\ H_3 &= 8\hat{y}^3 - 12\hat{y}, & H_4 &= 16\hat{y}^4 - 48\hat{y}^2 + 12. \end{aligned} \quad (8.33)$$

A Hermite polynomial is even or odd when m is even or odd, respectively; that is $H_m(-\hat{y}) = (-1)^m H_m(\hat{y})$. Note that we are using Hermite polynomials to describe the v field, so that mirror symmetry across the equator occurs when m is odd. The v field is then odd, but the u and ϕ fields are even, as shown in the appendix to this chapter.

Hermite polynomials multiplied by a Gaussian are a form of *parabolic cylinder function*,

$$V_m(y) = H_m(y) \exp(-y^2/2). \quad (8.34)$$

These functions are also orthogonal in the interval $[-\infty, +\infty]$; that is

$$\int_{-\infty}^{\infty} V_n V_m dy = \int_{-\infty}^{\infty} H_n(y) H_m(y) \exp(-y^2) dy = \sqrt{\pi} 2^n n! \delta_{nm}, \quad (8.35)$$

The appendix gives additional details. Given the Hermite solution for Ψ , the solutions for v are given by

$$v(\hat{y}) = V_m(\hat{y}) = H_m(\hat{y}) e^{-\hat{y}^2/2}, \quad m = 0, 1, 2, \dots \quad (8.36)$$

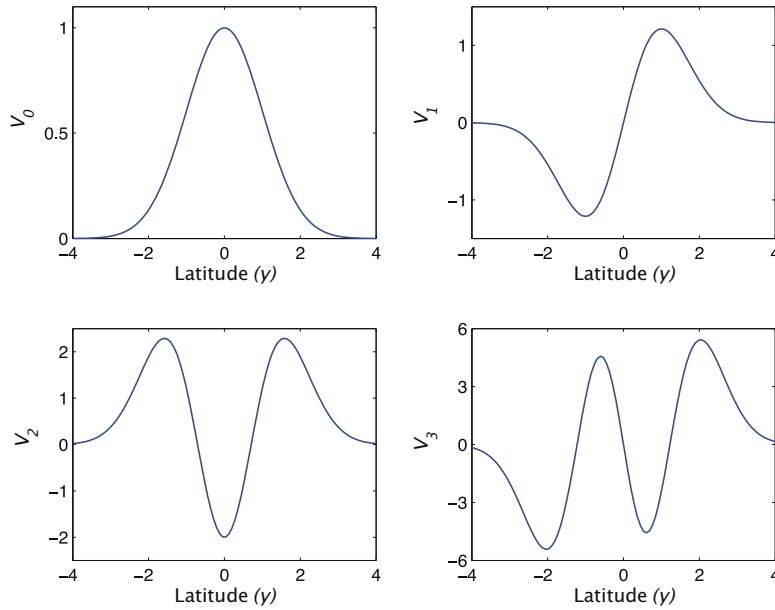


Fig. 8.5 Latitudinal variation of the wave amplitudes, $V_m(y)$, given by (8.36) as a function of non-dimensional latitude, $\hat{y}$ for $m = 0, 1, 2, 3$. The parameter m is analogous to a meridional wavenumber. The parabolic cylinder functions given by (8.43) have a similar form.

and so decay exponentially as $\hat{y} \rightarrow \pm\infty$ (as we require) with a decay scale of the equatorial deformation radius $\sqrt{c/\beta}$. The functions V_m are plotted in Fig. 8.5 for $m = 0$ to 3.

The dispersion relation follows from the quantization condition $\lambda = 2m$, which implies

$$\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{\hat{\omega}} = 2m + 1 \quad (8.37a)$$

or, using (8.29), the dimensional form,

$$\omega^2 - c^2 k^2 - \beta \frac{kc^2}{\omega} = (2m + 1)\beta c, \quad (8.37b)$$

This is a cubic equation in ω , and although a general solution is possible, it is easier to solve the quadratic equation for the wavenumber in terms of the frequency giving

$$\hat{k} = -\frac{1}{2\hat{\omega}} \pm \frac{1}{2} \left[\left(\frac{1}{\hat{\omega}} - 2\hat{\omega} \right)^2 - 8m \right]^{1/2}, \quad (8.37c)$$

or, in dimensional form,

$$k = -\frac{\beta}{2\omega} \pm \frac{1}{2} \left[\left(\frac{\beta}{\omega} - \frac{2\omega}{c} \right)^2 - \frac{8m\beta}{c} \right]^{1/2}. \quad (8.37d)$$

Equations (8.37) are forms of the dispersion relation for the shallow water equations on an equatorial beta-plane. Before exploring their properties we nondimensionalize in a different way.

Nondimensionalization II

We now scale time and distance with the quantities

$$T_{eq} = (2c\beta)^{-1/2}, \quad L_{eq} = (c/2\beta)^{1/2}. \quad (8.38a,b)$$

Velocity is still nondimensionalized by c . The nondimensional version of (8.28) becomes

$$\frac{d^2 \tilde{v}}{d\hat{y}^2} + \left(\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{2\hat{\omega}} - \frac{\hat{y}^2}{4} \right) \tilde{v} = 0, \quad (8.39)$$

which may be compared with (8.31). We now make the substitution

$$\tilde{v}(\hat{y}) = \Phi \exp(-\hat{y}^2/4), \quad (8.40)$$

which leads to

$$\frac{d^2 \Phi}{d\hat{y}^2} - \hat{y} \frac{d\Phi}{d\hat{y}} + \gamma \Phi = 0 \quad (8.41)$$

where $\gamma = \hat{\omega}^2 - \hat{k}^2 - \hat{k}/2\hat{\omega} - 1/2$. Equation (8.41) could be transformed into (8.32) by changing to the independent variable $y' = \hat{y}/\sqrt{2}$, and the dispersion relation then follows in the same way. More directly, solutions of (8.41) are given by the modified Hermite polynomials $\Phi(\hat{y}) = G_m(\hat{y})$ where

$$(G_0, G_1, G_2, G_3, G_4) = (1, \hat{y}, \hat{y}^2 - 1, \hat{y}^3 - 3\hat{y}, \hat{y}^4 - 6\hat{y}^2 + 3). \quad (8.42)$$

These are also known as the probabilists' Hermite polynomials, with (8.33) or (8.151) being the physicists' Hermite polynomials, reflecting historical use; the two sets of polynomials are connected by $H_n(y) = 2^{n/2} G_n(y\sqrt{2})$. The corresponding parabolic cylinder functions are given by

$$D_n(\hat{y}) = G_n(\hat{y}) \exp(-\hat{y}^2/4). \quad (8.43)$$

and these functions are solutions of (8.39). The orthonormality condition on the modified polynomials is that

$$\int_{-\infty}^{\infty} D_n(y) D_m(y) dy = \int_{-\infty}^{\infty} G_n(y) G_m(y) \exp(-y^2/2) dy = \sqrt{2\pi} n! \delta_{nm}, \quad (8.44a)$$

which may be compared to (8.35). The quantization condition on γ is that $\gamma = m$, where $m = 0, 1, 2, \dots$. Thus, the nondimensional dispersion relation is

$$\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{2\hat{\omega}} - \frac{1}{2} = m \quad (8.45)$$

and restoring the dimensions using (8.38) gives (8.37b). Later on, when dealing with the steady, forced-dissipative problem, the use of the probabilists' polynomials is a little more convenient.

8.2.2 Limiting and special cases

To further explore the wave case we stay with first nondimensionalization, namely (8.21), and with the goal of figuring out what's going on we'll consider various special cases of the dispersion relations (8.37). It is convenient to first partition the waves by frequency, and consider separately high frequency gravity waves and low frequency planetary waves. We need do this only for the case $m \geq 1$ because the $m = 0$ case (mixed Rossby-gravity waves) may be treated exactly. Then finally we look at the so-called $m = -1$ case, namely Kelvin waves.

High and low frequency waves

1. *High frequency waves.* The term $\beta k c^2 / \omega$ in (8.37) is small and may be neglected. The dispersion relation becomes

$$\hat{\omega}^2 = \hat{k}^2 + 2m + 1 \quad \text{or} \quad \omega^2 = c^2 k^2 + \beta c(2m + 1). \quad (8.46a,b)$$

This dispersion relation is similar to that of mid-latitude Poincaré waves, with βc replacing f_0^2 : recall the form of (3.122), namely $\omega^2 = c^2(k^2 + l^2) + f_0^2$. Waves satisfying (8.46) are thus sometimes called equatorially trapped Poincaré waves or equatorially trapped gravity waves.

The approximation requires that $\omega \gg \beta/|k|$, and is somewhat inaccurate for small k : note that (8.46) is symmetric around $k = 0$, whereas the full dispersion relation, plotted in Fig. 8.6, is offset. (Formally, the limit is valid for $\hat{k} \rightarrow \infty$, $\hat{\omega} \rightarrow \infty$ and $\hat{k}/\hat{\omega} = \text{constant}$.)

For finite m the limiting case at high wavenumber just $\hat{\omega} = \pm \hat{k}$, or, in dimensional form, $\omega = \pm ck$. This is just the dispersion relation for familiar conventional shallow water gravity waves, unaffected by rotation and the β -effect. However, in the rotating case the waves are trapped at the equator and propagate only in the zonal direction, albeit both eastward and westward.

2. *Low frequency waves.* For low frequency waves we neglect the term involving ω^2 in (8.37) and the dispersion relation becomes

$$\hat{\omega} = \frac{-\hat{k}}{2m+1+\hat{k}^2}, \quad \omega = \frac{-\beta k}{(2m+1)\beta/c+k^2}, \quad (8.47)$$

non-dimensionally and dimensionally, respectively. This is recognizable as the dispersion relation for a zonally propagating Rossby wave with large x -wavenumber, and these waves are called equatorially trapped Rossby waves, or equatorially trapped planetary waves. We may further consider two limits of these waves, as follows.

- (a) *Short, low frequency waves, with $\hat{k} \rightarrow \infty$, $\hat{\omega} \rightarrow 0$.* The dispersion relation becomes

$$\hat{\omega} = -\frac{1}{\hat{k}}, \quad \omega = -\frac{\beta}{k}. \quad (8.48)$$

The phase speed and group velocity in this limit are given by, dimensionally,

$$c_p = -\frac{\beta}{k^2}, \quad c_g = \frac{\beta}{k^2}, \quad (8.49)$$

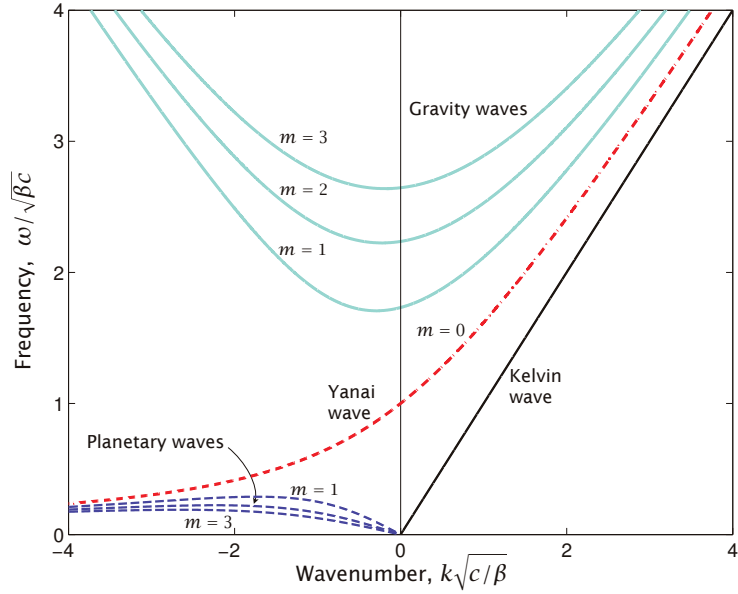
Thus, the phase speed is westward but the group velocity, and so the direction of energy propagation, is eastward.

- (b) *Long low frequency waves, with $\hat{k} \rightarrow 0$, $\hat{\omega} \rightarrow 0$.* The dispersion relation (8.37) becomes, in nondimensional and dimensional form,

$$\hat{\omega} = \frac{-\hat{k}}{2m+1}, \quad \omega = \frac{-ck}{2m+1}, \quad (8.50)$$

These represent westward propagating waves whose speed is given by $c/(2m+1)$. For $m = 1$ (the smallest allowable value for planetary waves) they have one-third the speed of a non-rotating gravity wave or of a Kelvin wave (discussed below). However, these waves propagate only westward, and they match with the westward propagating planetary waves derived above as wavenumber increases. They are conveniently nondispersive, and are also important near western boundaries where they superpose to create western boundary currents. The longwave approximation may be made from the outset, and is equivalent to assuming that the zonal flow is in geostrophic balance; that is, (8.20b) is replaced by $fu = -g'\partial\eta/\partial y$. Then, instead of solving (8.24) we solve (8.26). The only difference is in the value of λ in (8.32) — we find $\lambda = -\hat{k}/\hat{\omega} - 1$ — and so (8.50) immediately emerges. Short waves are filtered out of the system. This approximation will turn out to be particularly important when we consider the steady problem in section 8.5.

Fig. 8.6 Dispersion relation for equatorial waves, as given by (8.37), for $m = 0, 1, 2, 3$. The upper group of curves are gravity waves, given approximately by (8.46). The lower group with $k < 0$ are westward propagating planetary waves, given approximately by (8.47). Also shown are the Yanai wave with $m = 0$, satisfying (8.54), and the eastward propagating Kelvin wave (the ' $m = -1$ ' wave) satisfying $\omega = ck$ for $k \geq 0$.



There is a distinct gap in frequencies between the minimum frequency of the gravity waves, given by (8.46), and the maximum frequency of the planetary waves, given by (8.49) also with m small. The minimum gravity wave frequency occurs when $k = 0$ and is $\omega_{gmin}^2 = \beta c(2m + 1)$. From (8.47) the maximum planetary wave frequency occurs when $k^2 = (2m + 1)\beta/c$ and gives $\omega_{pmax}^2 = \beta c/[4(2m + 1)]$. The ratio of these two frequencies is

$$\frac{\omega_{gmin}}{\omega_{pmax}} = 2(2m + 1), \quad (8.51)$$

giving a value of six for $m = 1$ and two for $m = 0$ (a case we consider more below). Note that this ratio is *independent* of the values of the physical parameters β and c . Although the gap is distinct, it is not as large as the corresponding gap at midlatitudes, which may be an order of magnitude or more.

Special values of m

In addition to the limiting cases at low and high frequency, there are two other cases in which we can readily solve the dispersion relation, namely when $m = 0$ and the Kelvin wave case, as follows.

1. *The case with $m = 0$.* The resulting waves are known as *Yanai waves*,⁵ or *Rossby-gravity waves*, since they span the two types of waves. They are antisymmetric across the equator. From (8.37a) the dispersion relation simplifies to

$$\hat{k} = -\hat{\omega} \quad \text{or} \quad \hat{k} = -\frac{1}{\hat{\omega}} + \hat{\omega}. \quad (8.52a,b)$$

or dimensionally

$$k = -\frac{\omega}{c}, \quad k = -\frac{\beta}{\omega} + \frac{\omega}{c}. \quad (8.53a,b)$$

The case $k = -\omega/c$ is non-physical, for it represents a gravity wave moving westward. Such wave grows without bound as $|y|$ increases away from the equator, as we demonstrate explicitly in the discussion on Kelvin waves below. The physically realizable case, (8.53b) has the

explicit dispersion relation

$$\omega = \frac{kc}{2} \pm \frac{1}{2} \sqrt{k^2 c^2 + 4\beta c}. \quad (8.54)$$

Again it is useful to consider limiting cases, as follows.

- $k = 0$. In this case (8.54) gives $\omega = \sqrt{\beta c}$ and there is a balance between the two terms on the right-hand side of (8.53b). Note that in Fig. 8.6 the Yanai wave at $k = 0$ intercepts the ordinate at a value of nondimensional frequency of 1.
- $k \rightarrow +\infty$. In this case $\omega = ck$, with a balance between the left-hand side and the second term on the right-side of (8.53b). Evidently, this corresponds to eastward propagating gravity waves.
- $k \rightarrow -\infty$. In this case, because ω must be positive, we have $\omega = -\beta/k$, and a balance between the left-hand side and the first term on the right-side of (8.53b). The waves are westward propagating Rossby or planetary waves.

Yanai waves, therefore, are mixed Rossby-gravity waves: the phase of the Rossby wave propagates westward (like all Rossby waves) and has a low frequency, and the gravity wave propagates eastward (and only eastward, unlike conventional gravity waves). The group velocity of Yanai waves is positive in all cases, being given by, from (8.53b),

$$c_g^x \equiv \frac{\partial \omega}{\partial k} = \frac{\omega^2 c}{\beta c + \omega^2}. \quad (8.55)$$

The group velocity of the full problem is, from (8.37),

$$c_g^x = \frac{c^2 \omega (\beta + 2\omega k)}{2\omega^3 + \beta k c^2}. \quad (8.56)$$

This may be positive or negative, and vanishes when $\omega = -\beta/2k$.

2. *Kelvin waves, or the 'm = -1' case.* (This section may be considered to be an extension of section 3.8.3.) In general, Hermite's equation, (8.32), has solutions when m is a positive integer or zero. However, there is a class of waves that also satisfies the dispersion relation (8.37) with $m = -1$, namely equatorial Kelvin waves. These waves have identically zero meridional velocity and so their equations of motion are

$$\frac{\partial u}{\partial t} = -g' \frac{\partial \eta}{\partial x}, \quad f u = -g' \frac{\partial \eta}{\partial y}, \quad \frac{\partial \eta}{\partial t} + H \frac{\partial u}{\partial x} = 0. \quad (8.57a,b,c)$$

where $f = \beta y$. The zonal velocity is in geostrophic balance with the meridional pressure gradient, and (8.57a) and (8.57c) give the classic wave equation,

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0. \quad (8.58)$$

where $c = \sqrt{g'H}$ as before, and so the dispersion relation $\omega = \pm ck$. This is, in fact, a solution of (8.37) with $m = -1$, as may easily be checked.

The solution to (8.58), and the corresponding solution for η , is

$$u = F_1(x + ct, y) + F_2(x - ct, y), \quad \eta = \sqrt{\frac{H}{g'}} [-F_1(x + ct, y) + F_2(x - ct, y)] \quad (8.59)$$

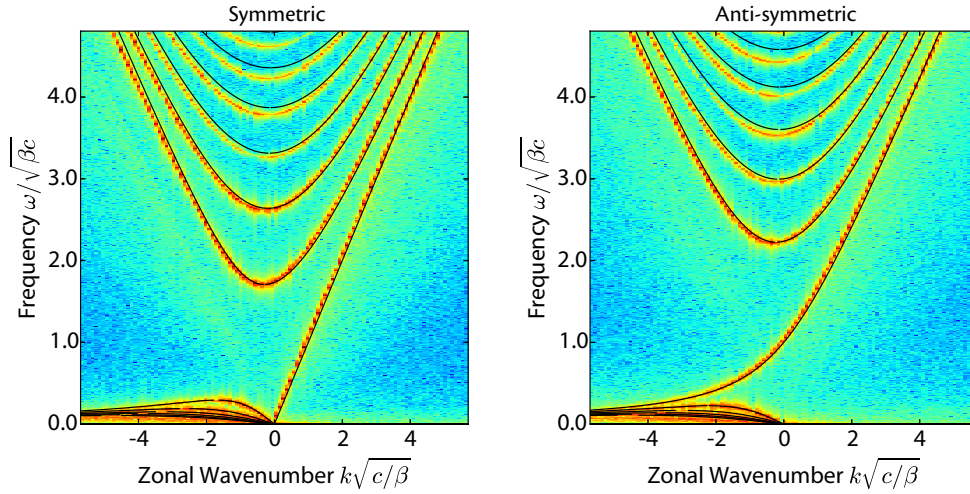


Fig. 8.7 Power spectrum from a numerical simulation of the shallow water equations (colour shading, with red the most intense), with the analytic dispersion relation for equatorial Rossby and gravity waves overlaid (solid black lines, as in Fig. 8.6). The left panel shows the symmetric component, obtained by adding Northern and Southern Hemispheres and with only the odd values of m plotted analytically, and the the right panel plots the anti-symmetric component and the even values of m .

where F_1 and F_2 are arbitrary functions, representing waves travelling westwards and eastwards, respectively. We obtain the y -dependence of these functions by using (8.57b) giving

$$\beta y F_1 = c \frac{\partial F_1}{\partial y}, \quad \beta y F_2 = -c \frac{\partial F_2}{\partial y}. \quad (8.60)$$

The solutions of these equations are

$$F_1 = F(x + ct) \exp[y^2/(2L_{eq}^2)], \quad F_2 = G(x - ct) \exp[-y^2/(2L_{eq}^2)] \quad (8.61a,b)$$

where F and G are the amplitudes at $y = 0$. Evidently, F_1 increases without bound away from the equator, and so this solution must be eliminated. The complete solution is thus:

$$u = G(x - ct) \exp[-y^2/(2L_{eq}^2)], \quad \eta = \frac{H}{c} u, \quad v = 0. \quad (8.62)$$

with dispersion relation

$$\omega = ck. \quad (8.63)$$

These waves are equatorially trapped Kelvin waves. They propagate eastward only, without dispersion, and their amplitude decays away from the equator in precisely the same way as the other equatorial waves considered above, and in a slightly different way from the Kelvin waves on the f -plane given by (3.136).

8.2.3 A numerical illustration

After the many mathematical manipulations above, the reader, if like the author, may well be skeptical that such waves do actually exist, especially on a sphere where Kelvin waves are not exactly realizable. To assuage this doubt, Fig. 8.7 shows the power spectrum from a numerical simulation

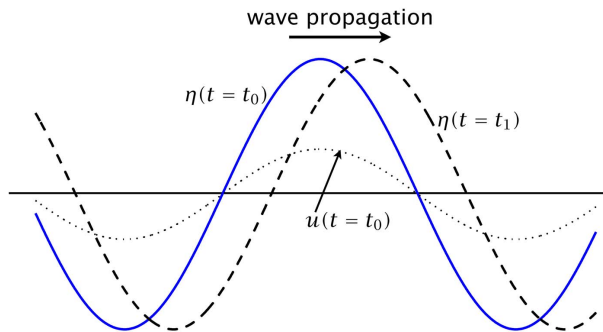


Fig. 8.8 A shallow water gravity wave, showing the fluid interface at an initial and later time $\eta(t_0)$ and $\eta(t_1)$, and the fluid velocity at the initial time, $u(t_0)$. The fluid flow is in the same direction as the phase speed (positive in this example) under the fluid crests, and is in the opposite direction under the troughs.

of the nonlinear shallow water equations over the full sphere. The height field is initialized with small random perturbations everywhere and the system allowed to freely evolve, with no damping except for that residing in the numerical scheme.⁶ The figure shows the resulting power spectrum, over a region from 15° S to 15° N, from the near-statistical equilibrium state that emerges. The equatorial waves emerge with beautiful clarity above the noise, with only small deviations for the highest modes due to resolution issues with the numerics that slow the waves. To see a simulation showing Rossby and Kelvin waves in physical space look ahead to Fig. 22.17. The real world is never quite so limpid, but Fig. 18.24 and Fig. 18.25 suggest that the waves are not merely figments of the imagination.

8.2.4 Why do Kelvin waves have a preferred direction of travel?

Both equatorial and coastal Kelvin waves have a preferred direction of travel: equatorial Kelvin waves move eastward and, consistently, coastal Kelvin waves travel such that they have a wall to their right in the Northern Hemisphere and to their left in the Southern Hemisphere. Why?

Consider the linear zonal momentum and mass continuity equations,

$$\frac{\partial u}{\partial t} = -g' \frac{\partial \eta}{\partial x}, \quad \frac{\partial \eta}{\partial t} = -H \frac{\partial u}{\partial x}. \quad (8.64)$$

Looking for wavelike solutions of the form $(u, h) = (\tilde{u}, \tilde{\eta})e^{i(kx - \omega t)}$ we obtain $\tilde{u} = g'\tilde{\eta}/c$ and $c\tilde{\eta} = H\tilde{u}$. This means that under the crests of fluid (i.e., positive values of η) u has the same sign as c ; the parcels of fluid are moving in the same direction as the phase of the wave. This property is also apparent if one considers how the fluid must move in order that the troughs and crests progress in a particular direction, as illustrated in Fig. 8.8. This property holds for shallow water waves quite generally, and is not restricted to Kelvin waves.

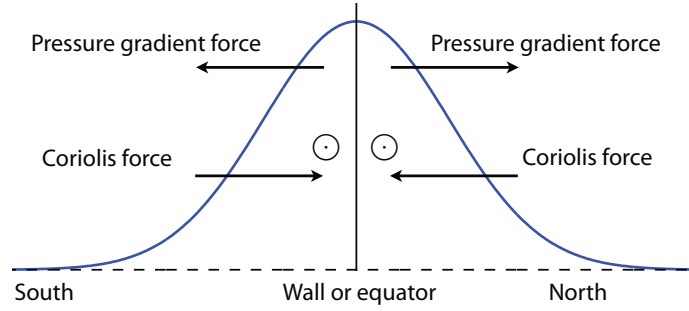
Now we add rotation and restrict attention to Kelvin waves. In the direction perpendicular to the direction of travel of the wave, the flow is in geostrophic balance:

$$fu = -g' \frac{\partial \eta}{\partial y}. \quad (8.65)$$

Consider the flow under a fluid crest in an equatorial Kelvin wave, as illustrated in Fig. 8.9. The pressure gradient force is directed away from the equator and, if the wave is travelling eastward the pressure force can be balanced by the Coriolis force directed toward the equator. Under a trough the fluid is flowing in the opposite direction to the wave itself, and both the pressure gradient force and the Coriolis force are reversed and geostrophic balance still holds. If the wave were to travel westwards, no such balances could be achieved.

Fig. 8.9 Balance of forces across a Kelvin wave. The solid line is the fluid surface and the phase line is directed out of the page.

Beneath a crest the fluid flow is in the direction of the phase speed and produces Coriolis forces as shown, so balancing the pressure gradient forces. If the wave were travelling in the opposite direction no such geostrophic balance could be achieved.



Very similar reasoning holds for coastal Kelvin waves, with a cross-wave pressure gradient supported by a wall. Geostrophic balance can now be maintained only if the wall is to the right of the direction of travel in the Northern Hemisphere (where $f > 0$) and to the left in the Southern Hemisphere (where $f < 0$).

8.2.5 Potential vorticity dynamics of equatorial Rossby waves

The Rossby waves and Rossby-gravity waves derived above are rather similar to their mid-latitude counterparts, which can be derived from a balanced potential vorticity equation without involving unbalanced dynamics at all. Can we do something similar for equatorial Rossby waves? The answer is yes, although the method is somewhat ad hoc.⁷ Kelvin waves and inertia-gravity waves are filtered out, but Rossby waves and Rossby-gravity waves are reproduced in a way that transparently illuminates their dynamics.

Let us begin with the unforced linearized potential vorticity equation which to remind ourselves, is

$$\frac{\partial}{\partial t} \left(\zeta - \frac{f\phi}{c^2} \right) + \beta v = 0, \quad (8.66)$$

where, as before, $f = \beta y$. Let us now suppose that the divergence is small and the flow close to geostrophic balance so that the velocity, vorticity and height fields can all be written in terms of a streamfunction,

$$u = -\frac{\partial \psi}{\partial y}, \quad v = \frac{\partial \psi}{\partial x}, \quad \zeta = \nabla^2 \psi, \quad \phi = f\psi. \quad (8.67)$$

This is similar to what is done in the quasi-geostrophic approximation, except that here the Coriolis parameter is allowed to vary, with $f = \beta y$. Equation (8.67) is best regarded as an ansatz — an approximation or assumption made for convenience — for it has not been rigorously justified.

Using (8.67) in (8.66) gives

$$\frac{\partial}{\partial t} \left(\nabla^2 \psi - \frac{f^2 \psi}{c^2} \right) + \beta \frac{\partial \psi}{\partial x} = 0. \quad (8.68)$$

We can seek wavelike solutions of this in the form

$$\psi = \tilde{\psi}(y) e^{i(kx - \omega t)}, \quad (8.69)$$

and (8.68) becomes

$$\frac{d^2 \tilde{\psi}}{dy^2} - \left(k^2 + \frac{\beta k}{\omega} + \frac{\beta^2 y^2}{c^2} \right) \tilde{\psi} = 0. \quad (8.70)$$

This is almost the same as (8.28) except for the replacement of $\tilde{v}$ by $\tilde{\psi}$ and the absence of the ω^2 term in the bracketed expression. Since meridional velocity is just $\partial \psi / \partial x \propto k \psi$ the meridional velocity

obeys the same equation as $\tilde{\psi}$, and the absence of ω^2 arises because we are in the low-frequency limit. We thus simply repeat the development following (8.28) and obtain a dispersion relation similar to (8.37b) but without the ω^2 term, to wit

$$\omega = \frac{-\beta k}{(2m+1)\beta/c + k^2}. \quad (8.71)$$

This is the same as the dispersion relation for low frequency waves discussed in section 8.2.2. The balanced system (8.68) thus *exactly* reproduces the Rossby waves and Rossby-gravity waves in the low frequency limit. We are not able to recover the behaviour of Kelvin waves by this methodology because such waves are essentially non-balanced: in the meridional direction the Coriolis force balances the height field, as in (8.65), but in the zonal direction there is a balance between the zonal acceleration and the pressure gradient.

8.3 RAY TRACING AND EQUATORIAL TRAPPING

We have seen that equatorial waves are trapped near the equator. What then happens to a wave that initially propagates in a direction away from the equator? The waves must either change their character completely, or be refracted back toward the equator. The former can only happen if there exists a class of midlatitude waves with similar frequency and wavenumber; otherwise no such waves can be excited and the waves must, if they are not absorbed, bend back if energy is to be conserved. Let us explore this using some ideas from ray theory, as discussed in section 6.3.

8.3.1 Dispersion relation and ray equations

Consider again the wave equation of motion for the meridional velocity, (8.24). We seek solutions of the form $v = \tilde{v}(y)e^{i(kx - \omega t)}$, giving

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta^2 y^2}{c^2} \right) \tilde{v} = 0. \quad (8.72)$$

If the term in brackets is positive then sinusoidal-like solutions in y are possible, but if the term is negative, which will occur for y larger than some critical value y_c , then the physically realizable solutions decay exponentially with y ; that is, wavelike solutions are trapped between two critical latitudes. Using the dispersion relation (8.37), equation (8.72) becomes

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{(2m+1)\beta}{c} - \frac{\beta^2 y^2}{c^2} \right) \tilde{v} = 0. \quad (8.73)$$

and therefore the critical latitudes are given by

$$y_c = \pm \left(\frac{\omega^2}{\beta^2} - \frac{c^2 k^2}{\beta^2} - \frac{c^2 k}{\beta \omega} \right)^{1/2} = \left((2m+1) \frac{c}{\beta} \right)^{1/2}, \quad (8.74)$$

For $k = 0$, and so for meridionally propagating waves, the critical latitudes are given by $y_c = \omega/\beta$, and at the critical latitude $\omega = f$. The waves are therefore trapped within their *inertial latitudes*, the latitudes at which their frequency is f . For larger k the critical latitudes are correspondingly smaller.

To explore this phenomenon using ray theory we assume that the medium is varying sufficiently slowly that it is possible to find wavelike solutions with spatially varying y -wavenumbers. We write (8.72) as

$$\frac{d^2 \tilde{v}}{dy^2} + l^2(y) \tilde{v} = 0, \quad (8.75)$$

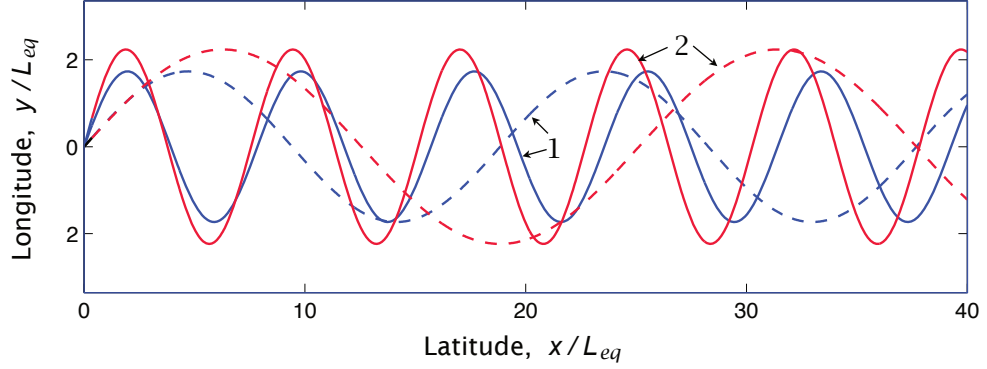


Fig. 8.10 Rays in the equatorial waveguide calculated using (8.83). The dashed lines show planetary wave trajectories and the solid lines are gravity wave trajectories, with $m = 1, 2$ (numbers marked on the graph) and $\hat{k} = 1$. The turning latitude for each wave is $(2m + 1)^{1/2} L_{eq}$, where $L_{eq} = \sqrt{c/\beta}$.

$$l^2(y) = \frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta^2 y^2}{c^2} = \frac{\beta^2}{c^2} (y_c^2 - y^2). \quad (8.76)$$

where $l(y)$ is assumed to vary slowly in the WKB sense (see the appendix on page 248). The WKB solution is

$$\bar{v}(y) = l^{-1/2} \exp\left(\pm i \int l dy\right). \quad (8.77)$$

The trajectory of the waves is determined by the rays paths — paths that are parallel to the direction of the group velocity. That is, their trajectory, $x(t)$, $y(t)$ is defined by

$$\frac{dx}{dt} = c_g^x, \quad \frac{dy}{dt} = c_g^y, \quad \text{so that} \quad \frac{dy}{dx} = \frac{c_g^y}{c_g^x}. \quad (8.78)$$

Using the dispersion relation (8.76) gives

$$\frac{\partial \omega}{\partial l} = \frac{2\omega^2 l c^2}{2\omega^3 + \beta k c^2}, \quad (8.79)$$

and using this and (8.56) gives the slope of the ray in the x - y plane,

$$\frac{dy}{dx} = \frac{c_g^y}{c_g^x} = \frac{l}{k + \beta/(2\omega)}. \quad (8.80)$$

Using the expression for l given by (8.76) we can write this in terms of y instead of l , so that

$$\frac{dy}{dx} = \frac{\beta(y_c^2 - y^2)^{1/2}}{kc + \beta c/(2\omega)}. \quad (8.81)$$

Using the standard result that

$$\int \frac{dy}{(y_c^2 - y^2)^{1/2}} = \sin^{-1} \frac{y}{y_c}, \quad (8.82)$$

we finally obtain

$$y = y_c \sin \left[\frac{\beta x}{ck + \beta c/(2\omega)} \right], \quad \hat{y} = (2m + 1)^{1/2} \sin \left[\frac{\hat{x}}{\hat{k} + 1/(2\hat{\omega})} \right]. \quad (8.83)$$

where the second expression is the nondimensional form. The ray path is therefore a sinusoid moving along the equator; the waves are confined to a *waveguide* centered at the equator and with a polewards extent of $y = \pm y_c$. Equation (8.83) holds for both planetary and gravity waves, and for the latter the term $\beta c/(2\omega)$ may be neglected.

8.4 ♦ FORCED-DISSIPATIVE WELIKE FLOW

We now consider linear equatorial dynamics in the presence of forcing and damping. (A somewhat simpler treatment of this issue is given in section 22.7 and some readers may wish to delay considering the problem until then. The important special case of the forced, steady problem is treated in section 8.5, and it is also possible to skip to that section and refer back here as needed.) If forcing is present then damping is needed so that a steady state can be reached, and the simplest form is a linear drag. From a physical perspective the presence of such a drag is the most unsatisfactory aspect of our treatment, for it has no real physical justification especially as, for mathematical reasons, the drag must be the same for momentum and height (implying a frictional spindown time equal to a radiative spindown time). Nevertheless, unresolved small scale processes often do act as some form of damping and a linear damping is the simplest form. We consider the full problem initially and then special cases.⁸

The linear forced-dissipative equations of motion are

$$\frac{\partial u}{\partial t} + \alpha u - f v + \frac{\partial \phi}{\partial x} = F^x, \quad \frac{\partial v}{\partial t} + \alpha v + f u + \frac{\partial \phi}{\partial y} = F^y \quad (8.84a,b)$$

$$\frac{\partial \phi}{\partial t} + \alpha \phi + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -Q. \quad (8.84c)$$

where F^x and F^y are the x and y components of the imposed forces, Q is a thermal or mass source and α is a damping coefficient, assumed the same for all three variables. If we interpret $\mathbf{F} = (F^x, F^y)$ as wind stress, $\boldsymbol{\tau}$, acting on a layer of fluid we might make the association of $\mathbf{F} = \boldsymbol{\tau}/H$. Still, for now we will treat this system simply as a problem in geophysical fluid dynamics. The potential vorticity equation corresponding to (8.84), obtained by cross differentiating (8.84a) and (8.84b), is

$$\left[\frac{\partial}{\partial t} + \alpha \right] \left(\zeta - \frac{f}{c^2} \phi \right) + \beta v = \text{curl}_z \mathbf{F} + \frac{fQ}{c^2}. \quad (8.85)$$

In much the same way as we derived (8.24) we can derive a single partial differential equation for v , namely

$$\begin{aligned} & \frac{1}{c^2} \left[\frac{\partial}{\partial t} + \alpha \right]^3 v + \frac{f^2}{c^2} \left[\frac{\partial}{\partial t} + \alpha \right] v - \left[\frac{\partial}{\partial t} + \alpha \right] \left(\frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} \right) - \beta \frac{\partial v}{\partial x} \\ &= \frac{1}{c^2} \left[\frac{\partial}{\partial t} + \alpha \right] \frac{\partial Q}{\partial y} - \frac{f}{c^2} \frac{\partial Q}{\partial x} \\ &+ \frac{1}{c^2} \left[\frac{\partial}{\partial t} + \alpha \right]^2 F^y - \frac{f}{c^2} \left[\frac{\partial}{\partial t} + \alpha \right] F^x - \frac{\partial}{\partial x} \left(\frac{\partial F^y}{\partial x} - \frac{\partial F^x}{\partial y} \right). \end{aligned} \quad (8.86)$$

The left-hand side is a minor variation of that of (8.24). This equation is obviously very complicated and perhaps not very attractive. Although the equation might be solved by similar methods to those used on (8.24) (or solved numerically) we will proceed in a slightly different and hopefully more informative way, the differences being twofold.

1. We consider only special cases of (8.84). For example, we simplify (8.84b) to geostrophic balance, $f u = -\partial \phi / \partial y$, and in section 8.5 we will pay particular attention to the steady version of the equations.

2. We will change variables from (u, v, ϕ) to a set denoted (q, r, v) , defined below, that allow an easier connection to be made between v and the variables u and ϕ .

8.4.1 Mathematical Development

For algebraic convenience we introduce the following linear combinations of u and ϕ ,

$$q \equiv \frac{\phi}{c} + u, \quad r \equiv \frac{\phi}{c} - u. \quad (8.87)$$

Note that u and ϕ have the same natural symmetry across the equator, with both symmetric unless forcing deems otherwise, whereas v tends to be antisymmetric. The equations for q and r are

$$\left(\frac{\partial}{\partial t} + \alpha \right) q + c \frac{\partial q}{\partial x} + c \frac{\partial v}{\partial y} - f v = F^x - Q/c, \quad (8.88a)$$

$$\left(\frac{\partial}{\partial t} + \alpha \right) r - c \frac{\partial r}{\partial x} + c \frac{\partial v}{\partial y} + f v = -F^x - Q/c \quad (8.88b)$$

and the v -momentum equation is

$$\left(\frac{\partial}{\partial t} + \alpha \right) v + \frac{f}{2}(q - r) = -\frac{c}{2} \frac{\partial}{\partial y}(q + r) + F^y. \quad (8.88c)$$

Nondimensionalization

We scale velocity by c and time and distance by

$$T_{eq} = (c/2\beta)^{-1/2}, \quad L_{eq} = (c/2\beta)^{1/2}. \quad (8.89a,b)$$

The nondimensional equations of motion are then

$$\left(\frac{\partial}{\partial t} + \alpha \right) q + \frac{\partial q}{\partial x} + \frac{\partial v}{\partial y} - \frac{1}{2} y v = F^x - Q, \quad (8.90a)$$

$$\left(\frac{\partial}{\partial t} + \alpha \right) r - \frac{\partial r}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{2} y v = -F^x - Q \quad (8.90b)$$

$$\left(\frac{\partial}{\partial t} + \alpha \right) v + \frac{y}{4}(q - r) = -\frac{1}{2} \frac{\partial}{\partial y}(q + r) + F^y. \quad (8.90c)$$

To avoid clutter here we do not use special notation for nondimensional variables.

The solutions of these equations may be expressed in terms of parabolic cylinder functions, $D_n(y)$. That is, we seek solutions of the form

$$(v, q, r) = \sum_{n=0}^{\infty} (v_n(x, t), q_n(x, t), r_n(x, t)) D_n(y). \quad (8.91)$$

with the forcing terms expanded in a similar fashion. The parabolic cylinder functions themselves have the form

$$(D_0, D_1, D_2, D_3) = (1, y, y^2 - 1, y^3 - 3y) \exp(-y^2/4), \quad (8.92)$$

and so on. The polynomial terms are just the modified Hermite polynomials $G_m(y)$ given by (8.42). The parabolic cylinder functions obey the ladder properties that

$$\frac{dD_n}{dy} + \frac{1}{2} y D_n = n D_{n-1}, \quad \frac{dD_n}{dy} - \frac{1}{2} y D_n = -D_{n+1}. \quad (8.93a,b)$$

If we substitute (8.91) into (8.90) we obtain ordinary differential equations for the amplitudes. From the q equation we obtain, after a little algebra,

$$\left(\frac{\partial}{\partial t} + \alpha\right)q_0 + \frac{\partial q_0}{\partial x} = F_0^x - Q_0, \quad (8.94a)$$

$$\left(\frac{\partial}{\partial t} + \alpha\right)q_{n+1} + \frac{\partial q_{n+1}}{\partial x} - v_n = F_{n+1}^x - Q_{n+1}, \quad n = 0, 1, 2, 3, \dots \quad (8.94b)$$

From the r equation we find

$$\left(\frac{\partial}{\partial t} + \alpha\right)r_{n-1} - \frac{\partial r_{n-1}}{\partial x} + nv_n = -(F_{n-1}^x + Q_{n-1}), \quad n = 1, 2, 3, \dots \quad (8.95)$$

and from the v equation we find

$$\left(\frac{\partial}{\partial t} + \alpha\right)v_0 + \frac{q_1}{2} = F_0^y, \quad (8.96a)$$

$$\left(\frac{\partial}{\partial t} + \alpha\right)v_n + \frac{(n+1)}{2}q_{n+1} - \frac{r_{n-1}}{2} = F_n^y, \quad n = 1, 2, 3, \dots \quad (8.96b)$$

Finally, we note without derivation that these equations may be combined to give

$$\left[\frac{\partial}{\partial t} + \alpha\right]^3 v_n + \left[\frac{\partial}{\partial t} + \alpha\right] \left((2n+1)v_n - \frac{\partial^2 v_n}{\partial x^2} \right) - \frac{\partial v_n}{\partial x} = G \quad (8.97)$$

where G is a combination of the various forcing terms. This equation most easily derived by substituting (8.91) into the nondimensional form of (8.86).

In principle, the above equations provide a means of solving the problem for almost any forcing. The equations have constant coefficients and may be solved by a superposition of harmonic functions in the x -direction, in conjunction with the variation in the y -direction given by the parabolic cylinder functions. In general this procedure would be somewhat tedious and uninformative. Thus, and to avoid being asphyxiated by an avalanche of algebra, we will consider some special cases. Enthusiasts may continue with the general development by themselves. (We might also note that modern geophysical fluid dynamics has advanced by way of using numerical methods to find solutions to complicated equations, in conjunction with using analytic methods to find solutions of simplified cases or to find general relations.) We will discuss two particularly important problems later, in sections 8.5 and 22.7, and for the rest of this section we content ourselves with some general comments about forced waves

8.4.2 ♦ Forced Waves

Consider the problem of forced waves in which we retain some of the forcing terms but neglect the damping. Our purpose is to show what kinds of waves might be excited and to help interpret (8.94)–(8.97). When $\alpha = 0$ equation (8.97) becomes, in dimensional form,

$$\frac{\partial}{\partial t} \left[\frac{1}{c^2} \frac{\partial^2 v_n}{\partial t^2} - \frac{\partial^2 v_n}{\partial x^2} + (2n+1) \frac{\beta}{c} v_n \right] - \beta \frac{\partial v_n}{\partial x} = G \quad (8.98)$$

From (8.98) the dispersion relation for free waves [that is, (8.37)] follows if we let $G = 0$ and seek harmonic solutions of the form $\exp(ikx - i\omega t)$.

Consider now a forcing, H say, that projects only onto the zeroth order parabolic function, D_0 . Equation (8.94a) becomes, in dimensional form,

$$\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right)q_0 = H_0. \quad (8.99)$$

The free solutions of this are Kelvin waves propagating eastwards at speeds $c = \omega/k$ for each k that might be excited; that is $q_0 = \text{Re } C \exp[ik(x - ct)]$, where C is a constant. Suppose that the forcing is harmonic in x and time,

$$G_0 = \text{Re } A \{ \exp[i(k_1 x - \omega_1 t)] + \exp[i(k_1 x + \omega_1 t)] \} = A [\cos(k_1 x - \omega_1 t) + \cos(k_1 x + \omega_1 t)], \quad (8.100)$$

if A is real. The solution to (8.99) with this forcing is given by

$$q_0 = -\frac{A \sin(k_1 x - \omega_1 t)}{\omega_1 - ck_1} + \frac{A \sin(k_1 x + \omega_1 t)}{\omega_1 + ck_1}. \quad (8.101)$$

All the parameters in the above equation, c, k_1, ω_1 , are positive. If the forcing is just one harmonic then, in general, $c \neq \omega_1/k_1$. However, if the forcing is a superposition of many harmonics then there may be one that is in resonance with the free mode, and this wave, an eastward propagating Kelvin wave represented by an expression like the first term on the right-hand side of (8.101), will be preferentially excited. Similar considerations apply to other waves too; that is, the forcing will excite waves that most resemble the forcing and can resonate with it. Sometimes, a forcing will resemble a delta function in both space and time. For example, a sudden and localized burst of wind over the ocean because of intense storm activity, and the forcing contains nearly all space and time scales (a Dirac delta function has equal representation of all Fourier modes). In this case, both eastward propagating Kelvin waves and westward propagating planetary waves will be excited, and to look at some of these it is useful to make a longwave approximation, as we now discuss.

Planetary waves, revisited

In the planetary wave, or longwave, approximation the highest time derivative in (8.98) is omitted, leaving

$$\frac{\partial}{\partial t} \left[\frac{\partial^2 v_n}{\partial x^2} - (2n+1) \frac{\beta}{c} v_n \right] + \beta \frac{\partial v_n}{\partial x} = G. \quad (8.102)$$

If $G = 0$ this equation gives the dispersion relation

$$\omega = \frac{-\beta k}{(2m+1)\beta/c + k^2}, \quad (8.103)$$

as in (8.47). Planetary waves will be excited when the forcing itself has a low frequency.

The longwave approximation, revisited

Many situations in low latitudes are characterized by having a longer zonal scale than meridional scale; thus, $|\partial\phi/\partial y| \gg |\partial\phi/\partial x|$. When this is the case, geostrophic balance will hold to a good approximation for the zonal flow even in the presence of forcing and dissipation, but not for the meridional flow, and to a good approximation the meridional momentum equation (8.84b) may be replaced by

$$fu = -\frac{\partial\phi}{\partial y}. \quad (8.104)$$

In this limit (8.98) simplifies to

$$\frac{\partial}{\partial t} \left[(2n+1) \frac{\beta}{c} v_n \right] - \beta \frac{\partial v_n}{\partial x} = G, \quad (8.105)$$

from which the dispersion relation,

$$\omega = \frac{-kc}{2n+1}, \quad (8.106)$$

immediately follows. The Kelvin wave is an eastward propagating wave with $n = -1$. When $n \geq 0$ the waves are westward propagating.

The amplitude equations, (8.94)–(8.96) then simplify as follows, also taking $\alpha = 0$. The q equations become

$$\frac{\partial q_0}{\partial t} + \frac{\partial q_0}{\partial x} = F_0^x - Q_0, \quad (8.107a)$$

$$\frac{\partial q_{n+1}}{\partial t} + \frac{\partial q_{n+1}}{\partial x} - v_n = F_{n+1}^x - Q_{n+1}, \quad n = 0, 1, 2, 3, \dots \quad (8.107b)$$

The r equation becomes

$$\frac{\partial r_{n-1}}{\partial t} - \frac{\partial r_{n-1}}{\partial x} + n v_n = -(F_{n-1}^x + Q_{n-1}), \quad n = 1, 2, 3, \dots \quad (8.108)$$

and from the v equation (geostrophic balance) we find

$$q_1 = 0, \quad (8.109a)$$

$$(n+1)q_{n+1} = r_{n-1}, \quad n = 1, 2, 3, \dots \quad (8.109b)$$

If we use (8.109b) to eliminate r_{n-1} in (8.108), and then use (8.107b) to eliminate v_n we obtain

$$(2n+1) \frac{\partial q_{n+1}}{\partial t} - \frac{\partial q_{n+1}}{\partial x} = n(F_{n+1}^x - Q_{n+1}) - (F_{n-1}^x + Q_{n-1}). \quad (8.110)$$

The above set of equations provide, in principle, a means for studying the response of the system to an imposed forcing, such as winds blowing over the ocean or a diabatic source in the atmosphere. Having neglected dissipation, wavelike solutions of constant amplitude will be found only if the forcing is oscillatory rather than steady. Solutions are found by solving the first-order wave equations (8.107a) and (8.110) for q_n , and then using (8.109b) to obtain r_n . A simple expression for v_n results if we add (8.107b) and (8.110).

Waves and adjustment

The wave described by (8.107a) is a Kelvin wave, moving eastwards with nondimensional speed unity, or dimensional speed c . The speed also follows from the dispersion relation, $\omega = -kc/(2n+1)$, with $n = -1$. In contrast, the waves described by (8.110) are westwards propagating, long, low frequency planetary waves. In dimensional form (8.110) becomes

$$(2n+1) \frac{\partial q_{n+1}}{\partial t} - c \frac{\partial q_{n+1}}{\partial x} = n(F_{n+1}^x - Q_{n+1}) - (F_{n-1}^x + Q_{n-1}). \quad (8.111)$$

and hence have a speed $-c/(2n+1)$, just as in (8.50). There are no short low frequency waves in this approximation.

As we noted above, an arbitrary forcing will in general excite both gravity waves and planetary waves and the initial flow will be out of geostrophic balance. In the midlatitude case (section 3.9) the gravity waves radiate to infinity (in the idealized problem) leaving behind an adjusted flow in geostrophic balance, determined by potential vorticity conservation. The process of adjustment is less efficient at low latitudes, because the waves are trapped between their inertial latitudes (section 8.3) and in the absence of dissipation the fluid will oscillate endlessly. In the zonal direction both planetary and Kelvin waves propagate. A gravity wave front moves away more quickly, with the eventual adjustment occurring by way of planetary waves.

Let us now turn our attention to a rather concrete problem, that of the steady response to a localised thermal forcing.

8.5 FORCED, STEADY FLOW: THE MATSUNO–GILL PROBLEM

We now consider the forced, steady version of the equatorial wave problem; that is to say, we seek steady solutions of (8.84), but with a mechanical or thermal forcing on the right-hand side.⁹ Because of its importance to the tropical circulation of the atmosphere this problem has become somewhat iconic and some readers may be tempted to begin reading this chapter here. However, the problem is really just the forced, steady version of the wave problems studied in sections 8.2 and 8.4, and the reader should have at least a passing familiarity with that material before proceeding. Those readers who have followed the previous sections closely will find the material that follows, namely the *Matsuno–Gill* problem, a pleasant stroll in the park.

8.5.1 Mathematical development

We begin with (8.84) and make two additional simplifications. First, that the flow is steady and second that the zonal wind is in geostrophic balance with the meridional pressure gradient. This ‘semi-geostrophic’ approximation is equivalent to the longwave approximation discussed in previous sections. The equations of motion become

$$\alpha u - f v + \frac{\partial \phi}{\partial x} = F^x, \quad f u + \frac{\partial \phi}{\partial y} = 0, \quad (8.112a,b)$$

$$\alpha \phi + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -Q. \quad (8.112c)$$

From these equations we may derive a single equation for v , namely

$$\frac{f^2}{c^2} \alpha v - \alpha \frac{\partial^2 v}{\partial y^2} - \beta \frac{\partial v}{\partial x} = \frac{\alpha}{c^2} \frac{\partial Q}{\partial y} - \frac{f}{c^2} \frac{\partial Q}{\partial x} - \frac{f}{c^2} \alpha F^x + \frac{\partial^2 F^x}{\partial x \partial y}. \quad (8.113)$$

This is just a simplification of (8.86) appropriate for a steady system with the zonal wind in geostrophic balance, obtained by omitting all the time derivatives, the term involving α^3 , and the F^y term on the right hand side. We nondimensionalize all the variables using the time and length scales

$$T_{eq} = (2c\beta)^{-1/2}, \quad L_{eq} = \left(\frac{c}{2\beta} \right)^{1/2}, \quad (8.114a,b)$$

so that the various dimensional and nondimensional (hatted) variables are related by

$$(u, v) = c(\hat{u}, \hat{v}), \quad \phi = c^2 \hat{\phi}, \quad (x, y) = L_{eq}(\hat{x}, \hat{y}), \quad (8.115)$$

$$\alpha = \frac{\hat{\alpha}}{T_{eq}}, \quad Q = \frac{c^3}{L_{eq}} \hat{Q}, \quad F^x = \frac{c^2}{L_{eq}} \hat{F}^x, \quad \hat{f} = \frac{1}{2} \hat{y}, \quad \hat{\beta} = \frac{1}{2}.$$

Unless specifically noted the variables from now on are nondimensional and we drop the hats. The equations of motion, (8.116), become

$$\alpha u - \frac{y}{2} v + \frac{\partial \phi}{\partial x} = F^x, \quad \frac{y}{2} u + \frac{\partial \phi}{\partial y} = 0, \quad \alpha \phi + \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -Q. \quad (8.116a,b,c)$$

and the v equation becomes

$$\frac{y^2}{4} \alpha v - \alpha \frac{\partial^2 v}{\partial y^2} - \frac{1}{2} \frac{\partial v}{\partial x} = \alpha \frac{\partial Q}{\partial y} - \frac{y}{2} \frac{\partial Q}{\partial x} - \frac{\alpha y}{2} \alpha F^x + \frac{\partial^2 F^x}{\partial x \partial y}. \quad (8.117)$$

As before when dealing with wave-like problems it is convenient to change variables to p and q where

$$q = \phi + u, \quad r = \phi - u. \quad (8.118a,b)$$

The equations of motion, (8.116), become

$$\begin{aligned} \alpha q + \frac{\partial q}{\partial x} + \frac{\partial v}{\partial y} - \frac{1}{2} y v &= F^x - Q, & \alpha r - \frac{\partial r}{\partial x} + \frac{\partial v}{\partial y} + \frac{1}{2} y v &= -F^x - Q, \\ \frac{y}{4} (q - r) + \frac{1}{2} \frac{\partial}{\partial y} (q + r) &= 0. \end{aligned} \quad (8.119a,b,c)$$

These are special cases of (8.90), with the first two equations being combinations of the u -momentum and pressure equations and the last one being the v -momentum equation (zonal geostrophic balance).

As in the general treatment given earlier we expand the variables and the forcing in terms of parabolic cylinder functions. Thus, for example,

$$Q(x) = \sum_{n=0}^{\infty} Q_n(x) D_n(y), \quad (8.120)$$

and similarly for the other variables. The resulting ordinary differential equations are special cases of (8.94)–(8.97), specifically

$$\alpha q_0 + \frac{\partial q_0}{\partial x} = F_0^x - Q_0, \quad (8.121a)$$

$$\alpha q_{n+1} + \frac{\partial q_{n+1}}{\partial x} - v_n = F_{n+1}^x - Q_{n+1}, \quad n = 0, 1, 2, 3, \dots \quad (8.121b)$$

$$\alpha r_{n-1} - \frac{\partial r_{n-1}}{\partial x} + n v_n = -(F_{n-1}^x + Q_{n-1}), \quad n = 1, 2, 3, \dots \quad (8.122)$$

$$q_1 = 0, \quad (8.123a)$$

$$(n+1)q_{n+1} - r_{n-1} = 0, \quad n = 1, 2, 3, \dots \quad (8.123b)$$

Using (8.121b), (8.122) and (8.123b) we obtain

$$\alpha(2n+1)q_{n+1} - \frac{\partial q_{n+1}}{\partial x} = n(F_{n+1}^x - Q_{n+1}) - (F_{n-1}^x + Q_{n-1}) \quad n = 1, 2, 3, \dots \quad (8.124)$$

Finally, although we shall not use it, the v equation (8.113) becomes

$$\alpha \left((2n+1)v_n - \frac{\partial^2 v_n}{\partial x^2} \right) - \frac{\partial v_n}{\partial x} = G, \quad (8.125)$$

where G represents the various forcing terms.

As in the wavelike case, the above equations provide, at least in principle, a means of solving for the response for any particular forcing. The procedure is to project the forcing onto parabolic cylinder functions, and then solve the amplitude equations (8.121)–(8.123) for the zonal dependence, and then finally to reconstruct the solutions using the $q_n(x)$, $r_n(x)$ and $v_n(x)$ and the parabolic cylinder functions. Naturally enough, this is easier said than done and we will go through the procedure in detail for just one important case.

8.5.2 Symmetric heating

A canonical case is that in which the system is forced by a heating that is confined in both the x - and y -directions, and is symmetric across the equator. Confinement in the y -direction may be achieved by supposing that the heating projects solely onto the first parabolic function, so that

$$Q(x) = Q_0(x)D_0(y) = G(x) \exp(-y^2/4), \quad (8.126)$$

and confinement in the x -direction may be achieved by supposing that the heating is of the form

$$G(x) = \begin{cases} A \cos kx & |x| < L \\ 0 & |x| > L, \end{cases} \quad (8.127)$$

where $k = \pi/2L$. This may seem an odd form to choose, but the harmonic variation for $|x| < L$ enables an analytic solution to be found in that region, and the absence of any forcing at all in the far field enables solutions to be found there in the form of decaying wavelike disturbances. Although this problem is clearly a special case the qualitative form of the solution transcends its precise details.

Kelvin wave contribution

We noted in section 8.4.2 that the equation for q_0 represents an eastwards propagating Kelvin wave, and this holds in the damped case also. That is to say, there will be a nonzero solution of (8.121) only in the forced region and eastward of it, where it will be progressively damped. Using this insight we can easily derive the solution in all three regions. First, for $X < -L$, we have

$$q_0 = 0, \quad x < -L. \quad (8.128a)$$

In the forcing region we have to solve (8.121) with a boundary condition of $q_0 = 0$ at $x = -L$. The solution is

$$q_0 = \frac{-A}{\alpha^2 + k^2} \left\{ \alpha \cos kx + k \left[\sin kx + e^{-\alpha(x+L)} \right] \right\} \quad |x| < -L. \quad (8.128b)$$

For $x > L$ we solve (8.121), but with a right-hand side equal to zero, with a boundary condition at $x = L$ given by (8.128b), namely $q_0 = -Ak(\alpha^2 + k^2)^{-1}[1 + \exp(-2\alpha L)]$. The solution is

$$q_0 = \frac{-Ak}{\alpha^2 + k^2} (1 + e^{-2\alpha L}) e^{\alpha(L-x)}. \quad x > L. \quad (8.128c)$$

Because the motion is a decaying Kelvin wave $v = 0$ and the nondimensional u and ϕ fields are equal to each other, with $r = 0$. Thus, from (8.118) and (8.128),

$$u = \phi = \frac{1}{2} q_0(x) \exp(-y^2/4), \quad v = 0. \quad (8.129)$$

This does not mean r_0 is zero; rather, it is associated with the planetary wave solution discussed below. The vertical velocity may be reconstructed from

$$w = - \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = \alpha \phi + Q, \quad (8.130)$$

whence

$$w = \frac{1}{2} [\alpha q_0(x) + Q_0(x)] \exp(-y^2/4). \quad (8.131)$$

We now complete the solution by finding a planetary wave contribution.

Planetary wave contribution

We now find the solution associated with q_2 and r_0 . From (8.124) we have

$$\frac{dq_2}{dx} - 3\alpha q_2 = Q_0. \quad (8.132a)$$

From (8.123b) and (8.121b) we have

$$r_0 = 2q_2, \quad v_1 = \alpha q_2 + \frac{dq_2}{dx}. \quad (8.132b,c)$$

These are planetary waves propagating westwards at a dimensional speed of $c/(2n+1) = c/3$. Thus, no signal is transmitted eastwards and we can find a solution to the above equations in an analogous fashion to how we found a solution for the Kelvin wave problem. After just a little algebra, the solution is found to be:

$$q_2 = 0, \quad x > L \quad (8.133a)$$

$$q_2 = \frac{A}{(3\alpha)^2 + k^2} \left[-3\alpha \cos kx + k \left(\sin kx - e^{3\alpha(x-L)} \right) \right], \quad |x| < L \quad (8.133b)$$

$$q_2 = \frac{-Ak}{(3\alpha)^2 + k^2} \left[1 + e^{-6\alpha L} \right] e^{3\alpha(x+L)}, \quad x < -L. \quad (8.133c)$$

The corresponding solutions for the pressure and velocity fields are

$$u = \frac{e^{-y^2/4}}{2} q_2(x)(y^2 - 3), \quad v = ye^{-y^2/4} [Q_0(x) + 4\alpha q_2(x)], \quad (8.134a,b)$$

$$\phi = \frac{e^{-y^2/4}}{2} q_2(x)(1 + y^2), \quad w = \frac{e^{-y^2/4}}{2} [Q_0(x) + \alpha q_2(x)(1 + y^2)]. \quad (8.135a,b)$$

The solutions appear complicated (they are complicated!), although still amenable to interpretation. But first we combine the Kelvin and planetary wave contributions and restore the dimensions, to give

$$u = \frac{c}{2} [q_0(x) + q_2(x)(2\beta y^2/c - 3)] e^{-\beta y^2/2c}, \quad (8.136a)$$

$$v = cy [Q_0(x) + (4\alpha/c)q_2(x)] e^{-\beta y^2/2c}, \quad (8.136b)$$

$$\phi = \frac{c^2}{2} [q_0(x) + q_2(x)(2\beta y^2/c + 1)] e^{-\beta y^2/2c}, \quad (8.136c)$$

$$w = w_0 \frac{e^{-\beta y^2/2c}}{2} [2Q_0(x) + \alpha q_0(x) + \alpha q_2(x)(1 + 2\beta y^2/c)] \quad (8.136d)$$

where $w_0 = (2c\beta)^{1/2}H$, and q_0 , q_2 and Q_0 are still nondimensional functions. The nondimensional forms are recovered by setting $w_0 = c = 1$ and $\beta = 1/2$. The solutions above are specific to the form of the forcing function we chose. However, a similar methodology could in principle be applied to forcing of any form, including forcing in the momentum equations, and, because the equations are linear, the solutions could be superposed. The solution above represents the physically important case of a localized heating, and the gross structure of the far field is largely independent of the details of the forcing: there is a rapidly decaying disturbance west and polewards of the forcing and a more slowly decaying disturbance east of the forcing close to the equator (Fig. 8.11).

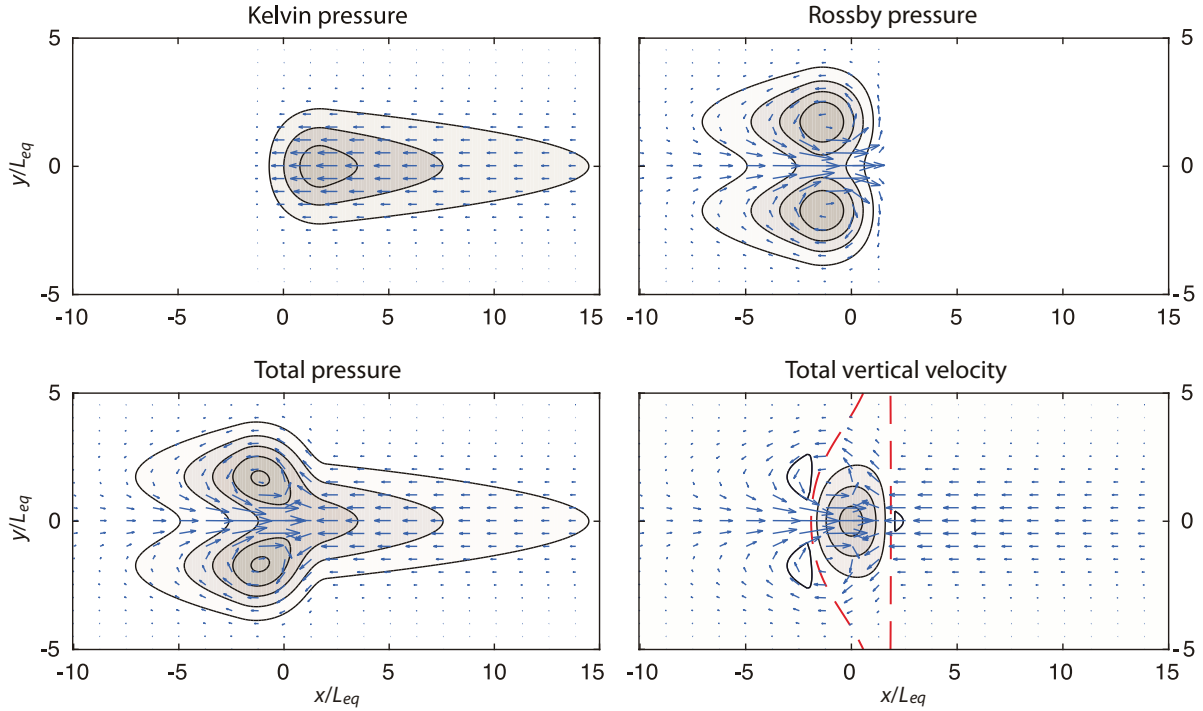


Fig. 8.11 Nondimensional solutions of the Matsuno–Gill model, with heating given by (8.127) with $L = 2$ and $\alpha = 0.1$. The shaded contours show the fields as indicated, and the arrows show the associated horizontal velocities. The ‘Kelvin’ and ‘Rossby’ designations indicate that just the Kelvin wave or Rossby (planetary) wave contributions are plotted as given by (8.129)–(8.131) and (8.134)–(8.135), respectively. For the pressure fields the contour interval is 0.3 and all fields are negative (so dark shading is low pressure) with the zero contour omitted. For vertical velocity the contour interval is 0.3 beginning at -0.1, and so is -0.1, 0.2, 0.5..., with an additional zero contour (red dashed) with upward motion within it.

Interpretation

Let’s now try to figure out what’s going on. A solution is illustrated in Fig. 8.11. The heating is confined to a region from $-2 < x < 2$ and exponentially falls away from the equator with an e-folding distance of 2, more-or-less corresponding to the shaded region of vertical velocity in the lower right panel, as intuitively expected and discussed more below.

Consider first the flow in the forcing region. Here the vertical velocity is positive, with the associated horizontal convergence being that of the zonal flow: the meridional flow is polewards, *away* from the maximum of the heating. To understand this, consider the limit $\alpha \rightarrow 0$. From (8.136) the vertical velocity field coincides with the heating and the (nondimensional) meridional velocity is given by

$$v = yQ_0 \exp(-y^2/4) = yw. \quad (8.137)$$

Thus, vertical motion is associated with poleward motion. To understand this, consider the inviscid vorticity equation

$$\beta v + f \nabla \cdot \mathbf{u} = 0, \quad \text{or} \quad \beta v = fw. \quad (8.138a,b)$$

which in nondimensional form is

$$v + y \nabla \cdot \mathbf{u} = 0, \quad \text{or} \quad v = yw. \quad (8.139a,b)$$

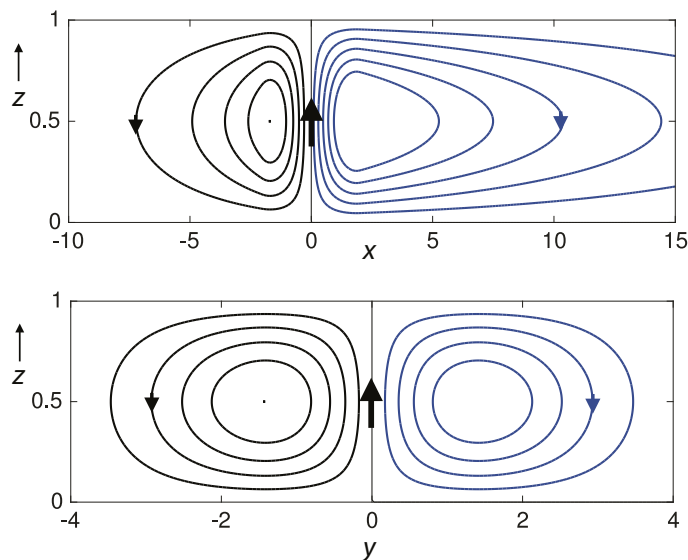


Fig. 8.12 The zonal overturning streamfunction of the meridionally averaged flow (top), and the meridional overturning streamfunction of the zonally-averaged flow (bottom) in the Matsuno–Gill problem with symmetric heating with a maximum at $y = 0$ and $x = 0$, as in Fig. 8.11.

There is rising motion around the location of the heating, sinking elsewhere. The contour interval in the top plot is about four times that of the bottom plot; that is, the Walker circulation is stronger than the Hadley circulation.

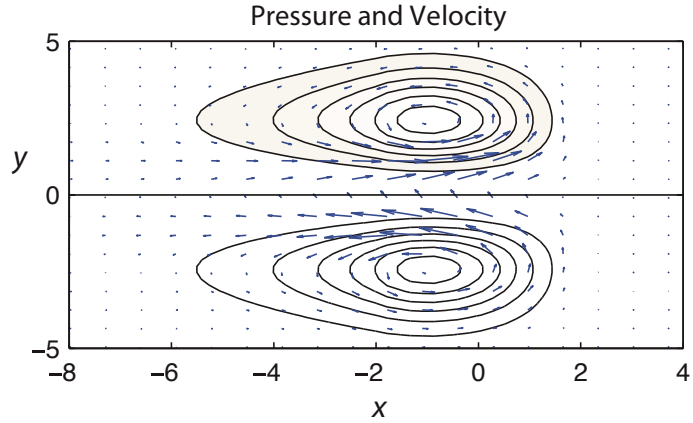
Evidently, (8.137) and (8.139b) are equivalent. Another way to think about this is to note that the rising motion in the region of the forcing causes vortex stretching, as discussed in chapter 4, and hence the generation of cyclonic vertical vorticity and a polewards migration. From the perspective of potential vorticity, then to the extent that the flow is adiabatic the quantity $Q = (f + \zeta)/h$ is conserved following the flow. The heating increases the value of h (the stretching), so that $f + \zeta$ also tends to increase in magnitude. The flow finds it easier to migrate polewards to increase its value of f than to increase its relative vorticity alone, for the latter would require more energy. If we interpret these equations as the lowest layer of a two-layer system, then the flow in the lower layer is away from the source, and toward the source in the upper layer.

Consider now the flow to the west of the heating, associated with $q_2(x)$. The disturbance here is produced by a decaying westwards propagating Rossby wave — a form of ‘Rossby plume’ that we will also encounter in chapter 19 (see Fig. 19.14 on page 754 and the associated discussion). the vertical velocity is negative, and the horizontal velocity is almost geostrophically balanced: the pressure perturbation is negative everywhere, and so circulating cyclonically around the centers of low pressure just to the west of heating. The flow converges to the equator, producing an eastward flow along the equator, converging in the heating zone. We may be tempted to interpret this in terms of the inviscid vorticity equation, as we did in the forcing region. This would suggest that, away from the forcing region, because the flow is divergent ($\nabla \cdot \mathbf{u} > 0$, $w < 0$) then from (8.138) the meridional velocity should be toward the equator in both hemispheres. However, this explanation is at best qualitative, because the vorticity equation above is not exactly satisfied by the solution (8.138), because non-zero solutions away from the forcing region depend entirely on the presence of dissipation.

The flow east of the forcing region motion is induced by an eastward propagating Kelvin wave, or more precisely the steady, eastward-decaying analogue of such a wave. Evidently, from Fig. 8.11, the pressure field extends further east of the source than west of the source, and this is because Kelvin waves decay more slowly than Rossby waves. Keeping both the time derivative and the damping, the unforced Kelvin wave satisfies, from (8.94),

$$\left[\alpha + \frac{\partial}{\partial t} \right] q_0 + \frac{\partial q_0}{\partial x} = 0, \quad (8.140)$$

Fig. 8.13 Pressure (contours) and horizontal velocity (arrows) in the Matsuno–Gill problem with an anti-symmetric heating given by (8.143) and decay factor $\alpha = 0.1$. The heating is in the Northern Hemisphere generating a low pressure region (shaded) with inflow and ascent, and cooling is in the Southern Hemisphere. The contour interval is 0.3 and the zero contour is along $y = 0$.



whereas the unforced Rossby wave satisfies, from (8.110) and (8.124) for $n = 1$

$$3 \left[\alpha + \frac{\partial}{\partial t} \right] q_1 - \frac{\partial q_1}{\partial x} = 0. \quad (8.141)$$

Thus, the effective damping rate of the Rossby wave is three times that of the Kelvin wave. Put another way, the Kelvin wave travels three times as fast as the Rossby wave so that if the damping rate, α , is the same the influence of the Kelvin wave spreads three times further east. The horizontal velocity in the Kelvin wave is purely zonal and near the surface it is directed toward the heating source.

Vertical structure

The zonal structure of the solution is a coarse representation of the Walker circulation in the equatorial Pacific. Here, the sea-surface temperature is high in the west, near Indonesia, and low in the east, near South America, because of the upwelling that brings deep, cold water to the surface. This distribution of sea-surface temperature effectively provides a heating in the western Pacific and induces westward winds along the equator, enhancing the westward trade winds that already exist as part of the general circulation. The overturning circulation in the zonal and meridional planes is illustrated in Fig. 8.12. This solution is obtained by supposing that the fields represent the first vertical mode, as discussed in section ?? . If the stratification is uniform then the modes are just sines and cosines and so we have

$$(u, v, \phi) = (\tilde{u}, \tilde{v}, \tilde{\phi}) \cos(\pi z/D), \quad w = \tilde{w} \sin(\pi z/D) \quad (8.142)$$

Now the modal form of the mass continuity equation, (3.67), is $\tilde{w} = -(c^2/g) \nabla \cdot \tilde{\mathbf{u}}$. If this is to be consistent with the usual form of $\partial w / \partial z = -\nabla \cdot \mathbf{u}$ then we make the association $\pi/D = g/c^2 = 1/H_1^*$, where H_1^* is the equivalent depth of the first mode. Given this vertical structure, we integrate the solutions meridionally so enabling a streamfunction to be defined (because $v = 0$ as $y \rightarrow \pm\infty$, so $\partial \tilde{u}^y / \partial x + \partial \tilde{w}^y / \partial z = 0$, with the overbar denoting meridional integration). The expressions for the meridional integrals are given in Appendix B to this chapter, with the streamfunctions given by (8.175) and (8.175).

8.5.3 Antisymmetric forcing

An analytic solution with asymmetric forcing may be obtained by using a forcing of the form

$$Q(x, y) = Q_1(x) D_1(y) = y \cos kx \exp(-y^2/4), \quad (8.143)$$

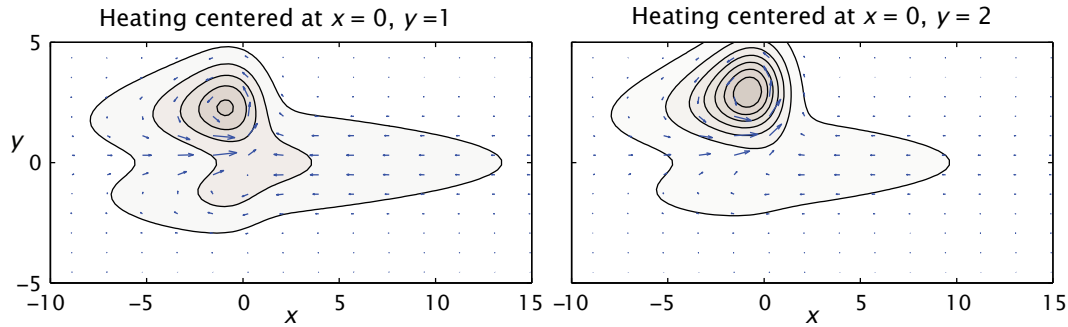


Fig. 8.14 Pressure (contours) and horizontal velocity (arrows) in the Matsuno–Gill model with the heating centered off the equator, as labelled, but otherwise similar to that of Fig. 8.11. As the heating moves to higher latitudes the Kelvin wave response weakens but the magnitude of the local response increases (the contour interval is the same in both panels).

using the same form of zonal localization as before. The algebra needed to obtain a solution is somewhat tedious but straightforward, of a very similar nature to that described above. One finds that there are, again, two parts to the response. One part corresponds to a long planetary wave with $n = 0$ and using (8.121)–(8.123) we find

$$q_1 = 0, \quad v_0 = Q_1. \quad (8.144)$$

There is no response outside the forcing region because long mixed waves have zero propagation velocity. The other part of the solution is obtained, again using (8.121)–(8.123), from

$$v_2 = \frac{dq_3}{dx} + \alpha q_3, \quad r_1 = 3q_3, \quad \frac{dq_3}{dx} - 5\alpha q_3 = Q_1 \quad (8.145a,b,c)$$

The solution of these equations is left as an exercise for the reader and is illustrated in Fig. 8.13. The solutions are zero east of the forcing region because there is no long wave so propagating. West of the forcing region there is eastward inflow into the heating region in the Northern Hemisphere (which is being heated), as well as a tendency for poleward flow for the reasons described earlier. Thus, there is a cyclone with upward motion somewhat west of the main heating region, and a corresponding anti-cyclone in the cooled region, as illustrated in Fig. 8.13. The zonally averaged solutions (not shown) resemble an asymmetric Hadley Cell, with the air rising in the Northern (summer) Hemisphere, moving southwards aloft into the winter hemisphere before sinking.

8.5.4 Other forcings

The solution to more general forcings can be constructed by using other forcing coefficients, or a superposition of forcing coefficients, and many solutions of interest to the tropical atmosphere and ocean may be so constructed. Solutions may also be constructed (and sometimes more easily) numerically, either by time-stepping the linear shallow water equations to equilibrium or by solving the elliptic equation (8.113) using standard techniques.¹⁰ We will present the solutions to two such cases: (i) a heating centered off the equator in the Northern Hemisphere, and (ii) a line source of heating, either centered at the equator or just north of it, mimicking the Inter-Tropical Convergence Zone (ITCZ).

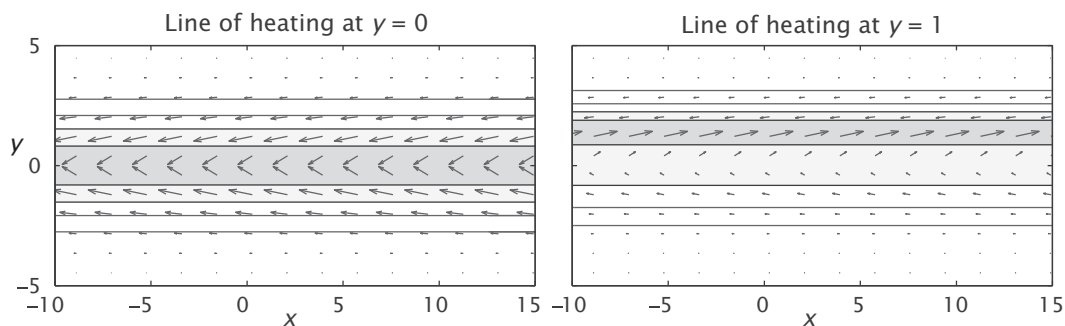


Fig. 8.15 As for Fig. 8.14 but with a line of heating at the equator (left panel) and at $y = 1$ (right panel). The heating generates a region of low pressure (shaded) where the flow converges. In the right panel the meridional velocity is larger on the equatorward side of the line than on the poleward side. See also Fig. 8.16.

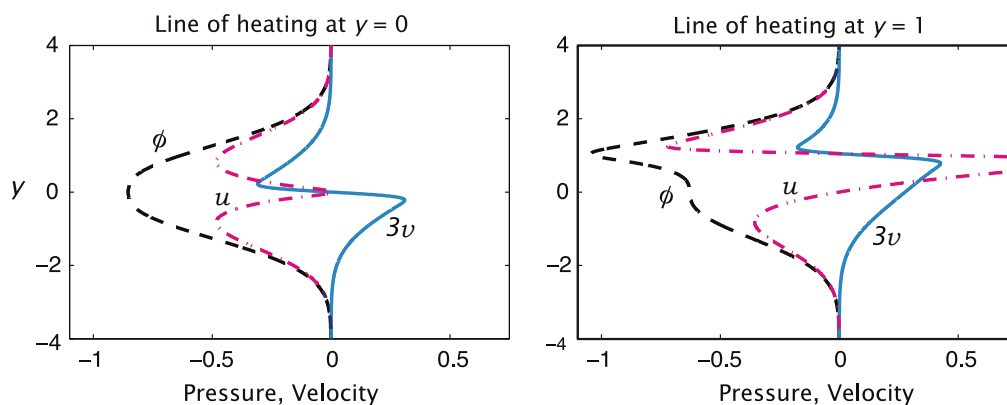


Fig. 8.16 As for Fig. 8.15, but showing line plots of pressure, ϕ , zonal velocity, u and three times (for presentational purposes) the meridional velocity v . Left panel is for a line of heating at the equator and the right panel for heating at $y = 1$. The heating creates a region of low pressure where the flow converges. Note that in the right panel the meridional velocity is larger on the equatorward side of the line than on the poleward side.

Heating off the equator

A heating off the equator may be constructed by adding the solutions for antisymmetric and symmetric heating presented above. In Fig. 8.14, we present a solution that has heating of a very similar form to that of the symmetric heating shown in Fig. 8.11, but centered off the equator at $y = 1$ and $y = 2$. The pattern is dominated by a low pressure region just to the west of the heating, with convergence and upward motion within it, and an eastward inflow between the equator and the center of the heating. In the solution with the heating centered at $y = 1$ there is also a response east of the heating region, largest at the equator, produced by the eastward propagating, damped Kelvin wave. As the heating moves further from the equator (in the right panel of Fig. 8.14), the pressure response becomes stronger but the flow around the heating is in near geostrophic balance.

A line of heating

Finally, let us consider the solutions when the heating is independent of x , and the solutions themselves are then independent of x . Two such solutions are presented in Fig. 8.15 and again, more

quantitatively, in Fig. 8.16, for a line of heating at the equator and at $y = 1$. As we noted above, these solutions might be thought of as a rather idealized versions of the ITCZ (although in the real ITCZ the location of the convective region is determined as part of the solution for the overall flow).

Consider first the solution with heating at the equator. A low pressure region develops over the heating and the flow converges there, producing equatorward and westward ‘trade winds’ and consequent upward motion at the equator, with the zonal velocity rapidly decreasing actually at the equator.

Now consider the what happens when the heating is off-equator, noting that the real ITCZ is generally situated a little north of the equator, especially in the Pacific Ocean. A low pressure region is formed along the line of the heating and the meridional velocity converges sharply there, with more inflow coming from the equatorial side of the line of heating (as can be seen in the right-hand panels of both Fig. 8.15 and Fig. 8.16). As regards the zonal velocity, there is an *eastward* jet along the line of the heating, with westward flow to either side. That is to say, there is a splitting of the westward trades caused by the line of sharp heating.

8.A APPENDIX: NONDIMENSIONALIZATION AND PARABOLIC CYLINDER FUNCTIONS

This appendix provides a brief discussion of the nondimensionalization used to derive the various dispersion relations in this chapter and some of the properties of the associated Hermite polynomials and parabolic cylinder functions. We do not provide proofs or detailed derivations.¹¹

In discussions of equatorial waves and their steady counterparts, one of two slightly different nondimensionalizations are often employed. They lead to the use of parabolic cylinder functions in two slightly different forms; they are essentially equivalent but one may be more convenient than the other depending on the setting. For definiteness, we begin with (8.28), namely

$$\frac{d^2 \tilde{v}}{dy^2} + \left(\frac{\omega^2}{c^2} - k^2 - \frac{\beta k}{\omega} - \frac{\beta^2 y^2}{c^2} \right) \tilde{v} = 0. \quad (8.146)$$

If we nondimensionalize time and distance using

$$T_{eq} = (c\beta)^{-1/2}, \quad L_{eq} = (c/\beta)^{1/2} \quad (8.147a,b)$$

we obtain

$$\frac{d^2 v}{d\hat{y}^2} + \left(\hat{\omega}^2 - \hat{k}^2 - \frac{\hat{k}}{\hat{\omega}} - \hat{y}^2 \right) v = 0. \quad (8.148)$$

The substitution

$$v(\hat{y}) = \Psi(\hat{y}) e^{-\hat{y}^2/2}, \quad (8.149)$$

leads to

$$\frac{d^2 \Psi}{d\hat{y}^2} - 2\hat{y} \frac{d\Psi}{d\hat{y}} + \lambda \Psi = 0 \quad (8.150)$$

where $\lambda = \hat{\omega}^2 - \hat{k}^2 - \hat{k}/\hat{\omega} - 1$. This is Hermite’s equation with solutions if and only if $\lambda = 2m$ for $m = 0, 1, 2, \dots$, and it is this quantization condition that gives the dispersion relation. The solutions are Hermite polynomials; that is, $\Psi(\hat{y}) = H_m(\hat{y})$, where

$$(H_0, H_1, H_2, H_3, H_4) = (1, 2\hat{y}, 4\hat{y}^2 - 2, 8\hat{y}^3 - 12\hat{y}, 16\hat{y}^4 - 48\hat{y}^2 + 12). \quad (8.151)$$

The Hermite polynomial multiplied by a Gaussian is a form of parabolic cylinder function, $V_m(y)$; that is

$$V_m(y) = H_m(y) \exp(-y^2/2). \quad (8.152)$$

The function $V_m(y)$ satisfies

$$\frac{d^2 V_m}{dy^2} + (2m + 1 - y^2)V_m = 0 \quad (8.153)$$

It is often useful to include the normalization coefficient in the definition of the cylinder function; that is, if

$$P_m = \frac{V_m}{\sqrt{2^m m!} \sqrt{\pi}}, \quad \text{then} \quad \int_{-\infty}^{\infty} P_m P_n = \delta_{mn}. \quad (8.154a,b)$$

As may be verified by direct manipulation, these forms of parabolic cylinder functions obey certain recurrence relations, namely

$$\frac{dP_m}{dy} = -\frac{(m+1)^{1/2}}{\sqrt{2}}P_{m+1} + \frac{m^{1/2}}{\sqrt{2}}P_{m-1} \quad \text{and} \quad yP_m = \frac{m^{1/2}}{\sqrt{2}}P_{m-1} + \frac{(m+1)^{1/2}}{\sqrt{2}}P_{m+1}. \quad (8.155a,b)$$

or equivalently

$$\frac{dP_m}{dy} + yP_m = (2m)^{1/2}P_{m-1} \quad \text{and} \quad \frac{dP_m}{dy} - yP_m = -\sqrt{2}(m+1)^{1/2}P_{m+1}. \quad (8.156)$$

When $m = 0$ the recurrence relations are

$$\frac{dP_0}{dy} = \frac{-1}{\sqrt{2}}P_1 \quad \text{and} \quad yP_0 = \frac{1}{\sqrt{2}}P_1. \quad (8.157a,b)$$

Had we developed the forced-dissipative problem using this form of cylinder functions these relations would have been used instead of (8.93).

The above relations may be used in conjunction with (8.20) and (8.36) to obtain the relations between u , v and ϕ in the equatorial wave problem. Using (8.20) the v and u , and the v and ϕ , fields are related by

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \beta y \frac{\partial v}{\partial t} + c^2 \frac{\partial^2 v}{\partial x \partial y}, \quad \frac{\partial^2 \phi}{\partial x^2} - c^{-2} \frac{\partial^2 \phi}{\partial t^2} = \beta y \frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial y \partial t}, \quad (8.158)$$

which in nondimensional form have $\beta = c = 1$. Suppose that the v field is a single Hermite mode of unit amplitude, meaning that

$$\tilde{v}(x, y, t) = \tilde{v}(y)e^{i(kx - \omega t)} = P_m(y)e^{i(kx - \omega t)}. \quad (8.159)$$

If $u(x, y, t) = \tilde{u}(y)e^{i(kx - \omega t)}$ then, using (8.158a) we obtain

$$\tilde{u} = \frac{-i}{(k^2 - \omega^2)} \left[\hat{y} \omega P_m - k \frac{dP_m}{d\hat{y}} \right] = \frac{i}{\sqrt{2}} \left[\frac{m^{1/2} P_{m-1}}{k + \omega} - \frac{(m+1)^{1/2} P_{m+1}}{k - \omega} \right] \quad (8.160)$$

where the rightmost expression uses the recurrence relations (8.155). Evidently, if m and $\tilde{v}$ are odd (even) then $\tilde{u}$ is an even (odd) function of y . The height field $\tilde{\phi}$ may similarly be related to P_m using (8.158b), giving

$$\tilde{\phi} = \frac{-i}{k^2 - \omega^2/c^2} \left[ykP_m + i\omega \frac{dP_m}{dy} \right] = \frac{-i}{\sqrt{2}} \left[\frac{m^{1/2} P_{m-1}}{k + \omega} + \frac{(m+1)^{1/2} P_{m+1}}{k - \omega} \right]. \quad (8.161)$$

Thus, for a given m , ϕ has the same symmetry across the equator as does u , the opposite of v .

Implications

As discussed in the main text, the gravest mode in y is the Kelvin wave, which has $v = 0$, and u and ϕ fields centred on the equator that decay away exponentially in y^2 , as for P_0 . The Yanai, or mixed Rossby-gravity mode, has $m = 0$ and the v field is even around the equator, meaning it is an antisymmetric mode. Using the recurrence relation (8.157) we see that this mode only generates the P_1 mode in u and ϕ . The gravest Rossby mode has $m = 1$, and from (8.160) and (8.161) we see that this generates $m = 0$ and $m = 2$ modes in u and ϕ . Thus, a symmetric disturbance centred at the equator will in general generate an eastward propagating Kelvin mode and a Rossby mode with both an equatorial signal and off equatorial modes with a structure similar to those of the stationary pattern in Fig. 8.11, and as we will see again in Fig. 22.17 when we study El Niño.

Other parabolic cylinder functions

The other commonly used form of parabolic cylinder functions, denoted $D_n(y)$, are the modified Hermite polynomials (8.42) multiplied by a Gaussian; that is

$$D_n(y) = G_n(y) \exp(-y^2/4). \quad (8.162)$$

and these functions are solutions of (8.39) which arises when we use the nondimensionalization

$$T_{eq} = (2c\beta)^{-1/2}, \quad L_{eq} = (c/2\beta)^{1/2}. \quad (8.163a,b)$$

These parabolic cylinder functions satisfy

$$\frac{d^2 D_m}{dy^2} + \frac{1}{2}(2m+1 - \frac{1}{2}y^2)D_m = 0, \quad (8.164)$$

which is sometimes called the Weber differential equation. The functions have the property that

$$\frac{dD_n}{dy} + \frac{1}{2}yD_n = nD_{n-1}, \quad \frac{dD_n}{dy} - \frac{1}{2}yD_n = -D_{n+1}. \quad (8.165a,b)$$

The above two equations may be combined to give (8.164), and by subtracting them we see that

$$D_{n+1} - yD_n + nD_{n-1} = 0. \quad (8.166)$$

The form of these particular ladder operators makes these parabolic cylinder functions convenient in our development of the forced, steady (i.e., Matsuno–Gill) problem, although the use of (8.156) would be equivalent.

8.B APPENDIX B: MATHEMATICAL RELATIONS IN THE MATSUNO–GILL PROBLEM

Here we provide various zonal and meridional integrals of the solutions given in section 8.5.2. The zonal integral of the forcing is given by

$$I = \int_{-\infty}^{\infty} Q_0(x) dx = \int_{-L}^L \cos(kx) dx = \frac{4L}{\pi}, \quad (8.167)$$

using $k = \pi/2L$. The zonal integrals of the various q , r and v fields are given as follows. Using (8.121) with $F_0 = 0$ we see that

$$\int_{-\infty}^{\infty} \alpha q_0 dx = -\left[q_0\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} Q_0(x) dx = -I. \quad (8.168)$$

Using (8.132) we obtain similar results for q_2, r_0 and v_1 , to wit

$$\int_{-\infty}^{\infty} (q_0, q_2, r_0, v_1) \, dx = \left(-1, -\frac{1}{3}, -\frac{2}{3}, -\frac{\alpha}{3} \right) \frac{I}{\alpha}. \quad (8.169)$$

The zonally integrated pressure and velocity fields are obtained using (8.167), (8.169) and the non-dimensional form of (8.136), giving

$$\int_{-\infty}^{\infty} (u, v, w, \phi) \, dx = \left(\frac{-y^2}{6\alpha}, \frac{-y}{3}, \frac{2-y^2}{6}, \frac{-4-y^2}{6\alpha} \right) \left(\frac{4L}{\pi} \right) \exp(-y^2/4). \quad (8.170)$$

The meridional integrals of the velocity fields may also be calculated. To do this we first note the integrals

$$\int_{-\infty}^{\infty} (1, y, y^2) \exp(-y^2/4) \, dy = (2, 0, 4) \sqrt{\pi}. \quad (8.171)$$

The first of these is a standard result, the second follows from considerations of symmetry and the third follows on integration by parts. Using (8.171) and the nondimensional form of (8.136) we obtain

$$\int_{-\infty}^{\infty} u \, dy = \sqrt{\pi} [q_0(x) - q_2(x)], \quad \int_{-\infty}^{\infty} v \, dy = 0, \quad (8.172a,b)$$

$$\int_{-\infty}^{\infty} w \, dy = \sqrt{\pi} [\alpha q_0(x) + 3\alpha q_2(x) + 2Q_0(x)], \quad \int_{-\infty}^{\infty} \phi \, dy = \sqrt{\pi} [q_0(x) + 3q_2(x)]. \quad (8.172c,d)$$

Equations (8.170) and (8.172) are useful because they allow us to define streamfunctions for the overturning circulation in the zonal and meridional plane, respectively. From (8.170) and (8.172), and using (8.121) and (8.132a), we find that

$$\overline{w}^x + \frac{\partial \overline{v}^x}{\partial y} = 0, \quad \overline{w}^y + \frac{\partial \overline{u}^y}{\partial x} = 0, \quad (8.173)$$

with the overbar denoting a zonal or meridional average, as indicated. These results are to be expected from the mass continuity equation, $w = -(\partial_x u + \partial_y v)$ on zonal and meridional integration, respectively, but the fact that the solutions show it so explicitly is a demonstration of the karma of mathematics.

A streamfunction may be constructed by supposing that, in a fluid of depth H , the horizontal and vertical velocities vary as

$$(u, v) = (\tilde{u}, \tilde{v}) \cos(\pi z/H), \quad w = \tilde{w} \sin(\pi z/H). \quad (8.174a,b)$$

Using (8.170) the streamfunction in the meridional plane, Ψ_M is given by

$$\Psi_M(y, z) = \frac{IH}{\pi} \frac{-y}{3} \exp(-y^2/4) \sin \pi z/H. \quad (8.175)$$

Using (8.172) the streamfunction in the zonal plane, Ψ_Z , is given by

$$\Psi_Z(x, z) = \frac{\sqrt{\pi}H}{\pi} [q_0(x) - q_2(x)] \sin \pi z/H. \quad (8.176)$$

Notes

- 1 Drawing from unpublished lecture notes of M. Hendershott in Chapman *et al.* (1989). For more, see Paldor *et al.* (2007) and Heifetz & Caballero (2014).
- 2 Waves of this type were deduced by Bjerknes (1937).
- 3 The first complete treatment of this problem seems to have been given by Matsuno (1966), with special cases to be found in Stern (1963) and Bretherton (1964). An analysis of the rotating linear shallow water equations on the sphere (as opposed to β -plane) was given by Longuet-Higgins (1968) with Paldor & Sigalov (2011) providing an extension to any rotating, smooth surface. A review of equatorial waves in an oceanic context provided by McCreary (1985).
- 4 Standard forms are in the eye of the beholder.
- 5 After M. Yanai, in particular Yanai & Maruyama (1966).
- 6 A semi-implicit, semi-Lagrangian scheme. I am grateful to James Penn for performing the simulation.
- 7 Verkley & van der Velde (2010).
- 8 For more on this type of problem see Gill & Clarke (1974).
- 9 This problem was considered by Matsuno (1966) and revisited by Gill (1980) in the context of understanding the response of the tropical atmosphere to diabatic heating. It is now commonly referred to as the *Matsuno–Gill* problem. The treatment given here is similar to that of Gill.
- 10 A code that solves the elliptic problem using Fourier transforms and a tridiagonal inversion was graciously provided by Chris Bretherton and Adam Sobel. Timestepping can also be a particularly simple way to obtain some solutions, as pointed out to me by Matthew Barlow. Both methods, as well as analytic solutions as a check, were used to obtain solutions.
- 11 For more information about Hermite polynomials and parabolic cylinder functions see, for example, Jeffreys & Jeffreys (1946), Abramowitz & Stegun (1965) or mathematical software such as Maple or Python.

Come with rain, O loud Southwester!
Bring the singer, bring the nester;
Give the buried flower a dream;
Make the settled snowbank steam.
Robert Frost, *To the Thawing Wind* 1915

CHAPTER 9

Barotropic and Baroclinic Instability

WHAT HYDRODYNAMIC STATES OCCUR IN NATURE? If we take as given the applicability of the Navier–Stokes equations, any flow must be a solution of these equations, subject to the relevant initial and boundary conditions. Most of the flows we experience are of course *time-dependent* solutions, not steady solutions. Why should this be? There are, in fact, many steady solutions to the equations of motion — certain purely zonal flows, for example. However, steady solutions do not abound in nature because, in order to persist, they must be stable to those small perturbations that inevitably arise. Indeed, all the steady solutions that are known for the large-scale flow in the Earth’s atmosphere and ocean have been found to be unstable. It is such instability that makes the subject an interesting one.

There are a myriad forms of hydrodynamic instability, but our focus in this chapter is on barotropic and baroclinic instability. *Baroclinic instability* (and we will define the term more precisely later on) is an instability that arises in rotating, stratified fluids that are subject to a horizontal temperature gradient. It is the instability that gives rise to the large- and mesoscale motion in the atmosphere and ocean — it produces atmospheric weather systems, for example — and so is, perhaps, the form of hydrodynamic instability that most affects the human condition. *Barotropic instability* is an instability that arises because of the shear in a flow, and may occur in fluids of constant density. It is important to us for two reasons: first, it is important in its own right as an instability mechanisms for jets and vortices, and as an important process in both two- and three-dimensional turbulence; second, many problems in barotropic and baroclinic instability are formally and dynamically similar, so that the solutions and insight we obtain in the often simpler problems in barotropic instability may be useful in the baroclinic problem.

9.1 KELVIN–HELMHOLTZ INSTABILITY

To introduce the issue, we will first consider, rather informally, perhaps the simplest physically interesting instance of a fluid-dynamical instability — that of a constant-density flow with a shear perpendicular to the fluid’s mean velocity, this being an example of a *Kelvin–Helmholtz instability*,¹ and of a barotropic instability. (More generally, Kelvin–Helmholtz instability can involve fluid with varying density.) Let us consider two fluid masses of equal density, with a common interface at $y = 0$, moving with velocities $-U$ and $+U$ in the x -direction, respectively (Fig. 9.1). There is no variation in the basic flow in the z -direction (normal to the page), and we will assume this is also true for the

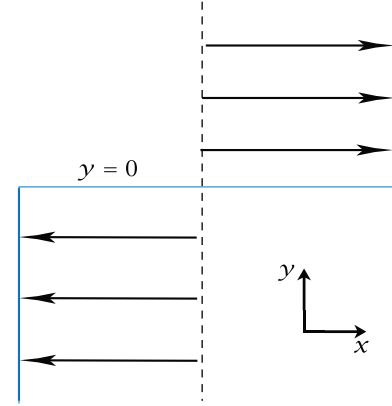


Fig. 9.1 A simple basic state giving rise to shear-flow instability. The velocity profile is discontinuous and the density is uniform.

instability (these restrictions are not essential). This flow is clearly a solution of the Euler equations. What happens if the flow is perturbed slightly? If the perturbation is initially small then even if it grows we can, for small times after the onset of instability, neglect the nonlinear interactions in the governing equations because these are the squares of small quantities. The equations determining the evolution of the initial perturbation are then the Euler equations linearized about the steady solution. Thus, denoting perturbation quantities with a prime and basic state variables with capital letters, for $y > 0$ the perturbation satisfies

$$\frac{\partial \mathbf{u}'}{\partial t} + U \frac{\partial \mathbf{u}'}{\partial x} = -\nabla p', \quad \nabla \cdot \mathbf{u}' = 0 \quad (9.1a,b)$$

and a similar equation holds for $y < 0$, but with U replaced by $-U$. Given periodic boundary conditions in the x -direction, we may seek solutions of the form

$$\phi'(x, y, t) = \text{Re} \sum_k \tilde{\phi}_k(y) \exp[ik(x - ct)], \quad (9.2)$$

where ϕ is any field variable (e.g., pressure or velocity), and Re denotes that only the real part should be taken. (Typically we use tildes over variables to denote Fourier-like modes, and we will often omit the marker 'Re'.) Because (9.1a) is linear, the Fourier modes do not interact and we may confine attention to just one. Taking the divergence of (9.1a), the left-hand side vanishes and the pressure satisfies Laplace's equation

$$\nabla^2 p' = 0. \quad (9.3)$$

This has solutions in the form

$$p' = \begin{cases} \text{Re} \tilde{p}_1 e^{ikx - ky} e^{\sigma t} & y > 0, \\ \text{Re} \tilde{p}_2 e^{ikx + ky} e^{\sigma t} & y < 0, \end{cases} \quad (9.4)$$

where, anticipating the possibility of growing solutions, we have written the time variation in terms of a growth rate, $\sigma = -ikc$. In general σ is complex: if it has a positive real component, the amplitude of the perturbation will grow and there is an instability; if σ has a non-zero imaginary component, then there will be oscillatory motion, and there may of course be both oscillatory motion *and* an instability. To obtain the dispersion relationship, we consider the y -component of (9.1a), namely (for $y > 0$)

$$\frac{\partial v'_1}{\partial t} + U \frac{\partial v'_1}{\partial x} = -\frac{\partial p'_1}{\partial y}. \quad (9.5)$$

Substituting a solution of the form $v'_1 = \tilde{v}_1 \exp(ikx + \sigma t)$ yields, with (9.4),

$$(\sigma + ikU)\tilde{v}_1 = k\tilde{p}_1. \quad (9.6)$$

But the velocity normal to the interface is, at the interface, nothing but the rate of change of the position of the interface itself; that is, at $y = +0$

$$v_1 = \frac{\partial \eta'}{\partial t} + U \frac{\partial \eta'}{\partial x}, \quad (9.7)$$

or

$$\tilde{v}_1 = (\sigma + ikU)\tilde{\eta}. \quad (9.8)$$

where η' is the displacement of the interface from its equilibrium position. Using this in (9.6) gives

$$(\sigma + ikU)^2 \tilde{\eta} = k\tilde{p}_1. \quad (9.9)$$

The above few equations pertain to motion on the $y > 0$ side of the interface. Similar reasoning on the other side gives (at $y = -0$)

$$(\sigma - ikU)^2 \tilde{\eta} = -k\tilde{p}_2. \quad (9.10)$$

But at the interface $p_1 = p_2$ (because pressure must be continuous). The dispersion relationship then emerges from (9.9) and (9.10), giving

$$\sigma^2 = k^2 U^2. \quad (9.11)$$

This equation has two roots, one of which is positive. Thus, the amplitude of the perturbation grows exponentially, like $e^{\sigma t}$, and the flow is *unstable*. The instability itself can be seen in the natural world when billow clouds appear wrapped up into spirals: the clouds are acting as tracers of fluid flow, and are a manifestation of the instability at finite amplitude, as seen later in Fig. 9.6.

9.2 INSTABILITY OF PARALLEL SHEAR FLOW

We now consider a little more systematically the instability of *parallel shear flows*, such as are illustrated in Fig. 9.2.² This is a classic problem in hydrodynamic stability theory, and there are two particular reasons for our own interest.

- (i) The instability is an example of *barotropic instability*, which abounds in the ocean and atmosphere. Loosely, barotropic instability arises when a flow is unstable by virtue of its horizontal shear, with gravitational and buoyancy effects being secondary.
- (ii) The instability is in many ways analogous to *baroclinic instability*, which is the main instability giving rise to weather systems in the atmosphere and similar phenomena in the ocean.

We will restrict attention to two-dimensional, incompressible flow; this illustrates the physical mechanisms in the most transparent way, in part because it allows for the introduction of a stream-function and the automatic satisfaction of the mass continuity equation. In fact, for parallel two-dimensional shear flows the most unstable disturbances are two-dimensional ones.³

The vorticity equation for incompressible two-dimensional flow is just

$$\frac{D\zeta}{Dt} = 0. \quad (9.12)$$

We suppose the basic state to be a parallel flow in the x -direction that may vary in the y -direction. That is

$$\bar{\mathbf{u}} = U(y)\mathbf{i}. \quad (9.13)$$

The linearized vorticity equation is then

$$\frac{\partial \zeta'}{\partial t} + U \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \mathcal{Z}}{\partial y} = 0 \quad (9.14)$$

where $\mathcal{Z} = -\partial_y U$. Because the mass continuity equation has the simple form $\partial u' / \partial x + \partial v' / \partial y = 0$, we may introduce a streamfunction ψ such that $u' = -\partial \psi' / \partial y$, $v' = \partial \psi' / \partial x$ and $\zeta' = \nabla^2 \psi'$. The linear vorticity equation becomes

$$\frac{\partial \nabla^2 \psi'}{\partial t} + U \frac{\partial \nabla^2 \psi'}{\partial x} + \frac{\partial \mathcal{Z}}{\partial y} \frac{\partial \psi'}{\partial x} = 0. \quad (9.15)$$

The coefficients of the x -derivatives are not themselves functions of x ; thus, we may seek solutions that are harmonic functions (sines and cosines) in the x -direction, but the y dependence must remain arbitrary at this stage and we write

$$\psi' = \text{Re } \tilde{\psi}(y) e^{ik(x-ct)}. \quad (9.16)$$

The full solution is a superposition of all wavenumbers, but since the problem is linear the waves do not interact and it suffices to consider them separately. If c is purely real then c is the phase speed of the wave; if c has a positive imaginary component then the wave will grow exponentially and is thus *unstable*.

From (9.16) we have

$$u' = \tilde{u}(y) e^{ik(x-ct)} = -\tilde{\psi}_y e^{ik(x-ct)}, \quad (9.17a)$$

$$v' = \tilde{v}(y) e^{ik(x-ct)} = ik \tilde{\psi} e^{ik(x-ct)}, \quad (9.17b)$$

$$\zeta' = \tilde{\zeta}(y) e^{ik(x-ct)} = (-k^2 \tilde{\psi} + \tilde{\psi}_{yy}) e^{ik(x-ct)}, \quad (9.17c)$$

where the y subscript denotes a derivative. Using (9.17) in (9.14) gives

$$(U - c)(\tilde{\psi}_{yy} - k^2 \tilde{\psi}) - U_{yy} \tilde{\psi} = 0, \quad (9.18)$$

which is known as *Rayleigh's equation*.⁴ It is the linear vorticity equation for disturbances to parallel shear flow, and in the presence of a β -effect it generalizes slightly to

$$(U - c)(\tilde{\psi}_{yy} - k^2 \tilde{\psi}) + (\beta - U_{yy}) \tilde{\psi} = 0. \quad (9.19)$$

9.2.1 Piecewise linear flows

Although Rayleigh's equation is linear and has a simple form, it is nevertheless quite difficult to analytically solve for an arbitrary smoothly varying profile. It is simpler to consider *piecewise linear* flows, in which U_y is a constant over some interval, with U or U_y changing abruptly to another value at a line of discontinuity, as illustrated in Fig. 9.2. The curvature, U_{yy} , is accounted for through the satisfaction of matching conditions, analogous to boundary conditions, at the lines of discontinuity (as in section 9.1), and solutions in each interval are then exponential functions.

Jump or matching conditions

The idea, then, is to solve the linearized vorticity equation separately in the continuous intervals in which vorticity is constant, matching the solution with that in the adjacent regions. The matching conditions arise from two physical conditions:

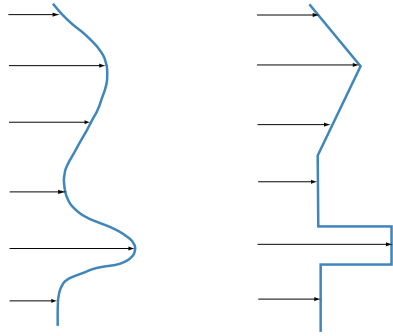


Fig. 9.2 Left: example of a smooth velocity profile, in which both the velocity and the vorticity are continuous. Right: example piecewise continuous profile, in which the velocity and vorticity may have finite discontinuities.

- (i) That normal stress should be continuous across the interface. For an inviscid fluid this implies that pressure be continuous.
- (ii) That the normal velocity of the fluid on either side of the interface should be consistent with the motion of the interface itself.

Let us consider the implications of these two conditions.

(i) *Continuity of pressure*

The linearized momentum equation in the direction along the interface is:

$$\frac{\partial u'}{\partial t} + U \frac{\partial u'}{\partial x} + v' \frac{\partial U}{\partial y} = -\frac{\partial p'}{\partial x}. \quad (9.20)$$

For normal modes, $u' = -\tilde{\psi}_y e^{ik(x-ct)}$, $v' = ik\tilde{\psi} e^{ik(x-ct)}$ and $p' = \tilde{p} e^{ik(x-ct)}$, and (9.20) becomes

$$ik(U - c)\tilde{\psi}_y - ik\tilde{\psi}U_y = -ik\tilde{p}. \quad (9.21)$$

Because pressure is continuous across the interface we have the first *matching or jump condition*,

$$\Delta[(U - c)\tilde{\psi}_y - \tilde{\psi}U_y] = 0, \quad (9.22)$$

where the operator Δ denotes the difference in the values of the argument (in square brackets) across the interface. That is, the quantity $(U - c)\tilde{\psi}_y - \tilde{\psi}U_y$ is continuous.

We can obtain this condition directly from Rayleigh's equation, (9.19), written in the form

$$[(U - c)\tilde{\psi}_y - U_y\tilde{\psi}]_y + [\beta - k^2(U - c)]\tilde{\psi} = 0. \quad (9.23)$$

Integrating across the interface gives (9.22).

(ii) *Material interface condition*

At the interface, the normal velocity v is given by the kinematic condition

$$v = \frac{D\eta}{Dt} \quad (9.24)$$

where η is the interface displacement. The linear version of (9.24) is

$$\frac{\partial \eta'}{\partial t} + U \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'}{\partial x}. \quad (9.25)$$

If the fluid itself is continuous then this equation must hold at either side of the interface, giving two equations and their normal-mode counterparts, namely,

$$\frac{\partial \eta'}{\partial t} + U_1 \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'_1}{\partial x} \longrightarrow (U_1 - c)\tilde{\eta} = \tilde{\psi}_1, \quad (9.26)$$

$$\frac{\partial \eta'}{\partial t} + U_2 \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'_2}{\partial x} \longrightarrow (U_2 - c)\tilde{\eta} = \tilde{\psi}_2. \quad (9.27)$$

Material continuity at the interface thus gives the second jump condition

$$\Delta \left[\frac{\tilde{\psi}}{U - c} \right] = 0. \quad (9.28)$$

That is, $\tilde{\psi}/(U - c)$ is continuous at the interface. Note that if U is continuous across the interface the condition becomes one of continuity of the normal velocity.

9.2.2 Kelvin–Helmholtz instability, revisited

We now use Rayleigh's equation and the jump conditions to consider the situation illustrated in Fig. 9.1; that is, vorticity is everywhere zero except in a thin sheet at $y = 0$. On either side of the interface, Rayleigh's equation is simply

$$(U_i - c)(\partial_{yy}\tilde{\psi}_i - k^2\tilde{\psi}_i) = 0 \quad i = 1, 2 \quad (9.29)$$

or, assuming that $U_i \neq c$, $\partial_{yy}\tilde{\psi}_i - k^2\tilde{\psi}_i = 0$. (This is just Laplace's equation, coming from $\nabla^2\psi' = \zeta'$, with $\zeta' = 0$ everywhere except at the interface.) Solutions of this that decay away on either side of the interface are

$$y > 0 : \quad \tilde{\psi}_1 = \Psi_1 e^{-ky}, \quad (9.30a)$$

$$y < 0 : \quad \tilde{\psi}_2 = \Psi_2 e^{ky}, \quad (9.30b)$$

where Ψ_1 and Ψ_2 are constants. The boundary condition (9.22) gives

$$(U_1 - c)(-k)\Psi_1 = (U_2 - c)(k)\Psi_2, \quad (9.31)$$

and (9.28) gives

$$\frac{\Psi_1}{(U_1 - c)} = \frac{\Psi_2}{(U_2 - c)}. \quad (9.32)$$

The last two equations combine to give $(U_1 - c)^2 = -(U_2 - c)^2$, which, supposing that $U = U_1 = -U_2$ gives $c^2 = -U^2$. Thus, since U is purely real, $c = \pm iU$, and the disturbance grows exponentially as $\exp(kU_1 t)$, just as we obtained in section 9.1. All wavelengths are unstable, and indeed the shorter the wavelength the greater the instability. In reality, viscosity will damp the smallest waves, although the presence of viscosity would also mean that the initial profile is not an exact, steady solution of the equations of motion.

9.2.3 Edge waves

We now consider a case sketched in Fig. 9.3 in which the velocity is continuous, but the vorticity is discontinuous. Since on either side of the interface $U_{yy} = 0$, Rayleigh's equation is just

$$(U(y) - c)(\tilde{\psi}_{yy} - k^2\tilde{\psi}) = 0. \quad (9.33)$$

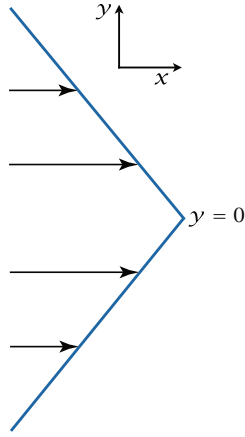


Fig. 9.3 Velocity profile of a point jet, in which vorticity is concentrated at a point. Although the vorticity is discontinuous, a small perturbation gives rise only to *edge waves* centred at $y = 0$, and so the jet is stable.

Provided $c \neq U$ this has solutions,

$$\tilde{\psi} = \begin{cases} \Phi_1 e^{-ky} & y > 0 \\ \Phi_2 e^{ky} & y < 0. \end{cases} \quad (9.34)$$

The value of c is found by applying the jump conditions (9.22) and (9.28) at $y = 0$. Using (9.34) these give

$$-k(U_0 - c)\Phi_1 - \Phi_1 \partial_y U_1 = k(U_0 - c)\Phi_2 - \Phi_2 \partial_y U_2 \quad (9.35a)$$

$$\Phi_1 = \Phi_2, \quad (9.35b)$$

where $U_1(y)$ and $U_2(y)$ the velocity either side of the interface, and both are equal to U_0 at the interface. (In Fig. 9.3 we illustrate the case with $U_1 = -Ay$ and $U_2 = Ay$, where A is a positive constant.) After a line of algebra the above equations give

$$c = U_0 + \frac{\partial_y U_1 - \partial_y U_2}{2k}. \quad (9.36)$$

This is the dispersion relationship for *edge waves* that propagate along the interface with speed equal to the sum of the fluid speed and a factor proportional to the difference in the vorticity between the two layers. No matter what the shear is on either side of the interface, the phase speed is purely real and there is no instability. Equation (9.36) is analogous to the Rossby wave dispersion relation $c = U_0 - \beta/K^2$, and reflects a similarity in the physics — β is a planetary vorticity gradient, which in (9.36) is collapsed to a front and represented by the difference $\partial_y U_1 - \partial_y U_2 = -(Z_1 - Z_2)$, where Z_1 and Z_2 are the basic-state vorticities on either side of the interface. One might even say that such edge waves *are* Rossby waves of a very simple form.

9.2.4 Interacting edge waves producing instability

Now we consider a slightly more complicated case in which edge waves may interact giving rise, as we shall see, to an instability. The physical situation is illustrated in Fig. 9.4. We consider the simplest case, that of a shear layer (which we denote as region 2) sandwiched between two semi-infinite layers, regions 1 and 3, as in the left-hand panel of the figure. Thus, the basic state is

$$y > a: \quad U = U_1 = U_0 \text{ (a constant)}, \quad (9.37a)$$

$$-a < y < a: \quad U = U_2 = \frac{U_0}{a} y, \quad (9.37b)$$

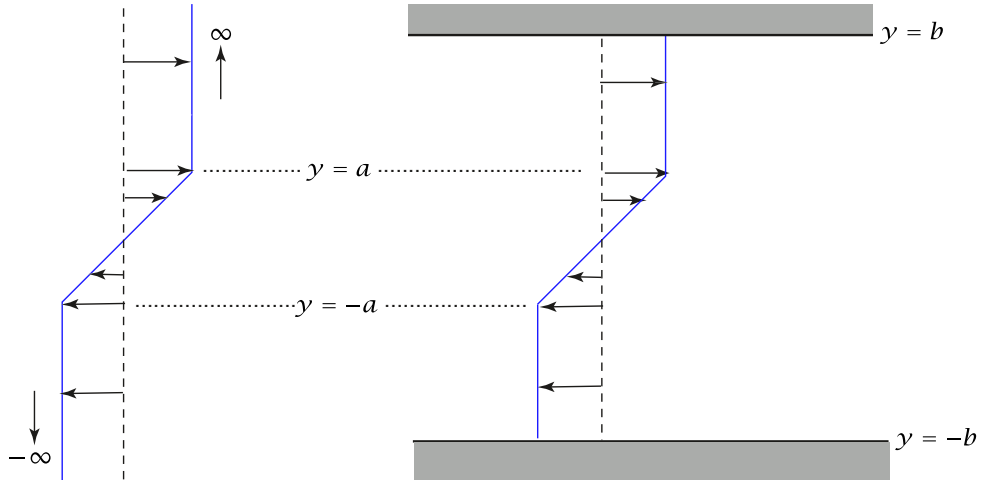


Fig. 9.4 Barotropically unstable velocity profiles. In the simplest case, on the left, a region of shear is sandwiched between two infinite regions of constant velocity. The edge waves at $y = \pm a$ interact to produce an instability. If $a = 0$, then the situation corresponds to that of Fig. 9.1, giving Kelvin–Helmholtz instability. In the case on the right, the flow is bounded at $y = \pm b$. The flow is unstable provided that b is sufficiently larger than a . If $b = a$ (plane Couette flow) the flow is stable to infinitesimal disturbances.

$$y < -a: \quad U = U_3 = -U_0. \quad (9.37c)$$

We assume a solution of Rayleigh's equation of the form:

$$y > a: \quad \tilde{\psi}_1 = A e^{-k(y-a)}, \quad (9.38a)$$

$$-a < y < a: \quad \tilde{\psi}_2 = B e^{k(y-a)} + C e^{-k(y+a)}, \quad (9.38b)$$

$$y < -a: \quad \tilde{\psi}_3 = D e^{k(y+a)}. \quad (9.38c)$$

These particular forms all decay away from the interfaces at the edges of domains in which the assumed solutions apply, as edge waves must do. Applying the jump conditions (9.22) and (9.28) at the interfaces at $y = a$ and $y = -a$ gives the following relations between the coefficients:

$$-A[(U_0 - c)k] = B \left[(U_0 - c)k - \frac{U_0}{a} \right] - C e^{-2ka} \left[\frac{U_0}{a} + (U_0 - c)k \right], \quad (9.39a)$$

$$A = B + C e^{-2ka}, \quad (9.39b)$$

$$D[(U_0 + c)k] = B e^{-2ka} \left[(U_0 + c)k + \frac{U_0}{a} \right] + C \left[\frac{U_0}{a} - (U_0 + c)k \right], \quad (9.39c)$$

$$D = B e^{-2ka} + C. \quad (9.39d)$$

These are a set of four homogeneous equations, with the unknown parameters A , B , C and D , which may be written in the form of a matrix equation,

$$\begin{pmatrix} k(U_0 - c) & k(U_0 - c) - U_0/a & -e^{-2ka}[U_0/a + k(U_0 - c)] & 0 \\ 1 & -1 & -e^{-2ka} & 0 \\ 0 & -e^{-2ka}[k(U_0 + c) + U_0/a] & k(U_0 + c) - (U_0/a) & k(U_0 + c) \\ 0 & e^{-2ka} & 1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0. \quad (9.40)$$

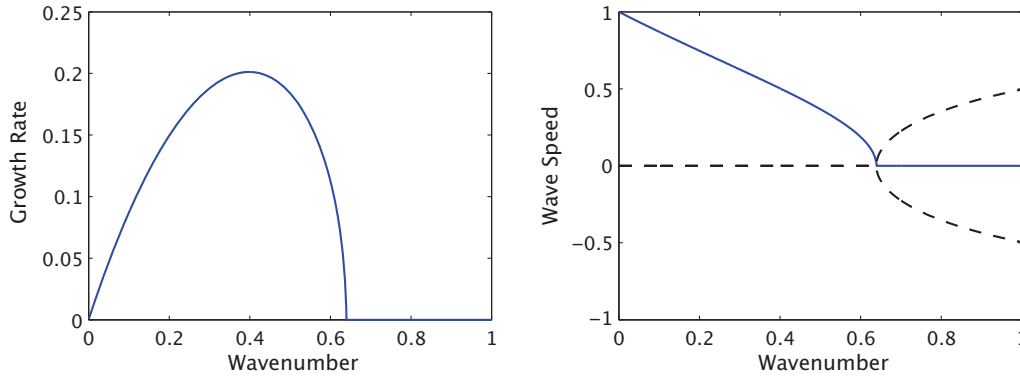


Fig. 9.5 Left: Growth rate ($\sigma = kc_i$) calculated from (9.41) with c non-dimensionalized by U_0 and k non-dimensionalized by $1/a$ (equivalent to setting $a = U_0 = 1$). Right: Real (c_r , dashed) and imaginary (c_i , solid) wave speeds. The flow is unstable for $k < 0.63$, with the maximum instability occurring at $k = 0.39$.

For non-trivial solutions the determinant of the matrix must be zero, and solving the ensuing equation gives, after some algebra, the dispersion relationship⁵

$$c^2 = \left(\frac{U_0}{2ka} \right)^2 \left[(1 - 2ka)^2 - e^{-4ka} \right], \quad (9.41)$$

and this is plotted in Fig. 9.5. The flow is unstable for sufficiently long wavelengths, since then the right-hand side of (9.41) is negative. The critical wavenumber below which instability occurs is found by solving $(1 - 2ka)^2 = e^{-4ka}$, which gives instability for $ka < 0.63293$. A numerical solution of the initial value problem is illustrated in Figs. 9.6 and 9.7. Here, the initial perturbation is small and random, containing components at all wavenumbers. All the modes in the unstable range grow exponentially, and the pattern is soon dominated by the mode that grows fastest — a horizontal wavenumber of 3 in this problem. Eventually, the perturbation grows sufficiently that the linear equations are no longer valid and, as is seen in the second column of Fig. 9.6, vortices form and pinch off. The vortices interact and the flow develops into two-dimensional turbulence, as considered in chapter 11.

The mechanism of the instability — an informal view

(A similar mechanism is discussed in section 9.7, and the reader may wish to read the two descriptions in tandem.) We have seen that an edge wave in isolation is stable, with the instability arising when two edge waves have sufficient cross-stream extent that they can interact with each other. This occurs for sufficiently long wavelengths because the cross-stream decay scale is proportional to the along-stream wavelength — hence the high-wavenumber cut-off. To see the mechanism of the instability transparently, let us first suppose that the interfaces are, in fact, sufficiently far away that the edge waves at each interface do not interact. Using (9.36) the edge waves at $y = -a$ and $y = +a$ have dispersion relationships

$$c_{+a} = U_0 - \frac{U_0/a}{2k}, \quad c_{-a} = -U_0 + \frac{U_0/a}{2k}. \quad (9.42a,b)$$

If the two waves are to interact these phase speeds must be equal, giving the condition

$$c = 0, \quad k = 1/(2a). \quad (9.43a,b)$$

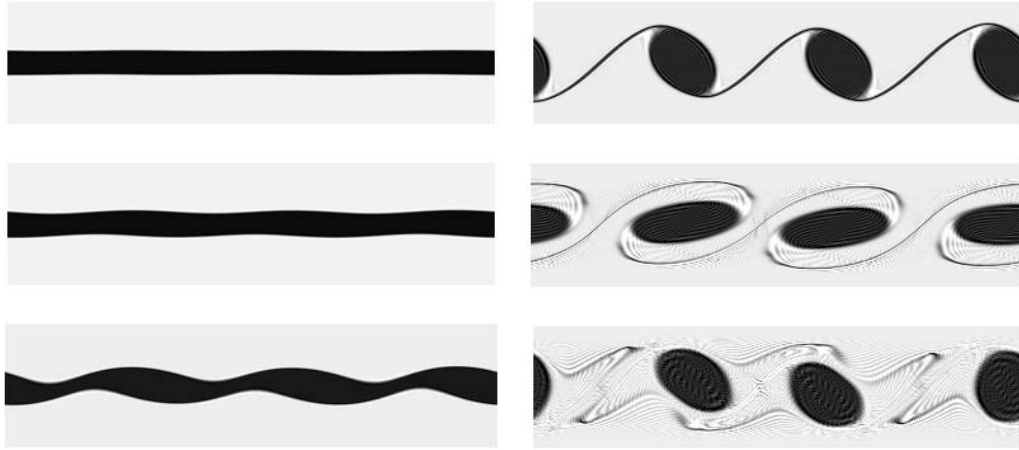


Fig. 9.6 A sequence of plots of the vorticity, at equal time intervals, from a numerical solution of the nonlinear vorticity equation (9.12), with initial conditions as in Fig. 9.4 with $a = 0.1$, plus a very small random perturbation. Time increases first down the left column and then down the right column. The solution is obtained in a rectangular (4×1) domain, with periodic conditions in the x -direction and slippery walls at $y = (0, 1)$. The maximum linear instability occurs for a wavelength of 1.57, which for a domain of length 4 corresponds to a wavenumber of 2.55. Since the periodic domain quantizes the allowable wavenumbers, the maximum instability is at wavenumber 3, and this is what emerges. Only in the first two or three frames is the linear approximation valid.

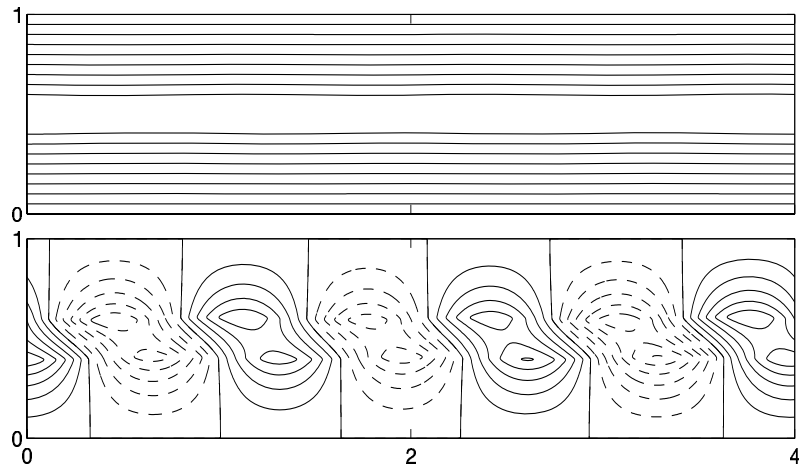


Fig. 9.7 The total streamfunction (top panel) and the perturbation streamfunction from the same numerical calculation as in Fig. 9.6, at a time corresponding to the second frame. Positive values are solid lines, and negative values are dashed. The perturbation pattern leans into the shear, and grows exponentially in place.

That is, the waves are stationary, and their wavelength is proportional to the separation of the two edges. In fact, (9.43) approximately characterizes the conditions at the critical wavenumber $k = 0.63/a$ (see Fig. 9.5). In the region of the shear the two waves have the form

$$\psi_{+a} = \text{Re } \tilde{\psi}_{+a}(t) e^{k(y-a)} e^{i\phi} e^{ikx}, \quad \psi_{-a} = \text{Re } \tilde{\psi}_{-a}(t) e^{-k(y+a)} e^{ikx} \quad (9.44a,b)$$

where ϕ is the phase shift between the waves; in the case of pure edge waves we have $\tilde{\psi}_{\pm a} = A_{\pm a} e^{-ikct}$ where we may take $A_{\pm a}$ to be real.

Now consider how the wave generated at $y = -a$ might affect the wave at $y = +a$ and vice versa. The contribution of ψ_{-a} to the acceleration of ψ_{+a} is given by applying the x -momentum equation, (9.20), at $y = +a$. Thus we take the kinematic solutions, (9.44), and use them in a dynamical equation, the momentum equation, to calculate the ensuing acceleration. At $y = +a$ we heuristically write

$$\frac{\partial u'_{+a}}{\partial t} = -[v'_{+a}(+a) + v'_{-a}(+a)] \frac{dU}{dy} \quad (9.45)$$

with a similar expression at $y = -a$, and omitting the pressure terms. Here $v'_{-a}(+a)$ denotes the value of v at $y = +a$ due to the edge wave generated at $-a$. It is the second term on the right-hand side that is necessary for any potential instability, as the first term gives only neutral edge waves. If the spatial dependence of the waves is given by (9.44), then at $y = +a$ we have, omitting the contribution from the edge waves,

$$\frac{\partial u'_{+a}}{\partial t} = -v'_{-a}(+a) \frac{dU}{dy} \quad \Rightarrow \quad -ke^{i\phi} e^{ikx} \frac{d\tilde{\psi}_{+a}}{dt} = -ike^{ikx} e^{-2ka} \frac{\partial U}{\partial y} \tilde{\psi}_{-a}. \quad (9.46)$$

The real part of this equation is: $\cos(kx + \phi) d\tilde{\psi}_{+a}/dt = -\sin kx e^{-2ka} dU/dy$. If the edge waves have the appropriate phase with respect to each other then the two edge waves can feed back on each other and couple to form a single growing mode. In particular if $\phi = \pi/2$ then the evolution equations for ψ_{+a} and ψ_{-a} become

$$\frac{d\tilde{\psi}_{+a}}{dt} = e^{-2ka} \frac{dU}{dy} \tilde{\psi}_{-a}, \quad \frac{d\tilde{\psi}_{-a}}{dt} = e^{-2ka} \frac{dU}{dy} \tilde{\psi}_{+a}, \quad (9.47a,b)$$

which give exponential growth. When $\phi = \pi/2$ the wave at $y = +a$ lags the wave at $y = -a$, and *the unstable perturbation tilts into the shear*; this property of the instability is seen in the full solution, Fig. 9.7.

9.3 NECESSARY CONDITIONS FOR INSTABILITY

9.3.1 Rayleigh's criterion

For simple profiles it may be possible to calculate, or even intuit, the instability properties, but for continuous profiles of $U(y)$ this is often impossible and it would be nice to have some general guidelines as to when a profile might be unstable. To this end, we will derive a couple of *necessary* conditions for instability, or *sufficient* conditions for stability, that will at least tell us if a flow *might* be unstable. We begin by writing Rayleigh's equation, (9.19), as

$$\tilde{\psi}_{yy} - k^2 \tilde{\psi} + \frac{\beta - U_{yy}}{U - c} \tilde{\psi} = 0. \quad (9.48)$$

Multiply by $\tilde{\psi}^*$ (the complex conjugate of $\tilde{\psi}$) and integrate over the domain of interest. After integrating the first term by parts we obtain

$$\int_{y_1}^{y_2} \left(\left| \frac{\partial \tilde{\psi}}{\partial y} \right|^2 + k^2 |\tilde{\psi}|^2 \right) dy - \int_{y_1}^{y_2} \frac{\beta - U_{yy}}{U - c} |\tilde{\psi}|^2 dy = 0, \quad (9.49)$$

assuming that $\tilde{\psi}$ vanishes at the boundaries. (The limits to the integral may be infinite, in which case it is assumed that $\tilde{\psi}$ decays to zero as $|y|$ approaches ∞ .) The only variable in this expression that is complex is c , and thus the first integral is real. The imaginary component of the second integral is

$$c_i \int \frac{\beta - U_{yy}}{|U - c|^2} |\tilde{\psi}|^2 dy = 0. \quad (9.50)$$

Thus, *either* c_i vanishes or the integral does. For there to be an instability c_i must be non-zero. [Note also that the eigenvalues of Rayleigh's equation come in pairs, for each decaying mode (negative c_i) there is a corresponding growing mode (positive c_i).] If c_i is to be non-zero, the integrand of (9.50) must be zero and therefore

A necessary condition for instability is that the expression

$$\beta - U_{yy}$$

change sign somewhere in the domain.

Equivalently, a sufficient criterion for stability is that $\beta - U_{yy}$ not vanish in the domain interior. This condition is known as Rayleigh's inflection-point criterion, or when $\beta \neq 0$, the Rayleigh–Kuo inflection point criterion.⁶

An alternate, more general, derivation

Consider again the vorticity equation, linearized about a parallel shear flow [cf. (9.14) with a β -term],

$$\frac{\partial \zeta}{\partial t} + U \frac{\partial \zeta}{\partial x} + v \left(\frac{\partial Z}{\partial y} + \beta \right) = 0, \quad (9.51)$$

(dropping the primes on the perturbation quantities). Multiply by ζ and divide by $\beta + Z_y$ to obtain

$$\frac{\partial}{\partial t} \left(\frac{\zeta^2}{\beta + Z_y} \right) + \frac{U}{\beta + Z_y} \frac{\partial \zeta^2}{\partial x} + 2v\zeta = 0, \quad (9.52)$$

and then integrate with respect to x to give

$$\frac{\partial}{\partial t} \int \left(\frac{\zeta^2}{\beta + Z_y} \right) dx = -2 \int v\zeta dx. \quad (9.53)$$

Now, using $\nabla \cdot \mathbf{u} = 0$, the vorticity flux may be written as

$$v\zeta = -\frac{\partial}{\partial y}(uv) + \frac{1}{2} \frac{\partial}{\partial x}(v^2 - u^2). \quad (9.54)$$

That is, *the flux of vorticity is the divergence of some quantity*. Its integral therefore vanishes provided there are no contributions from the boundary, and integrating (9.53) with respect to y gives

$$\frac{d}{dt} \int \left(\frac{\zeta^2}{\beta + Z_y} \right) dx dy = 0. \quad (9.55)$$

If there is to be an instability ζ must grow, but the integral is identically zero. These two conditions can only be simultaneously satisfied if $\beta + Z_y$, or equivalently $\beta - U_{yy}$, is zero somewhere in the domain.

This derivation shows that the inflection-point criterion applies even if disturbances are not of normal-mode form. The quantity $\zeta^2/(\beta + Z_y)$ is an example of a *wave-activity density* — a wave activity being a conserved quantity, quadratic in the amplitude of the wave. Such quantities play an important role in instabilities, and we consider then further in chapter 10.

9.3.2 Fjortoft's criterion

Another necessary condition for instability was obtained by Fjortoft.⁷ In this section we will derive his condition for normal-mode disturbances, and provide a more general derivation in section 10.7. From the real part of (9.49) we find

$$\int_{y_1}^{y_2} (\beta - U_{yy}) \frac{(U - c_r)}{|U - c|^2} |\tilde{\psi}|^2 dy = \int_{y_1}^{y_2} \left| \frac{\partial \tilde{\psi}}{\partial y} \right|^2 + k^2 |\tilde{\psi}|^2 dy > 0. \quad (9.56)$$

Now, from (9.50), we know that for an instability we must have

$$\int_{y_1}^{y_2} \frac{\beta - U_{yy}}{|U - c|^2} |\tilde{\psi}|^2 dy = 0. \quad (9.57)$$

Using this equation and (9.56) we see that, for an instability,

$$\int_{y_1}^{y_2} (\beta - U_{yy}) \frac{(U - U_s)}{|U - c|^2} |\tilde{\psi}|^2 dy > 0 \quad (9.58)$$

where U_s is *any* real constant. It is most useful to choose this constant to be the value of $U(y)$ at which $\beta - U_{yy}$ vanishes. This leads directly to the criterion

A necessary condition for instability is that the expression

$$(\beta - U_{yy})(U - U_s)$$

where U_s is the value of $U(y)$ at which $\beta - U_{yy}$ vanishes, be positive somewhere in the domain.

This criterion is satisfied if the magnitude of the vorticity has an extremum inside the domain, and not at the boundary or at infinity (Fig. 9.8). Why choose U_s in the manner we did? Suppose we chose U_s to have a large positive value, so that $U - U_s$ is negative everywhere. Then (9.58) just implies that $\beta - U_{yy}$ must be negative somewhere, and this is already known from Rayleigh's criterion. If we choose U_s to be large and negative, we simply find that $\beta - U_{yy}$ must be positive somewhere. The most stringent criterion is obtained by choosing U_s to be the value of $U(y)$ at which $\beta - U_{yy}$ vanishes.

Interestingly, the β -effect can be either stabilizing or destabilizing: It can stabilize the middle two profiles of Fig. 9.8, because if it is large enough $\beta - U_{yy}$ will be one-signed. However, the β -effect will destabilize a westward point jet, $U(y) = -(1 - |y|)$ (the negative of the jet in Fig. 9.3), because $\beta - U_{yy}$ is negative at $y = 0$ and positive elsewhere. An eastward point jet is stable, with or without β . Finally, we emphasize that both Fjortoft's and Rayleigh's criteria are necessary conditions for instability, and examples exist that do satisfy their criterion, yet which are stable to infinitesimal perturbations.

9.4 BAROCLINIC INSTABILITY

Baroclinic instability is a hydrodynamic instability that occurs in stably stratified, rotating fluids, and it is ubiquitous in the Earth's atmosphere and oceans, and almost certainly occurs in other planetary atmospheres. It gives rise to weather, and so is perhaps the form of hydrodynamic instability that affects us most, and that certainly we talk about most.

9.4.1 A physical picture

We will first draw a picture of baroclinic instability as a form of 'sloping convection' in which the fluid, although statically stable, is able to release available potential energy when parcels move along

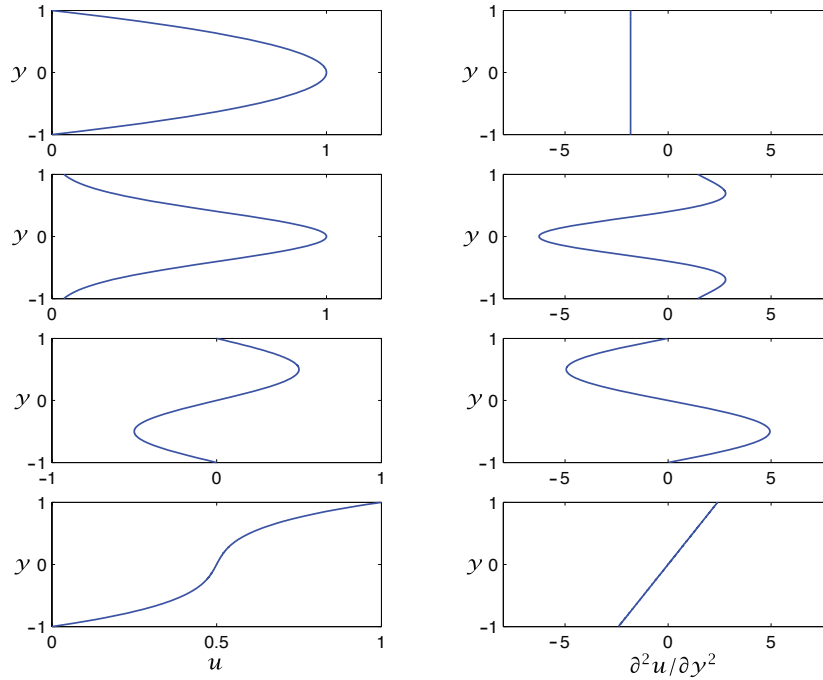


Fig. 9.8 Example parallel velocity profiles (left column) and their second derivatives (right column). From the top: Poiseuille flow ($u = 1 - y^2$); a Gaussian jet; a sinusoidal profile; a polynomial profile. By Rayleigh's criterion, the top profile is stable, whereas the lower three are potentially unstable. However, the bottom profile is stable by Fjørtoft's criterion (note that the vorticity maxima are at the boundaries). If the β -effect were present and large enough it would stabilize the middle two profiles.

a sloping path. To this end, let us first ask: what is the basic state that is baroclinically unstable? In a stably stratified fluid potential density decreases with height; we can also easily imagine a state in which the basic state temperature decreases, and the potential density increases, polewards. (We will couch most of our discussion in terms of the Boussinesq equations, and henceforth drop the qualifier 'potential' from density.) Can we construct a steady solution from these two conditions? The answer is yes, provided the fluid is also rotating; rotation is necessary because the meridional temperature gradient generally implies a meridional pressure gradient; there is nothing to balance this in the absence of rotation, and a fluid parcel would therefore accelerate. In a rotating fluid this pressure gradient can be balanced by the Coriolis force and a steady solution can be maintained even in the absence of viscosity. Consider a stably stratified Boussinesq fluid in geostrophic and hydrostatic balance on an f -plane, with buoyancy decreasing uniformly polewards. Then $fu = -\partial\phi/\partial y$ and $\partial\phi/\partial z = b$, where $b = -g\delta\rho/\rho_0$ is the buoyancy. These together give the thermal wind relation, $\partial u/\partial z = \partial b/\partial y$. If there is no variation of these fields in the zonal direction, then, for *any* variation of b with y , this is a steady solution to the primitive equations of motion, with $v = w = 0$.

The density structure corresponding to a uniform increase of density in the meridional direction is illustrated in Fig. 9.9. Is this structure stable to perturbations? The answer is no, although the perturbations must be a little special. Suppose the particle at 'A' is displaced upwards; then, since the fluid is (by assumption) stably stratified it will be denser than its surroundings and hence experience a restoring force, and similarly if displaced downwards. Suppose, however, we interchange the two parcels at positions 'A' and 'B'. Parcel 'A' finds itself surrounded by parcels of higher

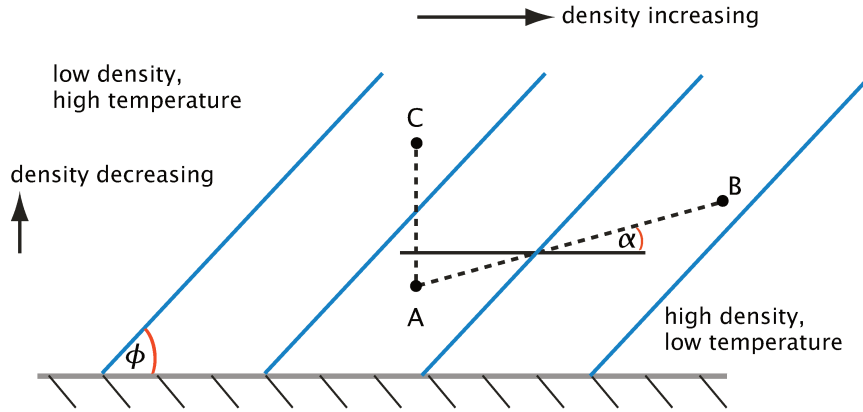


Fig. 9.9 A steady basic state giving rise to baroclinic instability. Potential density decreases upwards and equatorwards, and the associated horizontal pressure gradient is balanced by the Coriolis force. Parcel 'A' is heavier than 'C', and so statically stable, but it is lighter than 'B'. Hence, if 'A' and 'B' are interchanged there is a release of potential energy.

density that itself, and it is therefore buoyant; it is also higher than where it started. Parcel 'B' is negatively buoyant, and at a lower altitude than where it started. Thus, overall, the centre of gravity of the fluid has been lowered, and so its overall potential energy lowered. This loss in potential energy of the basic state must be accompanied by a gain in kinetic energy of the perturbation. Thus, the perturbation amplifies and converts potential energy into kinetic energy.

The loss of potential energy is easily calculated. Since

$$PE = \int \rho g dz \quad (9.59)$$

the change in potential energy due to the interchange is

$$\Delta PE = g(\rho_A z_A + \rho_B z_B - \rho_A z_B - \rho_B z_A) = g(z_A - z_B)(\rho_A - \rho_B) = g \Delta \rho \Delta z. \quad (9.60)$$

If both $\rho_B > \rho_A$ and $z_B > z_A$ then the initial potential energy is larger than the final one, energy is released and the state is unstable. If the slope of the isopycnals is ϕ [so that $\phi = -(\partial_y \rho)/(\partial_z \rho)$] and the slope of the displacements is α , then for a displacement of horizontal distance L the change in potential energy is given by

$$\Delta PE = g \Delta \rho \Delta z = g \left(L \frac{\partial \rho}{\partial y} + L \alpha \frac{\partial \rho}{\partial z} \right) \alpha L = g L^2 \alpha \frac{\partial \rho}{\partial y} \left(1 - \frac{\alpha}{\phi} \right), \quad (9.61)$$

if α and ϕ are small. If $0 < \alpha < \phi$ then energy is released by the perturbation, and it is maximized when $\alpha = \phi/2$. For the atmosphere the actual slope of the isotherms is about 10^{-3} , so that the slope and potential parcel trajectories are indeed shallow.

Although intuitively appealing, the thermodynamic arguments presented in this section pay no attention to satisfying the dynamical constraints of the equations of motion, and we now turn our attention to that.

9.4.2 Linearized quasi-geostrophic equations

To explore the dynamics of baroclinic instability we use the quasi-geostrophic equations, specifically a potential vorticity equation for the fluid interior and a buoyancy or temperature equation

at two vertical boundaries, one representing the ground and the other the tropopause. (The tropopause is the boundary between the troposphere and stratosphere at about 10 km; it is not a true rigid surface, but the higher static stability of the stratosphere inhibits vertical motion. We discuss the effects of that in section 9.9.) For a Boussinesq fluid the equations are

$$\begin{aligned}\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q &= 0, & 0 < z < H, \\ q &= \nabla^2 \psi + \beta y + \frac{\partial}{\partial z} \left(F \frac{\partial \psi}{\partial z} \right),\end{aligned}\tag{9.62}$$

where $F = f_0^2/N^2$, and the buoyancy equation with $w = 0$,

$$\begin{aligned}\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b &= 0, & z = 0, H, \\ b &= f_0 \frac{\partial \psi}{\partial z}.\end{aligned}\tag{9.63}$$

A solution of these equations is a purely zonal flow, $\mathbf{u} = U(y, z)\mathbf{i}$ with a corresponding temperature field given by thermal wind balance. The potential vorticity of this basic state is

$$Q = \beta y - \frac{\partial U}{\partial y} + \frac{\partial}{\partial z} F \frac{\partial \Psi}{\partial z} = \beta y + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial}{\partial z} F \frac{\partial \Psi}{\partial z},\tag{9.64}$$

where Ψ is the streamfunction of the basic state, related to U by $U = -\partial \Psi / \partial y$. Linearizing (9.62) about this zonal flow gives the potential vorticity equation for the interior,

$$\frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial Q}{\partial y} = 0, \quad 0 < z < H,\tag{9.65}$$

where $q' = \nabla^2 \psi' + \partial_z (F \partial_z \psi')$ and $v' = \partial_x \psi'$. Similarly, the linearized buoyancy equation at the boundary is

$$\frac{\partial b'}{\partial t} + U \frac{\partial b'}{\partial x} + v' \frac{\partial B}{\partial y} = 0, \quad z = 0, H,\tag{9.66}$$

where $b' = f_0 \partial_z \psi'$ and $\partial_y B = \partial_y (f_0 \partial_z \Psi) = -f_0 \partial U / \partial z$.

Just as for the barotropic problem, a standard way of proceeding is to seek normal-mode solutions. Since the coefficients of (9.65) and (9.66) are functions of y and z , but not of x , we seek solutions of the form

$$\psi'(x, y, z, t) = \text{Re } \tilde{\psi}(y, z) e^{ik(x-ct)},\tag{9.67}$$

and similarly for the derived quantities u' , v' , b' and q' . In particular,

$$\tilde{q} = \frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{\partial}{\partial z} F \frac{\partial \tilde{\psi}}{\partial z} - k^2 \tilde{\psi}.\tag{9.68}$$

Using (9.68) and (9.67) in (9.65) and (9.66) gives, with subscripts y and z denoting derivatives,

$$(U - c) (\tilde{\psi}_{yy} + (F \tilde{\psi}_z)_z - k^2 \tilde{\psi}) + Q_y \tilde{\psi} = 0 \quad 0 < z < H,\tag{9.69a}$$

$$(U - c) \tilde{\psi}_z - U_z \tilde{\psi} = 0 \quad z = 0, H.\tag{9.69b}$$

These equations are analogous to Rayleigh's equations for parallel shear flow, and illustrate the similarity between the baroclinic instability problem and that of a parallel shear flow.

9.4.3 Necessary conditions for baroclinic instability

Necessary conditions for instability may be obtained (as for parallel shear flows) by multiplying (9.69) by $\tilde{\psi}^*$ and integrating over the domain. Integrating by parts, we first note that

$$\int_{y_1}^{y_2} \tilde{\psi}^* \tilde{\psi}_{yy} dy = [\tilde{\psi}^* \tilde{\psi}_y]_{y_1}^{y_2} - \int_{y_1}^{y_2} |\tilde{\psi}_y|^2 dy. \quad (9.70)$$

If the integral is performed between two quiescent latitudes, or $\psi = 0$ at the meridional boundaries, then the first term on the right-hand side vanishes. Similarly,

$$\begin{aligned} \int_0^H \tilde{\psi}^* (F \tilde{\psi}_z)_z dz &= [F \tilde{\psi}^* \tilde{\psi}_z]_0^H - \int_0^H F |\tilde{\psi}_z|^2 dz \\ &= \left[\frac{F U_z |\tilde{\psi}|^2}{(U - c)} \right]_0^H - \int_0^H F |\tilde{\psi}_z|^2 dz, \end{aligned} \quad (9.71)$$

using (9.69b). Now, multiply (9.69a) by $\tilde{\psi}^*$ and integrate over y and z , and use (9.70) and (9.71) to obtain

$$\begin{aligned} \int_0^H \int_{y_1}^{y_2} |\psi_y|^2 + F |\tilde{\psi}_z|^2 + k^2 |\tilde{\psi}|^2 dy dz \\ - \int_{y_1}^{y_2} \left\{ \int_0^H \frac{Q_y}{U - c} |\tilde{\psi}|^2 dz + \left[\frac{F U_z |\tilde{\psi}|^2}{U - c} \right]_0^H \right\} dy = 0. \end{aligned} \quad (9.72)$$

The term on the first line is purely real. The term on the second line is complex, and its imaginary component is given by

$$-c_i \int_{y_1}^{y_2} \left\{ \int_0^H \frac{Q_y}{|U - c|^2} |\tilde{\psi}|^2 dz + \left[\frac{F U_z |\tilde{\psi}|^2}{|U - c|^2} \right]_0^H \right\} dy = 0. \quad (9.73)$$

If there is to be instability c_i must be non-zero, and the integrand must therefore vanish. This gives the *Charney–Stern–Pedlosky* (CSP) necessary condition for instability,⁸ namely that one of the following criteria must be satisfied:

- (i) Q_y changes sign in the interior.
- (ii) Q_y is the opposite sign to U_z at the upper boundary, $z = H$.
- (iii) Q_y is the same sign as U_z at the lower boundary, $z = 0$.
- (iv) U_z is the same sign at the upper and lower boundaries, a condition that differs from (ii) or (iii) if $Q_y = 0$.

In the Earth's atmosphere, Q_y is often dominated by β , and is positive everywhere, as, frequently, is the shear. The instability criterion is then normally satisfied through (iii): that is, both Q_y and $U_z(0)$ are positive. A more general derivation that does not rely on normal-mode disturbances is given in section 10.7.2.

9.5 THE EADY PROBLEM

We now proceed to explicitly calculate the stability properties of a particular configuration that has become known as the *Eady problem*. This was one of the first two mathematical descriptions of baroclinic instability, the other being the *Charney problem*.⁹ The two problems were formulated independently, each being the (largely unsupervised) Ph.D. thesis of its author, and although the Charney problem is in some respects more complete (for example in allowing a β -effect) the Eady problem displays the instability in a more transparent form. The Charney problem in its entirety is also quite mathematically opaque,¹⁰ and so we will first consider the Eady problem. The β -effect

can be incorporated relatively simply in the two-layer model of the next section, and in section 9.9.1 we look at some aspects of the Charney problem approximately. To begin, let us make the following simplifying assumptions.

- (i) The motion is on the f -plane ($\beta = 0$). This assumption, although not particularly realistic for the Earth's atmosphere, greatly simplifies the analysis.
- (ii) The fluid is uniformly stratified; that is, N^2 is a constant. This is a decent approximation for the atmosphere below the tropopause, but less so for the ocean where the stratification is quite non-uniform, being much larger in the upper ocean.
- (iii) The basic state has uniform shear; that is, $U_0(z) = \Lambda z = Uz/H$, where Λ is the (constant) shear and U is the zonal velocity at $z = H$, where H the domain depth. This profile is more appropriate for the atmosphere than the ocean — below the thermocline the ocean is relatively quiescent and the shear is small.
- (iv) The motion is contained between two rigid, flat horizontal surfaces. In the atmosphere this corresponds to the ground and a 'lid' at a constant-height tropopause.

Although, apart from (i), these assumptions are more appropriate for the atmosphere than the ocean, the qualitative nature of baroclinic instability is similar in both systems.

9.5.1 The linearized problem

With a basic state streamfunction of $\Psi = -\Lambda z y$, the basic state potential vorticity, Q , is

$$Q = \nabla^2 \Psi + \frac{H^2}{L_d^2} \frac{\partial}{\partial z} \left(\frac{\partial \Psi}{\partial z} \right) = 0. \quad (9.74)$$

The fact that $Q = 0$ makes the Eady problem a special case, albeit an illuminating one. The linearized potential vorticity equation is

$$\left(\frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \left(\nabla^2 \psi' + \frac{H^2}{L_d^2} \frac{\partial^2 \psi'}{\partial z^2} \right) = 0. \quad (9.75)$$

This equation has no x -dependent coefficients and in a periodic channel we may seek solutions of the form $\psi'(x, y, z, t) = \text{Re } \tilde{\psi}(y, z) e^{ik(x-ct)}$, yielding

$$(\Lambda z - c) \left(\frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{H^2}{L_d^2} \frac{\partial^2 \tilde{\psi}}{\partial z^2} - k^2 \tilde{\psi} \right) = 0. \quad (9.76)$$

This equation is (9.69a) applied to the Eady problem.

Boundary conditions

There are two sets of boundary conditions to satisfy, the vertical boundary conditions at $z = 0$ and $z = H$ and the lateral boundary conditions. In the horizontal plane we may either consider the flow to be confined to a channel, periodic in x and confined between two meridional walls, or, with a slightly greater degree of idealization but with little change to the essential dynamics, suppose that domain is doubly-periodic. Either case is dealt with easily enough by the choice of geometric basis function; we will choose a channel of width L and impose $\psi = 0$ at $y = +L/2$ and $y = -L/2$ and, to satisfy this, seek solutions of the form $\Psi = \Phi(z) \sin ly$ or, using (9.67)

$$\psi'(x, y, z, t) = \text{Re } \Phi(z) \sin ly e^{ik(x-ct)}, \quad (9.77)$$

where $l = n\pi/L$, with n being a positive integer.

The vertical boundary conditions are that $w = 0$ at $z = 0$ and $z = H$. We follow the procedure of section 9.4.2 and from (9.66) we obtain

$$\left(\frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \frac{\partial \psi'}{\partial z} - \Lambda \frac{\partial \psi'}{\partial x} = 0, \quad \text{at } z = 0, H. \quad (9.78)$$

Solutions

Substituting (9.77) into (9.76) gives the interior potential vorticity equation

$$(\Lambda z - c) \left[\frac{H^2}{L_d^2} \frac{\partial^2 \Phi}{\partial z^2} - (k^2 + l^2) \Phi \right] = 0, \quad (9.79)$$

and substituting (9.77) into (9.78) gives, at $z = 0$ and $z = H$,

$$c \frac{d\Phi}{dz} + \Lambda \Phi = 0 \quad \text{and} \quad (c - \Lambda H) \frac{d\Phi}{dz} + \Lambda \Phi = 0. \quad (9.80a,b)$$

These are equivalent to (9.69b) applied to the Eady problem. If $\Lambda z \neq c$ then (9.79) becomes¹¹

$$H^2 \frac{d^2 \Phi}{dz^2} - \mu^2 \Phi = 0, \quad (9.81)$$

where $\mu^2 = L_d^2(k^2 + l^2)$. The non-dimensional parameter μ is a horizontal wavenumber, scaled by the inverse of the Rossby radius of deformation. Solutions of (9.81) are

$$\Phi(z) = A \cosh \mu \hat{z} + B \sinh \mu \hat{z}, \quad (9.82)$$

where $\hat{z} = z/H$; thus, μ determines the vertical structure of the solution. The boundary conditions (9.80) are satisfied if

$$\begin{aligned} A [\Lambda H] + B [\mu c] &= 0, \\ A [(c - \Lambda H)\mu \sinh \mu + \Lambda H \cosh \mu] + B [(c - \Lambda H)\mu \cosh \mu + \Lambda H \sinh \mu] &= 0. \end{aligned} \quad (9.83)$$

Equations (9.83) are two coupled homogeneous equations in the two unknowns A and B . Non-trivial solutions will only exist if the determinant of their coefficients (the terms in square brackets) vanishes, and this leads to

$$c^2 - Uc + U^2(\mu^{-1} \coth \mu - \mu^{-2}) = 0, \quad (9.84)$$

where $U \equiv \Lambda H$ and $\coth \mu = \cosh \mu / \sinh \mu$. The solution of (9.84) is

$$c = \frac{U}{2} \pm \frac{U}{\mu} \left[\left(\frac{\mu}{2} - \coth \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (9.85)$$

The waves, being proportional to $\exp(-ikct)$, will grow exponentially if c has an imaginary part. Since $\mu/2 > \tanh(\mu/2)$ for all μ , for an instability we require that

$$\frac{\mu}{2} < \coth \frac{\mu}{2}, \quad (9.86)$$

which is satisfied when $\mu < \mu_c$, where $\mu_c = 2.399$. The growth rates of the instabilities themselves are given by the imaginary part of (9.85), multiplied by the x -wavenumber; that is

$$\sigma = kc_i = k \frac{U}{\mu} \left[\left(\coth \frac{\mu}{2} - \frac{\mu}{2} \right) \left(\frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (9.87)$$

These solutions suggest a natural non-dimensionalization: scale length by L_d , height by H and time by L_d/U . The *Eady growth rate* is the inverse of the time scaling, and is defined by

$$\sigma_E \equiv \frac{\Lambda H}{L_d} = \frac{U}{L_d}. \quad (9.88)$$

[xxx Needs rephrasing as the 0.31 factor now appears twice.] This growth rate is often multiplied by a factor 0.31 to give the growth rate that actually occurs in the instability problem, as discussed below. That is, $\sigma_E \equiv 0.31 \Lambda H / L_d = 0.31 U / L_d$. Its inverse, the Eady time scale, may also be written as

$$T_E = \frac{L_d}{U} = \frac{NH}{f_0 U} = \frac{1}{Fr f_0} = \frac{\sqrt{Ri}}{f_0}, \quad (9.89)$$

where $Fr = U/(NH)$ and $Ri = N^2/\Lambda^2$ are the Froude and Richardson numbers for this problem.

From (9.87) we may determine that the maximum growth rate occurs when $\mu = \mu_m = 1.61$, with an associated non-dimensional growth rate of $kc_i/\sigma_E = 0.31$, and phase speed $c_r/U = 0.5$. Note that for any given x -wavenumber, the most unstable wavenumber has $l = 0$, so that $L_d k = \mu$. [xxx Check, not true in channel. Need $l = 1$.] The unstable x -wavenumbers and corresponding wavelengths occur for

$$k < k_c = \frac{\mu_c}{L_d} = \frac{2.4}{L_d}, \quad \lambda > \lambda_c = \frac{2\pi L_d}{\mu_c} = 2.6 L_d. \quad (9.90a,b)$$

The wavenumber and wavelength at which the instability is greatest are:

$$k_m = \frac{1.6}{L_d}, \quad \lambda_m = \frac{2\pi L_d}{\mu_m} = 3.9 L_d. \quad (9.91a,b)$$

These properties are illustrated in the left-hand panels of Fig. 9.10 and in Fig. 9.11.

Given c , we may use (9.83) to determine the vertical structure of the Eady wave and this is, to within an arbitrary constant factor,

$$\Phi(\hat{z}) = \cosh \mu \hat{z} - \frac{U}{\mu c} \sinh \mu \hat{z} = \left(\cosh \mu \hat{z} - \frac{U c_r \sinh \mu \hat{z}}{\mu |c^2|} + \frac{i U c_i \sinh \mu \hat{z}}{\mu |c^2|} \right). \quad (9.92)$$

The wave therefore has a phase, $\theta(z)$, given by

$$\theta(\hat{z}) = \tan^{-1} \left(\frac{U c_i \sinh \mu \hat{z}}{\mu |c^2| \cosh \mu \hat{z} - U c_r \sinh \mu \hat{z}} \right). \quad (9.93)$$

The phase and amplitude of the Eady waves are plotted in the right panels of Fig. 9.10, and their overall structure in Fig. 9.12, where we see the unstable wave tilting into the shear.

9.5.2 Atmospheric and oceanic parameters

To get a qualitative sense of the nature of the instability we choose some typical parameters, as follows.

For the atmosphere

Let us choose

$$H \sim 10 \text{ km}, \quad U \sim 10 \text{ m s}^{-1}, \quad N \sim 10^{-2} \text{ s}^{-1}. \quad (9.94)$$

We then obtain:

$$\text{deformation radius: } L_d = \frac{NH}{f} \approx \frac{10^{-2} 10^4}{10^{-4}} \approx 1000 \text{ km}, \quad (9.95)$$

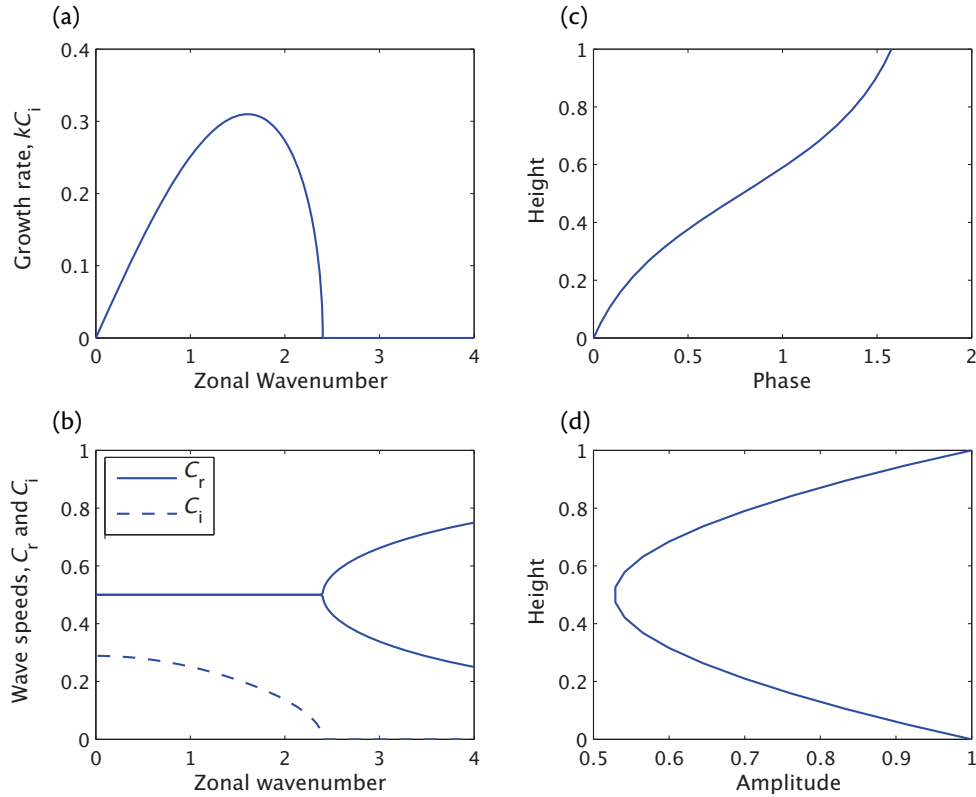


Fig. 9.10 Solution of the Eady problem, in non-dimensional units. (a) The growth rate, kc_i , as a function of scaled wavenumber μ , from (9.87) with $\Lambda = H = 1$ and for the gravest meridional mode (i.e., $l = \pi/L$). (b) The real (solid) and imaginary (dashed) wave speeds of those modes, as a function of horizontal wavenumber. (c) The phase of the single most unstable mode as a function of height. (d) The amplitude of that mode as a function of height. To obtain dimensional values, multiply the growth rate by $\Lambda H/L_d$ and the wavenumber by $1/L_d$.

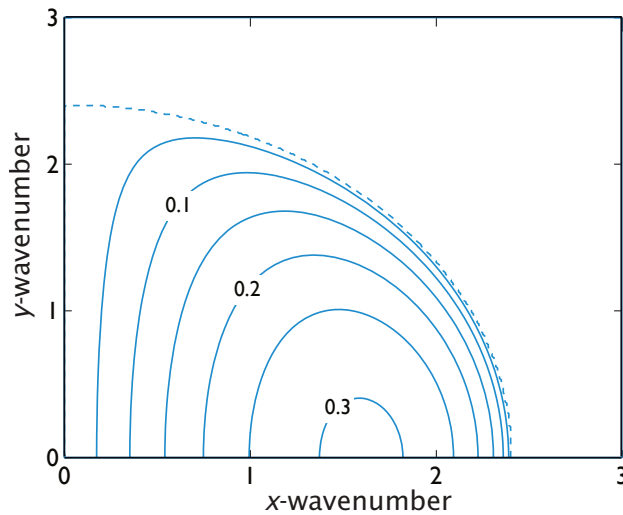


Fig. 9.11 Contours of the growth rate, σ , in the Eady problem, in the $k-l$ plane using (9.87), non-dimensionalized as in Fig. 9.10. The growth rate peaks near the deformation scale, and for any given zonal wavenumber the most unstable wavenumber is that with the gravest meridional scale.

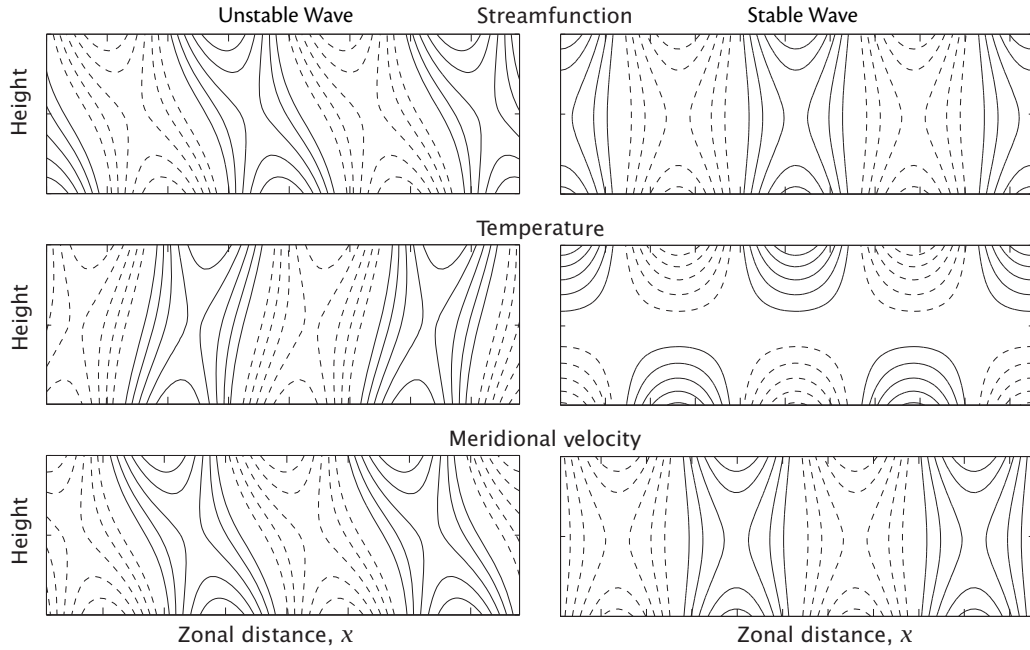


Fig. 9.12 Left column: vertical structure of the most unstable Eady mode. Top: contours of streamfunction. Middle: temperature, $\partial\psi/\partial z$. Bottom: meridional velocity, $\partial\psi/\partial x$. Negative contours are dashed, and two complete wavelengths are present in the horizontal direction. Polewards flowing (positive v) air is generally warmer than equatorwards flowing air. Right column: the same, but now for a wave just beyond the short-wave cut-off.

$$\text{scale of maximum instability: } L_{\max} \approx 3.9L_d \approx 4000 \text{ km}, \quad (9.96)$$

$$\text{growth rate: } \sigma \approx 0.3 \frac{U}{L_d} \approx \frac{0.3 \times 10}{10^6} \text{ s}^{-1} \approx 0.26 \text{ day}^{-1}. \quad (9.97)$$

For the ocean

For the main thermocline in the ocean let us choose

$$H \sim 1 \text{ km}, \quad U \approx 0.1 \text{ m s}^{-1}, \quad N \sim 10^{-2} \text{ s}^{-1}. \quad (9.98)$$

We then obtain:

$$\text{deformation radius: } L_d = \frac{NH}{f} \approx \frac{10^{-2} \times 1000}{10^{-4}} = 100 \text{ km}, \quad (9.99)$$

$$\text{scale of maximum instability: } L_{\max} \approx 3.9L_d \approx 400 \text{ km}, \quad (9.100)$$

$$\text{growth rate: } \sigma \approx 0.3 \frac{U}{L_d} \approx \frac{0.3 \times 0.1}{10^5} \text{ s}^{-1} \approx 0.026 \text{ day}^{-1}. \quad (9.101)$$

In the ocean, the Eady problem is not quantitatively applicable because of the non-uniformity of the stratification. Nevertheless, the above estimates give a qualitative sense of the scale and growth rate of the instability relative to the corresponding values in the atmosphere. A summary of the main points of the Eady problem is given in the shaded box on the next page.

9.6 TWO-LAYER BAROCLINIC INSTABILITY

Summary of Results from the Eady Problem

- The length scales of the instability are characterized by the deformation scale. The most unstable scale has a wavelength about four times the deformation radius L_d , where $L_d = NH/f_0$.
- The growth rate of the instability is approximately

$$\sigma_E \approx \frac{0.3U}{L_d} = 0.3\Lambda \frac{f_0}{N}. \quad (\text{BOX.1})$$

That is, it is proportional to the shear scaled by the Prandtl ratio f_0/N . The value σ_E is known as the *Eady growth rate*.

- The most unstable waves for a given zonal scale are those with the gravest meridional scale.
- There is a *short-wave cutoff* beyond which (i.e., at higher wavenumber than) there is no instability. This occurs near the deformation radius.
- The instability relies on an interaction between waves at the upper and lower boundaries. If either boundary is removed, the instability dies. This point is considered further in section 9.7.

The eigenfunctions displaying the largest growth rates in the Eady problem have a relatively simple vertical structure. This suggests that an even simpler mathematical model of baroclinic instability might be constructed in which the vertical structure is a priori restricted to a very simple form, namely the two-layer or two-level QG model of sections 5.3.2 and 5.4.5. This instability problem is often called the ‘Phillips problem’.¹² One notable advantage over the Eady model is that it is possible to include the β -effect in a simple way.

9.6.1 Posing the problem

For two layers or two levels of equal thickness, we write the potential vorticity equations in the dimensional form,

$$\frac{D}{Dt} \left[\zeta_i + \beta y + \frac{k_d^2}{2}(\psi_j - \psi_i) \right] = 0, \quad i = 1, 2, \quad j = 3 - i, \quad (9.102)$$

where, using two-level notation for definiteness,

$$\frac{k_d^2}{2} = \left(\frac{2f_0}{NH} \right)^2 \rightarrow k_d = \frac{\sqrt{8}}{L_d}, \quad (9.103)$$

where H is the total depth of the domain, as in the Eady problem. The basic state we choose is

$$\Psi_1 = -U_1 y, \quad \Psi_2 = -U_2 y = +U_1 y. \quad (9.104)$$

It is possible to choose $U_2 = -U_1$ without loss of generality because there is no topography and the system is Galilean invariant. The basic state potential vorticity gradient is then given by

$$Q_1 = \beta y + k_d^2 U y, \quad Q_2 = \beta y - k_d^2 U y \quad (9.105)$$

where $U = U_1$. (Note that U differs by a constant multiplicative factor from the U in the Eady problem.) Even in the absence of β there is a non-zero potential vorticity gradient. Why should this be different from the Eady problem? — after all, the shear is uniform in both problems. The difference arises from the vertical boundary conditions. In the standard layered formulation the temperature gradient at the boundary is absorbed into the definition of the potential vorticity in the interior. This results in a non-zero interior potential vorticity gradient at the two levels adjacent to the boundary (the only layers in the two-layer problem), but with isothermal boundary conditions $D(\partial\psi/\partial z)/Dt = 0$. In the Eady problem we have a zero interior gradient of potential vorticity but a temperature gradient at the boundary. The two formulations are physically equivalent — a finite-difference example of the Bretherton boundary layer, encountered in section 5.4.3.

The linearized potential vorticity equation is, for each layer,

$$\frac{\partial q'_i}{\partial t} + U_i \frac{\partial q'_i}{\partial x} + v'_i \frac{\partial Q_i}{\partial y} = 0, \quad i = 1, 2 \quad (9.106)$$

or, more explicitly,

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi'_1 + \frac{k_d^2}{2} (\psi'_2 - \psi'_1) \right] + \frac{\partial \psi'_1}{\partial x} (\beta + k_d^2 U) = 0, \quad (9.107a)$$

$$\left(\frac{\partial}{\partial t} - U \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi'_2 + \frac{k_d^2}{2} (\psi'_1 - \psi'_2) \right] + \frac{\partial \psi'_2}{\partial x} (\beta - k_d^2 U) = 0. \quad (9.107b)$$

For simplicity we will set the problem in a square, doubly periodic domain, and so seek solutions of the form,

$$\psi'_i = \text{Re } \tilde{\psi}_i e^{i(kx+ly-\omega t)} = \text{Re } \tilde{\psi}_i e^{ik(x-ct)} e^{ily}, \quad i = 1, 2. \quad (9.108)$$

Here, k and l are the x - and y -wavenumbers, and $(k, l) = (2\pi/L)(m, n)$, where L is the size of the domain, and m and n are integers. The constant $\tilde{\psi}_i$ is the complex amplitude.

9.6.2 The solution

Substituting (9.108) into (9.107) we obtain

$$[ik(U - c)] [-K^2 \tilde{\psi}_1 + k_d^2 (\tilde{\psi}_2 - \tilde{\psi}_1)/2] + ik \tilde{\psi}_1 (\beta + k_d^2 U) = 0, \quad (9.109a)$$

$$[-ik(U + c)] [-K^2 \tilde{\psi}_2 + k_d^2 (\tilde{\psi}_1 - \tilde{\psi}_2)/2] + ik \tilde{\psi}_2 (\beta - k_d^2 U) = 0, \quad (9.109b)$$

where $K^2 = k^2 + l^2$. Re-arranging these two equations gives

$$[(U - c)(k_d^2/2 + K^2) - (\beta + k_d^2 U)] \tilde{\psi}_1 - [k_d^2(U - c)/2] \tilde{\psi}_2 = 0, \quad (9.110a)$$

$$- [k_d^2(U + c)/2] \tilde{\psi}_1 + [(U + c)(k_d^2/2 + K^2) + (\beta - k_d^2 U)] \tilde{\psi}_2 = 0. \quad (9.110b)$$

These equations are of the form

$$[A] \tilde{\psi}_1 + [B] \tilde{\psi}_2 = 0, \quad [C] \tilde{\psi}_1 + [D] \tilde{\psi}_2 = 0, \quad (9.111)$$

where A, B, C, D correspond to the terms in square brackets in (9.110). For non-trivial solutions the determinant of coefficients must be zero, that is $AD - BC = 0$. This gives a quadratic equation in c and solving this we obtain

$$c = -\frac{\beta}{K^2 + k_d^2} \left\{ 1 + \frac{k_d^2}{2K^2} \pm \frac{k_d^2}{2K^2} \left[1 + \frac{4K^4(K^4 - k_d^4)}{k_\beta^4 k_d^4} \right]^{1/2} \right\}, \quad (9.112)$$

where $K^4 = (k^2 + l^2)^2$ and $k_\beta = \sqrt{\beta/U}$ (its inverse is known as the Kuo scale). We may non-dimensionalize this equation using the deformation radius L_d as the length scale and the shear velocity U as the velocity scale.¹³ Then, denoting non-dimensional parameters with hats, we have

$$k = \frac{\hat{k}}{L_d}, \quad c = \hat{c}U, \quad t = \frac{L_d}{U}\hat{t} \quad (9.113)$$

and the non-dimensional form of (9.112) is just

$$\hat{c} = -\frac{\hat{k}_\beta^2}{\hat{K}^2 + \hat{k}_d^2} \left\{ 1 + \frac{\hat{k}_d^2}{2\hat{K}^2} \pm \frac{\hat{k}_d^2}{2\hat{K}^2} \left[1 + \frac{4\hat{K}^4(\hat{K}^4 - \hat{k}_d^4)}{\hat{k}_\beta^4 \hat{k}_d^4} \right]^{1/2} \right\}, \quad (9.114)$$

where $\hat{k}_\beta = k_\beta L_d$ and $\hat{k}_d = \sqrt{8}$, as in (9.103). The non-dimensional parameter

$$\gamma = \frac{1}{4}\hat{k}_\beta^2 = \frac{\beta L_d^2}{4U}, \quad (9.115)$$

is often useful as a measure of the importance of β ; it is proportional to the square of the ratio of the deformation radius to the Kuo scale $\sqrt{U/\beta}$. (It is the two-layer version of the ‘Charney–Green number’ considered more in section 9.9.1.) Let us look at two special cases first, before considering the general solution to these equations.

I. Zero shear, non-zero β

If there is no shear (i.e., $U = 0$) then (9.110a) and (9.110b) are identical and two roots of the equation give the purely real phase speeds c

$$c = -\frac{\beta}{K^2}, \quad c = -\frac{\beta}{K^2 + k_d^2}. \quad (9.116)$$

The first of these is the dispersion relationship for Rossby waves in a purely barotropic flow, and corresponds to the eigenfunction $\tilde{\psi}_1 = \tilde{\psi}_2$. The second solution corresponds to the baroclinic eigenfunction $\tilde{\psi}_1 + \tilde{\psi}_2 = 0$.

II. Zero β , non-zero shear

If $\beta = 0$, then (9.110) yields, after a little algebra,

$$c = \pm U \left(\frac{K^2 - k_d^2}{K^2 + k_d^2} \right)^{1/2} \quad (9.117)$$

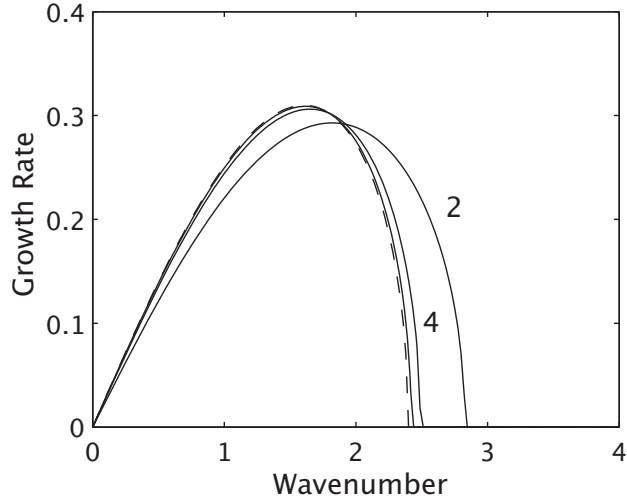
or, defining the growth rate σ by $\sigma = -i\omega$,

$$\sigma = Uk \left(\frac{k_d^2 - K^2}{K^2 + k_d^2} \right)^{1/2}. \quad (9.118)$$

These expressions are very similar to those in the Eady problem. Indeed, as we increase the number of layers (using a numerical method to perform the calculation) the growth rate converges to that of the Eady problem (Fig. 9.13). We note the following.

- There is an instability for *all* values of U .
- There is a high-wavenumber cut-off, at a scale proportional to the radius of deformation. For the two-layer model, if $K > k_d = 2.82/L_d$ there is no growth. For the Eady problem, the high wavenumber cut-off occurs at $2.4/L_d$.

Fig. 9.13 Growth rates for models with varying numbers of vertical layers, all with $\beta = 0$ and the same uniform stratification and shear. The dashed line is the analytic solution to the continuous (Eady) problem, and the solid lines are results obtained using two, four and eight layers. The two- and four-layer results are labelled, and the eight-layer result is almost coincident with the dashed line.



- There is no wavenumber cut-off.
- For any given k , the highest growth rate occurs for $l = 0$. In the two-layer model, from (9.118), for $l = 0$ the maximum growth rate occurs when $k = 0.634k_d = 1.79/L_d$. For the Eady problem, the maximum growth rate occurs at $1.61/L_d$.

Solution in the general case: non-zero shear and non-zero β

Using (9.114), the growth rate and wave speeds as function of wavenumber are plotted in Fig. 9.14. We observe that there still appears to be a high-wavenumber cut-off and, for $\beta = 0$, there is a low-wavenumber cut-off. A little analysis elucidates the origin of these features.

The neutral curve

For instability, there must be an imaginary component to the phase speed in (9.112); that is, we require

$$k_\beta^4 k_d^4 + 4K^4(K^4 - k_d^4) < 0. \quad (9.119)$$

This is a quadratic equation in K^4 for the value of K , K_c say, at which the growth rate is zero. Solving, we find

$$K_c^4 = \frac{1}{2}k_d^4 \left(1 \pm \sqrt{1 - k_\beta^4/k_d^4} \right), \quad (9.120)$$

and this is plotted in Fig. 9.15. From (9.119) useful approximate expressions can be obtained for the critical shear as a function of wavenumber in the limits of small K and $K \approx k_d$, and these are left as exercises for the reader.

Minimum shear for instability

From (9.119), instability arises when $\beta^2 k_d^4 / U^2 < 4K^4(k_d^4 - K^4)$. The maximum value of the right-hand side of this expression arises when $K^4 = \tilde{k}_d^4/2$; thus, instability arises only when

$$\frac{\beta^2 k_d^4}{U^2} < 4 \frac{k_d^4}{2} \frac{k_d^4}{2} \quad (9.121)$$

or

$$U_s > \frac{2\beta}{k_d^2} \quad (9.122)$$

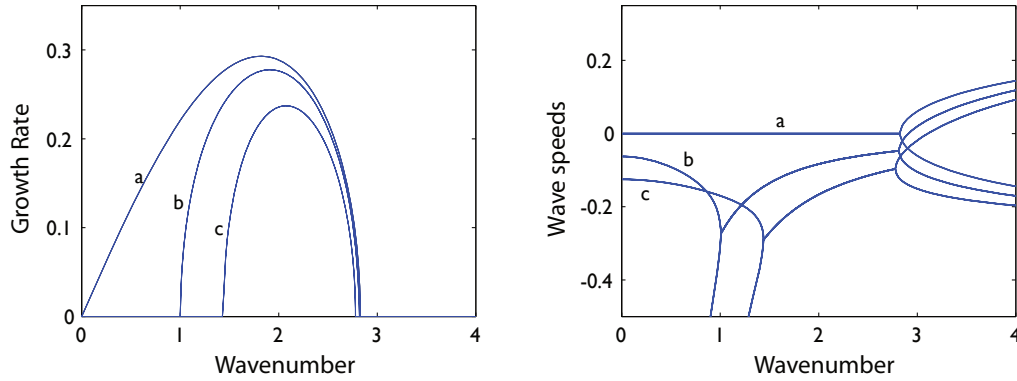


Fig. 9.14 Growth rates and wave speeds for the two-layer baroclinic instability problem, from (9.114), with three (non-dimensional) values of β : a, $\gamma = 0$ ($\bar{k}_\beta = 0$); b, $\gamma = 0.5$ ($\bar{k}_\beta = \sqrt{2}$); c, $\gamma = 1$ ($\bar{k}_\beta = 2$). As β increases, so does the low-wavenumber cut-off to instability, but the high-wavenumber cut-off is little changed. (The solutions are obtained from (9.114), with $\bar{k}_d = \sqrt{8}$ and $U_1 = -U_2 = 1/4$.)

where $U_s = U_1 - U_2 = 2U$. In terms of the deformation radius itself the minimum shear for instability is

$$U_s > \frac{1}{4} \beta L_d^2. \quad (9.123)$$

Figure 9.16 sketches how this might vary with latitude in the atmosphere and ocean. [In (9.123), the shear is the difference in the velocity between level 1 and level 2, whereas the deformation radius, NH/f_0 , is based on the total height of the fluid. If we were to use half the depth of the fluid in the definition of the deformation radius, the factor of 4 would disappear.] If the shear is just this critical value, the instability occurs at $k = 2^{-1/4} k_d = 0.84 k_d = 2.37/L_d$. As the shear increases, the wavenumber at which the growth rate is maximum decreases slightly (see Fig. 9.15), and for a sufficiently large shear the β -effect is negligible and the wavenumber of maximum instability is, as we saw earlier, $0.634 k_d$ or $1.79/L_d$.

Note the relationship of the minimum shear to the basic state potential vorticity gradient in the respective layers. In the upper and lower layers the potential vorticity gradients are given by, respectively,

$$\frac{\partial Q_1}{\partial y} = \beta + k_d^2 U, \quad \frac{\partial Q_2}{\partial y} = \beta - k_d^2 U. \quad (9.124a,b)$$

Thus, the requirement for instability is exactly that which causes the potential vorticity gradient to change sign somewhere in the domain, in this case becoming negative in the lower layer. This is an example of the general rule that potential vorticity (suitably generalized to include the surface boundary conditions) must change sign somewhere in order for there to be an instability.

High-wavenumber cut-off

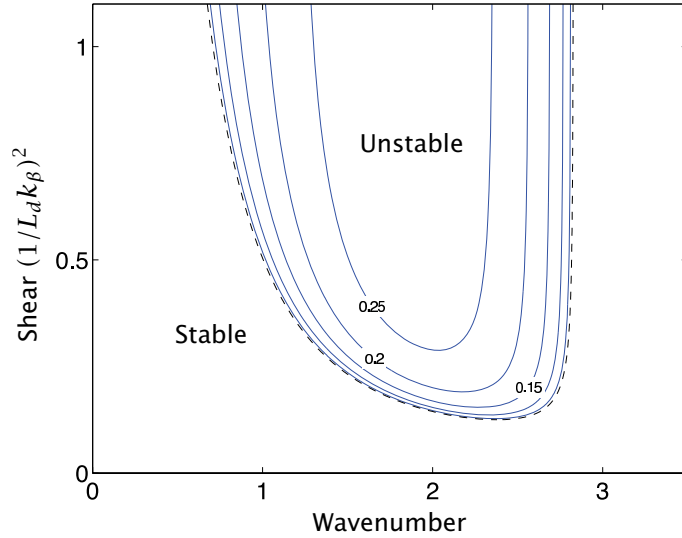
Instability can only arise when, from (9.119),

$$4K^4(k_d^4 - K^4) > k_\beta^4 k_d^4, \quad (9.125)$$

so that a necessary condition for instability is

$$k_d^2 > K^2. \quad (9.126)$$

Fig. 9.15 Contours of growth rate in the two-layer baroclinic instability problem. The dashed line is the neutral stability curve, (9.120), and the other curves are contours of growth rates obtained from (9.114). The wavenumber is scaled by $1/L_d$ (i.e., by $k_d/\sqrt{8}$) and growth rates are scaled by the inverse of the Eady time scale (i.e., by U/L_d). Thus, for $L_d = 1000$ km and $U = 10$ m s⁻¹, a non-dimensional growth rate of 0.25 corresponds to a dimensional growth rate of 0.25×10^{-5} s⁻¹ = 0.216 day⁻¹.



Thus, waves shorter than the deformation radius are always stable, no matter what the value of β . We also see from Fig. 9.14 and Fig. 9.15 that the high wavenumber cut-off in fact varies little with β if $k_d \gg k_\beta$. Note that the critical shear required for instability approaches infinity as K approaches k_d .

Low-wavenumber cut-off

Suppose that $K \ll k_d$. Then (9.119) simplifies to $k_\beta^4 < 4K^4$. That is, for instability we require

$$K^2 > \frac{1}{2} k_\beta^2 = \frac{\beta}{2U}. \quad (9.127)$$

Thus, using (9.126) and (9.127) the unstable waves lie approximately in the interval $\beta/(\sqrt{2}U) < K < k_d$.

9.7 AN INFORMAL VIEW OF THE MECHANISM OF BAROCLINIC INSTABILITY

In this section we take a more intuitive look at baroclinic instability, trying to understand the mechanism without treating the problem in full generality or exactness. We will do this by way of a semi-kinematic argument that shows how the waves in each layer of a two-layer model, or the waves on the top and bottom boundaries in the Eady model, can constructively interact to produce a growing instability. It is kinematic in the sense that we initially treat the waves independently, and only subsequently allow them to interact — but it is this dynamical interaction that gives the instability. We first revisit the two-layer model and simplify it to its bare essentials.

9.7.1 The two-layer model

A simple dynamical model

We first re-derive the instability ab initio from the equations of motion written in terms of the baroclinic streamfunction τ and the barotropic streamfunction ψ where

$$\tau \equiv \frac{1}{2}(\psi_1 - \psi_2), \quad \psi \equiv \frac{1}{2}(\psi_1 + \psi_2). \quad (9.128)$$

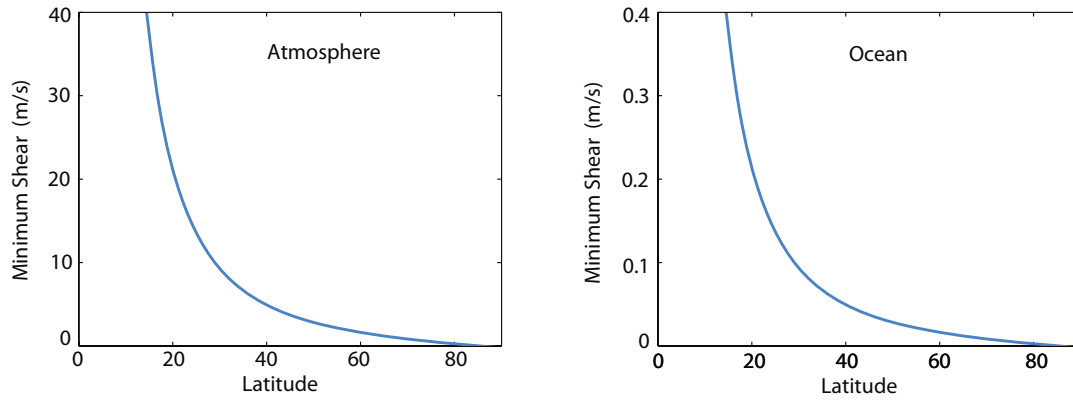


Fig. 9.16 The minimum shear ($U_s = U_1 - U_2$, in m s^{-1}) required for baroclinic instability in a two-layer model, calculated using (9.123), i.e., $U_s = \beta L_d^2/4$ where $\beta = 2\Omega a^{-1} \cos \vartheta$ and $L_d = NH/f$, with $f = 2\Omega \sin \vartheta$. The left panel uses $H = 10 \text{ km}$ and $N = 10^{-2} \text{ s}^{-1}$, and the right panel uses parameters representative of the main thermocline, $H = 1 \text{ km}$ and $N = 10^{-2} \text{ s}^{-1}$. The results are not quantitatively accurate, but the implications that the minimum shear is much less for the ocean, and that in both the atmosphere and the ocean the shear increases rapidly at low latitudes, are robust.

We linearize about a sheared basic state of with zero barotropic velocity and with $\beta = 0$. Thus, with $\psi = 0 + \psi'$ and $\tau = -Uy + \tau'$ the linearized equations of motion, equivalent to (9.107) with $\beta = 0$, are

$$\frac{\partial}{\partial t} \nabla^2 \psi' = -U \frac{\partial}{\partial x} \nabla^2 \tau', \quad (9.129a)$$

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \tau' = -U \frac{\partial}{\partial x} (\nabla^2 + k_d^2) \psi'. \quad (9.129b)$$

Neglecting the y -dependence for simplicity, we may seek solutions of the form $(\psi', \tau') = \text{Re}(\tilde{\psi}, \tilde{\tau}) \exp[ik(x - ct)]$, where $c = i c_i + c_r$, giving

$$c \tilde{\tau} - U \tilde{\tau} = 0, \quad (9.130a)$$

$$c(K^2 + k_d^2) \tilde{\tau} - U(K^2 - k_d^2) \tilde{\psi} = 0. \quad (9.130b)$$

These equations have non-trivial solutions if the determinant of the matrix of coefficients of $\tilde{\tau}$ and $\tilde{\psi}$ is zero, giving the quadratic equation $c^2(K^2 + k_d^2) - U^2(K^2 - k_d^2) = 0$. Solving this gives, reprising (9.117),

$$c = \pm U \left(\frac{K^2 - k_d^2}{K^2 + k_d^2} \right)^{1/2}. \quad (9.131)$$

Instabilities occur for $K^2 < k_d^2$, for which $c_r = 0$; that is, the wave speed is purely imaginary. From (9.130) unstable modes have

$$\tilde{\tau} = i \frac{c_i}{U} \tilde{\psi} = e^{i\pi/2} \frac{c_i}{U} \tilde{\psi}. \quad (9.132)$$

That is, τ lags ψ by 90° for a growing wave ($c_i > 0$). Similarly, τ leads ψ by 90° for a decaying wave. Now, the temperature is proportional to τ , and in the two-level model is advected by the vertically averaged perturbation meridional velocity, v say (with Fourier amplitude $\tilde{v}$), where $v = \partial\psi/\partial x$. Thus, for growing or decaying waves,

$$\tilde{v} = \tilde{\tau} \frac{kU}{c_i}, \quad (9.133)$$

and the meridional velocity is exactly *in phase* with the temperature for growing modes, and is *out of phase* with the temperature for decaying modes. That is, for unstable modes, poleward flow is correlated with high temperatures, and for decaying modes poleward flow is correlated with low temperatures. For neutral waves, $\tilde{\tau} = c_r \tilde{\psi}/U$ and so $\tilde{v} = ik\tilde{\tau}U/c_r$, and the meridional velocity and temperature are $\pi/2$ out of phase. Thus, to summarize:

- growing waves transport heat (or buoyancy) polewards;
- decaying waves transport heat equatorwards;
- neutral waves do not transport heat.

Further simplifications to the two-layer model

First consider (9.129) for waves much larger than the deformation radius, $K^2 \ll k_d^2$; we obtain

$$\frac{\partial}{\partial t} \nabla^2 \psi = -U \frac{\partial}{\partial x} \nabla^2 \tau, \quad \frac{\partial}{\partial t} \tau = U \frac{\partial}{\partial x} \psi. \quad (9.134a,b)$$

for which we obtain [either directly or from (9.131)] $c = \pm iU$; that is, the flow is unstable. To see the mechanism, suppose that the initial perturbation is barotropic and sinusoidal in x , with no y variation. Polewards flowing fluid (i.e., $\partial\psi/\partial x > 0$) will, by (9.134b), generate a positive τ , and the baroclinic flow will be out of phase with the barotropic flow. Then, by (9.134a), the advection of τ by the mean shear produces growth of ψ that is in phase with the original disturbance. Contrast this case with that for very small disturbances, for which $K^2 \gg k_d^2$ and (9.129) becomes

$$\frac{\partial}{\partial t} \nabla^2 \psi = -U \frac{\partial}{\partial x} \nabla^2 \tau, \quad \frac{\partial}{\partial t} \nabla^2 \tau = -U \frac{\partial}{\partial x} \nabla^2 \psi, \quad (9.135a,b)$$

or, in terms of the equations for each layer,

$$\frac{\partial}{\partial t} \nabla^2 \psi_1 = -U \frac{\partial}{\partial x} \nabla^2 \psi_1, \quad \frac{\partial}{\partial t} \nabla^2 \psi_2 = +U \frac{\partial}{\partial x} \nabla^2 \psi_2. \quad (9.136a,b)$$

That is, the layers are completely decoupled and no instability can arise. Motivated by this, consider waves that propagate independently in each layer on the potential vorticity gradient caused by β (if non-zero) and shear. Thus, in (9.107) we keep the potential vorticity gradients but neglect k_d^2 where it appears alongside ∇^2 and find

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \nabla^2 \psi'_1 + \frac{\partial \psi'_1}{\partial x} \frac{\partial Q_1}{\partial y} = 0, \quad (9.137a)$$

$$\left(\frac{\partial}{\partial t} - U \frac{\partial}{\partial x} \right) \nabla^2 \psi'_2 + \frac{\partial \psi'_2}{\partial x} \frac{\partial Q_2}{\partial y} = 0, \quad (9.137b)$$

where $\partial Q_1/\partial y = \beta + k_d^2 U$ and $\partial Q_2/\partial y = \beta - k_d^2 U$. Seeking solutions of the form (9.108), the phase speeds of the associated waves are

$$c_1 = U - \frac{\partial_y Q_1}{K^2}, \quad c_2 = -U - \frac{\partial_y Q_2}{K^2}. \quad (9.138a,b)$$

In the upper layer the phase speed is a combination of an eastward advection and a fast westward wave propagation due to a strong potential vorticity gradient. In the lower layer the phase speed is a combination of a westward advection and a slow eastward wave propagation due to the weak potential vorticity gradient. The two phase speeds are, in general, not equal, but they would need to be so if they were to combine to cause an instability. From (9.138) this occurs when $K^2 = k_d^2$ and $c_1 = c_2 = -\beta/k_d^2$. These conditions are just those occurring at the high-wavenumber cut-off

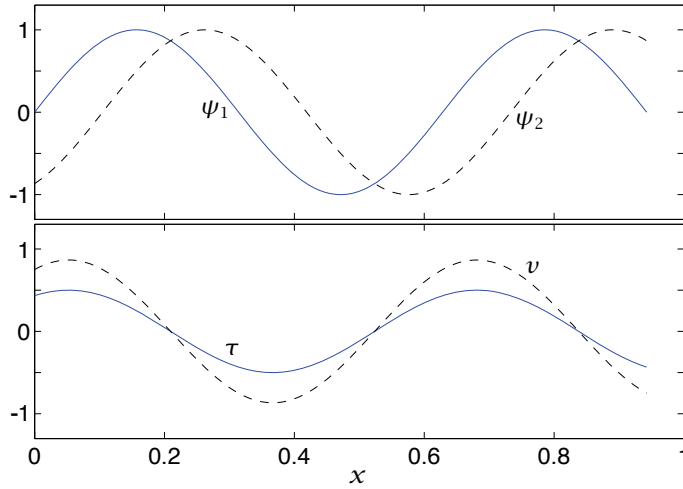


Fig. 9.17 Baroclinically unstable waves in a two layer model. The streamfunction is shown in the top panel, ψ_1 for the top layer and ψ_2 for the bottom layer. Given the westward tilt shown, the temperature, τ , and meridional velocity, v (bottom panel) are in phase, and the instability grows.

to instability in the two-level model. At higher wavenumbers, the waves are unable to synchronize, whereas at lower wavenumbers they may become inextricably coupled.

Let us suppose that the phase of the wave in the upper layer lags that (i.e., is westward of) that in the lower layer, as illustrated in the top panel Fig. 9.17. The lower panel shows the temperature field, $\tau = (\psi_1 - \psi_2)/2$, and the average meridional velocity, $v = \partial_x(\psi_1 + \psi_2)/2$. In this configuration, the temperature field is *in phase* with the meridional velocity, meaning that warm fluid is advected polewards. Now, let us allow the waves in the two layers to interact by adding one dynamical equation, the thermodynamic equation, which in its simplest form is

$$\frac{\partial \tau}{\partial t} = -v \frac{\partial \bar{\tau}}{\partial y} = vU, \quad (9.139)$$

where $\bar{\tau}$ is proportional to the basic state temperature field. The temperature field, τ , grows in proportion to v , which is proportional to τ if the waves tilt westwards with height, and an instability results. This dynamical mechanism is just that which is compactly described by (9.134). It is a straightforward matter to show that if the streamfunction tilts eastwards with height, v is out of phase with τ and the waves decay. A similar description applies to the Eady problem, as we now see.

9.7.2 Interacting edge waves in the Eady problem

We now explore how the edge waves at the top and bottom surfaces in the Eady problem give rise to an instability, by way of semi-kinematic description that is very similar to that of the barotropic problem described on page 347. Let us first consider the case in which the bottom and top surfaces are essentially uncoupled. Instead of solutions of (9.81) that have the structure (9.82) (which satisfies both boundary conditions) consider solutions that *separately* satisfy the bottom and top boundary conditions and that decay into the interior. These are

$$\psi_B = \text{Re } A_B e^{ik(x-c_T t)} e^{-\mu z/H}, \quad \psi_T = \text{Re } A_T e^{i\phi} e^{ik(x-c_B t)} e^{\mu(z-H)/H}, \quad (9.140a,b)$$

for the bottom and top surfaces, respectively, and ϕ is the phase shift, with A_B and A_T being real constants. The boundary conditions (9.80) then determine the phase speeds of the two systems and we find

$$c_B = \frac{\Lambda H}{\mu}, \quad c_T = \Lambda H \left(1 - \frac{1}{\mu} \right). \quad (9.141a,b)$$

These are the phase speeds of *edge waves* in the Eady problem; they are real and in general they are unequal. It must therefore be the *interaction* of the waves on the upper and lower boundaries that is necessary for instability, because the unstable wave has but a single phase speed. This interaction can occur when their phase speeds are equal and from (9.141) this occurs when $\mu = 2$, giving

$$k = \frac{2}{L_d} \quad \text{and} \quad c = \frac{\Lambda H}{2}. \quad (9.142a,b)$$

This phase speed is just that of the flow at mid-level, and at the critical wavenumber in the full Eady problem [$k_c = 2.4/L_d$, from (9.90)] the phase speed is purely real and equal to that of (9.142b) — see Fig. 9.10. Thus, (9.142) approximately characterizes the critical wavenumber in the full problem.

To turn this kinematic description into a dynamical instability, suppose that the two rigid surfaces are close enough so that the waves can interact, but still far enough so that their structure is approximately given by (9.140). (Note that if μ is too large, the waves decay rapidly away from the edges and will not intersect.) Specifically, let the buoyancy perturbation at a given boundary be advected by the total meridional velocity perturbation, including that arising from the perturbation at the other boundary, so that at the top and bottom boundaries

$$\frac{\partial b'_T}{\partial t} = -(v'_B + v'_T) \frac{\partial \bar{b}_T}{\partial y}, \quad \frac{\partial b'_B}{\partial t} = -(v'_B + v'_T) \frac{\partial \bar{b}_B}{\partial y}. \quad (9.143)$$

The waves will reinforce each other if v'_T is in phase with b'_B at the lower boundary, and if v'_B is in phase with b'_T at the upper boundary. Now, using (9.140), the velocity and buoyancy associated with the edge waves are given by

$$b_B = -\text{Re } k N A_B e^{-\mu z/H} e^{ikx}, \quad b_T = \text{Re } k N A_T e^{i\phi} e^{\mu(z-H)/H} e^{ikx}, \quad (9.144a)$$

$$v_B = \text{Re } i k A_B e^{-\mu z/H} e^{ikx}, \quad v_T = \text{Re } i k A_T e^{i\phi} e^{\mu(z-H)/H} e^{ikx}. \quad (9.144b)$$

The fields b_B and v_T , and b_T and v_B , will be positively correlated if $0 < \phi < \pi$, and will be exactly in phase if $\phi = \pi/2$, and this case is illustrated in Fig. 9.18. Just as in the two-layer case, this phase corresponds to a westward tilt with height, and it is this, in conjunction with geostrophic and hydrostatic balance, that allows warm fluid to move polewards and available potential energy to be released. From (9.143), the perturbation will grow and an instability will result. The analogy between baroclinic instability and barotropic instability should be evident from the similarity of this description and that of section 9.2.4, with z in the baroclinic problem playing the role of y in the barotropic problem, and b the role of u ; note that (9.143) is almost identical to (9.45). However, the analogy of the two problems in full is not perfect because the boundary condition that $w = 0$ does not have an exact correspondence in the barotropic problem. More importantly, the nonlinear development of the baroclinic problem, discussed in chapter 12, is generally three-dimensional, which need not be the case in the barotropic problem.

9.8 ♦ THE ENERGETICS OF LINEAR BAROCLINIC INSTABILITY

In baroclinic instability, warm parcels move polewards and cold parcels move equatorwards. This motion draws on the available potential energy of the mean state, because warm light parcels move upwards, and cold dense parcels move downwards and the height of the mean centre of gravity of the fluid falls, and the loss of potential energy is converted to kinetic energy of the perturbation. However, because the instability is growing, the energy of the perturbation is of course not conserved, and both the kinetic energy and the available potential energy of the perturbation will grow. However, we still expect a conversion of potential energy to kinetic energy, and the purpose of this section is to demonstrate that explicitly. For simplicity, we restrict attention to the flat-bottomed two-level model with $\beta = 0$.

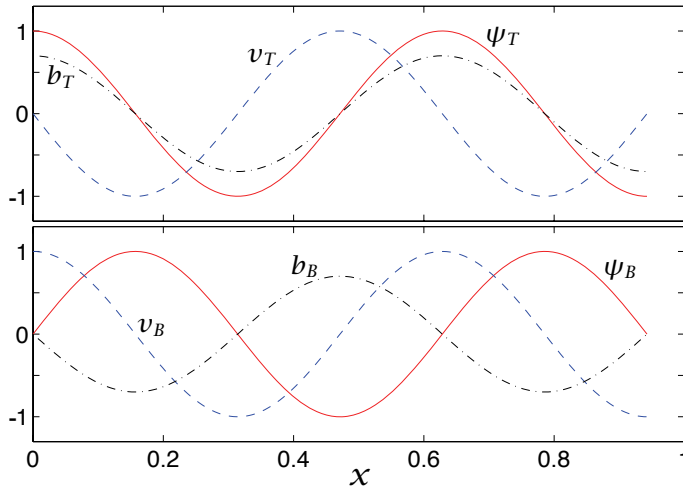


Fig. 9.18 Interacting edge waves in the Eady model. The upper panel shows waves on the top surface, and the lower panel show waves on the bottom. If the streamfunction tilts westwards with height, then the temperature on the top (bottom) is correlated with the meridional velocity on the bottom (top), the waves can reinforce each other. See also Fig. 9.12.

As in section 5.6, the energy may be partitioned into kinetic energy and available potential energy. In a three-dimensional quasi-geostrophic flow the kinetic energy is given by, in general,

$$KE = \frac{1}{2} \int (\nabla \psi)^2 dV \quad (9.145)$$

which, in the case of the two-layer model becomes

$$KE = \frac{1}{2} \int (\nabla \psi_1)^2 + (\nabla \psi_2)^2 dA = \int (\nabla \psi)^2 + (\nabla \tau)^2 dA. \quad (9.146)$$

Restricting attention to a single Fourier mode this becomes

$$KE = k^2 \tilde{\psi}^2 + k^2 \tilde{\tau}^2. \quad (9.147)$$

The available potential energy in the continuous case is given by

$$APE = \frac{1}{2} \int \left(\frac{f_0}{N} \right)^2 \left(\frac{\partial \psi}{\partial z} \right)^2 dV. \quad (9.148)$$

For a single Fourier mode in a two-layer model this becomes

$$APE = k_d^2 \tilde{\tau}^2. \quad (9.149)$$

Now, the nonlinear vorticity equations for each level is

$$\frac{\partial}{\partial t} \nabla^2 \psi_1 + J(\psi_1, \nabla^2 \psi_1) = -2 \frac{f_0 w}{H} \quad (9.150a)$$

$$\frac{\partial}{\partial t} \nabla^2 \psi_2 + J(\psi_2, \nabla^2 \psi_2) = 2 \frac{f_0 w}{H}, \quad (9.150b)$$

where w is the vertical velocity between the levels. (These equations are the two-level analogues of the continuous vorticity equation, with the right-hand sides being finite difference versions of $f_0 \partial w / \partial z$.) Multiplying the two equations of (9.150) by $-\psi_1$ and $-\psi_2$, respectively, and adding we readily find

$$\frac{d}{dt} KE = \frac{4f_0}{H} \int w \tau dA. \quad (9.151a)$$

For a single Fourier mode this becomes

$$\frac{d}{dt} KE = \text{Re} \frac{4f_0}{H} \tilde{w} \tilde{\tau}^*, \quad (9.151b)$$

where $w = \tilde{w} \exp[i(kx - ct)] + \text{c.c.}$, and the asterisk denotes a complex conjugate.

The continuous thermodynamic equation is

$$\frac{Db}{Dt} + wN^2 = 0. \quad (9.152)$$

Using $b = f_0 \partial \psi / \partial z$ and finite-differencing [with $\partial \psi / \partial z \rightarrow (\psi_1 - \psi_2)/(H/2) = 4\tau/H$], we obtain the two-level thermodynamic equation:

$$\frac{\partial \tau}{\partial t} + J(\psi, \tau) + \frac{wN^2 H}{4f_0} = 0. \quad (9.153)$$

The change of available potential energy is obtained from this by multiplying by $k_d^2 \tau$ and integrating, giving

$$\int \left(\frac{1}{2} \frac{d}{dt} k_d^2 \tau^2 + \tau w \frac{2f_0}{H} \right) dA = 0 \quad (9.154)$$

or

$$\frac{d}{dt} APE = -\frac{4f_0}{H} \int w \tau dA \quad (9.155a)$$

or, for a single Fourier mode,

$$\frac{d}{dt} APE = -\text{Re} \frac{4f_0}{H} \tilde{w} \tilde{\tau}^*. \quad (9.155b)$$

From (9.151) and (9.155) it is clear that in the nonlinear equations the sum of the kinetic energy and the available potential energy is conserved.

We now specialize by obtaining w from the linear baroclinic instability problem. Using this in (9.151) and (9.155) will give us the conversion between kinetic energy and potential energy in the growing baroclinic wave. It is important to realize that the total energy of the disturbance will not be conserved — both the potential and kinetic energy are growing, exponentially in this problem, because they are extracting energy from the mean state. To calculate w we use the linearized thermodynamic equation. From (9.153) this is

$$\frac{\partial \tau}{\partial t} - U \frac{\partial \psi}{\partial x} + \frac{HwN^2}{4f_0} = 0, \quad (9.156)$$

omitting the primes on perturbation quantities. For a single Fourier mode, this gives

$$\frac{HN^2}{4f_0} \tilde{w} = ik(c\tilde{\tau} + U\tilde{\psi}). \quad (9.157)$$

But, from (9.130), $c\tilde{\psi} = U\tilde{\tau}$ in two-layer f -plane baroclinic instability and so

$$\frac{HN^2}{4f_0} \tilde{w} = ikc\tilde{\tau} \left(1 + \frac{U^2}{c^2} \right) = ikc\tilde{\tau} \left(\frac{2K^2}{K^2 - k_d^2} \right), \quad (9.158)$$

using (9.131). For stable waves, $K^2 > k_d^2$ and $c = c_r$ and in that case the vertical velocity is $\pi/2$ out of phase with the temperature, and there is no conversion of APE to KE. For unstable waves $c = i c_i$ and $K^2 < k_d^2$, and the vertical velocity is in phase with the temperature; that is, warm air is rising

and so there is a conversion of APE to KE. To see this more formally, recall that the conversion from APE to KE is given by $4\tilde{\omega}\tilde{\tau}^*f_0/H$. Thus, using (9.158),

$$\frac{d}{dt}(APE \rightarrow KE) = \text{Re } 2ikck_d^2 \left(\frac{2K^2}{K^2 - k_d^2} \right) \tilde{\tau}^2, \quad (9.159)$$

using also the definition of k_d given in (9.103). If the wave is growing, then $K^2 < k_d^2$ and $c = ic_i$ and the right-hand side is real and positive. For neutral waves, if $c = c_r$ the right-hand side of (9.159) is pure imaginary, and so the conversion is zero. This completes our demonstration that baroclinic instability converts potential energy into kinetic energy.

9.9 ♦ BETA, SHEAR AND STRATIFICATION IN A CONTINUOUS MODEL

The two-layer model of section 9.6 indicates that β has a number of important effects on baroclinic instability. Do these carry over to the continuously stratified case? The answer by and large is yes, but with some important qualifications that generally concern weak or shallow instabilities. In particular, we will find that there is no short-wave cut-off in the continuous model with non-zero beta, and that the instability determines its own depth scale. We will illustrate these properties first by way of scaling arguments and then by way of numerical calculations.¹⁴

9.9.1 Scaling arguments for growth rates, scales and depth

With finite density scale height and non-zero β , the quasi-geostrophic potential vorticity equation, linearized about a mean zonal velocity $U(z)$, is

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) q' + \frac{\partial \psi'}{\partial x} \frac{\partial Q}{\partial y} = 0, \quad (9.160)$$

where

$$q' = \nabla^2 \psi' + \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial \psi'}{\partial z} \right), \quad \frac{\partial Q}{\partial y} = \beta - \frac{f_0^2}{\rho_R} \frac{\partial}{\partial z} \left(\frac{\rho_R}{N^2} \frac{\partial U}{\partial z} \right), \quad (9.161a,b)$$

and ρ_R is a specified density profile. If we assume that $U = \Lambda z$ where Λ is constant and that N is constant, and let $H_\rho^{-1} = -\rho_R^{-1} \partial \rho_R / \partial z$, then

$$\frac{\partial Q}{\partial y} = \beta + \frac{f_0^2 \Lambda}{N^2 H_\rho} = \beta(1 + \alpha), \quad \text{where} \quad \alpha = \left(\frac{f_0^2 \Lambda}{\beta N^2 H_\rho} \right). \quad (9.162)$$

The boundary conditions on (9.160) are

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \frac{\partial \psi'}{\partial z} - \frac{\partial \psi'}{\partial x} \frac{\partial U}{\partial z} = 0, \quad \text{at } z = 0 \quad (9.163)$$

and that $\psi \rightarrow 0$ as $z \rightarrow \infty$. The problem we have defined essentially constitutes the Charney problem. We can reduce this to the Eady problem by setting $\beta = 0$ and $H_\rho = \infty$, and providing a lid that is some finite height above the ground.

As in the Eady problem, we seek solutions of the form

$$\psi = \text{Re } \tilde{\psi}(z) e^{i(kx + ly - kct)}, \quad (9.164)$$

and substituting into (9.160) gives

$$\left(\frac{f_0^2}{H_\rho^2 N^2} \right) \left(H_\rho^2 \frac{d^2 \tilde{\psi}}{dz^2} - H_\rho \frac{d\tilde{\psi}}{dz} \right) - \left(K^2 - \frac{\beta + \Lambda f_0^2 / (N^2 H_\rho)}{\Lambda z - c} \right) \tilde{\psi} = 0. \quad (9.165)$$

The Boussinesq version of this expression, for a fluid contained between two horizontal surfaces, is obtained by letting $H_\rho = \infty$, giving

$$\left(\frac{f_0^2}{N^2}\right) \frac{d^2 \tilde{\psi}}{dz^2} - \left(K^2 - \frac{\beta}{\Lambda z - c}\right) \tilde{\psi} = 0. \quad (9.166)$$

It seems natural to non-dimensionalize (9.165) using:

$$z = H_\rho \hat{z}, \quad c = \Lambda H_\rho \hat{c}, \quad K = \left(\frac{f_0}{NH_\rho}\right) \hat{K}, \quad (9.167)$$

whence the equation becomes

$$\frac{d^2 \tilde{\psi}}{d\hat{z}^2} - \frac{d\tilde{\psi}}{d\hat{z}} - \left(\hat{K}^2 - \frac{\gamma + 1}{\hat{z} - \hat{c}}\right) \tilde{\psi} = 0 \quad (9.168)$$

where

$$\gamma = \alpha^{-1} = \frac{\beta N^2 H_\rho}{f_0^2 \Lambda} = \frac{\beta L_d^2}{H_\rho \Lambda} = \frac{H_\rho}{h}. \quad (9.169)$$

where $h \equiv \Lambda f_0^2 / (\beta N^2)$. The non-dimensional parameter γ is known as the Charney–Green number.¹⁵ The Boussinesq version, (9.166), may be non-dimensionalized using H_D in place of H_ρ , where H_D is the depth of the fluid between two rigid surfaces. In that case

$$\frac{d^2 \tilde{\psi}}{d\hat{z}^2} - \left(\hat{K}^2 - \frac{\gamma}{\hat{z} - \hat{c}}\right) \tilde{\psi} = 0, \quad (9.170)$$

where here the non-dimensional variables are scaled with H_D .

Now, suppose that γ is large, for example if β or the static stability are large or the shear is weak. Equation (9.168) admits no non-trivial balance, suggesting that we rescale the variables using h instead of H_ρ as the vertical scale in (9.167). The rescaled version of (9.168) is then

$$\frac{d^2 \tilde{\psi}}{d\hat{z}^2} - \frac{1}{\gamma} \frac{d\tilde{\psi}}{d\hat{z}} - \left(\hat{K}^2 - \frac{1 + \gamma^{-1}}{\hat{z} - \hat{c}}\right) \tilde{\psi} = 0, \quad (9.171)$$

or, approximately,

$$\frac{d^2 \tilde{\psi}}{d\hat{z}^2} - \left(\hat{K}^2 - \frac{1}{\hat{z} - \hat{c}}\right) \tilde{\psi} = 0. \quad (9.172)$$

This is exactly the same equation as results from a similar rescaling of the Boussinesq system, (9.170), as we might have expected because now the dynamical vertical scale, h , is much smaller than the scale height H_ρ (or H_D) and the system is essentially Boussinesq. Thus, noting that (9.172) has the same non-dimensional form as (9.170) save that γ is replaced by unity, and that (9.172) with $\gamma = 1$ must produce the same scales and growth rates as in the Eady problem, we may deduce that:

- (i) the wavelength of the instability is $\mathcal{O}(Nh/f_0)$;
- (ii) the growth rate of the instability is $\mathcal{O}(Kc) = \mathcal{O}(f_0 \Lambda / N)$.
- (iii) the vertical scale of the instability is $\mathcal{O}(h) = \mathcal{O}(f_0^2 \Lambda / (\beta N^2))$.

These are the same as for the Eady problem, except with the dynamical height h replacing the geometric or scale height H_D . Effectively, the dynamics has determined its own vertical scale, h , which is much less than the scale height or geometric height, producing ‘shallow modes’.

In the limit $\gamma \ll 1$ (strong shear, weak β), the Boussinesq and compressible problems differ. The Boussinesq problem reduces to the Eady problem, considered previously, whereas (9.168) becomes, approximately,

$$\frac{d^2 \tilde{\psi}}{d\tilde{z}^2} - \frac{d\tilde{\psi}}{d\tilde{z}} - \left(\tilde{K}^2 - \frac{1}{\tilde{z} - \tilde{c}} \right) \tilde{\psi} = 0, \quad (9.173)$$

and in this limit the appropriate vertical scale is the density scale height H_ρ . Because $H_\rho \gg h$ these are ‘deep modes’, occupying the entire vertical extent of the domain.

The scale h does not arise in the two-level model, but there is a connection between it and the critical shear for instability in the two-level model. The condition $\gamma \ll 1$, or $h \gg H$, may be written as

$$H\Lambda \gg \beta \left(\frac{NH}{f_0} \right)^2. \quad (9.174)$$

Compare this with the necessary condition for instability in a two-level model, (9.123), namely

$$(U_1 - U_2) > \beta \left(\frac{NH_\Delta}{f_0} \right)^2, \quad (9.175)$$

where H_Δ is the vertical distance between the two levels. Thus, essentially the same condition governs the onset of instability in the two-level model as governs the production of deep modes in the continuous model. This correspondence is a natural one, because in the two-level model *all* modes are ‘deep’, and the model fails (as it should) to capture the shallow modes of the continuous system. For similar reasons, there is a high-wavenumber cut-off in the two-level model: in the continuous model these modes are shallow and so cannot be captured by two-level dynamics. Somewhat counter-intuitively, for these modes the β -effect must be important, even though the modes have small horizontal scale: when $\beta = 0$ the instability arises via an interaction between edge waves at the top and bottom of the domain, whereas the shallow instability arises via an interaction of the edge waves at the surface with Rossby waves just above the surface.

9.9.2 Some numerical calculations

Adding β to the Eady model

Our first step is to add the β -effect to the Eady problem.¹⁶ That is, we suppose a Boussinesq fluid with uniform stratification, that the shear is zonal and constant and that the entire problem is sandwiched between two rigid surfaces. Growth rates and phase speeds of such an instability calculation are illustrated in Fig. 9.19 and the vertical structure is shown in Fig. 9.20. As in the two-layer problem, there is a low-wavenumber cut-off to the main instability, although there is now an additional weak instability at very large-scales. These so-called *Green modes* have no counterpart in the two-layer model — they are deep, slowly growing modes that will be dominated by faster growing modes in most real situations. (Also, the fact that the Green modes have a scale much larger than the deformation scale suggests that a degree of caution in the accuracy of the quasi-geostrophic calculation is warranted.) At high wavenumbers there is no cut-off to the instability in the continuous problem in the case of non-zero beta; the high-wavenumber modes are shallow and unstable via an interaction between edge waves at the lower boundary and Rossby waves in the lower atmosphere, and so have no counterpart in either the two-layer problem (where the modes are deep) or the Eady problem (which has no Rossby waves).

Effects of non-uniform shear and stratification

If the shear or stratification is non-uniform an analytic treatment is, even in problems without β , usually impossible and the resulting equations must be solved numerically. However, if we restrict attention to a *discontinuity* in the shear or the stratification, then the resulting problem is very

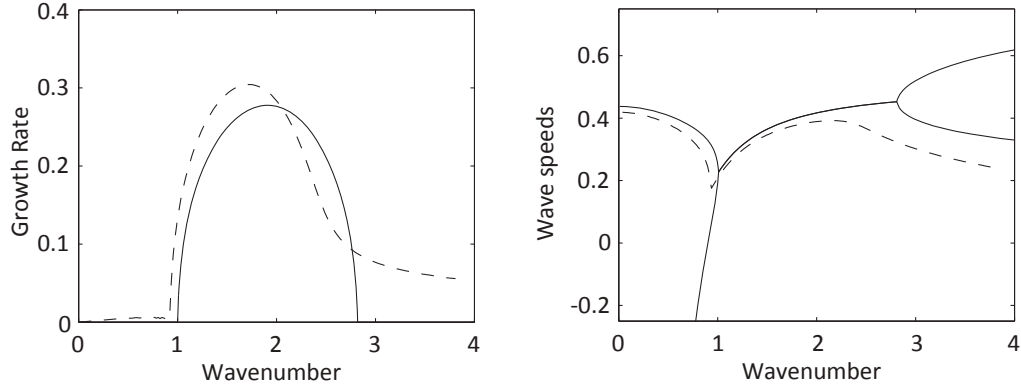


Fig. 9.19 Growth rates and wave speeds for the two-layer (solid) and continuous (dashed) models, with the same values of the Charney–Green number, γ , and uniform shear and stratification. (In the two-layer case $\gamma = \beta L_d^2 / [2(U_1 - U_2)] = 0.5$, and in the continuous case $\gamma = \beta L_d^2 / (H\Lambda) = 0.5$.) In the continuous case only the wave speed associated with the unstable mode is shown. In the two-layer case there are two real wave speeds which coalesce in the unstable region.

similar to the problem with rigid boundaries, and this property provides much of the justification for using the Eady problem to model instabilities in the Earth's atmosphere: in the troposphere the stratification is (approximately) constant, and the rapid increase in stratification in the stratosphere can be approximated by a lid at tropopause. Heuristically, we can see this from the form of the thermodynamic equation, namely

$$\frac{Db}{Dt} + N^2 w = 0. \quad (9.176)$$

If N^2 is high this suggests w will be small, and a lid is the limiting case of this. The oceanic problem is rather more involved, because although both the stratification and the shear are concentrated in the upper ocean, they vary relatively smoothly; furthermore, the shear is high where the stratification is high, and the two have opposing effects.

To go one step further, consider the Boussinesq potential vorticity equation, linearized about a zonally uniform state $\Psi(y, z)$, with a rigid surface at $z = 0$. The normal-mode evolution equations are similar to (9.69), namely

$$(U - c) \left[\frac{\partial^2}{\partial y^2} - k^2 + \frac{\partial}{\partial z} \left(F \frac{\partial}{\partial z} \right) \right] \tilde{\psi} + \frac{\partial Q}{\partial y} \tilde{\psi} = 0, \quad z > 0, \quad (9.177a)$$

$$(U - c) \frac{\partial \tilde{\psi}}{\partial z} - \frac{\partial U}{\partial z} \tilde{\psi} = 0, \quad \text{at } z = 0, \quad (9.177b)$$

where $\partial_y Q = \beta - \partial_{yy} U - \partial_z (F \partial_z U)$. Now suppose that there is a discontinuity in the shear and/or the stratification in the interior of the fluid, at some level $z = z_c$. Integrating (9.177a) across the discontinuity, noting that $\tilde{\psi}$ is continuous in z , gives

$$(U - c) \left[F \frac{\partial \tilde{\psi}}{\partial z} \right]_{z_c^-}^{z_c^+} - \tilde{\psi}(y, z_c) \left[F \frac{\partial U}{\partial z} \right]_{z_c^-}^{z_c^+} = 0. \quad (9.178)$$

which has similar form to (9.177b). This construction is evocative of the equivalence of a delta-function sheet of potential vorticity at a rigid boundary, except that now a discontinuity in the potential vorticity in the *interior* has a similarity with a rigid boundary.

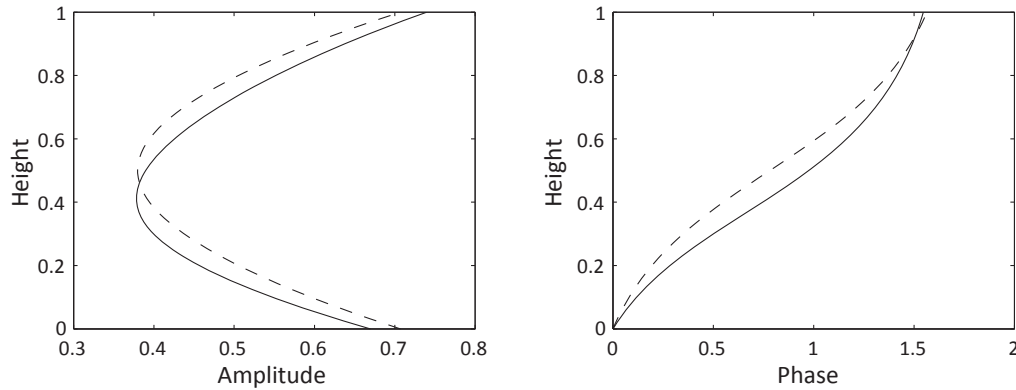


Fig. 9.20 Vertical structure of the most unstable modes in a continuously stratified instability calculation with $\beta = 0$ (dashed lines, the Eady problem) and $\beta \neq 0$ (solid lines), as in Fig. 9.19. The effect of beta is to depress the height of the maximum amplitude of the instability.

We can illustrate the effects of an interior discontinuity that crudely represents the tropopause by numerically solving the linear eigenvalue problem. We pose the problem on the f -plane, in a horizontally doubly-periodic domain, with no horizontal variation of shear, and between two horizontal rigid lids. The eigenvalue problem is defined by (9.69), and the numerical procedure then solves for the complex eigenvalue c and eigenfunction $\tilde{\psi}(z)$; various results are illustrated in Fig. 9.21. To parse this rather complex figure, first look at the solid curves in all the panels. These arise when the problem is solved with a uniform shear and a uniform stratification, with a lid at $z = 0$ and $z = 1$, hence simply giving the Eady problem. The familiar growth rates and vertical structure of the solution are given by the solid curves in panels (b), (c) and (d), and these are just the same as in Fig. 9.10. The various dotted and dashed curves show the results when the lid at $z = 1$ is replaced by a stratosphere stretching from $1 < z < 2$ either with high stratification, zero shear, or both, and in all of these cases the stratosphere acts qualitatively in the same way as a rigid lid. The vertical structure of the solution in the troposphere is, in all cases, quite similar, and the amplitude decays rapidly above the idealized tropopause, consistent with the almost uniform phase of the disturbance illustrated in panel (d) — recall that a tilting of the disturbance with height is necessary for instability. It is these properties that make the Eady problem of more general applicability to the Earth's atmosphere than might be first thought: the high stratification above the tropopause and consequent decay of the instability is mimicked by the imposition of a rigid lid.

In the ocean, the stratification is highest in the upper ocean where the shear is also strongest, and numerical calculations of the structure and growth rate of idealized profiles are illustrated in Fig. 9.22.¹⁷ The solid curve shows the Eady problem, and the various dashed curves show the phase speeds, growth rates and phase with combinations of the profiles illustrated in panel (a). Much of the ocean is characterized by having both a higher shear and a higher stratification in the upper 1 km or so, and this case is shown with the dotted line in Fig. 9.22. The amplitude of the instability is also largely confined to the upper ocean, and unlike the Eady problem it does not arise through the interaction of edge waves at the top and bottom: the potential vorticity changes sign because of the interior variations due to the non-uniform shear, mainly in the upper ocean. Consistently, the phase of the baroclinic waves is nearly constant in the lower ocean in those cases in which the shear is confined to the upper ocean. The real ocean is still more complicated, because the most unstable regions near intense western boundary currents are often also barotropically unstable, and the mean flow itself may be meridionally directed. Nevertheless, the result that linear baroclinic instability is primarily an *upper ocean* phenomenon is quite robust. However, we will

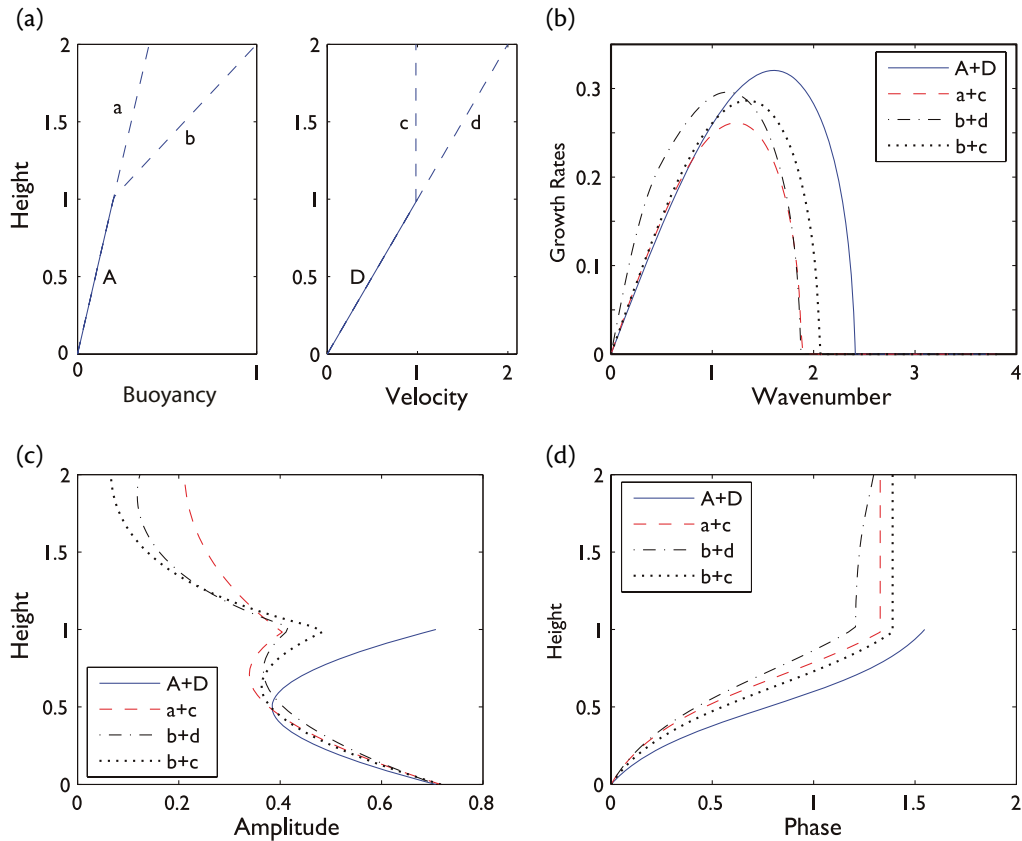


Fig. 9.21 The effect of a stratosphere on baroclinic instability: (a) the given profiles of shear and stratification; (b) the growth rate of the instabilities; (c) the amplitude of the most unstable mode as a function of height; (d) the phase of the most unstable mode. The instability problem is solved numerically with various profiles of stratification and shear. In each profile, in the idealized troposphere ($z < 1$) the shear and stratification are uniform and the same in each case. We consider four idealized stratospheres ($z \geq 1$): 1, A lid at $z = 1$, i.e., no stratosphere, giving the Eady problem itself (profiles A+D, solid lines); 2, stratospheric stratification as for the troposphere, but with zero shear (profiles a+c, dashed); 3, stratospheric shear as for troposphere, but with stratification (N^2) four times the tropospheric value (b+d, dot-dashed); 4, zero shear and high stratification in the stratosphere (b+c, dotted). In the troposphere the amplitude and structure of the instability are similar in all cases, illustrating the similarity of a rigid lid and abrupt changes in shear or stratification. Either a high stratification or a low shear (or both) will result in weak stratospheric instability.

find in chapter 12 that the nonlinear evolution of baroclinic instability leads to eddies throughout the water column.

Notes

- 1 Thomson (1871), Helmholtz (1868). Thomson later became Lord Kelvin.
- 2 See Drazin & Reid (1981) or Chandrasekhar (1961) for more detail.
- 3 This is Squire's theorem, which states that for every three-dimensional disturbance to a plane-parallel flow there exists a more unstable two-dimensional one (Squire 1933). This means there is no need to consider three dimensional effects to determine whether such a flow is unstable.

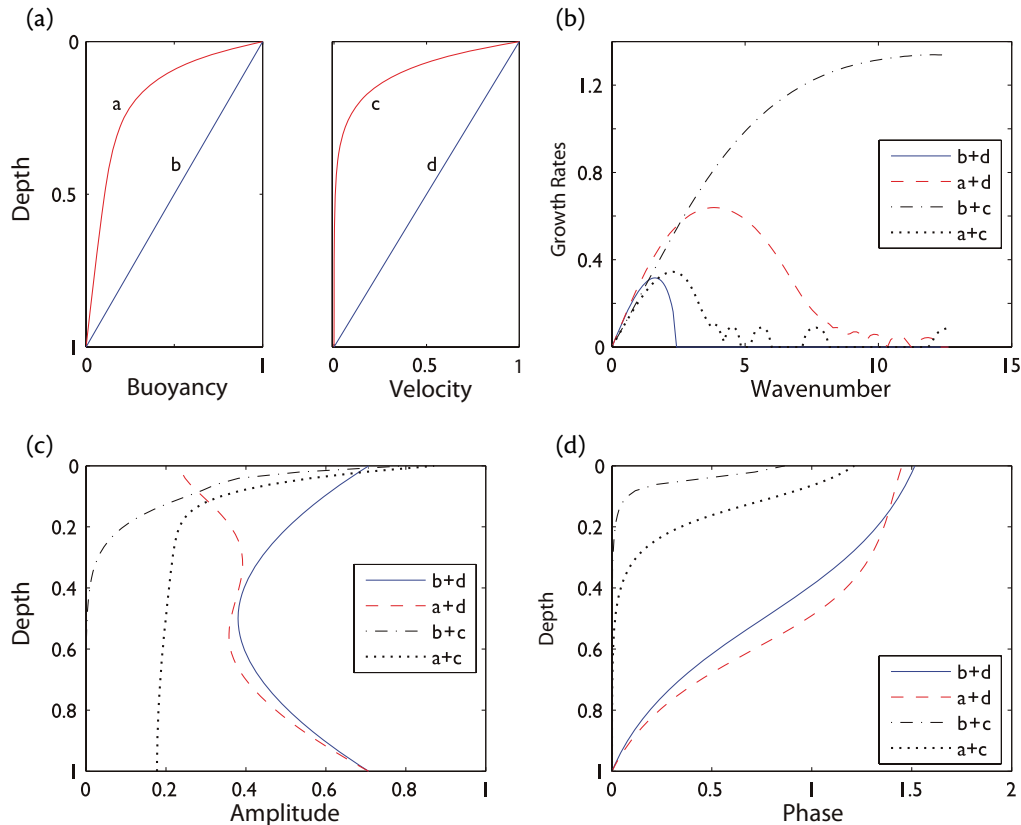


Fig. 9.22 The baroclinic instability in an idealized ocean, with four different profiles of shear or stratification. The panels are: (a) the profiles of velocity and buoyancy (and so N^2) used; (b) the growth rates of the various cases; (c) the vertical structure of the amplitude of the most unstable models; (d) the phase in the vertical of the most unstable modes. The instability is calculated numerically with four combinations of shear and stratification: 1, uniform stratification and shear i.e., the Eady problem, (profiles b+d, solid lines); 2, uniform shear, upper-ocean enhanced stratification (a+d, dashed); 3, uniform stratification, upper ocean enhanced shear (b+c, dot-dashed); 4, both stratification and shear enhanced in the upper ocean (a+c, dotted). Case 2 (a+d, dashed) is really more like an atmosphere with a stratosphere (see Fig. 9.21), and the amplitude of the disturbance falls off, rather unrealistically, in the upper ocean. Case 4 (a+c, dotted) is the most oceanographically relevant.

4 Rayleigh (1880).

5 First obtained by Rayleigh (1894). I also thank Adrian Matthews for comments and pointing out an error in an earlier derivation.

6 Rayleigh (1980) and, for the case with β , Kuo (1949).

7 Fjørtoft (1950).

8 Charney & Stern (1962), Pedlosky (1964).

9 Eady (1949), Charney (1947). Eric Eady (1915–1966) is best remembered today as the author of the iconic ‘Eady model’ of baroclinic instability, which describes the fundamental hydrodynamic instability mechanism that gives rise to weather systems. After an undergraduate education in mathematics he joined the U. K. Meteorological Office in 1937, becoming a forecaster and upper air analyst, in which capacity he served throughout the war. In 1946 he joined the Department of Mathematics at Imperial College, presenting his Ph.D. thesis in 1948 on ‘The theory of development

in dynamical meteorology', subsequently summarized in *Tellus* (Eady 1949). This work, masterly in its combination of austerity and relevance, provides a mathematical description of the essential aspects of cyclone development that stands to this day as a canonical model in the field. It also includes, rather obliquely, a derivation of the stratified quasi-geostrophic equations, albeit in a special form. The impact of the work was immediate and it led to visits to Bergen (in 1947 with J. Bjerknes), Stockholm (in 1952 with C.-G. Rossby) and Princeton (in 1953 with J. von Neumann and Charney). Eady followed his baroclinic instability work with prescient discussions of the general circulation of the atmosphere (Eady 1950, Eady & Sawyer 1951, Eady 1954). A perfectionist who sought to understand it all, Eady's subsequent published output was small and he later turned his attention to fundamental problems in other areas of fluid mechanics, the dynamics of the Sun and the Earth's interior, and biochemistry. He became increasingly divorced from normal scientific and human discourse, and finally took his own life. There is little published about him, save for the obituary by Charnock *et al.* (1966).

Jule Charney (1917–1981) played a defining role in dynamical meteorology in the second half of the twentieth century. He made seminal contributions in many areas including: the theory of baroclinic instability (Charney 1947); a systematic scaling theory for large-scale atmospheric motions and the derivation of the quasi-geostrophic equations (Charney 1948); a theory of stationary waves in the atmosphere (Charney & Eliassen 1949); the demonstration of the feasibility of numerical weather forecasts (Charney *et al.* 1950); planetary wave propagation into the stratosphere (Charney & Drazin 1961); a criterion for baroclinic instability (Charney & Stern 1962); a theory for hurricane growth (Charney & Eliassen 1964); and the concept of geostrophic turbulence (Charney 1971). His Ph.D. from UCLA in 1946 was entitled 'Dynamics of long waves in a baroclinic westerly current' and this became his well-known 1947 paper. After this he spent a year at Chicago and another at Oslo, and in 1948 joined the Institute of Advanced Study in Princeton where he stayed until 1956 (and where Eady visited for a while). He spent most of his subsequent career at MIT, interspersed with many visits to Europe, especially Norway. For a more complete picture of Charney, see Lindzen *et al.* (1990) and a brief biography by N. Phillips, available at <http://www.nap.edu/readingroom/books/biomems/jcharney.html>.

- 10 At least I find it so. My treatment of the Eady problem draws from unpublished notes by J. S. A. Green, as well as from Eady (1949) itself.
- 11 If c is real (and the waves are neutral), then there exists the possibility that $\Delta z - c = 0$, and the equation for Φ is

$$\frac{d^2 \Phi}{dz^2} - \mu^2 \Phi = C\delta(z - z_c), \quad z_c = c/\Lambda. \quad (9.179)$$

where C is a constant. Because z_c is continuous in the interval $[0, 1]$ so is c , and these solutions have a *continuous spectrum* of eigenvalues. The associated eigenfunctions provide formal completeness to the normal modes, enabling any function to be represented as their superposition.

- 12 After Phillips (1954).
- 13 Our non-dimensionalization of the two-layer system is such as to be in correspondence with that for the continuous system. Thus we choose H to be the total depth of the domain. This choice produces growth rates and wavenumbers that are equivalent to those in the Eady problem.
- 14 Green (1960) and Branscome (1983). Lindzen & Farrell (1980) also provide an approximate calculation of growth rates in the Charney problem.
- 15 After Charney (1947), in whose problem it appears, and Green (1960), who appreciated its importance.
- 16 Our numerical procedure is to assume a wave-like solution in the horizontal direction of the form $\tilde{\psi} \exp[i(kx + ly - \omega t)]$, and to finite difference the equations in the vertical direction. The resulting eigenvalue equations are solved by standard matrix methods, for each horizontal wavenumber. See Smith & Vallis (1998).
- 17 Gill *et al.* (1974) and Robinson & McWilliams (1974) were among the first to look at baroclinic instability in the ocean.

Strange as it may sound, the power of mathematics rests on its evasion of all unnecessary thought and on its wonderful saving of mental operations.
Ernst Mach (1838–1916).

CHAPTER 10

Waves, Mean-Flows, and their Interaction

WAVE-MEAN-FLOW INTERACTION is concerned with how some mean flow, perhaps a time or zonal average, interacts with a wave-like departure from that mean, and this chapter provides an elementary introduction to this topic. It is ‘elementary’ because our derivations and discussion are obtained by direct and straightforward manipulations of the equations of motion, often in the simplest case that will illustrate the relevant principle. It is implicit in what we do that it is a sensible thing to decompose the fields into a mean plus some departure, and one case when this is so is when the departure is of small amplitude. Departures from the mean — generically called *eddies* — are in reality not always small; for example, in the mid-latitude troposphere the eddies are often of similar amplitude to the mean flow, and chapters 12 and 13 will explore this from the standpoint of turbulence. However, in this chapter we will assume that eddies are indeed of small amplitude, and, in particular, that eddy-mean-flow interaction is larger than eddy-eddy interaction.

A *wave* is an eddy that satisfies, at least approximately, a dispersion relation. It is the presence of such a dispersion relation that enables a number of results to be obtained that would otherwise be out of our reach, and that gives rise to the appellation ‘wave-mean-flow’. It is implicit in defining waves this way that they are generally of small amplitude, for it is this that allows the equations of motion to be sensibly linearized and a dispersion relation to be obtained (although in some circumstances a wave may have finite amplitude and still satisfy a dispersion relation). However, having a small amplitude does not mean that the waves do not interact with each other and with the mean flow. Further, we may expect, or at least hope, that the qualitative nature of such interactions, as calculated by wave-mean-flow interaction theory, will carry over and provide insights into the finite-amplitude problem. Thus, one goal of wave-mean-flow theory is to provide a way of qualitatively understanding more realistic situations, and to suggest diagnostics that might be used to analyze both observations and numerical solutions of the fully nonlinear problem. In this chapter we will largely concern ourselves with a *zonal* mean, since this is the simplest and often most useful case because of the presence of simple — in particular periodic — boundary conditions. We will also be mainly concerned with quasi-geostrophic dynamics on a β -plane (and hence Rossby waves), using Boussinesq dynamics, since here the concepts are most clearly illustrated, but gravity waves interaction is certainly important, throughout the ocean and in the stratosphere. Readers who are anxious for real examples may wish to first look at chapters 15 and 17 and then come back to this chapter as needed.

10.1 QUASI-GEOSTROPHIC WAVE–MEAN-FLOW INTERACTION

10.1.1 Preliminaries

To fix our dynamical system and notation, we write down the Boussinesq quasi-geostrophic potential vorticity equation

$$\frac{\partial q}{\partial t} + J(\psi, q) = D, \quad (10.1)$$

where D represents any non-conservative terms and the potential vorticity in a Boussinesq system is

$$q = \beta y + \zeta + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} b \right), \quad (10.2)$$

where ζ is the relative vorticity and b is the buoyancy perturbation from the background state characterized by N^2 . [In an ideal gas $q = \beta y + \zeta + (f_0/\rho_R)\partial_z(\rho_R b/N^2)$, where ρ_R is a specified density profile, and most of our derivations can be extended to that case.] We will refer to lines of constant b as isentropes. In terms of the streamfunction, the variables are

$$\zeta = \nabla^2 \psi, \quad b = f_0 \frac{\partial \psi}{\partial z}, \quad q = \beta y + \left[\nabla^2 + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \psi. \quad (10.3)$$

where $\nabla^2 \equiv (\partial_x^2 + \partial_y^2)$. The potential vorticity equation holds in the fluid interior; the boundary conditions on (10.3) are provided by the thermodynamic equation

$$\frac{\partial b}{\partial t} + J(\psi, b) + wN^2 = H, \quad (10.4)$$

where H represents heating terms. The vertical velocity at the boundary, w , is zero in the absence of topography and Ekman friction, and if H is also zero the boundary condition is just

$$\frac{\partial b}{\partial t} + J(\psi, b) = 0. \quad (10.5)$$

Equations (10.1) and (10.5) are the evolution equations for the system and if both D and H are zero they conserve both the total energy, $\hat{E}$ and the total enstrophy, $\hat{Z}$:

$$\begin{aligned} \frac{d\hat{E}}{dt} &= 0, & \hat{E} &= \frac{1}{2} \int_V (\nabla \psi)^2 + \frac{f_0^2}{N^2} \left(\frac{\partial \psi}{\partial z} \right)^2 dV, \\ \frac{d\hat{Z}}{dt} &= 0, & \hat{Z} &= \frac{1}{2} \int_V q^2 dV. \end{aligned} \quad (10.6)$$

where V is a volume bounded by surfaces at which the normal velocity is zero, or that has periodic boundary conditions. The enstrophy is also conserved layerwise; that is, the horizontal integral of q^2 is conserved at every level.

10.1.2 Potential vorticity flux in the linear equations

Let us decompose the fields into a mean (to be denoted with an overbar) plus a perturbation (denoted with a prime), and let us suppose the perturbation fields are of small amplitude. (In linear problems, such as those considered in chapter 9, we decomposed the flow into a ‘basic state’ plus a perturbation, with the basic state fixed in time. Our approach here is similar, but soon we will allow the mean state to evolve.) The linearized quasi-geostrophic potential vorticity equation is then

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + u' \frac{\partial \bar{q}}{\partial x} + \bar{v} \frac{\partial q'}{\partial y} + v' \frac{\partial \bar{q}}{\partial y} = D'. \quad (10.7)$$

where D' represents eddy forcing and dissipation and, in terms of streamfunction,

$$(u'(x, y, z, t), v'(x, y, z, t)) = \left(-\frac{\partial \psi'}{\partial y}, \frac{\partial \psi'}{\partial x} \right), \quad (10.8a)$$

$$q'(x, y, z, t) = \nabla^2 \psi' + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial z} \right). \quad (10.8b)$$

If the mean is a zonal mean then $\partial \bar{q}/\partial x = 0$ and $\bar{v} = 0$ (because v is purely geostrophic) and (10.7) simplifies to

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = D', \quad (10.9)$$

where

$$\bar{q} = \beta y - \frac{\partial \bar{u}}{\partial y} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \bar{b} \right), \quad \text{and} \quad \frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right). \quad (10.10a,b)$$

using thermal wind, $f_0 \partial \bar{u} / \partial z = -\partial b / \partial y$.

Multiplying by q' and zonally averaging gives the enstrophy equation:

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} = -\overline{v' q' \frac{\partial \bar{q}}{\partial y}} + \overline{D' q'}. \quad (10.11)$$

The quantity $\overline{v' q'}$ is the meridional flux of potential vorticity; this is downgradient (by definition) when the first term on the right-hand side is positive (i.e., $\overline{v' q' \partial \bar{q} / \partial y} < 0$), and it then acts to increase the variance of the perturbation. (This occurs, for example, when the flux is diffusive so that $\overline{v' q'} = -\kappa \partial \bar{q} / \partial y$, where κ may vary but is everywhere positive.) This argument may be inverted: for inviscid flow ($D = 0$), if the waves are growing, as for example in the canonical models of baroclinic instability discussed in chapter 9, then *the potential vorticity flux is downgradient*.

If the second term on the right-hand side of (10.11) is negative, as it will be if D' is a dissipative process (e.g., if $D' = A \nabla^2 q'$ or if $D' = -r q'$, where A and r are positive) then a statistical balance can be achieved between enstrophy production via downgradient transport, and dissipation. If the waves are steady (by which we mean statistically steady, neither growing nor decaying in amplitude) and conservative (i.e., $D' = 0$) then we must have

$$\overline{v' q'} = 0. \quad (10.12)$$

Similar results follow for the buoyancy at the boundary; we start by linearizing the thermodynamic equation (10.5) to give

$$\frac{\partial b'}{\partial t} + \bar{u} \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = H', \quad (10.13)$$

where H' is a diabatic source term. Multiplying (10.13) by b' and averaging gives

$$\frac{1}{2} \frac{\partial \overline{b'^2}}{\partial t} = -\overline{v' b' \frac{\partial \bar{b}}{\partial y}} + \overline{H' b'}. \quad (10.14)$$

Thus growing adiabatic waves have a downgradient flux of buoyancy at the boundary. In the Eady problem there is no interior gradient of basic-state potential vorticity and all the terms in (10.11) are zero, but the perturbation grows at the boundary. If the waves are steady and adiabatic then, analogously to (10.12),

$$\overline{v' b'} = 0. \quad (10.15)$$

The boundary conditions and fluxes may be absorbed into the interior definition of potential vorticity and its fluxes by way of the delta-function boundary layer construction, described in section 5.4.3. In models with discrete vertical layers or a finite number of levels it is common practice to absorb the boundary conditions into the definition of potential vorticity at top and bottom.

10.1.3 Wave–mean-flow interaction

In linear problems we usually suppose that the mean flow is fixed and in that case in the zonal mean terms, $\bar{u}$ and $\bar{q}$ in (10.9), would be functions only of y and z . However, we in reality we might expect that the mean flow would change because of momentum and heat flux convergences arising from the eddy-eddy interactions. To calculate these changes we begin with the potential vorticity equation (10.1) and, in the usual way, express the variables as a zonal mean plus an eddy term and obtain

$$\frac{\partial \bar{q}}{\partial t} + \nabla \cdot (\bar{u} \bar{q}) + \nabla \cdot (\overline{u'q'}) = \bar{D}. \quad (10.16)$$

Now, since the mean flow is a zonal mean, and $\bar{v} = 0$, the first term is zero and the mean flow evolves according to

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial}{\partial y} \overline{v'q'} = \bar{D}. \quad (10.17)$$

Similarly, at the boundary the mean buoyancy evolution equation is

$$\frac{\partial \bar{b}}{\partial t} + \frac{\partial}{\partial y} \overline{v'b'} = \bar{H}. \quad (10.18)$$

To obtain $\bar{u}$ from $\bar{q}$ and $\bar{b}$ we use thermal wind balance to define a streamfunction Ψ . That is, since

$$f_0 \frac{\partial \bar{u}}{\partial z} = -\frac{\partial \bar{b}}{\partial y}, \quad \text{then} \quad \left(\bar{u}, \frac{1}{f_0} \bar{b} \right) = \left(-\frac{\partial \Psi}{\partial y}, \frac{\partial \Psi}{\partial z} \right) \quad (10.19a,b)$$

whence, using (10.10a), the potential vorticity is

$$\bar{q}(y, z, t) - \beta y = \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \Psi}{\partial z} \right) + \frac{\partial^2 \Psi}{\partial y^2}. \quad (10.20)$$

If $\bar{q}$ is known in the interior from (10.18), and $\bar{b}$ (i.e., $f_0 \partial \Psi / \partial z$) is known at the boundaries, then $\bar{u}$ and $\bar{b}$ in the interior may be obtained using (10.20) and (10.19b). The equations are summarized in the grey box on page 392.

To close the system we suppose that the eddy terms themselves evolve according to (10.9) and (10.13). If in those equations we were to include the eddy-eddy interaction terms we would simply recover the full system, so in neglecting those terms we have constructed an eddy–mean-flow system, commonly called a *wave–mean-flow* system because by eliminating the nonlinear terms in the perturbation equation the eddies will often be wavelike. Non quasi-geostrophic wave–mean-flow systems may be constructed in a similar fashion: for example, we could construct a system using the primitive equations with separate equations for eddy and zonal-mean temperature and velocity fields, and an example involving gravity waves is given in chapter 17.

It is important to realise that such systems do differ from linear ones. In constructing linear systems we posit that the eddy terms are small compared to the mean flow and thus neglect the eddy-eddy interaction terms and keep the mean flow fixed. In a wave–mean-flow problem we similarly suppose the eddy terms are small, and we neglect eddy-eddy interaction terms where they produce another eddy, because the terms involving the mean flow are larger. However, in the mean flow equation, (10.16), there are no larger mean flow terms and we keep the eddy-eddy terms and allow the mean flow to evolve. Such a justification is hardly a rigorous one, since if the eddy terms are small then the effects on the mean flow will be small, and so one might suppose that the mean flow should be held fixed. The wave–mean flow equations really can only be justified on a case-by-case basis with a detailed examination of the size of the terms and the rate at which they evolve, and that is the subject of weakly nonlinear theory. Another justification for wave–mean-flow problems is that they lead to insight into the behaviour of the full system.

We now consider some more properties of the waves themselves — how they propagate and what they conserve — beginning with a discussion of the potential vorticity flux and its relative, the Eliassen–Palm flux

10.2 THE ELIASSEN–PALM FLUX

The eddy flux of potential vorticity may be expressed in terms of vorticity and buoyancy fluxes as

$$v'q' = v'\zeta' + f_0v'\frac{\partial}{\partial z}\left(\frac{b'}{N^2}\right). \quad (10.21)$$

The second term on the right-hand side can be written as

$$\begin{aligned} f_0v'\frac{\partial}{\partial z}\left(\frac{b'}{N^2}\right) &= f_0\frac{\partial}{\partial z}\left(\frac{v'b'}{N^2}\right) - f_0\frac{\partial v'}{\partial z}\frac{b'}{N^2} \\ &= f_0\frac{\partial}{\partial z}\left(\frac{v'b'}{N^2}\right) - f_0\frac{\partial}{\partial x}\left(\frac{\partial\psi'}{\partial z}\right)\frac{b'}{N^2} \\ &= f_0\frac{\partial}{\partial z}\left(\frac{v'b'}{N^2}\right) - \frac{f_0^2}{2N^2}\frac{\partial}{\partial x}\left(\frac{\partial\psi'}{\partial z}\right)^2, \end{aligned} \quad (10.22)$$

using $b' = f_0\partial\psi'/\partial z$.

Similarly, the flux of relative vorticity can be written

$$v'\zeta' = -\frac{\partial}{\partial y}(u'v') + \frac{1}{2}\frac{\partial}{\partial x}(v'^2 - u'^2) \quad (10.23)$$

Using (10.22) and (10.23), (10.21) becomes

$$v'q' = -\frac{\partial}{\partial y}(u'v') + \frac{\partial}{\partial z}\left(\frac{f_0}{N^2}v'b'\right) + \frac{1}{2}\frac{\partial}{\partial x}\left((v'^2 - u'^2) - \frac{b'^2}{N^2}\right). \quad (10.24)$$

Thus the meridional potential vorticity flux, in the quasi-geostrophic approximation, can be written as the divergence of a vector: $v'q' = \nabla \cdot \mathcal{E}$ where

$$\mathcal{E} \equiv \frac{1}{2}\left((v'^2 - u'^2) - \frac{b'^2}{N^2}\right)\mathbf{i} - (u'v')\mathbf{j} + \left(\frac{f_0}{N^2}v'b'\right)\mathbf{k}. \quad (10.25)$$

A particularly useful form of this arises after zonally averaging, for then (10.24) becomes

$$\overline{v'q'} = -\frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\left(\frac{f_0}{N^2}\overline{v'b'}\right). \quad (10.26)$$

The vector defined by

$$\mathcal{F} \equiv -\overline{u'v'}\mathbf{j} + \frac{f_0}{N^2}\overline{v'b'}\mathbf{k} \quad (10.27)$$

is called the (quasi-geostrophic) *Eliassen–Palm (EP) flux*,¹ and its divergence, given by (10.26), gives the poleward flux of potential vorticity:

$$\overline{v'q'} = \nabla_x \cdot \mathcal{F}, \quad (10.28)$$

where $\nabla_x \cdot \equiv (\partial/\partial y, \partial/\partial z) \cdot$ is the divergence in the meridional plane. Unless the meaning is unclear, the subscript x on the meridional divergence will be dropped.

10.2.1 The Eliassen–Palm relation

On dividing by $\partial\bar{q}/\partial y$ and using (10.28), the enstrophy equation (10.11) becomes

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (10.29a)$$

where

$$\mathcal{A} = \frac{\overline{q'^2}}{2\partial\bar{q}/\partial y}, \quad \mathcal{D} = \frac{\overline{D'q'}}{\partial\bar{q}/\partial y}. \quad (10.29b)$$

and $\mathcal{F}$ is given by (10.27). Equation (10.29a) is known as the *Eliassen–Palm relation*, and it is a conservation law for the *wave activity density* $\mathcal{A}$. The conservation law is exact (in the linear approximation) if the mean flow is constant in time. It will be a good approximation if $\partial\bar{q}/\partial y$ varies slowly compared to the variation of $\overline{q'^2}$.

If we integrate (10.29b) over a meridional area A bounded by walls where the eddy activity vanishes, and if $\mathcal{D} = 0$, we obtain

$$\frac{d}{dt} \int_A \mathcal{A} dA = 0. \quad (10.30)$$

The integral is a wave activity — a quantity that is quadratic in the amplitude of the perturbation and that is conserved in the absence of forcing and dissipation. In this case $\mathcal{A}$ is the negative of the *pseudomomentum*, for reasons we will encounter later. (‘Wave action’ is a particular form of wave activity; it is the energy divided by the frequency and it is a conserved property in many wave problems.) Note that neither the perturbation energy nor the perturbation enstrophy are wave activities of the linearized equations, because there can be an exchange of energy or enstrophy between mean and perturbation — indeed, this is how a perturbation grows in baroclinic or barotropic instability! This is already evident from (10.11), or in general take (10.7) with $D' = 0$ and multiply by q' to give the enstrophy equation

$$\frac{1}{2} \frac{\partial \overline{q'^2}}{\partial t} + \frac{1}{2} \overline{\mathbf{u} \cdot \nabla q'^2} + \overline{\mathbf{u}' q' \cdot \nabla \bar{q}} = 0, \quad (10.31)$$

where here the overbar is an average (although it need not be a zonal average). Integrating this over a volume V gives

$$\frac{d\hat{Z}'}{dt} \equiv \frac{d}{dt} \int_V \frac{1}{2} \overline{q'^2} dV = - \int_V \overline{\mathbf{u}' q' \cdot \nabla \bar{q}} dV. \quad (10.32)$$

The right-hand side does not, in general, vanish and so $\hat{Z}'$ is not in general conserved.

10.2.2 The group velocity property for Rossby waves

The vector $\mathcal{F}$ describes how the wave activity propagates. We noted in chapter 6 that in the case in which the disturbance is composed of plane or almost plane waves that satisfy a dispersion relation, then $\mathcal{F} = \mathbf{c}_g \mathcal{A}$, where $\mathbf{c}_g$ is the group velocity and (10.29a) becomes

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot (\mathcal{A} \mathbf{c}_g) = 0. \quad (10.33)$$

This is a useful property, because if we can diagnose $\mathbf{c}_g$ from observations we can use (10.29a) to determine how wave activity density propagates. Let us demonstrate this explicitly for the pseudomomentum in Rossby waves.

The Boussinesq quasi-geostrophic equation on the β -plane, linearized around a uniform zonal flow and with constant static stability, is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (10.34)$$

where $q' = [\nabla^2 + (f_0^2/N^2)\partial^2/\partial z^2]\psi'$ and, if $\bar{u}$ is constant, $\partial \bar{q}/\partial y = \beta$. Thus we have

$$\left(\frac{\partial}{\partial t} + \bar{u} \frac{\partial}{\partial x} \right) \left[\nabla^2 \psi' + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \beta \frac{\partial \psi'}{\partial x} = 0. \quad (10.35)$$

Seeking solutions of the form

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly + mz - \omega t)}, \quad (10.36)$$

we find the dispersion relation,

$$\omega = \bar{u}k - \frac{\beta k}{\kappa^2}. \quad (10.37)$$

where $\kappa^2 = (k^2 + l^2 + m^2 f_0^2/N^2)$, and the group velocity components:

$$c_g^y = \frac{2\beta k l}{\kappa^4}, \quad c_g^z = \frac{2\beta k m f_0^2/N^2}{\kappa^4}. \quad (10.38)$$

Also, if $u' = \text{Re } \tilde{u} \exp[i(kx + ly + mz - \omega t)]$, and similarly for the other fields, then

$$\begin{aligned} \tilde{u} &= -\text{Re } i l \tilde{\psi}, & \tilde{v} &= \text{Re } i k \tilde{\psi}, \\ \tilde{b} &= \text{Re } i m f_0 \tilde{\psi}, & \tilde{q} &= -\text{Re } \kappa^2 \tilde{\psi}, \end{aligned} \quad (10.39)$$

The wave activity density is then

$$\mathcal{A} = \frac{1}{2} \frac{\overline{q'^2}}{\beta} = \frac{\kappa^4}{4\beta} |\tilde{\psi}|^2, \quad (10.40)$$

where the additional factor of 2 in the denominator arises from the averaging. Using (10.39) the EP flux, (10.27), is

$$\mathcal{F}^y = -\overline{u'v'} = \frac{1}{2} k l |\tilde{\psi}|^2, \quad \mathcal{F}^z = \frac{f_0}{N^2} \overline{v'b'} = \frac{f_0^2}{2N^2} k m |\tilde{\psi}|^2. \quad (10.41)$$

Using (10.38), (10.40) and (10.41) we obtain

$$\mathcal{F} = (\mathcal{F}^y, \mathcal{F}^z) = \mathbf{c}_g \mathcal{A}. \quad (10.42)$$

If the properties of the medium are slowly varying, so that a (spatially varying) group velocity can still be defined, then this is a useful expression to estimate how the wave activity propagates in the atmosphere and in numerical simulations.

10.2.3 ♦ The orthogonality of modes

It is a direct consequence of the conservation of wave activity that disturbance modes are orthogonal in the ‘wave activity norm’, defined later on, and thus are a useful measure of the amplitude of a particular mode.² To explore this, we start with the linearized potential vorticity equation,

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0. \quad (10.43)$$

Let us formally seek solutions of the form $\psi' = \text{Re } \Psi \exp(ikx)$ where Ψ is the sum of *modes*,

$$\Psi = \sum_n \tilde{\psi}_n(y, z) e^{-ikc_n t}, \quad (10.44)$$

where n is an identifier of the modes. The modes satisfy

$$(\bar{u}\Delta_k^2 + \bar{q}_y)\tilde{\psi}_n = c_n\Delta_k^2\tilde{\psi}_n, \quad (10.45)$$

where

$$\Delta_k^2 = \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) - k^2. \quad (10.46)$$

The upper and lower boundary conditions (at $z = 0, -H$) are given by the thermodynamic equation

$$\frac{\partial b'}{\partial t} + \bar{u} \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = 0, \quad (10.47)$$

and if we simplify further by supposing $\partial \bar{u} / \partial z = 0$ then the boundary condition becomes

$$\frac{\partial \psi'_z}{\partial t} + \bar{u} \frac{\partial \psi'_z}{\partial x} = 0. \quad (10.48)$$

There are no meridional buoyancy fluxes at the boundary. If N^2 is a constant (a simplifying but not essential assumption) then we can let $\tilde{\psi}_n(y, z) = \psi_n(y) \cos pz$, with $p = j\pi/H$ where j is an integer and the mode n now labels only the meridional modes. The corresponding potential vorticity modes are given by

$$q_n = \Delta_{k,m}^2 \psi_n, \quad \Delta_{k,m}^2 = \frac{\partial^2}{\partial y^2} - (f_0^2/N^2)m^2 - k^2, \quad (10.49)$$

and the boundary conditions are then built in to any solution we construct from (10.45) and (10.49).³ We may then consider a single zonal and a single vertical wavenumber. (If there is no horizontal variation of the shear, the meridional modes are harmonic functions, for example $\psi_n \propto \sin(n\pi y/L)$ for a channel of width L .)

For a given basic state we may imagine solving (10.45), numerically or analytically, and determining the modes. However, these modes are not orthogonal in the sense of either energy or enstrophy. That is, denoting the inner product by

$$\langle a, b \rangle \equiv (2L)^{-1} \int_L ab \, dy, \quad (10.50)$$

then, in general,

$$I_E = \langle \psi_n, q_m \rangle \neq 0, \quad I_Z = \langle q_n, q_m \rangle \neq 0, \quad (10.51a,b)$$

for $n \neq m$, where $q_n = \Delta_{k,p}^2 \psi_n$. Perturbation energy and enstrophy are thus not wave activities of the linearized equations, and it is not meaningful to talk about the energy or enstrophy of a particular mode. However, by the same token we may expect orthogonality in the wave activity norm. To prove this and understand what it means, suppose that at $t = 0$ the disturbance consists of two modes, n and m , so that at a later time $q = (q_n e^{-ikc_n t} + q_m e^{-ikc_m t} + \text{c.c.})$, where $c_m \neq c_n$ and we assume that both are real. The wave activity is

$$P \equiv \int \mathcal{A} \, dy \, dz = \langle q_n, q_m^* / \bar{q}_y \rangle e^{-ik(c_n - c_m)t} + \langle q_m, q_n^* / \bar{q}_y \rangle + \langle q_n, q_n^* / \bar{q}_y \rangle + \text{c.c.} \quad (10.52)$$

The second and third terms on the right-hand side are the wave activities of each mode, and these are constants (to see this, consider the case when the disturbance is just a single mode). Now, because $dP/dt = 0$ the first term must vanish if $c_n \neq c_m$, implying the modes are orthogonal and, in particular,

$$\text{Re} \int \frac{1}{\bar{q}_y} q_n q_m^* dy = 0, \quad (10.53)$$

for $n \neq m$. The inner product weighted by $1/\bar{q}_y$ defines the wave activity norm. Orthogonality is a useful result, for it means that the wave activity is a proper measure of the amplitude of a given mode unlike, for example, energy. The conservation of wave activity will lead to a particularly straightforward derivation of the necessary conditions for stability, given in section 10.6.

10.3 THE TRANSFORMED EULERIAN MEAN

The so-called *transformed Eulerian mean*, or TEM, is a transformation of the equations of motion that provides a useful framework for discussing eddy effects under a wide range of conditions.⁴ It is useful because, as we shall see, it is equivalent to a very natural form of averaging the equations that serves to eliminate eddy fluxes in the thermodynamic equation and collect them together, in a simple form, in the momentum equation and in so doing it highlights the role of potential vorticity fluxes. The TEM also provides a natural separation between diabatic and adiabatic effects or between advective and diffusive fluxes and, in the case in which the flow is adiabatic, a pleasing simplification of the equations. In later chapters we will use the TEM to better understand the mid-latitude troposphere and the dynamics of the Antarctic Circumpolar Current, and as a framework for the parameterization of eddy fluxes. Of course, there being no free lunch, the TEM brings with it its own difficulties, and in particular the implementation of boundary conditions can cause difficulties, especially in the actual numerical integration of the equations.

10.3.1 Quasi-geostrophic form

For simplicity we will use the Boussinesq equations on the beta-plane, and the zonally averaged Eulerian mean equations for the zonally averaged zonal velocity and the buoyancy may then be written as (see section 2.2.6)

$$\frac{\partial \bar{u}}{\partial t} - (f + \bar{\zeta})\bar{v} + \bar{w} \frac{\partial \bar{u}}{\partial z} = -\frac{\partial}{\partial y} \overline{u'v'} - \frac{\partial}{\partial z} \overline{u'w'} + \bar{F}, \quad (10.54a)$$

$$\frac{\partial \bar{b}}{\partial t} + \bar{v} \frac{\partial \bar{b}}{\partial y} + \bar{w} \frac{\partial \bar{b}}{\partial z} = -\frac{\partial}{\partial y} \overline{v'b'} - \frac{\partial}{\partial z} \overline{w'b'} + \bar{S}, \quad (10.54b)$$

where $\bar{F}$ and $\bar{S}$ represent frictional and heating terms, respectively. Note that the meridional velocity, $\bar{v}$, is purely ageostrophic. Using quasi-geostrophic scaling we neglect the vertical eddy flux divergences and all ageostrophic velocities except when multiplied by f_0 or N^2 . The above equations then become

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v} - \frac{\partial}{\partial y} \overline{u'v'} + \bar{F}, \quad (10.55a)$$

$$\frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w} - \frac{\partial}{\partial y} \overline{v'b'} + \bar{S}. \quad (10.55b)$$

These two equations are connected by the thermal wind relation,

$$f_0 \frac{\partial \bar{u}}{\partial z} = -\frac{\partial \bar{b}}{\partial y}, \quad (10.56)$$

which is a combination of the geostrophic v -momentum equation ($f_0 \bar{u} = -\partial \bar{\phi} / \partial y$) and hydrostasy ($\partial \bar{\phi} / \partial z = \bar{b}$). One less than ideal aspect of (10.55) is that in the extratropics the dominant balance is usually between the first two terms on the right-hand sides of each equation, even in time-dependent cases. Thus, the Coriolis force closely balances the divergence of the eddy momentum fluxes, and the advection of the mean stratification ($N^2 \bar{w}$, or ‘adiabatic cooling’) often balances the divergence of eddy heat flux, with heating being a small residual. This may lead to an underestimation of the importance of diabatic heating, as this is ultimately responsible for the mean meridional circulation. Furthermore, the link between $\bar{u}$ and $\bar{b}$ via thermal wind dynamically couples buoyancy and momentum, and obscures the understanding of how the eddy fluxes influence these fields — is it through the eddy heat fluxes or momentum fluxes, or some combination?

To address this issue we combine the terms $N^2 \bar{w}$ and the eddy flux in (10.55b) into a single total or *residual* (so recognizing the cancellation between the mean and eddy terms) heat transport term that in a steady state is balanced by the diabatic term $\bar{S}$. To do this, we first note that because $\bar{v}$ and $\bar{w}$ are related by mass conservation we can define a mean meridional streamfunction ψ_m such that

$$(\bar{v}, \bar{w}) = \left(-\frac{\partial \psi_m}{\partial z}, \frac{\partial \psi_m}{\partial y} \right). \quad (10.57)$$

The velocities then satisfy $\partial \bar{v} / \partial y + \partial \bar{w} / \partial z = 0$ automatically. If we define a *residual streamfunction* by

$$\psi^* \equiv \psi_m + \frac{1}{N^2} \overline{v' b'}, \quad (10.58a)$$

the components of the *residual mean meridional circulation* are then given by

$$(\bar{v}^*, \bar{w}^*) = \left(-\frac{\partial \psi^*}{\partial z}, \frac{\partial \psi^*}{\partial y} \right), \quad (10.58b)$$

and

$$\bar{v}^* = \bar{v} - \frac{\partial}{\partial z} \left(\frac{1}{N^2} \overline{v' b'} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left(\frac{1}{N^2} \overline{v' b'} \right). \quad (10.59)$$

Note that by construction, the residual overturning circulation satisfies

$$\frac{\partial \bar{v}^*}{\partial y} + \frac{\partial \bar{w}^*}{\partial z} = 0. \quad (10.60)$$

Substituting (10.59) into (10.55a) and (10.55b) the zonal momentum and buoyancy equations then take the simple forms

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v}^* + \overline{v' q'} + \bar{F}, \quad (10.61a)$$

$$\frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w}^* + \bar{S}. \quad (10.61b)$$

which are known as the (quasi-geostrophic) *transformed Eulerian mean equations*, or TEM equations. The potential vorticity flux, $\overline{v' q'}$, is given in terms of the heat and vorticity fluxes by (10.26), and is equal to the divergence of the Eliassen–Palm flux as in (10.28).

The TEM equations make it apparent that we may consider the potential vorticity fluxes, rather than the separate contributions of the vorticity and heat fluxes, to force the circulation. If we know the potential vorticity flux as well as $\bar{F}$ and $\bar{S}$, then (10.60) and (10.61b), along with thermal wind balance

$$f_0 \frac{\partial \bar{u}}{\partial z} = -\frac{\partial \bar{b}}{\partial y}, \quad (10.62)$$

form a complete set. The meridional overturning circulation is obtained by eliminating time derivatives from (10.61b) using (10.62), giving

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = f_0 \frac{\partial}{\partial z} \overline{v'q'} + f_0 \frac{\partial \bar{F}}{\partial z} + \frac{\partial \bar{S}}{\partial y}. \quad (10.63)$$

Thus, the residual or net overturning circulation is driven by the (vertical derivative of the) potential vorticity fluxes and the diabatic terms — ‘driven’ in the sense that if we know those terms we can calculate the overturning circulation, although of course the fluxes themselves depend on the circulation. Note that this equation applies at every instant, even if the equations are not in a steady state.

Use of the equations in TEM form is particularly useful when the eddy potential vorticity flux arises from wave activity, for example from Rossby waves. The potential vorticity flux is the convergence of the EP flux $\mathcal{F}$, as in (10.28), and if the eddies satisfy a dispersion relation the components of the EP flux are equal to the group velocity multiplied by the wave activity density $\mathcal{A}$, as in (10.42). Thus, knowing the group velocity tells us a great deal about how momentum is transported by waves. We’ll use the TEM to deduce the mean flow acceleration in sections 10.4, 10.5 and, in particular, in section 17.4.

Connection to potential vorticity and wave–mean-flow interaction

If we take the curl of (10.61b) — that is, cross differentiate its components — then, after using the residual mass continuity equation (10.60), we recover the zonally averaged potential vorticity equation namely

$$\frac{\partial \bar{q}}{\partial t} = -\frac{\partial}{\partial y} \overline{v'q'} - \frac{\partial \bar{F}}{\partial y}, \quad \text{where} \quad \bar{q}(y, t) = \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \bar{b} \right) - \frac{\partial \bar{u}}{\partial y}, \quad (10.64a,b)$$

which is essentially the same as (10.18) and (10.20), noting that we may add βy to the definition of zonally-averaged potential vorticity with no effect.

The corresponding equation for the evolution of eddy potential vorticity is, in its inviscid form,

$$\left(\frac{\partial}{\partial t} + \bar{u}(y, t) \frac{\partial}{\partial x} \right) q' + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (10.65)$$

as in (10.7). Equations (10.64) and (10.65) are a closed set of quasi-linear equations, and we have recovered the wave–mean-flow system described in section 10.1.3.

10.3.2 The TEM in isentropic coordinates

The residual circulation has an illuminating interpretation if we think of the fluid as comprising multiple layers of shallow water, or equivalently if we cast the problem in isentropic coordinates (section 3.10). Using the notation of a shallow water system, the momentum and mass conservation equation can then be written as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - f \mathbf{v} = \mathbf{F}, \quad \frac{\partial h}{\partial t} + \nabla \cdot (h \mathbf{u}) = S. \quad (10.66a,b)$$

The quantity h is the thickness between two isentropic surfaces and S is a thickness source term. (The field h plays the same role as σ in section 3.10.) With quasi-geostrophic scaling, so that variations in Coriolis parameter and layer thickness are small, zonally averaging in a conventional way gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v'\zeta'} + \bar{F}, \quad \frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}}{\partial y} = -\frac{\partial}{\partial y} \overline{v'h'} + \bar{S}. \quad (10.67a,b)$$

Quasi-Geostrophic Wave–Mean-Flow Interaction

The inviscid and unforced Boussinesq quasi-geostrophic set of wave–mean-flow equations is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (\text{WMF.1a})$$

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial}{\partial y} \overline{v' q'} = 0. \quad (\text{WMF.1b})$$

along with similar equations as needed for buoyancy at the boundary (see main text). The eddy terms are

$$q' = \left[\nabla^2 + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \psi', \quad (u', v') = \left(-\frac{\partial \psi'}{\partial y}, \frac{\partial \psi'}{\partial x} \right). \quad (\text{WMF.2a,b})$$

The mean flow terms are

$$\bar{q}(y, t) = \beta y - \frac{\partial \bar{u}}{\partial y} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \bar{b} \right). \quad (\text{WMF.3})$$

and

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \frac{\partial \bar{b}}{\partial y} \right) = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right), \quad (\text{WMF.4})$$

using thermal wind. To solve for the mean-flow we may define a streamfunction Ψ such that

$$\left(\bar{u}, \frac{1}{f_0} \bar{b} \right) = \left(-\frac{\partial \Psi}{\partial y}, \frac{\partial \Psi}{\partial z} \right) \quad (\text{WMF.5})$$

whence

$$\bar{q}(y, t) - \beta y = \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \Psi}{\partial z} \right) + \frac{\partial^2 \Psi}{\partial y^2}. \quad (\text{WMF.6})$$

Given $\bar{q}$ from (WMF.1b) we solve (WMF.6) to give $\bar{u}$ and $\bar{b}$. Equivalently, we may derive a single equation for the zonal wind by differentiating (WMF.1b) with respect to y and, using (WMF.4), we obtain

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \frac{\partial \bar{u}}{\partial t} = \frac{\partial^2}{\partial y^2} \overline{v' q'}. \quad (\text{WMF.7})$$

The evolution of the mean flow may also usefully be written in TEM form as

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* + \overline{v' q'} = 0, \quad (\text{WMF.8a})$$

$$\frac{\partial \bar{b}}{\partial t} + N^2 \bar{w}^* = 0, \quad (\text{WMF.8b})$$

where $\bar{v}^*$ and $\bar{w}^*$ are found by solving the elliptic equation (10.63), and the value of $\partial \bar{q} / \partial y$ [for use in (WMF.1a)] is obtained using (WMF.4).

The overbars in these equations denote averages taken along isentropes — i.e., they are averages for a given layer — but are otherwise conventional, and the meridional velocity is purely ageostrophic. By analogy to (10.59), we define the residual circulation by

$$\bar{v}^* \equiv \bar{v} + \frac{1}{H} \overline{v' h'}, \quad (10.68)$$

where H is the mean thickness of the layer. Using (10.68) in (10.67) gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \overline{v' q'} + \bar{F}, \quad \frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}^*}{\partial y} = \bar{S}, \quad (10.69a,b)$$

where

$$\overline{v' q'} = \overline{v' \zeta'} - \frac{f_0}{H} \overline{v' h'}, \quad (10.70)$$

is the meridional potential vorticity flux in a shallow water system. From (10.68) we see that the residual velocity is a measure of the *total meridional thickness flux*, eddy plus mean, in an isentropic layer. This is often a more useful quantity than the Eulerian velocity $\bar{v}$ because it is generally the former, not the latter, that is constrained by the external forcing. What we have done, of course, is to effectively use a thickness-weighted mean in (10.66b); to see this, define the thickness-weighted mean by

$$\bar{v}_* \equiv \frac{\overline{h v}}{\bar{h}}. \quad (10.71)$$

(We use $\bar{v}_*$ to denote a thickness- or mass-weighted mean, and $\bar{v}^*$ to denote a residual velocity; the quantities are closely related, as we will see.) From (10.71) we have

$$\bar{v}_* = \bar{v} + \frac{1}{\bar{h}} \overline{v' h'}, \quad (10.72)$$

then the zonal average of (10.66b) is just

$$\frac{\partial \bar{h}}{\partial t} + \frac{\partial}{\partial y} (\bar{h} \bar{v}_*) = \bar{S}, \quad (10.73)$$

which is the same as (10.69b) if we take $H = \bar{h}$. Similarly, if we use the thickness weighted velocity (10.72) in the momentum equation (10.67a) we obtain (10.69a).

Evidently, if the mass-weighted meridional velocity is used in the momentum and thickness equations then the eddy mass flux does not enter the equations explicitly: the only eddy flux in (10.69) is that of potential vorticity. That is, in isentropic coordinates the equations in TEM form are equivalent to the equations that arise from a particular form of averaging — thickness weighted averaging — rather than the conventional Eulerian averaging. A similar correspondence occurs in height coordinates, as we now see.

10.3.3 Connection between the residual and thickness-weighted circulation

It is evident from the above arguments that, in a shallow water system or in isentropic coordinates, the residual velocity is a measure of the total (i.e., mean plus eddy) thickness transport. In height coordinates, the definition of residual velocity, (10.58) does not lend itself so easily to such an interpretation. However, the residual velocity in height coordinates is, in fact, also a measure of the total thickness transport, or equivalently of the mass transport between two isentropic surfaces, as we now discover. Specifically, we show that averaging the total transport in isentropic layers

Aspects of the TEM Formulation

Properties and features

- The residual mean circulation is equivalent to the total mass-weighted (eddy plus Eulerian mean) circulation, and it is this circulation that is driven by the diabatic forcing.
- There are no explicit eddy fluxes in the buoyancy budget; the only eddy term is the flux of potential vorticity, and this is divergence of the Eliassen–Palm flux; that is $\overline{v'q'} = \nabla_x \cdot \mathcal{F}$.
- The residual circulation, $\bar{v}^*$, becomes part of the solution, just as $\bar{v}$ is part of the solution in an Eulerian mean formulation.

But note

- The TEM formulation does not solve the parameterization problem, and eddy fluxes are still present in the equations.
- The theory and practice are well developed for a zonal average, but less so for three-dimensional, non-zonal flow. This is because the geometry enforces simple boundary conditions in the zonal mean case.⁶
- The boundary conditions on the residual circulation are neither necessarily simple nor easily determined; for example, at a horizontal boundary $\bar{w}^*$ is not zero if there are horizontal buoyancy fluxes.

Examples of the use of the TEM and its relatives in the general circulation of the atmosphere and ocean arise in sections 15.2, 15.4, 17.4, 17.8 and 21.8.

is equivalent to the mass transport evaluated by the TEM formalism in height coordinates, and specifically that the thickness-weighted mean, $\bar{v}_*$ is equivalent to the residual velocity, $\bar{v}^*$ in height coordinates. Our demonstration is for a Boussinesq system, but the extension to a compressible gas is reasonably straightforward.⁵

Consider two isentropic surfaces, η_1 and η_2 with mean positions $\bar{\eta}_1$ and $\bar{\eta}_2$, as in Fig. 10.1. (We use z to denote the vertical coordinate, and η to denote the location of isentropic surfaces.) The meridional transport between these surfaces is given by

$$T = \int_{\eta_2}^{\eta_1} v \, dz. \quad (10.74)$$

If the velocity does not vary with height within the layer (and in the limit of layer thickness going to zero this is the case) then $T = v h$ where $h = \eta_1 - \eta_2$ is the thickness of the isentropic layer. The zonally averaged transport is then given by

$$\bar{T} = \frac{1}{L} \int_L T \, dx = \frac{1}{L} \int_L \left(\int_{\eta_2}^{\eta_1} v \, dz \right) dx = \overline{\int_{\eta_2}^{\eta_1} v \, dz} = \overline{v h} = \bar{v} h + \overline{v' h'}, \quad (10.75)$$

with obvious notation, and with an overbar denoting a zonal average. Letting the distance between isentropes shrink to zero this result allows us to write

$$\bar{v}_* \equiv \frac{\overline{v \sigma^b}}{\bar{\sigma}} = \bar{v}^b + \frac{\overline{v' \sigma'^b}}{\bar{\sigma}}, \quad (10.76)$$

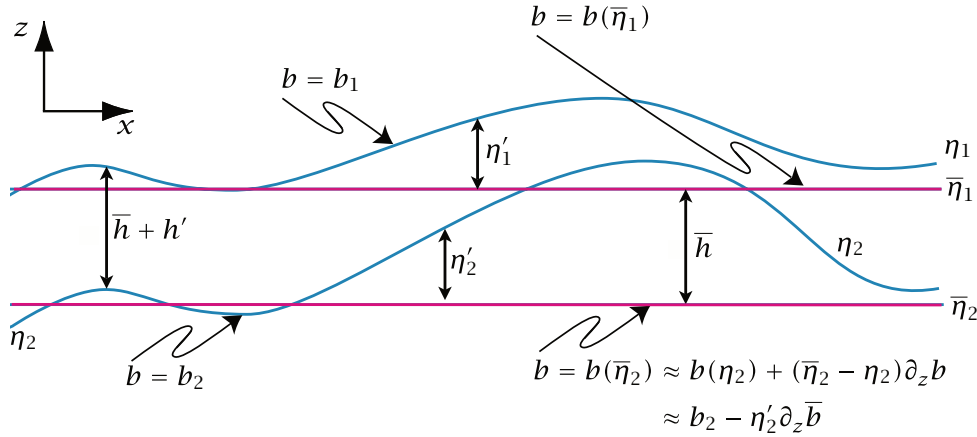


Fig. 10.1 Two isentropic surfaces, η_1 and η_2 , and their mean positions, $\bar{\eta}_1$ and $\bar{\eta}_2$. The departure of an isentrope from its mean position is proportional to the temperature perturbation at the mean position of the isentrope, and the variations in thickness (h') of the isentropic layer are proportional to the vertical derivative of this.

where $\overline{(\cdot)}^b$ denotes an average along an isentrope and $\bar{\sigma} = \overline{\partial z / \partial b}$ is the thickness density, a measure of the thickness between two isentropes. Equation (10.76) is analogous to (10.72), for a continuously stratified system. The averaged quantity $\bar{v}_*$ is not proportional to the average of the velocity at constant height, or even to the average along an isentrope; rather, it is the *thickness-weighted* zonal average of the velocity *between* two isentropic surfaces, Δb apart, of mean separation proportional to $\bar{\sigma} \Delta b$. Our goal is to express this transport in terms of Eulerian-averaged quantities, at a constant height z .

Let us first connect an average along an isentrope of some variable χ to its average at constant height by writing, for small isentropic displacements,

$$\bar{\chi}^b = \overline{\chi(z + \eta')^z} \approx \overline{\chi(z) + \eta' \partial_z \chi}^z, \quad (10.77)$$

where the superscript explicitly denotes how the zonal average is taken, and η' is the displacement of the isotherm from its mean position. This can be expressed in terms of the temperature perturbation at the location of the mean isentrope by Taylor expanding b around its value on that mean isentrope. That is,

$$b(\eta) = b(\bar{\eta}) + \left(\frac{\partial b}{\partial z} \right)_{z=\bar{\eta}} (\eta - \bar{\eta}) + \dots, \quad (10.78)$$

where $\bar{\eta} = \bar{\eta}(z)$, giving

$$\eta' \approx \frac{-b'}{\partial_z b(\bar{\eta})} \approx -\frac{b'}{\partial_z \bar{b}}, \quad (10.79)$$

where $\eta' = \eta - \bar{\eta}$ and $b' = b(\bar{\eta}) - b(\eta)$. Using (10.79) in (10.77) (and omitting the superscript z on $\partial_z \bar{b}$) we obtain, with $\chi = v$,

$$\bar{v}^b = \bar{v}^z - \frac{\bar{b}' \partial_z v'}{\partial_z \bar{b}}. \quad (10.80)$$

Note that if v is in thermal wind balance with b then the second term vanishes identically, but we will not invoke this.

We now transform the second term on the right-hand side (10.76) to an average at constant z . The variations in thickness of an isothermal layer are given by

$$\sigma' \approx \bar{\sigma} \frac{\partial \eta'}{\partial z} = -\bar{\sigma} \frac{\partial}{\partial z} \left(\frac{b'}{\partial_z \bar{b}} \right), \quad (10.81)$$

using (10.79). Thus, neglecting terms that are third-order in amplitude,

$$\overline{v' \sigma'^b} = -\bar{\sigma} \overline{v' \frac{\partial}{\partial z} \left(\frac{b'}{\partial_z \bar{b}} \right)^z}. \quad (10.82)$$

Using both (10.80) and (10.82), (10.76) becomes

$$\bar{v}_* = \bar{v}^z - \frac{\overline{b' \partial_z v'^z}}{\partial_z \bar{b}} - \overline{v' \frac{\partial}{\partial z} \left(\frac{b'}{\partial_z \bar{b}} \right)^z} = \bar{v}^z - \overline{\frac{\partial}{\partial z} \left(\frac{v' b'}{\partial_z \bar{b}} \right)^z}. \quad (10.83)$$

The right-hand side of the last equation is the TEM form of the residual velocity; thus, we have shown that

$$\bar{v}_* \equiv \frac{\overline{v \sigma}}{\bar{\sigma}} = \bar{v}^b + \frac{\overline{v' \sigma'^b}}{\bar{\sigma}} \approx \bar{v}^z - \overline{\frac{\partial}{\partial z} \left(\frac{v' b'}{\partial_z \bar{b}} \right)^z} \equiv \bar{v}^*. \quad (10.84)$$

We see the equivalence of the thickness-weighted mean velocity on the left-hand side and the residual velocity on the right-hand side. In the quasi-geostrophic limit $N^2 = \partial_z \bar{b}$ and $\bar{\sigma}$ is a reference thickness.

10.4 THE NON-ACCELERATION RESULT

We now consider further the interpretation and application of the potential vorticity flux and its relatives, using a quasi-geostrophic framework. We first derive an important result in wave-mean-flow dynamics, the non-acceleration condition.⁷ This result shows that under certain conditions, to be made precise below, waves have no net effect on the mean flow, an important and somewhat counter-intuitive result.

10.4.1 A derivation from the potential vorticity equation

Consider how the potential vorticity fluxes affect the mean fields. The unforced and inviscid zonally averaged potential vorticity equation is

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial \overline{v' q'}}{\partial y} = \bar{F}_q. \quad (10.85)$$

Now, in quasi-geostrophic theory the geostrophically balanced velocity and buoyancy can be determined from the potential vorticity via an elliptic equation, and in particular

$$\bar{q} - \beta y = \frac{\partial^2 \bar{\psi}}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \bar{\psi}}{\partial z} \right), \quad (10.86)$$

where $\bar{\psi}$ is such that $(\bar{u}, \bar{b}/f_0) = (-\partial \bar{\psi}/\partial y, \partial \bar{\psi}/\partial z)$. Differentiating (10.85) with respect to y we obtain

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \frac{\partial \bar{u}}{\partial t} = (\nabla \cdot \mathcal{F})_{yy}. \quad (10.87)$$

where $\nabla \cdot \mathcal{F} = \overline{v'q'}$ is the divergence of the EP flux. This is determined using the wave activity equation which, reprising (10.29a), is

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}. \quad (10.88)$$

If the waves are statistically steady (i.e., $\partial \mathcal{A} / \partial t = 0$) and have no dissipation ($\mathcal{D} = 0$) then evidently $\nabla \cdot \mathcal{F} = 0$. If there is no acceleration at the boundaries then the solution of (10.87) is

$$\frac{\partial \bar{u}}{\partial t} = 0. \quad (10.89)$$

This is a *non-acceleration result*. That is to say, under certain conditions the tendency of the mean fields, and in particular of the zonally-averaged zonal flow, are independent of the waves. To be explicit, those conditions are the following.

- (i) The waves are steady (so that, using the wave activity equation $\mathcal{A}$ does not vary).
- (ii) The waves are conservative [i.e., $\mathcal{D} = 0$ in (10.29a)]. Given this and item (i), the Eliassen–Palm relation implies that $\nabla \cdot \mathcal{F} = 0$; that is, the potential vorticity flux is zero.
- (iii) The waves are of small amplitude (all of our analysis has neglected terms that are cubic in perturbation amplitude).
- (iv) The waves do not affect the boundary conditions (so there are no boundary contributions to the acceleration).

Given the way we have derived it, the result does not seem too surprising; however, it can be powerful and counter-intuitive, for it means that steady waves (i.e., those whose amplitude does not vary) do not affect the zonal flow. However, they *do* affect the meridional overturning circulation, and the relative vorticity flux may also be non-zero. In fact, the non-acceleration theorem is telling us that the changes in the vorticity flux are exactly compensated for by changes in the meridional circulation, and there is no net effect on the zonally averaged zonal flow. It is *irreversibility*, often manifested by the breaking of waves, that leads to permanent changes in the mean flow.

The derivation of this result by way of the momentum equation, which one might expect to be more natural, is rather awkward because one must consider momentum and buoyancy fluxes separately. Furthermore, the zonally averaged meridional circulation comes into play: for example, the meridional velocity, $\bar{v}$, is small because it is purely ageostrophic, but it is not zero and we cannot neglect it because it is multiplied by the Coriolis parameter, which is large. Thus, the eddy vorticity fluxes can affect both the meridional circulation and the acceleration of the zonal mean flow, and it might seem impossible to disentangle the two effects without completely solving the equations of motion. Nevertheless, we *can* proceed by way of the momentum and buoyancy equations if we use the transformed Eulerian mean and this provides a useful alternate derivation, as follows.

10.4.2 Using TEM to give the non-acceleration result

We may use the TEM formalism to obtain the non-acceleration result. The explanation is largely equivalent to that given above, but the explication may be useful.

A two-dimensional case

Consider two-dimensional incompressible flow on the β -plane, for which there is no buoyancy flux. The linearized vorticity equation is

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \bar{\zeta}}{\partial y} = D', \quad (10.90)$$

from which we derive, analogously to (10.29a), the Eliassen–Palm relation

$$\frac{\partial \mathcal{A}}{\partial t} + \frac{\partial \mathcal{F}}{\partial y} = \mathcal{D}, \quad (10.91)$$

where $\mathcal{F} = -\overline{u'v'}$, $\mathcal{D}$ represents non-conservative forces, and

$$\mathcal{A} = \frac{\overline{\zeta'^2}}{2\overline{\partial_y \zeta}} = \frac{1}{2} \overline{\eta'^2} \frac{\partial \bar{\zeta}}{\partial y}. \quad (10.92)$$

The quantity $\eta' \equiv -\zeta'/\partial_y \bar{\zeta}$ is proportional to the meridional particle displacement in a disturbance. Now consider the x -momentum equation

$$\frac{\partial u}{\partial t} = -\frac{\partial u^2}{\partial x} - \frac{\partial uv}{\partial y} - \frac{\partial \phi}{\partial x} + fv. \quad (10.93)$$

Zonally averaging, noting that $\bar{v} = 0$, gives

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \bar{uv}}{\partial y} = \overline{v'\zeta'} = \frac{\partial \mathcal{F}}{\partial y}. \quad (10.94)$$

Finally, combining (10.91) and (10.94) gives

$$\frac{\partial}{\partial t} (\bar{u} + \mathcal{A}) = \mathcal{D}. \quad (10.95)$$

In the absence of non-conservative terms (i.e., if $\mathcal{D} = 0$) the quantity $\bar{u} + \mathcal{A}$ is constant.⁸ Further, if the waves are steady and conservative then $\mathcal{A}$ is constant and, therefore, so is $\bar{u}$. This is the non-acceleration result.

The stratified case

In the stratified case we can use the TEM form of the momentum equation to derive a similar result. The unforced zonally averaged zonal momentum equation can be written as

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \mathcal{F}, \quad (10.96)$$

and using the Eliassen–Palm relation, (10.29a), this may be written as

$$\frac{\partial}{\partial t} (\bar{u} + \mathcal{A}) - f_0 \bar{v}^* = \mathcal{D}, \quad (10.97)$$

and so again $\mathcal{A}$ is related to the momentum of the flow. If, furthermore, the waves are steady ($\partial \mathcal{A}/\partial t = 0$) and conservative ($\mathcal{D} = 0$), then $\partial \bar{u}/\partial t - f_0 \bar{v}^* = 0$. However, under these same conditions the residual circulation will also be zero. This is because the residual meridional circulation ($\bar{v}^*, \bar{w}^*$) arises via the necessity to keep the temperature and velocity fields in thermal wind balance, and is thus determined by an elliptic equation, namely (10.63). If the waves are steady and adiabatic then, since $\overline{v'q'} = 0$, the right-hand side of the equation is zero and it becomes

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = 0. \quad (10.98)$$

If $\psi^* = 0$ at the boundaries, then the unique solution of this is $\psi^* = 0$ everywhere. At the meridional boundaries we may certainly suppose that ψ^* vanishes if these are quiescent latitudes, and

at the horizontal boundaries the buoyancy flux will vanish if the waves there are steady, because from (10.14) we have

$$\overline{v'b'} \frac{\partial \bar{b}}{\partial y} = -\frac{1}{2} \frac{\partial}{\partial t} \overline{b'^2} = 0. \quad (10.99)$$

Under these circumstances, then, the residual meridional circulation vanishes in the interior and, from (10.96), the mean flow is steady, thus reprising the non-acceleration result.

Compare (10.96) with the momentum equation in conventional Eulerian form, namely

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v'\zeta'}. \quad (10.100)$$

There is no reason that the vorticity flux should vanish when waves are present, even if they are steady. However, such a flux is (under non-acceleration conditions) precisely compensated by the meridional circulation $f_0 \bar{v}$, something that is hard to infer or intuit directly from (10.100); even when non-acceleration conditions do not apply there will be a significant cancellation between the Coriolis and eddy terms. The difficulty boils down to the fact that, in contrast to $\overline{v'q'}$, $\overline{v'\zeta'}$ is not the flux of a wave activity.

Unlike the proof of the non-acceleration result given in section 10.4.1, the above argument does not use the invertibility property of potential vorticity directly, suggesting an extension to the primitive equations, and the reader may pursue that here.⁹ Various results regarding the TEM and non-acceleration are summarized in the shaded box on the next page.

10.4.3 The EP flux and form drag

It may seem a little magical that the zonal flow is driven by the Eliassen–Palm flux via (10.96). The poleward vorticity flux is clearly related to the momentum flux convergence, but why should a poleward buoyancy flux affect the momentum? The TEM form of the momentum equation may be written as

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v'b'} \right) + F_m, \quad (10.101)$$

where $F_m = \overline{v'\zeta'} + f_0 \bar{v}^*$ represents forces from the momentum flux and Coriolis force. The first term on the right-hand side certainly does not look like a force; however, it turns out to be directly proportional to the *form drag* between isentropic layers. Recall from section 3.6 that the form drag, τ_d , at an interface between two layers of shallow water is

$$\tau_d = -\overline{\eta' \frac{\partial p'}{\partial x}}, \quad (10.102)$$

where η is the interfacial displacement. But from (10.79) $\eta' = -b'/N^2$ and with this and geostrophic balance we have

$$\tau_d = \frac{\rho_0 f_0}{N^2} \overline{v'b'}. \quad (10.103)$$

Thus, the vertical component of the EP flux (i.e., the meridional buoyancy flux) is in fact a real stress acting on a fluid layer and equal to the momentum flux caused by the wavy interface. The net momentum convergence into an infinitesimal layer of mean thickness $\bar{h}$ is then [cf. (3.82)],

$$F_d = \bar{h} \frac{\partial \tau_d}{\partial z} = \bar{h} \rho_0 f_0 \frac{\partial}{\partial z} \left(\frac{\overline{v'b'}}{N^2} \right), \quad (10.104)$$

TEM, Residual Velocities, Non-Acceleration, and All That

For a Boussinesq quasi-geostrophic system, the TEM form of the unforced momentum equation and the thermodynamic equation are:

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \mathcal{F}, \quad \frac{\partial \bar{b}}{\partial t} + \bar{w}^* N^2 = \bar{S}, \quad (\text{T.1})$$

where $N^2 \partial \bar{b}_0 / \partial z$, $\bar{S}$ represents diabatic effects, $\mathcal{F}$ is the Eliassen–Palm (EP) flux and its divergence is the potential vorticity flux; that is $\nabla_x \cdot \mathcal{F} = v' q'$. The residual velocities are

$$\bar{v}^* = \bar{v} - \frac{\partial}{\partial z} \left(\frac{1}{N^2} \overline{v' b'} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left(\frac{1}{N^2} \overline{v' b'} \right). \quad (\text{T.2})$$

Spherical coordinate and ideal gas versions of these take a similar form. We may define a meridional overturning streamfunction such that $(\bar{v}^*, \bar{w}^*) = (-\partial \psi^* / \partial z, \partial \psi^* / \partial y)$, and using thermal wind to eliminate time-derivatives in (T.1) we obtain

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = f_0 \frac{\partial}{\partial z} \overline{v' q'} + \frac{\partial \bar{S}}{\partial y}. \quad (\text{T.3})$$

The manipulations (given in the main text) that lead to the above equations may seem formal, in that they simply transform the momentum and thermodynamic equation from one form to another. However, the resulting equations have two potential advantages over the untransformed ones.

- (i) The residual meridional velocity is approximately equal to the average thickness-weighted velocity between two neighbouring isentropic surfaces, and so is a measure of the total (Eulerian mean plus eddy) meridional transport of thickness or buoyancy.
- (ii) The EP flux is directly related to certain conservation properties of waves. The divergence of the EP flux is the meridional flux of potential vorticity:

$$\mathcal{F} = -(\overline{u' v'}) \mathbf{j} + \left(\frac{f_0}{N^2} \overline{v' b'} \right) \mathbf{k}, \quad \nabla \cdot \mathcal{F} = \overline{v' q'}. \quad (\text{T.4})$$

Furthermore, the EP flux satisfies, to second order in wave amplitude,

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad \text{where} \quad \mathcal{A} = \frac{\overline{q'^2}}{2 \partial \bar{q} / \partial y}, \quad \mathcal{D} = \frac{\overline{D' q'}}{\partial \bar{q} / \partial y}. \quad (\text{T.5})$$

The quantity $\mathcal{A}$ is a *wave activity density*, and $\mathcal{D}$ is its dissipation. For nearly plane waves, $\mathcal{A}$ and $\mathcal{F}$ are connected by the *group velocity property*,

$$\mathcal{F} = (\mathcal{F}^y, \mathcal{F}^z) = \mathbf{c}_g \mathcal{A}, \quad (\text{T.6})$$

where $\mathbf{c}_g$ is the group velocity of the waves. If the waves are steady ($\partial \mathcal{A} / \partial t = 0$) and dissipationless ($\mathcal{D} = 0$) then $\nabla \cdot \mathcal{F} = 0$ and using (T.1) and (T.3) there is no wave-induced acceleration of the mean flow; this is the ‘non-acceleration’ result. Commonly there is enstrophy dissipation, or wave-breaking, and $\nabla \cdot \mathcal{F} < 0$; such *wave drag* leads to flow deceleration and/or a poleward residual meridional velocity.

and a layer of mean thickness $\bar{h}$ is accelerated according to

$$\frac{\partial \bar{u}}{\partial t} = f_0 \frac{\partial}{\partial z} \left(\frac{\overline{v' b'}}{\partial_z \bar{b}} \right) + F_m. \quad (10.105)$$

The appearance of the buoyancy flux is really a consequence of the way we have chosen to average the equations: obtaining (10.105) involved averaging the forces over an isentropic layer, and given this it can only be the residual circulation that contributes to the Coriolis force. One might say that the vertical component of the EP flux is a force in drag, masquerading as a buoyancy flux.

10.5 ♦ INFLUENCE OF EDDIES ON THE MEAN FLOW IN THE EADY PROBLEM

We now consider the eddy fluxes in the Eady problem, and, in particular, how these might feed back on to the mean flow. Because of the simplicity of the setting the problem can be fully solved in both the Eulerian or residual frameworks and it is therefore a very instructive, albeit algebraically complex, example.¹⁰

10.5.1 Formulation

Let us first distinguish between the basic flow, the zonal mean fields, and the perturbation. The basic flow is the flow around which the equations of motion are linearized; this flow is unstable, and the perturbations, assumed to be small, grow exponentially with time. Because the perturbations are formally always small they do not affect the basic flow, but they do produce changes in the zonal mean velocity and buoyancy fields. In Eulerian form this is represented by,

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v} - \frac{\partial \overline{u' v'}}{\partial y}, \quad \frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w} - \frac{\partial \overline{b' v'}}{\partial y}, \quad (10.106)$$

and the TEM version of these equation is

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v}^* + \overline{v' q'}, \quad \frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w}^*, \quad (10.107)$$

where in the Eady problem $\partial_y(\overline{u' v'})$ and $\overline{v' q'}$ are both zero. We can calculate the perturbation quantities from the solution to the Eady problem (e.g., calculate $\overline{v' b'}$) and thus infer the structure of the mean flow tendencies $\partial \bar{u}/\partial t$ and $\partial \bar{b}/\partial t$ and the meridional circulation, $(\bar{v}, \bar{w})$ or $(\bar{v}^*, \bar{w}^*)$. All of these fields are perturbation quantities and all are exponentially growing, and so in reality they will eventually have a finite effect on the pre-existing zonal flow, but in the Eady problem, or any similar linear problem, such rectification is assumed to be small and is neglected.

Using the thermal wind relation, $f_0 \partial_z \bar{u} = -\partial_y \bar{b}$ to eliminate time derivatives in (10.106) gives an equation for the meridional streamfunction ψ_E , namely,

$$\frac{L^2}{L_d^2} \frac{\partial^2 \psi_E}{\partial z^2} + \frac{\partial^2 \psi_E}{\partial y^2} = -\frac{1}{N^2} \frac{\partial^2 \overline{b' v'}}{\partial y^2}, \quad (10.108)$$

where $(\bar{v}, \bar{w}) = (-\partial \psi_E / \partial z, \partial \psi_E / \partial y)$ and we have non-dimensionalized z with D and y with L . The boundary conditions are that $\psi_E = 0$ at $y = 0, L$ and $z = 0, D$. Similarly, and analogously to to (10.63), we obtain an equation for the residual streamfunction, ψ^* , namely

$$\frac{L^2}{L_d^2} \frac{\partial^2 \psi^*}{\partial z^2} + \frac{\partial^2 \psi^*}{\partial y^2} = 0, \quad (10.109)$$

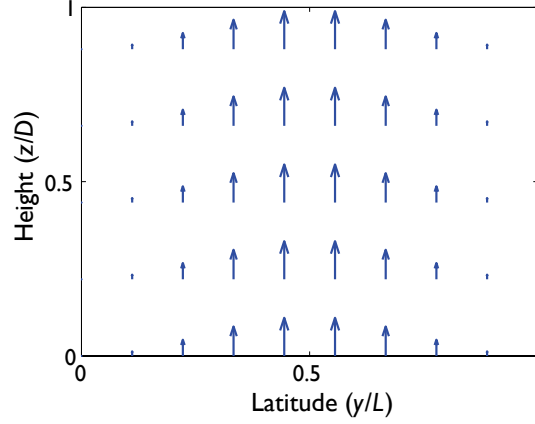


Fig. 10.2 The Eliassen–Palm vector in the Eady problem.

where now the boundary conditions are that $N^2 \bar{w}^* = \partial \bar{v}' b' / \partial y$ at the upper and lower boundaries, and $\bar{v} = 0$ at the lateral boundaries. In terms of the residual streamfunction this is

$$\psi^* = \frac{1}{N^2} \overline{v' b'}, \text{ at } z = 0, 1, \quad \psi^* = 0, \text{ at } y = 0, 1. \quad (10.110)$$

The residual and overturning circulations are related by (10.58a), and (10.108) and (10.109) are, at one level, simply different representations of the same problem, connected by a simple mathematical transformation. However, the residual streamfunction better represents the total transport of the fluid. Equation (10.109) is particularly simple, because of the absence of potential vorticity fluxes in the interior, and it is apparent that the residual circulation is driven by boundary sources. We care only about the spatial structure of the right-hand sides of (10.108) and of the boundary conditions of (10.110). The former is given by

$$-\frac{\partial^2 \overline{b' v'}}{\partial y^2} \propto -\frac{\partial^2}{\partial y^2} \sin^2 l y = -2l^2 \cos 2ly. \quad (10.111)$$

The eddy heat fluxes in the Eady problem are independent of height, as may be calculated explicitly from the solutions of chapter 9. In fact, the result follows without detailed calculation, by first noting that the eddy potential vorticity flux is zero because the basic state has zero QG potential vorticity and therefore none may be generated. Further, because the basic state does not vary in y there can be no momentum flux convergence in the y -direction, and so the momentum flux itself is zero if it is zero on the boundary. Thus [using for example (10.27) and (10.28)] the eddy heat flux is independent of height and the EP vectors are directed purely vertically (Fig. 10.2).

The boundary conditions for the residual circulation are

$$\psi^*(y, 0) = \psi^*(y, 1) \propto \sin^2 l y. \quad (10.112)$$

10.5.2 Solution

The solutions to (10.108) and (10.109) may be obtained either analytically or numerically. In a domain $0 < y < 1$ and $0 < z < 1$ the residual streamfunction for $l = \pi$ is given by:

$$\psi^* = \sum_{n=1}^{\infty} A_n \sin[(2n-1)ly] \frac{\cosh[L_d \pi(2n-1)(z-0.5)/L]}{\cosh[L_d \pi(2n-1)/2L]}, \quad (10.113)$$

$$A_n = \frac{2}{\pi(2n-1)} - \frac{1}{\pi(2n-1)-2l} - \frac{1}{\pi(2n-1)+2l}.$$

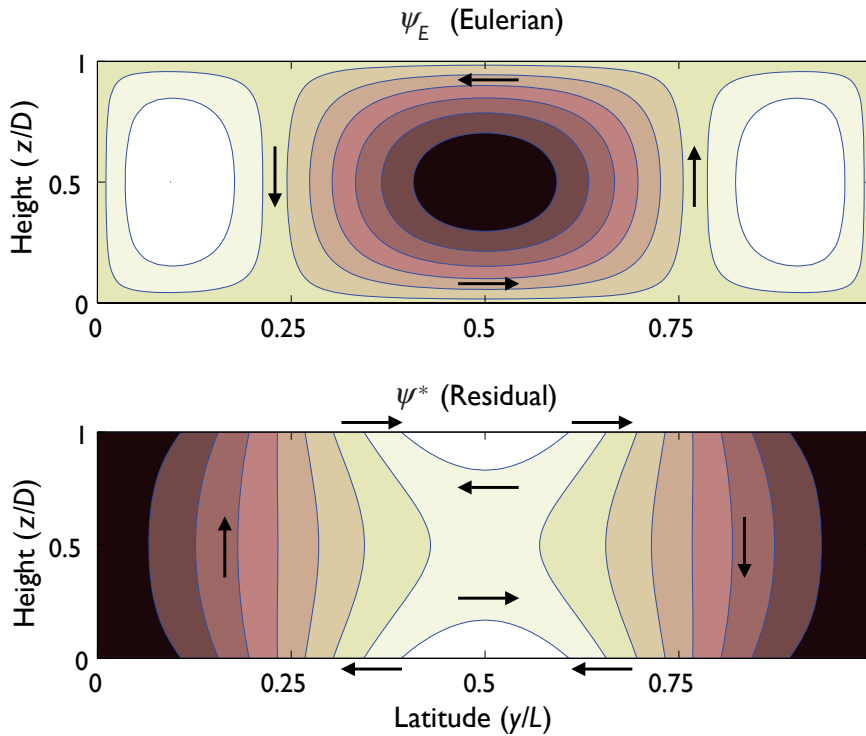


Fig. 10.3 The Eulerian streamfunction (top) and the residual streamfunction for the Eady problem, calculated using (10.108) and (10.109), with $L^2/L_d^2 = 9$.

The solution is obtained by first projecting the boundary conditions [proportional to $\sin^2 ly$, or $(1 - \cos 2ly)/2$] on to the eigenfunctions of the horizontal part of the Laplacian (i.e., sine functions), and this gives the coefficients of A_n . The vertical structure is then obtained by solving $(L/L_d)^2 \partial_z^2 \psi^* = -\partial_y^2 \psi^*$, which gives the cosh functions. The series converges very quickly, and the first term in the series captures the dominant structure of the solution, essentially because, for $l = \pi$, $\sin ly$ is not unlike $\sin^2 ly$ on the interval $[0, 1]$.

The Eulerian circulation is obtained from the residual circulation using (10.58a) and so by the addition of a field independent of z and proportional to $\sin^2 ly$. The resulting structure is dominated by this and the first term of (10.113) (proportional to $\sin ly$) and, noting that the circulation is symmetric about $z = 0.5$, we obtain a circulation dominated by a single cell, with equatorward motion aloft and poleward motion near the surface (Fig. 10.3). The heat flux convergence in high latitudes is leading to mean rising motion, with the precise shape of the streamfunction determined by the boundary conditions. Although this is true, the heat flux arises *because* of the motion of fluid parcels, so it may be a little misleading to infer, as one might from the Eulerian streamfunction, that the heat flux *causes* the individual parcels to rise or sink in this fashion. The residual streamfunction is a better indicator of the total mass transport and, perhaps as one might intuitively expect, these show parcels rising in the low latitudes and sinking in high latitudes, providing a tendency to flatten the isopycnals and to reduce the meridional temperature gradient.

The residual circulation also shows fluid entering or leaving the domain at the boundary — what does this represent? Suppose that instead of solving the continuous problem we had posed the problem in a finite number of layers (and we explicitly consider the two-layer problem below). As the number of layers increases the solutions to the linear baroclinic instability problem approaches that of the Eady problem (e.g., Fig. 9.13); however, as we saw in section 10.3 the residual circulation

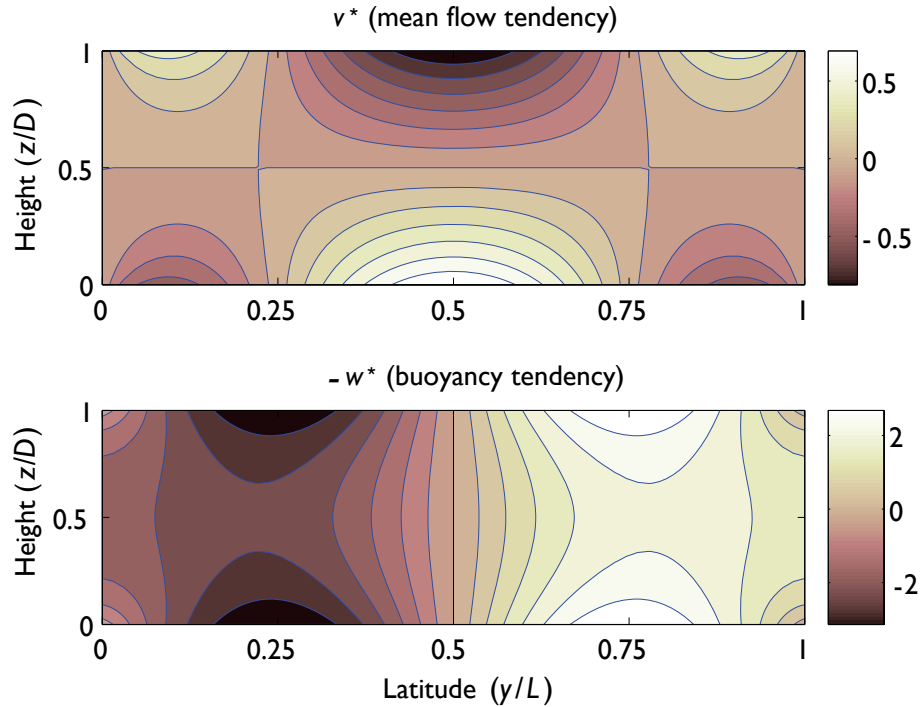


Fig. 10.4 The tendency of the zonal mean flow ($\partial \bar{u} / \partial t$) and the buoyancy ($\partial \bar{b} / \partial t$) for the Eady problem. Lighter (darker) shading means a positive (negative) tendency, but the units themselves are arbitrary.

is closed in the layered model, and the sum over all the layers of the meridional transport vanishes. Now, in the layered model the vertical boundary conditions are built in to the representation by way of a redefinition of the potential vorticity of the top and bottom layers, so that, in the layered version of the Eady problem there appears to be a potential vorticity gradient in these two layers, instead of a buoyancy gradient at the boundary. The residual circulation is then closed by a return flow that occurs only in the top and bottom layers, and as the number of layers increases this flow is confined to a thinner and thinner layer, and to a delta-function in the continuous limit. To indicate this we have placed arrows just above and below the domain in Fig. 10.3. (This equivalence between boundary conditions and delta-function sources is the same as that giving rise to the delta-function boundary layer of section 5.4.3.)

The effect on the mean flow is inferred directly from the residual circulation: the mean flow acceleration is proportional to $\bar{v}^*$ and the buoyancy tendency is proportional to $-\bar{w}^*$, and these are plotted in Figs. 10.4 and 10.5. Because there is no momentum flux convergence in the problem the zonal flow tendency is entirely baroclinic — its vertical integral is zero — and over most of the domain is such as to reduce the mean shear. Consistently (using thermal wind) the buoyancy tendency is such as to reduce the meridional temperature gradient; that is, the instabilities act to transport heat polewards and so reduce the instability of the mean flow.

10.5.3 The two-level problem

The residual circulation and mean-flow tendencies can also be calculated for the two-level (Phillips) problem, with the β -effect. The potential vorticity fluxes in each layer are non-zero and the mean

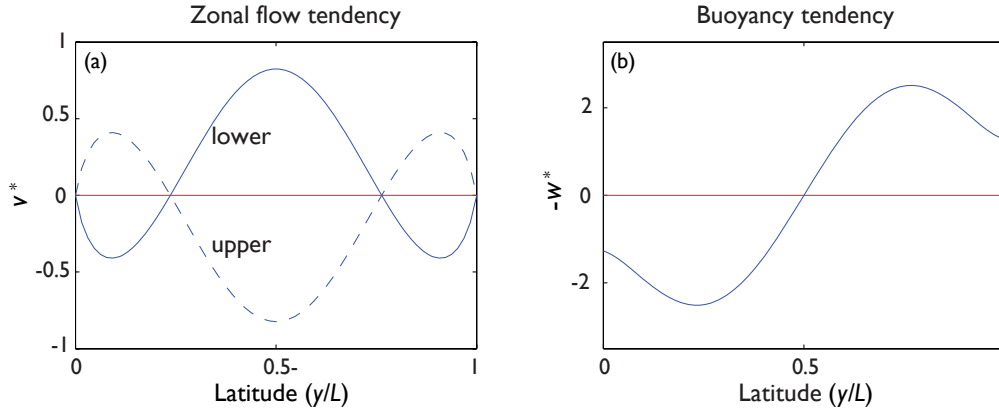


Fig. 10.5 (a) The tendency of the zonal mean flow ($\partial\bar{u}/\partial t$) just below the upper lid (dashed) and just above the surface (solid) in the Eady problem. The vertically integrated tendency is zero. (b) The vertically averaged buoyancy tendency.

flow equations are, for $i = 1, 2$,

$$\frac{\partial \bar{u}_i}{\partial t} = f_0 \bar{v}_i^* + \overline{v_i' q_i'}, \quad \frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w}^*. \quad (10.114)$$

The vertical velocity and buoyancy are evaluated at mid-depth, and the thermal wind equation is $\bar{u}_1 - \bar{u}_2 = -(D/2)\partial_y \bar{b}$ and, by mass conservation, $\bar{v}_1^* = -\bar{v}_2^*$. If we define a residual streamfunction ψ^* such that

$$\bar{v}_1^* = -\bar{v}_2^* = \psi^*, \quad \bar{w}^* = \frac{\partial \psi^*}{\partial y}, \quad (10.115)$$

then eliminating time derivatives in (10.114) gives an equation for the residual streamfunction,

$$\frac{\partial^2 \psi^*}{\partial y^2} - \frac{k_d^2}{2} \psi^* = \frac{2f_0 L^2}{N^2 D} (\overline{v_1' q_1'} - \overline{v_2' q_2'}), \quad (10.116)$$

where $k_d^2/2 = [2f_0/(NH)]^2$ and D is the total depth of the fluid, and we have non-dimensionalized vertical scales by D and horizontal scales by L . As in the Eady problem it is only the spatial structure of the terms on the right-hand side that are relevant, and these may be calculated from the solutions to the two-level instability problem. The main difference from the Eady problem is that the potential vorticity fluxes are non-zero, even in the case with $\beta = 0$: effectively, the boundary fluxes of the Eady problem are absorbed into the potential vorticity fluxes of the two layers. Solving for the residual circulation and interpreting the mean-flow tendencies is left as an exercise for the reader (problem 10.??).

10.6 ♦ NECESSARY CONDITIONS FOR INSTABILITY

Let's take a taxi to the finish line.

Chris Garrett, Ocean Science Meeting, Hawaii 2002.

As we noted in chapter 9, necessary conditions for instability, or sufficient conditions for stability, can be very useful because when satisfied they obviate the need to perform a detailed calculation. In the remainder of this chapter we use the conservation of wave activities — pseudomomentum and

pseudoenergy — to derive such conditions. In sections 9.3 and 9.4.3 we derived such conditions assuming the instability to be of normal-mode form. Here we give derivations that are both more general and, in some ways, simpler; they utilize the fact that the potential vorticity flux may be written as a divergence of a vector and therefore vanishes when integrated over a domain, aside from possible boundary contributions.

10.6.1 Stability conditions from pseudomomentum conservation

Consider the perturbation enstrophy equation,

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{q'^2} = -\frac{\partial \bar{q}}{\partial y} \nabla_x \cdot \mathcal{F}, \quad (10.117)$$

where $\mathcal{F}$ is the Eliassen–Palm flux given by (10.27), the overbar is a zonal mean and the divergence is in y - z plane. Dividing by $\partial \bar{q}/\partial y$ and integrating over a domain A which is such that the Eliassen–Palm flux vanishes at the boundaries gives the pseudomomentum conservation law,

$$\int_A \frac{\partial}{\partial t} \left(\frac{\overline{q'^2}}{\partial_y \bar{q}} \right) dy dz = 0. \quad (10.118)$$

Equation (10.118) implies that, in the *norm* $[\overline{q'^2}/\partial_y \bar{q}]$, the perturbation cannot grow unless $\partial \bar{q}/\partial y$ changes sign somewhere in the domain, or at the boundaries. This result does not depend upon the instability being of normal-mode form. The simplest result of all occurs in a barotropic problem with no vertical variation. Then $\partial \bar{q}/\partial y = \partial/\partial \zeta_a y = \beta - \partial^2 \bar{u}/\partial y^2$, and demanding that this must change sign for an instability reprises the inflection point (Rayleigh–Kuo) condition. In the more general case, if $\partial \bar{q}/\partial y$ changes sign along a vertical line then the instability is called a baroclinic instability, and if it changes sign along a horizontal line the instability is barotropic — these may be taken as the definitions of those terms. A mixed instability has a change of sign along both horizontal and vertical lines.

10.6.2 Inclusion of boundary terms

Suppose now the flow is contained between two flat boundaries, at $z = 0$ and $z = H$. The relevant equations of motion are the potential vorticity evolution in the interior, supplemented by the thermodynamic equation at the boundary. For unforced and inviscid flow these give [cf. (10.11) and (10.14)]

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \frac{\overline{q'^2}}{\partial_y \bar{q}} \right) = -\overline{v'q'}, \quad 0 < z < H, \quad (10.119)$$

and

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \frac{\overline{b'^2}}{\partial_y \bar{b}} \right) = -\overline{v'b'}, \quad z = 0, H. \quad (10.120)$$

The poleward flux of potential vorticity is

$$\overline{v'q'} = -\frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \left(\frac{f_0}{N^2} \overline{v'b'} \right), \quad (10.121)$$

and integrating this expression with respect to both y and z gives

$$\int_A \overline{v'q'} dy dz = \left[\frac{f_0}{N^2} \overline{v'b'} \right]_0^H, \quad (10.122)$$

assuming that the meridional boundaries are at quiescent latitudes. Integrating (10.119) over y and z , and using (10.122) gives

$$\frac{\partial}{\partial t} \iint \frac{1}{2} \frac{\bar{q}'^2}{\bar{q}} dy dz = - \left[\frac{f_0}{N^2} \bar{v}' \bar{b}' \right]_0^H. \quad (10.123)$$

Using (10.120) to eliminate $\bar{v}' \bar{b}'$ finally gives

$$\frac{\partial}{\partial t} \left\{ \iint \frac{1}{2} \frac{\bar{q}'^2}{\bar{q}} dy dz - \int \left[\frac{1}{2} \frac{f_0}{N^2} \frac{\bar{b}'^2}{\bar{b}} \right]_0^H dy \right\} = 0. \quad (10.124)$$

If this expression is positive or negative definite the perturbation cannot grow and therefore the basic state is stable. Stability thus depends on the meridional gradient of potential vorticity in the interior, and the meridional gradient of buoyancy at the boundary. If $\partial \bar{q} / \partial y$ changes sign in the interior, or $\partial \bar{b} / \partial y$ changes sign at the boundary, we have the potential for instability. If these are both one signed, then various possibilities exist, and using the thermal wind relation ($f_0 \partial \bar{u} / \partial z = -\partial \bar{b} / \partial y$) we obtain the following.

I. *A stable case:*

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \left. \frac{\partial u}{\partial z} \right|_{z=0} < 0 \text{ and } \left. \frac{\partial u}{\partial z} \right|_{z=H} > 0 \implies \text{stability.} \quad (10.125)$$

Stability also ensues if all inequalities are switched.

II. *Instability via interior-surface interactions:*

$$\frac{\partial \bar{q}}{\partial y} > 0 \text{ and } \left. \frac{\partial u}{\partial z} \right|_{z=0} > 0 \text{ or } \left. \frac{\partial u}{\partial z} \right|_{z=H} < 0 \implies \text{potential instability.} \quad (10.126)$$

The condition $\partial \bar{q} / \partial y > 0$ and $(\partial u / \partial z)_{z=0} > 0$ is the most common criterion for instability that is met in the atmosphere. In the troposphere we can sometimes ignore contributions of the buoyancy fluxes at the tropopause ($z = H$), and stability is then determined by the interior potential vorticity gradient and the surface buoyancy gradient. Similarly, in the ocean contributions from the ocean floor are normally very small.

III. *Instability via edge wave interaction:*

$$\left. \frac{\partial u}{\partial z} \right|_{z=0} > 0 \text{ and } \left. \frac{\partial u}{\partial z} \right|_{z=H} > 0 \implies \text{potential instability.} \quad (10.127)$$

(And similarly, with both inequalities switched.) Such an instability may occur where the troposphere acts like a lid, as for example in the Eady problem. If $\partial \bar{q} / \partial y = 0$ and there is no lid at $z = H$ (e.g., the Eady problem with no lid) then the instability disappears.

One consequence of the upper boundary condition is that it provides a condition on the depth of the disturbance. In the Eady problem the evolution of the system is determined by temperature evolution at the surface,

$$\frac{Db}{Dt} = 0 \quad \text{at } z = 0, H, \quad (10.128)$$

(where $b = f_0 \partial \psi / \partial z$) and zero potential vorticity in the interior, which implies that

$$\nabla^2 \psi + k_d^2 H^2 \frac{\partial^2 \psi}{\partial z^2} = 0 \quad 0 < z < H, \quad (10.129)$$

where $k_d = f_0/(HN)$. Assuming a solution of the form $b \sim \sin kx$ then the Poisson equation (10.129) becomes

$$H^2 k_d^2 \frac{\partial^2 \psi}{\partial z^2} = k^2 \psi, \quad (10.130)$$

with solutions $\psi = A \exp(-\alpha z) + B \exp(\alpha z)$, where $\alpha^2 = k^2 N^2 / f_0^2$. The scale height of the disturbance is thus

$$h \sim \frac{f_0 L}{2\pi N}. \quad (10.131)$$

where $L \sim 2\pi/k$ is the horizontal scale of the disturbance. If the upper boundary is higher than this, it cannot interact strongly with the surface, because the disturbances at either boundary decay before reaching the other. Put another way, if the structure of the disturbance is such that it is shallower than H , the presence of the upper boundary is not felt. In the Eady problem, we know that the upper boundary must be important, because it is only by its presence that the flow can be unstable. Thus, all unstable modes in the Eady problem must be ‘deep’ in this sense, which can be verified by direct calculation. This condition gives rise to a physical interpretation of the high-wavenumber cut-off: if L is too small, the modes are too shallow to span the full depth of the fluid, and from (10.131) the condition for stability is thus

$$L < L_c = 2\pi \frac{NH}{f_0} \quad \text{or} \quad K > K_c = \frac{f_0}{NH} = L_c^{-1}. \quad (10.132)$$

where L_c and K_c are the critical length scales and wavenumbers. Wavenumbers larger than the reciprocal of the deformation radius are stable in the Eady problem. If β is non-zero, this condition does not apply, because the necessary condition for instability can be satisfied by a combination of a surface temperature gradient and an interior gradient of potential vorticity provided by β , as in condition (II) in section 10.6.2. Thus, we may expect that, if $\beta \neq 0$, higher wavenumbers ($k > k_d$) may be unstable but if so they will be shallow, and this may be confirmed by explicit calculation (see Figs. 9.12 and 9.19). In the two-level model shallow modes are, by construction, not allowed so that high wavenumbers will be stable, with or without beta.

10.7 ♦ NECESSARY CONDITIONS FOR INSTABILITY: USE OF PSEUDOENERGY

In this section we derive another necessary condition for instability, sometimes called an ‘Arnold condition’, that is based on the conservation properties of energy and enstrophy. Such conditions can be derived more generally by variational methods, and these lead to somewhat stronger results (in particular, nonlinear results that do not require the perturbation to be small) but our derivations will be elementary and direct.¹¹

10.7.1 Two-dimensional flow

First consider inviscid, incompressible two-dimensional flow governed by the equation of motion

$$\frac{\partial q}{\partial t} + J(\psi, q) = 0, \quad (10.133)$$

where $q = \zeta + f = \nabla^2 \psi + f$ is the absolute vorticity and ψ is the streamfunction. In a steady state, the streamfunction and the potential vorticity are functions of each other so that

$$q = Q(\Psi) \quad \text{and} \quad \psi = \Psi(Q), \quad (10.134)$$

where Q is a differentiable but otherwise arbitrary function of its argument, and Ψ its functional inverse. Equation (10.133) is then

$$\frac{\partial q}{\partial t} = -\frac{dQ}{d\psi} J(\Psi, \Psi) = 0 \quad (10.135)$$

and all steady solutions are of the form (10.134). We shall prove that if $d\Psi/dQ > 0$ then the flow is stable, in a sense to be made explicit below. Consider the evolution of perturbations about such a steady state, so that

$$q = Q + q', \quad \psi = \Psi + \psi', \quad (10.136)$$

and we suppose that the perturbation vanishes at the domain boundary or that the boundary conditions are periodic. The potential vorticity perturbation satisfies, in the linear approximation,

$$\frac{\partial q'}{\partial t} + J(\psi', Q) + J(\Psi, q') = 0. \quad (10.137)$$

Now, because potential vorticity is conserved on parcels, any function of potential vorticity is also materially conserved, and in particular

$$\frac{D\Psi(q)}{Dt} = \frac{\partial \Psi}{\partial t} + J(\psi, \Psi) = 0. \quad (10.138)$$

Linearizing this using (10.136) gives

$$\frac{d\Psi}{dQ} \frac{\partial q'}{\partial t} + J(\psi', \Psi) + J\left(\Psi, \frac{d\Psi}{dQ} q'\right) = 0. \quad (10.139)$$

We now form an energy equation from (10.137) by multiplying by $-\psi'$ and integrating over the domain. Integrating the first term by parts we find

$$\frac{d}{dt} \int \frac{1}{2} (\nabla \psi')^2 dA = \int \psi' J(\Psi, q') dA. \quad (10.140)$$

Similarly, from (10.139) we obtain

$$\frac{d}{dt} \int \frac{1}{2} \frac{d\Psi}{dQ} q'^2 dA = - \int \left[q' J(\psi', \Psi) + q' J\left(\Psi, \frac{d\Psi}{dQ} q'\right) \right] dA. \quad (10.141)$$

The second term in square brackets vanishes. This follows using the property of Jacobians, obtained by integrating by parts, that

$$\langle aJ(b, c) \rangle = \langle bJ(c, a) \rangle = \langle cJ(a, b) \rangle = -\langle cJ(b, a) \rangle, \quad (10.142)$$

where the angle brackets denote horizontal integration. Using this we have

$$\begin{aligned} \left\langle q' J\left(\Psi, \frac{d\Psi}{dQ} q'\right) \right\rangle &= - \left\langle \frac{d\Psi}{dQ} q' J(\Psi, q') \right\rangle = -\frac{1}{2} \left\langle \frac{d\Psi}{dQ} J(\Psi, q'^2) \right\rangle \\ &= -\frac{1}{2} \left\langle q'^2 J\left(\frac{d\Psi}{dQ}, \Psi\right) \right\rangle = 0. \end{aligned} \quad (10.143)$$

Adding (10.140) and (10.141) the remaining nonlinear terms cancel and we obtain the conservation law,

$$\frac{d\hat{H}}{dt} = 0 \quad \text{where} \quad \hat{H} = \frac{1}{2} \int \left[(\nabla \psi')^2 + \frac{d\Psi}{dQ} q'^2 \right] dA. \quad (10.144)$$

The quantity $\widehat{H}$ is known as the *pseudoenergy* of the disturbance and because it is a conserved quantity, quadratic in the wave amplitude, it is (like pseudomomentum) a wave activity. Its conservation holds whether the disturbance is growing, decaying or neutral.

If $d\Psi/dQ$ is positive everywhere the pseudoenergy is a positive-definite quantity, and the growth of the disturbance is then largely prevented and the basic state is said to be *stable in the sense of Liapunov*. This means that the magnitude of the perturbation, as measured by some norm, is bounded by its initial magnitude. In the case here we define the norm

$$\|\psi\|^2 \equiv \int \left[(\nabla\psi)^2 + \frac{d\Psi}{dQ} (\nabla^2\psi)^2 \right] dA, \quad (10.145)$$

so that

$$\|\psi'(t)\|^2 = \|\psi'(0)\|^2. \quad (10.146)$$

If $d\Psi/dQ > 0$ then, although the energy of the disturbance can grow, its final amplitude is bounded by the initial value of the pseudoenergy, because if perturbation energy is to grow perturbation enstrophy must shrink but it cannot shrink past zero. Normal-mode instability, in which modes grow exponentially, is completely precluded.

If the pseudoenergy is *negative definite* then stability is also assured, but this is a less common situation for it demands that $d\Psi/dQ$ be sufficiently negative so that the (negative of the) enstrophy contribution is always larger than the energy contribution, and this can usually only be satisfied in a sufficiently small domain. To see this, suppose that $q' = \nabla^2\psi'$, and that in the domain under consideration the Laplacian operator has eigenvalues $-k^2$ where

$$\nabla^2\psi' = -k^2\psi' \quad (10.147)$$

and the smallest eigenvalue, by magnitude, is k_0^2 . Then, using Poincaré's inequality,

$$\int (\nabla^2\psi')^2 dA \geq k_0^2 \int (\nabla\psi')^2 dA, \quad (10.148)$$

a sufficient condition to make $\widehat{H}$ negative definite is that

$$\frac{d\Psi}{dQ} < -\frac{1}{k_0^2}. \quad (10.149)$$

As the domain gets bigger, k_0 diminishes and this condition becomes harder to satisfy.¹²

Parallel shear flow and Fjørtoft's condition

Consider the stability of a zonal flow (i.e., a flow in the x -direction), that varies only with y . The flow stability condition is then

$$\frac{d\Psi}{dQ} = \frac{d\Psi/dy}{dQ/dy} = -\frac{U - U_s}{\beta - U_{yy}} > 0, \quad (10.150)$$

where U_s is a constant, representing an arbitrary, constant, zonal flow. The last equality follows because the problem is Galilean invariant, and we are therefore at liberty to choose U_s arbitrarily. To connect this with Fjørtoft's condition (chapter 9) multiply the top and bottom by $(\beta - U_{yy})$, whence we see that a sufficient condition for stability is that $(U - U_s)(\beta - U_{yy})$ is everywhere negative. The derivation here, unlike our earlier one in section 9.3.2, makes it clear that the condition does not only apply to normal-mode instabilities.

10.7.2 ♦ Stratified quasi-geostrophic flow

The extension of the pseudoenergy arguments to quasi-geostrophic flow is mostly straightforward, but with a complication from the vertical boundary conditions at the surface and at an upper boundary, and the trusting reader may wish to skip straight to the results, (10.155)–(10.157).¹³ For definiteness, we consider Boussinesq, β -plane quasi-geostrophic flow confined between flat rigid surfaces at $z = 0$ and $z = H$. The interior flow is governed by the familiar potential vorticity equation $Dq/Dt = 0$ and the buoyancy equation $Db/Dt = 0$ at the two boundaries, where

$$q = \nabla^2 \psi + \beta y + \frac{\partial}{\partial z} \left(S(z) \frac{\partial \psi}{\partial z} \right), \quad b = f_0 \frac{\partial \psi}{\partial z}, \quad (10.151)$$

and $S(z) = f_0^2/N^2$ is positive. The basic state ($\psi = \Psi$, $q = Q$, $b = B_1, B_2$) satisfies

$$\begin{aligned} \psi &= \Psi(Q), & 0 < z < H, \\ \psi &= \Psi_1(B_1), & z = 0 \quad \text{and} \quad \psi = \Psi_2(B_2), & z = H. \end{aligned} \quad (10.152)$$

Analogous to the barotropic case, we obtain the equations of motion for the interior perturbation

$$\frac{\partial q'}{\partial t} + J(\psi', Q) + J(\Psi, q') = 0, \quad (10.153a)$$

$$\frac{d\Psi}{dQ} \frac{\partial q'}{\partial t} + J(\psi', \Psi) + J\left(\Psi, \frac{d\Psi}{dQ} q'\right) = 0, \quad (10.153b)$$

and at the two boundaries

$$\frac{\partial b'}{\partial t} + J(\psi', B_i) + J(\Psi_i, b') = 0, \quad (10.154a)$$

$$\frac{d\Psi_i}{dB_i} \frac{\partial b'}{\partial t} + J(\psi', \Psi_i) + J\left(\Psi_i, \frac{d\Psi_i}{dB_i} b'\right) = 0, \quad (10.154b)$$

for $i = 1, 2$. (By $d\Psi_i/dB_i$ we mean the derivative of Ψ_i with respect to its argument, evaluated at B_i .) From these equations, we form the pseudoenergy by multiplying (10.153a) by $-\psi'$, (10.153b) by q' , and (10.154a) by ψ' , (10.154b) by b' . After some manipulation we obtain the pseudoenergy conservation law:

$$\frac{d\widehat{H}}{dt} = 0 \quad \text{where} \quad \widehat{H} = \mathcal{E} + \mathcal{Z} + \mathcal{B}_1 + \mathcal{B}_2, \quad (10.155)$$

and

$$\begin{aligned} \mathcal{E} &= \frac{1}{2} \left\{ (\nabla \psi')^2 + S \left(\frac{\partial \psi'}{\partial z} \right)^2 \right\}, & \mathcal{Z} &= \frac{1}{2} \left\{ \frac{d\Psi}{dQ} q'^2 \right\}, \\ \mathcal{B}_1 &= \frac{1}{2} \left\langle \frac{S(0)}{f_0} \frac{d\Psi_1}{dB_1} b'(0)^2 \right\rangle, & \mathcal{B}_2 &= -\frac{1}{2} \left\langle \frac{S(H)}{f_0} \frac{d\Psi_2}{dB_2} b'(H)^2 \right\rangle. \end{aligned} \quad (10.156)$$

where the curly brackets denote a three-dimensional integration over the fluid interior, and the angle brackets denote a horizontal integration over the boundary surfaces at 0 and H . The pseudoenergy $\widehat{H}$ is positive-definite, and therefore stability is assured in that norm, if all of the following conditions are satisfied:

$$\frac{d\Psi}{dQ} > 0, \quad \frac{1}{f_0} \frac{d\Psi_1}{dB_1} > 0, \quad \frac{1}{f_0} \frac{d\Psi_2}{dB_2} < 0. \quad (10.157)$$

If the flow is compressible, the potential vorticity is $q = \nabla^2 \psi + \beta y + \rho_R^{-1} \partial_z (\rho_R S \partial_z \psi)$, where $\rho_R = \rho_R(z)$, but the final stability conditions are unaltered. If the upper boundary is then removed to

infinity where $\rho_R(z) = 0$ then only the lower boundary condition contributes to (10.157). In the layered form of the quasi-geostrophic equations the vertical boundary conditions are built in to the definitions of potential vorticity in the top and bottom layers. In this case, a sufficient condition for stability is that $d\Psi/dQ > 0$ in each layer. Indeed, an alternate derivation of (10.155)–(10.157) would be to incorporate the boundary conditions on buoyancy into the definition of potential vorticity by the delta-function construction of section 5.4.3.

Zonal shear flow

Consider now zonally uniform zonal flows, such as might give rise to baroclinic instability in a channel. The fields are then functions of y and z only, and the sufficient conditions for stability are:

$$\begin{aligned}\frac{d\Psi}{dQ} &= \frac{\partial\Psi/\partial y}{\partial Q/\partial y} = -\frac{U}{dQ/dy} > 0, \\ \frac{d\Psi_1}{dB_1} &= \frac{d\Psi_1/dy}{dB_1/dy} = \frac{U(0)}{dU(0)/dz} > 0, \\ \frac{d\Psi_2}{dB_2} &= \frac{d\Psi_2/dy}{dB_2/dy} = \frac{U(H)}{dU(H)/dz} < 0.\end{aligned}\tag{10.158}$$

using the thermal wind relation, and setting $f_0 = 1$ (its value is irrelevant). These results generalize Fjørtoft's condition to the stratified case,¹⁴ and as in that case we are at liberty to add a uniform zonal flow to all the velocities.

10.7.3 ♦ Applications to baroclinic instability

We may use the stability conditions derived above to provide a few more results about baroclinic instability, including an alternate derivation of the minimum shear criterion in two-layer flow, and a derivation of the high-wavenumber cut-off to instability. In what follows we do not derive any new criteria; rather, the derivations make it apparent that the criteria are not restricted to perturbations of normal-mode form.

Minimum shear in two-layer flow

We consider two layers of equal depth, on a flat-bottomed β -plane with basic state

$$\Psi_1 = -U_1 y, \quad \Psi_2 = -U_2 y \tag{10.159a}$$

$$Q_1 = \beta y - \frac{k_d^2}{2}(U_2 - U_1)y, \quad Q_2 = \beta y - \frac{k_d^2}{2}(U_1 - U_2)y. \tag{10.159b}$$

This state is characterized by $Q_i = \gamma_i \Psi_i$ where

$$\gamma_1 = -\frac{(\beta + k_d^2 \bar{U})}{(\bar{U} + \hat{U})}, \quad \gamma_2 = -\frac{(\beta - k_d^2 \hat{U})}{(\bar{U} - \hat{U})}, \tag{10.160}$$

with $\bar{U} = (U_1 + U_2)/2$ and $\hat{U} = (U_1 - U_2)/2$. The barotropic flow does not affect the stability properties, so without loss of generality we may choose $\bar{U} < -\hat{U}$, and this makes $\gamma_1 > 0$. Then γ_2 is also positive if $\beta > k_d^2 \hat{U}/2$. Thus, a sufficient condition for stability is that

$$\hat{U} < \frac{\beta}{k_d^2}, \tag{10.161}$$

as obtained in chapter 9. However, we now see that the stability condition does not apply only to normal-mode instabilities.¹⁵

Use of pseudomomentum conservation provides an alternative derivation of the same result. The flow will also be stable if in both layers $\partial Q/\partial y > 0$, for then the conserved pseudomomentum will be positive definite. If $U_1 > U_2$ then, from (10.159) $dQ_1/dy > 0$. The flow will be stable if $dQ_2/dy > 0$, and this gives

$$\widehat{U} = \frac{1}{2}(U_1 - U_2) < \frac{\beta}{k_d^2}, \quad (10.162)$$

as in (10.161).

The high-wavenumber cut-off in two-layer baroclinic instability

We can use a pseudoenergy argument to show that there is a high-wavenumber cut-off to two-layer baroclinic instability, with the basic state (10.159). The conserved pseudoenergy analogous to (10.155) and (10.156) is readily found to be

$$\widehat{H} = \left\langle (\nabla\psi'_1)^2 + (\nabla\psi'_2)^2 + \frac{1}{2}k_d^2(\psi'_1 - \psi'_2)^2 + \frac{q_1'^2}{\gamma_1} + \frac{q_2'^2}{\gamma_2} \right\rangle = 0. \quad (10.163)$$

Let us choose (without loss of generality) the barotropic flow to be $\overline{U} = \beta/k_d^2$. We then have $\gamma_1 = \gamma_2 = -1/k_d^2$, and the pseudoenergy is then just the actual energy minus k_d^{-2} times the total enstrophy. If we define $\psi = (\psi'_1 + \psi'_2)/2$ and $\tau = (\psi'_1 - \psi'_2)/2$ then, using (12.41a) and (12.44), (10.163) may be expressed as

$$\widehat{H} = \left\langle (\nabla\psi)^2 + (\nabla\tau)^2 + k_d^2\tau^2 - k_d^{-2} \left\{ (\nabla^2\psi)^2 + [(\nabla^2 - k_d^2)\tau]^2 \right\} \right\rangle. \quad (10.164)$$

Now, let us express the fields as Fourier sums,

$$(\tau, \psi) = \sum_{k,l} (\tilde{\tau}_{k,l}, \tilde{\psi}_{k,l}) e^{i(kx+ly)}. \quad (10.165)$$

(This expression assumes a doubly-periodic domain; essentially the same end-result is obtained in a channel.) The pseudoenergy may then be written as

$$\widehat{H} = \sum_{k,l} [K^2 \tilde{\psi}_{k,l}^2 (k_d^2 - K^2) + K'^2 \tilde{\tau}_{k,l}^2 (k_d^2 - K'^2)] \quad (10.166)$$

where $K^2 = k^2 + l^2$ and $K'^2 = K^2 + k_d^2$. If the deformation radius is sufficiently large (or the domain sufficiently small) that $K^2 > k_d^2$, then the pseudoenergy is *negative-definite*, so the flow is stable, no matter what the shear may be. Such a situation might arise on a planet whose circumference was less than the deformation radius, or in a small ocean basin. In the linear problem, in which perturbation modes do not interact, horizontal wavenumbers with $k^2 > k_d^2$ are stable and there is thus a high-wavenumber cut-off to instability, as was found in chapter 9 by direct calculation.

Notes

- 1 After Eliassen & Palm (1961).
- 2 Andrews & McIntyre (1976), Ripa (1981) and Held (1985).
- 3 These restrictions on the basic state are not necessary to prove orthogonality, but they make the algebra simpler. Also, we pay no attention here to the nature of the eigenvalues of (10.45), which, in general, consist of both a discrete and a continuous spectrum. See Farrell (1984) and McIntyre & Shepherd (1987).

- 4 The TEM was introduced by Andrews & McIntyre (1976, 1978) and Boyd (1976b). A precursor is the paper of Riehl & Fultz (1957), who noted the shortcomings of zonal averaging in uncovering the meaning of indirect cells in laboratory experiments, and by extension the atmosphere.
- 5 The main result of this subsection was originally obtained by McIntosh & McDougall (1996). I thank A. Plumb for a discussion about the derivation given here. It is in fact possible to write exact TEM-like equations wholly in terms of the thickness-weighted averaged quantities and without taking a zonal average. The literature is extensive and at times hard to follow, but the reader may usefully look to de Szoeke & Bennett (1993) and Young (2012), who show that thickness-weighted averaged equations may be derived that are identical to the unweighted equations except for the appearance, in the horizontal momentum equations, of an eddy forcing by the divergence of three-dimensional Eliassen–Palm vectors. The divergence of these EP vectors is related to the eddy flux of the full potential vorticity, in an analogous manner to (but more general than) the quasi-geostrophic result.
- 6 This problem can be worked around in some cases (Plumb 1990, Greatbatch 1998).
- 7 Non-acceleration arguments have a long history, with contributions from Charney & Drazin (1961), Eliassen & Palm (1961), Holton (1974) and, in particular, Boyd (1976b) and Andrews & McIntyre (1978). Dunkerton (1980) reviews and provides examples.
- 8 Conservation laws of this ilk, their connection to the underlying symmetries of the basic state and (relatedly) their finite-amplitude extension, are discussed by McIntyre & Shepherd (1987) and Shepherd (1990). Conservation of momentum is related to the translational invariance of the medium whereas conservation of $\mathcal{A}$ is related to the translational invariance of the basic state, and hence the appellation ‘pseudomomentum’.
- 9 Andrews & McIntyre (1978) and Young (2012).
- 10 Steve Garner and Raffaele Ferrari both provided very helpful input to this section.
- 11 The original papers are Arnold (1965, 1966), with a number of results being subsequently developed by Holm *et al.* (1985). See Shepherd (1990) for a review.
- 12 The stability criterion is sometimes referred to as ‘Arnold’s second condition’. More discussion, especially with regard to boundary conditions, is given in McIntyre & Shepherd (1987).
- 13 Blumen (1968), but the method we use is more direct.
- 14 Pedlosky (1964) derived these conditions by a normal-mode approach.
- 15 Pierini & Vulpiani (1981) and Vallis (1985) further consider the finite-amplitude case.

*Shootings of water thread down the slope
of the huge green stone —
The white eddy-rose that blossom'd up
against the stream in the scollop,
by fits and starts, obstinate in resurrection —
It is the life that we live.*

Samuel Taylor Coleridge, *The Eddy-Rose*, adapted from notebook, 1799.

CHAPTER 11

Basics of Incompressible Turbulence

TURBULENCE IS HIGH REYNOLDS NUMBER FLUID FLOW, dominated by nonlinearity, containing both spatial and temporal disorder. No definition is perfect, and it is hard to disentangle a definition from a property, but this statement captures the essential aspects. A turbulent flow has eddies with a spectrum of sizes between some upper and lower bounds, the former often determined by the forcing scale or the domain scale, and the latter usually by viscosity. The individual eddies come and go, and are inherently unpredictable.

The circulation of the atmosphere and ocean is, *inter alia*, the motion of a forced-dissipative fluid subject to various constraints such as rotation and stratification. The larger scales are orders of magnitude larger than the dissipation scale (the scale at which molecular viscosity becomes important) and at many if not all scales the motion is highly nonlinear and quite unpredictable. Thus, we can justifiably say that the atmosphere and ocean are turbulent fluids. Note that here we are not simply talking about the small-scale flows traditionally regarded as turbulent; rather, our main focus will be the large-scale flows associated with baroclinic instability and greatly influenced by rotation and stratification, a kind of turbulence known as *geostrophic turbulence*. However, before discussing turbulence in the atmosphere and ocean, in this chapter we consider from a fairly elementary standpoint the basic theory of two- and three-dimensional turbulence, and in particular the theory of inertial ranges. We do not provide a comprehensive discussion of turbulence; rather, we provide an introduction to those aspects of most interest or relevance to the dynamical oceanographer or meteorologist. In the next chapter we consider the effects of rotation and stratification and the problem of geostrophic turbulence.

11.1 THE FUNDAMENTAL PROBLEM OF TURBULENCE

Turbulence is a difficult subject because it is nonlinear, and because, and relatedly, there are interactions between scales of motion. Let us first see what difficulties these bring.¹

11.1.1 The closure problem

Although in a turbulent flow it may be virtually impossible to predict the detailed motion of each eddy, the statistical properties — time averages for example — are not necessarily changing and we might like to predict such averages. Effectively, we accept that we cannot predict the weather, but

we can try to predict the climate. Even though we know the equations that determine the system, this task proves to be very difficult because the equations are nonlinear, and we come up against the *closure problem*. To see what this is, let us decompose the velocity field into mean and fluctuating components,

$$\mathbf{v} = \bar{\mathbf{v}} + \mathbf{v}'. \quad (11.1)$$

Here $\bar{\mathbf{v}}$ is the mean velocity field, and $\mathbf{v}'$ is the deviation from that mean. The mean might be a time average, in which case $\bar{\mathbf{v}}$ is a function only of space and not of time, or it might be a time mean over a finite period (e.g., a season if we are dealing with the weather), or it might be some form of ensemble mean. The average of the deviation is, by definition, zero; that is $\bar{\mathbf{v}'} = 0$. The idea is to substitute (11.1) into the momentum equation and try to obtain a closed equation for the mean quantity $\bar{\mathbf{v}}$. Rather than dealing with the full Navier–Stokes equations, let us carry out this program for a model nonlinear system that obeys

$$\frac{du}{dt} + uu + ru = 0, \quad (11.2)$$

where r is a constant. The average of this equation is:

$$\frac{d\bar{u}}{dt} + \overline{uu} + r\bar{u} = 0. \quad (11.3)$$

The value of the term $\overline{uu}$ (i.e., $\overline{u^2}$) is not deducible simply by knowing $\bar{u}$, since it involves correlations between eddy quantities, namely $u'u'$. That is, $\overline{uu} = \bar{u}\bar{u} + \overline{u'u'} \neq \bar{u}\bar{u}$. We can go to the next order to try (vainly!) to obtain an equation for $\overline{u'u'}$. First multiply (11.2) by u to obtain an equation for u^2 , and then average it to yield

$$\frac{1}{2} \frac{d\overline{u^2}}{dt} + \overline{uuu} + r\overline{u^2} = 0. \quad (11.4)$$

This equation contains the undetermined cubic term $\overline{uuu}$. An equation determining this would contain a quartic term, and so on in an unclosed hierarchy. Many methods of closing the hierarchy make assumptions about the relationship of $(n+1)$ th order terms to n th order terms, for example by supposing that

$$\overline{uuuu} = \alpha \overline{uu}\overline{uu} + \beta \overline{uuuu}, \quad (11.5)$$

where α and β are some parameters, and closures set in physical space or in spectral space (i.e., acting on the Fourier transformed variables) have both been proposed. If we know that the variables are distributed normally then such closures can sometimes be made exact, but this is not generally the case in turbulence and all closures that have been proposed so far are, at best, approximations.

This same closure problem arises in the Navier–Stokes equations. If density is constant (say $\rho = 1$) the x -momentum equation for an averaged flow is

$$\frac{\partial \bar{u}}{\partial t} + (\bar{\mathbf{v}} \cdot \nabla) \bar{u} = -\frac{\partial \bar{p}}{\partial x} - \nabla \cdot \overline{\mathbf{v}'u'}. \quad (11.6)$$

Written out in full in Cartesian coordinates, the last term is

$$\nabla \cdot \overline{\mathbf{v}'u'} = \frac{\partial}{\partial x} \overline{u'u'} + \frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \overline{u'w'}. \quad (11.7)$$

These terms, and the similar ones in the y - and z -momentum equations, represent the effects of eddies on the mean flow and are known as *Reynolds stress* terms. The ‘closure problem’ of turbulence may be thought of as finding a representation of such Reynolds stress terms in terms of mean flow quantities. Nobody has been able to close the system, in any useful way, without introducing physical assumptions not directly deducible from the equations of motion themselves. Indeed, not only has the problem not been solved, it is not clear that in general a useful closed-form solution actually exists.

11.1.2 Triad interactions in turbulence

The nonlinear term in the equations of motion not only leads to difficulties in closing the equations, but it leads to interactions among different length scales, and in this section we write the equations of motion in a form that makes this explicit. For algebraic simplicity we will restrict our attention to two-dimensional flows, but very similar considerations also apply in three dimensions, and the details of the algebra following are not of themselves important to subsequent sections.²

The equation of motion for an incompressible fluid in two dimensions [see for example (4.67) or (5.119)] may be written as

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) = F + \nu \nabla^2 \zeta, \quad \zeta = \nabla^2 \psi. \quad (11.8)$$

We include a forcing and viscous term but no Coriolis term. Let us suppose that the fluid is contained in a square, doubly-periodic domain of side L , and let us expand the streamfunction and vorticity in Fourier series so that, with a tilde denoting a Fourier coefficient,

$$\psi(x, y, t) = \sum_{\mathbf{k}} \tilde{\psi}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}}, \quad \zeta(x, y, t) = \sum_{\mathbf{k}} \tilde{\zeta}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}}, \quad (11.9)$$

where $\mathbf{k} = i\mathbf{k}^x + j\mathbf{k}^y$, $\tilde{\zeta} = -k^2 \tilde{\psi}$ where $k^2 = k^x^2 + k^y^2$ and, to ensure that ψ is real, $\tilde{\psi}(k^x, k^y, t) = \tilde{\psi}^*(-k^x, -k^y, t)$, where $*$ denotes the complex conjugate, and this property is known as conjugate symmetry. The summations are over all positive and negative x - and y -wavenumbers, and $\tilde{\psi}(\mathbf{k}, t)$ is shorthand for $\tilde{\psi}(k^x, k^y, t)$. Substituting (11.9) in (11.8) gives, with (for the moment) F and ν both zero,

$$\begin{aligned} \frac{\partial}{\partial t} \sum_{\mathbf{k}} \tilde{\zeta}(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}} &= \sum_{\mathbf{p}} p^x \tilde{\psi}(\mathbf{p}, t) e^{i\mathbf{p} \cdot \mathbf{x}} \times \sum_{\mathbf{q}} q^y \tilde{\zeta}(\mathbf{q}, t) e^{i\mathbf{q} \cdot \mathbf{x}} \\ &\quad - \sum_{\mathbf{p}} p^y \tilde{\psi}(\mathbf{p}, t) e^{i\mathbf{p} \cdot \mathbf{x}} \times \sum_{\mathbf{q}} q^x \tilde{\zeta}(\mathbf{q}, t) e^{i\mathbf{q} \cdot \mathbf{x}}, \end{aligned} \quad (11.10)$$

where $\mathbf{p}$ and $\mathbf{q}$ are, like $\mathbf{k}$, horizontal wave vectors. We may obtain an evolution equation for the wavevector $\mathbf{k}$ by multiplying (11.10) by $\exp(-i\mathbf{k} \cdot \mathbf{x})$ and integrating over the domain, and using the fact that the Fourier modes are orthogonal; that is

$$\int e^{i\mathbf{p} \cdot \mathbf{x}} e^{i\mathbf{q} \cdot \mathbf{x}} dA = L^2 \delta(\mathbf{p} + \mathbf{q}), \quad (11.11)$$

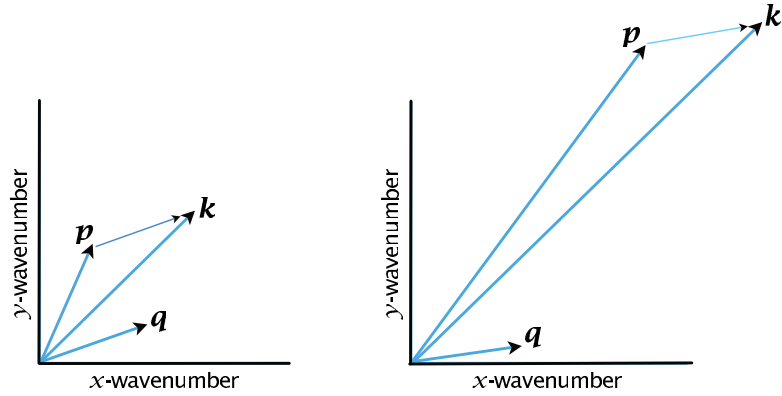
where $\delta(\mathbf{p} + \mathbf{q})$ equals unity if $\mathbf{p} = -\mathbf{q}$ and is zero otherwise. Using this, (11.10) becomes, restoring the forcing and dissipation terms,

$$\frac{\partial}{\partial t} \tilde{\psi}(\mathbf{k}, t) = \sum_{\mathbf{p}, \mathbf{q}} A(\mathbf{k}, \mathbf{p}, \mathbf{q}) \tilde{\psi}(\mathbf{p}, t) \tilde{\psi}(\mathbf{q}, t) + \tilde{F}(\mathbf{k}) - \nu k^2 \tilde{\psi}(\mathbf{k}, t), \quad (11.12)$$

where $A(\mathbf{k}, \mathbf{p}, \mathbf{q}) = (q^2/k^2)(p^x q^y - p^y q^x) \delta(\mathbf{p} + \mathbf{q} - \mathbf{k})$ is an ‘interaction coefficient’, and the summation is over all $\mathbf{p}$ and $\mathbf{q}$; however, note that only those wavevector triads with $\mathbf{p} + \mathbf{q} = \mathbf{k}$ make a non-zero contribution, because of presence of the delta function.

Consider, then, a fluid in which just two Fourier modes are initially excited, with wavevectors $\mathbf{p}$ and $\mathbf{q}$ say, along with their conjugate-symmetric partners at $-\mathbf{p}$ and $-\mathbf{q}$. These modes interact [obeying (11.12)] to generate third and fourth wavenumbers, $\mathbf{k} = \mathbf{p} + \mathbf{q}$ and $\mathbf{m} = \mathbf{p} - \mathbf{q}$ (along with their conjugate-symmetric partners). These four wavenumbers can interact among themselves to generate several additional wavenumbers, $\mathbf{k} + \mathbf{p}$, $\mathbf{k} + \mathbf{m}$ and so on, and these in turn lead to still more interactions so potentially filling out the entire spectrum of wavenumbers. The individual interactions are called *triad interactions*, and it is by way of such interactions that energy is transferred

Fig. 11.1 Two interacting triads, each with $\mathbf{k} = \mathbf{p} + \mathbf{q}$. On the left, a local triad with $k \sim p \sim q$. On the right, a non-local triad with $k \sim p \gg q$.



between scales in turbulent flows, in both two and three dimensions. The dissipation term does not lead to interactions between modes with different wavevectors; rather, it acts like a drag on each Fourier mode, with a coefficient that increases with wavenumber and therefore that preferentially affects small scales.

The selection rule for triad interactions — that $\mathbf{k} = \mathbf{p} + \mathbf{q}$ — does not restrict the scales of these interacting wavevectors, and the types of triad interactions fall between two extremes:

- (i) local interactions, in which $k \sim p \sim q$;
- (ii) non-local interactions, in which $k \sim p \gg q$.

These two kinds of triads are schematically illustrated in Fig. 11.1. Without very detailed analysis of the solutions of the equations of motion — an analysis that is in general impossible for fully-developed turbulence — we cannot say with certainty whether one particular kind of triad interaction dominates. The theory of Kolmogorov considered below, and its two-dimensional analogue, assume that it is the *local* triads that are most important in transferring energy; this is a reasonable assumption because, from the perspective of a small eddy, large eddies appear as a nearly-uniform flow and so simply translate the small eddies around without distorting them and thus without transferring energy between scales.

11.2 THE KOLMOGOROV THEORY

The foundation of many theories of turbulence is the spectral theory of Kolmogorov.³ This theory does not close the equations in quite as explicit a manner as (11.5), but it does provide a prediction for the energy spectrum of a turbulent flow (loosely speaking, how much energy is present at a particular spatial scale) and it does this by suggesting a relationship between the energy spectrum (a second-order quantity in velocity) and the spectral energy flux (a third-order quantity).

11.2.1 The physical picture

Consider high Reynolds number (Re) incompressible flow that is being maintained by some external force. Then the evolution of a system that has $\rho = 1$ is governed by

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + \mathbf{F} + \nu \nabla^2 \mathbf{v} \quad (11.13)$$

and

$$\nabla \cdot \mathbf{v} = 0. \quad (11.14)$$

Here, $\mathbf{F}$ is some force we apply to maintain fluid motion — for example, we stir the fluid with a spoon. (One might argue that such stirring is not a force, like gravity, but a continuous changing of the boundary conditions. Having noted this, we treat it as a force.) A simple scale analysis of these equations seems to indicate that the ratio of the size of the inertial terms on the left-hand side to the viscous term is the Reynolds number VL/ν , where V and L are velocity and length scales. To be explicit let us consider the ocean, and take $V = 0.1 \text{ m s}^{-1}$, $L = 1000 \text{ km}$ and $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$. Then $Re = VL/\nu \approx 10^{11}$, and it seems that we can neglect the viscous term on the right-hand side of (11.13). But this can lead to a paradox, as if the fluid is being forced this forcing is likely to put energy into the fluid. To see this, we obtain the energy budget for (11.13) by multiplying by $\mathbf{v}$ and integrating over a domain. If there is no flow into or out of our domain, the inertial terms in the momentum equation conserve energy and, recalling section 1.10, the energy equation is

$$\frac{d\hat{E}}{dt} = \frac{d}{dt} \int \frac{1}{2} \mathbf{v}^2 dV = \int (\mathbf{F} \cdot \mathbf{v} + \nu \mathbf{v} \cdot \nabla^2 \mathbf{v}) dV = \int (\mathbf{F} \cdot \mathbf{v} - \nu \omega^2) dV, \quad (11.15)$$

where $\hat{E}$ is the total energy. If we neglect the viscous term we are led to an inconsistency, since the forcing term is a source of energy ($\mathbf{F} \cdot \mathbf{v} > 0$), because a force will normally, on average, produce a velocity that is correlated with the force itself. Without viscosity, energy keeps on increasing.

What is amiss? It is true that for motion with a 1000 km length scale and a velocity of a few centimetres per second we can neglect viscosity when considering the balance of forces in the momentum equation. But this does not mean that there is no motion at much smaller length scales — indeed we seem to be led to the inescapable conclusion that there must be some motion at smaller scales in order to remove energy. Scale analysis of the momentum equation suggests that viscous terms will be comparable with the inertial terms at a scale L_ν , where the Reynolds number based on that scale is of order unity, giving

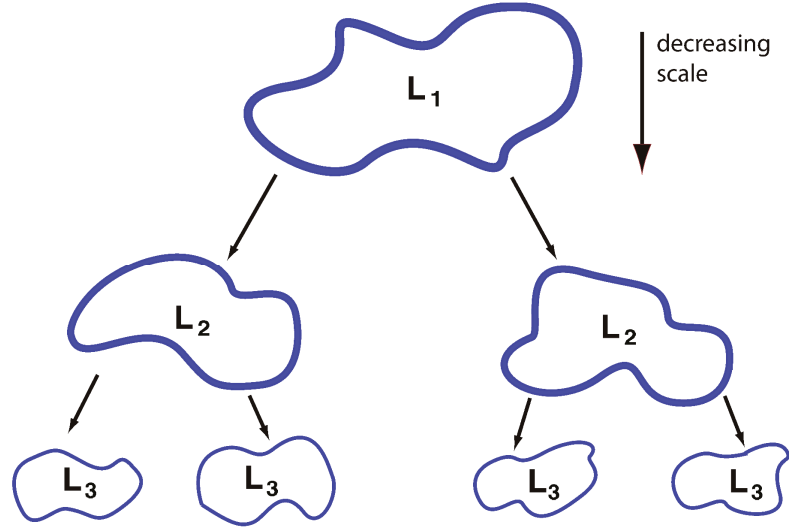
$$L_\nu \sim \frac{\nu}{V}. \quad (11.16)$$

This is a very small scale for geophysical flows, of order millimetres or less. Where and how are such small scales generated? Boundaries are one important region; if there is high Reynolds number flow above a solid boundary, for example the wind above the ground, then viscosity *must* become important in bringing the velocity to zero in order that it can satisfy the no-slip condition at the surface, as illustrated in Fig. 5.4.

Motion on very small scales may also be generated in the fluid interior. How might this happen? Suppose the forcing acts only at large scales, and its direct action is to set up some correspondingly large-scale flow, composed of eddies and shear flows and such-like. Then typically there will be an instability in the flow, and a smaller eddy will grow: initially, the large-scale flow may be treated as an unchanging shear flow, and the disturbance while small will obey linear equations of motion similar to those applicable in an idealized Kelvin–Helmholtz instability. This instability clearly must draw from the large scale quasi-stationary flow, and it will eventually saturate at some finite amplitude. Although it has grown in intensity, it is still typically smaller than the large scale flow that fostered it (remember how the growth rate of the shear instability gets larger as the wavelength of the perturbation decreased). As it reaches finite amplitude, the perturbation itself may become unstable, and smaller eddies will feed off its energy and grow, and so on.⁴ Vortex stretching plays an important role in all this, stretching line elements and creating eddies, and energy, at small scales. The general picture that emerges is of a large-scale flow that is unstable to eddies somewhat smaller in scale. These eddies grow, and develop still smaller eddies, and energy is transferred to smaller and smaller scales in a cascade-like process, sketched in Fig. 11.2. Finally, eddies are generated that are sufficiently small that they feel the effects of viscosity, and energy is drained away. Thus, there is a flux of kinetic energy from the large to the small scales, where it is dissipated into heat.

Fig. 11.2 The passage of energy to smaller scales: eddies at large scale break up into ones at smaller scale, thereby transferring energy to smaller scales. (The eddies in reality are embedded within each other.)

If the passage occurs between eddies of similar sizes (i.e., if it is spectrally local) the transfer is said to be a cascade.



11.2.2 Inertial-range theory

Given the above picture it becomes possible to predict what the energy spectrum is. Let us suppose that the flow is statistically isotropic (i.e., the same in all directions) and homogeneous (i.e., the same everywhere; all isotropic flows are homogeneous, but not vice versa). Homogeneity precludes the presence of solid boundaries but can be achieved in a periodic domain, and of course finite domain puts an upper limit, sometimes called the outer scale, on the size of eddies.

If we decompose the velocity field into Fourier components, then in a finite domain we may write

$$u(x, y, z, t) = \sum_{\mathbf{k}} \tilde{u}(\mathbf{k}, t) e^{i(k^x x + k^y y + k^z z)}, \quad (11.17)$$

where $\tilde{u}$ is the Fourier transformed field of u , with similar identities for v and w , and $\mathbf{k} = (k^x, k^y, k^z)$. The sum is a triple sum over all wavenumbers (k^x, k^y, k^z) , and in a finite domain these wavenumbers are quantized. Finally, to ensure that u is real we require that $\tilde{u}(-\mathbf{k}) = \tilde{u}^*(\mathbf{k})$, where the asterisk denotes the complex conjugate. Using Parseval's theorem (and assuming density is unity, as we shall throughout this chapter) the energy in the fluid is given by

$$\frac{1}{V} \int_V E dV = \frac{1}{2V} \int_V (u^2 + v^2 + w^2) dV = \frac{1}{2} \sum_{\mathbf{k}} (|\tilde{u}|^2 + |\tilde{v}|^2 + |\tilde{w}|^2) \equiv \sum_{\mathbf{k}} \mathcal{E}_{\mathbf{k}}, \quad (11.18)$$

where E is the energy density per unit mass, V is the volume of the domain, and the last equality serves to define the discrete energy spectrum $\mathcal{E}_{\mathbf{k}}$. We will now assume that the turbulence is isotropic, and that the domain is sufficiently large that the sums in the above equations may be replaced by integrals. We may then write

$$\bar{E} = \frac{1}{V} \hat{E} = \frac{1}{2V} \int_V \mathbf{v}^2 dV = \int \mathcal{E}(k) dk, \quad (11.19)$$

where $\bar{E}$ is the average energy, $\hat{E}$ is the total energy and $\mathcal{E}(k)$ is the energy spectral density, or the energy spectrum, so that $\mathcal{E}(k)\delta k$ is the energy in the small wavenumber interval δk . Because of the assumed isotropy, the energy is a function only of the scalar wavenumber k , where $k^2 = k^{x2} + k^{y2} + k^{z2}$. The units of $\mathcal{E}(k)$ are L^3/T^2 and the units of $\bar{E}$ are L^2/T^2 .

We now suppose that the fluid is stirred at large scales and, via the nonlinear terms in the momentum equation, that this energy is transferred to small scales where it is dissipated by viscosity.

Dimensions and the Kolmogorov Spectrum

Quantity	Dimension
Wavenumber, k	$1/L$
Energy per unit mass, E	$U^2 = L^2/T^2$
Energy spectrum, $\mathcal{E}(k)$	$EL = L^3/T^2$
Energy Flux, ε	$E/T = L^2/T^3$

If $\mathcal{E} = f(\varepsilon, k)$ then the only dimensionally consistent relation for the energy spectrum is

$$\mathcal{E} = \mathcal{K} \varepsilon^{2/3} k^{-5/3},$$

where $\mathcal{K}$ is a dimensionless constant.

The key assumption is to suppose that, if forcing scale is sufficiently larger than the dissipation scale, there exists a range of scales that is intermediate between the large scale and the dissipation scale and where neither forcing nor dissipation are explicitly important to the dynamics. This assumption, known as the *locality hypothesis*, depends on the nonlinear transfer of energy being sufficiently local (in spectral space). This intermediate range is known as the *inertial range*, because the inertial terms and not forcing or dissipation must dominate in the momentum balance. If the rate of energy input per unit volume by stirring is equal to ε , then if we are in a steady state there must be a flux of energy from large to small scales that is also equal to ε , and an energy dissipation rate, also ε .

Now, we have no general theory for the energy spectrum of a turbulent fluid, but we might suppose it takes the general form

$$\mathcal{E}(k) = g(\varepsilon, k, k_0, k_\nu), \quad (11.20)$$

where the right-hand side denotes a function of the spectral energy flux or cascade rate ε , the wavenumber k , the forcing wavenumber k_0 and the wavenumber at which dissipation acts, k_ν (and $k_\nu \sim L_\nu^{-1}$). The function g will of course depend on the particular nature of the forcing. Now, the locality hypothesis essentially says that at some scale within the inertial range the flux of energy to smaller scales depends only on processes occurring at or near that scale. That is to say, the energy flux is only a function of ε and k , or equivalently that the energy spectrum can be a function *only* of the energy flux ε and the wavenumber itself. From a physical point of view, as energy cascades to smaller scales the details of the forcing are forgotten but the effects of viscosity are not yet apparent, and the energy spectrum takes the form,

$$\mathcal{E}(k) = g(\varepsilon, k). \quad (11.21)$$

The function g is assumed to be *universal*, the same for every turbulent flow.

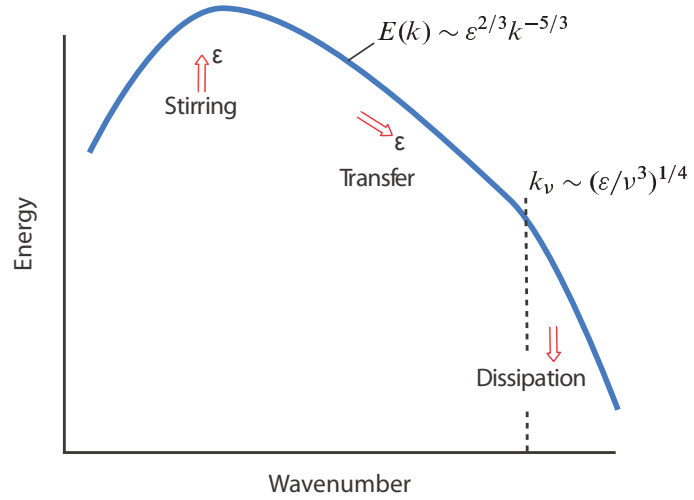
Let us now use dimensional analysis to give us the form of the function $g(\varepsilon, k)$ (see the shaded box above). In (11.21), the left-hand side has dimensions L^3/T^2 ; the factor T^{-2} can only be balanced by $\varepsilon^{2/3}$ because k has no time dependence; that is, (11.21), and its dimensions, must take the form

$$\mathcal{E}(k) = \varepsilon^{2/3} g(k), \quad (11.22a)$$

$$\frac{L^3}{T^2} \sim \frac{L^{4/3}}{T^2} g(k), \quad (11.22b)$$

where $g(k)$ is some function. Evidently $g(k)$ must have dimensions $L^{5/3}$, and the functional rela-

Fig. 11.3 Schema of energy spectrum in three-dimensional turbulence, in the theory of Kolmogorov. Energy is supplied at some rate ε ; it is cascaded to small scales, where it is ultimately dissipated by viscosity. There is no systematic energy transfer to scales larger than the forcing scale, so here the energy falls off.



tionship we must have, if the physical assumptions are right, is

$$\mathcal{E}(k) = \mathcal{K} \varepsilon^{2/3} k^{-5/3}. \quad (11.23)$$

This is the famous ‘Kolmogorov -5/3 spectrum’, enshrined as one of the cornerstones of turbulence theory. It is sketched in Fig. 11.3, and some experimental results are shown in Fig. 11.4. The parameter $\mathcal{K}$ is a dimensionless constant, undetermined by this theory; it is known as Kolmogorov’s constant and experimentally its value is found to be about 1.5.⁵

An equivalent, and revealing, way to derive this result is to first define an eddy turnover time τ_k , which is the time taken for a parcel with velocity v_k to move a distance $1/k$, v_k being the velocity associated with the (inverse) scale k . On dimensional considerations $v_k = [\mathcal{E}(k)k]^{1/2}$ so that

$$\tau_k = [k^3 \mathcal{E}(k)]^{-1/2}. \quad (11.24)$$

Kolmogorov’s assumptions are then equivalent to setting

$$\varepsilon \sim \frac{v_k^2}{\tau_k} = \frac{k \mathcal{E}(k)}{\tau_k}. \quad (11.25)$$

If we demand that ε be a constant then (11.24) and (11.25) yield (11.23).

The viscous scale and energy dissipation

At some small length scale we should expect viscosity to become important and the scaling theory we have just set up will fail. What is that scale? In the inertial range friction is unimportant because the time scales on which it acts are too long for it to be important and dynamical effects dominate. In the momentum equation the viscous term is $\nu \nabla^2 u$ so that a viscous or dissipation time scale at a scale k^{-1} , τ_k^ν , is

$$\tau_k^\nu \sim \frac{1}{k^2 \nu}, \quad (11.26)$$

so that the viscous time scale decreases with scale. The eddy turnover time, τ_k — that is, the inertial time scale — in the Kolmogorov spectrum is

$$\tau_k = \varepsilon^{-1/3} k^{-2/3}. \quad (11.27)$$

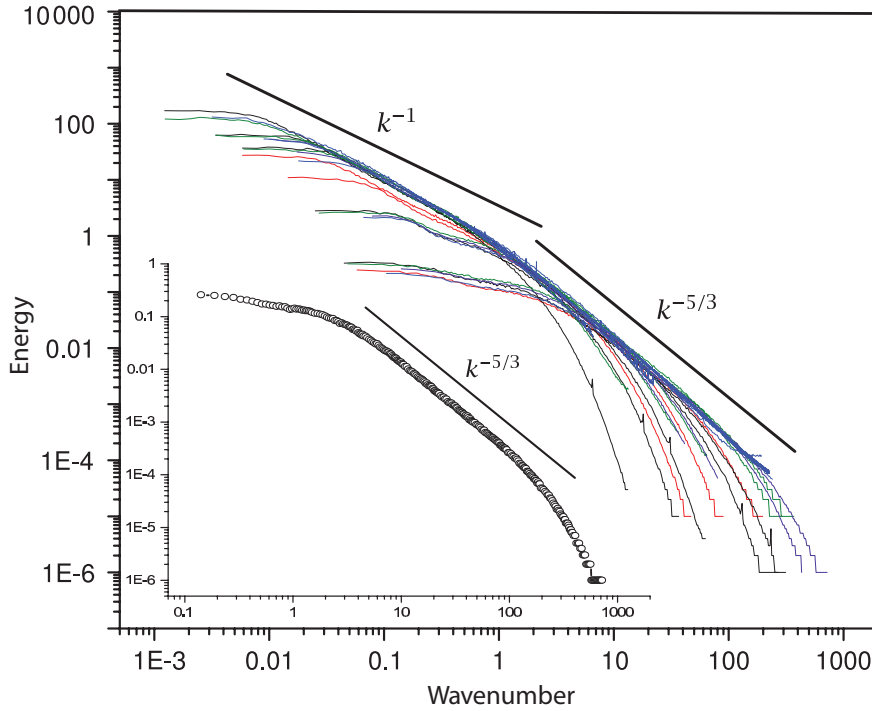


Fig. 11.4 The energy spectrum of 3D turbulence measured in some experiments at the Princeton Superpipe facility.⁶ The outer plot shows the spectra from a large number of experiments at different Reynolds numbers up to 10^6 , with the magnitude of their spectra appropriately rescaled. Smaller scales show a good $-5/3$ spectrum, whereas at larger scales the eddies feel the effects of the pipe wall and the spectra are a little shallower. The inner plot shows the spectrum in the centre of the pipe in a single experiment at $Re \approx 10^6$.

The wavenumber at which dissipation becomes important is then given by equating the above two time scales, yielding the dissipation wavenumber, k_v and the associated length scale, L_v ,

$$k_v \sim \left(\frac{\varepsilon}{\nu^3} \right)^{1/4}, \quad L_v \sim \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}. \quad (11.28a,b)$$

L_v is called the *Kolmogorov scale*. It is the *only* quantity which can be created from the quantities ν and ε that has the dimensions of length. (It is the same as the scale given by (11.16) provided that in that expression V is the velocity magnitude at the Kolmogorov scale.) Thus, for $L \gg L_v$, $\tau_k \ll \tau_k^\nu$ and inertial effects dominate. For $L \ll L_v$, $\tau_k^\nu \ll \tau_k$ and frictional effects dominate. In fact for length scales smaller than the dissipation scale, (11.27) is inaccurate; the energy spectrum falls off more rapidly than $k^{-5/3}$ and the inertial time scale falls off less rapidly than (11.27) implies, and dissipation dominates even more.

Given the dissipation scale, let us estimate the average energy dissipation rate, $\dot{\bar{E}}$. This is given by

$$\dot{\bar{E}} = \frac{1}{V} \int \nu \mathbf{v} \cdot \nabla^2 \mathbf{v} dV. \quad (11.29)$$

The length at which dissipation acts is the Kolmogorov scale and, noting that $v_k^2 \sim \varepsilon^{2/3} k^{-2/3}$ and

using (11.28a), the average energy dissipation rate scales as

$$\dot{\bar{E}} \sim \nu k_v^2 v_{k_v}^2 \sim \nu k_v^2 \frac{\varepsilon^{2/3}}{k_v^{2/3}} \sim \varepsilon. \quad (11.30)$$

That is, the energy dissipation rate is equal to the energy cascade rate. On the one hand this seems sensible, but on the other hand it is *independent of the viscosity*. In particular, in the limit of viscosity tending to zero the energy dissipation remains finite! Surely the energy dissipation rate must go to zero if viscosity goes to zero? To see that this is not the case, consider that energy is input at some large scales, and the magnitude of the stirring largely determines the energy input and cascade rate. The scale at which viscous effects then become important is determined by the viscous scale, L_v , given by (11.28b). *As viscosity tends to zero L_v becomes smaller in just such a way as to preserve the constancy of the energy dissipation.* This is one of the most important results in three-dimensional turbulence. Now, we established in section 1.10 that the Euler equations (i.e., the fluid equations with the viscous term omitted from the outset) do conserve energy. This means that the Euler equations are a *singular limit* of the Navier–Stokes equations: the behaviour of the Navier–Stokes equations as viscosity tends to zero is different from the behaviour resulting from simply omitting the viscous term from the equations ab initio.

How big is L_v in the atmosphere? A crude estimate, perhaps wrong by an order of magnitude, comes from noting that ε has units of U^3/L , and that at length scales of the order of 100 m in the atmospheric boundary layer (where there might be a three-dimensional energy cascade to small scales) velocity fluctuations are of the order of 1 cm s^{-1} , giving $\varepsilon \approx 10^{-8} \text{ m}^2 \text{ s}^{-3}$. Using (11.28b) we then find the dissipation scale to be of the order of a millimetre or so. In the ocean the dissipation scale is also of the order of millimetres. Various inertial range properties, in both three and two dimensions, are summarized in the shaded box on the next page.

Degrees of freedom

How many degrees of freedom does a turbulent fluid like the atmosphere potentially have? We might estimate this number, N say, by the expression

$$N \sim \left(\frac{L}{L_v} \right)^3, \quad (11.31)$$

where L is the length scale of the energy-containing eddies at the large scale. If we take $L = 1000 \text{ km}$ and $L_v = 1 \text{ mm}$ this gives about 10^{27} ! On a rather more general basis, we can obtain an expression for N using (11.28b), to give

$$N \sim L^3 \left(\frac{\varepsilon}{\nu^3} \right)^{3/4}, \quad (11.32)$$

or, using $\varepsilon \sim U^3/L$,

$$N \sim \left(\frac{UL}{\nu} \right)^{9/4} = Re^{9/4}, \quad (11.33)$$

where Re is the Reynolds number based on the large-scale flow. For typical large-scale atmospheric flows with $U \sim 10 \text{ m s}^{-1}$, $L \sim 10^6 \text{ m}$ and $\nu = 10^{-5} \text{ m}^2 \text{ s}^{-1}$, $Re \sim 10^{12}$ and again $N \sim 10^{27}$. Obviously, this number is very approximate, but nevertheless the number of potential degrees of freedom in the atmosphere is truly enormous, greater than Avogadro's number. Thus trying to model the turbulent atmosphere explicitly is akin to trying to model the gas in a room by following the motion of each individual molecule, and it seems unnecessary. How *should* we model it? That, in a nutshell, is the (unsolved) problem of turbulence.

Inertial Range Properties in 3D and 2D Turbulence

For reference, a few inertial range properties are listed below, omitting non-dimensional constants.

	3D Energy Range	2D Enstrophy Range	
Energy spectrum	$\varepsilon^{2/3} k^{-5/3}$	$\eta^{2/3} k^{-3}$	(T.1)
Turnover time	$\varepsilon^{-1/3} k^{-2/3}$	$\eta^{-1/3}$	(T.2)
Viscous scale, L_ν	$(\nu^3/\varepsilon)^{1/4}$	$(\nu^3/\eta)^{1/6}$	(T.3)
Passive tracer spectrum	$\chi \varepsilon^{-1/3} k^{-5/3}$	$\chi \eta^{-1/3} k^{-1}$	(T.4)

In these expressions:

ν = viscosity, k = wavenumber, ε = energy cascade rate,
 η = enstrophy cascade rate, χ = tracer variance cascade rate.

11.2.3 A final note on our assumptions

The assumptions of homogeneity and isotropy that are made in the Kolmogorov theory are ansatzes, in that we make them because we want to have a tractable model of turbulence (and certainly we can conceive of an experiment in which turbulence is for most practical purposes homogeneous and isotropic). The essential *physical* assumptions are: (i) that there exists an inertial range in which the energy flux is constant, and (ii) that the energy is cascaded from large to small scales in a series of small steps, as the energy spectra will then be determined by spectrally local quantities. The second assumption is the locality assumption and without it we could have, instead of (11.23),

$$\mathcal{E}(k) = C\varepsilon^{2/3} k^{-5/3} g(k/k_0) h(k/k_\nu), \quad (11.34)$$

where g and h are unknown functions. We essentially postulate that there exists a range of intermediate wavenumbers over which $g(k/k_0) = h(k/k_\nu) = 1$.

The first, and less obvious, assumption might be called the *non-intermittency* assumption, and it demands that rare events (in time or space) with large amplitudes do not dominate the energy flux or the dissipation rate. If they were to do so, then the flux would fluctuate strongly, the turbulent statistics would not be completely characterized by ε and Kolmogorov's theory would not be exactly right. Note that in the theory ε is the mean energy cascade rate, and $\varepsilon^{2/3}$ is the two-thirds power of the mean, which is not equal to the mean of the two-thirds power. In fact, in high Reynolds turbulence the $-5/3$ spectra is often observed to a fairly high degree of accuracy (e.g., as in Fig. 11.4), although the higher-order statistics (e.g., higher-order structure functions) predicted by the theory are often found to be in error, and it is generally believed that Kolmogorov's theory is not exact.⁷

11.3 TWO-DIMENSIONAL TURBULENCE

Two-dimensional turbulence behaves in a profoundly different way from three-dimensional turbulence, largely because of the presence of another quadratic invariant, the enstrophy (defined below;

see also section 5.6.3). In two dimensions, the vorticity equation for incompressible flow is:

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta = F + \nu \nabla^2 \zeta \quad (11.35)$$

where $\mathbf{u} = ui + vj$ and $\zeta = k \cdot \nabla \times \mathbf{u}$ and F is a stirring term. In terms of a streamfunction, $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$, and $\zeta = \nabla^2\psi$, and (11.35) may be written as

$$\frac{\partial \nabla^2 \psi}{\partial t} + J(\psi, \nabla^2 \psi) = F + \nu \nabla^4 \psi. \quad (11.36)$$

We obtain an energy equation by multiplying by $-\psi$ and integrating over the domain, and an enstrophy equation by multiplying by ζ and integrating. When $F = \nu = 0$ we find

$$\hat{E} = \frac{1}{2} \int_A (u^2 + v^2) dA = \frac{1}{2} \int_A (\nabla \psi)^2 dA, \quad \frac{d\hat{E}}{dt} = 0, \quad (11.37a)$$

$$\hat{Z} = \frac{1}{2} \int_A \zeta^2 dA = \frac{1}{2} \int_A (\nabla^2 \psi)^2 dA, \quad \frac{d\hat{Z}}{dt} = 0, \quad (11.37b)$$

where the integral is over a finite area with either no-normal flow or periodic boundary conditions. The quantity $\hat{E}$ is the energy, and $\hat{Z}$ is known as the *enstrophy*. The enstrophy invariant arises because the vortex stretching term, so important in three-dimensional turbulence, vanishes identically in two dimensions. In fact, because vorticity is conserved on parcels it is clear that the integral of *any* function of vorticity is zero when integrated over A ; that is, from (11.35)

$$\frac{Dg(\zeta)}{Dt} = 0 \quad \text{and} \quad \frac{d}{dt} \int_A g(\zeta) dA = 0. \quad (11.38)$$

where $g(\zeta)$ is an arbitrary function. Of this infinity of conservation properties, enstrophy conservation (with $g(\zeta) = \zeta^2$) in particular has been found to have enormous consequences to the flow of energy between scales, as we will soon discover.⁸

11.3.1 Energy and enstrophy transfer

In three-dimensional turbulence we posited that energy is cascaded to small scales via vortex stretching. In two dimensions that mechanism is absent, and it turns out that it is more reasonable to expect energy to be transferred to *larger scales*. This counter-intuitive behaviour arises from the twin integral constraints of energy and enstrophy conservation, and the following three arguments illustrate why this should be so.

1. Vorticity elongation

Consider a band or a patch of vorticity, as in Fig. 11.5, in a nearly inviscid fluid. The vorticity of each element of fluid is conserved as the fluid moves. Now, we should expect that the quasi-random motion of the fluid will act to elongate the band but, as its area must be preserved, the band narrows and so vorticity gradients will increase. This is equivalent to the enstrophy moving to smaller scales. Now, the energy in the fluid is

$$\hat{E} = -\frac{1}{2} \int \psi \zeta dA, \quad (11.39)$$

where the streamfunction is obtained by solving the Poisson equation $\nabla^2 \psi = \zeta$. If the vorticity is locally elongated primarily only in one direction (as it must be to preserve area), the integration involved in solving the Poisson equation will lead to the scale of the streamfunction becoming

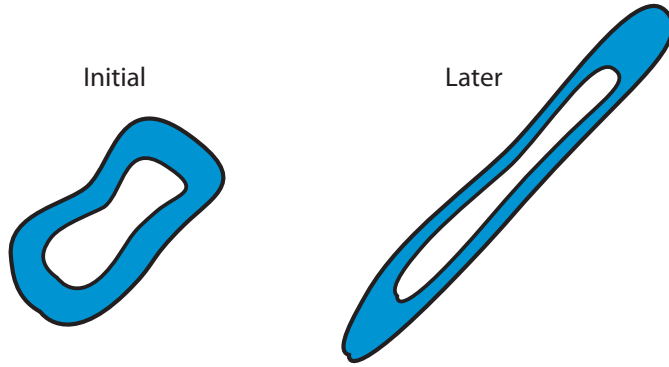


Fig. 11.5 In incompressible two-dimensional flow, a band of fluid is elongated, but its area is preserved. Elongation is followed by folding and more elongation, producing filaments as in Fig. 11.8. As vorticity is tied to fluid parcels, the values of the vorticity in the shaded area (and in the hole) are maintained; thus, vorticity gradients increase and the enstrophy is thereby, on average, moved to smaller scales.

larger in the direction of stretching, but virtually no smaller in the perpendicular direction. Because stretching occurs, on average, in all directions, the overall scale of the streamfunction will increase in all directions, and the cascade of enstrophy to small scales will be accompanied by a transfer of energy to large scales.

II. An energy-enstrophy conservation argument

A moment's thought will reveal that the distribution of energy and enstrophy in wavenumber space are respectively analogous to the distribution of mass and moment of inertia of a lever, with wavenumber playing the role of distance from the fulcrum. Any re-arrangement of mass such that its distribution also becomes wider must be such that the centre of mass moves toward the fulcrum. Thus, analogously, any rearrangement of a flow that preserves both energy and enstrophy, and that causes the distribution to spread out in wavenumber space, will tend to move energy to small wavenumbers and enstrophy to large. To prove this we begin with expressions for the average energy and enstrophy:

$$\bar{E} = \int \mathcal{E}(k) dk, \quad \bar{Z} = \int \mathcal{Z}(k) dk = \int k^2 \mathcal{E}(k) dk, \quad (11.40)$$

where $\mathcal{E}(k)$ and $\mathcal{Z}(k)$ are the energy and enstrophy spectra. A wavenumber characterizing the spectral location of the energy is the centroid,

$$k_e = \frac{\int k \mathcal{E}(k) dk}{\int \mathcal{E}(k) dk} \quad (11.41)$$

and, for simplicity, we normalize units so that the denominator is unity. The spreading out of the energy distribution is formalized by setting

$$I \equiv \int (k - k_e)^2 \mathcal{E}(k) dk, \quad \frac{dI}{dt} > 0. \quad (11.42)$$

Here, I measures the width of the energy distribution, and this is assumed to increase. Expanding out the integral gives

$$\begin{aligned} I &= \int k^2 \mathcal{E}(k) dk - 2k_e \int k \mathcal{E}(k) dk + k_e^2 \int \mathcal{E}(k) dk \\ &= \int k^2 \mathcal{E}(k) dk - k_e^2 \int \mathcal{E}(k) dk, \end{aligned} \quad (11.43)$$

where the last equation follows because $k_e = \int k \mathcal{E}(k) dk$ is, from (11.41), the energy-weighted centroid. Because both energy and enstrophy are conserved, (11.43) gives

$$\frac{dk_e^2}{dt} = -\frac{1}{\bar{E}} \frac{dI}{dt} < 0. \quad (11.44)$$

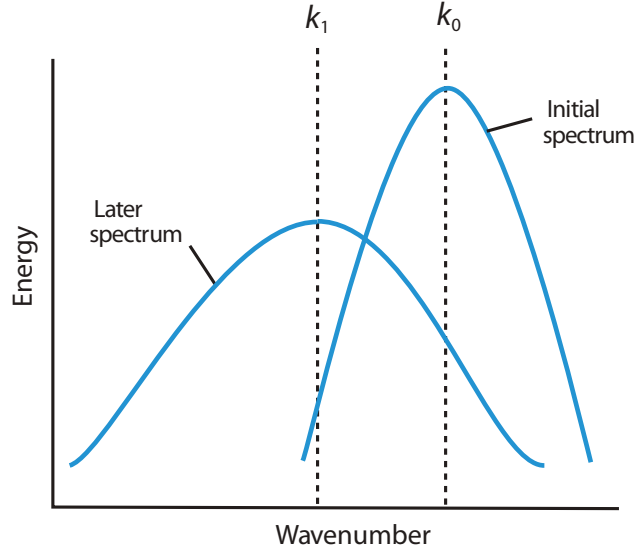


Fig. 11.6 In two-dimensional flow, the centroid of the energy spectrum will move to large scales (smaller wavenumber) provided that the width of the distribution increases — as can be expected in a nonlinear, eddying flow.

Thus, the centroid of the distribution moves to smaller wavenumbers and to larger scales (see Fig. 11.6).

An appropriately defined measure of the centre of the enstrophy distribution, on the other hand, moves to higher wavenumbers. The demonstration follows easily if we work with the inverse wavenumber, which is a direct measure of length. Let $q = 1/k$ and assume that the enstrophy distribution spreads out by nonlinear interactions, so that, analogously to (11.42),

$$J \equiv \int (q - q_e)^2 \mathcal{Y}(q) dq, \quad \frac{dJ}{dt} > 0, \quad (11.45)$$

where $\mathcal{Y}(q)$ is such that the enstrophy is $\int \mathcal{Y}(q) dq$ and

$$q_e = \frac{\int q \mathcal{Y}(q) dq}{\int \mathcal{Y}(q) dq}. \quad (11.46)$$

Expanding the integrand in (11.45) and using (11.46) gives

$$J = \int q^2 \mathcal{Y}(q) dq - q_e^2 \int \mathcal{Y}(q) dq, \quad (11.47)$$

But $\int q^2 \mathcal{Y}(q) dq$ is conserved, because this is the energy. Thus,

$$\frac{dJ}{dt} = -\frac{d}{dt} q_e^2 \int \mathcal{Y}(q) dq \quad (11.48)$$

whence

$$\frac{dq_e^2}{dt} = -\frac{1}{Z} \frac{dJ}{dt} < 0. \quad (11.49)$$

Thus, the length scale characterizing the enstrophy distribution gets smaller, and the corresponding wavenumber gets larger.

III. A similarity argument

Consider an initial value problem, in which a fluid with some initial distribution of energy is allowed to freely evolve, unencumbered by boundaries. We note two aspects of the problem:

- (i) there is no externally imposed length scale (because of the way the problem is posed);
- (ii) the energy is conserved (this being an assumption).

It is the second condition that limits the argument to two dimensions, for in three dimensions energy is quickly cascaded to small scales and dissipated, but let us here posit that this does not occur. These two assumptions are then sufficient to infer the general direction of transfer of energy, using a rather general similarity argument. To begin, write the energy per unit mass of the fluid as

$$\bar{E} = U^2 = \int \mathcal{E}(k, t) dk, \quad (11.50)$$

where $\mathcal{E}(k, t)$ is the energy spectrum and U is proportional to the square root of the total energy and has units of velocity. On dimensional considerations we could write

$$\mathcal{E}(k, t) = U^2 L \hat{\mathcal{E}}(\hat{k}, \hat{t}), \quad (11.51)$$

where $\hat{\mathcal{E}}$, and its arguments, are non-dimensional quantities, and L is some length scale. However, if, over time, the initial conditions are forgotten then there is no length scale in the problem and the only parameters available to determine the energy spectrum are energy, time, and wavenumber, that is U , t and k . A little thought reveals that the most general form for the energy spectrum is then

$$\mathcal{E}(k, t) = U^3 t \hat{\mathcal{E}} = U^3 t g(Ukt), \quad (11.52)$$

where g is an arbitrary function of its arguments. The argument of g is the only non-dimensional grouping of U , t and k , and $U^3 t$ provides the proper dimensions for $\mathcal{E}$. Conservation of energy now implies that the integral

$$I = \int_0^\infty t g(Ukt) dk \quad (11.53)$$

is *not* a function of time. Defining $\vartheta = Ukt$, this requirement is met if

$$\int_0^\infty g(\vartheta) d\vartheta = \text{constant}. \quad (11.54)$$

Now, the spectrum is a function of k only through the combination $\vartheta = Ukt$. Thus, as time proceeds features in the spectrum move to smaller k . Suppose, for example, that the energy is initially peaked at some wavenumber k_p ; the product tk_p is preserved, so k_p must diminish with time and the energy must move to larger scales. Similarly, the energy weighted mean wavenumber, k_e , moves to smaller wavenumbers, or larger scales. To see this explicitly, we have

$$k_e = \frac{\int k \mathcal{E} dk}{\int \mathcal{E} dk} = \frac{\int k \mathcal{E} dk}{U^2} = \int k U t g(Ukt) dk = \int \frac{\vartheta g(\vartheta)}{U t} d\vartheta = \frac{C}{U t} \quad (11.55)$$

where all the integrals are over the interval $(0, \infty)$ and $C = \int \vartheta g(\vartheta) d\vartheta$ is a constant. Thus, the wavenumber centroid of the energy distribution decreases with time, and the characteristic scale of the flow, $1/k_e$, increases with time. Interestingly, the enstrophy does not explicitly enter this argument, and in general it is not conserved; rather, it is the requirement that energy be conserved that limits the argument to two dimensions; if we accept ab initio that energy *is* conserved, it must be transferred to larger scales.⁹

11.3.2 Inertial ranges in two-dimensional turbulence

If, unlike the case in three dimensions, energy is transferred to larger scales in inviscid, nonlinear, two-dimensional flow then we might expect the inertial ranges of two-dimensional turbulence to be quite different from their three-dimensional counterparts. But before looking in detail at the inertial ranges themselves, let us establish a couple of general properties of forced-dissipative flow in two dimensions.

Some properties of forced-dissipative flow

We will first show that, unlike the case in three dimensions, energy dissipation goes to zero as the Reynolds number rises. In the absence of forcing terms, the total dissipation of energy is, from (11.35)),

$$\frac{d\hat{E}}{dt} = -\nu \int \zeta^2 dA. \quad (11.56)$$

Energy dissipation can only remain finite as $\nu \rightarrow 0$ if vorticity becomes infinite. However, this cannot happen because vorticity is conserved on parcels except for the action of viscosity, meaning that $D\zeta/Dt = \nu \nabla^2 \zeta$. However, the viscous term can only *reduce* the value of vorticity on a parcel, and so vorticity can never become infinite if it is not so initially, and therefore using (11.56) energy dissipation goes to zero with ν . (In three dimensions vorticity becomes infinite as viscosity goes to zero because of the effect of vortex stretching.) This conservation of energy is related to the fact that energy is trapped at large scales, even in forced-dissipative flow. On the other hand, enstrophy is transferred to small scales and therefore we expect it to be dissipated at large wavenumbers, even as the Reynolds number becomes very large.

We can show that energy is trapped at large scales in forced-dissipative two-dimensional flow (in a sense that will be made explicit) by the following argument.¹⁰ Suppose that the forcing of the fluid is confined to a particular scale, characterized by the wavenumber k_f , and that dissipation is effected by a linear drag and a small viscosity. The equation of motion is

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) = F - r\zeta + \nu \nabla^2 \zeta, \quad (11.57)$$

where F is the stirring and r and ν are positive constants. This leads to the following energy and enstrophy equations:

$$\frac{d\hat{E}}{dt} = -2r\hat{E} - \int \psi F dA - \int \nu \zeta^2 dA \approx -2r\hat{E} - \int \psi F dA, \quad (11.58a)$$

$$\frac{d\hat{Z}}{dt} = -2r\hat{Z} + \int \zeta F dA - D_Z \approx -2r\hat{Z} - k_f^2 \int \psi F dA - D_Z, \quad (11.58b)$$

where $D_Z = \int \nu (\nabla \zeta)^2 dA$ is the enstrophy dissipation. To obtain the right-most expressions, in (11.58a) we assume there is no dissipation of energy by the viscous term, and in (11.58b) we assume that the forcing is confined to wavenumbers near k_f . In a statistically steady state we may eliminate the integral involving ψF between (11.58a) and (11.58b), and writing $\hat{E} = \int E(k) dk$ and $\hat{Z} = \int k^2 \mathcal{E}(k) dk$ (for a domain of unit area) we obtain

$$\int k^2 \mathcal{E}(k) dk + \frac{D_Z}{2r} = \int k_f^2 \mathcal{E}(k) dk, \quad (11.59)$$

where the integrations are over all wavenumbers. Now, from the obvious inequality $\int (k - k_e)^2 \mathcal{E}(k) dk \geq 0$, where k_e is the energy centroid defined in (11.41), we obtain

$$\int (k^2 - k_e^2) \mathcal{E}(k) dk \geq 0. \quad (11.60)$$

Combining (11.59) and (11.60) gives

$$\int (k_f^2 - k_e^2) \mathcal{E}(k) dk \geq \frac{D_Z}{2r} > 0. \quad (11.61)$$

Thus, in a statistically steady state, the energy containing scale, as characterized by k_e^{-1} , must be larger than the forcing scale k_f^{-1} . This demonstration (like argument II in section 11.3.1) relies both on the conservation of energy and enstrophy by the nonlinear terms and on the particular relationship between the energy and enstrophy spectra.

This result, and (especially) the arguments of section 11.3.1, suggest that in a forced-dissipative two-dimensional fluid, energy is transferred to larger scales and enstrophy is transferred to small scales. To obtain a statistically steady state friction [such as the Rayleigh drag of (11.57)] is necessary to remove energy at large scales, and enstrophy must be removed at small scales, but if the forcing scale is sufficiently well separated in spectral space from such frictional effects then two inertial ranges may form — an *energy inertial range* carrying energy to larger scales, and an *enstrophy inertial range* carrying enstrophy to small scales (Fig. 11.7). These ranges are analogous to the three-dimensional inertial range of section 11.2, and similar conditions must apply if the ranges are to be truly inertial — in particular we must assume spectral locality of the energy or enstrophy transfer. Given that, we can then calculate their properties, as follows.

The enstrophy inertial range

In the enstrophy inertial range the enstrophy cascade rate η , equal to the rate at which enstrophy is supplied by stirring, is assumed constant. By analogy with (11.25) we may assume that this rate is given by

$$\eta \sim \frac{k^3 \mathcal{E}(k)}{\tau_k}. \quad (11.62)$$

With τ_k (still) given by (11.24) we obtain

$$\mathcal{E}(k) = \mathcal{K}_\eta \eta^{2/3} k^{-3}, \quad (11.63)$$

where $\mathcal{K}_\eta$ is, we presume, a universal constant, analogous to the Kolmogorov constant of (11.23). This, and various other properties in two- and three-dimensional turbulence, are summarized in the shaded box on page 425.

It is also possible to obtain (11.63) from scaling arguments similar to those in section ???. The scaling transformation (??) still holds, but now instead of (??) we assume that the enstrophy flux is constant with wavenumber. Dimensionally, and analogously to (??), we have

$$\eta \sim \frac{v_k^3}{l_k^3} \sim \lambda^{3r-3}, \quad (11.64)$$

and the constancy of η gives $r = 1$ for the scaling exponent. The exponent n determining the slope of the inertial range is given, as before, by $n = -(2r + 1)$ yielding the -3 spectra of (11.63). The velocity at a particular wavenumber then scales as

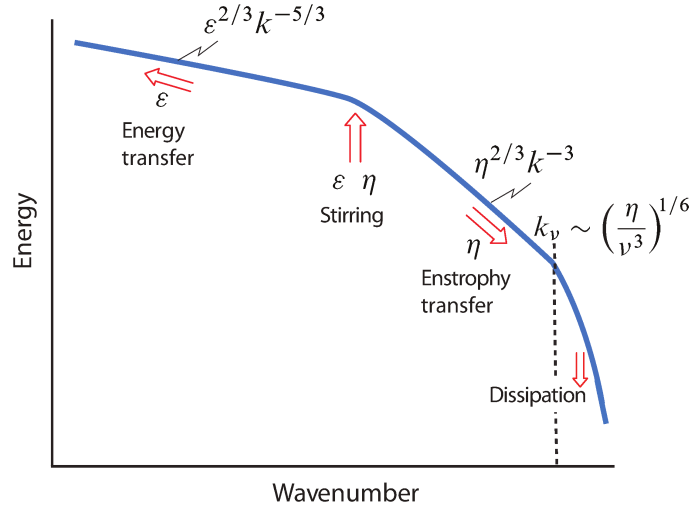
$$v_k \sim \eta^{1/3} k^{-1}, \quad (11.65)$$

and the time scales as

$$t_k \sim l_k/v_k \sim \eta^{-1/3}. \quad (11.66)$$

We may also obtain (11.66) by substituting (11.63) into (11.24). Thus, *the eddy turnover time in the enstrophy range of two-dimensional turbulence is length-scale invariant*. The appropriate viscous scale is given by equating the inertial and viscous terms in (11.35). Using (11.65) we obtain,

Fig. 11.7 The energy spectrum of two-dimensional turbulence. (Compare with Fig. 11.3.) Energy supplied at some rate ε is transferred to large scales, whereas enstrophy supplied at some rate η is transferred to small scales, where it may be dissipated by viscosity. If the forcing is localized at a scale k_f^{-1} then $\eta \approx k_f^2 \varepsilon$.



analogously to (11.28a),

$$k_v \sim \left(\frac{\eta^{1/3}}{\nu} \right)^{1/2}. \quad (11.67)$$

The enstrophy dissipation, analogously to (11.30) goes to a finite limit given by

$$\dot{\bar{Z}} = \nu \int_A \zeta \nabla^2 \zeta \, dA \sim \nu k_v^4 \nu_{k_v}^2 \sim \eta, \quad (11.68)$$

using (11.65) and (11.67). Thus, the enstrophy dissipation in two-dimensional turbulence is (at least according to this theory) independent of the viscosity.

Energy inertial range

The energy inertial range of two-dimensional turbulence is quite similar to that of three-dimensional turbulence, except in one major respect: the energy flows from smaller to larger scales! Because the atmosphere and ocean behave in some ways as two-dimensional fluids, this has profound consequences on their behaviour, and is something we return to in the next chapter. The upscale energy flow is known as the *inverse cascade*, and the associated energy spectrum is, as in the three-dimensional case,

$$\mathcal{E}(k) = \mathcal{K}_\varepsilon \varepsilon^{2/3} k^{-5/3}, \quad (11.69)$$

where $\mathcal{K}_\varepsilon$ is a non-dimensional constant [sometimes called the Kolmogorov–Kraichnan constant, and not necessarily equal to $\mathcal{K}$ in (11.23)], and ε is the rate of energy transfer to larger scales. Of course we now need a mechanism to remove energy at large scales, otherwise it will pile up at the scale of the domain and a statistical steady state will not be achieved. Introducing a linear drag, $-r\zeta$, into the vorticity equation, as in (11.57), is one means of removing energy, and such a term may be physically justified by appeal to Ekman layer theory (section 5.7). Although such a term appears to be scale invariant, its effects will be felt primarily at large scales because at smaller scales the time scale of the turbulence is much shorter than that of the friction, and we may estimate the scale at which the drag becomes important by equating the two time scales. The turbulent timescale is given by (11.27), and equating this to the frictional time scale r^{-1} gives $r^{-1} = \varepsilon^{-1/3} k_r^{-2/3}$, or

$$k_r = \left(\frac{r^3}{\varepsilon} \right)^{1/2}, \quad \text{or} \quad L_r = \left(\frac{\varepsilon}{r^3} \right)^{1/2} \quad (11.70)$$

where k_r is the frictional wavenumber, and frictional effects are important at scales *larger* than the frictional scale L_r .

11.3.3 † More about the phenomenology

The phenomenology of two-dimensional turbulence is not quite as settled as the above arguments imply. Note, for example, that time scale (11.66) is independent of length scale, whereas in three-dimensional turbulence the time scale decreases with length scale, which seems more physical and more conducive to spectrally local interactions. A useful heuristic measure of this locality is given by estimating the contributions to the straining rate, $S(k)$, from motions at all scales larger than k^{-1} .¹¹ The strain rate scales like the shear, so that an estimate of the total strain rate is given by

$$S(k) = \left[\int_{k_0}^k \mathcal{E}(p) p^2 dp \right]^{1/2}, \quad (11.71)$$

where k_0 is the wavenumber of the largest scale present. The contributions to the integrand from a given wavenumber octave are given by

$$\int_p^{2p} \mathcal{E}(p') p'^3 d \log p' \sim \mathcal{E}(p) p^3. \quad (11.72)$$

In three dimensions, use of the $-5/3$ spectrum indicates that the contributions from each octave below a given wavenumber k increase with wavenumber, being a maximum close to k , and this is a posteriori consistent with the locality hypothesis. However, in two-dimensional turbulence with a -3 spectrum each octave makes the same contribution. That is to say, the contributions to the strain rate at a given wavenumber, as defined by (11.71), are not spectrally local. This does not prove that the enstrophy transfer is spectrally non-local, but nor does it build confidence in the theory.

Dimensionally the strain rate is the inverse of a time, and if this is a spectrally non-local quantity then, instead of (11.24), we might use the inverse of the strain rate as an eddy turnover time giving

$$\tau_k = \left[\int_{k_0}^k p^2 \mathcal{E}(p) dp \right]^{-1/2}. \quad (11.73)$$

This has the advantage over (11.24) in that it is a non-increasing function of wavenumber, whereas if the spectrum is steeper than k^{-3} (11.24) implies a time scale increasing with wavenumber. Using this in (11.62) gives a prediction for the enstrophy inertial range, namely

$$\mathcal{E}(k) = \mathcal{K}_\eta \eta^{2/3} [\log(k/k_0)]^{-1/3} k^{-3}, \quad (11.74)$$

which is similar to (11.63) except for a logarithmic correction. This expression is, of course, spectrally non-local, in contradiction to our original assumption: it has arisen by noting the spectral locality inherent in (11.71), and proposing a reasonable, although ad hoc, solution.

The discussion above suggests that the phenomenology of the forward enstrophy cascade is on the verge of being internally inconsistent, and that the k^{-3} spectral slope might be the shallowest limit that is likely to be actually achieved in nature or in any particular computer simulation or laboratory experiment rather than a robust, universal slope. To see this, suppose the detailed fluid dynamics strives in some way to produce a slope shallower than k^{-3} ; then, using (11.72), the strain is local and the shallow slope is forbidden by the Kolmogorovian scaling results. However, if the dynamics organizes itself into structures with a slope steeper than k^{-3} the strain is quite non-local. The fundamental assumption of Kolmogorov scaling is not satisfied, and there is no internal inconsistency. The theory then simply does not apply, and a slope steeper than k^{-3} is not theoretically inconsistent.

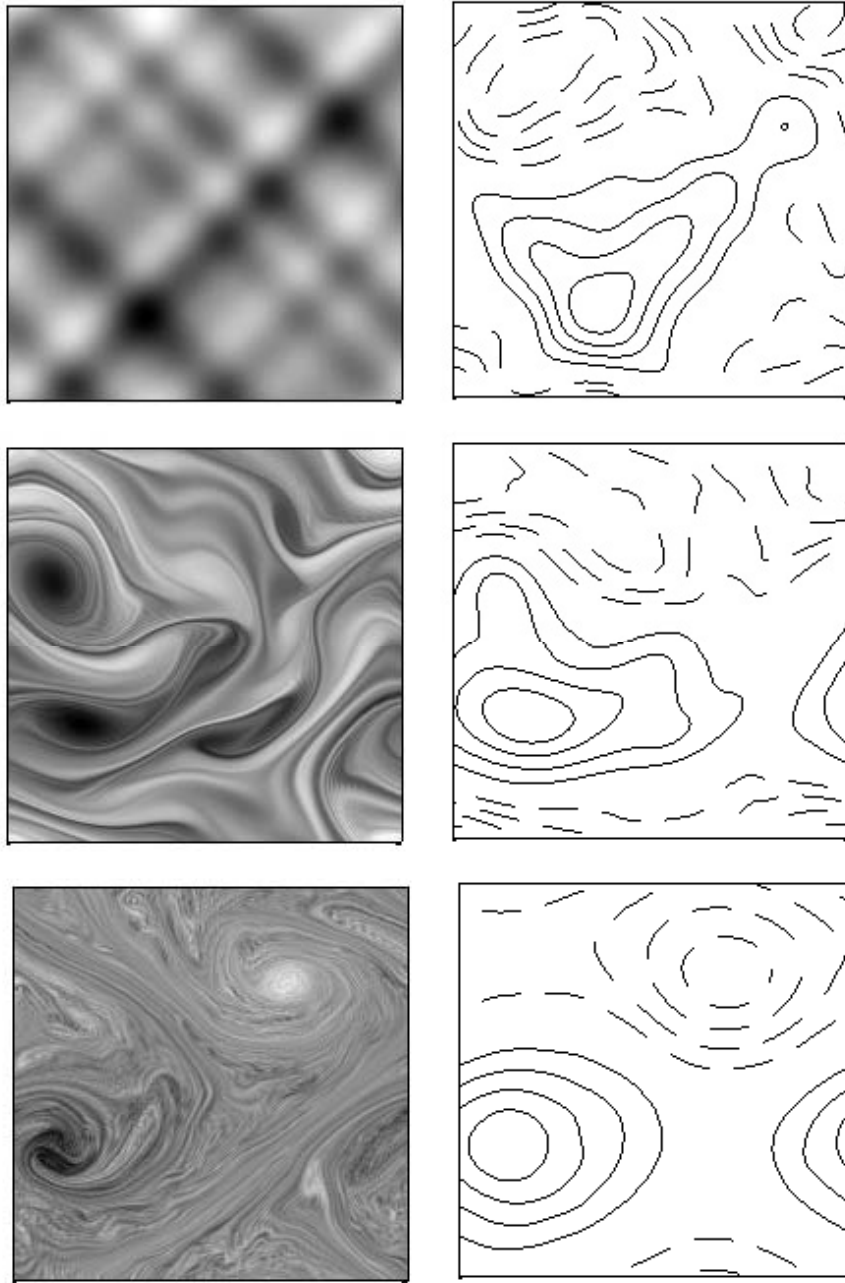


Fig. 11.8 Nearly-free evolution of vorticity (left column) and streamfunction (right column) in a doubly-periodic square domain (of length 2π) at times (from the top, and in units of inverse vorticity) $t = 0, 50$ and 260 , obeying the two-dimensional vorticity equation with no forcing but with a weak viscous term. The initial conditions have just a few non-zero Fourier modes with randomly generated phases, producing a maximum value of vorticity of about three. Kelvin–Helmholtz instability leads to vortex formation and roll-up (as in Fig. 9.6), and like-signed vortices merge, ultimately leading to a state of just two oppositely-signed vortices. Between the vortices, enstrophy cascades to smaller scales. The scale of the streamfunction grows larger, reflecting the transfer of energy to larger scales.

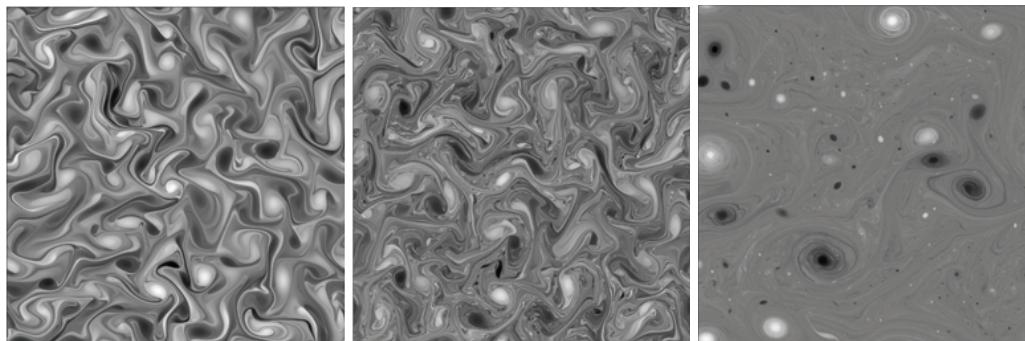


Fig. 11.9 Snapshots of the vorticity field in decaying two-dimensional turbulence, similar to Fig. 11.8, with time increasing left to right. The flow ultimately consists of a small number of vortices whose trajectories are similar to that of interacting point vortices, with occasional close encounters leading to vortex merger.

There are two other potential issues with the theory of two-dimensional turbulence. One is that enstrophy is only one of an infinity of invariants of inviscid two-dimensional flow, and the theory takes no account of the presence of others. The second is that, as in three-dimensional turbulence, if there is strong intermittency the flow cannot be fully characterized by single enstrophy and energy cascade rates. In two-dimensional turbulence the main form of intermittency may be the formation of coherent vortices, discussed more below. In spite of these problems, the notions of a forward transfer of enstrophy and an inverse transfer of energy are quite robust, and have considerable numerical support. Indeed, the realization that in two-dimensional turbulence energy is transferred to larger scales was arguably one of the most important developments in fluid mechanics in the second half of the twentieth century, with important ramifications in rotating and stratified *three-dimensional* fluids.¹²

11.3.4 Numerical illustrations

Numerical simulations nicely illustrate both the classical phenomenology and its shortcomings. In the simulations shown in Figs. 11.8 and 11.9 the vorticity field is initialized ‘randomly’, with little structure in the initial field, and with only a few non-zero Fourier components; the flow then freely evolves, save for the effects of a weak viscosity.

Vortices soon form, and between them enstrophy is cascaded to small scales where it is dissipated, producing a flat and nearly featureless landscape. The energy cascade to larger scales is reflected in the streamfunction field (right column of Fig. 11.8), the length scale of which slowly grows larger with time. The vortices themselves form through a roll-up mechanism, similar to that illustrated in Fig. 9.6, and their presence provides problems to the phenomenology. Because circular vortices are nearly exact, stable solutions of the inviscid equations they can ‘store’ enstrophy, disrupting the relationship between enstrophy flux and enstrophy itself that is assumed in the Kolmogorov–Kraichnan phenomenology. When vortices merge, the enstrophy dissipation rate increases rapidly for a short period of time, so providing a source of intermittency.

Nevertheless, some forced-dissipative numerical simulations suggest that the presence of vortices may be confined to scales close to that of the forcing, and if the resolution is sufficiently high then the $-5/3$ inverse cascade and -3 forward enstrophy cascade appear. Certainly, if the forcing is spectrally localized, then a well-defined $-5/3$ spectrum robustly forms, as illustrated in Fig. 11.10. The forward k^{-3} spectrum is typically more delicate, being influenced by the presence of coherent vortices, but it does arise in some numerical simulations when the resolution is sufficiently high.¹⁴ The atmosphere itself is not observed to have an inverse $-5/3$ spectrum at large scales; indeed there

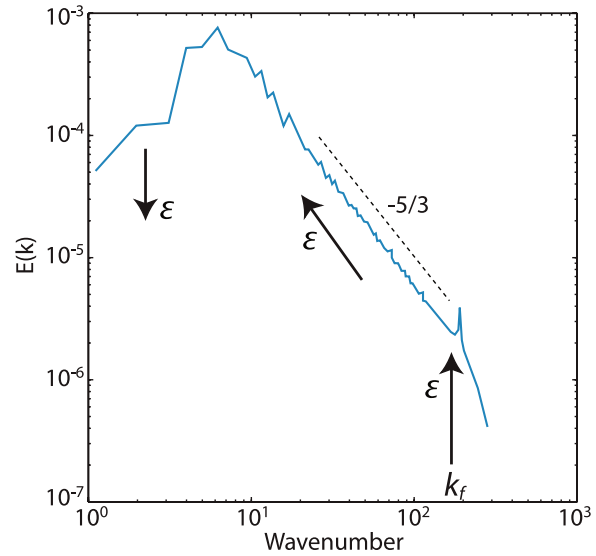


Fig. 11.10 The energy spectrum in a numerical simulation of forced-dissipative two-dimensional turbulence. The fluid is stirred at wavenumber k_f and dissipated at large scales with a linear drag, and there is an $k^{-5/3}$ spectrum at intermediate scales. The arrows indicate the direction of the energy flux, ε .¹³

is no well-defined inverse energy cascade in the sense described above. The atmosphere does have a -3 cascade (see Fig. 12.9 in the next chapter), but whether we should attribute this to a classical forward enstrophy cascade is not settled.

11.4 PREDICTABILITY OF TURBULENCE

Small differences in the initial conditions may produce very great ones in the final phenomenon... Prediction becomes impossible... A tenth of a degree more or less at any given point, and the cyclone will burst here and not there, and extend its ravages over districts it might otherwise have spared.

Henri Poincaré, *Science and Method*, 1908.

Forecasting the weather is hard. That this is so stems from the fact that the atmosphere is chaotic, and chaotic systems are unpredictable virtually by definition. However, the atmosphere (and turbulence in general) is certainly not a *low-dimensional* chaotic system (meaning a system with only a few degrees of freedom), and the connection between atmospheric unpredictability and the so-called ‘sensitive dependence on initial conditions’ of low-dimensional systems is not as straightforward as it might seem. In this section we expand on and clarify these issues, beginning with an informal discussion of a few aspects of low-dimensional dynamical systems.

11.4.1 Low-dimensional chaos and unpredictability

Chaos, or temporal disorder leading to effective indeterminism, is a ubiquitous property of nonlinear dynamical systems. Much of this was known to Poincaré, but in its modern reincarnation it stems in part from the ‘Lorenz equations’.¹⁵ These are a set of three coupled nonlinear ordinary differential equations, originally derived by way of a rather *ad hoc* truncation of the fluid equations governing a two-dimensional convective system: the streamfunction of a convective roll is written as $\psi(x, z, t) = X(t) \sin kx \sin \pi z$, and the temperature perturbation as $\theta(x, z, t) = Y(t) \cos kx \sin \pi z + Z(t) \sin 2\pi z$, where x and z are the horizontal and vertical coordinates in physical space, k is a wavenumber, and X, Y, Z are amplitudes (not coordinates). Thus, X represents the rotational speed of the convection roll, Y the temperature difference horizontally across the roll, and Z the deviation temperature from a background vertical stratification. The resulting equations

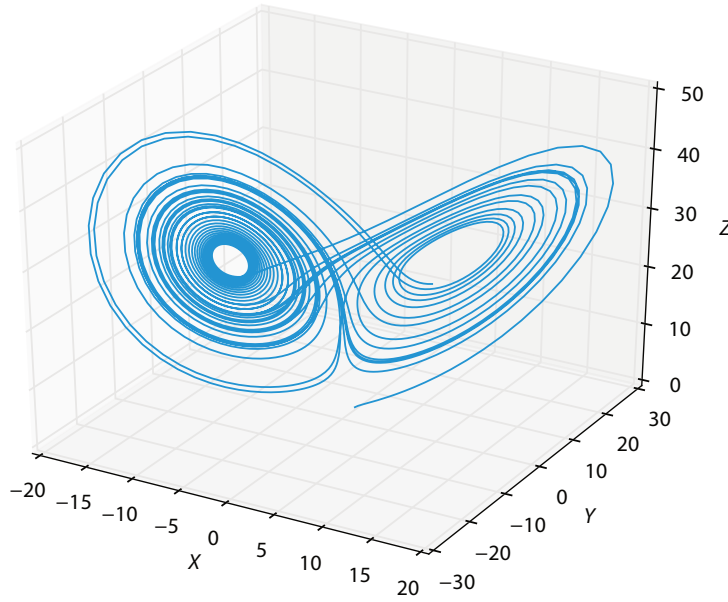


Fig. 11.11 A solution of the Lorenz equations, with $\sigma = 10$, $r = 28$ and $b = 8/3$. The plot shows a trajectory of a solution in phase space, with initial conditions $X = Y = Z = 1$.

are, in notation standard for them,

$$\frac{dX}{dt} = \sigma(Y - X), \quad \frac{dY}{dt} = rX - Y - XZ, \quad \frac{dZ}{dt} = XY - bZ, \quad (11.75a,b,c)$$

where the parameters are σ , the Prandtl number; r , proportional to the Rayleigh number; and b , a wavenumber dependent dissipation coefficient. The behaviour of the system varies with the parameters, and a well studied set uses $\sigma = 10$, $r = 28$ and $b = 8/3$. A typical solution of the system, a ‘flow’, is given in Fig. ??, and evidently the behaviour is quite complex. It is aperiodic, and the frequency spectrum (not shown) is quite broad. Now, suppose that at any given instant the flow is perturbed slightly. Or to put it another way, suppose that we are trying to predict the future behaviour of the system by integrating the equations of motion but that we have inaccurate knowledge about the system at some particular time. We find that the evolution of the original flow and that of the perturbed flow diverge from each other, and after a little while the two systems are completely different (Fig. 11.12). Because we can never expect to have completely accurate information about the state of the system, the system is thus unpredictable; that is, the details of the evolution depend sensitively on the initial conditions, and small errors in the initial conditions grow to finite amplitude. Let us make two other points.

1. The time taken for the trajectories to diverge depends on the magnitude of the initial perturbation. Small perturbations grow exponentially at first, and at any given point in the trajectory, the smaller the perturbation the longer the predictability.
2. Once the perturbation has reached finite amplitude, the predictability time — the time for the error to become as large as the solution itself — will typically be of the order of the characteristic advective time of the system, the time for a convective roll to overturn.

Deterministic unpredictability is in fact a common feature of nonlinear dynamical systems, and may be taken as an informal definition of a chaotic system.

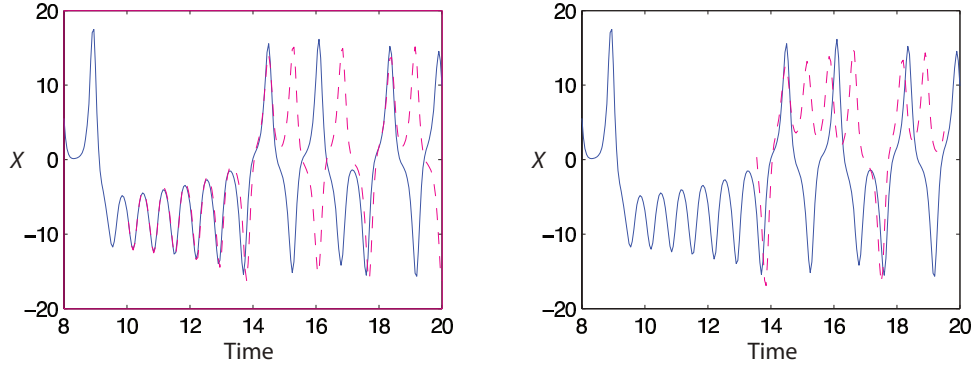


Fig. 11.12 Examples of the evolution of the variable X in the Lorenz model subject to a small perturbation at time 10 (left panel) and time 13.5 (right panel). The original and perturbed systems are the solid and dashed lines, respectively.

11.4.2 ♦ Predictability of a turbulent flow

A turbulent flow will be unpredictable if turbulence is chaotic. Because it is our common experience is that turbulence is both spatially and temporally disordered, it seems, perhaps with the benefit of hindsight, that turbulence must be chaotic and unpredictable. However, this has not always been evident, and it relies on the non-trivial recognition that chaotic systems exist.¹⁶

Presuming that a turbulent flow *is* unpredictable, can we estimate its predictability time, namely the time taken for an initially small error to completely contaminate the system? Let us first note that turbulent flows in general contain multiple scales of motion, and let us suppose that the initial error is confined to small scales. The predictability time of the system may be taken as the time taken for the error to contaminate all scales of motion. There are two possible routes that the error may take in affecting the larger scales. In the first we suppose, following classical turbulence phenomenology, that errors on a small scale will mostly contaminate the motion on the next larger scale (in a logarithmic sense), and that this contamination occurs on the local eddy turnover time. Eddies on this larger scale then grow and affect the next larger scale, and the error field is so cascaded upscale via local triad interactions finally reaching the largest scales of the fluid. Note that this mechanism does not rely on there being an inverse cascade of energy — it is only the error, or the contamination, that is cascaded upscale. In the second route we suppose that errors occurring on the small scale immediately contaminate the largest scales, with an initial error equal to the amplitude of the small scale, and that the large-scale error then grows exponentially.

I. Error growth via a local cascade

Let us suppose that the error is initially confined to some small scale characterized by the (inverse of) the wavenumber k_1 , as determined by the resolution of our observing network. For modes at that scale the error may be considered finite rather than infinitesimal, and it will saturate and contaminate the next largest scale in a time scale comparable to the eddy turnover time at that scale. Thus, in general, errors initially confined to a scale k will contaminate the scale $2k$ after a time τ_k , with τ_k given by (11.24). The total time taken for errors to propagate from the small scale k_1 to the largest scale k_0 is then given by

$$T = \int_{k_0}^{k_1} \tau_k d(\ln k) = \int_{k_0}^{k_1} [k^3 \mathcal{E}(k)]^{-1/2} d(\ln k), \quad (11.76)$$

treating the wavenumber spectrum as continuous. The logarithmic integral arises because the cascade proceeds logarithmically — error cascades from k to $2k$ in a time τ_k . For an energy spectrum

of the form $E = Ak^{-n}$ this becomes

$$T = \frac{2}{A^{1/2}(n-3)} \left[k^{(n-3)/2} \right]_{k_0}^{k_1}, \quad (11.77)$$

for $n \neq 3$, and $T = A^{-1/2} \ln(k_1/k_0)$ for $n = 3$. If in two-dimensional turbulence we have $n = 3$ and $A = \eta^{2/3}$, and if in three-dimensional turbulence we have $n = 5/3$ and $A = \varepsilon^{2/3}$, then the respective predictability times are given by

$$T_{2d} \sim \eta^{-1/3} \ln(k_1/k_0), \quad T_{3d} \sim \varepsilon^{-1/3} k_0^{-2/3}. \quad (11.78a,b)$$

As $k_1 \rightarrow \infty$, that is as the initial error is confined to smaller and smaller scales, predictability time grows larger for two-dimensional turbulence (and for $n \geq 3$ in general), but remains finite for three-dimensional turbulence.

II. Error growth via a direct interaction

Let us now assume that the small scale error directly affects the large scales, where the error then grows exponentially until it saturates. That is, if ϕ is a measure of the amplitude of the large-scale error then

$$\phi \sim \phi_0 \exp(\sigma t), \quad (11.79)$$

where σ is the inverse of the eddy turnover time at the large scale, and ϕ_0 is the amplitude of its initial error and this, we assume, is equal to the amplitude of the motion at the poorly-observed small scales at wavenumber k_1 . In the enstrophy cascade of two-dimensional turbulence the eddy turnover time is given by $\tau_k \sim \eta^{-1/3}$ and $A \sim \eta^{2/3}$, and so we take

$$\sigma = \eta^{1/3} = A^{1/2}, \quad \phi_0 = Ak_1^{-n}. \quad (11.80)$$

where $n = 3$. The time, T'_{2d} , needed for the error to saturate the large scales, k_0 , is then approximately given by the solution of

$$Ak_0^{-n} = Ak_1^{-n} \exp(A^{1/2} T'_{2d}), \quad (11.81)$$

giving

$$T'_{2d} \sim \eta^{-1/3} \ln(k_1/k_0). \quad (11.82)$$

In three-dimensional turbulence the eddy turnover time is given by $\tau_k \sim \varepsilon^{-1/3} k^{-2/3}$ and $A \sim \varepsilon^{2/3}$, and so we take

$$\sigma = k_0^{2/3} A^{1/2}, \quad \phi_0 = Ak_1^{-n} \quad (11.83)$$

where $n = 5/3$. The time, T'_{3d} , needed for an error to saturate the large scale is then approximately given by the solution of

$$Ak_0^{-n} = Ak_1^{-n} \exp(k_0^{2/3} A^{1/2} T'_{3d}), \quad (11.84)$$

giving

$$T'_{3d} \sim k_0^{-2/3} \varepsilon^{-1/3} \ln(k_1/k_0). \quad (11.85)$$

The estimates (11.82) and (11.85) are to be compared with (11.78). For two-dimensional turbulence, the estimates are equal (reflecting the scale independence of the eddy turnover time), whereas for three-dimensional turbulence the estimate from (11.78) is much shorter than that from (11.85) if $k_1 \gg k_0$, meaning that the local cascade mechanism of error growth will dominate.

11.4.3 Implications and weather predictability

In two-dimensional flow the predictability time, (11.78a) increases without bound as the scale of the initial error decreases. This is consistent with what has been rigorously proven about the two-dimensional Navier–Stokes equations, namely that provided the initial conditions are sufficiently smooth, the solutions have a continuous dependence on the initial conditions, and a change in solution at some later time may be bounded by reducing the magnitude of the change in the initial conditions. This does not mean that two-dimensional flow is in practice necessarily predictable: a small error or small amount of noise in the system will still render a flow truly unpredictable sometime in the future, but we can put off that time indefinitely if we know the initial conditions well enough.

In three dimensions, with a spectrum of $k^{-5/3}$, the predictability-time estimate from (11.78b) is not dependent on the scale of the initial error. Thus, even if the initial error is confined to smaller and smaller scales, the predictability time is bounded. The time it takes for such errors to spread to the largest scales is simply a few large eddy turnover times, essentially because the eddy turnover times of the small scales are so small. For such a fluid, there is no unique error doubling time, because the error growth rate is a function of scale.

In the troposphere the large-scale flow behaves more like a two-dimensional fluid than a three-dimensional fluid and from scales from a few hundred to a few thousand kilometres it has, roughly, a k^{-3} spectrum (look ahead to Fig. 12.9). If this spectrum extended indefinitely to small scales the predictability time would be correspondingly large, but at scales smaller than about 100 km or so, the atmosphere starts to behave more three dimensionally and the predictability time cannot be significantly extended by making observations at still finer scales. That is, the effective limit to predictability is governed by the horizontal scale at which the atmosphere turns three dimensional. (Hypothetically, we might be able to increase the predictability time if we could observe scales well into the viscous regime where the spectrum steepens again, but this is not a practicable proposition.) Putting in the numbers gives a predictability limit of about 12 days (but there is at least a factor of two uncertainty in such a calculation), and small perturbations that are impossible to observe will change the course of the large-scale weather systems on this time scale. The ‘butterfly effect’ has its origins in this argument: a butterfly flapping its wings is, so it goes, able to change the course of the weather a week or so later.¹⁷

As regards our attempts to predict weather, as the atmosphere becomes observed more and more accurately, the initial error will become concentrated at smaller and smaller scales, eventually reaching the scale at which the atmosphere ceases to behave as a quasi-two-dimensional fluid and at which its spectrum flattens. The initial error growth rate will then increase, indicating the unavoidable onset of diminishing returns in adding resolution to our observing systems and models. Note that, unlike the situation in a low-order chaotic system, the growth rate of errors in a turbulent flow is not, in general, exponential (except for a pure -3 spectrum) even for a small initial error. This is because the initial error will never be properly infinitesimal, in that a given error will nearly always project on to some scale at finite amplitude.¹⁸

11.5 ♦ SPECTRA OF PASSIVE TRACERS

In fluid dynamics we are often concerned with the transport of tracers by turbulent flows. An *active tracer* is one that affects the flow itself (potential vorticity and salt are examples), whereas a *passive tracer* does not affect the flow field (a neutrally buoyant dye, for example), and its dynamics are often simpler. In some circumstances an active tracer can be considered to be approximately passive; the atmospheric temperature field at very large scales turns out to be one example, and its transport is obviously a key determinant of our climate. Let us therefore spend some time discussing the dynamics of a passive tracer; in this section we will consider their spectra, and in chapter 13 we will consider their diffusive transport.

We consider a tracer that obeys

$$\frac{D\phi}{Dt} = F[\phi] + \kappa \nabla^2 \phi, \quad (11.86)$$

where $F[\phi]$ is the stirring of the dye and κ is its diffusivity; κ in general differs from the kinematic molecular viscosity ν . If ϕ is temperature, the ratio of viscosity to diffusivity is called the *Prandtl number* and denoted σ , so that $\sigma \equiv \nu/\kappa$. If ϕ is a passive tracer, the ratio is sometimes called the *Schmidt number*, but we shall call it the Prandtl number in all cases. We assume that the tracer variance is created at some well-defined scale k_0 , and that κ is sufficiently small that dissipation only occurs at very small scales. (Note that dissipation only reduces the tracer *variance*, not the amount of tracer itself.) The turbulent flow will generically tend to stretch patches of dye into elongated filaments, in much the same way as vorticity in two-dimensional turbulence is filamented — note that Fig. 11.5 applies just as well to a passive tracer in either two or three dimensions as it does to vorticity in two dimensions. Thus we expect a transfer of tracer variance from large to small scales. If the dye is stirred at a rate χ then, by analogy with our treatment of the cascade of energy, we posit that

$$\mathcal{K}_\chi \chi \propto \frac{\mathcal{P}(k)k}{\tau_k}, \quad (11.87)$$

where $\mathcal{P}(k)$ is the spectrum of the tracer, k is the wavenumber, τ_k is an eddy time scale and $\mathcal{K}_\chi$ is a constant, not necessarily the same constant in all cases. (In the rest of the section, Kolmogorov-like constants will be denoted by $\mathcal{K}$, differentiated by miscellaneous superscripts or subscripts.) We will also assume that τ_k is given by

$$\tau_k = [k^3 \mathcal{E}(k)]^{-1/2}. \quad (11.88)$$

Suppose that the turbulent spectrum is given by $\mathcal{E}(k) = Ak^{-n}$. Using (11.88), (11.87) becomes

$$\mathcal{K}_\chi \chi = \frac{\mathcal{P}(k)k}{[Ak^{3-n}]^{-1/2}}, \quad (11.89)$$

and

$$\mathcal{P}(k) = \mathcal{K}_\chi A^{-1/2} \chi k^{(n-5)/2}. \quad (11.90)$$

Note that the steeper the energy spectrum the shallower the tracer spectrum. If the energy spectrum is steeper than -3 then (11.88) may not be a good estimate of the eddy turnover time, and we use instead

$$\tau_k = \left[\int_{k_0}^k p^2 \mathcal{E}(p) dp \right]^{-1/2}, \quad (11.91)$$

where k_0 is the low-wavenumber limit of the spectrum. If the energy spectrum is shallower than -3 , then the integrand is dominated by the contributions from high wavenumbers and (11.91) effectively reduces to (11.88). If the energy spectrum is steeper than -3 , then the integrand is dominated by contributions from low wavenumbers. For $k \gg k_0$ we can approximate the integral by $[k_0^3 \mathcal{E}(k_0)]^{-1/2}$, that is the eddy-turnover time at large scales, τ_{k_0} , given by (11.88). The tracer spectrum then becomes

$$\mathcal{P}(k) = \mathcal{K}'_\chi \chi \tau_{k_0} k^{-1}, \quad (11.92)$$

where $\mathcal{K}'_\chi$ is a constant. In all these cases the tracer cascade is to smaller scales even if, as may happen in two-dimensional turbulence, energy is cascading to larger scales.

The scale at which diffusion becomes important is given by equating the turbulent time scale τ_k to the diffusive time scale $(\kappa k^2)^{-1}$. This is independent of the flux of tracer, χ , essentially because the equation for the tracer is linear. Determination of expressions for these scales in two and three dimensions are left as problems for the reader.

11.5.1 Examples of tracer spectra

Energy inertial range flow in three dimensions

Consider a range of wavenumbers over which neither viscosity nor diffusivity directly influence the turbulent motion and the tracer. Then, in (11.90), $A = \mathcal{K}\varepsilon^{2/3}$ where ε is the rate of energy transfer to small scales, $\mathcal{K}$ is the Kolmogorov constant, and $n = 5/3$. The tracer spectrum becomes¹⁹

$$\mathcal{P}(k) = \mathcal{K}_\chi^{3d} \varepsilon^{-1/3} \chi k^{-5/3}. \quad (11.93)$$

where $\mathcal{K}_\chi^{3d}$ is a (putatively universal) constant. It is interesting that the $-5/3$ exponent appears in both the energy spectrum and the passive tracer spectrum. Using (11.88), this is the only spectral slope for which this occurs. Experiments show that this range does, at least approximately, exist with a value of $\mathcal{K}_\chi^{3d}$ of about 0.5–0.6 in three dimensions.

Inverse energy-cascade range in two-dimensional turbulence

Suppose that the energy injection occurs at a smaller scale than the tracer injection, so that there exists a range of wavenumbers over which energy is cascading to larger scales while tracer variance is simultaneously cascading to smaller scales. The tracer spectrum is then

$$\mathcal{P}(k) = \mathcal{K}_\chi^{2d} \varepsilon^{-1/3} \chi k^{-5/3}, \quad (11.94)$$

the same as (11.93), although ε is now the energy cascade rate to larger scales and the constant $\mathcal{K}_\chi^{2d}$ does not necessarily equal $\mathcal{K}_\chi^{3d}$.

Enstrophy inertial range in two-dimensional turbulence

In the forward enstrophy inertial range the eddy time scale is $\tau_k = \eta^{-1/3}$ (assuming of course that the classical phenomenology holds). Directly from (11.87) the corresponding tracer spectrum is then

$$\mathcal{P}(k) = \mathcal{K}_\chi^{2d*} \eta^{-1/3} \chi k^{-1}. \quad (11.95)$$

The passive tracer spectrum now has the same slope as the spectrum of vorticity variance (i.e., the enstrophy spectrum), which is perhaps reassuring since the tracer and vorticity obey similar equations in two dimensions.

The viscous-advective range of large Prandtl number flow

If the Prandtl number $\sigma = \nu/\kappa \gg 1$ (and in seawater $\sigma \approx 7$) then there may exist a range of wavenumbers in which viscosity is important, but not tracer diffusion. The energy spectrum is then very steep, and (11.92) will apply. The straining then comes from wavenumbers near the viscous scale, so that for three dimensional flow the appropriate k_0 to use in (11.92) is the viscous wavenumber, and $k_0 = k_\nu = (\varepsilon/\nu^3)^{1/4}$. The dynamical time scale at this wavenumber is given by

$$\tau_{k_\nu} = \left(\frac{\nu}{\varepsilon} \right)^{1/2}, \quad (11.96)$$

and using this and (11.92) the tracer spectrum in this viscous-advective range becomes

$$\mathcal{P}(k) = \mathcal{K}'_B \left(\frac{\nu}{\varepsilon} \right)^{1/2} \chi k^{-1}. \quad (11.97)$$

This spectral form applies for $k_\nu < k < k_\kappa$, where k_κ is the wavenumber at which diffusion becomes important, found by equating the eddy turnover time given by (11.96) with the diffusive time scale $(\kappa k^2)^{-1}$. This gives

$$k_\kappa = \left(\frac{\varepsilon}{\nu \kappa^2} \right)^{1/4}, \quad (11.98)$$

and k_κ is known as the Batchelor wavenumber (and its inverse is the Batchelor scale). Beyond k_κ , the diffusive flux is not constant and the tracer spectrum can be expected to decay as wavenumber increases. A heuristic way to calculate the spectrum in the diffusive range is to first note that the flux of the tracer is not constant but diminishes according to

$$\frac{d\chi'(k)}{dk} = -2\kappa k^2 \mathcal{P}(k), \quad (11.99)$$

where χ' is the wavenumber-dependent rate of tracer transfer. Let us still assume that χ' and $\mathcal{P}(k)$ are related by an equation of the form (11.87), where now τ_k is a constant, given by (11.96). Thus,

$$\mathcal{K}_B \chi' = \frac{\mathcal{P}(k)k}{\tau_{k_\kappa}} = \frac{\mathcal{P}(k)k}{(\nu/\varepsilon)^{1/2}}, \quad (11.100)$$

where $\mathcal{K}_B$ is a constant. Using (11.99) and (11.100) we obtain

$$\frac{d\chi'}{dk} = -2\mathcal{K}_B \kappa k \left(\frac{\nu}{\varepsilon}\right)^{1/2} \chi'. \quad (11.101)$$

Solving this, using $\chi' = \chi$ (where χ is a constant) for small k , gives

$$\mathcal{P}(k) = \mathcal{K}_B \left(\frac{\nu}{\varepsilon}\right)^{1/2} \chi k^{-1} \exp[-\mathcal{K}_B(k/k_\kappa)^2]. \quad (11.102)$$

This reduces to (11.97) if $k \ll k_\kappa$, and is known as the Batchelor spectrum.²⁰ Its high-wavenumber part $k > k_\kappa$ is known as the viscous-diffusive subrange. The spectrum, and its two-dimensional analogue, are illustrated in Fig. 11.13

In two dimensions the viscous-advective range occurs for wavenumbers greater than $k_\nu = (\eta/\nu^3)^{1/6}$. The appropriate time scale within this subrange is $\eta^{-1/3}$, and therefore gives a spectrum with the precisely the same form as (11.95). At sufficiently high wavenumbers tracer diffusion becomes important, with the diffusive scale now given by equating the eddy turnover time $\eta^{-1/3}$ with the viscous time scale $(\kappa k^2)^{-1}$. This gives the diffusive wavenumber, analogous to (11.98), of $k_\kappa = (\eta/\kappa^3)^{1/6}$. Using (11.101) and the procedure above we then obtain an expression for the spectrum in the region $k > k_\nu$, that is a two-dimensional analogue of (11.102), namely

$$\mathcal{P}(k) = \mathcal{K}'_B \eta^{-1/3} \chi k^{-1} \exp[-\mathcal{K}'_B(k/k_\kappa)^2]. \quad (11.103)$$

For $k \ll k_\kappa$ this reduces to (11.95), possibly with a different value of the Kolmogorov-like constant.

† The inertial-diffusive range of small Prandtl number flow

For small Prandtl number ($\nu/\kappa \ll 1$) the energy inertial range may coexist with a range over which tracer variance is being dissipated, giving us the so-called inertial-diffusive range. The tracer will begin to be dissipated at a wavenumber obtained by equating a dynamical eddy turnover time with a diffusive time, and this gives a diffusive wavenumber

$$k'_\kappa = \begin{cases} (\varepsilon/\kappa^3)^{1/4} & \text{in three dimensions,} \\ (\eta/\kappa^3)^{1/6} & \text{in two dimensions.} \end{cases} \quad (11.104)$$

Beyond the diffusive wavenumber the flux of the tracer is no longer constant but diminishes according to (11.99).

Given a non-constant flux and an eddy-turnover time that varies with wavenumber there is no self-evidently correct way to proceed. One way is to assume that χ and $\mathcal{P}(k)$ are related by $\mathcal{K}''_\chi \chi = \mathcal{P}(k)k/\tau_k$ [as in (11.100), but with a potentially different proportionality constant] and

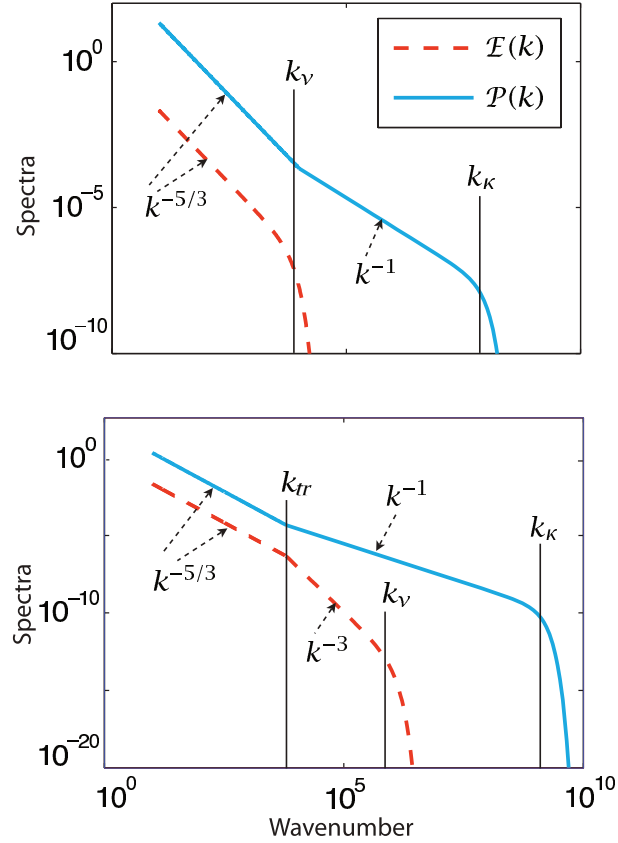


Fig. 11.13 The energy spectra, $\mathcal{E}(k)$ and passive tracer spectra $\mathcal{P}(k)$ in large Prandtl number three-dimensional (top) and two-dimensional (bottom) turbulence.

In three dimensions $\mathcal{P}(k)$ is given by (11.93) for $k < k_v$ and by (11.102) for $k > k_v$. In two dimensions, if k_{tr} marks the transition between a $k^{-5/3}$ inverse energy cascade and a k^{-1} forward enstrophy cascade, then $\mathcal{P}(k)$ is given by (11.94) for $k < k_{tr}$ and by (11.103) for $k > k_{tr}$. In both two and three dimensions the tracer spectra falls off rapidly for $k > k_k$.

with τ_k given by (11.88); that is, $\tau_k = \varepsilon^{-1/3} k^{-2/3}$ in three dimensional turbulence. Using this in (11.99) leads to

$$\mathcal{P}(k) = \mathcal{K}_\chi'' \chi \varepsilon^{-1/3} k^{-5/3} \exp[-(\mathcal{K}_\chi''/2)(k/k_\kappa')^{4/3}]. \quad (11.105)$$

where χ is the tracer flux at the beginning of the tracer dissipation range. (A similar expression emerges in two-dimensional turbulence.) However, given such a steep spectrum an argument based on spectral locality is likely to be suspect. Another argument posits a particular relationship between the tracer spectrum and energy spectrum in the inertial-diffusive range, and this leads to

$$\mathcal{P}(k) = \frac{\mathcal{K}_B''}{3} \chi_0 \varepsilon^{2/3} \kappa^{-3} k^{-17/3} = \mathcal{K}_B'' \chi_0 \varepsilon^{-1/3} k^{-5/3} g(k/k_\kappa), \quad (11.106)$$

where $g(\alpha) = \alpha^{-4}/3$ and $\mathcal{K}_B''$ is a constant.²¹

Notes

- 1 Horace Lamb has been quoted as saying that when he died and went to Heaven he hoped for enlightenment on two things, quantum electrodynamics and turbulence, although he was only optimistic about the former. Werner Heisenberg expressed a similar sentiment. But Heaven may be the wrong place to seek such enlightenment, for turbulence is the invention of the Devil, put on Earth to torment us.
- 2 The algebra of the three-dimensional case is more complicated because of the pressure term and because the momentum equation is a vector equation. Nevertheless, in incompressible flow we

can take the divergence of the momentum equation to obtain an elliptic equation for pressure of the form $\nabla^2 p = Q(v)$ where the right-hand side is a quadratic function of velocity and its derivatives, Fourier transform this and then proceed much as in the two-dimensional case.

- 3 A. N. Kolmogorov (1903–1987) was a Russian mathematician and theoretical physicist, who made seminal contributions to turbulence (with famous papers in 1941 and in 1962), to probability and statistics, and to classical mechanics (e.g., the Kolmogorov–Arnold–Moser theorem). Yaglom (1994) provides more details on both the man and his scientific contributions.
- 4 The process has been encapsulated in the following ditty by Lewis Fry (L.F.) Richardson (1881–1953), his own summary of Richardson (1920):
*Big whorls have little whorls, that feed on their velocity;
 And little whorls have lesser whorls, and so on to viscosity.*
 The verse follows a well-known one by the mathematician Augustus De Morgan (in *A Budget of Paradoxes* 1872), ‘*Great fleas have little fleas upon their backs to bite ‘em...*’, which in turn is a parody on a poem by Jonathan Swift. Richardson himself was a British scientist best known as the person who (following earlier work by Cleveland Abbe, in 1901, and Vilhelm Bjerknes, in 1904) envisioned weather forecasting in its current form — that is, numerical weather prediction. However, as described in his 1922 book, instead of an electronic computer performing the calculations, he imagined, perhaps fancifully, a hall full of people performing calculations in unison all directed by a conductor at the front. His first numerical forecast, calculated by hand, was wildly inaccurate both because he failed to initialize his atmosphere properly and because his timestep was too long and did not satisfy the CFL condition, and unrealistic gravity waves dominated the solution. However, it was a prescient and important effort. He also worked on turbulence, and seems to have envisioned the turbulent cascade prior to Kolmogorov (to wit the verse above), and the ‘Richardson number’, a measure of fluid stratification, is named after him. He also made contributions to the theory of war and was known as a pacifist — he was a conscientious objector and drove ambulances in the first World War, and resigned from the U. K. Meteorological Office because it became part of the Air Ministry.
- 5 Kolmogorov (1941) obtained the result in a slightly different way, using distances in real space rather than wavenumber and deriving the equivalent result for the longitudinal structure function, $D(r) \equiv \langle [u_i(\mathbf{x} + \mathbf{r}) - u_i(\mathbf{x})]^2 \rangle \sim r^{2/3}$. It was Obukhov (1941) who gave an argument in spectral space and first wrote down that $E(k) \sim k^{-5/3}$. Kolmogorov’s argument is regarded as more general and hence the 5/3 spectrum is usually named for him, but sometimes it is called the ‘Kolmogorov–Obukhov’ 5/3 spectrum.
- 6 Results kindly provided by R. Zhao.
- 7 The first observations confirming the -5/3 predictions were from a tidal channel (Grant et al. 1962). These results were initially presented at a turbulence conference in Marseille in 1961, ironically at the same time as Kolmogorov presented a modification of his original theory that incorporated a local mean dissipation rate, to try to take intermittency into account, recognizing that his first theory was incomplete (Kolmogorov 1962). It is said to have been L. D. Landau who pointed out the consequences of intermittency to Kolmogorov, soon after the K41 theory first appeared.
- 8 Two early papers on two-dimensional turbulence are those of Lee (1951) and Fjørtoft (1953), the former noting the incompatibility of the material conservation of vorticity with Kolmogorov’s energy inertial range, and the latter recognizing the two-dimensional nature of large-scale atmospheric motion. Batchelor (1953b) noted the tendency of energy to concentrate in small wavenumbers as a consequence of energy and enstrophy conservation. The theory was developed by Kraichnan (1967) (who predicted the spectral slopes of the two-dimensional cascades), Leith (1968) and Batchelor (1969). Lilly (1969) performed some early numerical integrations. For reviews see Kraichnan & Montgomery (1980), Danilov & Gurarie (2001) and Boffetta & Ecke (2012).
- 9 This similarity argument is due to Batchelor (1969), and its validity was explored by Bartello & Warn (1996) using numerical simulations of decaying two-dimensional turbulence. They found that the similarity hypothesis is not quantitatively accurate, and in particular that higher-order moments of the vorticity do not obey the predictions of the theory. This failure may be ascribed to the fact that

in two-dimensional flow vorticity is conserved on parcels, and in the presence of coherent vortices this is an effective constraint that is not included in the theory. In flow with a finite deformation radius coherent vortices are found to be less important and an analogous similarity hypothesis appears to work better (Iwayama *et al.* 2002).

- 10 Arbic *et al.* (2007), Colin de Verdière (1980). See also Scott (2001).
- 11 To read more about strain rate, see Lesieur (1997).
- 12 For example, Peltier & Stuhne (2002) and Smith & Waleffe (1999).
- 13 Adapted from Maltrud & Vallis (1991).
- 14 For numerical simulations illustrating these and other properties of two-dimensional turbulence see, among others, McWilliams (1984), Maltrud & Vallis (1991), Oetzel & Vallis (1997), Lindborg & Alvelius (2000) and Smith *et al.* (2002). Also look at Jupiter through a telescope! A statistical-mechanical argument that there is tendency for a small number of vortices to form in two-dimensional turbulence was given by Onsager (1949).
- 15 The Lorenz equations were written down by Lorenz (1963), based on some earlier work of Saltzman (1962), and inspired a veritable industry of study. The field of chaos, or more generally nonlinear dynamics, has grown enormously since then, prompted also by work in mathematics occurring at about the same time, and its development is sometimes regarded as one of the true revolutions of science in the twentieth century. Aubin & Dahan Dalmedico (2002) write an interesting history of it. The correspondence of the Lorenz equations to a real fluid system is tenuous, but the importance of the properties they demonstrate transcends this; we regard the equations simply as an example of a chaotic system with some fluid relevance. The equations and variations about them have reappeared in studies of, among other things, lasers, dynamos, chemical reactions, mechanical waterwheels and El Niño.
- 16 That a turbulent flow is, *inter alia*, chaotic and unpredictable follows from the work of Lorenz (1963), Ruelle & Takens (1971) and others who showed that fluid turbulence was generically a consequence of a small number of bifurcations as some controlling parameter (such as the Reynolds number) is changed. Prior to this, turbulence was sometimes thought, following Landau (1944), to be a large collection of periodic motions with incommensurate frequencies that would have complex and non-repeating but presumably predictable behaviour. Notwithstanding the Landau picture, it seems to have been known, well before the development of nonlinear dynamical systems theory in the 1960s and 1970s, that the weather was inherently unpredictable. Poincaré was perhaps the first to properly understand this at the turn of the twentieth century, although in the review of a book by the thermodynamicist P. Duhem, W. S. Franklin wrote in 1898 that 'An infinitesimal cause produces a finite effect. Long range detailed weather prediction is therefore impossible... the flight of a grasshopper in Montana may turn a storm aside from Philadelphia to New York!' Weather forecasters themselves long seem to have intuited that the atmosphere was intrinsically, and not just practically, difficult to forecast. In the 1941 novel *Storm* by G. R. Stewart (1895–1980, a professor of English at UC Berkeley) we find a forecaster recalling his old professor's saying that 'A Chinaman sneezing in Shen-si may set men to shoveling snow in New York City'. (*Storm* is also notable because it used female names for intense storms, a practice that became common among forecasters in World War II and that was used by the US Weather Service from 1953–1978, after which gender equity pertained.) In a more academic setting, the predictability problem is mentioned in the book by Godske *et al.* (1957) (much of which was written in the 1930s and 1940s), and Thompson (1957) and Novikov (1959) studied the unpredictability of atmospheric flows from the perspective of turbulence, evidently unaware of, or at least uninfluenced by, either Poincaré or Landau. Phillip Thompson himself was in the Joint Numerical Weather Prediction Unit of the U.S. government in the 1950s, whose task was to numerically produce weather forecasts, and this practical experience undoubtedly confronted and guided his theoretical thinking. These various strands came together and were clarified by the dynamical systems viewpoint coupled with the view of the atmosphere as a geostrophically turbulent fluid, and this led to the viewpoint we describe here, and to estimates of the limit to predictability of the atmospheric weather of about two weeks, although at any given time the predictability may certainly be shorter or longer than that.

- 17 The more technical phrase ‘sensitive dependence on initial conditions’ is a paraphrase of one of Poincaré, but the catchier one ‘butterfly effect’ is more recent. It seems to have first appeared in a meteorological context in Smagorinsky (1969), where we find ‘Would the flutter of a butterfly’s wings ultimately amplify to the point where the numerical simulation departed from reality...? If not the flutter of the butterfly’s wings, the disturbance might be the result of instrumental errors in the initial conditions...’ The phrase became more well known following a lecture by Lorenz to the American Association for the Advancement of Science (AAAS) in 1972 entitled ‘Predictability: does the flap of a butterfly’s wings in Brazil set off a tornado in Texas?’ The shape of the Lorenz attractor in phase space also resembles a butterfly (Fig. 11.11), and the two (not unrelated) phenomena are sometimes conflated. The death of a butterfly changes the course of history in a science fiction story by Ray Bradbury, *A Sound of Thunder* (1952). See also endnote 16 above and the brief history by Hilborn (2004).
- 18 A related point is that the predictability of a turbulent system is not well characterized by its spectrum of Lyapunov exponents: in a turbulent system in three dimensions the largest Lyapunov exponent is likely to be associated with very small scales of motion, and the error growth associated with this effectively saturates at small scales.
- 19 First derived by Obukhov (1949). See also Corrsin (1951).
- 20 Batchelor (1959) also suggests that the constant $\mathcal{K}_B$ in (11.102) should have the value 2. There is some observational support for the k^{-1} viscous-advective range in the temperature spectra of the ocean, one of the first measurements being that of Grant *et al.* (1968). Aside from their intrinsic interest, the viscous and diffusive scales are used in microstructure theory and measurements that lead to estimates of the ocean’s energy dissipation rate (Gregg 1998, Stips 2005).
- 21 Batchelor *et al.* (1959). There is some numerical support for the $-17/3$ spectrum using a large-eddy simulation (LES) model (Chasnov 1991). See also O’Gorman & Pullin (2005).

CHAPTER 12

Geostrophic Turbulence and Baroclinic Eddies

GEOSTROPHIC TURBULENCE is turbulence in flows that are stably stratified and in near-geostrophic balance. Like any problem in turbulence it is difficult, and a real ‘solution’ — meaning an accurate, informative statement about average states, without computation of the detailed evolution — may be out of our reach, and may not exist. Nevertheless, and ironically, it is sometimes easier to say something interesting about geostrophic turbulence than about incompressible isotropic two- or three-dimensional turbulence. In the latter class of problems there is nothing else to understand other than the problem of turbulence itself; on the other hand, rotation and stratification give one something else to grasp, a nettle though it may be, and it becomes possible to address geophysically interesting phenomena without having to solve the whole turbulence problem. Furthermore, in inhomogeneous geostrophic turbulence, asking questions about the *mean fields* is meaningful and useful, whereas this is trivial in isotropic turbulence.

Geostrophic turbulence is not restricted to quasi-geostrophic flow; indeed, the large scale turbulence of the Earth’s ocean and atmosphere is sometimes simply called ‘macro-turbulence’. Nevertheless, the quasi-geostrophic equations retain advective nonlinearity in the vorticity equation, and they capture the constraining effects of rotation and stratification that are so important in geophysical flows in a simple and direct way; for these reasons the quasi-geostrophic equations will be our main tool. Let us consider the effects of rotation first, then stratification.

12.1 EFFECTS OF DIFFERENTIAL ROTATION IN TWO-DIMENSIONAL TURBULENCE

In the limit of motion of a scale much shorter than the deformation radius, and with no topography, the quasi-geostrophic potential vorticity equation, (5.118), reduces to the two-dimensional equation,

$$\frac{Dq}{Dt} = 0, \quad (12.1)$$

where $q = \zeta + f$. This is the perhaps the simplest equation with which to study the effects of rotation on turbulence. Suppose first that the Coriolis parameter is constant, so that $f = f_0$. Then (12.1)

becomes simply the two-dimensional vorticity equation

$$\frac{D\zeta}{Dt} = 0. \quad (12.2)$$

Thus constant rotation has *no* effect on purely two-dimensional motion. Flow that is already two-dimensional — flow on a soap film, for example — is unaffected by rotation. (In the ocean and atmosphere, or in a rotating tank, it is of course the effects of rotation that lead to the flow being quasi-two dimensional in the first instance.)

Suppose, though, that the Coriolis parameter is variable, as in $f = f_0 + \beta y$. Then we have

$$\frac{D}{Dt}(\zeta + \beta y) = 0 \quad \text{or} \quad \frac{D\zeta}{Dt} + \beta v = 0. \quad (12.3a,b)$$

If the asymptotically dominant term in these equations is the one involving β , then we obtain $v = 0$. That is, if β is very large, then the meridional flow v must be correspondingly small to ensure that the equation can balance, and this argument holds in the presence of forcing and dissipation. Any flow must then be predominantly *zonal*. On the sphere the constraint may be interpreted as a consequence of angular momentum as follows. A ring of fluid encircling the Earth at a velocity u has an angular momentum per unit mass $a \cos \theta(u + \Omega a \cos \theta)$, where θ is the latitude and a is the radius of the Earth. Moving this ring of air polewards while conserving its angular momentum requires that its zonal velocity and hence energy must increase, so unless there is a source for that energy the flow is constrained to remain zonal.

In the following sections we look at the mechanism of jet formation in a little more detail. We first give some fairly general scaling arguments, and then proceed in two complimentary ways — first with an argument couched in spectral space using the language of turbulent cascades, and second with an argument in physical space. In section 15.1 we come back to the problem from a third perspective, one more appropriate for Earth's atmosphere.

12.1.1 The wave–turbulence cross-over

Scaling

Let us now consider how turbulent flow might interact with Rossby waves. We write (12.1) in full as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0. \quad (12.4)$$

If $\zeta \sim U/L$ and if $t \sim T$ then the respective terms in this equation scale as

$$\frac{U}{LT} \quad \frac{U^2}{L^2} \quad \beta U. \quad (12.5)$$

The way that time scales (i.e., advectively or with a Rossby wave frequency scaling) is determined by which of the other two terms dominates, and this in turn is scale dependent. For large scales the β -term will be dominant, and at smaller scales the advective term is dominant. The cross-over scale, denoted L_R , is called the *Rhines scale* and is given by¹

$$L_R \sim \left(\frac{U}{\beta} \right)^{1/2}. \quad (12.6)$$

The U in (12.6) should be interpreted as the root-mean-square velocity at the energy containing scales, not a mean or translational velocity. We refer to the specific scale $\sqrt{U/\beta}$ as the Rhines scale,

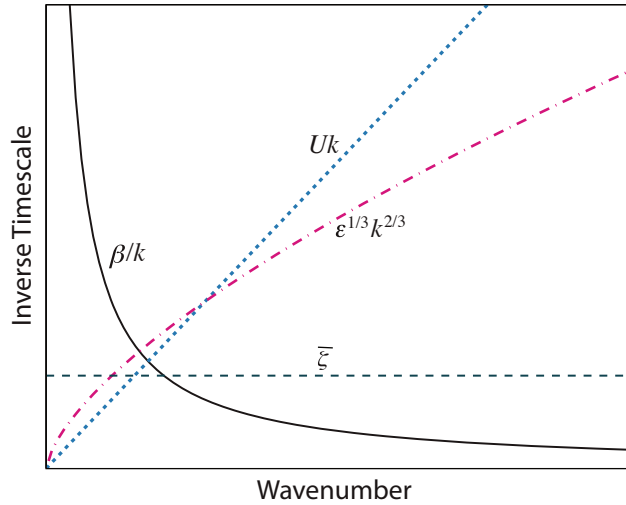


Fig. 12.1 Inverse timescales, or frequencies, in wavenumber space in beta-plane turbulence. The solid curve is the frequency of Rossby waves, proportional to β/k , and the other three curves are various estimates of the inverse turbulence time scale. These are the turbulent eddy transfer rate, proportional to $\epsilon^{1/3} k^{2/3}$ in a $k^{-5/3}$ spectrum; the estimate Uk where U is an root-mean-squared velocity; and the mean vorticity, which is constant.

Where the Rossby wave frequency is larger (smaller) than the turbulent frequency, i.e., at large (small) scales, Rossby waves (turbulence) dominate the dynamics.

L_R , and more general scales involving a balance between nonlinearity and β as the β -scale, denoted L_β .

This is not a unique way to arrive at a β -scale, since we have chosen the length scale that connects vorticity to velocity to also be the β -scale, and it is not obvious a priori that this should be so. If the two scales are different, the three terms in (12.4) scale as

$$\frac{Z}{T} : \frac{UZ}{L} : \beta U, \quad (12.7)$$

respectively, where Z is the scaling for vorticity [i.e., $\zeta = \mathcal{O}(Z)$]. Equating the second and third terms gives the scale

$$L_{\beta Z} = \frac{Z}{\beta}. \quad (12.8)$$

Although (12.6) and (12.8) differ in detail, both indicate that at some *large* scale Rossby waves are likely to dominate whereas at small scales advection, and turbulence, dominates.

Another heuristic way to derive (12.6) is by a direct consideration of time scales. Ignoring anisotropy, Rossby wave frequency is β/k and an inverse advective time scale is Uk , where k is the wavenumber. Equating these two gives an equation for the Rhines wavenumber

$$k_R \sim \left(\frac{\beta}{U} \right)^{1/2}. \quad (12.9)$$

This equation is the inverse of (12.6), but factors of order unity (e.g., π) cannot be revealed by simple scaling arguments such as these. We denote the Rhines wavenumber as k_R and a more generic cross-over wavenumbers as k_β . The cross-over between waves and turbulence is reasonably sharp, as indicated in Fig. 12.1.

Turbulent phenomenology

Let us now move beyond elementary scaling arguments and examine wave–turbulence cross-over using the phenomenology of two-dimensional turbulence. We will suppose that the fluid is stirred at some well-defined scale k_f , producing an energy input ϵ . Then (assuming no energy is lost to smaller scales) energy cascades to large scales at that same rate. At some scale, the β -term in the vorticity equation will start to make its presence felt. By analogy with the procedure for finding the viscous dissipation scale in turbulence, we can find the scale at which linear Rossby waves dominate

by equating the inverse of the turbulent eddy turnover time to the Rossby wave frequency. The eddy-turnover time is

$$\tau_k = \varepsilon^{-1/3} k^{-2/3}, \quad (12.10)$$

and equating this to the inverse Rossby wave frequency k/β gives estimates for the β -wavenumber and its inverse, the β -scale, namely

$$k_\varepsilon = \left(\frac{\beta^3}{\varepsilon} \right)^{1/5}, \quad L_\varepsilon = \left(\frac{\varepsilon}{\beta^3} \right)^{1/5}. \quad (12.11a,b)$$

We denote these scales with a subscript ε because of the appearance of the energy cascade rate. In a real fluid these expressions are harder to evaluate than (12.9), since it is generally much easier to measure velocities than energy transfer rates, or even vorticity. On the other hand, (12.11) is more satisfactory from the point of view of turbulence theory as ε may be determined by processes largely independent of β , whereas the magnitude of the eddies at the energy containing scales is likely to be a function of β . We also note that the scale given by (12.11b) is not necessarily the energy-containing scale, and may in principle differ considerably from the scale given by (12.9). This is because the inverse cascade is not necessarily *halted* at the scale (12.11b) — this is just the scale at which Rossby waves become important. Energy may continue to cascade to larger scales, albeit anisotropically as discussed below, and so the energy containing scale may be larger.

12.1.2 Generation of zonal flows and jets

None of the effects discussed so far takes into account the anisotropy inherent in Rossby waves, and such anisotropy can give rise to predominantly zonal flows and jets. To understand this, let us first note that energy transfer will be relatively inefficient at those scales where linear Rossby waves dominate the dynamics. But the wave–turbulence boundary is not isotropic; the Rossby wave frequency is quite anisotropic, being given by

$$\omega = -\frac{\beta k^x}{k^{x2} + k^{y2}}. \quad (12.12)$$

If, albeit a little crudely, we suppose that the turbulent part of the flow remains isotropic, the wave–turbulence boundary is then given by equating the inverse of (12.10) with (12.12); that is solution of

$$\varepsilon^{1/3} k^{2/3} = \frac{\beta k^x}{k^2}, \quad (12.13)$$

where k is the isotropic wavenumber. Solving this gives expressions for the x - and y -wavenumber components of the wave–turbulence boundary, namely

$$k_\varepsilon^x = \left(\frac{\beta^3}{\varepsilon} \right)^{1/5} \cos^{8/5} \theta, \quad k_\varepsilon^y = \left(\frac{\beta^3}{\varepsilon} \right)^{1/5} \sin \theta \cos^{3/5} \theta, \quad (12.14)$$

where the polar coordinate is parameterized by the angle $\theta = \tan^{-1}(k^y/k^x)$. This odd-looking formula is illustrated in Fig. 12.2, and it defines the anisotropic wave–turbulence boundary. (Slight variations on this theme are produced by using different expressions for the turbulence time scale.)

What occurs physically? The region inside the dumbbell shapes in Fig. 12.2 is dominated by Rossby waves, where the natural frequency of the oscillation is *higher* than the turbulent frequency. If the flow is stirred at a wavenumber higher than this the energy will cascade to larger scales, but because of the frequency mismatch the turbulent flow will be unable to efficiently excite modes within the dumbbell. Nevertheless, there is still a natural tendency of the energy to seek the gravest

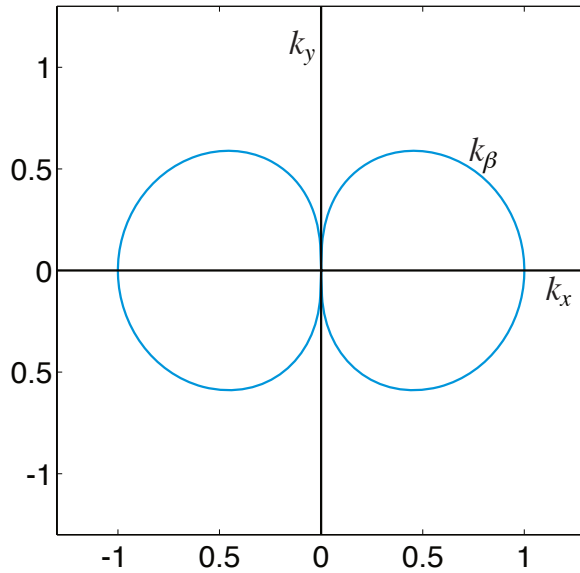


Fig. 12.2 The anisotropic wave–turbulence boundary k_e , in wave-vector space calculated by equating the turbulent eddy transfer rate, proportional to $k^{2/3}$ in a $k^{-5/3}$ spectrum, to the Rossby wave frequency $\beta k^x/k^2$, as in (12.14).

Within the dumbbell Rossby waves dominate and energy transfer is inhibited. The inverse cascade plus Rossby waves thus leads to a generation of zonal flow. [The wavenumbers are scaled such that $\beta^3/\varepsilon = 1$.]

mode, and it will do this by cascading toward the $k^x = 0$ axis; that is, toward zonal flow. Thus, the combination of Rossby waves and turbulence will lead to the formation of zonal flow and, potentially, zonal jets.³

Figure 12.3 illustrates this mechanism; it shows the freely evolving (unforced, inviscid) energy spectrum in a simulation on a β -plane, with an initially isotropic spectrum. The energy implodes, cascading to larger scales but avoiding the region inside the dumbbell and piling up at $k^x = 0$. In physical space the flow organizes itself into zonally elongated structures and jets, in both freely-decaying and forced-dissipative simulations (Fig. 12.4 and Fig. 12.7).

† Joint effect of beta and friction

The β term does not remove energy from a fluid. Thus, if energy is being added to a fluid at some small scales, and the energy is cascading to larger scales, then the β -effect does not of itself halt the inverse cascade, it merely deflects the cascade such that the flow becomes more zonal. Suppose that the fluid obeys the barotropic vorticity equation,

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = F - r\zeta + \nu \nabla^2 \zeta, \quad (12.15)$$

where the viscosity, ν , is small and acts only to remove enstrophy, and not energy, at very small scales. The forcing, F , supplies energy at a rate ε and this is cascaded upscale and removed by the linear drag term $-r\zeta$, where the drag coefficient r is a constant. If the friction is sufficiently large then the energy is removed before it feels the effect of β at the scale (11.70), namely $L_r = (\varepsilon/r^3)^{1/2}$. However, if friction is small, such that $L_\varepsilon = (\varepsilon/\beta^3)^{1/5}$ is smaller than L_r , then the cascade feels the beta effect before it feels frictional effects, and L_r is then unlikely to be a relevant parameter.

If the forcing scale is small and an inverse cascade exists, the turbulence is characterized by three parameters, β , ε , and r . There is only one way to make a non-dimensional parameter from these, and this is

$$\gamma = \beta^2 \varepsilon r^{-5}, \quad (12.16)$$

along with powers of γ . There is no unique way to make a length scale from these parameters, but various scales arise phenomenologically. One is L_ε , noted above, and another arises from energetic

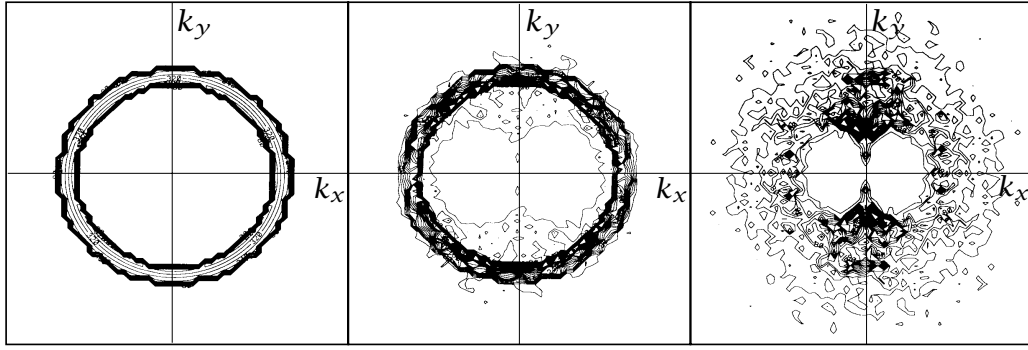


Fig. 12.3 Evolution of the energy spectrum in a freely evolving two-dimensional simulation on the β -plane. The panels show contours of energy in wavenumber (k_x, k_y) space at successive times. The initial spectrum is isotropic. The energy ‘implodes’, but its passage to large scales is impeded by the β -effect, and second and third panels show the spectrum at later times, illustrating the dumbbell predicted by (12.14) and Fig. 12.2.²

considerations. If we form an energy equation from (12.15) by multiplying by $-\psi$ and spatially integrating (and neglecting viscosity) we find the energy balance

$$\varepsilon = -\frac{1}{A} \int_A \psi F dA = \frac{r}{A} \int_A (\nabla \psi)^2 dA = 2r\bar{E}, \quad (12.17)$$

where $\bar{E}$ is the average energy of the fluid per unit mass. This expression holds whether or not β is nonzero. Using (12.6) with $U = \sqrt{2\bar{E}}$, where $\bar{E}$ is obtained from (12.17), we obtain the scale

$$L_{\beta r} = \left(\frac{\varepsilon}{r\beta^2} \right)^{1/4}, \quad (12.18)$$

which may be thought of as a prediction for the Rhines scale, L_R . The ratio of the scales $L_{\beta r}$ to L_ε is

$$\frac{L_{\beta r}}{L_\varepsilon} = \left(\frac{\beta^2 \varepsilon}{r^5} \right)^{1/20} = \gamma^{1/20}. \quad (12.19)$$

What do the two scales, L_ε and $L_{\beta r}$, represent? The first is the scale at which the β -effect is first felt by the inverse cascade, and it is relevant for large-scale meridional mixing, and so for the meridional heat transport in the atmosphere. If friction is small (i.e., if $\gamma \gg 1$) the inverse energy transfer will continue past this scale, with the energy largely being at very small values of k_x , and (12.18) then characterize the largest scale reached by the inverse cascade, and so the meridional scale of the jets. In this region one might hypothesize that the spectrum is determined by β and k alone, and if so it is given by

$$\mathcal{E}(k) = \mathcal{C}_z \beta^2 k^{-5}, \quad (12.20)$$

where $\mathcal{C}_z$ is a constant, and most of the energy is in the zonal flow. Only the extent (and not the amplitude) of this region is then determined by the forcing strength. Turbulence of this character is called *zonostrophic*, and is illustrated in Fig. 12.5. However, we might also expect the drag coefficient, r , to be a factor in at least part of spectral slope because, with such a steep slope, the timescale of the flow diminishes as the wavenumber falls, so there is no scale, L_{fric} say, such that friction acts only at scales larger than L_{fric} .⁴ But rather than pursuing more arguments in spectral space, let us turn to a physical space argument.

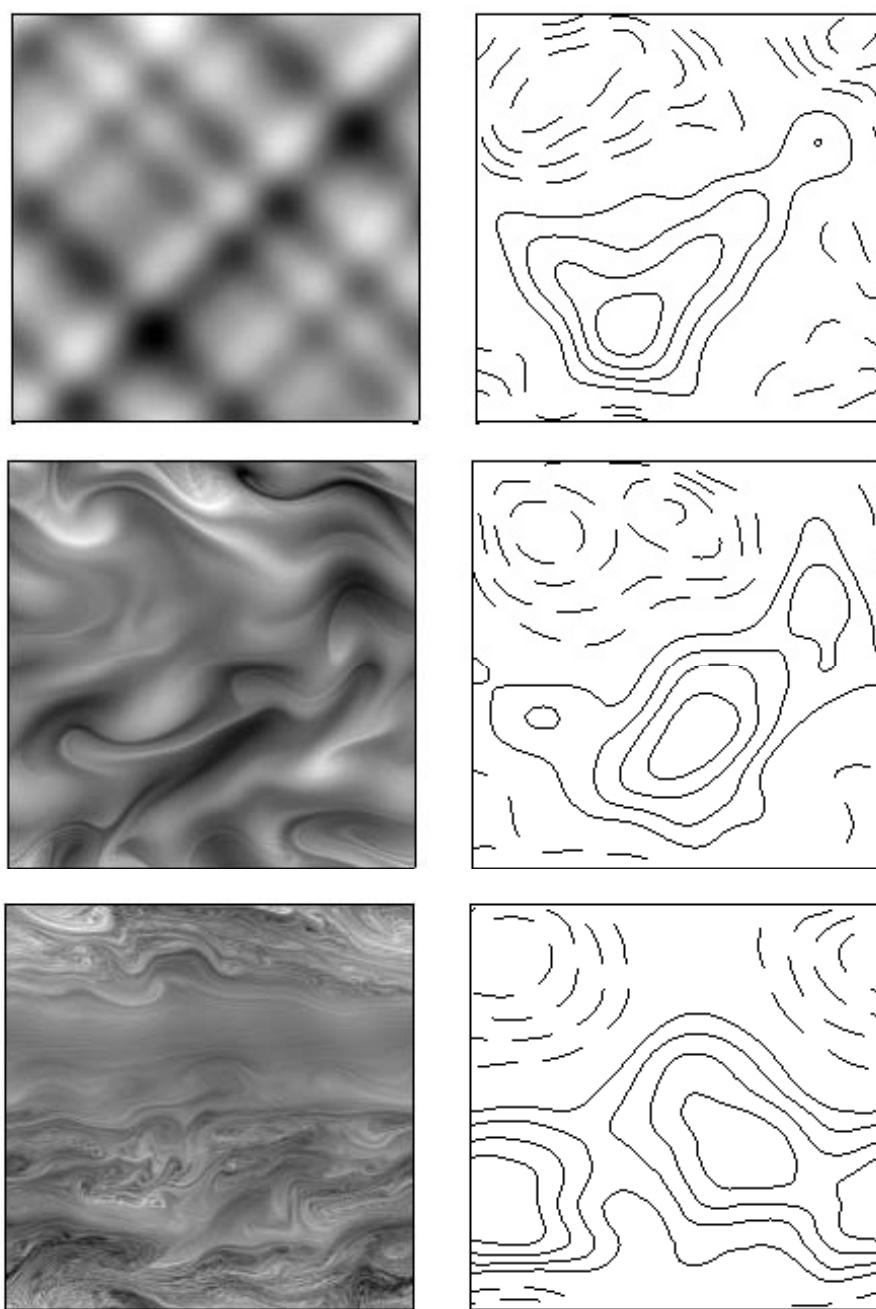
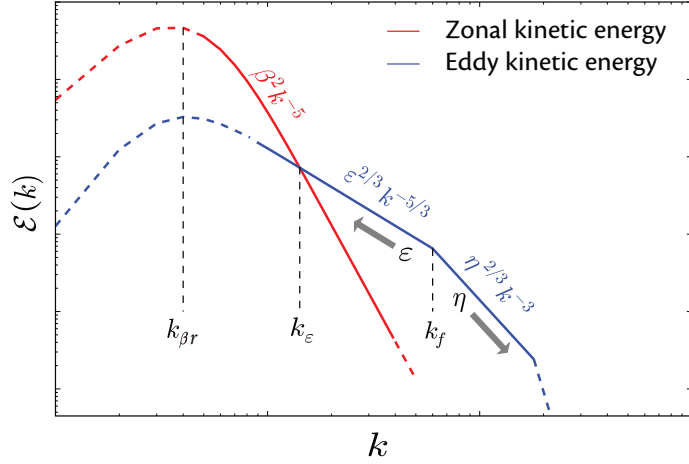


Fig. 12.4 Evolution of vorticity (greyscale, left column) and streamfunction (contour plots, right column) in a doubly periodic square domain (of length 2π) at times $t = 0, 50$ and 260 (in units of inverse vorticity), obeying (12.4) with the addition of a weak viscous term on the right-hand side. The initial conditions are the same as for Fig. 11.8, with a maximum value of vorticity of about 3. As $\beta = 3$, the β -Rossby number, $|\zeta|/\beta L$ is about unity. Compared to Fig. 11.8, vortex formation is inhibited and there is a tendency toward zonal flow.

Fig. 12.5 Putative kinetic spectra, $\mathcal{E}(k)$, in beta-plane turbulence. Energy injected at wavenumber k_f cascades to larger scales until it reaches the turbulent beta scale, k_e . The energy in the zonal modes then forms a zonostrophic k^{-5} spectrum which extends as far as the frictional Rhines scale, $k_{\beta r}$.



12.1.3 Potential vorticity staircases

Let us now consider a complimentary approach in physical space, with the essential idea being the following. Suppose that a turbulent flow exists on a background state that has a potential vorticity gradient, such as might be caused by the beta-effect. The eddies will seek to homogenize the potential vorticity but, in general, they will be unable to do so completely. Instead, the flow becomes partially homogenized, with regions of homogeneous potential vorticity separated by jumps, as illustrated in Fig. 12.6. Such a structure is called a *potential vorticity staircase*, and the staircase implies the existence of zonal jets.⁵

The homogenization of a scalar is discussed further in section 13.5, but it can be imagined that, in a turbulent flow, the gradients of a scalar that is both advected and diffused will become smeared out as much as possible: diffusion will dissipate any extrema and advection cannot recreate them. Potential vorticity is a scalar, so we might expect it too to become homogenized, but it is not passive and therefore the process of homogenization will affect the flow itself, preventing complete homogenization. One reason for that is the boundary conditions, another is that the process of homogenization requires more energy than the flow contains. To see this, consider a freely-evolving flow on a beta-plane obeying the barotropic vorticity equation

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta + \beta y) = \nu \nabla^2 \zeta, \quad (12.21)$$

where ν is sufficiently small that energy is well conserved over the timescales of interest. If potential vorticity, $q = \zeta + \beta y$, is to be homogenized over some meridional scale L then, in the homogenized region, $\zeta \approx -\beta y$. If the flow is predominantly zonal then this gives the estimate $U \sim \beta y^2$ and the energy in the region is, very approximately,

$$\frac{1}{2} \int u^2 dx dy \approx \int (\beta y^2)^2 dx dy \quad (12.22)$$

giving the estimate $U^2 L \sim \beta^2 L^5$. If we suppose that the initial root-mean square velocity is also U , and that the flow becomes predominantly zonal, then solving the above estimate for L suggests that the potential vorticity will become homogenized over the scale

$$L_h \sim \left(\frac{U}{\beta} \right)^{1/2}. \quad (12.23)$$

This is the same as the Rhines scale given in (12.6), as it has to be by dimensional analysis, although now it is supposed that much of the energy lies in the zonal flow. This way of looking at the problem

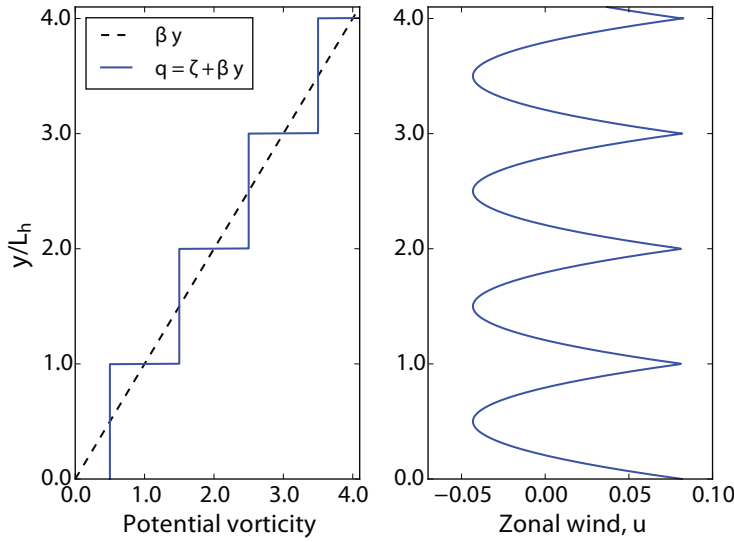


Fig. 12.6 An idealized potential vorticity staircase on a β -plane with $\beta = 1$. The left panel shows homogeneous regions of potential vorticity of meridional extent L_h separated by jumps, as calculated using (12.24). The right-hand panel shows the corresponding zonal flow, calculated using (12.28) with u_0 chosen to be such that the average zonal flow is zero.

also adds something to the physical picture — *an asymmetry between eastward and westward flow*, as we now discuss.

In the idealized staircase of Fig. 12.6 potential vorticity is piecewise continuous and given by

$$q = \zeta + \beta y = q_0, \quad 0 < y < L_h, \quad (12.24a)$$

$$q = \zeta + \beta y = q_0 + \beta L_h \equiv q_1, \quad L_h < y < 2L_h, \quad \text{and so on.} \quad (12.24b)$$

In any one of the homogenized regions the flow is given by solving

$$\frac{\partial u}{\partial y} = \beta y - q_n \quad \text{giving} \quad u = \frac{1}{2}\beta y^2 - q_n y + \text{constant} \quad (12.25)$$

where $q_n = \beta L_h/2 + n\beta L_h$. The constant may be determined by requiring continuity of u across the regions, and we then obtain

$$u = \frac{1}{2}\beta(y - L_h/2)^2 + u_0, \quad 0 < y < L_h, \quad (12.26)$$

$$u = \frac{1}{2}\beta(y - 3L_h/2)^2 + u_0, \quad 0 < y < L_h, \quad (12.27)$$

or in general,

$$u = \frac{1}{2}\beta(y - (n - 1/2)L_h)^2 + u_0, \quad nL_h < y < (n + 1)L_h. \quad (12.28)$$

where u_0 is a constant. Relative to u_0 , the flow is weakly westward in the homogenized regions, whereas in the transition region the flow has a sharp eastward peak, as shown in Fig. 12.6. The spectrum of such a flow is k^{-4} , similar to the k^{-5} spectrum of zonostrophic flow.

Finally, let us make a few remarks. First, the potential vorticity gradient does not change sign, so the flow is (just) barotropically stable — one is supposing that the eddies arise by some external mechanism (e.g., some external stirring, or baroclinic instability) and they are creating the jet, not vice versa. Second, the staircase structure implies that potential vorticity is (obviously) well mixed in the regions of constant potential vorticity, but that there is no mixing across the jumps. The jumps are known as *mixing barriers* and, once formed, are found to be very persistent in numerical

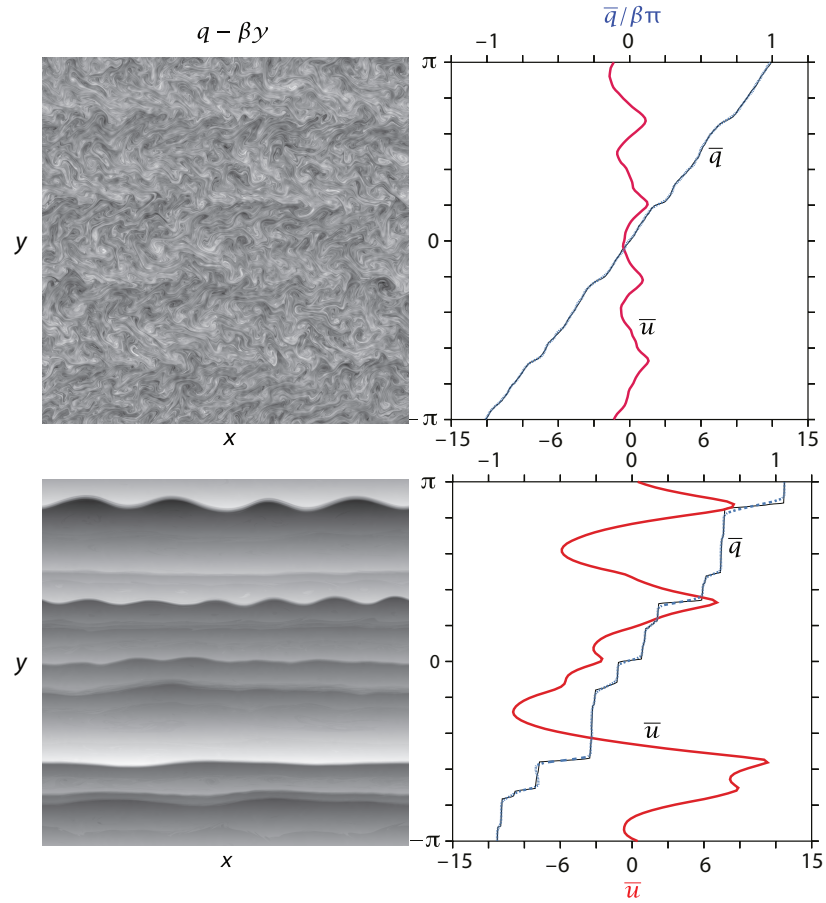


Fig. 12.7 Simulations of two-dimensional turbulence on a beta-plane.⁶

The left panels show snapshots of the relative vorticity ζ , and right panels show zonally-averaged potential vorticity and zonal wind. The top panels show an integration with $L_R/L_e \approx 3$ and the bottom panels have $L_R/L_e \approx 11$.

simulations. The edge of the stratospheric polar vortex, and the boundaries of ocean gyres, may also be examples. A fluid parcel is energetically incapable of jumping across the barrier, and the barrier's existence makes clear that parameterizing turbulent mixing as a downgradient flux of a conserved quantity cannot always work, at least with a constant diffusivity, since here the gradient is strong but the mixing is weak.

Numerical simulations do reproduce the staircase in some, but not all, circumstances (Fig. 12.7). The simulations suggest that there needs to be a good separation between the turbulent beta scale and the Rhines scales, in particular with $L_R/L_e = \gamma \gg 1$, and that this is best achieved with weak friction and/or forcing that is at large scales, perhaps even larger than L_e . However, the field continues to evolve and the reader should consult the primary literature to learn more.

12.1.4† Zonal jets — other mechanisms and final remarks

In the above we have described how zonal jets may be created from two distinct points of view — one in spectral space and the other in physical space. We described the spectral approach in terms of a cascade — a local energy transfer from one wavenumber to another. However, in many circumstances the energy transfer from eddies to zonal flow may be much more direct, with no spectral intermediaries. That is, a zonally asymmetric flow may be unstable (even if not to a normal-mode perturbation) and, even in the absence of eddy-eddy interactions, produce a zonal jet. The interaction is then deemed ‘quasi-linear’ and the triad interaction is highly nonlocal, as in the right-hand panel of Fig. 11.1.⁷ There are still other ways in which we may describe jet formation — see

section 15.1 — and we can only remark that the degree to which all the various descriptions are truly different explanations, or merely different explications, remains tenebrous.⁸

12.2 STRATIFIED GEOSTROPHIC TURBULENCE

12.2.1 An analogue to two-dimensional flow

Now let us consider stratified effects in a simple setting, using the quasi-geostrophic equations with constant Coriolis parameter and constant stratification.⁹ The (dimensional) unforced and inviscid governing equation may then be written as

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + Pr^2 \frac{\partial^2 \psi}{\partial z^2}, \quad (12.29a)$$

where $Pr = f_0/N$ is the *Prandtl ratio* (and Pr/H is the inverse of the deformation radius) and $D/Dt = \partial/\partial t + \mathbf{u} \cdot \nabla$ is the two-dimensional material derivative. The vertical boundary conditions are

$$\frac{D}{Dt} \left(\frac{\partial \psi}{\partial z} \right) = 0, \quad \text{at } z = 0, H. \quad (12.29b)$$

These equations are analogous to the equations of motion for purely two-dimensional flow. In particular, with periodic lateral boundary conditions, or conditions of no-normal flow, there are two quadratic invariants of the motion, the energy and the enstrophy, which are obtained by multiplying (12.29a) by $-\psi$ and q and integrating over the domain, as in chapter 5. The conserved energy is

$$\frac{d\hat{E}}{dt} = 0, \quad \hat{E} = \frac{1}{2} \int_V \left[(\nabla \psi)^2 + Pr^2 \left(\frac{\partial \psi}{\partial z} \right)^2 \right] dV, \quad (12.30)$$

where the integral is over a *three-dimensional* domain. The enstrophy is conserved at each vertical level, and of course the volume integral is also conserved, namely

$$\frac{d\hat{Z}}{dt} = 0, \quad \hat{Z} = \frac{1}{2} \int_V q^2 dV = \frac{1}{2} \int_V \left[\nabla^2 \psi + Pr^2 \left(\frac{\partial^2 \psi}{\partial z^2} \right) \right]^2 dV. \quad (12.31)$$

The analogy with two-dimensional flow is even more transparent if we further rescale the vertical coordinate by $1/Pr$, and so let $z' = z/Pr$. Then the energy and enstrophy invariants are:

$$\hat{E} = \int (\nabla_3 \psi)^2 dV, \quad \hat{Z} = \int q^2 dV = \int (\nabla_3^2 \psi)^2 dV, \quad (12.32)$$

where $\nabla_3 = \mathbf{i} \partial/\partial x + \mathbf{j} \partial/\partial y + \mathbf{k} \partial/\partial z'$. The invariants then have almost the same form as the two-dimensional invariants, but with a three-dimensional Laplacian operator instead of a two-dimensional one.

Given these invariants, we should expect that any dynamical behaviour that occurs in the two-dimensional equations *that depends solely on the energy/enstrophy constraints* should have an analogue in quasi-geostrophic flow. In particular, the transfer of energy to large-scales and enstrophy to small scales will also occur in quasi-geostrophic flow with, in so far as these transfers are effected by a local cascade, corresponding spectra of $k^{-5/3}$ and k^{-3} . However, in the quasi-geostrophic case, it is the *three-dimensional* wavenumber that is relevant, with the vertical component scaled by the Prandtl ratio. As a consequence, the energy cascade to larger horizontal scales is generally accompanied by a cascade to larger vertical scales — a *barotropization* of the flow, and this is an important and robust process in geostrophic turbulence. Still, the analogy between two-dimensional and quasi-geostrophic cascades should not be taken too far, because in the latter the potential vorticity is advected only by the horizontal flow. Thus, the dynamics of quasi-geostrophic turbulence will *not* in general be isotropic in three-dimensional wavenumber. To examine these dynamics more fully we first turn to a simpler model, that of two-layer flow.

12.2.2 Two-layer geostrophic turbulence

Let us consider flow in two layers of equal depth, governed by the quasi-geostrophic equations with (for now) $\beta = 0$, namely

$$\frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0, \quad i = 1, 2, \quad (12.33)$$

where

$$q_1 = \nabla^2 \psi_1 + \frac{1}{2} k_d^2 (\psi_2 - \psi_1), \quad q_2 = \nabla^2 \psi_2 + \frac{1}{2} k_d^2 (\psi_1 - \psi_2), \quad (12.34a)$$

$$J(a, b) = \frac{\partial a}{\partial x} \frac{\partial b}{\partial y} - \frac{\partial a}{\partial y} \frac{\partial b}{\partial x}, \quad \frac{1}{2} k_d^2 = \frac{2f_0^2}{g'H} \equiv \frac{4f_0^2}{N^2 H^2}. \quad (12.34b)$$

The wavenumber k_d is inversely proportional to the baroclinic radius of deformation, and the two equivalent expressions given are appropriate in a layered model and a level model, respectively. The equations conserve the total energy,

$$\frac{d\hat{E}}{dt} = 0, \quad \hat{E} = \frac{1}{2} \int_A \left[(\nabla \psi_1)^2 + (\nabla \psi_2)^2 + \frac{1}{2} k_d^2 (\psi_1 - \psi_2)^2 \right] dA, \quad (12.35)$$

and the enstrophy in each layer

$$\frac{d\hat{Z}_1}{dt} = 0, \quad \hat{Z}_1 = \int_A q_1^2 dA, \quad (12.36a)$$

$$\frac{d\hat{Z}_2}{dt} = 0, \quad \hat{Z}_2 = \int_A q_2^2 dA. \quad (12.36b)$$

The first two terms in the energy expression, (12.35), represent the kinetic energy, and the last term is the available potential energy, proportional to the variance of temperature.

Baroclinic and barotropic decomposition

Define the barotropic and baroclinic streamfunctions by

$$\psi \equiv \frac{1}{2} (\psi_1 + \psi_2), \quad \tau \equiv \frac{1}{2} (\psi_1 - \psi_2). \quad (12.37)$$

Then the potential vorticities for each layer may be written as

$$q_1 = \nabla^2 \psi + (\nabla^2 - k_d^2) \tau, \quad q_2 = \nabla^2 \psi - (\nabla^2 - k_d^2) \tau \quad (12.38a,b)$$

and the equations of motion may be rewritten as evolution equations for ψ and τ as follows:

$$\frac{\partial}{\partial t} \nabla^2 \psi + J(\psi, \nabla^2 \psi) + J(\tau, (\nabla^2 - k_d^2) \tau) = 0 \quad (12.39a)$$

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \tau + J(\tau, \nabla^2 \psi) + J(\psi, (\nabla^2 - k_d^2) \tau) = 0. \quad (12.39b)$$

We note the following about the nonlinear interactions in (12.39).

- (i) ψ and τ are like vertical modes. That is, ψ is the barotropic mode with a ‘vertical wavenumber’, k^z , of zero, and τ is a baroclinic mode with an effective vertical wavenumber of one.

- (ii) Just as purely two-dimensional turbulence can be considered to be a plethora of interacting triads, whose two-dimensional vector wavenumbers sum to zero, it is clear from (12.39b) that geostrophic turbulence may be considered to be similarly composed of a sum of interacting triads, although now there are two distinct kinds, namely

$$(i) \quad (\psi, \psi) \rightarrow \psi, \quad (ii) \quad (\tau, \tau) \rightarrow \psi \quad \text{or} \quad (\psi, \tau) \rightarrow \tau. \quad (12.40)$$

The first kind is a *barotropic triad*, for it involves only the barotropic mode. The other two are examples of a *baroclinic triad*. If a barotropic mode has a vertical wavenumber of zero, and a baroclinic mode has a vertical wavenumber of plus or minus one, then the three vertical wavenumbers of the triad interactions must sum to zero. There is no triad that involves only the baroclinic mode, as we may see from the form of (12.39). (If the layers are of unequal depths, then purely baroclinic triads do exist.)

- (iii) Wherever the Laplacian operator acts on τ , it is accompanied by $-k_d^2$. That is, it is *as if* the effective horizontal wavenumber (squared) of τ is shifted, so that $k^2 \rightarrow k^2 + k_d^2$.

Conservation properties

Multiplying (12.39a) by ψ and (12.39b) by τ and horizontally integrating over the domain of area A , assuming once again that this is either periodic or has solid walls, gives

$$\hat{T} = \int_A (\nabla \psi)^2 dA, \quad \frac{d\hat{T}}{dt} = \int_A \psi J(\tau, (\nabla^2 - k_d^2)\tau) dA \quad (12.41a)$$

$$\hat{C} = \int_A [(\nabla \tau)^2 + k_d^2 \tau^2] dA, \quad \frac{d\hat{C}}{dt} = \int_A \tau J(\psi, (\nabla^2 - k_d^2)\tau) dA. \quad (12.41b)$$

Here, $\hat{T}$ is the energy associated with the barotropic flow and $\hat{C}$ is the energy of the baroclinic flow. An integration by parts shows that

$$\int_A \psi J(\tau, (\nabla^2 - k_d^2)\tau) dA = - \int_A \tau J(\psi, (\nabla^2 - k_d^2)\tau) dA, \quad (12.42)$$

and therefore

$$\frac{d\hat{E}}{dt} = \frac{d}{dt}(\hat{T} + \hat{C}) = 0. \quad (12.43)$$

That is, total energy is conserved.

An enstrophy invariant is obtained by multiplying (12.39a) by $\nabla^2 \psi$ and (12.39b) by $(\nabla^2 - k_d^2)\tau$ and integrating over the domain and adding the two expressions. The result is

$$\frac{d\hat{Z}}{dt} = 0, \quad \hat{Z} = \int_A (\nabla^2 \psi)^2 + [(\nabla^2 - k_d^2)\tau]^2 dA. \quad (12.44)$$

This also follows from (12.36).

Just as for two-dimensional turbulence, we may define the spectra of the energy and enstrophy. Then, with obvious notation, for the energy we have

$$\frac{1}{A} \hat{T} = \int \mathcal{T}(k) dk \quad \text{and} \quad \frac{1}{A} \hat{C} = \int \mathcal{C}(k) dk, \quad (12.45)$$

The enstrophy spectrum $\mathcal{Z}(k)$ is related to the energy spectra by

$$\frac{1}{A} \hat{Z} = \int \mathcal{Z}(k) dk = \int [k^2 \mathcal{T}(k) + (k^2 + k_d^2) \mathcal{C}(k)] dk. \quad (12.46)$$

which is analogous to the relationship between energy and enstrophy in two-dimensional flow. We thus begin to suspect that the phenomenology of two-layer turbulence is closely related to, but richer than, that of two-dimensional turbulence.

12.2.3 Phenomenology of two-layer turbulence

In this section we explore the phenomenology of two-layer geostrophic turbulence, by examining how the equations of motion simplify if we suppose that the nonlinear interactions occur at particular scales, and by looking at the forms of the triad interactions.

Triad interactions and scales of interactions

Two types of triad interactions are possible.

Barotropic triads. An interaction that is purely barotropic (i.e., as if $\tau = 0$) conserves $\hat{T}$, the barotropic energy, and the associated enstrophy $\int k^2 \mathcal{T}(k) dk$, and a barotropic triad behaves as purely two-dimensional flow. Explicitly, the conserved quantities are

$$\text{energy:} \quad \frac{d}{dt} [\mathcal{T}(k) + \mathcal{T}(p) + \mathcal{T}(q)] = 0, \quad (12.47a)$$

$$\text{enstrophy:} \quad \frac{d}{dt} [k^2 \mathcal{T}(k) + p^2 \mathcal{T}(p) + q^2 \mathcal{T}(q)] = 0. \quad (12.47b)$$

Baroclinic triads. Baroclinic triads involve two baroclinic wavenumbers (say p, q) interacting with a barotropic wavenumber (say k). The energy and enstrophy conservation laws for this triad are

$$\text{energy:} \quad \frac{d}{dt} [\mathcal{T}(k) + \mathcal{C}(p) + \mathcal{C}(q)] = 0, \quad (12.48a)$$

$$\text{enstrophy:} \quad \frac{d}{dt} [k^2 \mathcal{T}(k) + (p^2 + k_d^2) \mathcal{C}(p) + (q^2 + k_d^2) \mathcal{C}(q)] = 0. \quad (12.48b)$$

To the extent that the triad interactions are local, they may be divided into three general types:

- (i) interactions only involving scales much smaller than the deformation radius;
- (ii) interactions only involving scales much larger than the deformation radius;
- (iii) interactions involving scales comparable to the deformation radius.

We consider each in turn.

(i) Interactions at small scales

For small scales the potential vorticity of each layer is given by

$$q_i = \nabla^2 \psi_i + \frac{1}{2} k_d^2 (\psi_j - \psi_i) \approx \nabla^2 \psi_i \quad (12.49)$$

where $i = 1, 2$ and $j = 3 - i$. Thus, each layer is decoupled from the other and obeys the equations of purely two-dimensional turbulence. Enstrophy is cascaded to small scales and, were there to be an energy source at small scales, energy would be transferred upscale until it reached a scale comparable to the deformation scale. As regards triad interactions, interactions at small scales correspond to the case $(p, q) \gg k_d$ in (12.48). If we neglect k_d^2 then we see that a baroclinic triad behaves like a barotropic triad, for (12.48) has the same form as (12.47). In reality, the small scales of a continuously stratified flow may not be representable by a two-layer model, because in a continuously stratified quasi-geostrophic model the enstrophy cascade occurs in *three-dimensional* wavenumber space. Thus, as the horizontal scales become smaller, so does the vertical scale and higher deformation radii [e.g., $n > 1$ in (6.88)] may play a role in reality.

(ii) *Interactions at large scales*

At scales much larger than the deformation radius we take $|\nabla^2| \sim k^2 \ll k_d^2$, and neglect terms in (12.39) that involve ∇^2 if they appear alongside terms involving k_d^2 . Noting that $J(\tau, k_d^2 \tau) = 0$, and assuming that $|J(\tau, \nabla^2 \psi)| \sim |J(\psi, \nabla^2 \tau)|$, we obtain

$$\frac{\partial}{\partial t} \nabla^2 \psi + J(\psi, \nabla^2 \psi) = -J(\tau, \nabla^2 \tau), \quad \frac{\partial \tau}{\partial t} + J(\psi, \tau) = 0. \quad (12.50a,b)$$

Thus, the baroclinic streamfunction obeys the equation of a passive tracer, although because of the term on the right-hand side of (12.50a) it is not truly passive. Nevertheless, and given the discussion of passive tracers in section 11.5, these equations suggest that the variance of the baroclinic streamfunction, and thus the energy in the baroclinic mode, will be transferred to smaller scales.

In terms of triad interactions, this case corresponds to $(p, q, k) \ll k_d$. The energy and enstrophy conservation triad interaction laws, (12.48), collapse to

$$\frac{d}{dt} [\mathcal{E}(p) + \mathcal{E}(q)] = 0. \quad (12.51)$$

That is to say, energy is conserved among the baroclinic modes alone, with the barotropic mode k mediating the interaction. Consistent with the analysis in physical space, there is no constraint preventing the transfer of baroclinic energy to smaller scales, and no production of barotropic energy at $k \ll k_d$.

(iii) *Interactions at scales comparable to the deformation radius*

In this case there is, in general, no simplification of the equations in physical space, or of the conservation laws governing triad interactions, and both baroclinic and barotropic modes are important. However, we can write the triad interaction conservation laws in an instructive form if we define the pseudowavenumber k' by $k'^2 \equiv k^2 + k_d^2$ for a baroclinic mode and $k'^2 \equiv k^2$ for a barotropic mode, and similarly for p' and q' . Then (12.47) and (12.48) can both be written as

$$\frac{d}{dt} [\mathcal{E}(k) + \mathcal{E}(p) + \mathcal{E}(q)] = 0, \quad (12.52a)$$

$$\frac{d}{dt} [k'^2 \mathcal{E}(k) + p'^2 \mathcal{E}(p) + q'^2 \mathcal{E}(q)] = 0, \quad (12.52b)$$

where $\mathcal{E}(k)$ is the energy (barotropic or baroclinic) of the particular mode. These are formally identical with the conservation laws for purely two-dimensional flow and so we expect energy to seek the gravest mode (i.e., smallest pseudowavenumber). Since the gravest mode has $k_d = 0$ this implies a tendency of energy to be transferred into the barotropic mode — a *barotropization* of the flow.

Baroclinic instability in the classic two-layer problem concerns the instability of a flow with vertical but no horizontal shear. This is like a triad interaction for which $p \ll (k, q, k_d)$, where p is the wavenumber of a baroclinic mode, and thus $k^2 \sim q^2$. The triad conservation laws then become

$$\begin{aligned} \frac{d}{dt} [\mathcal{T}(k) + \mathcal{C}(p) + \mathcal{C}(q)] &= 0, \\ \frac{d}{dt} [k^2 \mathcal{T}(k) + k_d^2 \mathcal{C}(p) + (q^2 + k_d^2) \mathcal{C}(q)] &= 0. \end{aligned} \quad (12.53)$$

From these two equations, and with $k^2 \approx q^2$, we derive

$$q^2 \dot{\mathcal{C}}(q) = (k_d^2 - k^2) \mathcal{T}(k). \quad (12.54)$$

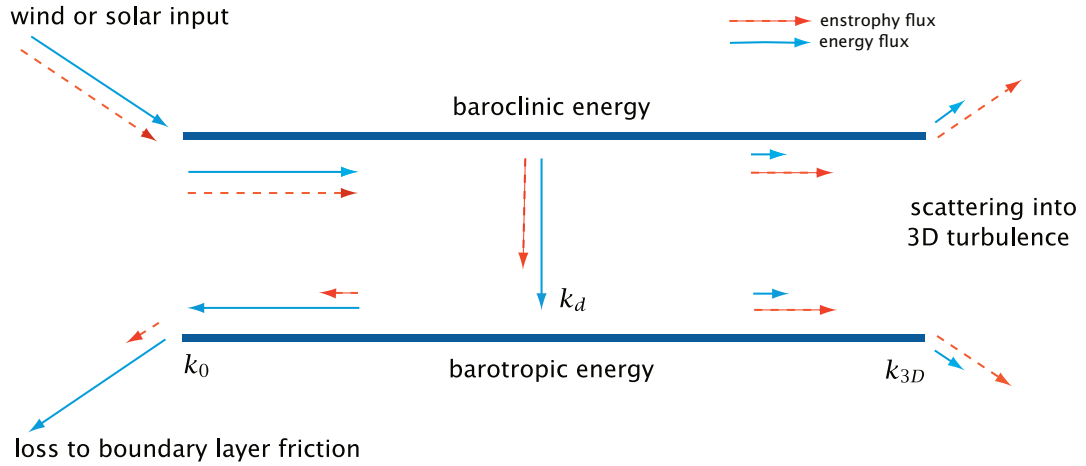


Fig. 12.8 Schema of idealized two-layer baroclinic turbulence.¹⁰ The horizontal axis represents horizontal wavenumber, and the vertical variation is decomposed into two vertical modes — the barotropic and first baroclinic. Energy transfer is shown by solid arrows and enstrophy transfer by dashed arrows.

Large-scale forcing maintains the available potential energy, and so provides energy to the baroclinic mode at very large scales. Energy is transferred to smaller baroclinic scales, and then into barotropic energy at horizontal scales comparable to and larger than the deformation radius (this is baroclinic instability), and then transferred to larger barotropic scales in an upscale cascade. The entire process of baroclinic instability and energy transfer may be thought of as a generalized inverse cascade in which the energy passes to smaller pseudowavenumber $k'^2 \equiv k^2 + k_d^2$.

Baroclinic instability requires that both $\hat{C}(q)$ and $\hat{T}(k)$ be positive. This can only occur if

$$k^2 < k_d^2. \quad (12.55)$$

Thus, there is a *high-wavenumber cut-off* for baroclinic instability, as we previously found in section 9.6 by direct calculation. We can see that the cut-off arises by virtue of energy and enstrophy conservation, and is not dependent on linearizing the equations and looking for exponentially growing normal-mode instabilities.

For small scales, we previously noted that each layer is decoupled from the other and that enstrophy, but not energy, may cascade to smaller scales. Now, baroclinic instability (of the large-scale flow) occurs at scales comparable to or larger than the deformation radius. Thus, the energy that is extracted from the mean flow is essentially trapped at scales larger than the deformation scale: it cannot be transferred to smaller scales in a cascade; rather, it is transferred to barotropic flow at scales comparable to the deformation radius, from which it cascades upwards to larger barotropic scales.

Summary of two-layer phenomenology

Putting together the considerations above leads to the following picture of geostrophic turbulence in a two-layer system (Fig. 12.8). At large horizontal scales we imagine some source of baroclinic energy, which in the atmosphere might be the differential heating between pole and equator, or in the ocean might be the wind and surface heat fluxes. Baroclinic instability effects a non-local transfer of energy to the deformation scale, where both baroclinic and barotropic modes are excited. From here there is an enstrophy cascade in each layer to smaller and smaller scales, until eventually the wavenumber is large enough (denoted by k_{3D} in Fig. 12.8) that non-geostrophic effects become important and enstrophy is scattered by three-dimensional effects, and dissipated. At scales larger than the deformation radius, there is an inverse barotropic cascade of energy to larger scales, where

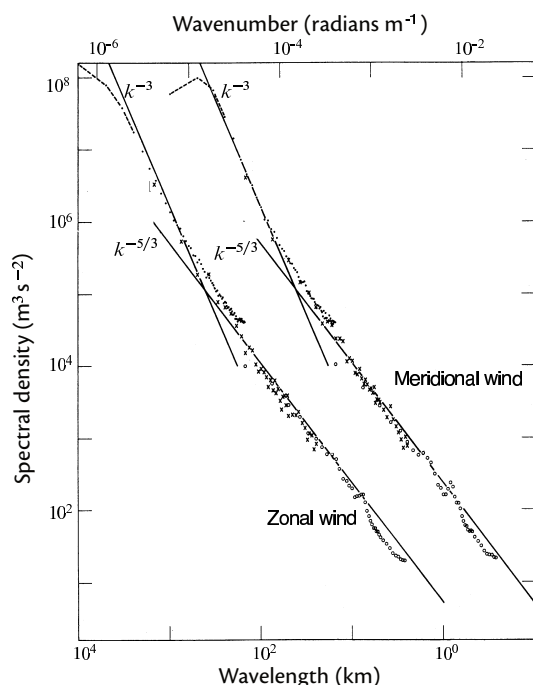


Fig. 12.9 Energy spectra of the zonal and meridional wind near the tropopause, from thousands of commercial aircraft measurements between 1975 and 1979. The meridional spectrum is shifted one decade to the right.

The -3 spectrum may well be associated with a forward enstrophy cascade, but the origin of the $-5/3$ spectrum at smaller scales is not definitively known.¹²

it is modified by the β -effect and halted by the effects of friction, with the latter being responsible for the dissipation of the large-scale barotropic energy.

There are two aspects of this phenomenology that prevent it from being a complete theory of baroclinic eddies in the atmosphere or ocean.

- (i) Even as a model of two-layer quasi-geostrophic turbulence the model is not ideal, because our assumptions are questionable. For example, we have largely neglected the effects of friction and we have assumed the baroclinic mode to be passive at large scales. Numerical simulations tend to support the model qualitatively, but they also show a significant transfer of energy from the baroclinic mode to the barotropic mode at scales larger than the deformation radius.
- (ii) The ideas do not perfectly apply to either the atmosphere or ocean. In the latter, the turbulence is quite inhomogeneous except perhaps in the Antarctic Circumpolar Current. In the atmosphere, observations indicate that the deformation radius is almost as large as the Rhines scale, and is only a factor of a few smaller than the pole-equator scale, leaving little room for isotropic inverse cascade to fully develop. On the other hand, the atmosphere does display k^{-3} spectra at scales similar to and somewhat smaller than the deformation radius (Fig. 12.9), and analysis of this indicates that it may indeed be associated with a forward cascade of enstrophy.¹¹

12.3 † A SCALING THEORY FOR GEOSTROPHIC TURBULENCE

Let us now build on the phenomenological model of the previous section and construct a quantitative theory of two-layer forced-dissipative geostrophic turbulence.¹³ Although, as we have noted, the underlying model is imperfect, some of its qualitative properties transcend its limitations and are quite revealing about the real system. We will consider a system in which the basic state is a purely zonal flow, with constant vertical shear and no horizontal variation; our goal is to predict

the amplitude and the scale of the eddies that result from the baroclinic instability of this flow. The assumptions that the mean flow and the stratification are constants are quite severe and thus the model, even if correct within its own terms of reference, is certainly not a complete theory of the baroclinic eddies in mid-latitude flow.

12.3.1 Preliminaries

We shall suppose that, as in Fig. 12.8, baroclinic instability at large scales leads to a transfer of energy to the barotropic mode at a scale comparable to the deformation radius, followed by an inverse cascade of energy within the barotropic mode, and the energy is finally dissipated in the Ekman layer. At scales smaller than the deformation radius the layers are largely uncoupled, and in each layer there is an enstrophy cascade to small scales. The equations of motion describing all of this are (12.39), but we will explicitly recognize the effect of a shear flow by replacing τ by $\tau - Uy$, where U is constant, this being equivalent to supposing there is a constant shear in the flow. The equations of motion, (12.39), become

$$\frac{\partial}{\partial t} \nabla^2 \psi + J(\psi, \nabla^2 \psi) + J(\tau, (\nabla^2 - k_d^2) \tau) + U \frac{\partial}{\partial x} \nabla^2 \tau = D[\psi], \quad (12.56a)$$

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2) \tau + J(\tau, \nabla^2 \psi) + J(\psi, (\nabla^2 - k_d^2) \tau) + U \frac{\partial}{\partial x} (\nabla^2 \psi + k_d^2 \psi) = D[\tau]. \quad (12.56b)$$

These are similar to the equations used for studying two-layer baroclinic instability in chapter 9, but we now retain the nonlinear terms and include dissipation, represented by $D[\cdot]$.

For scales much larger than the deformation radius (12.56) become, analogously to (12.50),

$$\frac{\partial}{\partial t} \nabla^2 \psi + J(\psi, \nabla^2 \psi) = -J(\tau, \nabla^2 \tau) - U \frac{\partial}{\partial x} \nabla^2 \tau + D[\psi], \quad (12.57)$$

$$\frac{\partial \tau}{\partial t} + J(\psi, \tau) = U \frac{\partial \psi}{\partial x} - k_d^{-2} D[\tau]. \quad (12.58)$$

The equation for ψ is just the barotropic vorticity equation, ‘forced’ via its interaction with the baroclinic mode, namely the terms on the right-hand side of (12.57). The equation for the baroclinic streamfunction is the same as the equation for a passive scalar, except for the forcing term $U \partial \psi / \partial x$.

12.3.2 Scaling properties

In the barotropic equation, we may argue that in the energy-containing scales, $k^2 \ll k_d^2$, the magnitude of the barotropic streamfunction is much larger than that of the baroclinic streamfunction; that is, $|\psi| \gg |\tau|$, as follows. We may reasonably suppose that the forcing of the barotropic vorticity equation occurs at wavenumbers close to k_d , as in baroclinic instability. At larger scales the barotropic streamfunction obeys the two-dimensional vorticity equation, and we may expect an energy cascade to large scales with energy spectrum given by

$$\mathcal{E}_\psi(k) = \mathcal{K}_1 \varepsilon^{2/3} k^{-5/3}, \quad (12.59)$$

where $\mathcal{K}_1$ is a constant and ε is the as yet undetermined energy flux through the system. We may suppose that this cascade holds for wavenumbers $k_0 < k \ll k_d$, where the wavenumber k_0 is the halting scale of the inverse cascade, determined by one or more of: friction, the β -effect, and the domain size. Now, in this same wavenumber regime the baroclinic streamfunction is being advected as a passive tracer — it is being stirred by ψ . Thus, any baroclinic energy that is put in at large scales by the interaction with the mean flow (via the term proportional to $U \psi_x$) will be

cascaded to smaller scales. Thus, we expect the baroclinic energy spectrum to be that of a passive tracer whose variance is cascading to smaller scales in a forward cascade, with a spectrum given by [cf. (11.94)]

$$\mathcal{E}_\tau(k) = \mathcal{K}_2 \varepsilon_\tau \varepsilon^{-1/3} k^{-5/3}, \quad (12.60)$$

where $\mathcal{K}_2$ is a constant, ε_τ is the transfer rate of baroclinic energy and ε is the same quantity appearing in (12.59). Now, because energy is not lost to small scales we have $\varepsilon_\tau = \varepsilon$, both being equal to the energy flux in the system. Thus, using (12.59) and (12.60), the energy in the barotropic and baroclinic modes are comparable at sufficiently large scales. Since the energy density in the former is $(\nabla\psi)^2$ and in the latter $(\nabla\tau)^2 + k_d^2 \tau^2 \sim k_d^2 \tau^2$, the magnitude of ψ must then be much larger than that of τ , and specifically

$$|\psi| \sim \frac{k_d |\tau|}{k_0} \gg |\tau|. \quad (12.61)$$

Let us write the baroclinic equation (12.58) in the form

$$\frac{\partial \tau}{\partial t} + J(\psi, \tau - Uy) = 0, \quad (12.62)$$

which is the equation for a passive tracer (τ) in a mean gradient (U), stirred by the flow (ψ), and we omit dissipation. Because there is a large scale separation (in fact, an infinite one) between the scale of the mean gradient of τ (i.e., the scale of variations of U) and the scale of its fluctuations, we can write

$$\tau \sim l' \frac{\partial \bar{\tau}}{\partial y} = -l' U, \quad (12.63)$$

where l' is the scale of the fluctuation. Thus, at the scale k_0^{-1} , the magnitude of τ , and the associated baroclinic velocity v_τ , are given by

$$\tau \sim \frac{U}{k_0}, \quad v_\tau \sim U. \quad (12.64)$$

At this scale, and using (12.61), the magnitude of the barotropic streamfunction and its associated velocity are given by

$$\psi \sim \frac{k_d U}{k_0^2}, \quad v_\psi \sim \frac{k_d U}{k_0}. \quad (12.65a,b)$$

This is an important result, regardless of the other details in the calculation. In particular, (12.65b) implies that the barotropic velocity, v_ψ , scales like the mean velocity U multiplied by the ratio of the length scale of the eddies to the deformation radius. In the Earth's atmosphere, this ratio is an $\mathcal{O}(1)$ number; in the ocean it is somewhat larger.

How much energy flows through the system? The mean shear is the ultimate source of energy here, and for simplicity this shear is kept constant in time, analogous to an infinite heat bath supplying energy to a smaller system without its own temperature changing. The conversion of energy from the mean shear to the eddy flow is given by multiplying (12.58) by $k_d^2 \tau$ and integrating over the domain. This gives an expression for the rate of increase of the available potential energy of the system, namely

$$\frac{d}{dt} APE = \frac{d}{dt} \int_A k_d^2 \tau^2 dA = \int_A 2U k_d^2 \tau \frac{\partial \psi}{\partial x} dA. \quad (12.66)$$

(Note that the energy input to the system equals the poleward heat flux.) From this we estimate the average energy flux as

$$\varepsilon = 2U k_d^2 \overline{\psi_x \tau} \sim \frac{U^3 k_d^3}{k_0^2}. \quad (12.67)$$

The correlation between ψ_x and τ cannot be determined by this argument. This aside, we have produced a physically based closure for the flux of energy through the system in terms only of the mean shear, the halting scale k_0 (discussed below) and the deformation scale k_d .

Finally, we calculate the eddy diffusivity defined by

$$\kappa \equiv -\frac{\overline{v_\psi \tau}}{\partial_y \overline{\tau}}. \quad (12.68)$$

(Eddy diffusion is considered at greater length in chapter 13.) Using (12.64) and (12.65) gives

$$\kappa \sim \frac{k_d U}{k_0^2} \quad (12.69)$$

which, if the mixing velocity is the barotropic stirring velocity, implies a mixing length of k_0^{-1} . (Note also that the eddy diffusivity is just the magnitude of the barotropic streamfunction at the energy-containing scales.)

12.3.3 The halting scale and the β effect

Let us suppose that, as discussed in section 12.1, the β effect provides a barrier for the inverse cascade at the scale (12.11), so that $k_0 = k_\varepsilon \sim (\beta^3/\varepsilon)^{1/5}$. Using (12.67) we find that the stopping scale is given by

$$k_0 = k_\varepsilon \sim \frac{\beta}{U k_d}, \quad (12.70)$$

and using (12.69) and (12.70) we obtain for the energy flux and the eddy diffusivity,

$$\varepsilon \sim \frac{U^5 k_d^5}{\beta^2}, \quad \kappa \sim \frac{U^3 k_d^3}{\beta^2}. \quad (12.71)$$

The magnitudes of the eddies themselves are easily given using (12.65) and (12.64), whence

$$\tau \sim \frac{U^2 k_d}{\beta}, \quad v_\tau \sim U, \quad \psi \sim \frac{U^3 k_d^3}{\beta^2}, \quad v_\psi \sim \frac{U^2 k_d^2}{\beta}. \quad (12.72)$$

Evidently, in this model (in which the mean shear and deformation radius are fixed), the eddies become more energetic with decreasing β , and the eddy amplitudes and poleward heat flux increase very rapidly with the mean shear, more so than in a model in which the energy-containing scale is fixed. This is because as β decreases, the inverse cascade can extend to larger scales, thereby increasing the overall energy of the flow. Similarly, as U increases not only does the eddy amplitude increase as a direct consequence [as in (12.64) and (12.65)] but also k_ε falls [see (12.70)], and these effects combine to give a rapid increase of the eddy magnitudes with U .

Frictional effects

Whether β is present or not, friction is necessary to ultimately remove the energy flowing through the system, as well as to remove enstrophy at small scales. Friction provides another mechanism for halting the inverse cascade, and the simplest case is that of a linear drag representing Ekman friction, in which case we write

$$D[\psi] = -r \nabla^2 \psi. \quad (12.73)$$

and the stopping wavenumber for a given ε is given, as in (11.70), by $k_r = (r^3/\varepsilon)^{1/2}$. However, the use of this in (12.67) fails to give a result for ε (problem 12.??). From a physical perspective, a linear drag that is weak enough to allow an inverse cascade to form is, *ipso facto*, too weak to equilibrate

the flow. A friction that becomes larger at larger scales, for example an ‘inverse Laplacian’, has no such problems. More physically, a nonlinear drag, proportional to the square of the amplitude of the flow, also leads to a well-posed problem with the cascade halting at a well-defined scale. Finally, we should point out that in neither the atmosphere nor ocean is there an extended inverse cascade, because the deformation scale and the beta scale are not asymptotically well separated (although the β -effect does not prevent an inverse cascade to large zonal scales).

12.4† PHENOMENOLOGY OF BAROCLINIC EDDIES IN THE ATMOSPHERE AND OCEAN

In the remaining sections of this chapter we take a more informal approach, illustrated by numerical experiments, to the problem of baroclinic eddies in the atmosphere and ocean. We draw from our treatment of geostrophic turbulence but by being a little less formal we are able to travel farther, for we spend less time looking at the map (but with a concomitant danger that we lose our way).

12.4.1 The magnitude and scale of baroclinic eddies

How big, in both amplitude and scale, do baroclinic eddies become? Suppose that the time-mean flow is given, and that it is baroclinically unstable. Eddies will grow, initially according to the linear theory of chapter 9, but they cannot and do not continue to amplify: they ultimately equilibrate, and this by way of nonlinear mechanisms. The eddies will extract energy from the mean flow, but at the same time the available energy of the mean flow is being replenished by external forcing (i.e., the maintenance of an equator–pole temperature gradient by radiative forcing in the atmosphere, and wind and buoyancy forcing at the surface in the ocean). Thus, we cannot a priori determine the amplitude of baroclinic eddies by simply assuming that all of the available potential energy in the mean flow is converted to eddying motion. To close the problem we find we need to make three, not necessarily independent, assumptions:

- (i) an assumption about the magnitude of the baroclinic eddies;
- (ii) an assumption relating eddy kinetic energy to eddy available potential energy;
- (iii) an assumption about the horizontal scale of the eddies.

Baroclinic eddies extract available potential energy (APE) from the mean flow, and it is reasonable to suppose that an eddy of horizontal scale L_e can extract, as an upper bound, the APE of the mean flow contained within that scale. The APE is proportional to the variation of the buoyancy field so that

$$(\Delta b')^2 \sim |\Delta \bar{b}|^2 \sim L_e^2 |\nabla \bar{b}|^2, \quad (12.74)$$

where Δb is the variation in the buoyancy over the horizontal scale L_e . (For simplicity we stay with the Boussinesq equations, and $b = -g\delta\rho/\rho_0$. However, we could extend the arguments to an ideal-gas atmosphere with $b = g\delta\theta/\theta_0$.) Equivalently, we might simply write

$$b' \sim L_e |\nabla \bar{b}|, \quad (12.75)$$

which arises from a mixing-length approach. Supposing that the temperature gradient is mainly in the y -direction then, using thermal wind, we have

$$b' \sim L_e f \frac{\partial \bar{u}}{\partial z} \quad \text{and} \quad v'_\tau \sim \bar{u}, \quad (12.76a,b)$$

where v'_τ is an estimate of the shear (multiplied by the depth scale) of the eddying flow. [These estimates are the same as (12.64), with $\bar{u}$ replacing U .]

Our second assumption is to relate the barotropic eddy kinetic energy to the eddy available potential energy, and the most straightforward one to make is that there is a rough equipartition between the two. This assumption is reasonable because in the baroclinic lifecycle (or baroclinic inverse cascade) energy is continuously transferred from eddy available potential energy to eddy kinetic energy, and the assumption is then equivalent to supposing that the relevant eddy magnitude is always proportional to this rate of transfer. Thus we assume $v'_\psi \sim (b'/N)^2$ or

$$v'_\psi \sim \frac{b'}{N}. \quad (12.77)$$

Finally, the scale of the eddies is determined by the extent to which the eddies might grow through nonlinear interactions. As we discussed earlier, possibilities for this scale include the deformation radius itself (if the inverse cascade is weak) or the Rhines scale (if the inverse cascade is slowed by the β -effect), or even the domain scale if neither of these apply.

Some consequences

The simple manipulations above have some very interesting consequences. Using (12.76) and (12.77) we find

$$v'_\psi \sim \frac{fL_e}{NH} \bar{u} = \frac{L_e}{L_d} \bar{u} \quad (12.78)$$

where $L_d = NH/f_0$ is the deformation radius and $\bar{u}$ is the amplitude of the mean baroclinic velocity, that is the mean shear multiplied by the height scale. This important relationship relates the magnitude of the eddy kinetic energy to that of the mean. In the atmosphere the scale of the motion is not much larger than the deformation radius (which is about 1000 km) and the eddy and mean kinetic energies are, consistently, comparable to each other. In the ocean the deformation radius (about 50 km over large areas) is significantly smaller than the scale of mesoscale eddies (which is more like 200 km or more), and observations consistently reveal that the eddy kinetic energy is an order of magnitude larger than the mean kinetic energy.¹⁴

One other important and slightly counter-intuitive result concerns the timescale of eddies. From (12.78) we have

$$T_e \sim \frac{L_e}{v'_\psi} \sim \frac{L_d}{\bar{u}} \equiv T_E, \quad (12.79)$$

where T_E is the Eady time scale. That is, the eddy time scale (at the scale of the largest eddies) is independent of the process that ultimately determines the spatial scale of those eddies; if the eddy length scale increases somehow, perhaps because friction or β are decreased, the velocity scale increases in proportion.

12.4.2 Baroclinic eddies and their lifecycle in the atmosphere

Amplitude and Scale

We saw in section 9.9.2 that baroclinic instability in the atmosphere occurs predominantly in the troposphere, i.e., in the lowest 10 km or so of the atmosphere, with the higher stratification of the eponymous stratosphere inhibiting instability. In the mid-latitude troposphere the vertical shear and the stratification are also relatively uniform which is why fairly simple models, such as the two-layer model or the Eady model (with the addition of the β -effect) are reasonable first-order models.

The mean pole–equator temperature gradient is about 40 K and the deformation radius NH/f is about 1000 km. The Rhines scale, $\sqrt{\bar{U}/\beta}$ is a little larger than the deformation radius, being perhaps 2000 km, and is similar to the width of the main mid-latitude baroclinic zone which lies

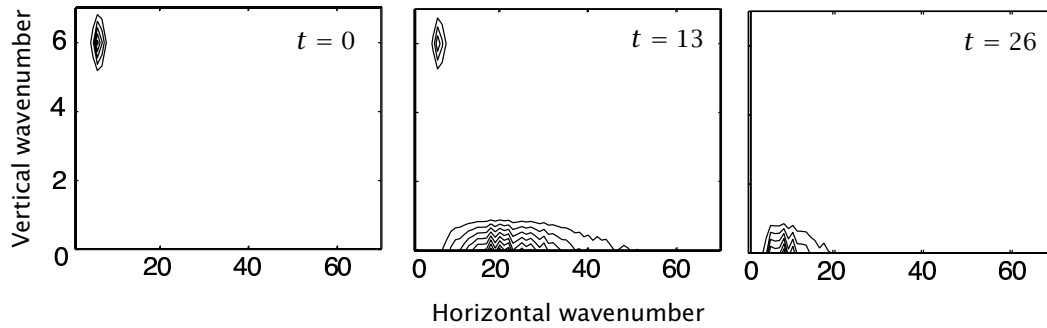


Fig. 12.10 A numerical simulation of an idealized baroclinic lifecycle, showing contours of energy in spectral space at successive times. Initially, there is baroclinic energy at low horizontal wavenumber, as in a large-scale shear. Baroclinic instability transfers this energy to barotropic flow at the scale of the deformation radius, and this is followed by a barotropic inverse cascade to large scales. Most of the transfer to the barotropic mode in fact occurs quite quickly, between times 11 and 14, but the ensuing barotropic inverse cascade is slower. The entire process may be thought of as a generalized inverse cascade. The stratification (N^2) is uniform, and the first deformation radius is at about wavenumber 15. Times are in units of the eddy turnover time.¹⁶

between about 40° and 65° , in either hemisphere. Given these, and especially given that the maximum wavelength for instability occurs at scales somewhat larger than the deformation radius, there is little prospect of an extended upscale cascade, and for this reason the Earth's atmosphere has comparable eddy kinetic and mean kinetic energies.¹⁵

The baroclinic lifecycle

The baroclinic lifecycle of geostrophic turbulence, sketched schematically in Fig. 12.8, can be nicely illustrated by way of numerical initial value problems, and we describe two such. The first is extremely idealized: take a doubly-periodic quasi-geostrophic model on the f -plane, initialize it with baroclinic energy at large horizontal scales, and then let the flow freely evolve. Figure 12.10 shows the results. The flow, initially concentrated in high vertical wavenumbers to best illustrate the energy transfer, is baroclinically unstable, and energy is transferred to barotropic flow at wavenumbers close to the first radius of deformation, here at about wavenumber 15. Energy then slowly cascades back to large scales in a predominantly barotropic inverse cascade, piling up at the largest scales much as in decaying, two-dimensional turbulence. Nearly all of the initial baroclinic energy is converted to barotropic, eddy kinetic energy and, even without any surface friction, the flow evolves to a baroclinically stable state. Couched in these terms, it is easy to see that the baroclinic lifecycle is a form of baroclinic inverse cascade, with an energy transfer to large total wavenumber, K_{tot} , that is made up of contributions from both horizontal and vertical wavenumbers:

$$K_{tot}^2 = K_h^2 + k_d^2 m^2, \quad (12.80)$$

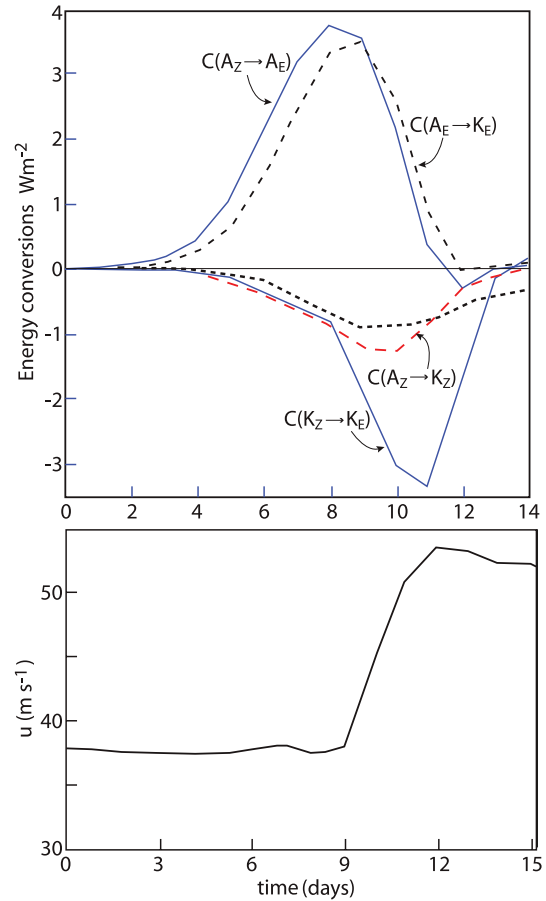
where m is the vertical wavenumber and K_h is the horizontal wavenumber. As we noted earlier, the twin constraints of energy and enstrophy conservation prevent the excitation of horizontal scales with very large horizontal wavenumbers, and so the lifecycle proceeds through wavenumbers at the deformation scale.

The results of the second, and more realistic, initial value problem are illustrated in Fig. 12.11.

Here, the atmospheric primitive equations on a sphere are integrated forward, beginning from a baroclinically unstable zonal flow, plus a small-amplitude disturbance at zonal wavenumber 6. The disturbance grows rapidly through baroclinic instability, accompanied by a conversion of energy

Fig. 12.11 Top: energy conversion and dissipation processes in a numerical simulation of an idealized atmospheric baroclinic lifecycle, simulated with a GCM. Bottom: evolution of the maximum zonal-mean velocity. Top: energy conversions. A_Z and A_E are zonal and eddy available potential energies (APEs), and K_Z and K_E are the corresponding kinetic energies, and $C(\cdot \rightarrow \cdot)$ represents the corresponding conversion.

Initially baroclinic processes dominate, with conversions from zonal to eddy APE, $C(A_Z \rightarrow A_E)$, and then eddy APE to eddy kinetic energy, $C(A_E \rightarrow K_E)$, followed by the barotropic conversion of eddy kinetic to zonal kinetic energy [a negative value of $C(K_Z \rightarrow K_E)$]. The latter process is reflected in the *increase* of the maximum zonal-mean velocity at about day 10, shown in the lower panel.¹⁷



initially from the zonal mean potential energy to eddy available potential energy (EAPE), and then from EAPE to eddy kinetic energy (EKE), and finally from EKE to zonal kinetic energy (ZKE). The last stage of this roughly corresponds to the barotropic inverse cascade of quasi-geostrophic theory, and because of the presence of a β -effect the flow becomes organized into a zonal jet. The parameters in the Earth's atmosphere are such that there is only one such jet, and in the lower panel of Fig. 12.11 we see its amplitude increase quickly from days 10 to 12, associated with the conversion of EKE to ZKE. (In other lifecycle experiments, the end state is found to be more akin to a barotropic vortex or, in meteorological parlance, a 'cut-off cyclone'.¹⁸) A commonality between all the lifecycles is the evolution toward a generally barotropic flow, with variable degrees of zonality in the final state, depending on the importance of the β -effect.

Of course, the real atmosphere is never in the zonally uniform state that is used in idealized baroclinic instability or lifecycle studies. Rather, at any given time, finite-amplitude eddies exist and these provide a finite amplitude perturbation to the baroclinically unstable zonal flow. For this reason we rarely, if ever, see an exponentially growing normal mode. Furthermore, given any instantaneous atmospheric state, zonally symmetric or otherwise, the fastest growing (linear) instability is not necessarily exponential but may be 'non-modal', with a secular or linear growth that, over some finite time period and in some given norm, is much more rapid than exponential.¹⁹

12.4.3 Baroclinic eddies and their lifecycle in the ocean

Basic ideas

Baroclinic instability was first developed as a theory for mid-latitude synoptic-scale instabilities in the atmosphere and the original problems were accordingly set in a zonally re-entrant channel. The ocean, apart from the Antarctic Circumpolar Current (ACC), is not zonally re-entrant. However, the ocean is driven by buoyancy and wind-forcing at the surface, and these combine to produce a region of enhanced stratification and associated shear in the ocean in the upper 500–1000 m or so — that is, in the ‘thermocline’ — as discussed more fully in chapter 20. The sloping isopycnals indicate that there is a pool of available potential energy that might be converted to kinetic energy, and so the ocean is potentially baroclinically unstable. Satellite observations indicate that baroclinic eddies are in fact almost ubiquitous in the mid- and high-latitude oceans, two particularly eddy-rich regions being the areas in and surrounding intense western boundary currents, such as the Gulf Stream, and the ACC. The ocean is, literally, a sea of eddies.²⁰

In addition to the geometry, the main differences between the oceanic and atmospheric problems are twofold.

- (i) In the ocean, the shear and the stratification are not uniform between two rigid lids, nor even uniform between one rigid lid and a structure like the tropopause. Instead, both stratification and shear are largest in the upper ocean, decaying into a quiescent and nearly unstratified abyss.
- (ii) In the ocean, the first radius of deformation is much smaller than the scale of the large-scale flow; that is, of the gyres or the large-scale overturning circulation. On dimensional grounds we have, using a height scale and stratification representative of the upper ocean, $L_d \sim NH/f \sim 10^{-2} \times 10^3/10^{-4} = 100$ km.

A consequence of the enhanced shear in the upper ocean is that the amplitude of the growing waves is also largely concentrated in the upper ocean, as we saw in Fig. 9.22. Regarding the stratification, in quasi-geostrophic theory we may, as in section 6.5.2, define the deformation radii by solution of the eigenvalue problem

$$\frac{\partial}{\partial z} \frac{f^2}{N^2} \frac{\partial \phi_n}{\partial z} + \Gamma_n \phi_n = 0, \quad \text{with} \quad \frac{\partial \phi_n}{\partial z} = 0 \quad \text{at } z = 0, H. \quad (12.81a,b)$$

The successive eigenvalues, Γ_n , are related to the successive deformation radii, L_n , by $L_n^2 = 1/\Gamma_n$. If the stratification (N^2) is uniform the resulting eigenfunctions are cosines with corresponding eigenvalues and deformation radii given by

$$\Gamma_n = n^2 \frac{f^2 \pi^2}{N^2 H^2}, \quad L_n = \frac{1}{\sqrt{\Gamma_n}} = \frac{NH}{n\pi f_0}. \quad (12.82a,b)$$

If the stratification is non-uniform, we must in most cases solve the eigenproblem numerically (or by WKB methods, see section 3.4.2), and the results of one such calculation are given in Fig. 12.12. The case with uniform stratification reproduces cosine modes, whereas in the more realistic case the modes tend to have highest amplitude in the upper ocean, where the stratification is strongest — a result that is typical of oceanic profiles.²¹ The results of a calculation of the first deformation radius using observed oceanic profiles are given in Fig. 12.13, and values of 50–100 km emerge in mid-latitudes.²³ We may therefore expect oceanic baroclinic instability to occur on a scale much smaller than that in the atmosphere, and much smaller than the scale of an ocean basin. (We should, however, note that the scale of baroclinic instability will typically be larger than the first deformation radius, L_1 , shown in Fig. 12.13, because of two compounding effects. First, in uniform stratification the first deformation radius, L_1 , as given by (12.82b) is a factor of π smaller than the simple definition NH/f . Second, in simple baroclinic instability problems like the Eady problem the wavelength of maximum instability is a few times NH/f . Thus, the wavelength of maximum instability will be an order of magnitude larger than L_1 .)

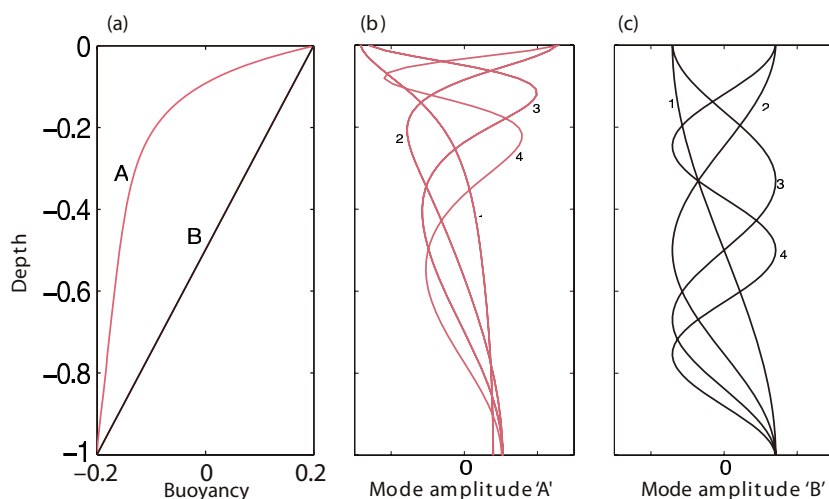


Fig. 12.12 (a) Two buoyancy profiles, $-g\delta\rho/\rho_0$, one being a fairly realistic oceanic case with enhanced stratification in the upper ocean (profile A), and the other with uniform stratification (profile B). (b) and (c) The first four baroclinic modes [eigenfunctions of (12.81)]. With profile B the eigenmodes are cosines, whereas in profile A they have a larger amplitude and shorter local wavelength in the upper ocean. The number of zero crossings is equal to the mode number.

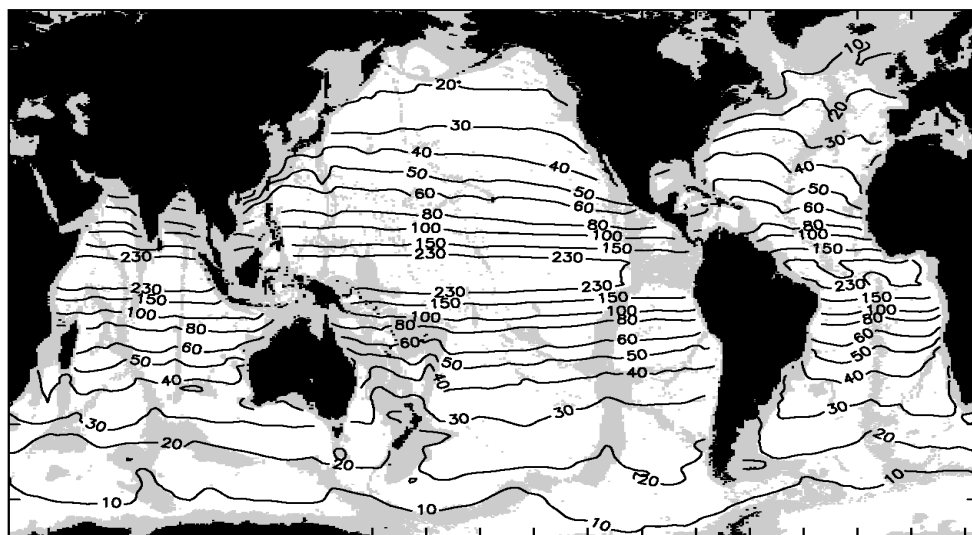


Fig. 12.13 The oceanic first deformation radius L_d in kilometres, calculated using the observed stratification and an equation similar to (12.81). Near equatorial regions are excluded, and regions of ocean shallower than 3500 m are shaded. Variations in Coriolis parameter are responsible for much of large-scale variability, although weak stratification also reduces the deformation radius at high latitudes.²²

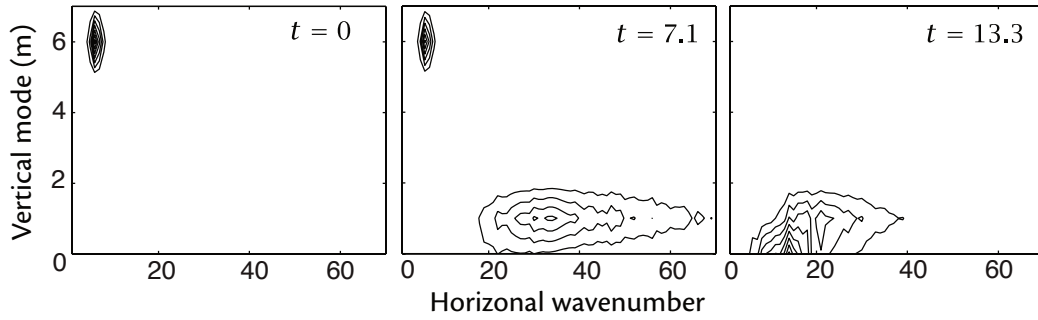


Fig. 12.14 Idealized baroclinic lifecycle, similar to that in Fig. 12.10, but with enhanced stratification of the basic state in the upper domain, representing the oceanic thermocline.

Eddy amplitudes and scales

The consequences of this small deformation radius on the lifecycle and finite-amplitude equilibration of oceanic baroclinic eddies are far-reaching, one being is that there is more scope for an inverse cascade than in the atmosphere, and indeed observations indicate that the horizontal scale of the eddies is typically a few to several times larger than the local deformation radius itself. The situation is not clear cut, however, as some observations²⁴ indicate that the eddy size nevertheless scales with the local deformation radius, suggesting that the eddy scale may be set by the instability scale and not an inverse cascade.

In any case, suppose that an ocean eddy has a horizontal scale 200 km, and that it sits in the subtropical gyre where the mean temperature gradient is 10^{-5} K m^{-1} , that the mean shear and ensuing baroclinic activity is mainly confined to the upper 1000 m of the ocean, and that the deformation radius is 50 km. The temperature gradient corresponds to a temperature difference of about 20 K across 2000 km, a horizontal buoyancy gradient of about $2 \times 10^{-9} \text{ s}^{-2}$ [using the simple equation of state $\rho = \rho(1 - \beta_T \Delta T)$ where $\beta_T = 2 \times 10^{-4} \text{ K}^{-1}$] and a shear of about 2 cm s^{-1} over the upper 1 km of ocean. Then, using (12.78), we can estimate a typical eddy velocity scale as

$$v'_\psi \sim \frac{L_e}{L_d} \bar{u} \approx 4\bar{u} \approx 8 \text{ cm s}^{-1}, \quad (12.83)$$

implying, as we noted earlier, an EKE that is an order of magnitude larger than the mean kinetic energy. Associated with this are typical temperature perturbations whose magnitude we can estimate using (12.75) or (12.76) as being about 2 K. These estimates are comparable to those observed in mid-ocean, with more energetic eddies forming near intense western boundary currents where gradients are large and barotropic instability also provides a source of energy for the eddies. There is least a factor of a few uncertainty, but it is noteworthy that they are roughly comparable to the values observed.

Eddy lifecycles

The lifecycle of a mid-oceanic baroclinic eddy will differ from its atmospheric counterpart in two main respects:

- (i) baroclinic eddies may be advected by the mean flow into regions with quite different properties from where they initially formed;
- (ii) the non-uniformity of the stratification affects the passage to barotropic flow.

Both of these can best be studied by numerical means. Regarding the first, eddies will often form in or near intense western boundary currents, but then will be advected by that current into the potentially less unstable open ocean before completing their lifecycle. Regarding the second, an

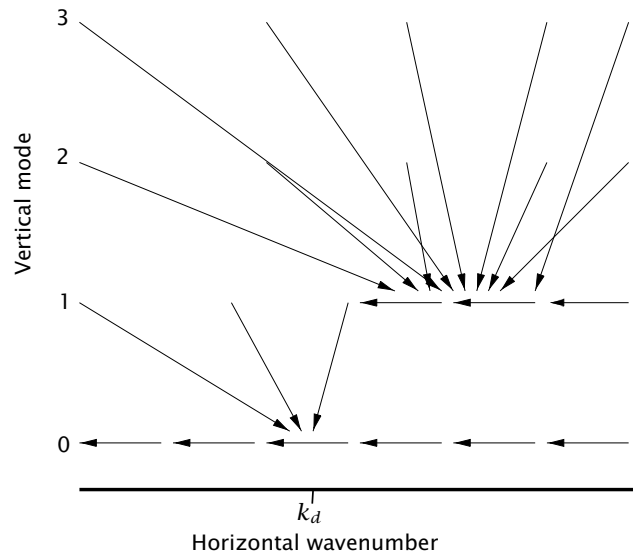


Fig. 12.15 Energy transfer paths as a function of vertical mode and horizontal wavenumber, in a fluid with an oceanic stratification; i.e., with a thermocline. Vertical mode 0 is the barotropic mode.

oceanic analogue of the lifecycle illustrated in Fig. 12.10 is shown in Fig. 12.14. The main difference between this case and the atmospheric one is that baroclinic instability initially leads to the transfer of energy to vertical mode one, followed by a transfer to larger horizontal scales in the barotropic mode, as illustrated schematically in Fig. 12.15.²⁵ If the energy is initially solely in the first baroclinic mode the cycle is more similar to the atmospheric one, but higher baroclinic modes may be more readily excited in the ocean than the atmosphere.

Notes

- 1 Rhines (1975). See also Holloway & Hendershott (1977) and Vallis & Maltrud (1993).
- 2 Adapted from Vallis & Maltrud (1993).
- 3 The mechanism described here follows Vallis & Maltrud (1993). Interactions between Rossby waves will also give rise to zonal flow, as described by Newell (1969) and Rhines (1975). Williams (1978) was one of the first numerical simulation to show the production of jets.
- 4 More discussion on these matters is variously given by Smith *et al.* (2002), Danilov & Gryanik (2002), Galperin *et al.* (2006, 2010), Sukoriansky *et al.* (2007), Scott & Dritschel (2012), Chai (2016) and others.
- 5 A PV staircase was proposed for the jets of Jupiter by Marcus (1993) and further discussed by Peltier & Stuhne (2002), and a review is to be found in Dritschel & McIntyre (2008). Staircases in turbulent flow can and have been found more generally, in particular in stratified flow in which an initially smooth density gradient may break down into steps and layers when stirred (Phillips 1972, Ruddick *et al.* 1989).
- 6 From Scott & Dritschel (2012).
- 7 See Farrell & Ioannou (2008) and Srinivasan & Young (2012). Tobias & Marston (2013) describe the mechanisms in a more general context at the price of some analytic accessibility, discussing the circumstances in which the jet formation proceeds by way of a cascade or a direct interaction with the zonal flow.
- 8 Readers whose native language is French may be confused by the difference. An explication is a particular way of presenting an explanation, so there can be different explications of precisely the same mechanism; the explications are equivalent but expressed in different ways.

- 9 Quasi-geostrophic turbulence was introduced by Charney (1971). Salmon (1980) and Rhines (1977) provided much of the two-layer phenomenology. Various laboratory experiments are discussed by Read (2001).
- 10 Adapted from Salmon (1980)
- 11 Lindborg (1999) concluded that the data were consistent with a forward cascade of enstrophy between wavelengths of a few thousand kilometres and a few hundred kilometres. Boer & Shepherd (1983) and Shepherd (1987) also found a k^{-3} spectrum at similar scales, with the latter noting the importance of interactions involving stationary waves. At scales smaller than 100 km or so the spectrum is shallower than -3 , and more like $-5/3$. This may be due to non-geostrophic effects, for example a forward cascade of energy associated with gravity wave breaking, or it may be due to a two-dimensional inverse cascade of energy with an energy source at very small scales associated with convection, or it may be due to effects associated with 'surface quasi-geostrophic' dynamics.
- 12 Adapted from Gage & Nastrom (1986).
- 13 Larichev & Held (1995) and Held & Larichev (1996). See Spall (2000) for an oceanic extension, and Thompson & Young (2006) for evidence that the theory fails to account for coherent structures.
- 14 Related arguments concerning eddy magnitudes were given by Gill *et al.* (1974). Atmospheric energetics, and atmospheric observations in general, are described by Peixoto & Oort (1992). For the oceanic case, see Wyrtki *et al.* (1976), Richardson (1983), and Stammer (1997).
- 15 This does not address the issue as to *why* the Rhines scale and deformation radius are similar. See also chapter 15.
- 16 Modified from Smith & Vallis (2001).
- 17 Adapted from Simmons & Hoskins (1978).
- 18 See Thorncroft *et al.* (1993). These authors identify two classes of lifecycles, which they call LC1 and LC2, which have differing degrees of decay of eddy kinetic energy in the later parts of the lifecycle, and with LC2 producing cut-off cyclones. Initial conditions and spatial inhomogeneities, including the horizontal shear of the flow and the presence of critical layers, play an important role in guiding the location of the wave breaking, and hence the final state that is reached.
- 19 The theory of this has been developed by Farrell (1984), Farrell & Ioannou (1996) and other papers by these authors. These authors have rightly emphasized that exponential growth is the exception, not the rule, in the real world. Indeed, if the basic flow is oscillatory, baroclinic instability can arise even if the basic flow is never unstable via the CSP criterion.
- 20 This realization came to fruition as a result of the bilateral US–USSR POLYMODE project in the 1970s. See Robinson (1984).
- 21 For example, Kundu *et al.* (1975).
- 22 From Chelton *et al.* (1998).
- 23 The eigenproblem actually solved was

$$\partial^2 \phi / \partial z^2 + (N^2(z)/c^2) \phi = 0, \quad \phi = 0 \quad \text{at } z = 0, H, \quad (12.84)$$

where H is the ocean depth and N is the observed buoyancy frequency. The deformation radius is given by $L_d = c/f$ where c is the first eigenvalue and f is the latitudinally varying Coriolis parameter.

- 24 Stammer (1997).
- 25 Fu & Flierl (1980) and Smith & Vallis (2001) examined this issue in more detail, both analytically and numerically. Figure 12.15 is adapted from these papers. Barnier *et al.* (1991) and Hua & Haidvogel (1986) looked at the role of high baroclinic modes in oceanic eddies and geostrophic turbulence.

CHAPTER 13

Turbulent Diffusion and Eddy Transport

THE TRANSPORT OF FLUID PROPERTIES BY UNSTEADY MOTION — that is, the way in which the properties of a fluid may be carried from one location to another by waves and turbulence — is one of the most important topics in geophysical fluid dynamics. It may be the dominant transport in a fluid, greatly exceeding that of the mean flow — in the atmosphere, for example, heat is transferred polewards primarily by the action of unsteady weather systems, not by the much weaker time-mean flow. However, we are often not interested in the details of the turbulent eddies and hence we might seek to *parameterize* the turbulent transport in terms of the mean flow; unfortunately, no general theory exists for such transport, for indeed such a theory would amount to a theory of turbulence. In the absence of this, we focus our attention in this chapter on the theory (such as it is) and practice of *turbulent diffusion*. In models of turbulent diffusion, the turbulent transport is generally related to the gradient of the mean flow, and it is the simplicity of the resulting expressions that has led to their wide adoption in areas as different as turbulent pipe flow, atmospheric boundary layer transport and large-scale ocean modelling. Diffusive models are, or aim to be, rational, simple and tractable — a blend of heuristic reasoning and elementary mathematics, the latter needed to ensure that certain basic requirements (conservation laws, for example) of a physical process are captured by a parameterization. However, just like other turbulent closures, they rely on physical assumptions that cannot be rigorously justified. In the first part of the chapter we consider turbulent diffusion from a general standpoint, and then specialize our discussion to geofluids, and in particular to large-scale transport by baroclinic eddies. Those readers with some prior knowledge of turbulent diffusion may choose to skip ahead to section 13.6.

13.1 DIFFUSIVE TRANSPORT

We begin with a brief discussion of the diffusion equation itself, to wit

$$\frac{\partial \phi}{\partial t} = \kappa \nabla^2 \phi, \quad (13.1)$$

where κ is a constant, positive, scalar diffusivity and the tracer ϕ is a scalar field. We expect that an initially concentrated blob of tracer would spread out — it would diffuse — and thus small parcels of tracer are transported. How quickly does this occur, or, put another way, is there an effective diffusive transport velocity?

If the rate of spreading becomes independent of the initial conditions then, purely from dimensional considerations, the spreading can depend only on the diffusivity and time itself and we can write

$$\overline{X^2} = \alpha \kappa t, \quad (13.2)$$

where $\overline{X^2}$ is the mean-square displacement, t is time and α is a non-dimensional constant. Let us quantify this with an explicit calculation. If ϕ is interpreted as the density of markers of fluid parcels, then the mean-square displacement of the markers is given by (in three dimensions)

$$\overline{X^2} = \frac{\int_0^\infty r^2 \phi r^2 dr}{\int_0^\infty \phi r^2 dr}, \quad (13.3)$$

where the denominator, the total amount of tracer present, is a constant and we have assumed a spherically symmetric distribution of tracer. Using (13.1) we find

$$\frac{d}{dt} \int_0^\infty r^2 \phi r^2 dr = \kappa \int_0^\infty r^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) r^2 dr = 6\kappa \int_0^\infty \phi r^2 dr \quad (13.4)$$

after a couple of integrations by parts. Thus

$$\frac{d}{dt} \overline{X^2} = 6\kappa \quad (13.5)$$

and because κ is a constant we have the important result that

$$\overline{X^2} = 6\kappa t. \quad (13.6)$$

In two dimensions the equivalent calculation begins with

$$\overline{X^2} = \frac{\int_0^\infty r^2 \phi r dr}{\int_0^\infty \phi r dr} \quad (13.7)$$

and using the diffusion equation we find

$$\frac{d}{dt} \int_0^\infty r^2 \phi r dr = \kappa \int_0^\infty r^2 \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) r dr = 4\kappa \int_0^\infty \phi r dr. \quad (13.8)$$

Thus we obtain

$$\overline{X^2} = 4\kappa t. \quad (13.9)$$

Finally, in one dimension (i.e., spreading along a line) it is easy to show that

$$\overline{X^2} = 2\kappa t. \quad (13.10)$$

13.1.1 An explicit example

We gain a little more intuition about what the above calculations mean by considering the case in which the initial tracer distribution is a delta function at the origin. If the total amount of tracer is unity, then in three dimensions at subsequent times the tracer is given by the distribution

$$\phi(r, t) = \frac{1}{8(\pi \kappa t)^{3/2}} \exp(-r^2/4\kappa t), \quad (13.11)$$

as may be checked by substitution back into the equation of motion. (See problem 13.??.) The distribution clearly broadens with time, and the mean-square distance from the origin is given by

$$\overline{X^2} = \int_0^\infty \frac{4\pi r^2}{8(\pi\kappa t)^{3/2}} \exp(-r^2/4\kappa t) dr = 6\kappa t, \quad (13.12)$$

as in (13.6). The important point is that the mean distance travelled by a particle during a time interval t is proportional to the square root of that time interval. This is, of course, redolent of a random walk (see the shaded box on the next page), which brings us to the subject of turbulent diffusion.

13.2 TURBULENT DIFFUSION

Fluids differ from solids in that they can transport properties by advection — thus, heat is primarily transferred polewards in the atmosphere by means of air movement and not by molecular diffusion. Turbulent fluid motion differs from laminar fluid motion in that such advective transport may be greatly enhanced by the seemingly random motion of the fluid, the net transport being much larger than that which would be effected by the time-mean fluid motion alone. Indeed, to continue the atmospheric example, away from the tropics the poleward transport of heat in the atmosphere is largely effected by way of the (large-scale) turbulent transfer of heat in mid-latitude weather systems. Of course, such transfer *is* simply by advection, and if we could explicitly calculate the motion of the fluid parcels we could explicitly calculate the transport. However, turbulent transport is both very complicated and very sensitive to the initial conditions, so that any hope of performing such a calculation exactly in a real situation is often a forlorn one.

Turbulent transport is most important in *inhomogeneous* situations, because it is the divergence of the transport that is important and the mean divergence is non-zero only if there is inhomogeneity. The theories of chapters 11 and 12 do not lend themselves to an easy extension to inhomogeneous flow, and we turn to a slightly more empirical approach.¹

13.2.1 Simple theory

Let us consider how fluid markers are transported in a statistically steady, homogeneous, turbulent flow. The markers are introduced at the origin $x = y = z = 0$ at $t = 0$; we may create an ensemble of such markers by performing many such tracer release experiments on different realizations of the turbulent flow, but with each flow having the same statistical properties. The question is, what is the average rate of dispersion of a single particle of fluid?

The displacement of a marker at a time t is given by

$$\mathbf{X}(t) = \int_0^t \mathbf{V}(t') dt', \quad (13.13)$$

where $\mathbf{V}$ is the velocity of the fluid parcel — a material velocity. [We will use uppercase variables to denote material ('Lagrangian') quantities.] The mean-square displacement is

$$\overline{X^2(t)} = \int_0^t dt_1 \int_0^t \overline{\mathbf{V}(t_1) \cdot \mathbf{V}(t_2)} dt_2, \quad (13.14)$$

where the overbar denotes an ensemble average, and thus $\overline{\mathbf{V}(t_1) \cdot \mathbf{V}(t_2)}$ is a measure of the velocity correlation between the velocities of the fluid parcels at times t_1 and t_2 . That is,

$$\overline{\mathbf{V}(t_1) \cdot \mathbf{V}(t_2)} = \overline{v^2} R(t_2 - t_1) = \overline{v^2} R(\tau), \quad (13.15)$$

A Random Walk

Here we give an elementary derivation of the most basic result in random walk theory, the relationship of the mean-square displacement to the number of steps taken.² A loose analogy is that of drunkards staggering randomly from here to there, with no correlation between their successive steps. After any number of steps, the *mean* displacement of the drunkards is zero, but we expect their *root-mean-square* displacement to increase: this is because drunkards independently thrown out of the same bar will generally wander off in different directions (which is why the mean displacement is zero), but after some time most of them will indeed end up some distance away.

For simplicity consider steps, $\mathbf{s}_n$, each with random orientation but equal magnitude, s . The displacement after n steps is related to the displacement after $n - 1$ steps by

$$\mathbf{D}_n = \mathbf{D}_{n-1} + \mathbf{s}_n, \quad (\text{R.1})$$

so that the amplitude of $\mathbf{D}_n$, namely D_n , is given by

$$\begin{aligned} D_n^2 &= (\mathbf{D}_{n-1} + \mathbf{s}_n) \cdot (\mathbf{D}_{n-1} + \mathbf{s}_n) \\ &= D_{n-1}^2 + s^2 + 2\mathbf{D}_{n-1} \cdot \mathbf{s}_n. \end{aligned} \quad (\text{R.2})$$

Taking an ensemble average over many realizations gives

$$\overline{D_n^2} = \overline{D_{n-1}^2} + s^2 \quad (\text{R.3})$$

having used $\overline{\mathbf{D}_{n-1} \cdot \mathbf{s}_n} = 0$, because each step is random.

Now, $D_0 = 0$, so that $\overline{D_1^2} = s^2$, $\overline{D_2^2} = 2s^2$ and so on. Thus, using (R.3) to proceed inductively, we have

$$\overline{D_n^2} = ns^2, \quad (\text{R.4})$$

or

$$\overline{D_n^2}^{1/2} = \sqrt{ns}. \quad (\text{R.5})$$

Thus, in a random walk the *root-mean-square displacement increases with the half-power of the number of steps taken*. More work is required to calculate the distribution of the random walkers, but it may be shown that in the limit of infinitesimally small steps the random walk becomes a Wiener process and the distribution becomes Gaussian, as in (13.11) (with the exact form depending on the dimensionality of the problem), indicating a diffusive process. Diffusion may be thought of as a continuous random walk, with the displacement proportional to the square root of time.

where $R(t_2 - t_1)$ is the velocity correlation function and, because the turbulence is statistically steady, this depends only on the time difference $\tau = t_2 - t_1$. Furthermore, $R(-\tau) = R(\tau)$. Thus,

$$\overline{X^2(t)} = \int_0^t dt_1 \overline{v^2} \int_0^t R(t_2 - t_1) dt_2 = \int_0^t d\hat{t} \overline{v^2} \int_{-\hat{t}}^{t-\hat{t}} R(\tau) d\tau, \quad (13.16)$$

changing variables to τ and $\hat{t} = t_1$ (Fig. 13.1). We expect the velocity correlation function to fall monotonically from its initial value of unity to a value approaching zero as $\tau \rightarrow \infty$, as in Fig. 13.2,

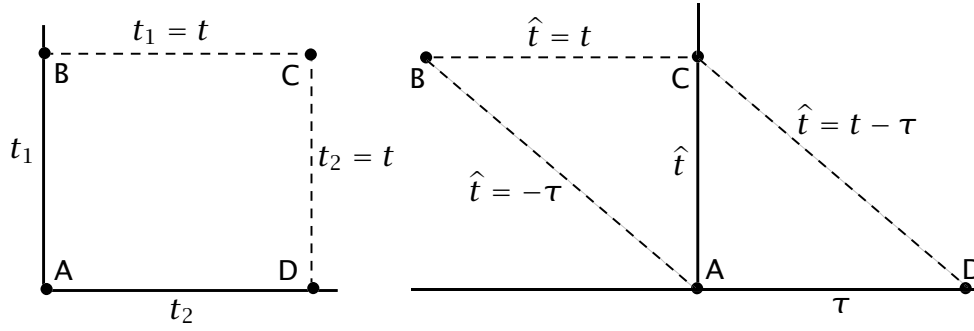


Fig. 13.1 Changes of time variables involved in (13.16) and (13.32). The original two-dimensional integral is over the rectangle ABCD. Defining $\tau = t_2 - t_1$ and $\hat{t} = t_1$, then the area is spanned by $[\hat{t} = (0, t), \tau = (-\hat{t}, t - \hat{t})]$ as in (13.16), or by $[\tau = (0, t), \hat{t} = (0, t - \tau)]$ (i.e., ACD) plus $[\tau = (-t, 0), \hat{t} = (-\tau, t)]$ (i.e., ABC) in (13.32).

and typically, there will be some characteristic time τ_{corr} that parameterizes the behaviour of the function. In general, we cannot obtain explicit general solutions without detailed knowledge of this correlation function, but there are two interesting limits.

- (i) *The short-time limit.* For small times, i.e., for $t \ll \tau_{corr}$ (and so $t_1, t_2 \ll \tau_{corr}$) the correlation function will be approximately unity and so (13.16) becomes

$$\overline{X^2(t)} \approx \int_0^t d\hat{t} \overline{v^2} \int_0^t d\hat{t} = \overline{v^2} t^2. \quad (13.17)$$

Thus, the root-mean-square displacement increases linearly with time, and linearly with the root-mean-square velocity of the flow. For small times, the fluid parcel's behaviour is well correlated with that at the initial time, and so the displacement increases linearly in the direction it was initially going. Indeed, directly from (13.13) we have, for small times, $X(t) \approx Vt$, which leads directly to (13.17).

- (ii) *The long-time limit.* We are now concerned with the case $t \gg \tau_{corr}$. Because the correlation function falls with time, most of the contributions to the second integrand [involving $R(\tau)$] in (13.16) are from $\tau \leq \tau_{corr}$. Thus, without much loss in accuracy, we can replace the limits of integration by $-\infty$ and $+\infty$; that is

$$\overline{X^2(t)} \approx \int_0^t d\hat{t} \overline{v^2} \int_{-\infty}^{\infty} R(\tau) d\tau. \quad (13.18)$$

Assuming the second integral converges, it is just a number; in fact, noting that $R(\tau) = R(-\tau)$, we may use it to *define* the correlation time τ_{corr} by

$$\tau_{corr} \equiv \int_0^{\infty} R(\tau) d\tau. \quad (13.19)$$

We then have the important result that

$$\overline{X^2(t)} \approx 2\overline{v^2} t \int_0^{\infty} R(\tau) d\tau = 2\overline{v^2} \tau_{corr} t. \quad (13.20)$$

That is, for times that are long compared with the turbulence correlation time, the distance travelled by a fluid parcel in some time interval is proportional to the square-root of that

time interval, just as for a diffusive process; this is because the fluid parcels are essentially undergoing random walks. The equation (13.20) connects two quite different fluid properties: the left-hand side tells us how tracers are dispersed in a turbulent flow, a material property, whereas the right-hand side can be evaluated from the Eulerian velocity field at different times. Both the left- and right-hand sides can be directly measured, by looking at the dispersion of a dye and by measuring the velocity at successive times.

We may define a *coefficient of turbulent diffusivity* by

$$K_{turb} = \frac{1}{3} \overline{v^2} \tau_{corr} \quad (13.21)$$

and then we have the result that

$$\overline{X^2(t)} = 6K_{turb}t. \quad (13.22)$$

Comparison of (13.22) with (13.6) indicates that the transport of turbulent flow, under these conditions, is like a diffusive transport, with a coefficient of diffusivity given by (13.21). [Sometimes, the numerical factors are neglected, and a diffusivity is defined by the expressions

$$K_{turb} \equiv \frac{d\overline{X^2(t)}}{dt} \quad \text{or} \quad K_{turb} \equiv \frac{1}{2} \frac{d\overline{X^2(t)}}{dt}. \quad (13.23)$$

These lose the exact connection with a true diffusion coefficient, but usually the turbulent diffusivity can only be estimated, anyway.]

We may define a correlation length scale to be the approximate distance that a parcel moves, on average, in a material (i.e., ‘Lagrangian’) correlation time. Thus

$$l_{corr} \equiv v_{rms} \tau_{corr}, \quad (13.24)$$

where $v_{rms} = (\overline{v^2})^{1/2}$, whence

$$K_{turb} = \frac{1}{3} v_{rms} l_{corr}. \quad (13.25)$$

In most situations, the numerical coefficient (1/3 here) cannot be trusted because a real turbulent flow is unlikely to satisfy the restrictions of stationarity and homogeneity that we have imposed. Nevertheless, a relationship similar to (13.25) — that a turbulent diffusivity is proportional to an r.m.s. turbulent velocity and a correlation length scale, is the foundation for semi-empirical *mixing length* theories that we discuss in section 13.4.

The simple relationships between the mean-square displacement, the Lagrangian time scale, the mean-square velocity and the eddy diffusivity allows the diffusivity to be computed from the statistics of particle trajectories. Thus, suppose that a cluster of floats is released into the ocean, or some balloons are released in the atmosphere. If neutrally buoyant, these instruments then essentially become labelled fluid particles, and one may compute K_{turb} directly from the dispersion of the cluster using (13.22). If it is possible to measure their root-mean-square velocity, then one may use (13.21) to estimate the diffusivity from this and the material correlation time scale.

13.2.2 ♦ An anisotropic generalization

We now consider the correlation between the different components of the displacement in anisotropic, but still homogeneous, flow.³ The displacement of a fluid particle is given by (13.13), and this is a

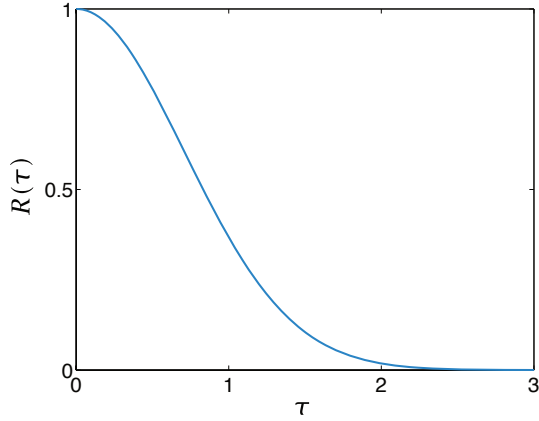


Fig. 13.2 Idealized velocity correlation function in turbulent flow with correlation time $\tau_{corr} = \mathcal{O}(1)$. For small times, $\tau \ll \tau_{corr}$, $R(\tau) \approx 1$. For large times, $\tau \gg \tau_{corr}$, $R(\tau) \ll 1$. We may define the correlation time by $\tau_{corr} = \int_0^\infty R(\tau) d\tau$.

random vector. Thus, generalizing (13.14), we may define the fluid particle displacement covariance tensor by

$$D_{ij}(t) = \overline{X_i(t)X_j(t)} = \int_0^t \int_0^t \overline{V_i(t_1)V_j(t_2)} dt_1 dt_2, \quad (13.26)$$

where the velocity denoted by V_i is the i th component of the velocity of a fluid element. For small times, $X_i(t) \approx v_i(\mathbf{a}, 0)t$, where $v_i(\mathbf{a}, 0)$ is the fluid velocity at the parcel's initial position, $\mathbf{a}$, and we obtain

$$D_{ij}(t) \approx \overline{v_i(\mathbf{a}, 0)v_j(\mathbf{a}, 0)}t^2. \quad (13.27)$$

If the flow is statistically steady and homogeneous the average of any quantity has no spatial or temporal dependence and so

$$D_{ij}(t) = A_{ij}t^2, \quad (13.28)$$

where the tensor $A_{ij} = \overline{v_i v_j}$ has constant entries, and this is a slight generalization of (13.17).

The velocity covariance of a fluid parcel at times t_1 and t_2 is, as before, a function only of the time difference $t_1 - t_2$ and so it must have the form

$$\overline{V_i(t_1)V_j(t_2)} = \left(\overline{v_i^2} \overline{v_j^2}\right)^{1/2} R_{ij}(t_2 - t_1). \quad (13.29)$$

Except in the case of isotropic flow $R_{ij}(\tau) \neq R_{ij}(-\tau)$, but we do have, in general,

$$R_{ij}(\tau) = R_{ji}(-\tau). \quad (13.30)$$

Now, to obtain a generalization of (13.20), we first use (13.29) in (13.26) to obtain

$$D_{ij}(t) = \left(\overline{v_i^2} \overline{v_j^2}\right)^{1/2} \int_0^t \int_0^t R_{ij}(t_2 - t_1) dt_1 dt_2. \quad (13.31)$$

If we change variables to $\tau = t_2 - t_1$ and $\hat{t} = t_1$ (see Fig. 13.1) we obtain⁴

$$D_{ij}(t) = \left(\overline{v_i^2} \overline{v_j^2}\right)^{1/2} \left(\int_0^t d\tau \int_0^{t-\tau} d\hat{t} R_{ij}(\tau) + \int_{-t}^0 d\tau \int_{-\tau}^t d\hat{t} R_{ij}(\tau) \right), \quad (13.32)$$

and using (13.30) this becomes

$$D_{ij}(t) = 2 \left(\overline{v_i^2} \overline{v_j^2}\right)^{1/2} \int_0^t d\tau \int_0^{t-\tau} d\hat{t} \hat{R}_{ij}(\tau), \quad (13.33)$$

where $\hat{R}_{ij} = (R_{ij} + R_{ji})/2$. This order of integration enables us to perform the integration over $\hat{t}$, giving

$$D_{ij}(t) = 2 \left(\overline{v_i^2} \overline{v_j^2} \right)^{1/2} \int_0^t (t - \tau) \hat{R}_{ij}(\tau) d\tau. \quad (13.34)$$

For long times, i.e., for $t \gg \tau_{corr}$, the upper limit of the integration may be taken to be infinity, again because the contributions to the integrand from $\hat{R}_{ij}(\tau)$ all come from small τ . Furthermore, we expect that for large t ,

$$\int_0^\infty t \hat{R}_{ij}(\tau) d\tau \gg \int_0^\infty \tau \hat{R}_{ij}(\tau) d\tau, \quad (13.35)$$

because $R_{ij}(\tau)$ is only non-negligible for small τ , and $t \gg \tau$ in this range. Thus, we finally obtain a generalization of (13.20) for the displacement covariance of two components of the displacement, namely

$$D_{ij} = 2 \left(\overline{v_i^2} \overline{v_j^2} \right)^{1/2} t \int_0^\infty \hat{R}_{ij}(\tau) d\tau. \quad (13.36)$$

The integral is a tensor with constant entries, analogous to the turbulent decorrelation time scale of (13.19). Then, with $\tau_{ij} \equiv \int_0^\infty \hat{R}_{ij}(\tau) d\tau$, the corresponding turbulent diffusivity is

$$K_{ij} = \frac{1}{3} \left(\overline{v_i^2} \overline{v_j^2} \right)^{1/2} \tau_{ij}. \quad (13.37)$$

13.2.3 Discussion

We have shown that, for sufficiently long times, the distance travelled by a fluid parcel in some time is proportional to the square root of that time, just as for a diffusive process and just as for a random walk. The motion of our fluid parcel is analogous to that of a dust particle undergoing Brownian motion — both are continually buffeted and undergo random walks as a result. Still, it may appear that the usefulness of our results is limited by the assumptions of stationarity and homogeneity — it is well-nigh impossible in nature to produce a statistically stationary, homogeneous turbulent flow, because statistical stationarity implies there must be an energy source and this, as well as the presence of boundaries, militate against homogeneity. However, we should not be so pessimistic, on two counts.

- (i) Similar ideas may be directly applied to flows that are homogeneous in one direction.⁵
- (ii) Often, a flow will *not* be homogeneous in any direction. However, if the *statistics* of the eddy motion vary on a space scale that is longer than $v_{rms} \tau_{corr}$, then the eddy transport properties may be determined by a local theory. For example, the size of the eddy diffusivity is then determined by $D_t \sim v_{rms} l_{corr}$ where the parameters, and hence the diffusion coefficient, vary, but only on a scale longer than the energy-containing scale.

The essential results of this section thus lie in (13.20), (13.21) and (13.25): that the dispersion of a fluid particle in a turbulent flow is *diffusive* in nature, and that the turbulent diffusivity is proportional to the product of the root-mean-square velocity and the correlation length.

13.3 TWO-PARTICLE DIFFUSIVITY

Let us now consider the problem of determining the mutual separation of two fluid parcels; the problem is relevant to geofluids because by tracking the separation of floats in the ocean, or balloons in the atmosphere, we can learn much about the nature of large-scale turbulence in those systems. The problem differs from the one-particle problem, because the separation of the particles itself will affect the rate of increase of the separation. In the one-particle problem in homogeneous flow, the position of the particular tagged fluid particle plays no direct role in determining its rate

of spreading from its initial condition — any one position is the same as any other. But if two particles are close together, they may be swept away together by some large eddy, without affecting their mutual separation whereas two particles that are widely separated will undergo largely uncorrelated motion. Thus, we identify two regimes.

- (i) A regime in which the separation of the particles is greater than the scale of the largest eddies. In this case, each particle is undergoing a random walk that is effectively uncorrelated with that of the other particle.
- (ii) A regime in which the separation of the particles is less than the energy-containing scale of the motion. In this case, the eddies that contribute most to the two-particle separation are those that are comparable in scale to the separation itself.

If we attempt to apply the Taylor analysis ab initio we evidently have, by analogy to (13.13)

$$\mathbf{Y}(t) = \mathbf{X}_1(0) - \mathbf{X}_2(0) + \int_0^t [\mathbf{V}_1(t') - \mathbf{V}_2(t')] dt', \quad (13.38)$$

and a mean-square separation of

$$\overline{Y^2(t)} = \overline{(\mathbf{X}_1(0) - \mathbf{X}_2(0))^2} + \int_0^t dt_1 \int_0^t dt_2 [\mathbf{W}_1(t_1) \cdot \mathbf{W}_2(t_2)], \quad (13.39)$$

where $\mathbf{W}(t) = \mathbf{V}_1(t) - \mathbf{V}_2(t)$. However, it is now difficult to proceed much further. The problem is that we cannot write

$$\overline{\mathbf{W}(t_1) \cdot \mathbf{W}(t_2)} = \overline{w^2} R(t_2 - t_1), \quad (13.40)$$

because the correlation will depend on the initial separation of the particles as well as the time since then. Thus, the diffusivity itself will depend on both time and the initial particle separation, and the results analogous to those of the single-particle diffusivity cannot easily be recovered. However, we can make some progress by separately considering the two above-mentioned regimes.

13.3.1 Large particle separation

This case is analogous to the single-particle case. The particle separation is given by

$$\mathbf{Y}(t) = \mathbf{X}_1(t) - \mathbf{X}_2(t), \quad (13.41)$$

so the mean-square separation is

$$\overline{Y^2(t)} = \overline{X_1(t)^2} + \overline{X_2(t)^2} - 2\overline{\mathbf{X}_1(t) \cdot \mathbf{X}_2(t)}. \quad (13.42)$$

For long times, the last term is zero because the motion of the two particles is uncorrelated. Furthermore, each of the first two terms is given by (13.20) or (13.22), so that the mean separation varies as

$$\overline{Y^2(t)} = 4\overline{v^2} \tau_{corr} t \quad (13.43)$$

and the rate of separation, for large t , is given by

$$\frac{d\overline{Y^2(t)}}{dt} = 4\overline{v^2} \tau_{corr} = 12K_{turb}. \quad (13.44)$$

Thus, the relative diffusion is twice that of the single-particle process, in the limit that the particles are separated by an amount larger than the largest eddies.

13.3.2 Separation within the inertial range

How do fluid parcels whose separation is at inertial scales behave relative to each other?⁶ Suppose that two particles are tagged, and that their separation is greater than the viscous scale but smaller than the scales of the largest eddies — that is, the separation lies within the inertial range of the flow. Then, the rate of separation of the two particles can depend only on two quantities, the separation itself and properties of the inertial range, meaning (in three dimensions) the energy flux, ε , through the system. It cannot depend on the time, because this would imply that the subsequent rate of particle separation depends on the history of how the particles came to their current positions. Thus we can write

$$\frac{d\bar{L}^2}{dt} = g(\bar{L}, \varepsilon), \quad (13.45)$$

where $\bar{L} \equiv \overline{Y(t)^2}^{1/2}$. Dimensional analysis then gives

$$\frac{d\bar{L}^2}{dt} = A\varepsilon^{1/3}\bar{L}^{4/3}, \quad (13.46)$$

where A is a non-dimensional constant, and this is known as ‘Richardson’s four-thirds law’. We can integrate (13.46) to give

$$\bar{L}^2 \sim \varepsilon t^3. \quad (13.47)$$

Another way of deriving (13.46) is to suppose that the separation obeys the diffusive law

$$\frac{d\bar{L}^2}{dt} = K_{turb}, \quad (13.48)$$

where K_{turb} is a turbulent diffusivity that is a *function of the separation itself*. This is because the farther apart the eddies are, the larger the scale of the eddies that can move the two particles independently, rather than just sweeping them along together. An estimate of the diffusivity is then

$$\mathcal{K}_{turb} \sim vl, \quad (13.49)$$

where v is the characteristic velocity of an eddy of scale l , and $l \sim \bar{L} = \overline{Y^2}^{1/2}$. Using the inertial range scaling $v \sim (l\varepsilon)^{1/3}$ this is

$$\mathcal{K}_{turb} \sim \varepsilon^{1/3}\bar{L}^{4/3}, \quad (13.50)$$

and so (13.48) becomes

$$\frac{d\bar{L}^2}{dt} \sim \varepsilon^{1/3}\bar{L}^{4/3}, \quad (13.51)$$

as before. Of course, dimensional consistency demands that we obtain the same result, but the derivation is intuitive and the estimate of the two-particle diffusivity (i.e., (13.50), that the eddy diffusivity governing the separation of two fluid parcels goes as the 4/3 power of their root-mean-square separation) is useful. If the particle separation is greater than the scale of the largest eddies in the system, l_{max} , then

$$K_{turb} \sim v(l_{max})l_{max} \sim \varepsilon^{1/3}l_{max}^{4/3} = \text{constant}. \quad (13.52)$$

The two-particle separation then proceeds as a conventional random walk or diffusive process, with the mean-square separation increasing linearly with time.

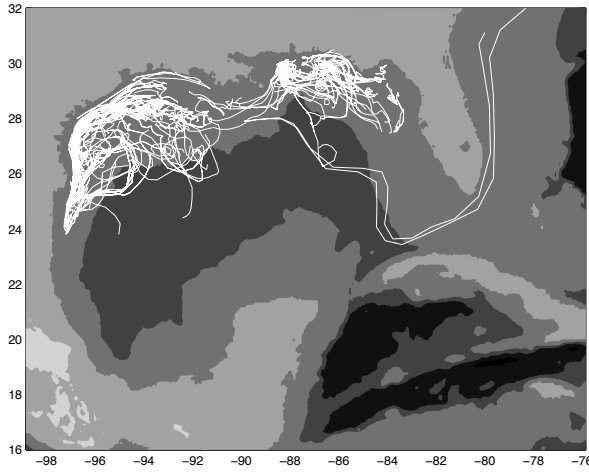


Fig. 13.3 Trajectories of surface drifters in the Gulf of Mexico, each truncated to produce paths of just 25 days. The drifters were released as part of ‘SCULP’ — the Surface Current and Lagrangian drift Program.⁷

Diffusion in two-dimensional flow

In two dimensions the turbulent diffusivity will differ depending on whether the two-particle separation is in the energy inertial range or in the enstrophy inertial range. In the energy inertial range the scaling is the same as in the three-dimensional case, but in the enstrophy inertial range the rate of separation will depend on the enstrophy cascade rate, η , and the separation itself. Dimensional analysis then leads to

$$\frac{d\bar{L}^2}{dt} = B\eta^{1/3}\bar{L}^2, \quad (13.53)$$

where B is a non-dimensional constant. This integrates to

$$\bar{L}^2 = \bar{L}(0)^2 \exp(B\eta^{1/3}t), \quad (13.54)$$

or $\bar{Y}^2(t) = \bar{Y}^2(0) \exp(B\eta^{1/3}t)$. Thus, the rate of separation is *exponential* in the enstrophy inertial range, a result unique to two-dimensional turbulence. Similarly, using $v \sim \eta^{1/3}l$, the turbulent diffusivity is given by

$$K_{turb} \sim \eta^{1/3}\bar{L}^2. \quad (13.55)$$

A geophysical example

The above ideas are well illustrated by analysing the trajectories of surface drifters in the Gulf of Mexico. The drifters are free-moving buoys which float about a half metre below the surface and which thus act as imperfect fluid markers — imperfect because they cannot follow the full three-dimensional motion of water parcels. Nevertheless, the motion at these scales can be expected to be quasi-geostrophic and nearly horizontal, so the associated error will be small. The drifters are tracked by satellite and their trajectories, proxies for the motion of fluid parcels, are shown in Fig. 13.3. The two-particle, or two-drifter, separation is illustrated in Fig. 13.4 and two regimes may be discerned. In the first, the pair separations grow approximately exponentially in time, with an e-folding time of 2 days, consistent with motion within an enstrophy inertial range using (13.54). The second regime is characterized by a power-law growth, proportional to $t^{2.2}$, somewhat slower than the t^3 separation expected for an energy inverse cascade using (13.47).⁸ The boundary for the two regimes occurs at about 75 km, which is similar to the first deformation radius. Note that no late-time diffusive regime (where the dispersion goes like t) is observed, suggesting that there exist long-time drifter correlations; these correlations arise because the separation of the drifters is never significantly larger than the energy-containing scale of the eddies themselves.

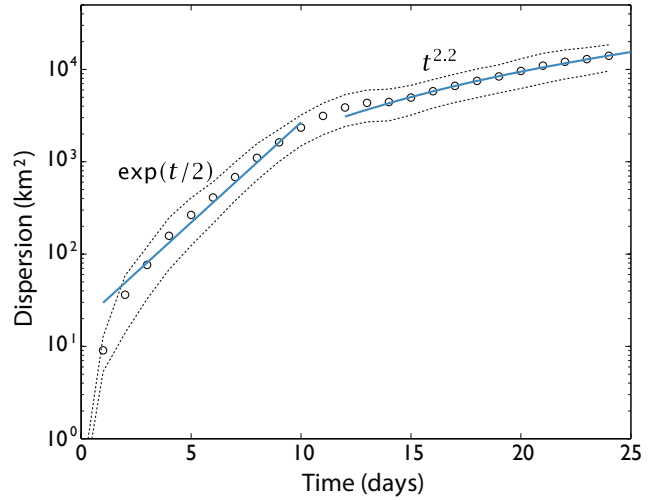


Fig. 13.4 Relative dispersion (the mean-square separation) for 140 drifter pairs as a function of time. The analysis utilizes all drifter pairs which come within 1 km of each other during their lifetimes.

In the atmosphere similar exponential separation of pairs of drifting balloons in the stratosphere at scales of less than 1000 km has been seen, consistent with an enstrophy inertial range. Evidence of a t^3 separation at larger scales, consistent with an energy inverse cascade, has been less forthcoming.⁹

13.4 MIXING LENGTH THEORY

The discussion of the previous two sections deals with the dispersion of marked fluid parcels. However, both for practical and fundamental reasons, we would like to be able to represent the turbulent transport of a fluid property in an Eulerian form. Thus, consider the equation for a conserved quantity ϕ in an incompressible turbulent flow

$$\frac{D\bar{\phi}}{Dt} = -\nabla \cdot (\overline{v'\phi'}) + \kappa \nabla^2 \bar{\phi}, \quad (13.56)$$

where κ is the molecular diffusivity and the overbar denotes some kind of averaging, perhaps an ensemble average over realizations with different initial conditions. (We also adopt the convention that, unless noted, whenever the material derivative written as D/Dt is applied to an averaged field, the advection is by the averaged velocity only.) We expect the transport of ϕ to be enhanced by the turbulent flow and, as we saw in the previous sections, in some circumstances this transport will have a diffusive nature, completely overwhelming the molecular diffusivity. Let us consider this from an Eulerian angle, and by analogy with molecular mixing.

Given the mean distribution $\bar{\phi}(x, y, z)$, let a fluid parcel be displaced from its mean position by a turbulent fluctuation. Suppose that the displaced parcel of fluid is able to carry its initial properties a distance l' before mixing with its surroundings. Then just prior to mixing with the environment the fluctuation of ϕ is given by, in the one-dimensional case,

$$\phi' = -l' \frac{\partial \bar{\phi}}{\partial x} - \frac{1}{2} l'^2 \frac{\partial^2 \bar{\phi}}{\partial x^2} + O(l'^3). \quad (13.57)$$

If the mean gradient is varying on a space-scale, L , that is larger than the mixing length l' , that is if

$$L \equiv \frac{|\partial \bar{\phi} / \partial x|}{|\partial^2 \bar{\phi} / \partial x^2|} \gg l', \quad (13.58)$$

then we can neglect terms in l'^2 and higher. The turbulent flux of ϕ -stuff is then given by

$$F = \overline{u' \phi'} = -\overline{u' l'} \frac{\partial \bar{\phi}}{\partial x}. \quad (13.59)$$

In more than one dimension, we have

$$\mathbf{F} = F_i = -\overline{v'_i l'_j} \partial_j \bar{\phi} = -K_{ij} \partial_j \bar{\phi}, \quad (13.60)$$

with summation over repeated indices, where $K_{ij} \equiv \overline{v'_i l'_j}$. The quantity K_{ij} (which we also write as $\mathbf{K}$) is known as the eddy (or turbulent) diffusivity tensor. At high Reynolds number it is a property of the flow rather than the fluid itself but, supposing that it can somehow be determined, the equation for the mean value of ϕ becomes

$$\frac{D\bar{\phi}}{Dt} = \nabla \cdot (\mathbf{K} \nabla \bar{\phi}) = \partial_i (K_{ij} \partial_j \bar{\phi}), \quad (13.61)$$

neglecting molecular diffusion.

Suppose that there exists a coordinate system in which the displacements in one direction ($l'^{\hat{x}}$, displacements in the $\hat{x}$ -direction) are not correlated with the fluctuating velocity in another, orthogonal, direction (v' , the velocity in the $\hat{y}$ -direction) and for simplicity we restrict ourselves to two dimensions. Then, in that coordinate system $\mathbf{K}$ is symmetric and

$$\mathbf{K} = \begin{pmatrix} \overline{u' l'^{\hat{x}}} & \overline{u' l'^{\hat{y}}} \\ \overline{v' l'^{\hat{x}}} & \overline{v' l'^{\hat{y}}} \end{pmatrix} = \begin{pmatrix} \overline{u' l'^{\hat{x}}} & 0 \\ 0 & \overline{v' l'^{\hat{y}}} \end{pmatrix}. \quad (13.62)$$

The tensor may then, if needs be, be rotated to some other Cartesian coordinate system, but it will remain a symmetric tensor. In isotropic flow the two diagonal entries are equal and the equation of motion is,

$$\frac{D\bar{\phi}}{Dt} = \nabla \cdot (K \nabla \bar{\phi}), \quad (13.63)$$

which is identical to the equation with molecular diffusion, save that the eddy diffusivity scalar, K , is different from the molecular diffusivity. To the extent, then, that K_{ij} is a symmetric tensor with constant entries, then the turbulence acts like an enhanced diffusion. If the flow is homogeneous, then K does not vary spatially.

13.4.1 Requirements for turbulent diffusion

Turbulent diffusion evidently is a tractable and rational approach for parameterizing the effects of turbulent transport.¹⁰ However, the underlying assumptions are not always satisfied and the derivation itself is rather heuristic, and turbulent diffusion is in no way a fundamental solution to the turbulence closure problem. Nonetheless, it can be an extremely useful parameterization in the appropriate circumstances, these being that:

- (i) there should be a scale separation between the mean gradient and the maximum mixing length, and the mixing length and decorrelation time scale should be well-defined.
- (ii) the diffused property ϕ should be a semi-materially conserved quantity; that is, materially conserved except for the effects of mixing (the next item).
- (iii) The diffused property ϕ should be able to *mix* with its environment.

These are all largely self-evident from the derivation, but items ((ii)) and ((iii)) deserve more discussion, as follows.

(ii) Material conservation of tracer

We assumed that a parcel of fluid carries its value of ϕ a distance, on average, equal to its mixing length before irreversibly mixing with its environment; this assumption is necessary in order that one may write $\phi' = -l' \partial \bar{\phi} / \partial x$. If ϕ is not materially-conserved over this scale other terms enter this formula. In particular, momentum is affected by the pressure force, and so is not normally a good candidate for turbulent diffusion. Potential vorticity is a better candidate, because it is a true material invariant, save for dissipative terms, and in large-scale geophysical flows potential vorticity also contains much of the information about the flow.

(iii) Tracer mixing and turbulent cascades

If a parcel cannot mix with its surroundings, then turbulent mixing cannot take place at all. Instead, we have what might be called turbulent stirring and if ϕ were, say, a dye then it would merely become threaded through the environment, producing streaks and swirls of colour rather than a truly mixed fluid. As another example, let ϕ be temperature and suppose that it has a mean gradient, so that temperature falls in the direction of increasing y . Now, we intuitively expect that turbulence can be a much more efficient transporter of heat than molecular diffusion. However, if a displaced parcel of fluid does not mix with or assume the value of its new environment at some stage, then there will be no correlation between the velocity producing the displacement and the value of the fluctuating quantity ϕ' . Suppose, for example, that an eddy causes parcels to be displaced from their mean positions. If a displaced parcel mixes with its surroundings, then a correlation will develop between v' and ϕ' , and we would have $\overline{v' \phi'} \neq 0$. However, if no mixing occurs, then the eddy simply recirculates with eddies retaining their initial values, and $\overline{v' \phi'}$ is zero because of a lack of correlation between the two quantities. Thus, it is essential that there be a degree of *irreversibility* to the flow in order for turbulent diffusion to be appropriate.

Molecular diffusion is not the only process that enables an eddy to assume the value of its surroundings — a Newtonian or other relaxation back to a specified temperature may have much the same effect. Indeed, in the atmosphere a displaced parcel will be subject to a radiation field that acts in qualitatively this way. For example, suppose that the temperature equation is

$$\frac{DT}{Dt} = -\lambda(T - T^*(y, z)), \quad (13.64)$$

where the right-hand side crudely represents radiative effects via a relaxation back to a specified profile. Then a displaced parcel will be subject to a radiative damping that is different from that at its initial position, and this will allow the parcel to take on the value of its surroundings, and so potentially enable turbulent diffusion to occur (provided λ is small so that T is approximately materially conserved).

For molecular diffusion to be the mechanism whereby a parcel mixes with its surroundings, the turbulence must create scales that are small enough for diffusion to act. This means that turbulence must create a cascade of ϕ -stuff to small scales. This is quite consistent with the notion that ϕ is a materially conserved quantity, because a scalar field ϕ that satisfies

$$\frac{D\phi}{Dt} = F + \kappa \nabla^2 \phi, \quad (13.65)$$

where F might represent a spectrally local source of variance of ϕ , is certainly cascaded to smaller scales. The presence of a molecular diffusion does not substantially affect the requirement that ϕ be conserved on parcels, because on scales comparable to the eddy mixing length the effect of molecular diffusion is negligible. (And if it were not, perhaps because κ were extremely large or because the turbulence were anaemic, we would not be particularly interested in the turbulent transport.)

13.4.2 A macroscopic perspective

Consider turbulent diffusion from a more macroscopic point of view, and in particular consider the transport of a nearly materially conserved tracer obeying

$$\frac{D\phi}{Dt} = D, \quad (13.66)$$

where the advecting flow is incompressible and D is a dissipative process such that $\overline{D\phi} \leq 0$ (a conventional harmonic diffusion has this property). By decomposing the fields into mean and eddy components in the usual way an equation for the evolution of the tracer variance can be straightforwardly derived, namely

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{\phi'^2} + \overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} + \frac{1}{2} \bar{\mathbf{v}} \cdot \nabla \overline{\phi'^2} + \frac{1}{2} \nabla \cdot \overline{\mathbf{v}'\phi'^2} = \overline{D'\phi'}, \quad (13.67)$$

and where we may assume $\overline{D'\phi'} < 0$. If the mean flow is small and if the third-order term may be neglected then in a statistically steady state we have

$$\overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} \approx \overline{D'\phi'} < 0. \quad (13.68)$$

Therefore, on average, the flux of ϕ is downgradient in regions of dissipation, implying a positive average eddy diffusivity, and a balance is maintained between the downgradient flux of ϕ (which increases the variance) and dissipation. However, it should also be clear from (13.67) that if the turbulence is not statistically stationary, or if there is a mean flow, then downgradient transport cannot necessarily be expected. Indeed, the transport may be upgradient in regions where the eddy variance is falling, for then we may have the balance

$$\overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} \approx -\frac{1}{2} \frac{\partial}{\partial t} \overline{\phi'^2} > 0. \quad (13.69)$$

13.5 HOMOGENIZATION OF A SCALAR THAT IS ADVECTED AND DIFFUSED

Let us now assume that the effects of turbulence on a tracer are indeed diffusive. An important consequence of this is that, in the absence of additional forcing, there can be no extreme values of the tracer in the interior of the fluid and, in some circumstances, the diffusion will *homogenize* values of the tracer in broad regions. In this section we demonstrate and explore these properties.

13.5.1 Non-existence of extrema

Consider a tracer that obeys the equation

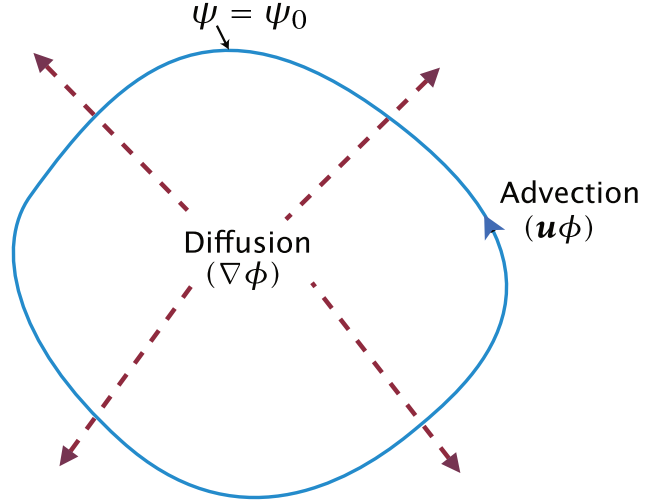
$$\frac{D\phi}{Dt} = \nabla \cdot (\kappa \nabla \phi) + S, \quad (13.70)$$

where $\kappa > 0$ and the advecting velocity is divergence-free. We now show that in regions where the source term, S , is zero there can be no interior extrema of ϕ if the flow is steady. The proof is in the form of a *reductio ad absurdum* argument — we first suppose there is an extrema of ϕ in the fluid, and prove a contradiction.

Given an extremum, there will then be a surrounding surface (in three dimensions), or a surrounding contour (in two), connecting constant values of ϕ . For definiteness consider two-dimensional incompressible flow for which the steady flow satisfies

$$\nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (\kappa \nabla \phi). \quad (13.71)$$

Fig. 13.5 If an extremum of a tracer ϕ exists in the fluid interior, then diffusion will provide a downgradient tracer flux. But over an area bounded by a streamline, or by an isoline of ϕ , the net advective flux is zero. Thus, the diffusion cannot be balanced by advection and so in a steady state no extrema can exist.



Integrating the left-hand side over the area, A , enclosed by the iso-line of ϕ , and applying the divergence theorem, gives

$$\iint_A \nabla \cdot (\mathbf{u}\phi) dA = \oint (\mathbf{u}\phi) \cdot \mathbf{n} dl = \phi \oint \mathbf{u} \cdot \mathbf{n} dl = \phi \iint_A \nabla \cdot \mathbf{u} dA = 0. \quad (13.72)$$

where $\mathbf{n}$ is a unit vector normal to the contour. (If we were to integrate over an area bounded by a velocity contour, then $\mathbf{u} \cdot \mathbf{n} = 0$ and the integral would similarly vanish.) But the integral of the right-hand side of (13.71) over the same area is non-zero; that is

$$\iint_A \nabla \cdot (\kappa \nabla \phi) dA = \oint \kappa \nabla \phi \cdot \mathbf{n} dl \neq 0, \quad (13.73)$$

if the integral surrounds an extremum. This is a contradiction for steady flow. Hence, there can be no isolated extrema of a conserved quantity in the interior of a fluid, if there is any diffusion at all. The result (which applies in two or three dimensions) is kinematic, in that ϕ can be any tracer at all, active or passive. The physical essence of the result is the integrated effects of diffusion are non-zero surrounding an extremum, and cannot be balanced by advection. Thus, if the initial conditions contain an extremum, diffusion will smooth away the extremum until it no longer exists. This process, and the homogenization discussed below, are illustrated in Fig. 13.5.

13.5.2 Homogenization in two-dimensional flow

For two-dimensional flow we can obtain a still stronger result if we allow ourselves to make more assumptions about the strength and nature of the diffusion. The steady distribution of a scalar quantity being advected by an incompressible flow is governed by

$$J(\psi, \phi) = \nabla \cdot (\kappa \nabla \phi) + S, \quad (13.74)$$

where the terms on the right-hand side represent diffusion and source terms. Suppose that these terms are small, in the sense that the individual terms on the left-hand side nearly balance each other, so that

$$|J(\psi, \phi)| \ll \frac{U\phi}{L}. \quad (13.75)$$

This means we are in the high Peclet number limit ($P = UL/\kappa \gg 1$), and the dominance of advection suggests that any steady solution to (13.74) is of the form

$$\phi = G(\psi) + \mathcal{O}(P^{-1}), \quad (13.76)$$

where G is (for the moment) any function of its argument. Thus, isolines of ϕ are nearly coincident with streamlines, and

$$\nabla\phi \approx \nabla\psi \frac{d\phi}{d\psi}. \quad (13.77)$$

On integrating (13.74) over the area, A , bounded by some closed streamline, $\psi = \psi_0$ say, the left-hand side vanishes and we obtain

$$0 = \iint_A S dA + \oint_{\psi_0} \kappa \nabla\phi \cdot \mathbf{n} d\mathbf{l}. \quad (13.78)$$

Using (13.77) then gives

$$\iint_A S dA = - \oint_{\psi_0} \kappa \frac{d\phi}{d\psi} \nabla\psi \cdot \mathbf{n} d\mathbf{l}. \quad (13.79)$$

Since $d\phi/d\psi$ is constant along streamlines, and using $\mathbf{u} = \nabla^\perp\psi$, we have

$$\frac{d\phi}{d\psi} = - \frac{\iint_A S dA}{\oint_{\psi_0} \kappa \mathbf{u} \cdot d\mathbf{l}}. \quad (13.80)$$

This relationship determines ϕ as a function of ψ — that is, it determines $G(\psi)$ — in terms of the forcing and dissipation acting on the fluid. If the fluid is both unforced and inviscid, then a steady solution pertains when ϕ is an *arbitrary* function of ψ . If the source term S is zero, but dissipation is non-zero then the denominator of (13.80) is non-zero and therefore ϕ must be uniform: ϕ has been *homogenized*. The homogenization result also follows if we choose to integrate over an area surrounded by an isoline of ϕ , ϕ_0 say, but we leave this as a problem for the reader.

Interpretation

The homogenization result applies to a statistically steady flow in which the eddy transport of ϕ -stuff by the eddying motion may be parameterized diffusively, and in which there is an approximate functional relationship between mean ϕ and mean ψ . The first of these assumptions we have discussed at length in previous sections. The second requires that the diffusion must not be too strong, so that locally the tracer is conserved on fluid parcels. In the steady state the tracer is then a function of the streamfunction, the same function everywhere within the closed region.

Given these assumptions, the dynamics giving rise to homogenization is transparent: integrating round a contour of ψ or ϕ the effect of the advective terms vanish; the source (S) and the diffusion must balance each other, and if there is no source term there can be no tracer gradient. Put another way, the flow will circulate endlessly and steadily around the contours of ψ , which nearly coincide with contours of ϕ . Advection cannot alter the mean value of ϕ , so diffusion smooths out gradients within the closed contours, effectively *expelling* gradients of ϕ to the boundaries and forming a plateau of ϕ -values. Because extrema of ϕ are forbidden, the value of ϕ on the plateau cannot be a maximum or minimum: at the edge of the plateau the values of ϕ must fall somewhere, and rise somewhere else. The plateau can be a flat region etched out of a hillside, but a plateau on top of a butte is forbidden, for in that case diffusion would erode the butte down to the level of the surrounding land. Our derivation makes no distinction between a passive scalar like a dye and an active scalar, like potential vorticity. In reality, in the latter case the dynamics will further constrain the flow because the scalar distribution must be consistent with the velocity field that advects it, and this is particularly important in the dynamics of ocean gyres.

13.6† TRANSPORT BY BAROCLINIC EDDIES

In the next few sections we discuss the transport of fluid properties by large-scale eddies typically generated by baroclinic instability — mesoscale eddies in the ocean, and weather systems in the atmosphere. Our motivation is twofold.

1. Mesoscale eddies in the ocean cannot be easily resolved in numerical models of its large-scale circulation, especially those used for climate simulations involving integrations of the global ocean over decades and centuries. In such models, the effects of eddies must be *parameterized* in terms of properties of the mean flow.
2. Eddy–mean flow interaction is simply a key part of the dynamics of the atmosphere and the ocean.

The first of these will, before too long, be solved for us by the increasing power of computers, and of course in the atmosphere numerical models of the general circulation already resolve most of the effects of baroclinic eddies. However, as computer power increases and we resolve mesoscale eddies, item (i) is replaced with the hardly less difficult problem of understanding those massive, turbulent, numerical integrations, and for that task a theory of turbulent transport is a *sine qua non*.

13.6.1 Symmetric and antisymmetric diffusivity tensors

A tracer evolving freely save for the effects of molecular diffusivity obeys the equation

$$\frac{D\phi}{Dt} = \nabla \cdot (\kappa_m \nabla \phi), \quad (13.81)$$

where κ_m is the molecular diffusivity, a positive scalar quantity. In the more general case we have

$$\frac{D\phi}{Dt} = -\nabla \cdot \mathbf{F} = \nabla \cdot \mathbf{K} \nabla \phi, \quad (13.82)$$

where $\mathbf{K}$ is (if ϕ is a scalar) a second-rank tensor and $\mathbf{F} = -\mathbf{K} \nabla \phi$ is the diffusive flux of ϕ . (We will omit the identifying field in square brackets if no ambiguity is so caused.) The flux has a component across the isosurfaces of ϕ , called the diffusive flux, and a component along the isosurfaces, called the skew flux [cf. (??) and (??)]. We will see below that these fluxes are associated with the symmetric and antisymmetric components of the diffusivity tensor, respectively, where

$$\mathbf{K} = \mathbf{S} + \mathbf{A}, \quad (13.83)$$

and, using component notation,

$$S_{mn} = \frac{1}{2}(K_{mn} + K_{nm}), \quad A_{mn} = \frac{1}{2}(K_{mn} - K_{nm}). \quad (13.84)$$

The diagonal elements of the antisymmetric tensor are zero. The transport that is effected by these two tensors has different physical characteristics, as we now discuss.

13.6.2 ♦ Diffusion with the symmetric tensor

In the simplest case of all, with an isotropic medium, $\mathbf{K}$ is diagonal with equal entries,

$$\mathbf{K} = \mathbf{S} = \begin{pmatrix} \kappa & 0 & 0 \\ 0 & \kappa & 0 \\ 0 & 0 & \kappa \end{pmatrix}, \quad (13.85)$$

and we have the familiar $\mathbf{F} = -\kappa \nabla \phi$, and (13.82) has the same form as (13.81). If κ is positive, then the flux is *downgradient*, meaning that

$$\mathbf{F} \cdot \nabla \phi < 0, \quad (13.86)$$

even if κ is spatially non-uniform. Furthermore, such a diffusion is variance-dissipating; to see this, suppose we have the equation of motion

$$\frac{D\phi}{Dt} = \nabla \cdot (\kappa \nabla \phi). \quad (13.87)$$

Multiplying by ϕ and integrating over the domain V gives

$$\frac{1}{2} \frac{d}{dt} \int_V \phi^2 dV = \int_V \mathbf{F} \cdot \nabla \phi dV = - \int_V \kappa (\nabla \phi)^2 dV \leq 0, \quad (13.88)$$

after an integration by parts and assuming that the normal derivative of ϕ vanishes at the boundaries; that is, there is no flux of ϕ -stuff through the boundary. However, diffusion does preserve the first moment of the field; that is

$$\frac{d}{dt} \int_V \phi dV = \int_V \nabla \cdot (\kappa \nabla \phi) dV = 0, \quad (13.89)$$

again assuming no flux through the boundaries.

The transport that is effected by the symmetric diffusion tensor is the diffusive flux, $\mathbf{F}_d$, where

$$\mathbf{F}_d = -\mathbf{S} \nabla \phi = -S_{mn} \partial_n \phi, \quad (13.90)$$

where we employ the common convention that repeated indices are summed. In general, the flux has a component that is parallel to the tracer gradient; that is, $\mathbf{F}_d \cdot \nabla \phi \neq 0$. Suppose, for simplicity, we have the equation of motion

$$\frac{\partial \phi}{\partial t} = -\nabla \cdot \mathbf{F}_d = \nabla \cdot (\mathbf{S} \nabla \phi). \quad (13.91)$$

This equation preserves the first moment of ϕ , provided there is no flux through the boundary. Tracer variance evolves according to

$$\frac{1}{2} \frac{\partial}{\partial t} \int_V \phi^2 dV = \int_V \phi \nabla \cdot (\mathbf{S} \nabla \phi) = - \int_V (\mathbf{S} \nabla \phi) \cdot \nabla \phi dV. \quad (13.92)$$

This can be shown to be negative or zero, provided that $\mathbf{S}$ is positive semi-definite, meaning that

$$\nabla \phi \mathbf{S} \nabla \phi = \partial_m \phi S_{mn} \partial_n \phi \geq 0. \quad (13.93)$$

The flux effected by such a diffusivity is then downgradient in the sense that

$$\mathbf{F}_d \cdot \nabla \phi = -\mathbf{S} \nabla \phi \cdot \nabla \phi \leq 0. \quad (13.94)$$

13.6.3 ♦ The skew flux

The transport associated with the antisymmetric transport tensor is perpendicular to the gradient of ϕ , and so is neither upgradient nor downgradient. The flux is

$$\mathbf{F}_{sk} = -\mathbf{A} \nabla \phi = -A_{mn} \partial_n \phi \quad (13.95)$$

and thus

$$\mathbf{F}_{sk} \cdot \nabla \phi = -\mathbf{A} \nabla \phi \cdot \nabla \phi = -A_{mn} \partial_n \phi \partial_m \phi = 0, \quad (13.96)$$

where the final result follows because of the antisymmetry of $\mathbf{A}$ — the contraction of a symmetric tensor and an antisymmetric tensor is zero.¹¹ For this reason, the associated transport is known as a *skew flux* (a term applying in general to fluxes that are perpendicular to the tracer gradient) or a *skew diffusion* (when those fluxes are parameterized using an antisymmetric diffusivity). It follows from this that if a tracer obeys

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (\mathbf{A} \nabla \phi), \quad (13.97)$$

then the tracer variance is conserved. This may be verified by multiplying this equation by ϕ and integrating by parts, assuming that the flux vanishes at the boundaries. That is, *a skew diffusion has no effect on the variance of the skew diffused variable*. One other familiar physical process shares these properties, and that is advection by a divergence-free flow. A skew diffusion is physically equivalent to such an advection in that the divergence of a skew diffusive flux is the same as the divergence of an appropriately chosen advective flux. We proved this in section ??, but let us look at it again from the point of view of diffusion.

We define an advective flux of a tracer ϕ to be a flux of the form

$$\mathbf{F}_{ad} \equiv \tilde{\mathbf{v}} \phi, \quad (13.98)$$

where $\tilde{\mathbf{v}}$ is a divergence-free vector field. The divergence of the flux is just

$$\nabla \cdot \mathbf{F}_{ad} = \nabla \cdot (\tilde{\mathbf{v}} \phi) = \tilde{\mathbf{v}} \cdot \nabla \phi. \quad (13.99)$$

The field $\tilde{\mathbf{v}}$ might be called a *pseudoveLOCITY* or a *quasi-velocity* — it acts like a velocity but is not necessarily the velocity of any fluid particle. Because $\tilde{\mathbf{v}}$ is divergence-free, we may, as in section ??, define a vector streamfunction $\boldsymbol{\psi}$ such that

$$\tilde{\mathbf{v}} = \nabla \times \boldsymbol{\psi}. \quad (13.100)$$

The field $\boldsymbol{\psi}$ is not unique: the gradient of an arbitrary function may be added to it, this gradient vanishing upon taking the curl, resulting in the same velocity field. That is, if $\boldsymbol{\psi}' = \boldsymbol{\psi} + \nabla \gamma$, then $\tilde{\mathbf{v}} = \nabla \times \boldsymbol{\psi} = \nabla \times \boldsymbol{\psi}'$. The scalar field γ is known as the *gauge*, and the freedom to choose it is the gauge freedom.

The advective flux $\mathbf{F}_{ad}$ is related to the skew flux $\mathbf{F}_{sk}$ by

$$\phi \tilde{\mathbf{v}} = \phi \nabla \times \boldsymbol{\psi} = \nabla \times (\phi \boldsymbol{\psi}) - \nabla \phi \times \boldsymbol{\psi}, \quad (13.101)$$

or

$$\mathbf{F}_{ad} = \mathbf{F}_r + \mathbf{F}_{sk}, \quad (13.102)$$

where $\mathbf{F}_r = \nabla \times (\phi \boldsymbol{\psi})$ is a rotational flux with no divergence, and

$$\mathbf{F}_{sk} = -\nabla \phi \times \boldsymbol{\psi} \quad (13.103)$$

is a skew flux — ‘skewed’ because it is manifestly orthogonal to the gradient of ϕ , i.e., $\nabla \phi \cdot \mathbf{F}_{sk} = 0$. Because $\nabla \cdot \mathbf{F}_r = 0$ we have

$$\nabla \cdot \mathbf{F}_{ad} = \nabla \cdot \mathbf{F}_{sk}. \quad (13.104)$$

However, the skew flux, $-\nabla \phi \times \boldsymbol{\psi}$, and the advective flux, $\phi \nabla \times \boldsymbol{\psi}$, may have, and in general do have, different magnitudes and directions; only their divergences are equal. If the divergence of the skew

fluxes given by (13.95) and (13.103) are to be the same then ψ must be related to the antisymmetric tensor $\mathbf{A}$. Using (13.95) we have

$$\begin{aligned}\nabla \cdot \mathbf{F}_{sk} &= -\partial_m (A_{mn} \partial_n \phi) \\ &= -(\partial_n \phi)(\partial_m A_{mn}) - [A_{mn} \partial_n \partial_m \phi] \\ &= -\partial_n (\phi \partial_m A_{mn}) + [\phi \partial_n \partial_m A_{mn}],\end{aligned}\tag{13.105}$$

where the quantities in square brackets are zero as a consequence of the antisymmetry of $\mathbf{A}$ — a symmetric operator acting on an antisymmetric tensor is zero. But the skew flux divergence is equal to the advective flux divergence

$$\nabla \cdot \mathbf{F}_{sk} = \nabla \cdot \mathbf{F}_{ad} = \partial_n (\phi \tilde{v}_n),\tag{13.106}$$

so the associated skew velocity is given by

$$\tilde{v}_n = -\partial_m A_{mn},\tag{13.107}$$

and this is divergence-free because $\partial_n \partial_m A_{mn} = 0$. The streamfunction and the antisymmetric tensor are thus related, because

$$\tilde{v}_n = \epsilon_{lmn} \partial_l \psi_m.\tag{13.108}$$

This expression is equivalent to $\tilde{\mathbf{v}} = \nabla \times \boldsymbol{\psi}$, as can be verified by expanding the expression out in Cartesian coordinates. The Levi-Civita symbol ϵ_{mnp} is such that $\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$, $\epsilon_{132} = \epsilon_{321} = \epsilon_{213} = -1$, and $\epsilon_{mnp} = 0$ for other combinations.

Using (13.107) and (13.108), and just a little algebra, gives

$$A_{mn} = \epsilon_{mnp} \psi_p = \begin{pmatrix} 0 & \psi_3 & -\psi_2 \\ -\psi_3 & 0 & \psi_1 \\ \psi_2 & -\psi_1 & 0 \end{pmatrix}.\tag{13.109}$$

This expression provides a connection between the antisymmetric tensor A_{mn} and the streamfunction for the skew velocity $\tilde{\mathbf{v}}$.

13.6.4 The story so far

In this chapter we are making a rather large assumption — that we can parameterize turbulent fluxes via some form of diffusive parameterization, but given that let us be reasonably careful how we do it. Thus far, we have established the following.

- Any flux can be decomposed into a component across iso-surfaces of a scalar (the along-gradient or diffusive flux) and a component along isosurfaces and so perpendicular to the gradient (the skew flux).
- The along-gradient (usually downgradient) flux is effected by a diffusion using a symmetric diffusivity tensor.
- The skew flux is effected by a diffusion using an antisymmetric diffusivity tensor, and this is equivalent to an advection by some divergence-free velocity.
- The diffusive flux reduces tracer variance if the diffusivity is positive (in which case the diffusion is downgradient), whereas the skew diffusion has no effect on variance.

Let us now consider how all of this is relevant to the large-scale flow in the atmosphere and the ocean.

13.7 † EDDY DIFFUSION IN THE ATMOSPHERE AND OCEAN

Ah, but a man's reach should exceed his grasp, Or what's a heaven for?

Robert Browning, *Andrea del Sarto*, 1855.

13.7.1 Preliminaries

Consider a tracer that obeys the equation

$$\frac{D\phi}{Dt} = \nabla \cdot (\kappa_m \nabla \phi), \quad (13.110)$$

where κ_m is the molecular diffusivity. If the advecting flow is divergence-free then the ensemble average flow obeys, neglecting the molecular diffusion,

$$\frac{D\bar{\phi}}{Dt} = -\nabla \cdot \bar{\mathbf{v}'\phi'}. \quad (13.111)$$

We can prepare the ensemble by, for example, varying the initial conditions slightly; if the flow is chaotic, the realizations will differ from each other. If we parameterize the eddy transport by a diffusion then

$$\overline{\mathbf{v}'\phi'} = -\mathbf{K}\nabla\bar{\phi}, \quad (13.112)$$

where $\mathbf{K}$ is, in general, a second-rank tensor. If the average is a zonal average then

$$\frac{D\bar{\phi}}{Dt} = -\frac{\partial \overline{v'\phi'}}{\partial y} - \frac{\partial \overline{w'\phi'}}{\partial z}. \quad (13.113)$$

If we are to employ a diffusive parameterization for the eddy terms in these equations, the issues that then arise fall into two general camps:

- (i) the overall *magnitude* of the eddy diffusivity, possibly as a function of the mean flow;
- (ii) the *structure* of the diffusivity tensor, and in particular the separate structure of its symmetric and antisymmetric parts.

13.7.2 Magnitude of the eddy diffusivity

The magnitude and scale of eddies were previously considered in sections 12.3 and 12.4. Here we see how these give rise to corresponding estimates of the magnitude of an eddy diffusivity. If we restrict attention for the moment to the meridional transfer of tracer properties, then we might write

$$\overline{v'\phi'} = -\kappa^{vy} \frac{\partial \bar{\phi}}{\partial y} - \kappa^{vz} \frac{\partial \bar{\phi}}{\partial z}, \quad (13.114)$$

where κ^{vy} and κ^{vz} are components of the eddy diffusivity tensor. These components have the dimensions of a length times a velocity and, to the extent that the diffusion represents the eddying motion we expect that κ^{vy} has an approximate magnitude of

$$\kappa^{vy} \sim v'l', \quad (13.115)$$

where v' is a typical magnitude of the horizontal eddy velocity, and l' is *mixing length* of the eddies, generally taken to be a typical length scale of the eddies. Larger and more energetic eddies thus have a larger effect on the mean flow. We can estimate v' and l' in a number of reasonable ways depending on the flow conditions, and we consider a few such below.¹² The magnitude of the

component κ^{vz} may then be estimated by making choices about the plane of parcel displacements, and this is considered in the next section.

Perhaps the simplest assumption to make follows from the fact that the eddies are a consequence of baroclinic instability, and so one might suppose that the eddy length scale is the scale of the instability — the first deformation radius. One might also suppose that the eddy velocity is of the same approximate magnitude as the mean flow, $\bar{u}$, thus giving

$$\kappa^{vy} \sim L_d \bar{u} = \frac{NH\bar{u}}{f}. \quad (13.116)$$

Another way of deriving this result is by noting that $\kappa^{vy} \sim l'^2/T_e$, where T_e is a characteristic eddy time scale. If T_e is the Eady time scale, $L_d/\bar{u}$, and if $l' \sim L_d$, we reproduce (13.116). Note also that (13.116) may be written as

$$\kappa^{vy} \sim L_d \bar{u} \sim \frac{L_d^2 f}{\sqrt{Ri}} \sim L_d^2 Fr f, \quad (13.117)$$

where $Ri \equiv N^2/\Lambda^2 = N^2 H^2/\bar{u}^2$ and $Fr \equiv U/(NH)$ are the Richardson and Froude numbers for this problem, respectively.

A little more generally, if there is a cascade to larger scales then the eddy scale, L_e say, may be larger than the deformation scale. Depending on circumstances, L_e might be the domain scale (if eddies grow to the size of the domain), the β -scale (if the β -effect halts the cascade), or some scale determined by frictional effects (possibly in conjunction with β). However, the arguments of section 12.3 suggest that the eddy time scale is the Eady time scale in all cases. We therefore have:

$$\text{eddy length scale} \sim L_e \quad (13.118a)$$

$$\text{eddy time scale, } T_e \sim L_d/\bar{u} \quad (13.118b)$$

$$\text{eddy velocity scale, } U_e \sim \bar{u}(L_e/L_d). \quad (13.118c)$$

These give the general estimate for the horizontal diffusivity of

$$\kappa^{vy} \sim \bar{u} \left(\frac{L_e^2}{L_d} \right). \quad (13.119)$$

The estimate (13.116) is a special case of this, with $L_e = L_d$; the two estimates will thus differ if the eddy scale is much larger than the deformation radius.

In the case in which the inverse cascade is modified by the Rossby waves we might (and neglecting friction effects; see section 12.1.2) suppose that the eddy scale is the β -scale, (12.6), and we have

$$L_e \sim L_\beta = \left(\frac{U_e}{\beta} \right)^{1/2} = \frac{\bar{u}}{\beta L_d}, \quad (13.120)$$

using (13.118c). The eddy velocity scale is, using (13.118b),

$$U_e \sim \bar{u} \frac{L_e}{L_d} = \frac{\bar{u}^2}{\beta L_d^2}, \quad (13.121)$$

and combining (13.120) and (13.121) gives the estimate for the eddy diffusivity,

$$\kappa^{vy} \sim \frac{\bar{u}^3}{\beta^2 L_d^3}. \quad (13.122)$$

A similar estimate be written in terms of the inverse energy cascade rate, ε , giving

$$\kappa^{\nu y} \sim \left(\frac{\varepsilon^3}{\beta^4} \right)^{1/5}. \quad (13.123)$$

This expression may be obtained purely by dimensional analysis, if it is assumed that the only factors determining κ are ε and β . The estimate may be useful if ε is known independently, for example by calculating the energy throughput in the atmosphere.

To summarize: the magnitude of any eddy diffusion may be estimated as the product of the velocity scale and the energy-containing length scale of the eddies. If we assume that the time scale is the Eady time scale we obtain (13.119), where L_e is undetermined. If the eddy scale is the β -scale, then (13.119) becomes (13.122). However, in neither the atmosphere nor the ocean is the β -scale significantly (e.g., an order of magnitude) larger than the deformation scale, but nor, complicating matters, does the inverse cascade necessarily halt at the β -scale (see section 12.1.2). From an observational standpoint, the atmosphere has no $-5/3$ inverse cascade, although there is some evidence for one in some regions of ocean.¹³ In other oceanic regions the eddies may be advected away from each other and away from the unstable zone, or dispersed by Rossby waves, before an inverse cascade can be organized, and energy will remain at the deformation scale. These arguments suggest that although we can make sensible estimates, we cannot determine with certainty what the magnitude of an eddy diffusivity should be, in either the atmosphere or ocean.

13.7.3 ♦ Structure: the symmetric transport tensor

Along-gradient transport, or diffusion, is transport by a symmetric transport tensor as in (13.90). To explore its structure, let us consider transfer in a re-entrant channel with zonally homogeneous eddy statistics so that the averaging operator is the zonal average; the meridional and upward transport of tracer are then given by

$$\overline{v'\phi'} = -\kappa^{\nu y} \frac{\partial \bar{\phi}}{\partial y} - \kappa^{\nu z} \frac{\partial \bar{\phi}}{\partial z} \quad (13.124)$$

$$\overline{w'\phi'} = -\kappa^{\omega y} \frac{\partial \bar{\phi}}{\partial y} - \kappa^{\omega z} \frac{\partial \bar{\phi}}{\partial z}, \quad (13.125)$$

where $\kappa^{\omega y} = \kappa^{\nu z}$ by the posited symmetry. The relationship between the various transfer coefficients will be determined by the trajectories of the fluid parcels in the eddying motion. In the Cartesian y - z frame the transport tensor is not necessarily diagonal (i.e., $\kappa^{\nu z}$ and $\kappa^{\omega y}$ may be non-zero) but locally there is always a natural coordinate system in which the diffusivity tensor is diagonal. (A symmetric matrix may always be diagonalized by a suitable rotation of axes.) In that diagonal frame we can write

$$S' = \kappa_s \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}, \quad (13.126)$$

where κ_s determines the overall size of the transfer coefficients (as estimated in the previous section), and α is the ratio of sizes of the components in the two orthogonal directions. Now, fluid displacements in large-scale baroclinic eddies are nearly, but not exactly, horizontal — they may be along isopycnals, for example, or at an angle between the horizontal and the isopycnals. We may argue that the coordinate system in which the tensor is diagonal is the coordinate system defined by the plane along which fluid displacements occur. This is sensible because the transfers along and orthogonal to the fluid paths are each a consequence of different physical phenomena, and so we may expect the transfer tensor to be diagonal in these coordinates.

Because eddy displacements are predominantly horizontal, the diagonal coordinate system has a small slope, s , at an angle θ with respect to the horizontal, where $s = \tan \theta \approx \theta \ll 1$. Furthermore,

we expect the parameter α to be small (i.e., $\alpha \ll 1$), because this represents transfer in a direction orthogonal to the eddy fluid motion. We rotate the tensor $\mathbf{S}'$ through an angle θ to move into the usual y - z frame; that is

$$\mathbf{S} = \kappa_s \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \quad (13.127a)$$

$$\approx \kappa_s \begin{pmatrix} 1 + s^2 \alpha & s(1 - \alpha) \\ s(1 - \alpha) & s^2 + \alpha \end{pmatrix} \quad (\text{for small } s) \quad (13.127b)$$

$$\approx \kappa_s \begin{pmatrix} 1 & s \\ s & s^2 + \alpha \end{pmatrix} \quad (\text{for small } s \text{ and small } \alpha). \quad (13.127c)$$

We can follow the same procedure in three dimensions. Then, if the eddy transport is isotropic in the plane of eddy displacements, the three-dimensional transport tensor is

$$\mathbf{S}' = \kappa_s \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \alpha \end{pmatrix}, \quad (13.128)$$

and the slope of the motion is a two-dimensional vector $\mathbf{s} = (s^x, s^y)$, with the superscripts denoting components. If we rotate the transport tensor into physical space then we obtain, analogously to (13.127),

$$\mathbf{S} = \kappa_s \begin{pmatrix} 1 + s^{y2} + \alpha s^{x2} & (\alpha - 1)s^x s^y & (1 - \alpha)s^x \\ (\alpha - 1)s^x s^y & 1 + s^{x2} + \alpha s^{y2} & (1 - \alpha)s^y \\ (1 - \alpha)s^x & (1 - \alpha)s^y & \alpha + s^2 \end{pmatrix} \quad (13.129a)$$

$$\approx \kappa_s \begin{pmatrix} 1 & 0 & s^x \\ 0 & 1 & s^y \\ s^x & s^y & \alpha + s^2 \end{pmatrix}, \quad (13.129b)$$

for small s and small α , where $s^2 = s^{x2} + s^{y2}$.

The plane of eddy displacements

We are now in a position to make heuristic choices about the transfer coefficients, and we will consider two bases for this.

1. *Using linear baroclinic instability theory.*¹⁴ In a simple model of a growing baroclinic (Eady) wave, parcel trajectories that are along half the slope of the mean isopycnals are able to release the most potential energy. We thus suppose that $s = s_\rho/2$, where s_ρ is the isopycnal slope, and that $\alpha = 0$ in (13.127c) or (13.129b). In two dimensions this gives

$$\mathbf{S} = \kappa_s \begin{pmatrix} 1 & s_\rho/2 \\ s_\rho/2 & s_\rho^2/4 \end{pmatrix} \quad (13.130)$$

and so

$$\overline{v'\phi'} = -\kappa_s \left(\frac{\partial \bar{\phi}}{\partial y} + \frac{1}{2} s_\rho \frac{\partial \bar{\phi}}{\partial z} \right) \quad (13.131a)$$

$$\overline{w'\phi'} = -\frac{1}{2} \kappa_s s_\rho \left(\frac{\partial \bar{\phi}}{\partial y} + \frac{1}{2} s_\rho \frac{\partial \bar{\phi}}{\partial z} \right). \quad (13.131b)$$

If the tracer ϕ is potential temperature (and not just a passive tracer) then (13.131), along with one of the estimates for the size of κ_s given in section 13.7.2, constitutes a parameterization for the diffusive poleward and upward heat flux in the atmosphere.

II. *Flow along neutral surfaces.* If the fluid interior is adiabatic and steady, then fluid trajectories are along neutral surfaces; that is, along surfaces of potential density or potential temperature. One might therefore be inclined to assume that the eddy fluxes are aligned along the mean neutral surfaces and choose $s = s_\rho$. However, even in the adiabatic case, this is not always a good choice. From the adiabatic thermodynamic equation $Db/Dt = 0$ we may derive the equation for the eddy buoyancy variance, namely

$$\begin{aligned} \frac{1}{2} \frac{\partial \overline{b'^2}}{\partial t} + \frac{1}{2} \overline{\mathbf{u}} \cdot \nabla_z \overline{b'^2} + \frac{1}{2} \overline{w} \frac{\partial \overline{b'^2}}{\partial z} + \overline{\mathbf{u}'b'} \cdot \nabla_z \overline{b} + \overline{w'b'} \frac{\partial \overline{b}}{\partial z} \\ + \frac{1}{2} \nabla_z \cdot \overline{\mathbf{u}'b'^2} + \frac{1}{2} \frac{\partial}{\partial z} \overline{w'b'^2} = 0, \end{aligned} \quad (13.132)$$

and specialize to the case of a zonally uniform basic state and small-amplitude wave. In that case

$$\frac{1}{2} \frac{\partial \overline{b'^2}}{\partial t} = -\overline{v'b'} \frac{\partial \overline{b}}{\partial y} - \overline{w'b'} \frac{\partial \overline{b}}{\partial z}. \quad (13.133)$$

If the wave is statistically steady then the left-hand side is zero and

$$\overline{\mathbf{v}'_m b'} \cdot \nabla_x \overline{b} = 0, \quad (13.134)$$

where the subscript x indicates that the vectors are in the meridional plane, with no variation in x . In this case there is indeed no along-gradient flux. However, if the wave is growing then $\overline{\mathbf{v}'_m b'} \cdot \nabla_x \overline{b} < 0$ and if $\partial \overline{b}/\partial y < 0$ and $\overline{v'b'} > 0$, as in the Northern Hemisphere, then

$$\frac{\overline{w'b'}}{\overline{v'b'}} > -\frac{\partial \overline{b}/\partial y}{\partial \overline{b}/\partial z}, \quad (13.135)$$

and so the mixing slope is *less* steep than the mean isopycnal slope, even though the flow may be adiabatic. (The mixing slope result also holds if $\overline{v'b'} < 0$, although the sign of the inequality is different.) Similarly, if the wave is decaying the mixing slope is steeper than that of the mean isopycnals. In an inhomogeneous flow, the advection by the mean flow in (13.132) plays a similar role to time dependence: the advection of eddy variance by the mean flow into a region of larger variance will give rise to a mixing slope that is less steep than the isopycnal slope, and conversely for a flow entering a region of less variance. Only for a statistically steady, adiabatic, linear wave field is the mixing slope guaranteed to be along the isopycnals.

Having said all this, let us proceed by assuming that the fluid trajectories are indeed along neutral surfaces. If there is no diffusion orthogonal to this $\alpha = 0$, and the transport tensor is

$$\mathbf{S} = \kappa_s \begin{pmatrix} 1 & 0 & s_\rho^x \\ 0 & 1 & s_\rho^y \\ s_\rho^x & s_\rho^y & |s_\rho|^2 \end{pmatrix}, \quad (13.136)$$

or in two dimensions

$$\mathbf{S} = \kappa_s \begin{pmatrix} 1 & s_\rho^2 \\ s_\rho & s_\rho^2 \end{pmatrix}, \quad (13.137)$$

and in this case

$$\overline{v'\phi'} = -\kappa_s \left(\frac{\partial \overline{\phi}}{\partial y} + s_\rho \frac{\partial \overline{\phi}}{\partial z} \right), \quad \overline{w'\phi'} = -\kappa_s s_\rho \left(\frac{\partial \overline{\phi}}{\partial y} + s_\rho \frac{\partial \overline{\phi}}{\partial z} \right). \quad (13.138a,b)$$

Suppose that ϕ is potential temperature θ , and that surfaces of potential temperature do define neutral surfaces. Then plainly eddy motion along potential temperature surfaces does not transfer potential temperature, and the diffusion defined by (13.138) should have no effect. The equations themselves respect this, for then

$$s_\rho = -\frac{\partial_y \bar{\theta}}{\partial_z \bar{\theta}} \quad (13.139)$$

and using this in (13.138) gives

$$\overline{v'\theta'} = 0, \quad \overline{w'\theta'} = 0. \quad (13.140a,b)$$

In the real ocean the presence of salinity means that the potential temperature, potential density and salinity surfaces are, in general, not parallel and there will be eddy diffusion of θ and S (salinity) along neutral surfaces. However, this does *not* provide a parameterization for the heat flux by baroclinic eddies, because we cannot expect such a flux to depend for its existence on the presence of a second tracer, salinity. For such a parameterization, we turn to the antisymmetric transport tensor.

13.7.4 ♦ Structure: the antisymmetric transport tensor

The antisymmetric transport tensor gives rise to the skew flux, or the pseudoadvection. In two dimensions (one horizontal, one vertical) we can immediately write down its form, namely

$$\mathbf{A} = \begin{pmatrix} 0 & -\kappa'_a \\ \kappa'_a & 0 \end{pmatrix}, \quad (13.141)$$

where the transfer coefficient κ'_a , which may vary in space and time depending on the flow itself, determines the overall strength of the transport. In three dimensions then by inspection we can write

$$\mathbf{A} = A_{ij} = \begin{pmatrix} 0 & 0 & -\kappa_a'^{ix} \\ 0 & 0 & -\kappa_a'^{iy} \\ \kappa_a'^{ix} & \kappa_a'^{iy} & 0 \end{pmatrix}, \quad (13.142)$$

where we have used our gauge freedom to choose that $A_{21} = -A_{12} = 0$. Note that (13.142) preserves the form of (13.141) if one of the horizontal dimensions is absent — that is, if either row one and column one, or row two and column two, is eliminated. Our remaining choice is to determine the sign and magnitude of the transport coefficients.

An adiabatic, potential-energy diminishing, eddy transport scheme

A very useful parameterization for the transport of tracers in ocean models by baroclinic eddy fluxes, commonly known as the Gent–McWilliams or GM scheme,¹⁵ can be constructed using the antisymmetric transport tensor. The maintenance of two properties is the foundation of the scheme.

- (i) Moments of the tracer should be preserved; in particular, the amount of fluid between two isopycnal surfaces should be preserved. This suggests the scheme should not diffuse buoyancy across constant-buoyancy surfaces.
- (ii) The amount of available potential energy in the flow should be reduced, so mimicking the effects of baroclinic instability, which transfers available potential energy to kinetic energy.

The first of these is automatically satisfied by using an antisymmetric diffusivity tensor. The second property can be satisfied by choosing the transfer coefficients to be proportional to the slope of the isopycnals, in which case we may write (13.142) as

$$\mathbf{A} = \kappa_a \begin{pmatrix} 0 & 0 & -s^x \\ 0 & 0 & -s^y \\ s^x & s^y & 0 \end{pmatrix}, \quad (13.143)$$

where $\mathbf{s} = (s^x, s^y) = \nabla_\rho z = -\nabla_z \rho / (\partial \rho / \partial z)$ is the isopycnal slope (recall that s^x denotes a component of a vector, and s_x a derivative) and κ_a determines the overall magnitude of the diffusivity. In an ocean model separately carrying temperature and salinity fields, then (13.143) would be applied to each of these, with the isopycnal slope being determined using the equation of state. To more easily see what properties are implied by the transport, let us specialize to the salt-free case, with buoyancy, b , the only thermodynamic variable. The isopycnal slope is then $\mathbf{s} = -(b_x/b_z, b_y/b_z)$ and the horizontal eddy buoyancy transfer $\mathbf{F}_h = (F^x, F^y)$ is given by

$$\mathbf{F}_h = - \left(-\kappa_a \mathbf{s} \frac{\partial b}{\partial z} \right) = -\kappa_a \left(\frac{\partial b}{\partial x}, \frac{\partial b}{\partial y} \right) = -\kappa_a \nabla_z b, \quad (13.144a)$$

which for positive κ_a is the same as conventional downgradient diffusion.

The vertical transfer is given by

$$F^z = -\kappa_a \left(s^x \frac{\partial b}{\partial x} + s^y \frac{\partial b}{\partial y} \right) = \kappa_a s^2 \frac{\partial b}{\partial z}, \quad (13.144b)$$

where $s^2 = \mathbf{s} \cdot \mathbf{s}$. This flux is *up* the vertical gradient; however, by construction, the total skew flux is neither upgradient nor downgradient.

The combination of the downgradient horizontal flux and the upgradient vertical flux acts to reduce the potential energy of the flow at the same time as preserving the volume of fluid within each density interval. The upgradient flux in the vertical is a consequence of the need to reduce the available potential energy: suppose warm light fluid overlays cold dense fluid in a statically stable configuration, then a downgradient vertical diffusion would raise the centre of gravity of the fluid, increasing its potential energy — just the opposite of the action of baroclinic instability. Thus, the sign on the vertical diffusivity must be negative and this, in combination with the structure of (13.143) (and so a positive horizontal diffusivity) allows both properties (i) and (ii) above to be satisfied. The parameterization does not preserve total energy; the loss of potential energy is not balanced by a corresponding gain of kinetic energy, rather it is assumed to be lost to dissipation. Finally, to determine the magnitude of the (skew) eddy diffusivity we may turn again to the phenomenological estimates of section 13.7.2.

The eddy transport velocity

Applying (13.107) to (13.143) gives the eddy transport velocities,

$$\begin{aligned} \tilde{\mathbf{u}} &= -\frac{\partial}{\partial z} (\kappa_a \mathbf{s}), \\ \tilde{w} &= \nabla_z \cdot (\kappa_a \mathbf{s}). \end{aligned} \quad (13.145)$$

The streamfunction associated with $\mathbf{A}$ is found using (13.109) and (13.143) giving

$$\psi = (-\kappa_a s^y, \kappa_a s^x, 0) = \mathbf{k} \times \kappa_a \mathbf{s}. \quad (13.146)$$

Two equivalent ways of implementing the GM parameterization are thus as a skew flux, as in (13.144), or as an advection by the pseudovelocities (13.145). The vanishing of the normal

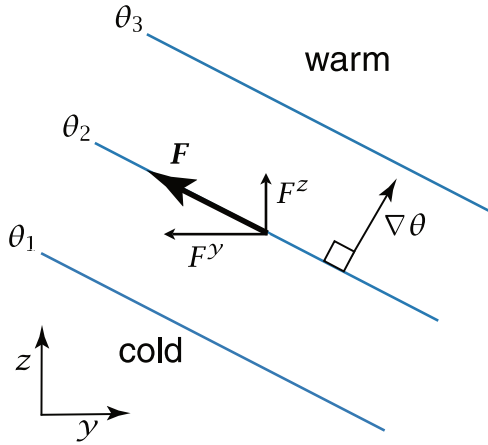


Fig. 13.6 The GM skew fluxes arising from sloping isotherms. The flux itself, F , is parallel to the isotherms, with the horizontal flux being directed down the horizontal gradient but the vertical flux being upgradient.

The effect of the vertical flux is to lower the centre of gravity of the fluid, and reduce the potential energy. The horizontal flux tries to make the temperature more uniform in the horizontal direction. The net effect of the skew flux is to *flatten* the isotherms.

component of the velocity is equivalent to the vanishing of the normal component of the flux at the boundary, and ensures that the scheme conserves tracer moments. The advective flux of buoyancy is just

$$F_{ad} = b\tilde{v} = b\nabla \times \psi = b\nabla \times (\mathbf{k} \times \kappa_a \mathbf{s}), \quad (13.147)$$

whereas using (13.103) the skew flux is given by

$$F_{sk} = -\nabla b \times \psi = -\nabla b \times (\mathbf{k} \times \kappa_a \mathbf{s}). \quad (13.148)$$

Vector manipulation readily shows that the divergences of these two fluxes are equal.

13.7.5 Examples

Consider a situation with sloping isotherms (and with the density determined solely by temperature) as illustrated in Fig. 13.6. The vertical flux attempts to tighten the temperature distribution, whereas the horizontal flux, being downgradient, attempts to smooth out horizontal inhomogeneities. Taken together, their net effect is to preserve the amount of fluid between any two isotherms, but at the same time to rotate and flatten the isotherms, so reducing the available potential energy of the flow. This is different from a conventional downgradient diffusion. A purely horizontal diffusion would, in principle, act to equalize values at each level, and a three-dimensional downgradient diffusion would try to equalize all values. Thus, a skew flux behaves quite differently from the usual downgradient diffusion, which merely acts to reduce gradients without caring much about other fluid properties.

To illustrate this consider a very simple example, that of a two-dimensional (y - z) fluid in which the initial density field is a 3×3 grid, with initial conditions

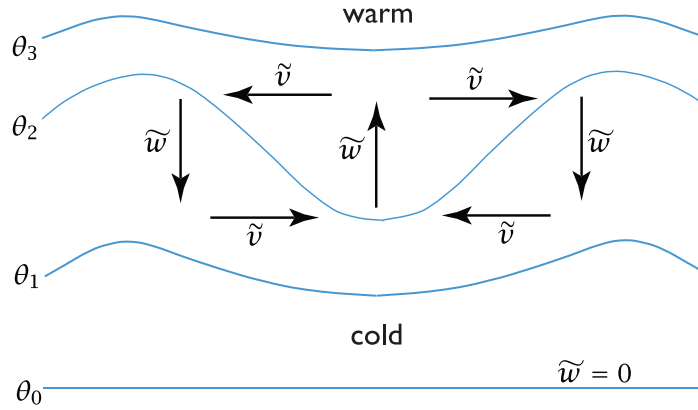
$$\rho_{init} = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 3 & 3 & 2 \end{bmatrix}. \quad (13.149)$$

The isopycnals are sloping, much as in Fig. 13.6, and the flow is statically stable everywhere.

A purely horizontal diffusion would lead to, in the absence of other processes and with zero normal flux at the boundaries, a final state of

$$\rho_{hd} = \begin{bmatrix} 1.33 & 1.33 & 1.33 \\ 2 & 2 & 2 \\ 2.66 & 2.66 & 2.66 \end{bmatrix}, \quad (13.150)$$

Fig. 13.7 The eddy-induced velocities in the Gent–McWilliams parameterization. The induced circulation attempts to flatten the sloping isopycnals. The induced vertical velocity, $\tilde{w}$, is zero on flat isopycnals.



and a full (vertical and horizontal) diffusion would give

$$\rho_{hvd} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}. \quad (13.151)$$

Neither of the above two final states preserves the density census (i.e., its distribution) and both imply strong diabatic effects — the fluid has been *mixed*, and the density variance has been reduced.

In contrast, a skew diffusion or eddy-transport advection will rotate the density surfaces until the isopycnal slope is zero, at which point the value of the transfer coefficients becomes zero and the process stops. The final state is then

$$\rho_{GM} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}. \quad (13.152)$$

This both preserves the density census and reduces the available potential energy.

We can equally well interpret these effects in terms of eddy-transport velocities, so emphasizing that it is not the eddy flux itself that is important; rather, it is the flux divergence. If the slopes of Fig. 13.6 extended uniformly everywhere, then the associated fluxes would have zero divergence, and the eddy-induced velocities, given by (13.145), would be zero. On the other hand, consider the case illustrated in Fig. 13.7, with variously sloping isotherms. For a constant value of the eddy diffusivity κ the slope of the isopycnals, $s = -(\partial\rho/\partial y)/(\partial\rho/\partial z)$, provides the streamfunction for the eddy-induced velocity:

$$\tilde{v} = -\frac{\partial\psi}{\partial z} = -\frac{\partial}{\partial z}(\kappa_a s), \quad \tilde{w} = \frac{\partial\psi}{\partial y} = \frac{\partial}{\partial y}(\kappa_a s). \quad (13.153)$$

This induces the velocities illustrated in Fig. 13.7, which evidently serve to flatten the isopycnals. (If the isopycnals steepen as they approach the surface, the effects of the scheme may be harder to interpret.)

Finally, we note that downgradient (symmetric) diffusion and skew diffusion would normally be used together, and the transport tensor will then have both symmetric and antisymmetric components. In ocean models the mixing of temperature and salinity is often chosen to be along isopycnal surfaces. On the other hand, in the atmosphere (especially in the troposphere) diabatic effects are quite important and the symmetric tensor needs to be chosen appropriately.

13.8† THICKNESS DIFFUSION

In the previous section, we considered the structure of the diffusivity tensor, and then chose the entries by physical reasoning to mimic the effects of baroclinic instability. An alternative approach is to choose, a priori, a quantity to be diffused downgradient (i.e., not skew diffused), and to then transform the equations appropriately to represent the effect in the equations of motion as commonly used. In this section we explore the use of thickness as a ‘diffusee’. Thickness is the vertical distance between two isotherms or isopycnals, and it is a candidate for diffusion because a down-gradient thickness transfer within an isopycnal layer satisfies two conditions.

- (i) the total mass contained between two isopycnals is preserved, provided there are no boundary fluxes, so the effect is adiabatic.
- (ii) If there is no orography a thickness flux serves to flatten isopycnals, and hence to reduce the available potential energy of the flow, mimicking baroclinic instability.

These are similar to the two properties listed on page 505, and indeed we will find that thickness diffusion is very similar to the GM scheme. However, thickness is *not* a materially conserved quantity; thus, the arguments of section 13.2 do not apply and so turbulent diffusion of thickness is somewhat ad hoc. Let us explore all these issues further.

13.8.1 Equations of motion

Thickness and its variance

Recalling the results of section 3.10, in a Boussinesq fluid the distance between two surfaces of constant buoyancy is given by

$$\text{thickness} = \Sigma = \int_{z(b_1)}^{z(b_2)} dz = \int_{b_1}^{b_2} \frac{\partial z}{\partial b} db, \quad (13.154)$$

and thus we may define the thickness field (strictly a ‘thickness density’ field), $\sigma \equiv \partial z / \partial b$. The volume of fluid between two isopycnal surfaces is proportional to $\sigma \Delta A$, where ΔA is an infinitesimal area, and in the absence of diabatic processes this is conserved. Thus, we have $D(\sigma \Delta A) / Dt = 0$ and using $D\Delta A / Dt = \Delta A \nabla_b \cdot \mathbf{u}$ we obtain the equation of motion for thickness [cf. (3.183)]

$$\frac{D\sigma}{Dt} + \sigma \nabla_b \cdot \mathbf{u} = D_\sigma \quad \text{or} \quad \frac{\partial \sigma}{\partial t} + \nabla_b \cdot (\mathbf{u} \sigma) = D_\sigma, \quad (13.155)$$

now including a term D_σ to represent any diabatic terms. From (13.155) we obtain, after a little algebra, the variance equation

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{\sigma'^2} + \overline{\mathbf{u}' \sigma'} \cdot \nabla_b \bar{\sigma} + \frac{1}{2} \bar{\mathbf{u}} \cdot \nabla_b \overline{\sigma'^2} + \frac{1}{2} \overline{\mathbf{u}' \cdot \nabla_b \sigma'^2} = -\overline{w' \sigma'} + \overline{D'_\sigma \sigma'}, \quad (13.156)$$

where we have written $w' \equiv (\sigma \nabla_b \cdot \mathbf{u})'$. This equation is to be compared with the corresponding equation for a conserved tracer, (13.67). If the mean flow is small and the third-order correlations may be neglected then (13.156) becomes, in a statistically steady state,

$$\overline{\mathbf{u}' \sigma'} \cdot \nabla_b \bar{\sigma} \approx -\overline{w' \sigma'} + \overline{D'_\sigma \sigma'}. \quad (13.157)$$

Unlike the case for a materially conserved tracer [cf. (13.68)], the transport of thickness is not necessarily downgradient in regions of dissipation. However, in regions of baroclinic instability where there may be conversion of available potential energy to kinetic energy $\overline{w' \sigma'}$ is positive and we do then expect thickness to be transferred downgradient, suggesting a diffusive parameterization.

The eddy-induced and residual velocities

Now, let us decompose these variables in the usual manner into a mean component (say an ensemble mean, denoted with an overbar), and an eddy component, denoted with a prime. Then the averaged thickness equation is

$$\frac{\partial \bar{\sigma}}{\partial t} + \nabla_b \cdot (\bar{\sigma} \bar{\mathbf{u}} + \overline{\sigma' \mathbf{u}'}) = 0, \quad (13.158)$$

where $\overline{\sigma' \mathbf{u}'}$ is the eddy thickness flux. This equation may be written as

$$\frac{\partial \bar{\sigma}}{\partial t} + \nabla_b \cdot [(\bar{\mathbf{u}} + \tilde{\mathbf{u}}) \bar{\sigma}] = 0, \quad (13.159)$$

where

$$\tilde{\mathbf{u}} \equiv \frac{\overline{\sigma' \mathbf{u}'}}{\bar{\sigma}} \quad (13.160)$$

is the ‘eddy-induced velocity’, sometimes referred to as the ‘bolus velocity’, so-called because the thickness flux is said to be evocative of a peristaltic transfer along a passage bounded by impermeable but elastic walls. The quantity

$$\bar{\mathbf{u}}^* = \bar{\mathbf{u}} + \tilde{\mathbf{u}} \quad (13.161)$$

is the residual velocity we encountered in chapter 10, and it accounts for the total transport of thickness, including both eddy and Eulerian means.

In adiabatic flow, the evolution of a materially conserved tracer τ is given by

$$\frac{D}{Dt}(\tau \sigma \Delta A) = 0, \quad (13.162)$$

whence, because $\sigma \Delta A$ is a constant,

$$\frac{\partial \tau}{\partial t} + \mathbf{u} \cdot \nabla_b \tau = 0. \quad (13.163)$$

Combining this with the thickness equation, (13.155) with $D_\sigma = 0$, gives

$$\frac{\partial}{\partial t}(\sigma \tau) + \nabla_b \cdot (\sigma \mathbf{u} \tau) = 0, \quad (13.164)$$

which in turn leads to

$$\frac{\partial}{\partial t}(\bar{\sigma} \bar{\tau} + \overline{\sigma' \tau'}) + \nabla_b \cdot (\bar{\sigma} \bar{\mathbf{u}} \bar{\tau}) + \nabla_b \cdot \overline{\sigma' \mathbf{u}' \tau} + \nabla_b \cdot [(\overline{\sigma \mathbf{u}})' \tau'] = 0, \quad (13.165)$$

or, using (13.158),

$$\frac{\partial \bar{\tau}}{\partial t} + \frac{1}{\bar{\sigma}} \frac{\partial}{\partial t}(\overline{\sigma' \tau'}) + \left[\bar{\mathbf{u}} + \frac{\overline{\sigma' \mathbf{u}'}}{\bar{\sigma}} \right] \cdot \nabla_b \bar{\tau} = -\frac{1}{\bar{\sigma}} \nabla_b \cdot [(\overline{\sigma \mathbf{u}})' \tau']. \quad (13.166)$$

(To derive these, first let $\overline{\sigma \tau \mathbf{u}} = \overline{(\sigma \mathbf{u})' \tau'} + \overline{\sigma \mathbf{u}} \bar{\tau}$.) If we neglect the correlation between σ' and τ' , then (13.166) has the form

$$\frac{\partial \bar{\tau}}{\partial t} + (\bar{\mathbf{u}} + \tilde{\mathbf{u}}) \cdot \nabla_b \bar{\tau} = -\frac{1}{\bar{\sigma}} \nabla_b \cdot [(\overline{\sigma \mathbf{u}})' \tau']. \quad (13.167)$$

Thus, the averaged tracer evolves as if it were advected by two velocity fields: the large-scale field itself, $\bar{\mathbf{u}}$, and the eddy-induced velocity $\tilde{\mathbf{u}}$, their sum being the residual velocity. The term on the right-hand side of (13.167) is the divergence of the transport of the tracer along the isopycnals by the eddy transport $(\sigma \mathbf{u})'$.

13.8.2 Diffusive thickness transport

A downgradient diffusion of thickness parameterizes the eddy transport velocity by

$$\tilde{\mathbf{u}} \equiv \frac{\overline{\sigma' \mathbf{u}'}}{\bar{\sigma}} = -\frac{1}{\bar{\sigma}} \kappa \nabla_b \bar{\sigma}, \quad (13.168)$$

where κ is an eddy diffusivity. Similarly, we might parameterize the right-hand side of (13.166) by

$$-\frac{1}{\bar{\sigma}} \nabla_b \cdot [(\overline{\sigma \mathbf{u}})' \tau'] = \frac{1}{\bar{\sigma}} \nabla_b \cdot (\kappa \bar{\sigma} \nabla_b \bar{\tau}), \quad (13.169)$$

that is, as a diffusion of the tracer along isopycnals.

In height coordinates the eddy transport velocity will be a three-dimensional field, obtained by appropriately transforming $\tilde{\mathbf{u}}$. We have

$$\tilde{\mathbf{u}} \approx -\frac{1}{\bar{\sigma}} \kappa \nabla_b \bar{\sigma} = -\kappa \frac{\partial b}{\partial z} \nabla_b \left(\frac{\partial \bar{z}}{\partial b} \right) = -\kappa \frac{\partial b}{\partial z} \frac{\partial \bar{s}}{\partial b} = -\kappa \frac{\partial \bar{s}}{\partial z}, \quad (13.170)$$

where the third equality uses $\nabla_b z = \mathbf{s}$, the isopycnal slope. The final result is not quite the same as (13.145), because the diffusivity is now outside the z -derivative. It is a subtle but important distinction, because it means that if κ varies the vertical velocity can no longer be obtained easily as a local function. That is to say, given (13.170), we no longer have $\tilde{w} = \nabla_z \cdot \kappa \bar{s}$ as in (13.145). Rather, $\tilde{w}$ must be evaluated by a non-local integration of the mass conservation requirement, so that

$$\tilde{w} = \int \nabla_z \cdot \left(\kappa \frac{\partial \bar{s}}{\partial z} \right) dz. \quad (13.171)$$

This should not be disconcerting from a physical standpoint, because the baroclinic activity of eddies certainly involves vertical communication — recall the tendency toward barotropic flow in baroclinic lifecycles. From a computational standpoint, it is a little less convenient. A less satisfactory feature of thickness diffusion arises when the ocean floor is not flat; a strict thickness diffusion might then increase the slope of the isopycnals and increase the available potential energy, which would be an unwanted effect. Nevertheless, overall the GM scheme is evidently similar to a thickness diffusion.

13.9 † EDDY TRANSPORT AND THE TRANSFORMED EULERIAN MEAN

A natural framework to discuss how eddy fluxes interact with the mean flow is the transformed Eulerian mean (TEM), discussed in chapter 10. It provides a simplification of the formalism (at least if one is already familiar with TEM theory!) and a means to consider more general cases, such as the flux of potential vorticity and other variables.

From section ?? the zonally averaged thermodynamic equation may be written as

$$\frac{\partial \bar{b}}{\partial t} + (\bar{\mathbf{u}} + \tilde{\mathbf{u}}) \cdot \nabla \bar{b} + (\bar{w} + \tilde{w}) \frac{\partial \bar{b}}{\partial z} = \frac{\partial G}{\partial z}, \quad (13.172)$$

where

$$G = \frac{\overline{\mathbf{v}' b'} \cdot \nabla \bar{b}}{\partial_z \bar{b}} = \frac{1}{\partial_z \bar{b}} \left(\overline{\mathbf{u}' b'} \cdot \nabla_z \bar{b} + \overline{w' b'} \frac{\partial \bar{b}}{\partial z} \right), \quad (13.173)$$

and

$$\tilde{\mathbf{u}} = -\frac{\partial \tilde{\phi}}{\partial z}, \quad \tilde{w} = \nabla \cdot \tilde{\phi}, \quad \tilde{\phi} = \frac{F[b]}{\partial_z \bar{b}} = \frac{\overline{\mathbf{u}' b'}}{\partial_z \bar{b}}. \quad (13.174)$$

If the eddy fluxes are parallel to the isopycnals, then $G = 0$ and the sole effect of the eddies on the mean flow is via the eddy-induced velocity. Using the results of section 10.3.3 to connect buoyancy transport to thickness transport, and then diffusing thickness, we can express these velocities as

$$\tilde{\mathbf{u}} = -\frac{\partial}{\partial z} \left(\frac{\overline{\mathbf{u}'b'}}{\partial_z \bar{b}} \right) \approx \left(\frac{\overline{\mathbf{u}'\sigma'}}{\bar{\sigma}} \right) \approx -\frac{1}{\bar{\sigma}} \kappa \nabla_b \bar{\sigma} \approx -\kappa_a \frac{\partial \mathbf{s}}{\partial z}, \quad (13.175)$$

using (13.170), so recovering, for constant κ , the GM scheme. Note that if we write the residual velocity in standard TEM fashion as $\bar{\mathbf{v}}^* = \bar{\mathbf{v}} + \nabla \times \psi$ then with the GM parameterization the TEM streamfunction may be written as

$$\psi = -\kappa \frac{\mathbf{k} \times \nabla_z \bar{b}}{\partial_z \bar{b}}, \quad (13.176)$$

which reprises (13.146).

13.9.1 Potential vorticity diffusion

Preliminaries

From a more fundamental perspective, potential vorticity, Q , is a better candidate for diffusion than thickness because it is a materially conserved quantity.¹⁶ It is not the only variable that is materially conserved — potential temperature is also. However, potential temperature is advected by the *three-dimensional* velocity field, and the vertical advection complicates matters, since any diffusion tensor certainly cannot be isotropic and is probably not symmetric. On the other hand, in isentropic coordinates the adiabatic potential vorticity advection occurs in the isentropic plane (and in quasi-geostrophic flow the advection is purely horizontal). Thus, only the two-dimensional diffusion need be considered, and the diffusion tensor will be much simplified. Indeed, if the effects of differential rotation are unimportant, the turbulence may be nearly isotropic and the diffusivity becomes a simple scalar.

However, near the upper and lower boundaries buoyancy may in fact be the appropriate field to diffuse. This is because (for flat boundaries) $w = 0$ and buoyancy is conserved on parcels when advected by the horizontal flow:

$$\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla_z b = 0. \quad (13.177)$$

Potential vorticity diffusion, on the other hand, may be difficult at boundaries where outcropping occurs: the thickness between two isentropic surfaces goes to zero and $(\zeta + f)/\sigma$ is ill-defined. This has an analogue in quasi-geostrophic dynamics, where the equations of motion are horizontal advection of potential vorticity in the fluid interior with horizontal buoyancy advection at the boundaries providing the boundary conditions. Horizontal diffusion of buoyancy is, of course, not an adiabatic parameterization, but diabatic effects do occur at the surface. These considerations suggest that downgradient potential vorticity diffusion on isentropic surfaces in the fluid interior, combined with downgradient buoyancy diffusion at the upper and lower boundaries, is as rational a parameterization of eddy transfer effects as any simple diffusion scheme can be.

Implementations and approximations

When the equations of motion are written in conventional form the potential vorticity flux does not directly appear, and the equations must be manipulated in order to implement a potential vorticity scheme. Suppose that the large-scale fields obey planetary-geostrophic scaling, and for simplicity we remain on the β -plane. Then, in isentropic coordinates the potential vorticity is

$$Q = \frac{f + \zeta}{\sigma} \approx \frac{f}{\sigma}, \quad (13.178)$$

which gives

$$\nabla_b \bar{Q} = \frac{1}{\bar{\sigma}} \nabla_b f - \frac{f}{\bar{\sigma}^2} \nabla_b \bar{\sigma} = \frac{\beta}{\bar{\sigma}} \mathbf{j} - \frac{f}{\bar{\sigma}^2} \nabla_b \bar{\sigma}. \quad (13.179)$$

Potential vorticity diffusion is then just

$$\overline{\mathbf{u}'Q'} = -K \nabla_b \bar{Q} = -K \left(\frac{\beta}{\bar{\sigma}} \mathbf{j} - \frac{f}{\bar{\sigma}^2} \nabla_b \bar{\sigma} \right), \quad (13.180)$$

where K is the diffusivity (a two-dimensional symmetric tensor in the isentropic plane, or a scalar). If the eddy flux of potential vorticity also satisfies planetary-geostrophic scaling then

$$\overline{\mathbf{u}'Q'} \approx -\frac{f}{\bar{\sigma}^2} \overline{\mathbf{u}'\sigma'}. \quad (13.181)$$

Using (13.160) and (13.181) gives

$$\tilde{\mathbf{u}} = \frac{1}{\bar{\sigma}} \overline{\mathbf{u}'\sigma'} \approx -\frac{\bar{\sigma}}{f} \overline{\mathbf{u}'Q'}, \quad (13.182)$$

and using (13.180) yields

$$\tilde{\mathbf{u}} = K \left(\frac{\beta}{f} \mathbf{j} - \frac{1}{\bar{\sigma}} \nabla_b \bar{\sigma} \right). \quad (13.183)$$

This differs from (13.168) mainly in the existence of the term involving β on the right-hand side. The expression is singular at the equator, a consequence of ignoring the relative vorticity term in the expression for potential vorticity. For use in z -coordinates, (13.183) can be transformed in the same way as (13.170), giving

$$\tilde{\mathbf{u}} \approx K \left(\frac{\beta}{f} \mathbf{j} - \frac{\partial \bar{s}}{\partial z} \right). \quad (13.184)$$

Other approaches involving the diffusion of potential vorticity may be possible; for example, one could in principle choose the transformed Eulerian mean to be taken with respect to potential vorticity rather than potential temperature. One might also write the entire system of equations in TEM form, whence potential vorticity fluxes appear directly in the momentum equation.

Notes

- 1 Much of the theory of turbulent diffusion stems from G. I. (Geoffrey Ingram) Taylor (1886–1975), who made important contributions to both fluid and solid mechanics, in the former to meteorology, oceanography, and aerodynamics. In addition to his work in turbulence, Taylor is known for his work on the theory of rotating fluids (the ‘Taylor–Proudman’ effect, for example) and on hydrodynamic stability (analysis of stability of Couette flow, for example), and for his clear and simple laboratory experiments. The main results of this section were first derived by Taylor (1921a). Ludwig Prandtl (1875–1953) was the other great pioneer of turbulent diffusion; he is also famous for his work in boundary-layer theory and aerodynamics.
- 2 A number of textbooks in both fluid dynamics and stochastic processes give more detail on this topic. Gardiner (1985) is one.
- 3 Batchelor & Townsend (1956).
- 4 The way variables are changed between (13.31) and (13.32) could also have been used to derive (13.20), but in that case a simpler transformation sufficed.
- 5 Batchelor & Townsend (1956).

- 6 This topic was first addressed empirically by Richardson (1926), although it was Obukhov (1941) who first theoretically obtained the '4/3 power law' describing how the eddy diffusivity varies with separation for parcels in the inertial range. Our treatment takes advantage of Kolmogorov scaling.
- 7 This figure and Fig. 13.4 were kindly provided by Joe LaCasce; see LaCasce & Ohlmann (2003).
- 8 In the open ocean, Ollitrault *et al.* (2005) do find float separation that increases with t^3 , consistent with a $-5/3$ inverse cascade range.
- 9 Morel & Larcheveque (1974) and Er-El & Peskin (1981). Earlier dispersion calculations were made by Richardson (1926) who measured smoke spreading from chimneys, finding results that are consistent with a three-dimensional energy inertial range at small scales.
- 10 Turbulent diffusion is both widely used and widely criticized. If there is a scale-separation between a well-defined mean flow and the eddies then turbulent diffusion can be a very useful parameterization. However, this condition is often *not* satisfied, because it is unusual in fluid mechanics for the turbulent eddies to be significantly smaller than the mean flow. Baroclinic turbulence is something of an exception because there is a natural scale of the turbulence — the deformation radius — that is in general different from the scale of the mean flow, although even this scale separation may be lost if there is an inverse cascade or if the deformation scale is sufficiently large, as in the Earth's atmosphere. Furthermore, properly choosing what variable is to be diffused, and ensuring that various fluid conservation properties remain respected by the diffusion, remain difficult problems.
- 11 If S_{ij} and A_{ij} are symmetric and antisymmetric tensors respectively, then, summing over repeated indices, their contraction is $A_{ij}S_{ij} = -A_{ji}S_{ij} = -A_{ji}S_{ji} = -A_{ij}S_{ij}$, where the last equality follows because the indices are dummy. Thus, the contraction must equal zero.
- 12 Green (1970) and Stone (1972), in the context of the meridional transport of heat in the Earth's atmosphere, suggested that the magnitude of the turbulent diffusivity coefficients could be obtained by dynamical arguments using such things as baroclinic instability theory and the amount of available potential energy in the atmosphere, although their suggestions differ in such important details as the eddy mixing length. Other efforts have drawn on geostrophic turbulence theory, for example Larichev & Held (1995) and Smith & Vallis (2002).
- 13 Ollitrault *et al.* (2005).
- 14 Green (1970).
- 15 The Gent-McWilliams (GM) scheme originated in Gent & McWilliams (1990) and was clarified by Gent *et al.* (1995). Previously, Plumb (1979) had noted the connection between symmetric and antisymmetric diffusivities and diffusive and advective fluxes, and Griffies (1998) explicitly showed how the GM bolus velocities are related to a skew flux and can be calculated using an antisymmetric diffusivity tensor. See McDougall (1998) and (Griffies 2004) for reviews and more discussion. Visbeck *et al.* (1997) suggested that the values of eddy diffusivities in the GM scheme might be determined by dynamical arguments similar to those of Green (1970) and Stone (1972).
- 16 Potential vorticity diffusion was suggested by Green (1970) as a parameterization for large-scale eddies in the atmosphere, and used in ocean contexts by Welander (1973), Marshall (1981) and Rhines & Young (1982a). Tréguier *et al.* (1997) and Greatbatch (1998) explored the use of potential vorticity based parameterizations for ocean models. Lee *et al.* (1997), Marshall *et al.* (1999), Drijfhout & Hazeleger (2001), and others, have explored numerically whether the eddy transfer of tracers in the ocean is in fact diffusive, and whether potential vorticity or thickness is a better quantity to diffuse. For more about the construction of parameterizations for ocean models see Killworth (1997), Smith & Vallis (2002) and Ferrari & McWilliams (2006).

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

Part IV

LARGE-SCALE OCEANIC CIRCULATION

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